UltraScan III Documentation

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Ultrascan-III is a Analytical Ultracentrifugation *Data acquisition* and *analysis software*.

Note: For more information on AUC and how to install Ultrascan-III visit Ultrascan-III website.

CHAPTER

ONE

GETTING STARTED

Analysis Steps:

- *Overview of Ultrascan-III*
- *UltraScan-III Installation:*

1.1 Overview of Ultrascan-III

UltraScan is a comprehensive data analysis software package for hydrodynamic data from analytical ultracentrifugation experiments. The software features an integrated data editing and analysis environment with a portable graphical user interface. UltraScan provides the highest possible resolution for sedimentation velocity experiments using high-performance computing modules for 2-dimensional spectrum analysis, genetic algorithm, and for Monte Carlo analysis. A MySQL database backend is integrated for data management, and the UltraScan LIMS system provides web-based access to the database and supercomputing functionality. Supercomputing resources are provided through the NSF-funded UltraScan Science Gateway, which currently is supported by the NSF-ACCESS infrastructure, which provides cloud storage and high performance supercomputing resources to the UltraScan project. Special care has been taken in the design of the software to guarantee an optimum in flexibility, user-friendliness, stability and convenience. The software is written in C++.

The software features popular sedimentation velocity and equilibrium analysis methods as well as simulation modules, utilities, and editing modules with support for velocity and equilibrium experiments, single and multichannel centerpieces, absorbance and interference optics. All analysis methods generate ASCII formatted data suitable for import into third-party plotting programs, and contains a built-in postscript printer engine to generate publication - quality graphical representations of analysis data.

The LIMS component of UltraScan, coupled with the high-performance computing backend dramatically improves analysis speed and throughput and allows simultaneous analysis of a virtually unlimited number of experiments. Any institution, academic or for profit can obtain free data storage and supercomputing analysis resources through an NSF TeraGrid allocation grant issued to Borries Demeler. Please contact us for assistance with setting up a secure database and supercomputer access.

1.2 UltraScan-III Installation:

Download the software. UltraScan-III is available for free download in binary or source-code format for three major operating systems: Windows, Macintosh, and Linux.

The UltraScan is multi-platform and can be installed on Windows, Macintosh, and Linux computers. For Linux, several popular distributions are supported. Each operating system has operating system specific instructions:

- Microsoft Windows: Download the installation file and execute it on your computer. When prompted for an installation directory, please choose C:Program FilesUltraScan, this is the expected path. If your international version of Windows has a different path, please create this subdirectory first before installation. On Windows 7, also install in C:Program FilesUltraScan, not in C:Program Files (x86)UltraScan. You may have to run the program as Administrator.
- Mac OS-X: Download and Double-click the installation package. If you have not previously installed XQuartz, you will be requested to do so. XQuartz can be downloaded from https://www.xquartz.org.
- Linux, Unix/X11: To install the tar.xz package for any distribution, please follow these steps:

After downloading the tar.xz package, you need to uncompress and extract it in a location of your choice, for example, /usr/lib or /usr/local or /usr/local/lib. For the purpose of this document, it is assumed you are going to install UltraScan in /usr/lib and /path is the path to your downloaded file.

- cd /usr/lib
- tar Jxf /path/ultrascan-version.tar.xz

At this point, a new directory will be created (you can rename it without issue) Under that directory will be a bin subdirectory. In the bin subdirectory are two scripts:

- us.sh
- us_somo.sh

You can run these to start the UltraScan GUI and the US-SOMO GUI respectively.

Register the software. UltraScan development is funded by NIH and your registration helps us document usage of the software to NIH. Only operating system, and the number of active users for any given year are shared with the NIH. Registrations are free, and valid for one year. After that they have to be renewed (also free).

Triple========== Flowchart for the Analysis of Sedimentation Velocity Data ==========

Analysis Steps:

- *Step 1: Import Experimental data into UltraScan-III OpenAUC format*
- *Step 2: Edit experimental data.*
- *Step 3: Perform a Time Derivative analysis and find the limits of the s-value range.*
- *Step 4: Submit a 2DSA analysis request in USLIMS.*
- *Step 5: Perform Meniscus fit.*

- *Step 6: Perform a final 2DSA refinement*
- *Step 7: Genetic Algorithm analysis - (optional)*
- *Step 8: Perform 2DSA Monte Carlo analysis - (recommended)*
- *Step 9: Perform Monte Carlo GA analysis - (optional)*
- *Step 10: Perform van Holde-Weischet analysis - (recommended)*
- *Step 11: Overlay combined distributions - (recommended)*
- *Step 12: Perform Parametrically Constrained Spectrum Analysis (PCSA) - (optional)*
  - *Reference*

CHAPTER

TWO

STEP 1: IMPORT EXPERIMENTAL DATA INTO ULTRASCAN-III OPENAUC FORMAT

Note: Required for data from XLA or XLI, this step is performed automatically in the Optima AUC. Sedimentation velocity data should not be measured in absorbance mode, use intensity mode instead.

1. Import the experimental data: Utilities: *Import Experimental Data*
2. Confirm Investigator setting and local/database selection
3. Import Experimental Data from local disk
4. Edit Run Information, Select Lab/Rotor/Calibration
5. Enter a Label (verbose description for the run)
6. Select the corresponding project by clicking on the Project button. If there is no project, create a new project.
7. Confirm experiment type and optical detection system
8. Enter any comments, if applicable
9. Select instrument, rotor, rotor calibration and operator
10. **Click on Accept.**
    - Edit the Description field if necessary
11. Navigate to the first channel and select the centerpiece type.
12. **Select the proper solution - Make sure that the solution contains at least one analyte and a buffer**
    - If you have more than one Triple, you can click on Apply to All but verify centerpiece and solution for each Triple first. Also, check the Description field again to make sure the appropriate information is saved.
13. If data were collected in intensity mode, you will need to Define Reference Scans by selecting a short region from the air-to-air interface portion of the data.
14. For equilibrium data from 6-channel centerpieces you should separate each channel with the Define Subsets/Process Subsets functions.

15. Failed Triples or empty Triples can be excluded from the run by clicking on Drop Selected Triples.
16. When everything has been set you can Save the scans to database or disk.

CHAPTER

THREE

STEP 2: EDIT EXPERIMENTAL DATA.

1. Edit the data into Edit Data: *Edit Data*.
2. Load data from the database or a local data directory that contains the UltraScan 3 data files previously converted from the Beckman raw data.
3. Select the Cell / Channel / Wavelength Triple to be edited.
4. Specify the meniscus of the data by holding down the Control key and using the left mouse button. The meniscus value may be manually adjusted with the keyboard.
5. If the data were collected with the interference detector, specify the left and right edges of the air gap area of the data.
6. Specify the left and right edges of the data to be analyzed.

> **Note:** Please note: Do not pick the left data edge too close to the meniscus. During meniscus fitting, the evaluated meniscus positions may reach inside of the data range and violate the boundary conditions of the finite element solution. This will cause the meniscus fit to fail.

7. Specify the location of the scan plateau. This is the radial position where most scans have a stable plateau, but the selected position should not reach into the back-diffusion region. The most appropriate point tends to be close to the right edge of the data range, but not so far to the right that it extends into the region where the concentration of the later scans curves upward at the bottom of the cell due to back-diffusion.
8. Make any other optional adjustments to the data that are necessary and save the edit profile. When saving, a pop-up message is presented asking for an edit ID. The default for this ID is the current date and time in the form of YYMMddhhmm (Year / Month / Day / Hour / Minute), but this default can be supplemented with a suffix of your own choice.

CHAPTER

FOUR

STEP 3: PERFORM A TIME DERIVATIVE ANALYSIS AND FIND THE LIMITS OF THE S-VALUE RANGE.

1. Select Velocity: *Time Derivative* and Load Experiment
2. Set Data Smoothing to ~ 10
3. Set the Boundary Pos (%) to zero
4. Exclude scans that do not exhibit a stable upper plateau.
5. Select Average S to plot the dc/dt S-value distribution (default setting).
6. Adjust S-value cutoff as needed.
7. The correct S limits to choose are the left and right limits of the S-value distribution where all signal returns to baseline. The minimum S-value allowed is 0.2 S.

CHAPTER

FIVE

STEP 4: SUBMIT A 2DSA ANALYSIS REQUEST IN USLIMS.

1. Log in and select the run and triples in USLIMS.
2. Locate the dataset and set the s-value limits to values obtained in step 3, and set f/f0 limits to 1-4 or adjust the upper limit based on prior knowledge of the sample (in case of DNA for example).
3. Set the resolution for S and f/f0 to the desired value. 64 is the default for both S and f/f0, the resolution is the number of points into which this variable will be discretized. For example, if the range is selected from 1-5, and the resolution is set to 40, there will be 10 grid points/S-value, resulting in an increment of 0.1 S.
4. Set the grid resolution. The default resolution of 64 will be sufficient for most situations. When you have a very polydisperse or heterogeneous sample you may have to go higher. Larger values will increase compute time. A good strategy is to check if there are noticeable jumps in the fit. If there are, increase the resolution. If you are trying to fit a bi- or tri-modal system, you may be able to get a higher resolution by using the custom grid method, and use the appropriate resolution for each grid.
5. Turn on time invariant noise and leave all other settings at default values.
6. Submit the job to your desired cluster or calculate locally using the 2DSA GUI method
7. Check queue viewer for job completion
8. Confirm results with Velocity: *FE Model Viewer* and *FE Model Module*, load newly generated time invariant noise file. If random residuals are obtained, proceed with step 5, but if residuals do not look random, and a strong diagonal line in the residual bitmap is apparent, investigate range settings for S and f/f0 settings and repeat 2DSA with improved ranges.

Note: Do not set the lower s-value limit too low, as this could create artificially low-s species signal if a baseline or slowly changing baseline exists. These artifacts are better handled in the time invariant noise.

CHAPTER

SIX

STEP 5: PERFORM MENISCUS FIT.

1. Repeat submission by also fitting the meniscus over 0.03 cm with 10 points, turning on both time invariant and radially invariant noise. Make sure to load the previously generated time invariant noise when loading the dataset. Use the same range settings as in Step 4.
2. Check queue or e-mail and once results are available, load Utilities: *Fit Meniscus*. Click on Scan Database if using the database, and check the Status line for new results. Load the desired meniscus fit (check the UltraScan manual for details).
3. After updating the meniscus, confirm the deletion of the scans that resulted in non-optimal RMSD values.

CHAPTER

SEVEN

STEP 6: PERFORM A FINAL 2DSA REFINEMENT

1. **Refine the analysis by re-fitting the data with the improved meniscus value by repeating the analysis with identical range settings as before fitted in Step 5, except this time selecting the time- and radially invariant noise that was generated in Step 5, and also do not select meniscus fitting.**
    - Instead, select Iterative Refinement and set the refinement level to 10 iterations. Also refit time and radially invariant noise.
2. Visualize the final results in Velocity: *FE Model Viewer* and save results to database.
3. All subsequent analyses methods should now be based on the model generated in this final 2DSA refinement step.

At this point, multiple analysis options exist depending on the properties of the analyte distribution. If a polydisperse solution is obtained, parsimonious regularization with the genetic algorithm method is appropriate. Otherwise, the data should be analyzed only by the 2DSA analysis in conjunction with a 50-iteration Monte Carlo analysis. Both options are explained below.

CHAPTER

EIGHT

STEP 7: GENETIC ALGORITHM ANALYSIS - (OPTIONAL)

1. If the refined 2DSA data are appropriate for genetic algorithm analysis, select Velocity: *Initialize Genetic Algorithm* and load the model from Step 6 into the initialization program. See the corresponding UltraScan Manual section for details.
2. Assign initialization and save to disk.
3. Log into USLIMS and submit data to Genetic Algorithm analysis, but do not refit time or radially invariant noise. Make sure to select time and radially invariant noise generated in Step 6, which is set to occur by default unless overridden by the user.
4. Select the gadistro file from the UltraScan/results/run-id directory for the correct Triple.
5. Visualize results by using the Velocity: *FE Model Viewer*.

CHAPTER

NINE

STEP 8: PERFORM 2DSA MONTE CARLO ANALYSIS - (RECOMMENDED)

1. Use the same limits as before, but do not refit time or radially invariant noise, instead, load the noise corrections from Step 6 (which will be done by default in the LIMS and GUI).
2. Select 50 Monte Carlo iterations.
3. If the 2DSA distribution appears to be a sparse solute situation, and not a smooth continuous distribution of many species, you can further refine the data with a parsimonious regularization using the GA analysis.
4. When using 2DSA Monte Carlo distributions for the GA initialization, make sure to use the manual GA initialization method in Velocity: *Initialize Genetic Algorithm <ga_initialize>*.
5. Proceed as described in Step 7.

CHAPTER

TEN

STEP 9: PERFORM MONTE CARLO GA ANALYSIS - (OPTIONAL)

1. Using the results from Step 7, initialize the genetic algorithm - Monte Carlo analysis as described in Step 7.
2. In USLIMS, load the data, using the noise files generated in Step 6 (will be done by default if settings are not changed).
3. Select a multiple of 8 Monte Carlo iterations (48, 56 or 64 are good choices)
4. Select parallel processing with 8 program groups.
5. Repeat Step 7 by loading the gadistro file generated from the GA distribution model.
6. Submit to desired cluster and visualize results by using the Velocity: *FE Model Viewer*.

CHAPTER

ELEVEN

STEP 10: PERFORM VAN HOLDE-WEISCHET ANALYSIS - (RECOMMENDED)

1. Open Velocity: *Enhanced van Holde-Weischet*.
2. Load the desired experiment, applying the noise files from Step 6 (the latest model).
3. Check Plateaus from 2DSA and Use Enhanced vHW.
4. Adjust Beck Diffusion Tolerance, Divisions, Data Smoothing, % of Boundary, and Boundary Position to desired values.
5. If appropriate, delete early scans to improve resolution and reduce noise. Only keep scans and boundary portions that contribute to well correlated line fits in the linear extrapolations.
6. Select groups, if appropriate, to generate weight averaged s-values for discrete species.
7. Display *Distribution Plot and histogram*.
8. Save Data and distributions.

Note: Refer to the *van Holde-Weischet manual* page for additional details.

CHAPTER

TWELVE

STEP 11: OVERLAY COMBINED DISTRIBUTIONS - (RECOMMENDED)

1. All van Holde-Weischet distributions and finite element models can be combined into a single plot for easy comparison.
2. Use Velocity: *Combine Distribution Plots* (vHW) for van Holde-Weischet plots.
3. Use Velocity: *Combine Discrete Distributions* for all finite element models (2DSA, GA, Monte Carlo).

CHAPTER

THIRTEEN

STEP 12: PERFORM PARAMETRICALLY CONSTRAINED SPECTRUM ANALYSIS (PCSA) - (OPTIONAL)

1. Inspect the results from the 2DSA-Monte Carlo analysis to determine an appropriate parameterization constraint for the PCSA analysis.
2. **Open Velocity:** ***Parametrically Constrained Spectrum Analysis*****.**
    - Select a parametrization (straight line, increasing sigmoid, decreasing sigmoid, power law or horizontal line (equivalent to C(s), not recommended) that is consistent with the pattern displayed in the pseudo-3D plot of the 2DSA-Monte Carlo analysis, i.e., a line shape that best describes the distribution of most solute points. You can select the PCSA either from the desktop GUI (Velocity:PCSA) or from the corresponding LIMS option after loading the data.
3. Decide on a parameter combination to be fitted and set appropriate limits. If fitting s and f/f0, use the same limits as were used in the 2DSA.
4. **Optionally, regularize PCSA result either by Monte Carlo or Tikhonov regularization.**
    - Tikhonov regularization can be performed with a specified regularization level, or by asking the program to auto-compute the most appropriate regularization level based on the L-curve criterion, see:

13.1 Reference

2DSA

Brookes E, Cao W, Demeler B. A two-dimensional spectrum analysis for sedimentation velocity experiments of mixtures with heterogeneity in molecular weight and shape. Eur Biophys J. (2010) 39(3):405-14.

Kim H, Brookes E, Cao W, Demeler B. Two-dimensional grid optimization for sedimentation velocity analysis in the analytical ultracentrifuge. Eur Biophys J. 2018 Oct;47(7):837-844. doi: 10.1007/s00249-018-1309-z. PMID: 29777290

Experimental Design

Demeler, B. (2024). Methods for the design and analysis of analytical ultracentrifugation experiments. Current Protocols, 4, e974. doi: 10.1002/cpz1.974

PCSA

Gorbet G., T. Devlin, B. Hernandez Uribe, A. K. Demeler, Z. Lindsey, S. Ganji, S. Breton, L. Weise-Cross, E.M. Lafer, E.H. Brookes, B. Demeler. A parametrically constrained optimization method for fitting sedimentation velocity experiments. Biophys. J. (2014) vol 106, 1741-50.

Genetic Algorithm

Brookes E, Demeler B. Parsimonious Regularization using Genetic Algorithms Applied to the Analysis of Analytical Ultracentrifugation Experiments. GECCO '07: Proceedings of the 9th annual conference on Genetic and evolutionary computation, London, July 7-11, 2007, 361–368, https://doi.org/10.1145/1276958.1277035, ACM 978-1-59593-697-4/07/0007

Monte Carlo

Demeler B and E. Brookes. Monte Carlo analysis of sedimentation experiments. Colloid Polym Sci (2008) 286(2) 129-137

Adaptive Space-Time Finite Element Solution

Cao, W and Demeler B. Modeling Analytical Ultracentrifugation Experiments with an Adaptive Space-Time Finite Element Solution for Multi-Component Reacting Systems. Biophys. J. (2008) 95(1):54-65

CHAPTER

FOURTEEN

FLOWCHART FOR THE ANALYSIS OF MULTI-WAVELENGTH SEDIMENTATION VELOCITY DATA

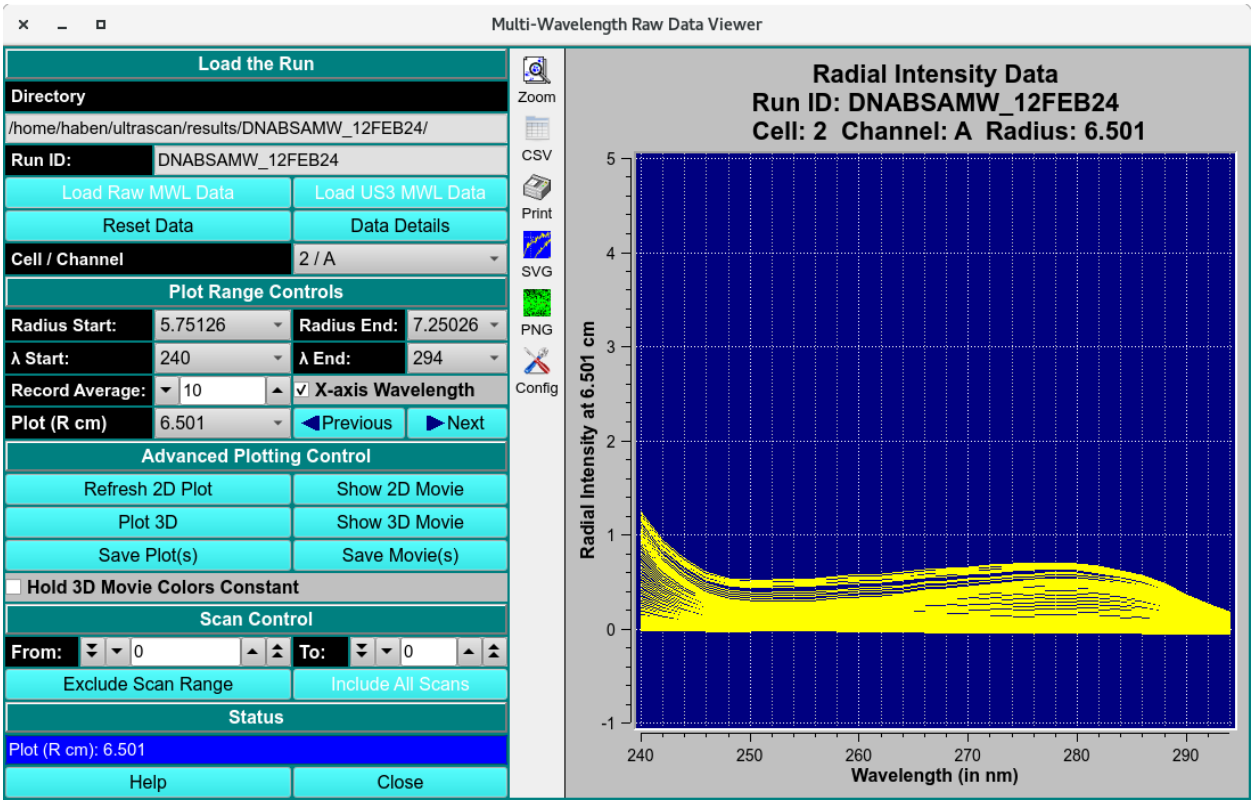
Analysis Steps:

- *Step 1: Import Experimental data into UltraScan-III OpenAUC format*
- *Step 2: Edit experimental data*
- *Step 3: Inspect and plot the raw data*
- *Step 4: Perform a 2DSA refinement following the* 2DSA Flowchart
- *Step 5: Simulated a time-grid synchronized model based on 2DSA-IT MWL models*
- *Step 6: Create basis spectra for each species*
- *Step 7: Import and Edit the **ISSF-** data file*
- *Step 8: Deconvolute MW data using species basis spectra*
- *Step 9: Import and Edit the SSF-ISSF- data files*
- *Step 10: Perform a 2DSA with refinement only*
- *Step 11: Perform van Holde-Weischet analysis - (recommended)*
- *Step 12: Overlay combined distributions - (recommended)*
  - *Reference*

14.1 Step 1: Import Experimental data into UltraScan-III OpenAUC format

Note: Required for data from XLA or XLI, this step is performed automatically in the Optima AUC. Sedimentation velocity data should not be measured in absorbance mode, use intensity mode instead.

1. Import the experimental data: Utilities: *Import Experimental Data*
2. Confirm Investigator setting and local/database selection
3. Import Experimental Data from local disk
4. Edit Run Information, Select Lab/Rotor/Calibration
5. Enter a Label (verbose description for the run)
6. Select the corresponding project by clicking on the Project button. If there is no project, create a new project.
7. Confirm experiment type and optical detection system
8. Enter any comments, if applicable
9. Select instrument, rotor, rotor calibration and operator
10. **Click on Accept.**
    - Edit the Description field if necessary
11. Navigate to the first channel and select the centerpiece type.
12. **Select the proper solution - Make sure that the solution contains at least one analyte and a buffer**
    - If you have more than one Triple, you can click on Apply to All but verify centerpiece and solution for each Triple first. Also, check the Description field again to make sure the appropriate information is saved.
13. If data were collected in intensity mode, you will need to Define Reference Scans by selecting a short region from the air-to-air interface portion of the data.
14. For equilibrium data from 6-channel centerpieces you should separate each channel with the Define Subsets/Process Subsets functions.
15. Failed Triples or empty Triples can be excluded from the run by clicking on Drop Selected Triples.
16. When everything has been set you can Save the scans to database or disk.



Alternate Record and X-axis

14.2 Step 2: Edit experimental data

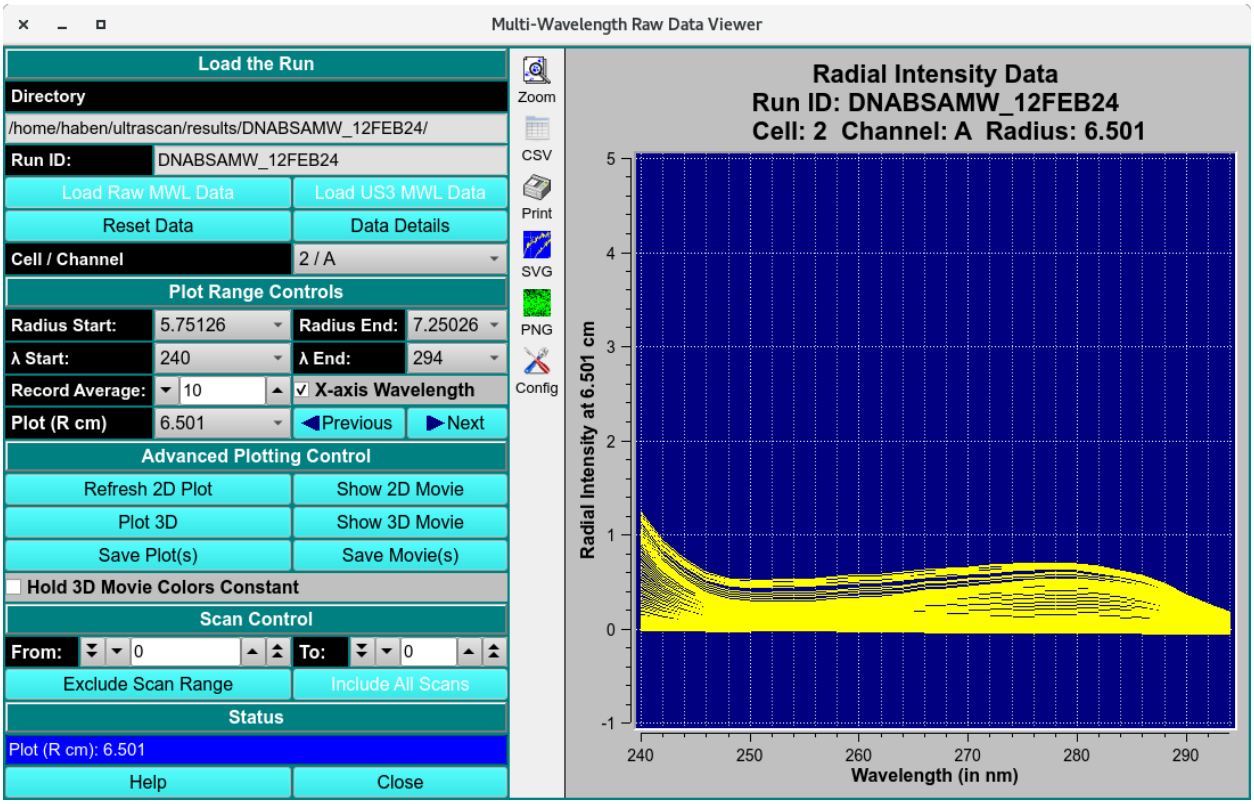
1. Edit the data in the *Edit Data* module.
2. Load data from the database or a local data directory containing the UltraScan-III openAUC data files.data.
3. Select the Cell / Channel / Wavelength Triple to be edited.
4. Specify the meniscus of the data by holding down the Control key and using the left mouse button. The meniscus value may be manually adjusted with the keyboard.
5. Specify the left and right edges of the data to be analyzed.

> **Note:** Please note: Do not pick the left data edge too close to the meniscus. During meniscus fitting, the evaluated meniscus positions may reach inside of the data range and violate the boundary conditions of the finite element solution. This will cause the meniscus fit to fail.

6. Specify the location of the scan plateau. This is the radial position where most scans have a stable plateau, but the selected position should not reach into the back-diffusion region. The most appropriate point tends to be close to the right edge of the data range, but not so far to the right that it extends into

the region where the concentration of the later scans curves upward at the bottom of the cell due to back-diffusion.

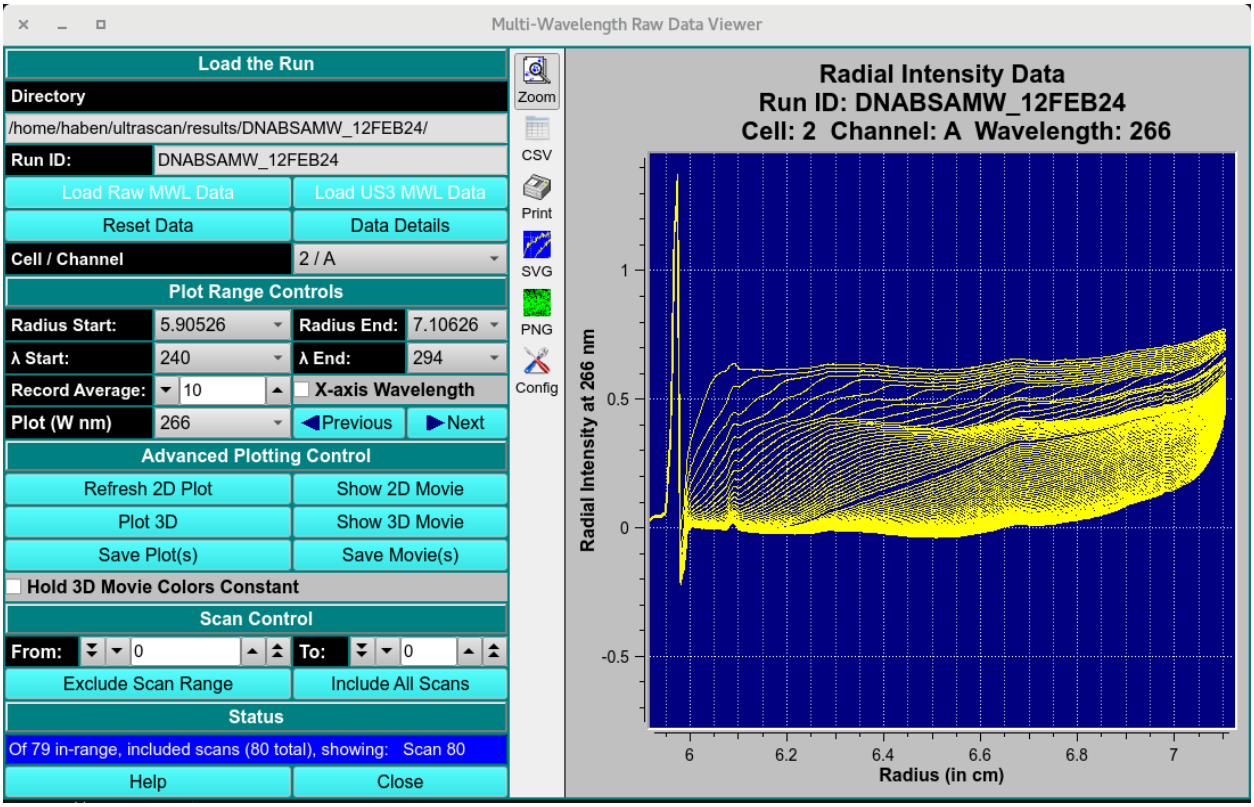
7. Make any other optional adjustments to the data that are necessary and save the edit profile. When saving, a pop-up message is presented asking for an edit ID. The default for this ID is the current date and time in the form of YYMMdd hhmm (Year / Month / Day / Hour / Minute), but this default can be supplemented with a suffix of your own choice.



Alternate Record and X-axis

14.3 Step 3: Inspect and plot the raw data

1. Inspect the raw MWL data using the *Multi-Wavelength Raw Data Viewer* in the Multi-wavelength modules.
2. Load the data from the local disk or the database.
3. Select the Cell / Channel / Wavelength Triple to be inspected.
4. Generate and save the 3-D plot and movie.



Raw MWL Data Inspection and Plotting

14.4 Step 4: Perform a 2DSA refinement following the 2DSA Flowchart

1. Follow Steps 3-6 in the *2DSA Flowchart* to perform 2DSA analysis and fit the associated Time- and Radially- invariant noises, the meniscus and bottom boundary positions and rotor stretching and final iterative refinements.

14.5 Step 5: Simulated a time-grid synchronized model based on 2DSA-IT MWL models

1. Simulate a time synchronized **ISSF_** model based on the MWL 2DSA-IT models using the *MWL Pre-Fit Species Simulation*.
2. Click **PreFilter Models** to define the MWL pre-filter for the simulation and
3. Click **Select Models** and load all related 2DSA-IT models from one MWL dataset channel.
4. **Define Buffer** and open the **:doc:`Buffer Management <buffer/index>`_** dialog to define the experimental parameters of the dataset
5. Define the simulation parameters

6. Click **Define Rotor** and open the **:doc:`Rotor Management <buffer/index>`_** dialog to select the Default 1003 (Simulation) rotor.
7. Start the simulation and save the open-auc file into $HOME/ultrascan/imports folder.

14.6 Step 6: Create basis spectra for each species

1. Generate the extinction coefficient basis spectrum of an analyte in the *Spectrum Fitter* dialog.
2. Using a UV-visible spectrophotometer, scan the serial dilutions of analyte - spectra.
3. load csv files, define the column separator character and click **Apply**.
4. Add the extinction coefficient of the analyte at 260 or 280 nm, and click Perform a global fit if known.
5. A second fitting dialog will appear to perform the fit. Typical users don't have to change any setting. Click **Fit**.
6. Once the fit is complete, the result will automatically populate the main window. On the main window, click **Save**. Give the .dat file a unique name.

14.7 Step 7: Import and Edit the ISSF- data file

1. Repeat steps 1 and 2 for the newly generated **ISSF-** open auc file.

14.8 Step 8: Deconvolute MW data using species basis spectra

1. Deconvolute the analytes in *MWL Species Fit Analysis*.
2. Load the time-synchronized **ISSF-** simulation generated using *Optima MWL Fit Simulations* using **:doc:`Load Run Data Dialog <common_dialogs.html#data-loader>`_**.
3. Click **Load Species Fits** and load the species extinction coefficient basis spectra from local disk.
4. Perform the deconvolution by clicking **Species Fit Data**.
5. Deconvoluted sedimentation velocity data of each species will by generated with a **SSF-ISSF-** pre-fix.

14.9 Step 9: Import and Edit the SSF-ISSF- data files

1. Repeat steps 1 and 2 for the newly generated **SSF-ISSF-** open auc files.

14.10 Step 10: Perform a 2DSA with refinement only

1. Perform 2DSA refinement of the **SSF-ISSF-** datasets.

14.11 Step 11: Perform van Holde-Weischet analysis - (recommended)

1. Open Velocity: *Enhanced van Holde-Weischet*.
2. Load the desired experiment, applying the noise files from Step 6 (the latest model).
3. Check Plateaus from 2DSA and Use Enhanced vHW.
4. Adjust Beck Diffusion Tolerance, Divisions, Data Smoothing, % of Boundary, and Boundary Position to desired values.
5. If appropriate, delete early scans to improve resolution and reduce noise. Only keep scans and boundary portions that contribute to well correlated line fits in the linear extrapolations.
6. Select groups, if appropriate, to generate weight averaged s-values for discrete species.
7. Display *Distribution Plot and histogram*.
8. Save Data and distributions.

Note: Refer to the *van Holde-Weischet manual* page for additional details.

14.12 Step 12: Overlay combined distributions - (recommended)

1. All van Holde-Weischet distributions and finite element models can be combined into a single plot for easy comparison.
2. Use Velocity: *Combine Distribution Plots* (vHW) for van Holde-Weischet plots.
3. Use Velocity: *Combine Discrete Distributions* for all finite element models (2DSA, GA, Monte Carlo).
4. Use Velocity: *Combine Integral Distributions* for all finite element models (2DSA, GA, Monte Carlo).
5. Use Velocity: *Combine pseudo-3D Distributions* for all finite element models (2DSA, GA, Monte Carlo).

14.12.1 Reference

2DSA

Brookes E, Cao W, Demeler B. A two-dimensional spectrum analysis for sedimentation velocity experiments of mixtures with heterogeneity in molecular weight and shape. Eur Biophys J. (2010) 39(3):405-14.

Kim H, Brookes E, Cao W, Demeler B. Two-dimensional grid optimization for sedimentation velocity analysis in the analytical ultracentrifuge. Eur Biophys J. 2018 Oct;47(7):837-844. doi: 10.1007/s00249-018-1309-z. PMID: 29777290

Experimental Design

Demeler, B. (2024). Methods for the design and analysis of analytical ultracentrifugation experiments. Current Protocols, 4, e974. doi: 10.1002/cpz1.974

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Gorbet G., T. Devlin, B. Hernandez Uribe, A. K. Demeler, Z. Lindsey, S. Ganji, S. Breton, L. Weise-Cross, E.M. Lafer, E.H. Brookes, B. Demeler. A parametrically constrained optimization method for fitting sedimentation velocity experiments. Biophys. J. (2014) vol 106, 1741-50.

Genetic Algorithm

Brookes E, Demeler B. Parsimonious Regularization using Genetic Algorithms Applied to the Analysis of Analytical Ultracentrifugation Experiments. GECCO '07: Proceedings of the 9th annual conference on Genetic and evolutionary computation, London, July 7-11, 2007, 361–368, https://doi.org/10.1145/1276958.1277035, ACM 978-1-59593-697-4/07/0007

Monte Carlo

Demeler B and E. Brookes. Monte Carlo analysis of sedimentation experiments. Colloid Polym Sci (2008) 286(2) 129-137

Adaptive Space-Time Finite Element Solution

Cao, W and Demeler B. Modeling Analytical Ultracentrifugation Experiments with an Adaptive Space-Time Finite Element Solution for Multi-Component Reacting Systems. Biophys. J. (2008) 95(1):54-65

Triple========== Reporting SV-Data ==========

Index

- *Ultrascan-III Results from Sedimentation Velocity Data*
  - *1. Assessment of the Sedimentation Velocity Meta-data*
  - *2. Simulation Quality Assessment*
  - *3. Model Assessment*
    - *A. 2-D and 3-D Plots*
    - *B. 2-D and 3-D Assessments*
  - *Summary*

CHAPTER

FIFTEEN

ULTRASCAN-III RESULTS FROM SEDIMENTATION VELOCITY DATA

Sedimentation velocity (SV) experiments in UltraScan-III produce a broad range of results that support data validation, simulation quality control, and model interpretation. The outputs range from basic run metadata and diagnostic residual analyses to visual, statistical, and confidence-based assessments of derived models. This document summarizes the principal result categories that an be obtained from sedimentation velocity data and identifies the modules used to access them.

15.1 1. Assessment of the Sedimentation Velocity Meta-data

Modules responsible:

- *Run Details*
- *Time State Viewer*

The first level of result assessment in sedimentation velocity analysis is the verification and interpretation of experiment metadata. These values describe the conditions under which the experiment was acquired and are essential for judging whether the run is internally consistent and suitable for downstream analysis.

The metadata assessment includes the following items:

Speed	The measured rotor speed used during the experiment. This is one of the most important experimental descriptors because sedimentation behavior depends directly on angular velocity.
Temperature	The recorded temperature during the run. Since solvent density, viscosity, and sedimentation behavior are temperature dependent, this value is critical for accurate physical interpretation.
Run length	The total duration of the experiment. The run length determines whether sufficient sedimentation progression was captured to support robust model fitting.
Time correction	A corrected experimental time axis derived from the acquisition records. This is important when instrument timing and recorded scan timing differ, and it ensures that the analysis uses physically accurate elapsed times.
Run ID	A unique identifier assigned to the experiment. The run ID links the data to a specific acquisition and is essential for traceability and reproducibility.
Directory of raw data	The file system location of the raw experimental scans. This information supports provenance tracking and facilitates review or reprocessing of the original data.
Time interval	The interval between scans. This value is useful for determining temporal resolution and whether the acquisition frequency is appropriate for the sedimentation process being observed.
Time	The individual scan times or cumulative elapsed times used in the run. These values are required for precise reconstruction of the sedimentation process.
Raw speed and set-speed	Both the measured rotor speed and the target rotor speed are reported. Comparing these values helps determine whether the rotor operated as intended and whether speed deviations could influence interpretation.
$\omega^2 t$	The integrated centrifugal field term, often expressed as $\omega^2 t$. This quantity is fundamental in sedimentation analysis because it combines rotor speed and elapsed time into a single physically meaningful parameter.
Steps	The sequence of acquisition or speed steps recorded during the run. These values help identify acceleration phases, speed transitions, or segmented un structures that may affect data interpretation.

Taken together, these metadata outputs provide the foundation for determining whether a sedimentation velocity experiment is complete, physically plausible, and ready for quantitative analysis.

15.2 2. Simulation Quality Assessment

Modules responsible:

- *Finite Element Model Viewer*
- *RMSD Model Query*

Once metadata have been verified, the next level of assessment concerns the quality of the fit between the experimental data and the simulation or model. These outputs quantify how well the selected model reproduces the observed sedimentation process.

The simulation quality assessment includes the following results:

root-mean-square deviation	The root-mean-square deviation (RMSD) is reported together with the edit profile, analysis method, and Triple designation. This allows the user to evaluate fit quality in the exact context in which the model was generated. RMSD is a primary indicator of how closely the simulation matches the experimental scans.
:doc:`Bitmap <fe_match/fe_match_simul: bit-map>`_	A bitmap representation of fit quality or residual structure may be generated to provide a rapid visual diagnostic of agreement between the model and the data. Such displays can help reveal systematic artifacts, scan-specific problems, or localized fitting errors.
:doc:`Time- and radially invariant noise profiles <fe_match/fe_match_simul: data-analysis>`_	Noise decomposition is a central element of SV analysis. UltraScan3 can report both time-invariant noise and radially invariant noise, allowing the user to determine whether systematic baseline distortions are contributing to the apparent signal. These noise profiles are important for judging whether the fit quality is limited by model inadequacy or by non-ideal data structure.
Residuals	Residuals represent the difference between experimental data and simulated values. Inspection of residuals is essential for identifying systematic deviations, concentration-dependent effects, convection artifacts, optical distortions, or incomplete modeling assumptions. Ideally, residuals should be random and centered around zero.

Together, these outputs form the core quality-control layer for determining whether a given fit is acceptable and whether further refinement, editing, or model adjustment is required.

15.3 3. Model Assessment

Model assessment focuses on the interpretation of the fitted sedimentation velocity solution. In UltraScan3, model assessment includes both visual and statistical outputs. These results help the user decide whether a model is physically meaningful, whether multiple species are resolved, and how stable the fitted parameters are across simulations.

15.3.1 A. 2-D and 3-D Plots

The visual assessment tools provide direct graphical insight into the fitted distributions and related spectral or structural properties.

Modules and outputs include:

Combine Distribution Plot (vHw) A visualization of the van Holde-Weischet-style interpretation or related transformed sedimentation boundaries, useful for examining heterogeneity and distributional spread.

Combine Integral Distribution An integral representation of the distribution, emphasizing cumulative contributions across the modeled sedimentation range.

Combine Discrete Distribution A discrete-species representation in which resolved components are shown as separate populations with distinct parameter values.

Combine Pseudo-3D Distribution A pseudo-distribution view that provides an approximate or smoothed representation of model content for easier interpretation.

Multi-Wavelength S Spectra Viewer This module supports inspection of multiwavelength-related results in a model context. It can help connect hydrodynamic outcomes to wavelength- dependent behavior, especially in experiments involving spectrally resolved data.

3-D plot Three-dimensional plots provide a richer visualization of model structure, allowing distributions or signal relationships to be viewed across multiple axes. These plots are especially useful for complex systems, multi-parameter interpretations, and multiwavelength data exploration.

Visual outputs are often the first means by which a user identifies sample heterogeneity, discrete species, broad distributions, or model artifacts from improper grid ranges.

15.3.2 B. 2-D and 3-D Assessments

Modules responsible:

- *Finite Element Model Statistics and Metrics*
- *Initialize Genetic Algorithm*

Beyond visual inspection, UltraScan3 provides higher-order statistical assessment of finite element and genetic algorithm-derived models. These tools support quantitative evaluation of model robustness and parameter certainty.

Finite Element Model Statistics and Metrics This module provides summary statistics for finite element-based models. The results support evaluation of model quality across parameters and may include descriptors of central tendency, spread, and agreement among simulated solutions.

Initialize Genetic Algorithm Genetic algorithm analysis provides population-based model exploration and permits statistical characterization of fitted attributes across many solutions. This is particularly useful for determining whether a derived solution is stable, unique, and supported by repeated simulations.

The principal results from these assessments include:

95% confidence interval of every simulated attribute For each simulated parameter or attribute, UltraScan-III can estimate the range within which the true value is expected to lie with 95% confidence. These intervals are critical for distinguishing well-resolved parameters from poorly constrained ones.

Standard deviation of every simulated attribute The standard deviation provides a direct measure of variability among the simulated solutions. Small standard deviations indicate stable and reproducible parameter estimates, while large standard deviations suggest uncertainty, parameter coupling, or insufficient information in the data.

These 2-D and 3-D statistical assessments are especially important when comparing alternative models, evaluating fit robustness, and reporting quantitative confidence in the derived sedimentation results.

15.4 Summary

Sedimentation velocity data in UltraScan3 can yield results in three broad categories:

1. Metadata assessment, which verifies the integrity and physical context of the experiment.
2. Simulation quality assessment, which measures how well the model reproduces the experimental data.
3. Model assessment, which provides graphical and statistical interpretation of the derived solution.

Together, these outputs allow the user to move from simple run verification to deep physical interpretation of macromolecular behavior in solution.

CHAPTER

SIXTEEN

ANALYSIS OF INTERACTING SOLUTES

Index

- *Step 1: 2DSA refinement*
- *Step 2: Define the model*
- *Step 4: Perform DMGA Monte Carlo Analysis*
- *Step 5: Review parameter results*
- *Step 6: Refine constraints*

16.1 Step 1: 2DSA refinement

#. Follow steps 1-6 in the **:doc:`2DSA Flowchart <start_page>`_** to perform a 2DSA analysis and fit the boundary conditions and the associated time- and radially-invariant noises using iterative refinements.

16.2 Step 2: Define the model

1. Open the Discrete Model Genetic Algorithm (DMGA) function from the velocity menu.
2. Select **Define Base Model** to create a new model to be fitted.
3. The Model Editor window will open an existing model, or allow the user to create a new model.
4. To create a new model, select **Create New Model** and change the name in the **Model Description** text box.
5. Select **Manage Components** to open the Component window and to define the analytes in the model.
6. Click on **New Analyte** to select a previously entered analyte from the analyte management window and **Accept**.

#. Update the signal concentration from the #. #. #. #. #. Step 3: Define the constraints ———————————- Create a model with Analytes or TAke GA components - define them in components -

select attribute to fit

16.3 Step 4: Perform DMGA Monte Carlo Analysis

with 8-10 iterations (LIMS)

16.4 Step 5: Review parameter results

16.5 Step 6: Refine constraints

CHAPTER

SEVENTEEN

ANALYSIS OF ANALYTICAL BUOYANCY DENSITY EQUILIBRIUM DATA

Modules are incomplete - Protocol on Hold

CHAPTER

EIGHTEEN

FLOWCHART FOR THE ANALYSIS OF DENSITY MATCHING SEDIMENTATION VELOCITY DATA

Need more information on the density difference of buffers - Protocol on Hold

CHAPTER

NINETEEN

TIPS FOR INITIAL 3,000 RPM CHECKS:

It is common practice in some laboratories to measure initial 3,000 rpm scans from each cell before reaching full experimental speed, to make sure that the absorbance in each cell is appropriate and to make sure there are no leaks. Making sure there are no leaks is a very good idea, since centerpieces (especially epon-charcoal filled centerpieces) can easily bend or break the septum if one channel leaks and the other does not because the hydrostatic pressure difference between the channels can be very large, especially at high speeds, if different volumes are present on either sector. However, it is quite possible that leaks are missed if the sample doesn't start leaking until high speed has been reached, when the pressure is maximal. Hence, checking for leaks at 3,000 rpm may miss this crucial information, and leak checking really should be performed at high speed by comparing the menisci from the reference and sample channels from the first 2-3 scans and making sure they have not moved. In the XLA data acquisition program, when measuring in intensity mode, the sample and reference channels are recorded in blue and black traces and can easily be distinguished.

Optimal experimental designs strongly favor intensity measurements over absorbance experiments because of the reduced noise and doubled sample capacity, hence, when measuring with the UV/vis optics it is recommended to only use intensity measurements. In such cases, the reported signal consists of the counts of the intensity from each channel, not the absorbance, since there may also be a sample in the reference channel and no reference subtraction is performed. Intensity counts may or may not be easily interpretable to some to figure out if the desired concentration has been loaded, hence, it is much better to simply measure the absorbance of each sample with a benchtop spectrophotometer before cell loading. Please also read the "Tips for successful Absorbance/Intensity Experiments" about this topic.

A third complication comes from a recent addition in the UltraScan software that more precisely models the acceleration profile of the instrument with a timestate object. During acceleration, all samples will sediment and diffuse, even at slow speed, if just slightly. This change in sedimentation and diffusion can be modeled precisely, if the speed is known at each instant through the experiment. This profile is certainly affected by initial 3,000 rpm runs, and changes the expected omega2t profile if the 3,000 rpm scans are not considered in the acceleration period. In the new Optima AUC instrument, this is not a concern since the instrument can be programmed to collect speed/time/omega2t profiles during the entire run in 1 second increments, which is

exactly what UltraScan will use to predict the acceleration profile. In the XLA this information needs to be deduced from the available run information, which is stored in the file header from each recorded scan. The problem is that the XLA data acquisition software does NOT record the 3,000 rpm data in the same folder as the final experimental speed scans. Therefore, UltraScan is deprived of the needed information to predict accurate acceleration profiles, and this compromises the calculation of good fits.

So, how can you address this problem? There are multiple options that work:

1. eliminate 3,000 rpm runs altogether (the easiest solution)
2. Collect 3,000 rpm runs as before, but move the 3,000 rpm files into the full speed directory, overwriting the earlier scans in your full speed directory, if you only take a couple of scans this is acceptable, and you only lose info you would probably delete anyway. Import all files, exclude the 3krpm traces during editing *before* you pick the meniscus, since the meniscus position will change and move to the right as the rotor stretches
3. move them into the directory after renaming all your files in the full speed directory so the sequence # is simply higher by as many scans as you add 3,000 rpm traces.
4. program the run as an equilibrium method scan with multiple speed steps, this has the disadvantage that you have to program multiple steps for each speed with more than 99 scans, since 99 is the max per step, but if you want 495 scans you can program 5 speed steps with the same speed, this works too.
5. Stop after the 3,000 rpm run, shake up the cell, and restart.

Our preferred option is (1), and waiting a few minutes until you have scanned all cells 2-3 times so you can see if there are leaks. If there are, stop the run immediately to avoid damaging the cells.

CHAPTER

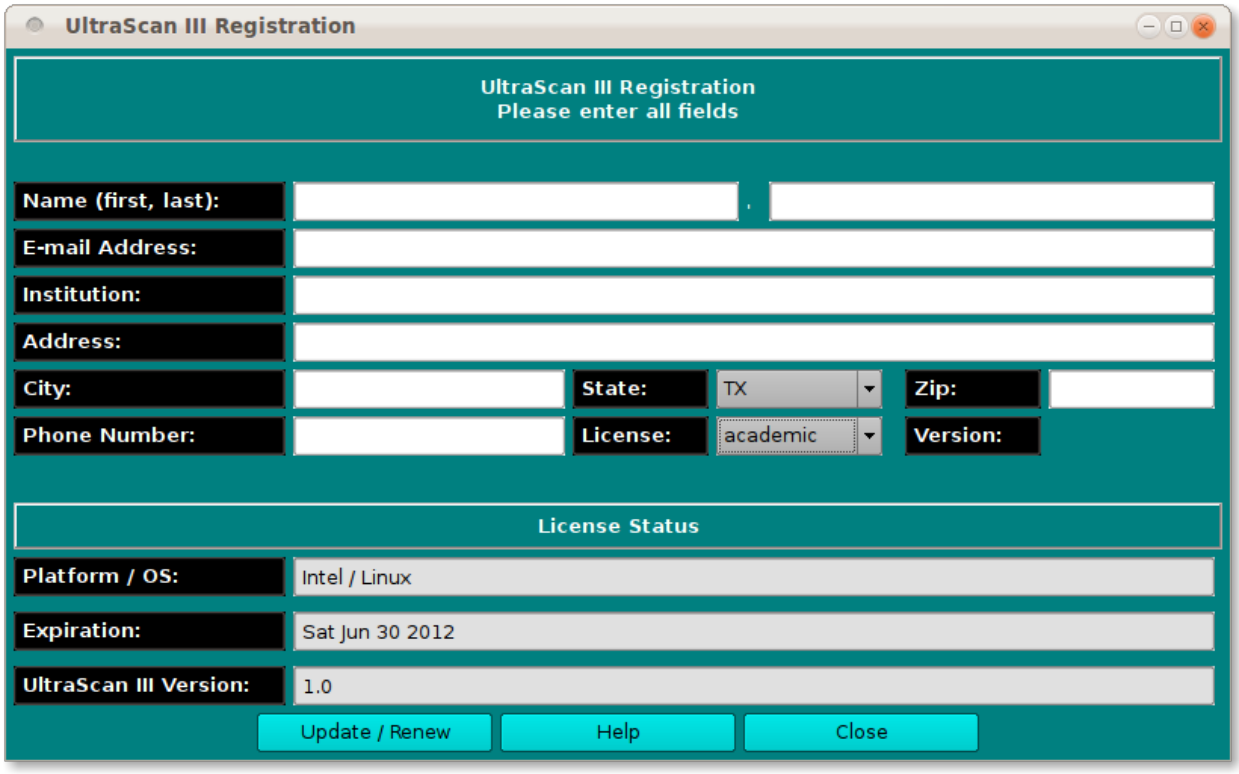
TWENTY

REGISTRATION / LICENSE UPDATE PANEL

Index

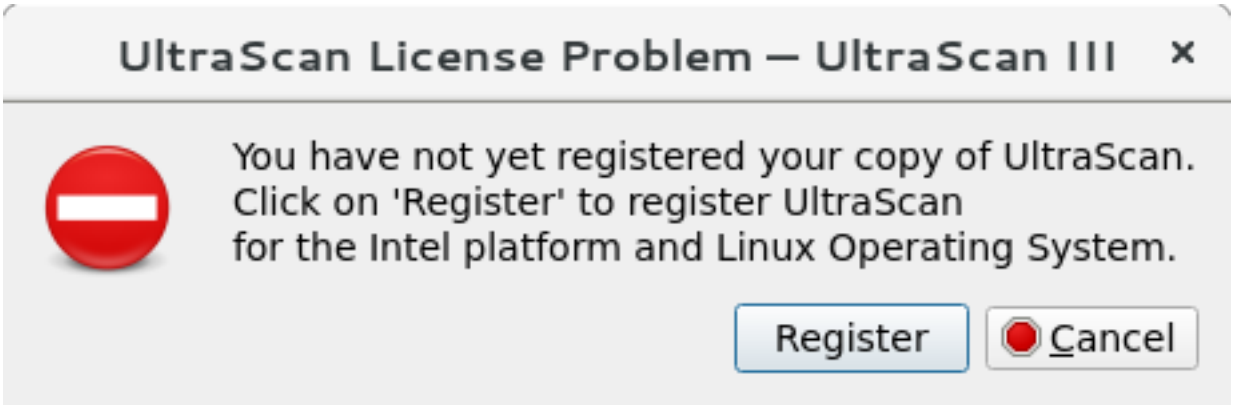
- *Process:*
- *License Functions:*

The registration control panel allows Ultrascan-III registration or update of a license. The contact fields should be filled out completely and then the Update / Renew button clicked. If you are registering for the first time, you will be sent an e-mail with instructions for completing the registration process. If you are simply updating information, the process is complete after Update / Renew.

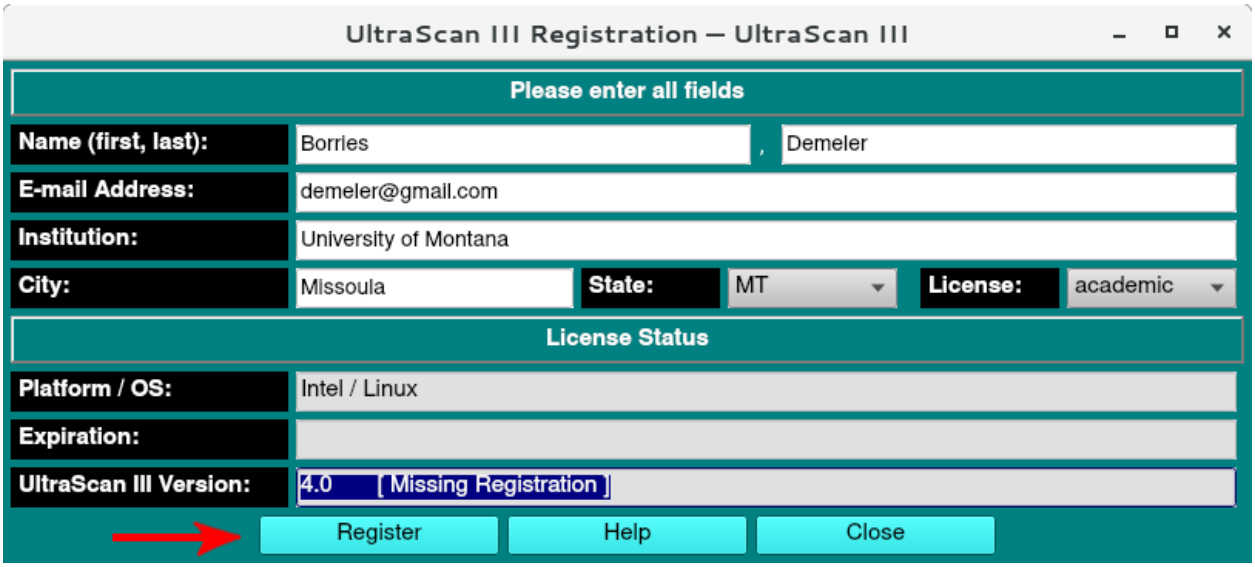


20.1 Process:

Each user must register their Ultrascan-III installation. If you have multiple computers, your registration must be repeated on each device and each account sued, but the associated license is automatically transferred, as long as the same e-mail address is used. Registration is invoked automatically the first time UltraScan is started, or when the current registration is expired. If UltraScan has not previously been registered, a dialog pops up. The registration program will auto-detect your hardware platform and operating system. Click on "Register":



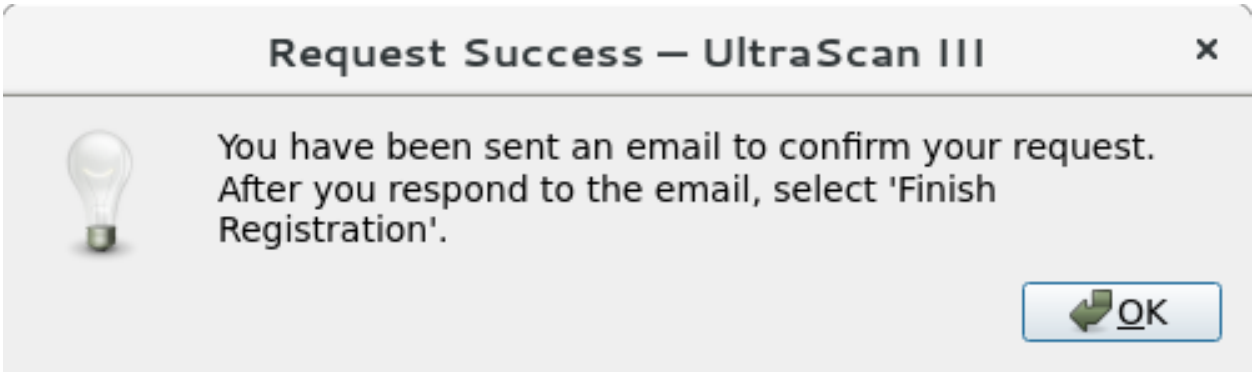
After clicking on "Register", the registration dialog opens up:



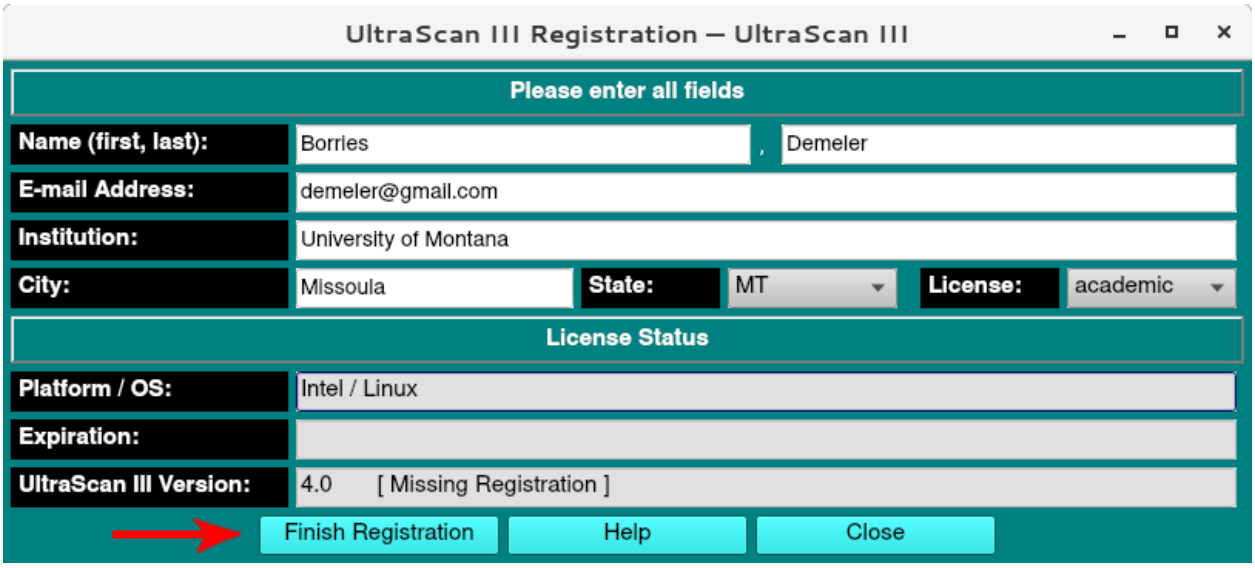
Fill in your contact information and click on "Register" to submit your registration (red arrow).

Note: Your computer must be connected to the network for the registration to succeed!

A 1 year license for the free use of UltraScan will be generated remotely. An email will be sent to the registered email address with an activation link, which must be confirmed:



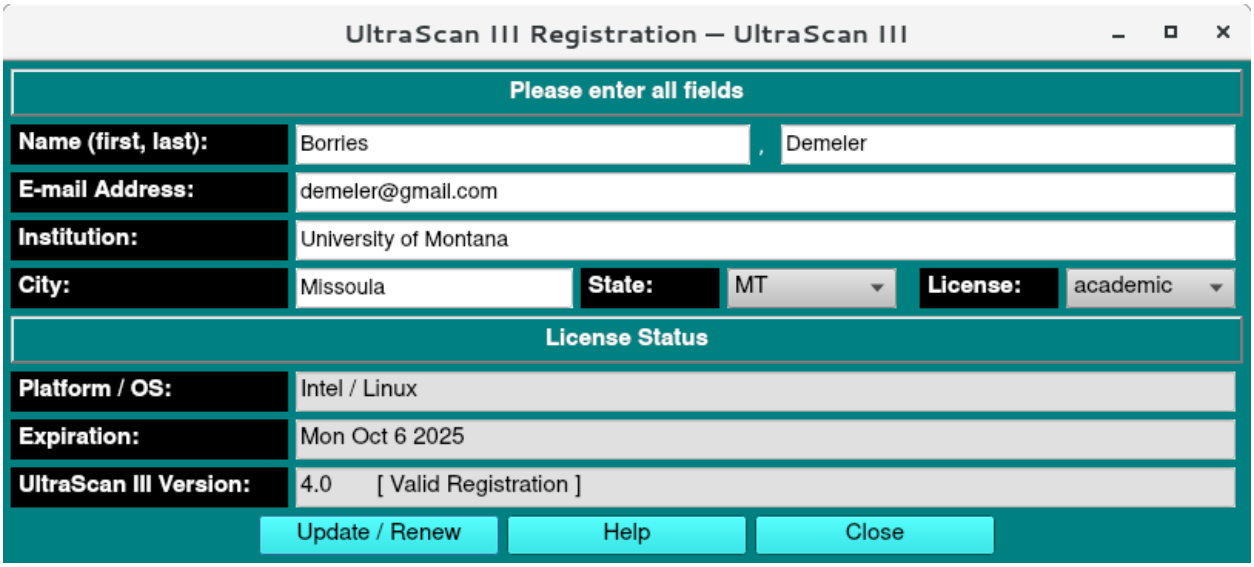
After confirming your UltraScan registration by clicking on the link in the email that was sent, the last step is to click on "Finish Registration" (red arrow) to confirm receipt of the license. If the registration was successful, you can finish the registration process by clicking on "Finish Registration" (red arrow):



Note: Again, your computer must be connected to the network for this step of the registration to succeed.

A 1 year license for the free use of UltraScan is now associated with the user's e-mail address, and a dialog should indicate the success of this step:

The leftmost button now should read "Update/Renew": The license will expire after one year and can be renewed by simply clicking on the **Update/Renew** button:



20.2 License Functions:

This panel is self-explanatory. For completeness, the GUI options are detailed below. Be sure to fill out all text boxes.

- **Name (first, last):** - Fill in your first and last name.
- **E-mail address:** - Give your best contact e-mail address.
- **Institution:** - Fill in the institution or company at which you will be using Ultrascan-III.
- **Address:** - Fill in address number and street.
- **City:** - Give your city name.
- **State:** - Select your state.
- **Zip:** - Fill in the postal zip code.
- **Phone Number:** - Give your best contact telephone number.
- **License:** - Enter the license type you desire:** academic or commercial.
- **Update / Renew** - When all the above fields are complete, click this button to complete the registration or update.
- **Help** - See this documentation.
- **Close** - Exit the registration dialog.

CHAPTER

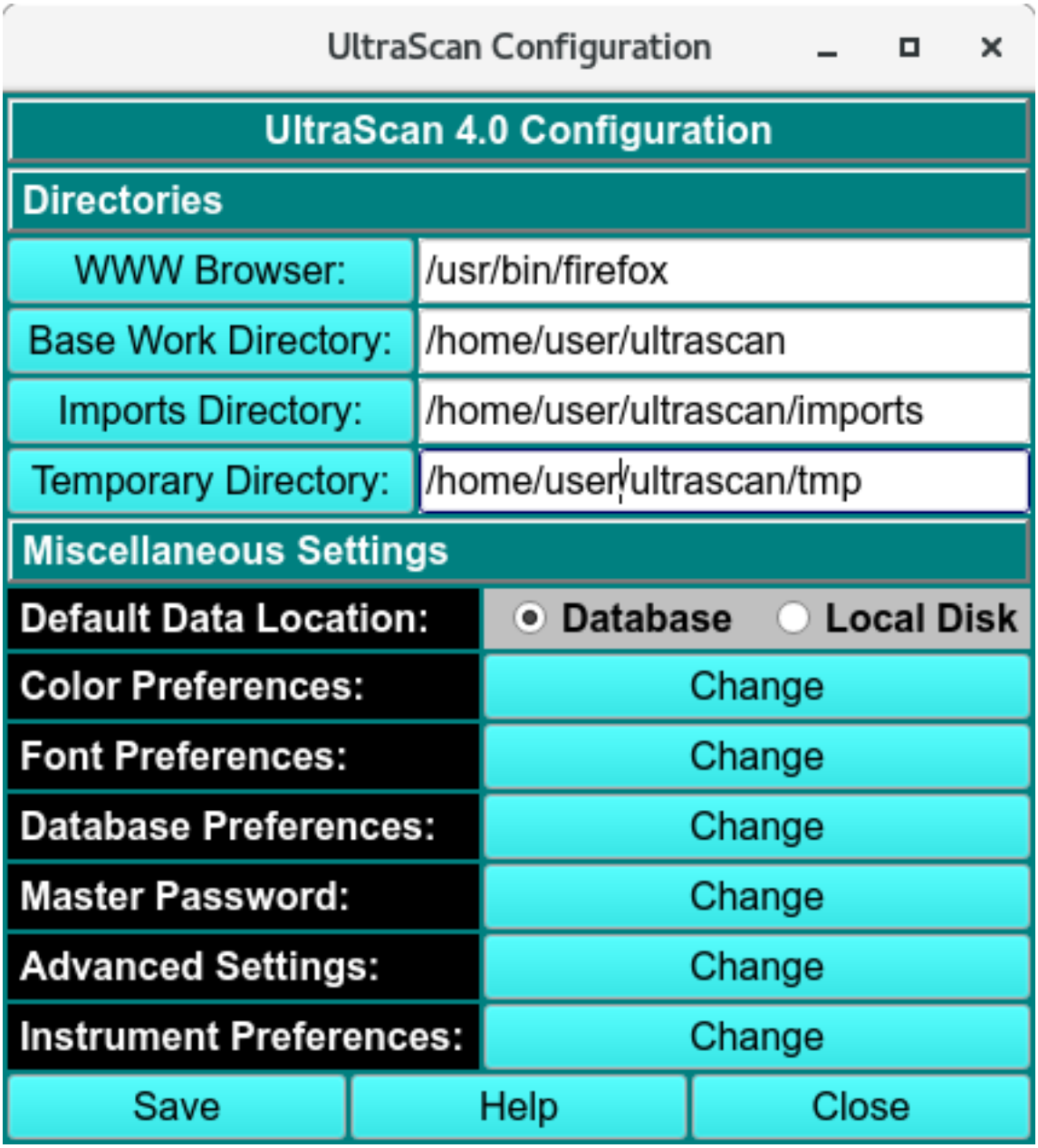
TWENTYONE

ULTRASCAN-III CONFIGURATION AND PREFERENCE

Index

- *Configuration Functions*
  - *Color Configuration Panel*
  - *Font Configuration Panel*
  - *Database Configuration Panel*
  - *Master Password Panel*
  - *Advanced Configuration Panel*
    - *Adv. Config. Functions*
  - *Instrument Preference Panel*

There are a number of parameters that can be set to customize UltraScan-III according to personal preferences. The configuration panel is accessed from the main menu under the Edit menu:



UltraScan-III Configuration

If you start UltraScan-III for the first time, the configuration settings will be set to the UltraScan default settings. After starting the configuration panel, you can edit the items in the list or click on the respective pushbutton for the file selection window to select your preferred web browser or set data paths. When a path is required, select the desired directory name. For the browser, select a filename of a preferred browser.

21.1 Configuration Functions

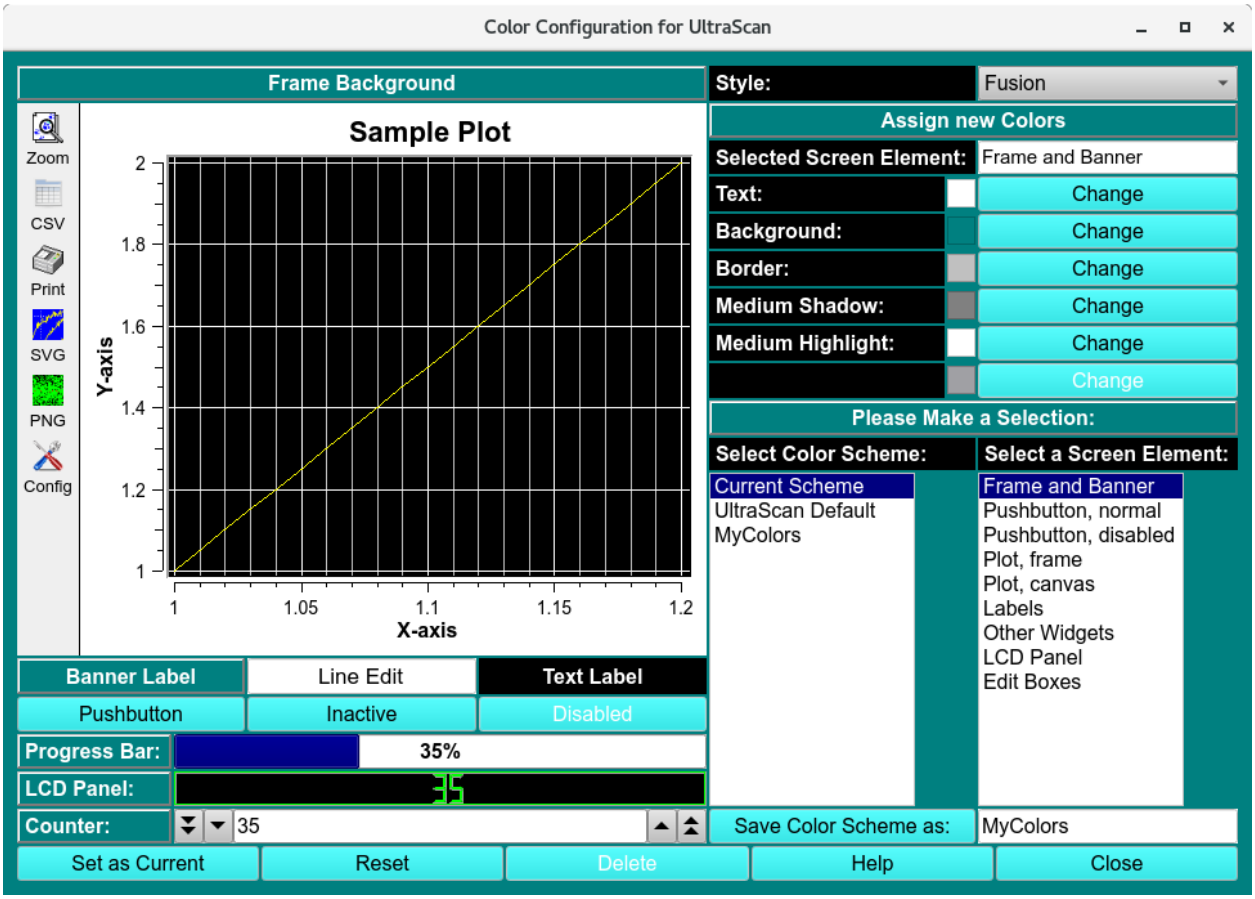
WWW Browser:	This is the path to the executable of your html browser. If your preferred browser (for example, firefox) is in your search path, the single word 'firefox' will suffice here, otherwise enter the complete path.
Base Work Directory:	The default location for all UltraScan related files and folders.
Import Directory:	The location where all data to be.
Temporary Directory	The location where UltraScan stores temporary files.

Miscellaneous Settings:

Default Data Location	The default location of the data storage and retrieval.
Color Preferences:	See the Color Configuration below on the *Change* button if you want to load or define a different color scheme.
Font Preferences:	See the Font Configuration below on the *Change* button if you want to set a new default font for all of UltraScan's applications.
Database Preferences:	See the Database Configuration below on the *Change* button if you want to set the database username, password, database name and server host name.
Master Password:	See the Database Configuration below on the *Change* button to set or reset your personal **UltraScan-III** password. Be sure to remember the password. If it is lost, you will need to reset all saved database information in the *Database Preferences* panel.
Advanced Settings:	See the Database Configuration below on the *Change* button to set advanced settings related to debug, advanced level, and number of threads.

Once all settings are correct, click on **Save** to make your settings permanent.

21.1.1 Color Configuration Panel

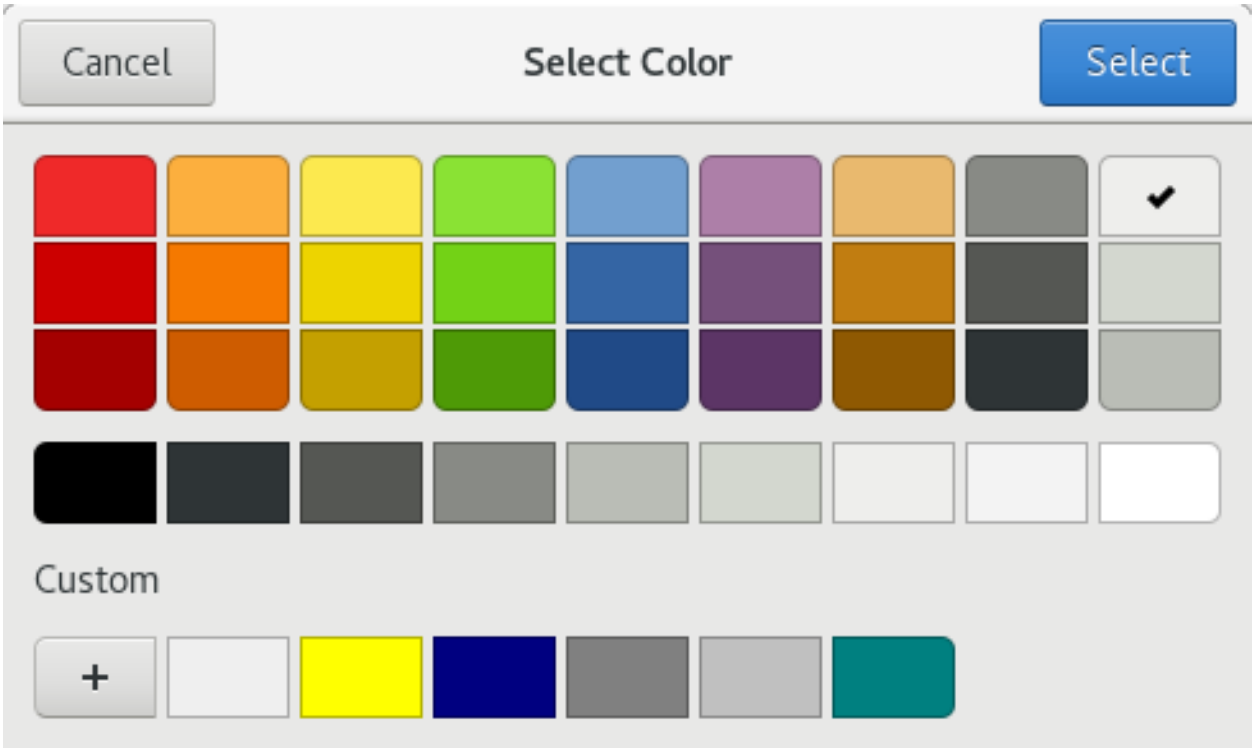


Color Configuration Panel

The color configuration control can be invoked from the configuration panel by clicking on the *Change* button from the color preferences option.

The panel has two sections, a *preview* section (on the left) and a *control* section (on the right). With this module you can select custom color combinations for all screen elements (widgets) used by UltraScan and save them as custom color schemes. You can also set the desired margin around graphs.

To change a widget's color, select the widget you would like to change from the screen element listbox. All applicable colors for that widget will be shown in the Assign new Colors section. Click on the Change button for the desired color definition of the selected widget. A **Color Select** dialog will be displayed that lets you select and define custom colors for the selected color definition. After you click on the "OK" button you will be returned to the color configuration control window and the new color will be applied to the preview section of the screen. Continue to define new colors for each widget until all color changes you want to make have been applied.



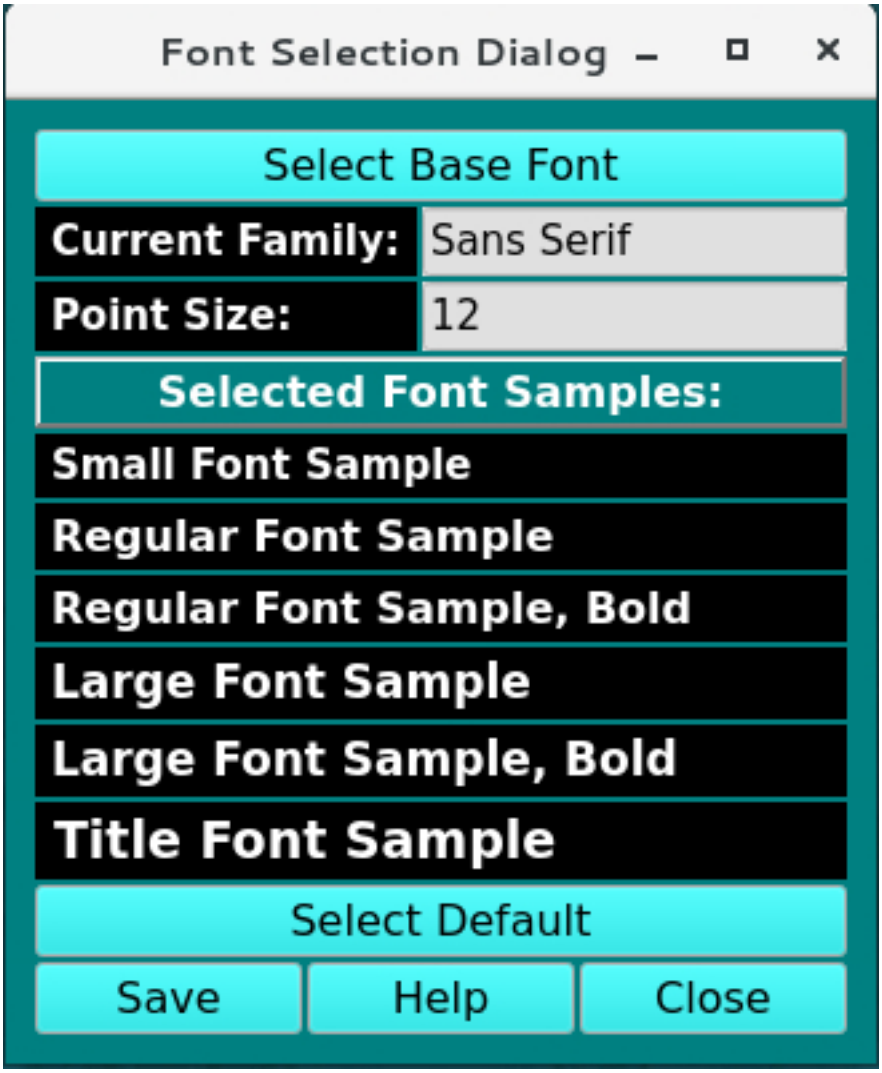
Color Select Dialog

To reset to the previous color definition, click on the *Reset* button. Click on *Set as Current* to apply the colors to the current scheme. If you want to save the scheme, define a name (without spaces) under Save Color Scheme as: and click on the Save Color Scheme as: button. Keep in mind that certain names are reserved for predefined UltraScan color schemes, and they cannot be overridden.

The number of colors that can be simultaneously displayed on the screen depends on your display adapter's memory and resolution. Common display settings are 16 bit (65536 colors), and 24 bit (~16.8 million colors).

If you want to delete a custom scheme, select the scheme and click on the Delete button. Predefined UltraScan schemes cannot be deleted. To select a different scheme from the list of available color schemes, simply double click on the desired scheme and click on the Save as Current button.

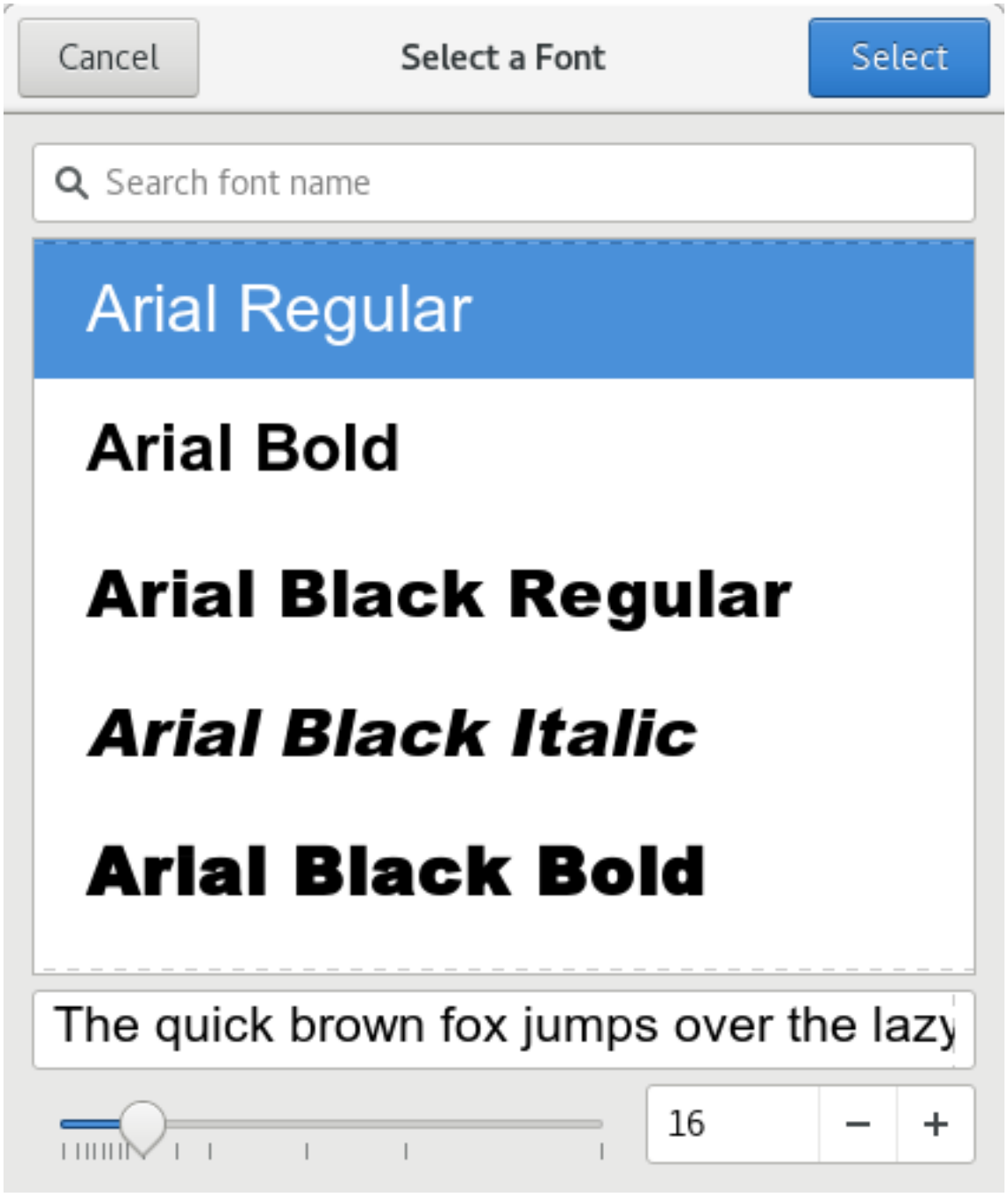
21.1.2 Font Configuration Panel



Current Font Dialog

Using this dialog, you can change the default font for all of UltraScan's applications. All fonts used in UltraScan are listed in this dialogue, and you should pick a font that displays well in the allotted space. When changing the font, a **Font Selection Dialog** will be displayed that lets you select and define custom fonts for UltraScan. After you click on the OK button you will be returned to the font configuration control window and the new font will be applied to the preview section of the screen.

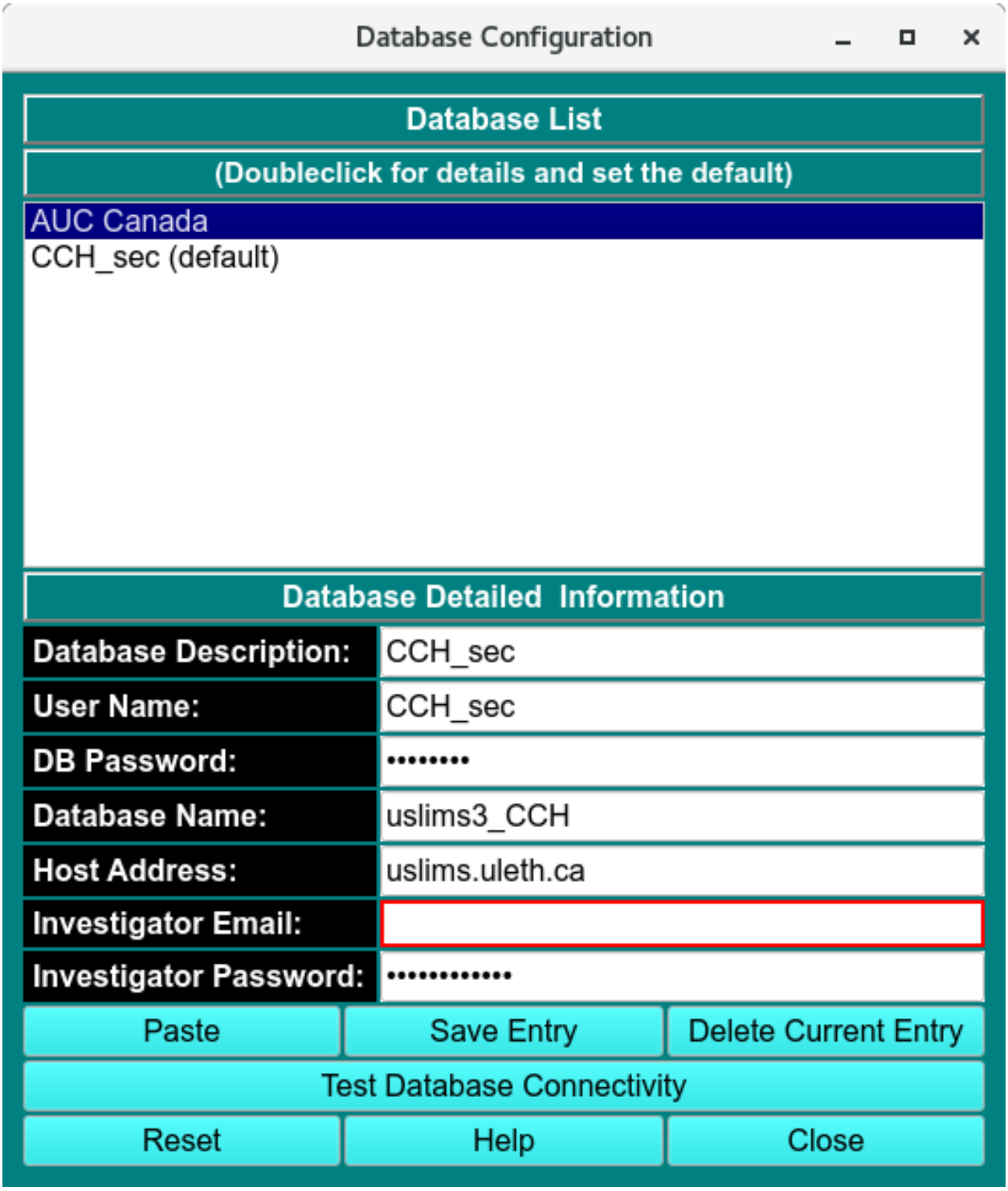
Selecting *Save* will save the currently displayed font as the UltraScan *default* font.



Font Selection Dialog

21.1.3 Database Configuration Panel

Note: Whenever UltraScan needs to use a database entry, it needs the user's master password. The first time it is needed, it will prompt for it and save it in memory, but not on disk storage. When needed again, UltraScan uses the stored password. Before this can be done, the master password needs to be set up and a cryptographic hash of the password stored. Use the Master Password panel, accessed form the main configuration panel, to set up or change the master password.



Database Configuration

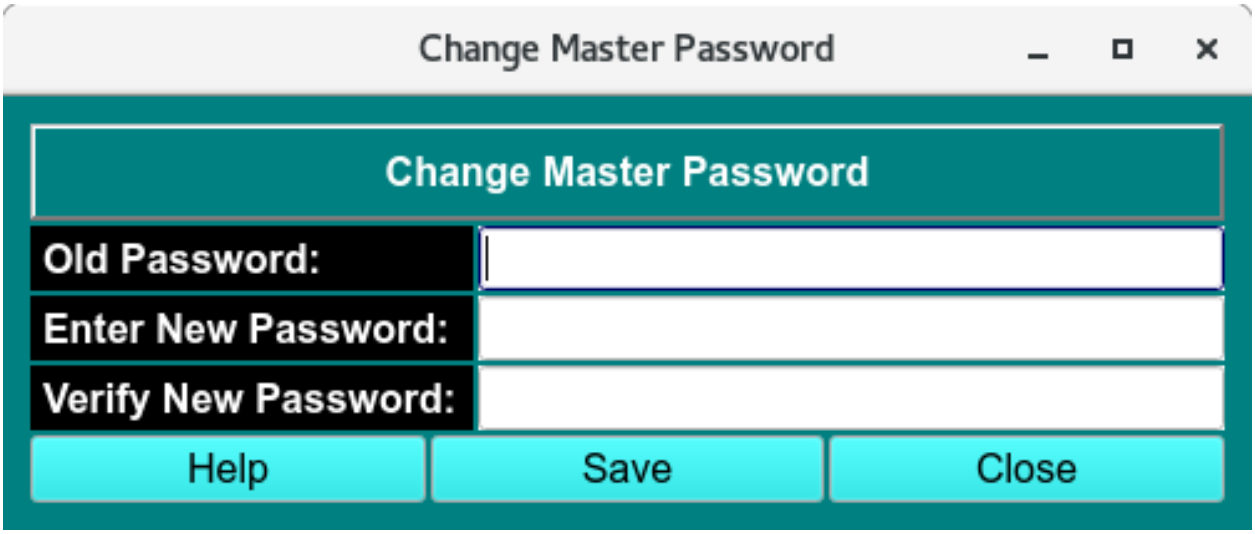
The database configuration panel allows you to configure the connection to the database that holds your experimental data. Here you can enter the name of the database, the username you use to connect to the database engine as well as your database login password, and finally the name of the host on which the database engine is running. UltraScan offers the option to configure multiple databases for facilities that may use multiple databases from different sites. The currently selected database is listed with the "(default)" tag. To change

the default database, simply double-click on the desired database in the "Database List" section to select the database that needs to be set as default. Now click on the Save as Default button. The name in the database list with the "(default)" tag will then reflect the new default database.

When Reset is selected, the fields are cleared and a new entry can be entered. The Test DB Connectivity button will try to connect to the database in the Detailed Information fields to provide a confirmation that the data is correct.

To delete an entry, select it and click on Delete Current Entry.

21.1.4 Master Password Panel



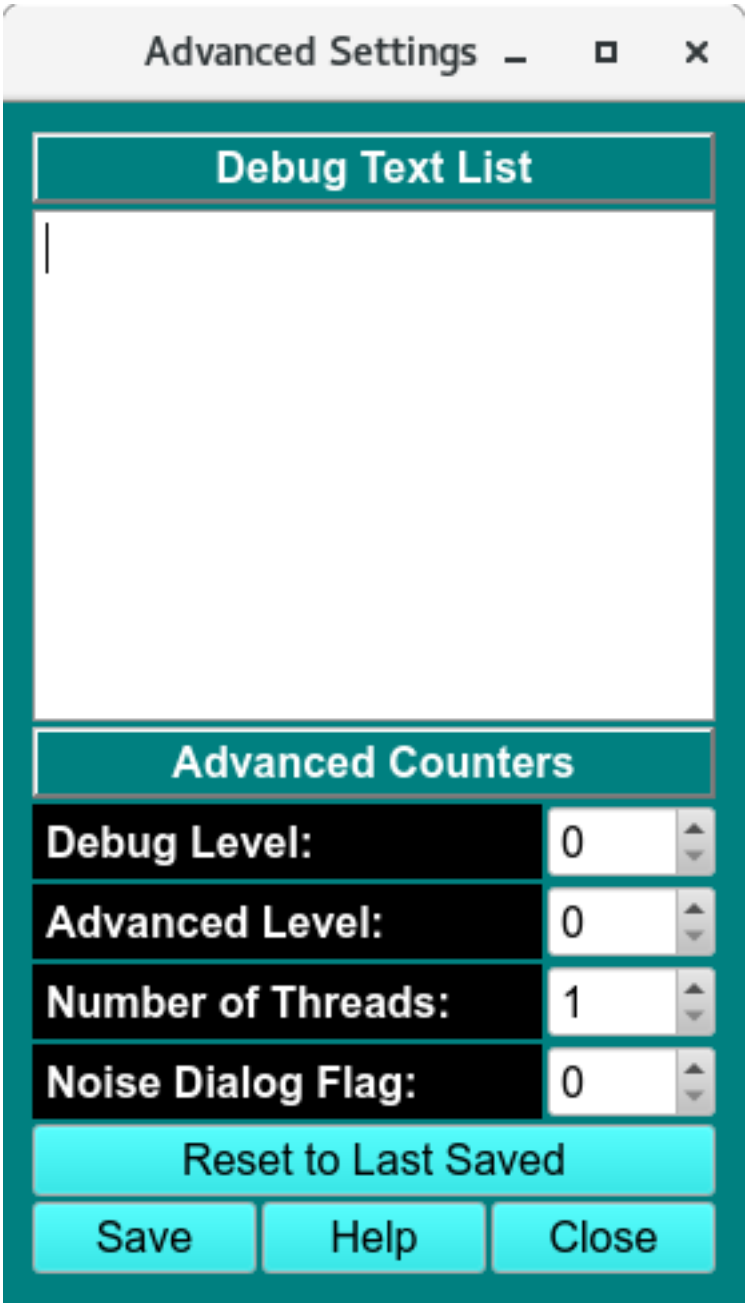
Change Master Password

The user's master password needs to be entered in order to access encrypted database information. Enter the **old password** and the **new password** *twice* to change the password. When entering the password for the first time, the Old Password entry will not be shown.

Caution: There is no way to recover the master password so it must be remembered. If the password is lost, UltraScan can be reset, but all database login settings will have to be manually reentered.

21.1.5 Advanced Configuration Panel

Note: Some **UltraScan-III** applications output debug text to STDOUT or a log file. Some only include certain GUI elements when in an advanced mode. Certain applications can benefit from use of more than one thread. These variations are governed by the configuration values that are set in this panel. Actual values should be set only under the guidance of software developers, since meanings of settings vary considerably from application to application.



Advanced Settings

Adv. Config. Functions

Debug Text List	Additional debugging to aid in trouble-shooting can be turned on or off based on a debug text phrase. One or more of such phrases may be entered in this list. Leading or trailing blanks and multiple consecutive blanks are ignored.
Debug Level	The quantity of debug outputs may be determined by the magnitude of the value selected here. The greater the count,the greater the number of debug lines output for applications that take advantage of the debug level value.
Advanced Level	For certain applications, there are options that should only be visible and changeable by a highly knowledgeable user working in advanced mode. The level of advanced mode may be set with the counter here.
Number of Threads	On multi-core computers, a thread count greater than one is advantageous in certain applications that are thread-aware. The number depends on the application and the available resources.
Noise Dialog Level	This function enables user to select from multiple noise profiles that are associated with a model. When the level flag is 1 (enabled), in *Finite Element Viewer*, user is prompted to select from a list of noise profiles, where the earlier profiles will have higher Model Sibling 000#. See *Finite Element Viewer* help documentation for further instructions.
Reset to Last Saved	All of the above settings can be made to revert to last-saved settings by clicking this button.

Table 1: **Debug Text Options**

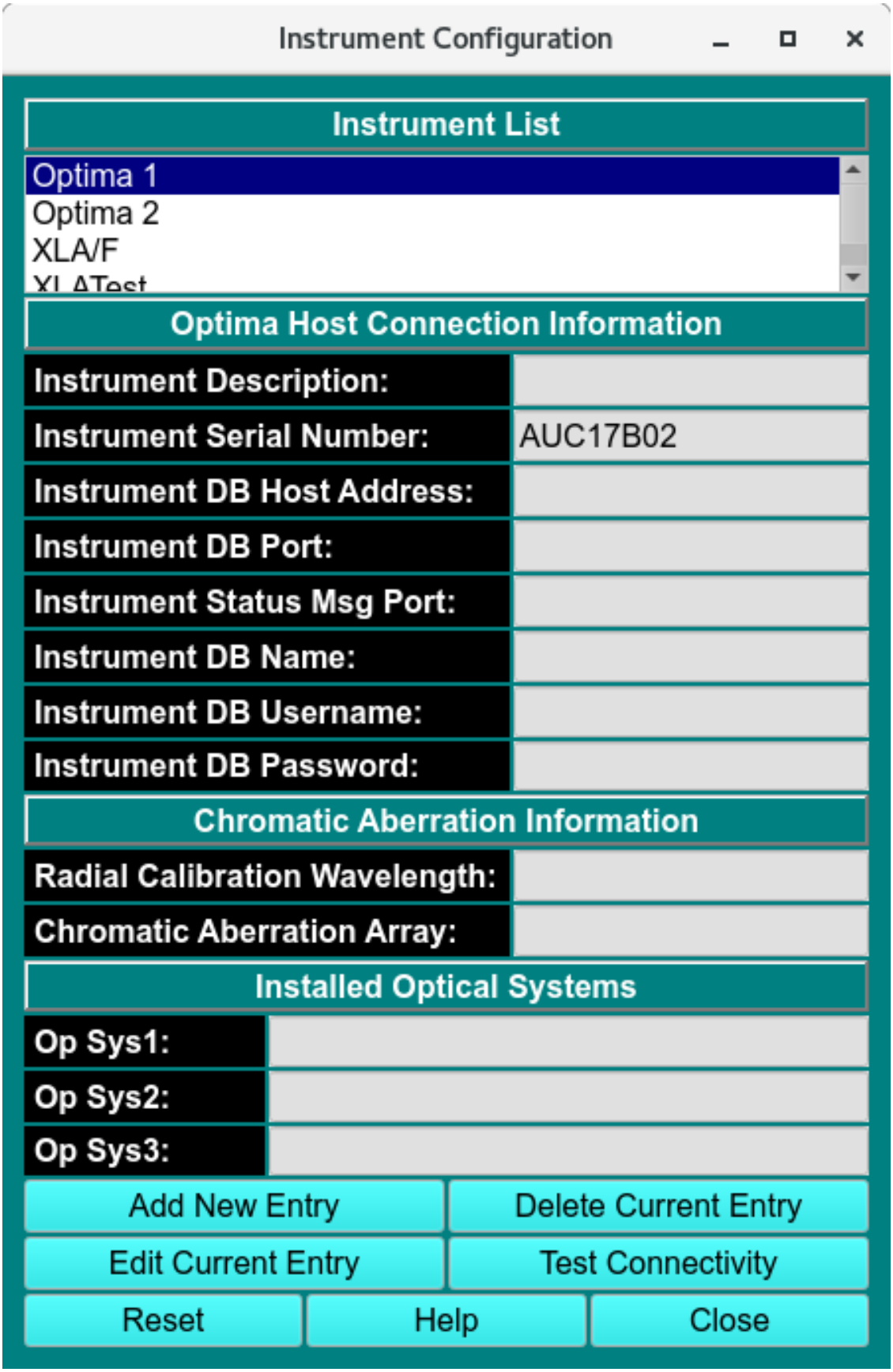
Debug Text Option	Description
SetSpeedResolution	The resolution of the speed (raw_speed/speed_resolution)*speed_resolution. Default: *SetSpeedResolution=100*
SetSpeedLowAcc	The threshold for the acceleration rate to be considered low acceleration. Default: *SetSpeedLowAcc=250*
SetSpeedLowR	The threshold for eligible experimental speed. Default: *SetSpeedLowR=1500*
SpeedStepLowSec	Minimum time in seconds for a constant speed zone without scans to be treated as distinct speed step. Default: *SpeedStepLowSec=20*
normCutoff	The norm cut off, value for which a solute gets removed from the grid space. Default: *normCutoff=1.0*
2dsaFinalNoiseOnly	Flag to apply any noise correction only on the final iteration.
MC-GaussianSmooth	Flag to construct MC sigmas from residuals smoothed for each scan instead from iteration 1 residuals.
2dsaTiming	Flag to print debug timings in us_2dsa.
protocolFromDisk	Flag to load protocols from disk and not from database.
men2click	Flag to require user to specify right and left border of the meniscus in us_edit instead of specifying one position.
pcsaTiming	Flag to print debug timings in us_pcsa.
hsclogo	Flag to include the University of Texas Health Science Center logo in the generated reports.
becklogo	Flag to include the Beckman Coulter logo in the generated reports.
us3logo	Flag to include the UltraScan-III logo in the generated reports.
DB_ENFORCE_DEFAULT_	Flag to enforce the use of the OS configured default SSL security level for database connections, instead of lowering it to for UltraScan database connections to zero.
auto_positioning	Flag to enable the legacy window positioning behavior from UltraScan instead of relying on the OS window manager.

Table 2: **Window Controls**

Reset	Indicate that parameters are to be reset and the plots re-displayed based on original parameters.
Help	Display this detailed help.
Close	Close all windows and exit.

21.1.6 Instrument Preference Panel

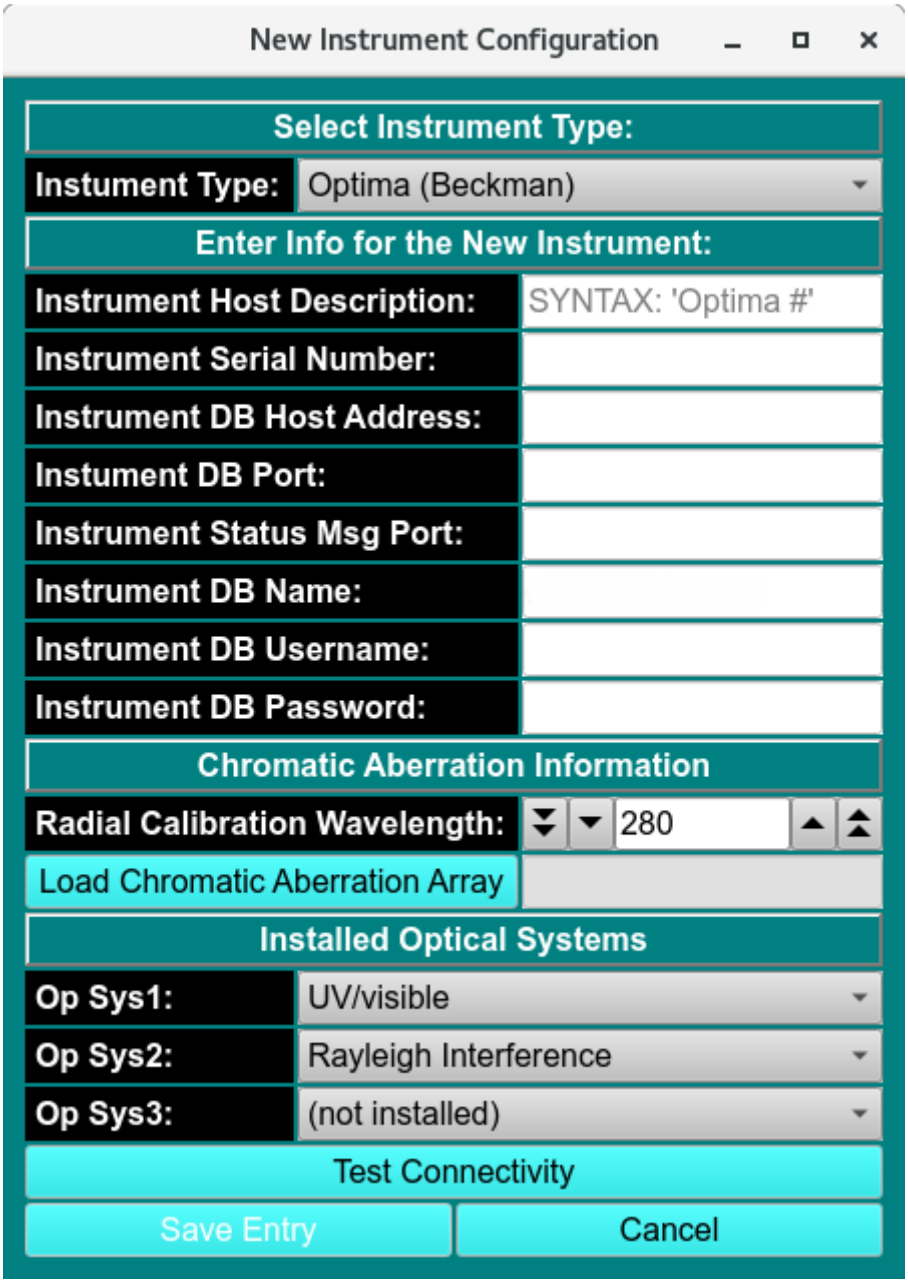
Note: Whenever UltraScan needs to use an Optima host entry, it needs the user's master password. The first time it is needed, it will prompt for it and save it in memory, but not on disk storage. When needed again, UltraScan uses the stored password. Before this can be done, the master password needs to be set up and a cryptographic hash of the password stored. Use the *Master Password* panel, accessed from the main configuration panel, to set up or change the master password.


Instrument Configuration
Instrument List
Optima 1
Optima 2
XLA/F
XLATest
Optima Host Connection Information
Instrument Description:
Instrument Serial Number:
AUC17B02
Instrument DB Host Address:
Instrument DB Port:
Instrument Status Msg Port:
Instrument DB Name:
Instrument DB Username:
Instrument DB Password:
Chromatic Aberration Information
Radial Calibration Wavelength:
Chromatic Aberration Array:
Installed Optical Systems
Op Sys1:
Op Sys2:
Op Sys3:
Add New Entry
Delete Current Entry
Edit Current Entry
Test Connectivity
Reset
Help
Close

Instrument Configuration

The Optima host configuration panel allows you to configure the connection to one or more Optima instruments in your lab. Here you can enter the description, host address and port of the Optima instrument, as well as Optima database name, user name, and password. Additional information includes a radial calibration wavelength, a file from which to upload a chromatic aberration array, and installed optical systems.

When Add *New Entry* or *Edit Current Entry* is selected, one of the following dialogs is opened:



New Entry Dialog

Optima Host Connection Information

Instrument Host Description	An Optima host description, always given in a strict "Optima (number)" form, such as "Optima 1".
Instrument Serial Number	The instrument's serial number.
Instrument DB Host Address	IP address to the Optima, generally in a non-routable subnet (example: 192.168.*.*).
Instrument DB Port	Port to the Optima database.
Instrument Status Msg Port:	Port used for the UltraScan data acquisition software (ssl-encrypted data exchange)
Instrument DB Name	Name of the database on the Optima AUC system.
Instrument DB Username	Optima database login name.
Instrument DB Password	Optima database login password.

Chromatic Aberration Information

Radial Calibration Wavelength	Wavelength at which radial calibration was performed.
(Load) Chromatic Aberration Array	A file (specified in a file dialog) containing an array of wavelength/value pairs that specify radial position corrections for each wavelength.
Save Entry	Update instrument settings in the database.
Cancel	Close the dialog without updating the database.
Installed Optical Systems	Up to three optical systems available on the instrument where choices are: • UV/Visible • Rayleigh interference • (not installed) - Available optical slot

Installed Optical Systems: Text description of which optical system is installed (UV/Visible, Rayleigh interference or available slot)

Test the connection to the new or modified instrument by clicking *Test Connectivity*

For appropriate Optima Host values, consult with your administrator or with the authors of this software.

Note: To delete an entry, select it and click on Delete Current Entry.

CHAPTER

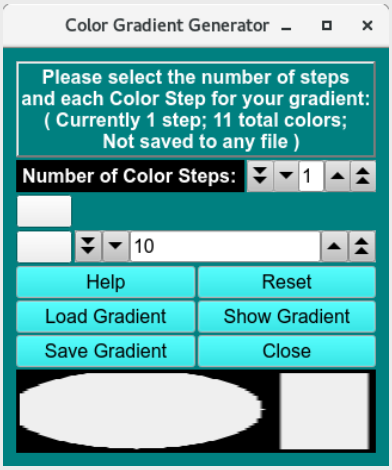
TWENTYTWO

COLOR GRADIENT GENERATOR

Index

- *Example Color Schemes and Plots*

Color gradients can be generated for the purpose of providing a third dimension to selected Ultrascan-III program graphs. Color gradients are specified with a start color and a number of color steps, each one having an end color and a number of points. The total number of colors in the gradient is the sum of step points plus one for the start color. The steps are saved in an XML file, with an option to save a PNG file with the color legend.

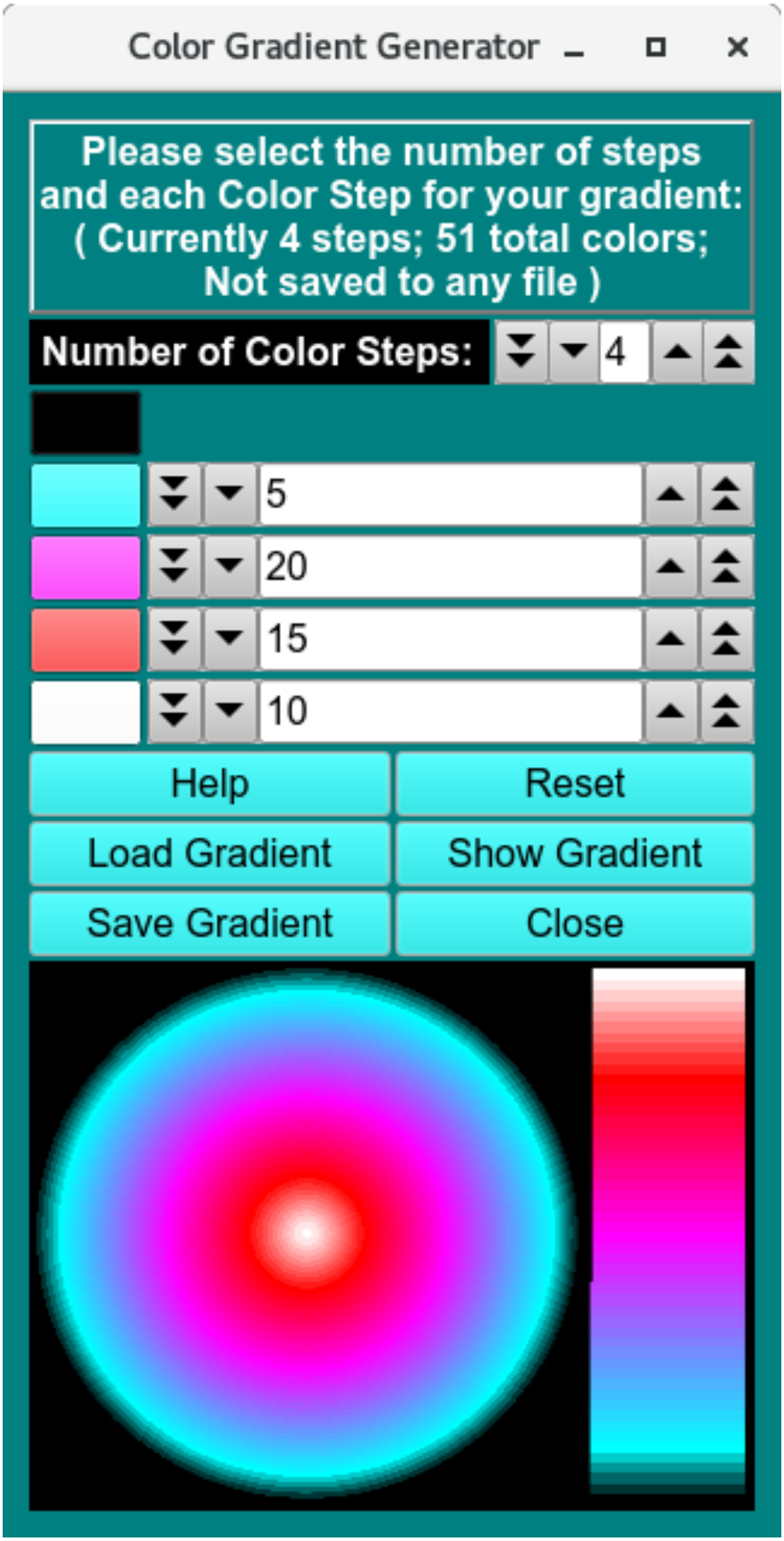


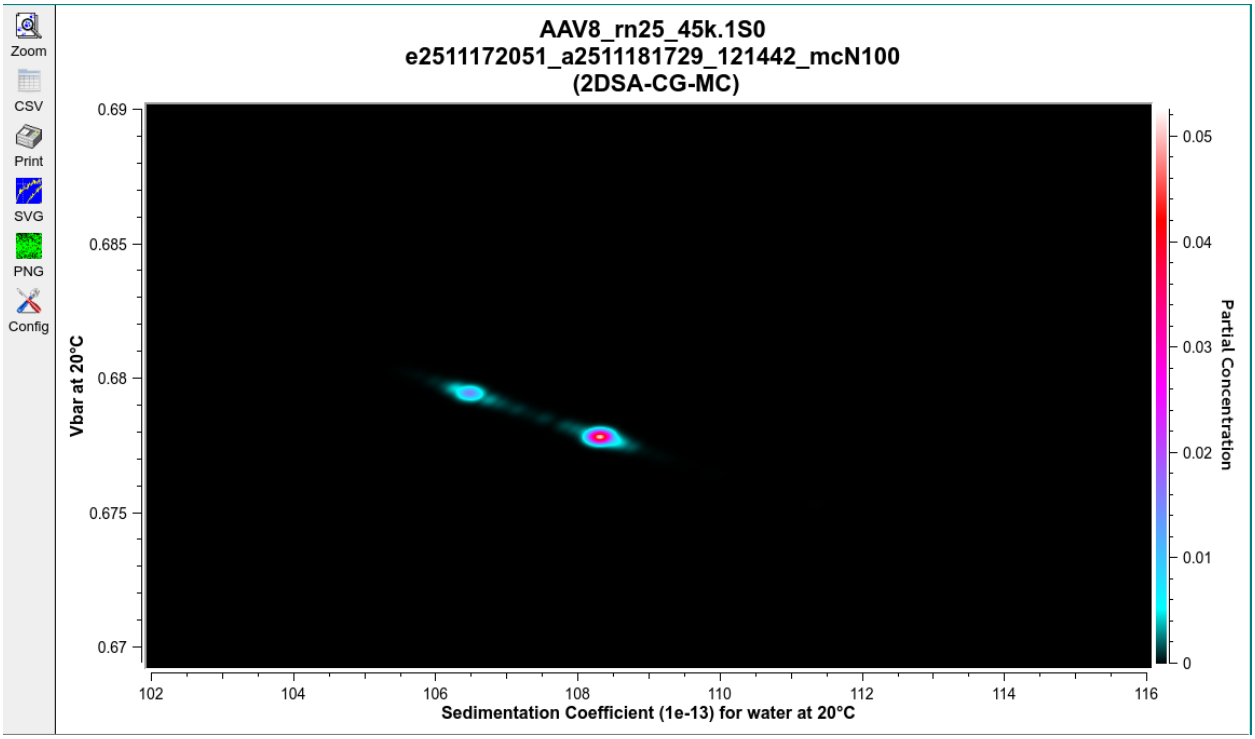
Functions:

- **Number of Color Steps:** Choose the total number of color steps.
- **Select Starting Color:** Click this button to choose a start gradient color. The chosen color appears in the button and the gradient displays below.
- **Select Step Color:** Click any step color button to choose an end color for the step.
- **Number of Points in this Step:** Set the number of points for each step using the counter.
- **Help:** Display detailed color gradient generator help.
- **Reset:** Remove all selections and revert to a single one-point step with color black.
- **Load Gradient:** Load gradient selections from a previously saved color gradient XML file.
- **Show Gradient:** Draw a representation of the currently specified color gradient.
- **Save Gradient:** Save the current color gradient to an XML file, optionally also saving a PNG legend.
- **Close:** Close the session (prompts to save if you haven't).

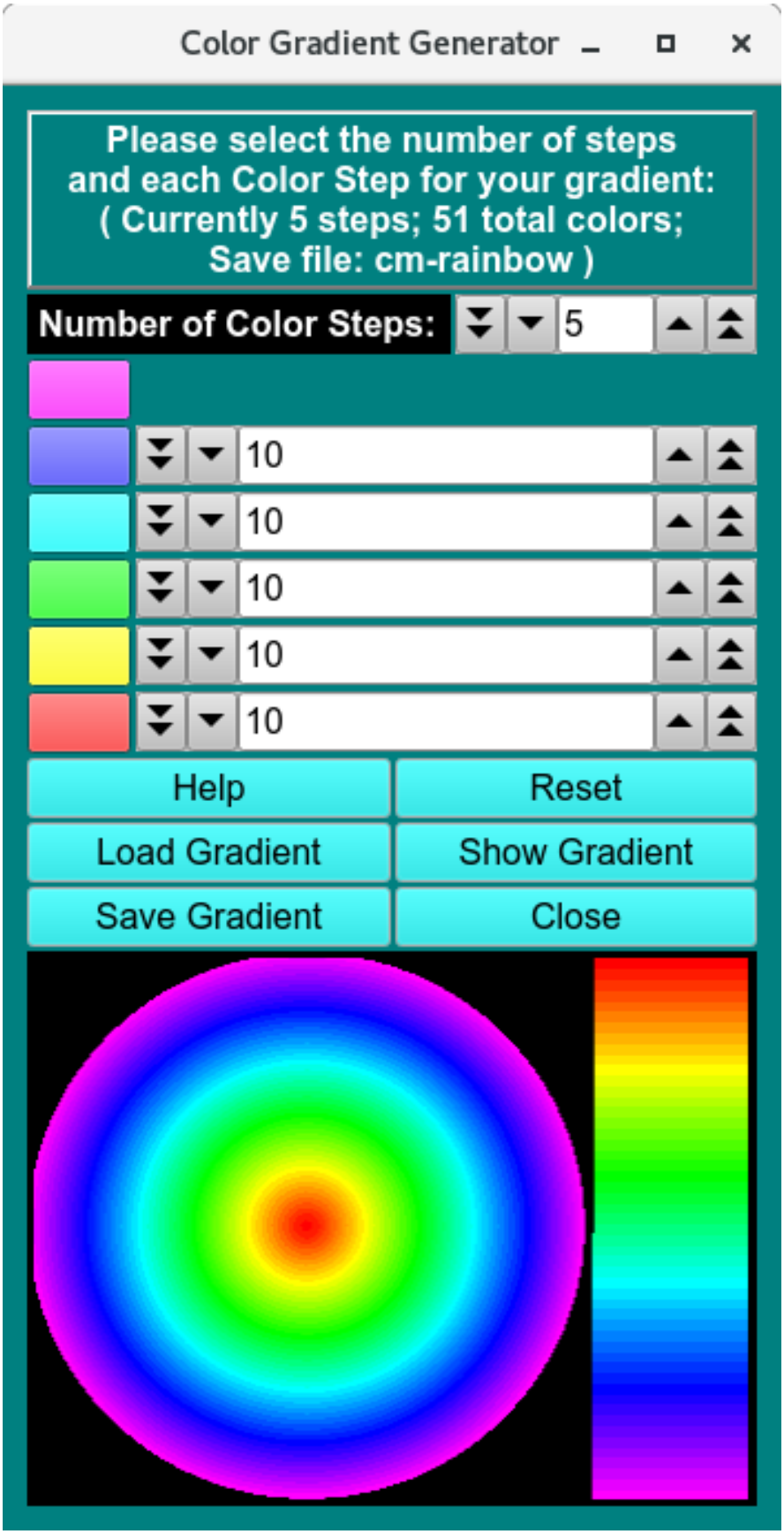
22.1 Example Color Schemes and Plots

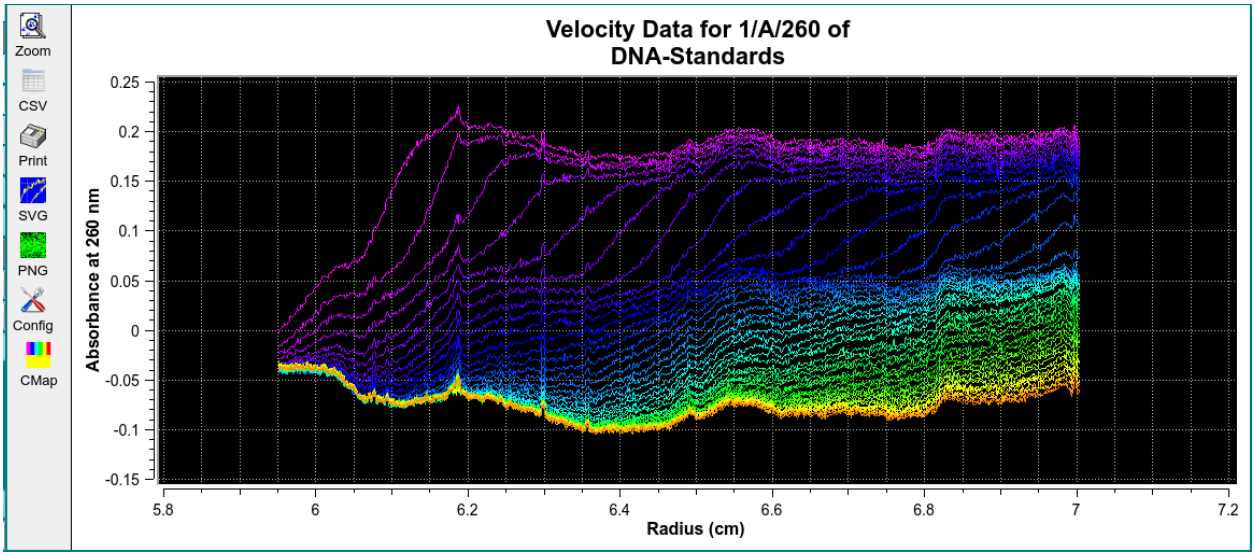
User generated color schemes can be applied to multiple windows and plots using **load color** or CMap buttons in applicable windows.


Color Gradient Generator
Please select the number of steps and each Color Step for your gradient: (Currently 4 steps; 51 total colors; Not saved to any file)
Number of Color Steps:
4
5
20
15
10
Help
Reset
Load Gradient
Show Gradient
Save Gradient
Close



Example Color scheme applied to Pseudo3D and GA plots


Color Gradient Generator
Please select the number of steps
and each Color Step for your gradient:
(Currently 5 steps; 51 total colors;
Save file: cm-rainbow)
Number of Color Steps:
5
10
10
10
10
10
Help
Reset
Load Gradient
Show Gradient
Save Gradient
Close



Example Color scheme applied to 2DSA, PCSA or Finite Element viewer data plots

CHAPTER

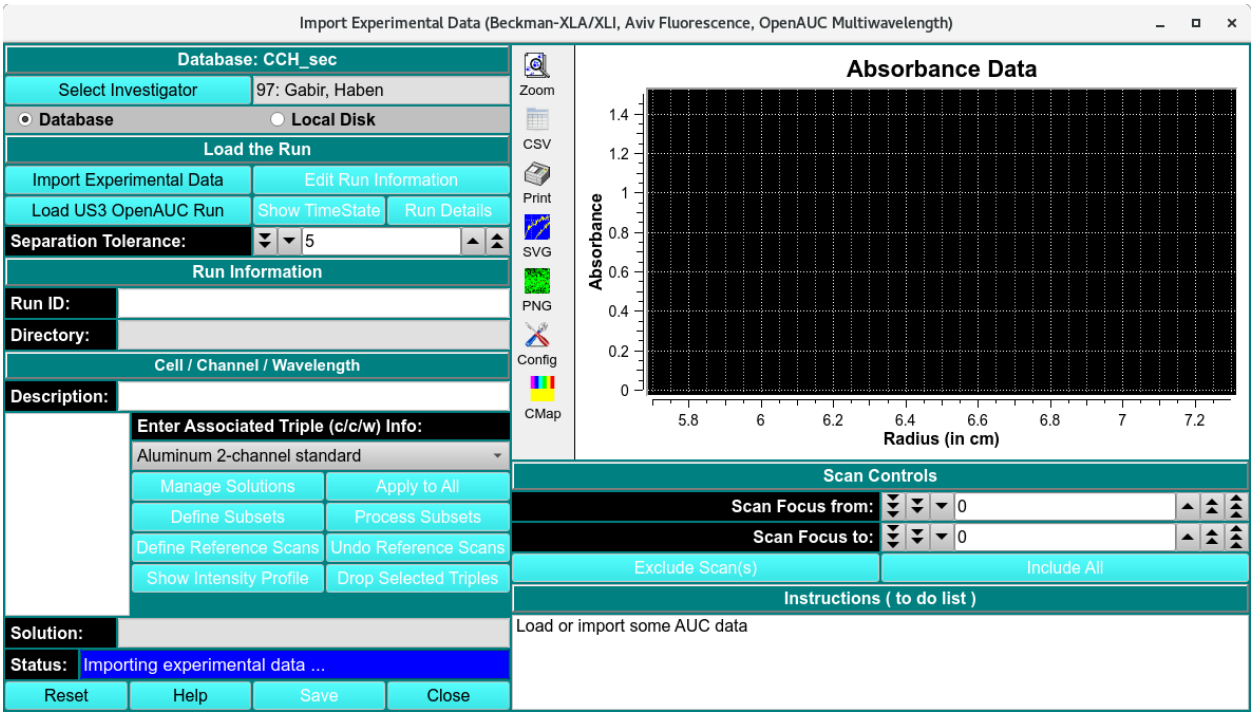
TWENTYTHREE

IMPORT EXPERIMENTAL DATA

Index

- *Import and Convert Single and Multi-Wavelength Experimental Data*
  - ***STEP 1:** Loading Data:*
    - *Multi-Wavelength data:*
  - ***STEP 2:** Modify Run ID (Optional):*
  - ***STEP 3:** Edit Run Information:*
  - ***STEP 4:** Adjust Separation Tolerance (Optional):*
  - ***STEP 5:** Modify Cell/Channel/Wavelength (c/c/w) Info:*
  - ***STEP 6:** Saving the data:*
- *Processing different data types and datasets*
  - *Radial intensity (RI) data:*
  - *6- and 8- channel centerpieces:*
  - *Wavelength Data:*

Import and convert raw data into the UltraScan-III OpenAUC format.

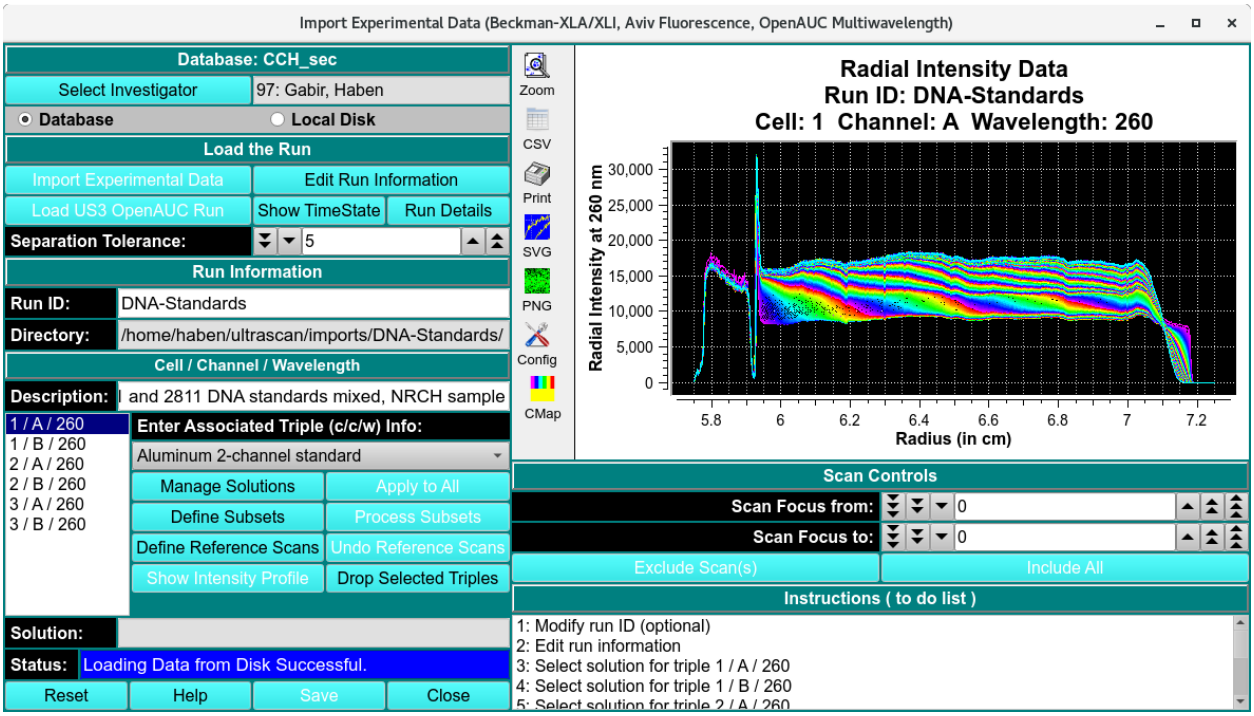


Import Experimental Data

23.1 Import and Convert Single and Multi-Wavelength Experimental Data

23.1.1 STEP 1: Loading Data:

Raw data can be loaded into the window by clicking **Import Experimental Data**.



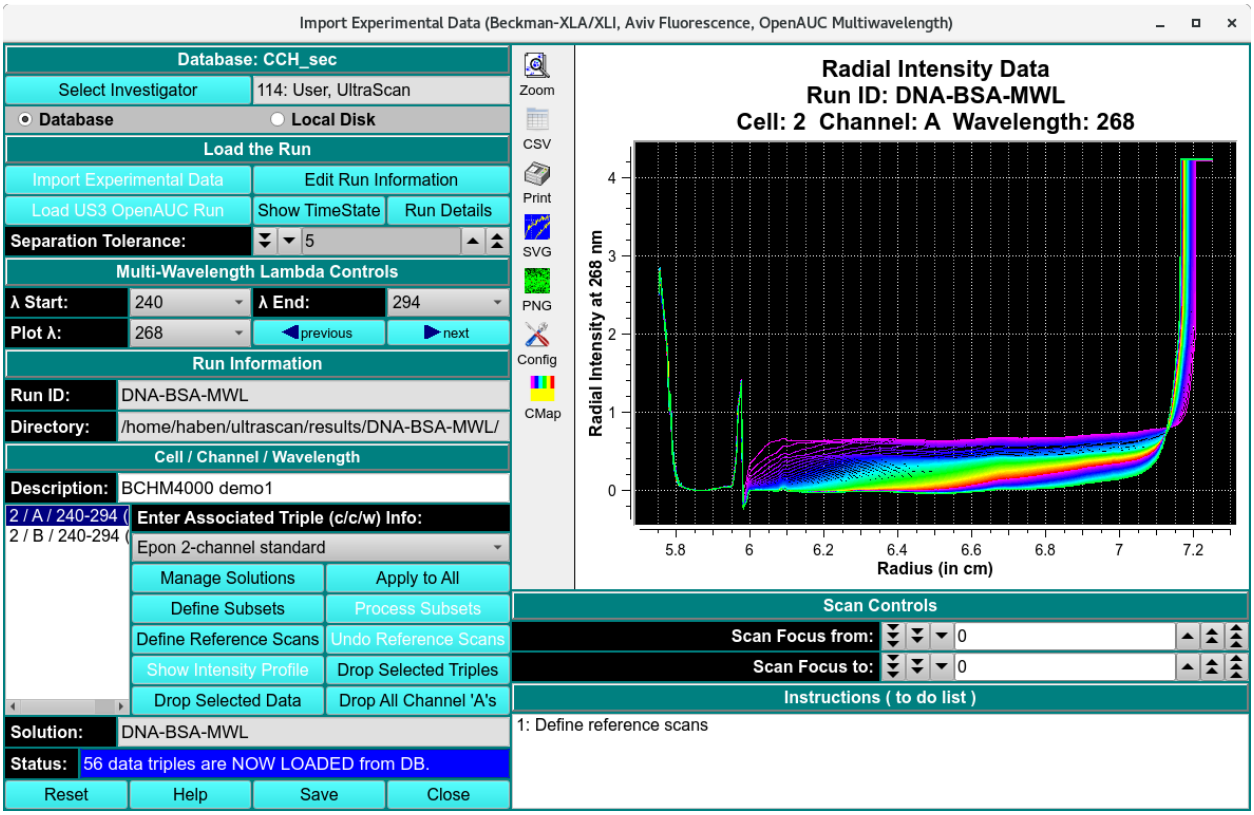
Single wavelength data is listed in the Cell/Channel/Wavelength (c/c/w) format.

Multi-Wavelength data:

Multi-wavelength data is listed in the Cell/Channel/Wavelength range (# of wavelengths) format. The different wavelengths in the dataset can be navigated through using the **Plot λ** drop-down menu and the 'Previous' and 'Next' buttons.

Data can be loaded into the window using one of the following:

- **Import Experimental Data:** Load raw data located on the local disk
- **Load US3 OpenAUC Run:** Load data that has already been converted to the UltraScan-III OpenAUC format using *Data load dialog*. This allows for steps 3 and 5 to be skipped as the information defined in these steps is populated based upon the experiment submission.

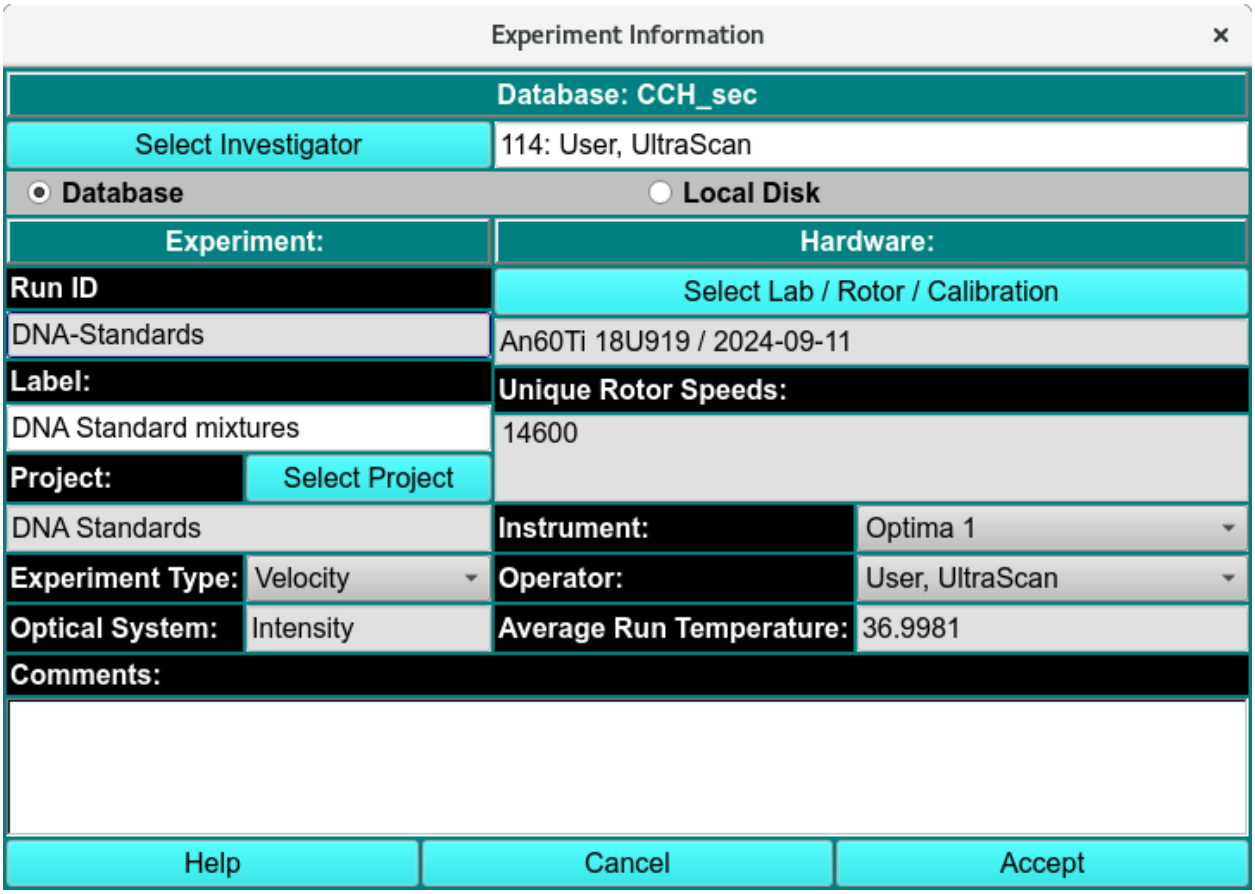


Multi-wavelength Experimental data

23.1.2 STEP 2: Modify Run ID (Optional):

If the run is being reimported, the Run ID must be made unique to avoid deleting the existing imported data, associated edit profiles and models.

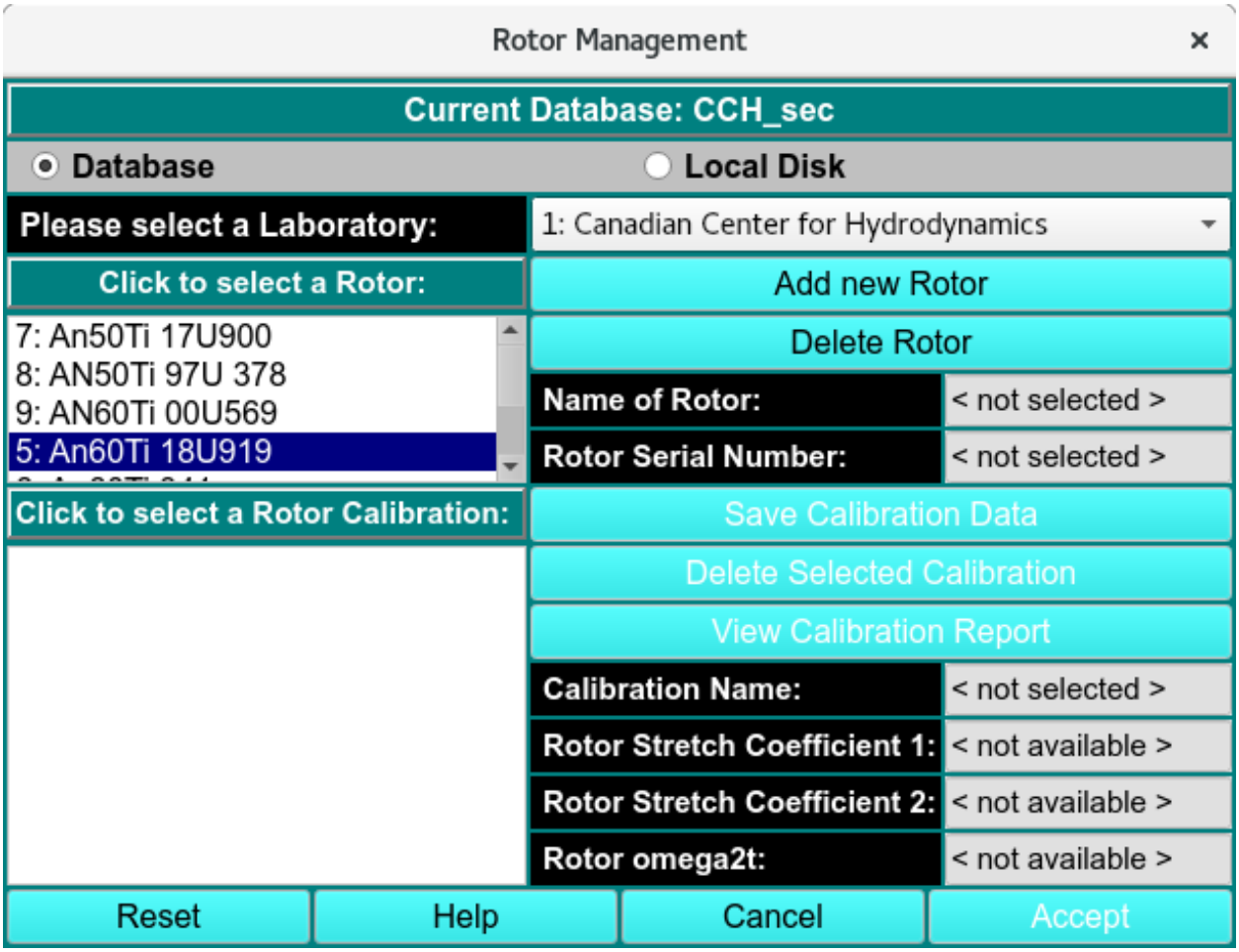
23.1.3 STEP 3: Edit Run Information:



Modify Experiment Information

In the 'Experiment Information' window, a label can be created for the run, the *'Select Project'* button assigns the data to a pre-existing project, the 'Experiment Type' drop-down menu specifies the type of experiment (Velocity, Equilibrium, Diffusion, Buoyancy, Calibration or Other), the 'Instrument' drop-down menu associates the data to a laboratory instrument, and the 'Operator' drop-down menu records the individual who conducted the experiment.

- **Select Lab/Rotor/Calibration:** Allows for the selection of the rotor and associated rotor (stretch) calibration profile.



Edit Run Information

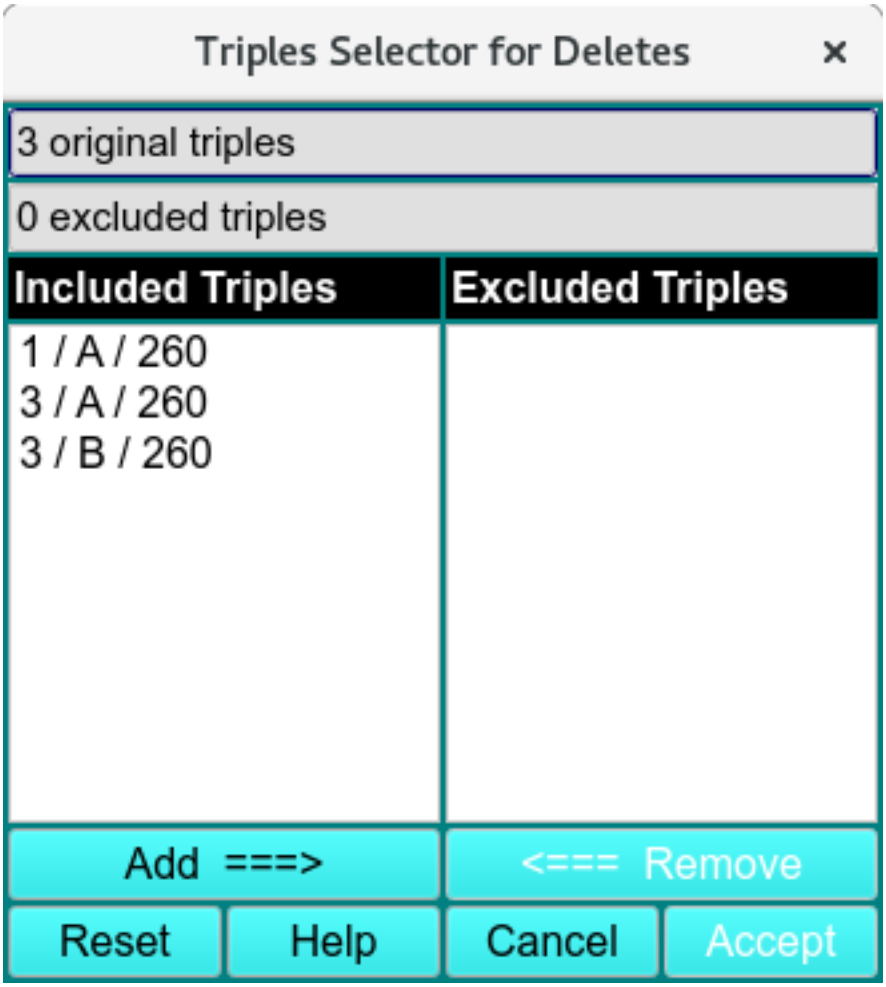
23.1.4 STEP 4: Adjust Separation Tolerance (Optional):

The separation tolerance (0-100) defines the threshold for distinguishing closely-spaced features during analysis. A larger value merges similar features to reduce noise sensitivity, while a smaller value improves resolution but increases noise sensitivity. The default value of 5 is typically sufficient for most datasets.

23.1.5 STEP 5: Modify Cell/Channel/Wavelength (c/c/w) Info:

The **Description** text box allows for notes to be assigned to each channel. Next, the **centerpiece** drop-down menu can be used to assign the centerpiece to each channel. The **Manage Solutions** button opens the *'Solutions Management' <solution/solution_select>* window, where new or pre-existing solutions can be assigned to each channel. **Apply to All** assigns the centerpiece and solution information of the currently channel to all channels.

Dropping Data: 'Drop Selected Data' removes all of the wavelengths for a particular cell and channel. 'Drop All Channel 'A's removes all of the listed A channels when an 'A' channel is selected in the list. 'Drop All Channel 'B's removes all of the listed B channels when a 'B' channel is selected in the list. 'Drop Selected Triples' opens the 'Triples Selector for Deletes' window:



Triples Selector for Deletes

Triples are highlighted on the **Included Triples** list and added to the **Excluded Triples** list using the **Add ===>** button. Triples can be removed from the **Excluded Triples** list using the **<=== Remove** button, and both lists can be set back to default using the **Reset** button. The excluded Triples selection can be confirmed using the **Accept** button.

23.1.6 STEP 6: Saving the data:

Finally, the data can be saved to the database or disk using the **Save** button. If the **Save** button is unavailable, some of the tasks in the **Instructions (to do list)** have not been completed. Completing these tasks will activate the button to allow for saving.

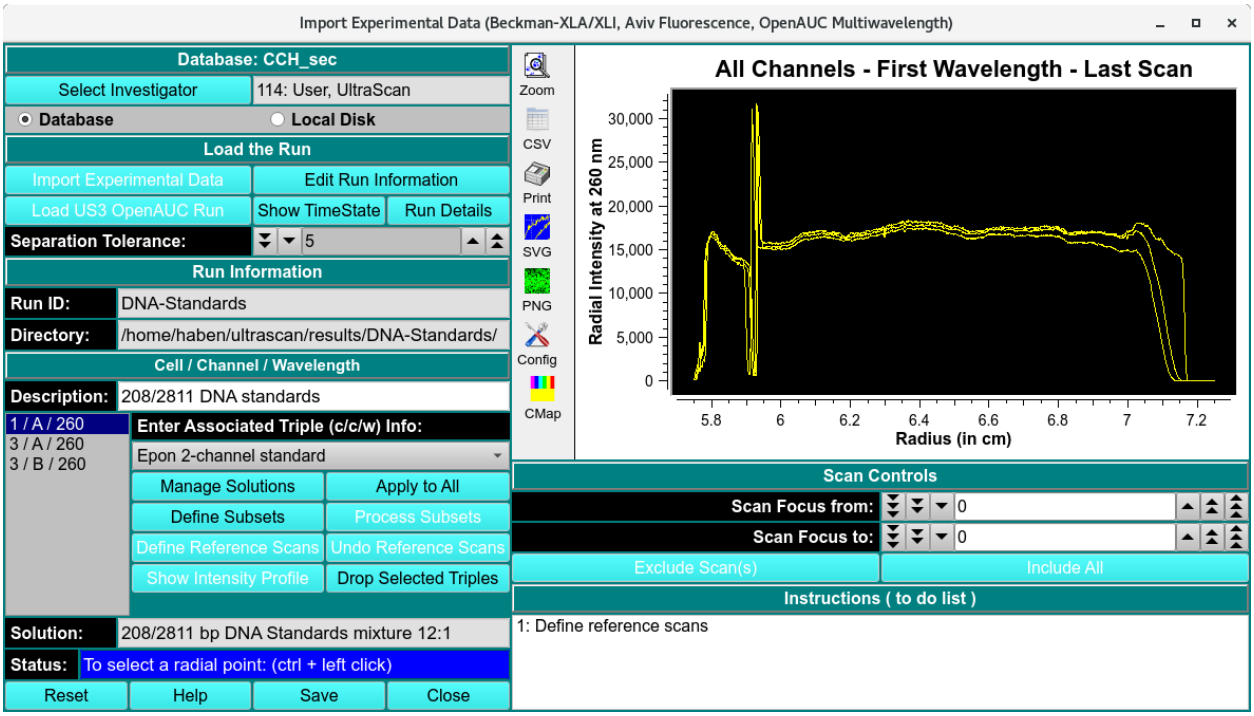
23.2 Processing different data types and datasets

1. Radial intensity (RI) data
2. 6- and 8- channel centerpieces
3. Wavelength data

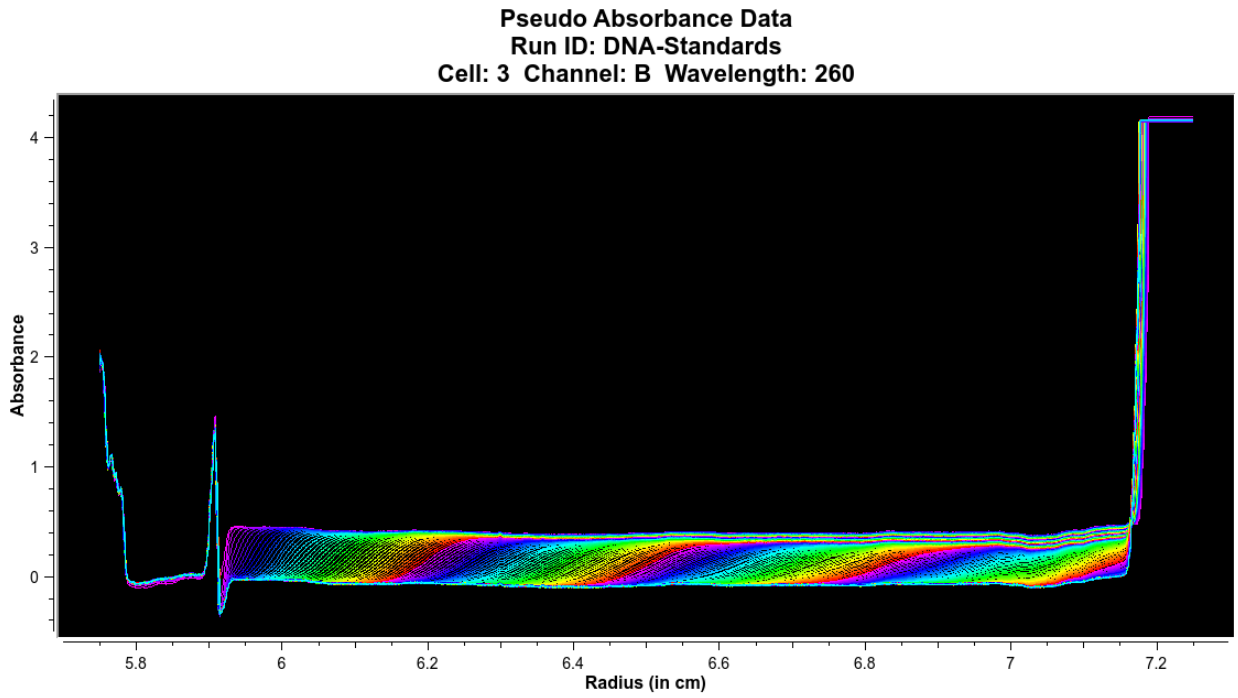
23.2.1 Radial intensity (RI) data:

The **Define Reference Scans** button displays the last scan for the first wavelength of all channels:

CTRL-clicking the centre of the air-gap region above the meniscus defines the reference scans and converts the intensity data for all channels to pseudo-absorbance data:



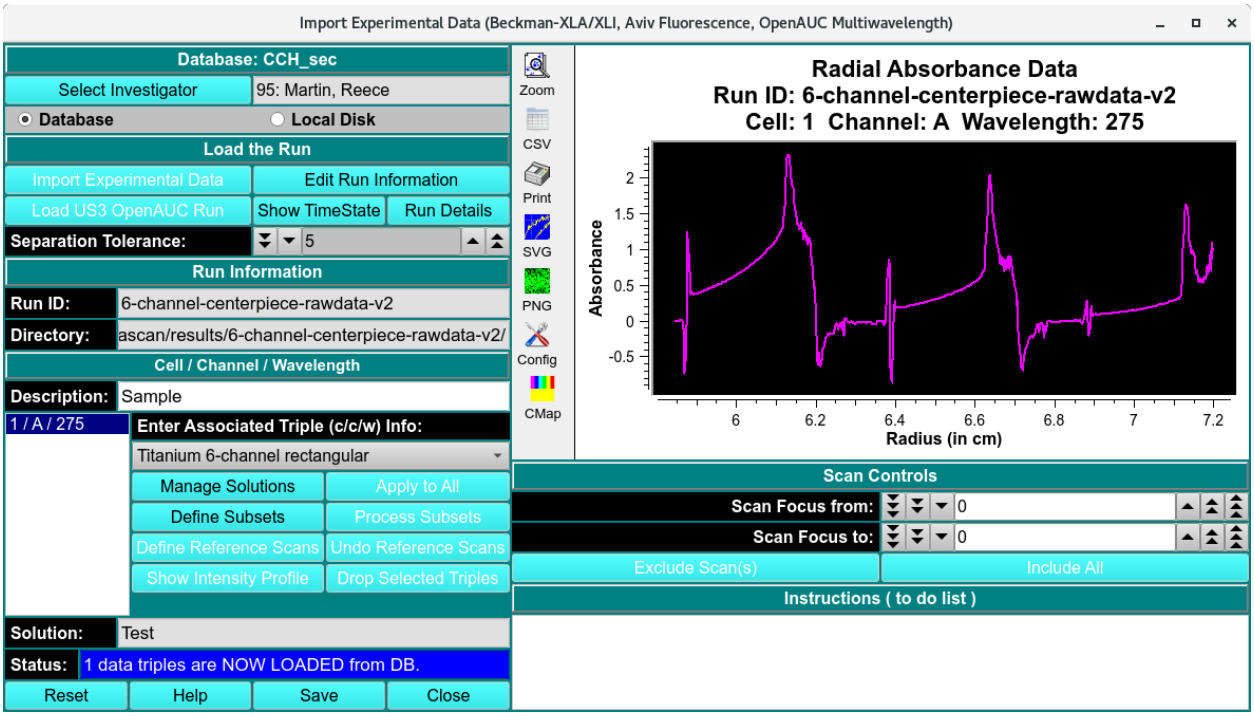
Last Scans of all Channel defined as Reference scans



Pseudo-Absorbance data with Defined Reference Scans

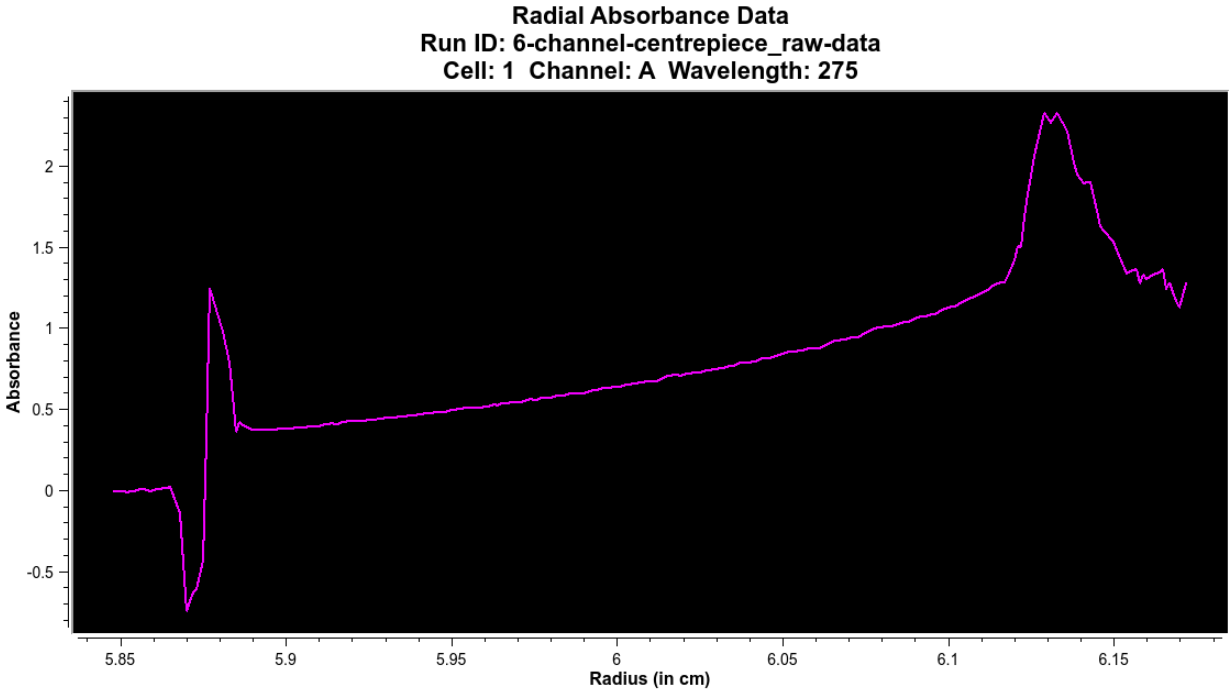
23.2.2 6- and 8- channel centerpieces:

When a 6- or 8- channel centerpiece is used, there will be multiple channels in a single measured channel that need to be separated into individual c/c/w Triples. The 'Define Subsets' button allows for these c/c/w Triples to be defined by CTRL-clicking at the radius that separates the channels. The starting and ending radius for the entire dataset does not be specified as these values are assumed.



Raw and Unprocessed 6- Channel Dataset

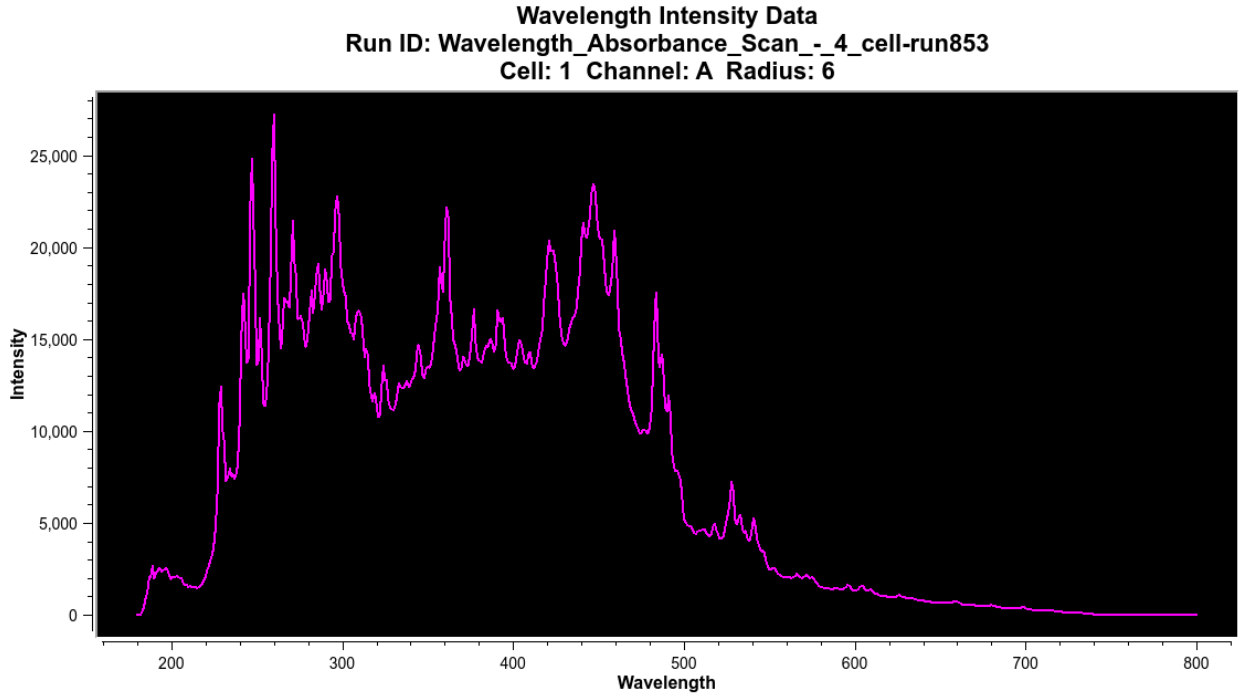
The **Process Subsets** button separates the c/c/w Triples.



Single Separated Channel from 6- Channel Dataset

23.2.3 Wavelength Data:

Wavelength scan data is listed in the Cell/Channel/radial position (c/c/r) format.



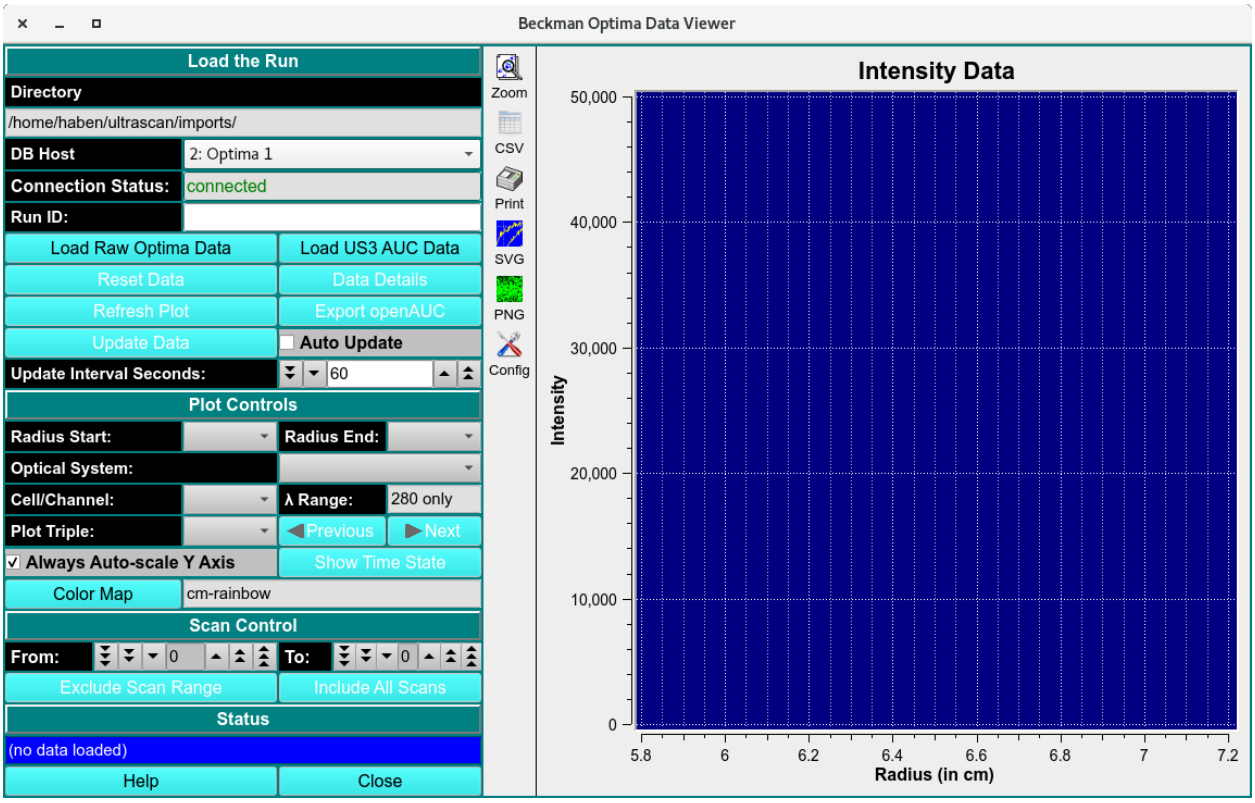
Wavelength Dataset

Triple========== Beckman Optima Data Viewer
==========

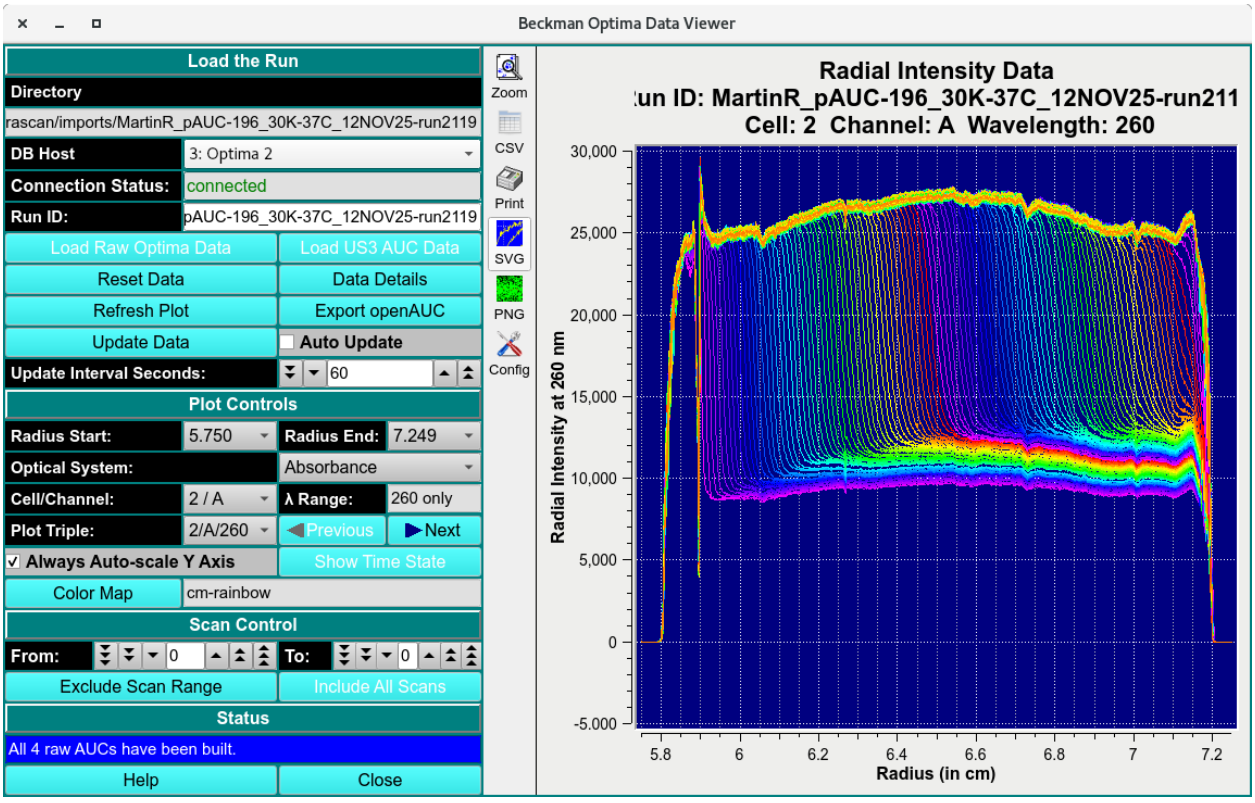
Index

- *Data Viewer Process:*
- *Data Viewer Functions:*
  - *Load the Run*
  - *Plot Controls:*
  - *Scan Controls*
  - *Status*

The **Load Raw Optima Data** allows the user to visualize and load raw data collected from connected Optima instruments via Data Acquisition or GMP modules and export as OpenAUC format. This module functions during and after data collection.



Beckman Optima Data Viewer



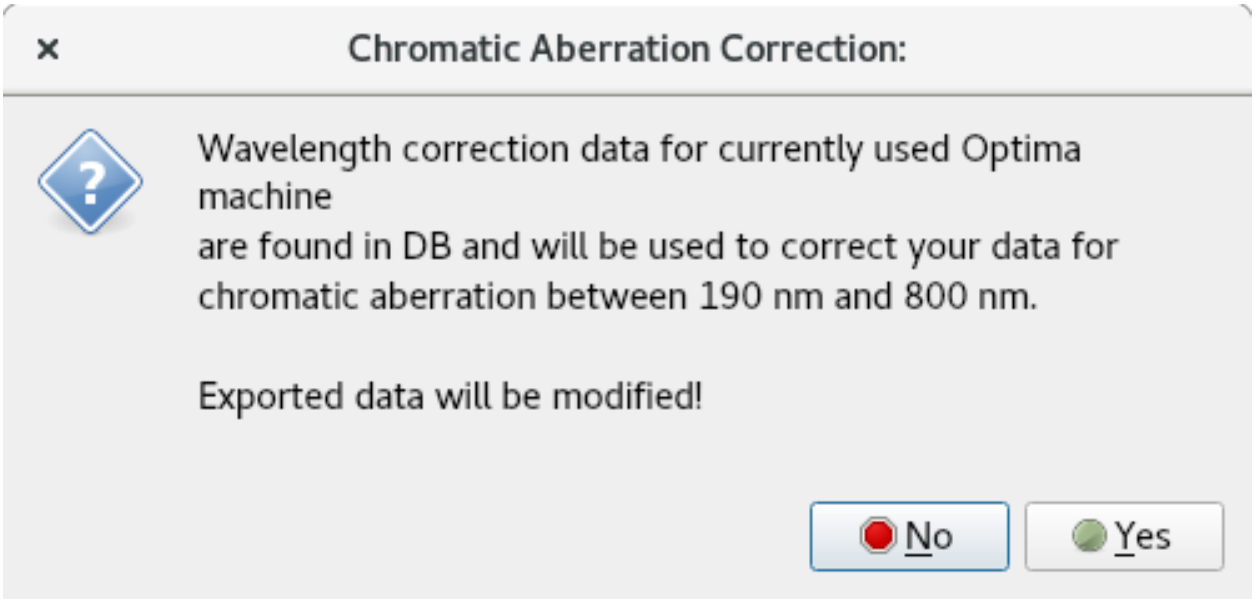
Loaded Data Viewer

CHAPTER

TWENTYFOUR

DATA VIEWER PROCESS:

1. Ensure the instrument is connected by reading a green **connected** text in the **Connection Status:** textbox.
2. Select the instrument from the **DB Host** pulldown menu.
3. Click **Load Raw Optima Data** for raw Optima data or **Load US3 AUC Data** to call a previously imported dataset.



Saved Data

CHAPTER

TWENTYFIVE

DATA VIEWER FUNCTIONS:

25.1 Load the Run

Directory Text Box	The location of the directory file the scans will be saved in.
DB Host	Name of the Optima instrument.
Connection Status:	Status of connection to Optima instrument(s).
Run ID:	Name of the data run id.
Load Raw Optima Data:	open the Raw Optima PostgrSQL database runs to load a run.
Load US3 AUC Data	Load local disk US3 files using **US3 Directories with Optima-derived .auc Files** dialog.
Reset Data	Reset module and remove loaded data.
Data Details	This button brings up a report file *Optima Raw Data Statistics* to that describes additional information associated with the data.
Refresh Plot	Refreshes the Plot settings to default.
Export OpenAUC	Exports .auc files to the directory location.
Update Data	Scans the connected instrument for updated data.
Auto Update	Set the module to automatically Update.
Update Interval Seconds:	Set the time interval of the update.

25.2 Plot Controls:

Radius Start:	Start of data range.
Radius End:	End of data range.
Optical Systems:	Select the type of data collected.
Cell/Channel:	Name of cell and channel.
Lambda Range:	Wavelength range of the data collected.
Plot Triple:	Select the Triple to plot.
Color Map	Load a color map generated to change the default color schema.

25.3 Scan Controls

From:	Start of scans
To:	End of scans
Exclude Scan Range:	Set range of scans to exclude.
Include All scans	Include all excluded scans.

25.4 Status

Description textbox	Module action status box.
Help	The Help Documentation of this module.
Close	Close window.

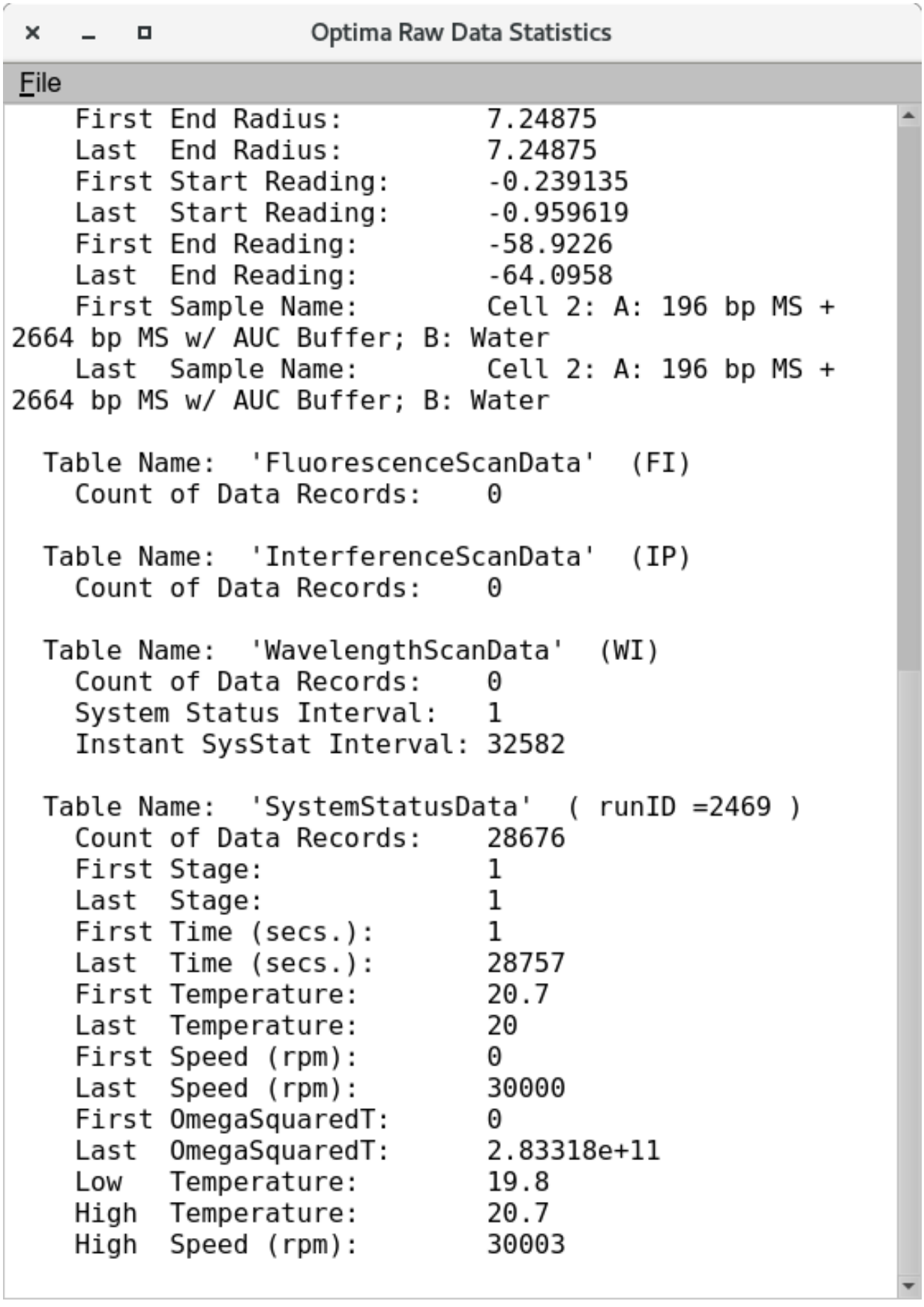
Optima Raw Data Statistics

File

```
Optima Data Statistics for RunID "MartinR_OPT1-Res-
Test-30K_20C_19FEB26-run2469",
 from Database at Host:Port demeler5.uleth.ca:5551 .
Data Summary.
    Triples Count:            6
    Cell/Channel Count:       2
    Radius Points:            1500
    Max. Scan/Triple Count:   200
    Total Scans Count:        1200
    Wavelengths Count:        3
    System Status Interval:   1

ScanData Tables.

  Table Name:  'AbsorbanceScanData'  (RI)
    Count of Data Records:    1200
    First Seconds:            133
    Last  Seconds:            28756
    First Stage:              1
    Last  Stage:              1
    First Scan Number:        1
    Last  Scan Number:        200
    First Wavelength:         255
    Last  Wavelength:         265
    First Temperature:        20.3
    Last  Temperature:        20
    First Speed (RPM):        30000
    Last  Speed (RPM):        30000
    First OmegaSquaredT:      8.09205e+08
    Last  OmegaSquaredT:      2.83308e+11
    First Value Count:        1500
    Last  Value Count:        1500
    First Start Radius:       5.75
    Last  Start Radius:       5.7502
    First End Radius:         7.24875
    Last  End Radius:         7.24875
    First Start Reading:      -0.239135
    Last  Start Reading:      -0.959619
```


Optima Raw Data Statistics
File
First End Radius: 7.24875
Last End Radius: 7.24875
First Start Reading: -0.239135
Last Start Reading: -0.959619
First End Reading: -58.9226
Last End Reading: -64.0958
First Sample Name: Cell 2: A: 196 bp MS + 2664 bp MS w/ AUC Buffer; B: Water
Last Sample Name: Cell 2: A: 196 bp MS + 2664 bp MS w/ AUC Buffer; B: Water
Table Name: 'FluorescenceScanData' (FI)
Count of Data Records: 0
Table Name: 'InterferenceScanData' (IP)
Count of Data Records: 0
Table Name: 'WavelengthScanData' (WI)
Count of Data Records: 0
System Status Interval: 1
Instant SysStat Interval: 32582
Table Name: 'SystemStatusData' (runID =2469)
Count of Data Records: 28676
First Stage: 1
Last Stage: 1
First Time (secs.): 1
Last Time (secs.): 28757
First Temperature: 20.7
Last Temperature: 20
First Speed (rpm): 0
Last Speed (rpm): 30000
First OmegaSquaredT: 0
Last OmegaSquaredT: 2.83318e+11
Low Temperature: 19.8
High Temperature: 20.7
High Speed (rpm): 30003

Data Details

CHAPTER

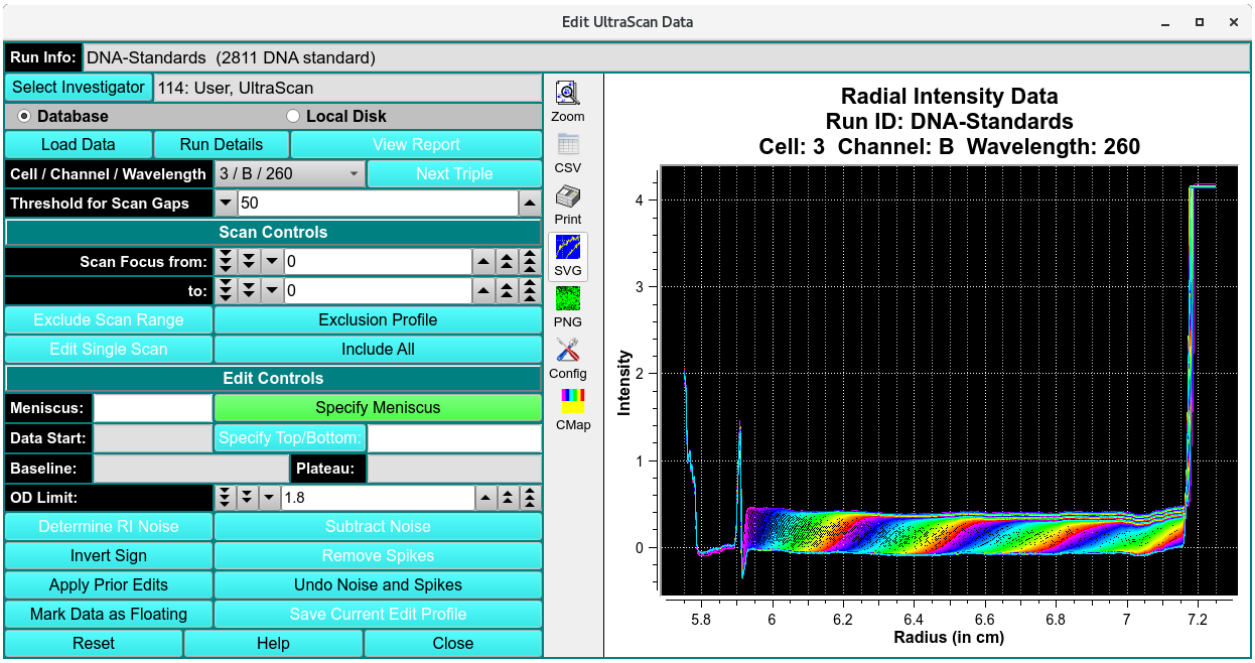
TWENTYSIX

EDIT ULTRASCAN DATA

Index

- *Edit Functions:*
- *Editing Steps*
- *Edit Multi-Wavelength (MWL) Data*
  - *Functions specific to MWL:*
  - *Editing steps specific to MWL*
- *Edit Specific Scans*
- *Exclusion Profile*

All experimental data acquired with the analytical ultracentrifuge must be processed into an edit profile before analysis can be performed using UltraScan-III modules. For velocity experiment, the edit profiles define key analysis parameters, including the meniscus and data range, and scans selected. A user can choose to create multiple edit profiles for a particular data set, for example, to explore the effect of various editing steps on the analysis results. For example, a user may wish to exclude either early or late scans to track time dependent changes in the sample. Each profile will receive a unique name automatically based on the time at which the editing process was conducted or a user-defined name. For data analysis, any associated edit profile may be selected at run time to provide a custom data selection.



Edit Data Main Window

26.1 Edit Functions:

Run Info	This text box displays run information for the currently selected run.
Select Investigator	This button brings up a window that allows selecting the current investigator for input from the database.
Database	Check to specify data input from the database.
Local Disk	Check to specify data input from local disk.
Load Data	Click here and, in the resulting Load Data Dialog, select raw data set(s) to load for editing.
Run Details	This button brings up a child window *Details for Raw Data* to that describes additional information associated with the data. The data description may be updated in this window.
Cell / Channel / Wavelength	Select the Triple for which to perform editing.
Threshold for Scan Gaps	This spin button controls the check for consecutive interpolated data. If more than the specified number of consecutive points on a scan have been interpolated, a warning is displayed when the data is loaded. When editing wavelength data, this control specifies the gap in wavelengths

Scan Control

Scans to be included in analysis, exluded scans are note deleted.

Scan Focus from:	Set the counter to the lower bound of scans to exclude.
Scan Focus to:	Set the counter to the upper bound of scans to exclude.
Exclude Scan Range	Exclude the scans indicated by the two preceding scan bounding counters.
Exclusion Profile	This button displays a window where additional options for scan exclusion are available. Here it is possible to save only every 2nd, 3rd or nth scan.
Edit Single Scan	This button displays a window where individual points may be changed by a Control-Left_Click-Drag operation.
Include All	This button removes all data and sets the program to a known starting point.

Edit Control

Specify Meniscus	This button enables the edit range selection step to be set to specify the meniscus via Ctrl-Click on the meniscus position. The radius value of the selected point is displayed in the text box tothe right of the button.
Specify Data Range	This button enables the edit range selection step to be set to specify the data range via Ctrl-Click's on the start and end data positions. The radius values of the selected points are displayed in the text box to the right of the button.
Specify Plateau	This button enables the edit range selection stepto be set to specify the plateau via Ctrl-Click on the plateau position.The radius value of the selected point is displayed in the text box to the right of the button.
Baseline	The text box displays the radius value and average OD value of a baseline automatically computed after the plateau is chosen.
OD Limit:	Select an upper limit on OD values in the data above which data is excluded from 2-Dimensional Spectrum Analysis computations.
Determine RI Noise	This button analyzes all scans to determine any radial invariant noise. It brings up a window that shows polynomial coefficients for a best fit of the scan integral. Residuals to this integral represent radially invariant noise. Subtracting the residual values from each data point in a scan will correct radially invariant noise. It is important that the loss of mass in the data as a function of time is properly matched to an appropriate polynomial, so select a polynomial degree thatprovides the best overall fit to all integral values.
Subtract Noise	This button removes the calculated noise from the current data set.
Invert Sign	This button inverts the measured data.
Remove Spikes	This button determines if data readings are beyond acceptable limits and adjusts outlying data points to an interpolated
Apply Prior Edits	This button requests the user to specify a saved set of edit operations and applies them to the current data.
Undo Noise and Spikes	This button removes any corrections made to the data by noise or spike calculations.
Mark Data as Floating	This button sets a flag to mark the data as floating.
Save Current Edit Profile	This button initiates a save of the edited data to local file(s) and/or database.

Window Controls

Reset	This button removes all data and sets the program to a known starting point.
Help	Display this help.
Close	Close the editing session.

26.2 Editing Steps

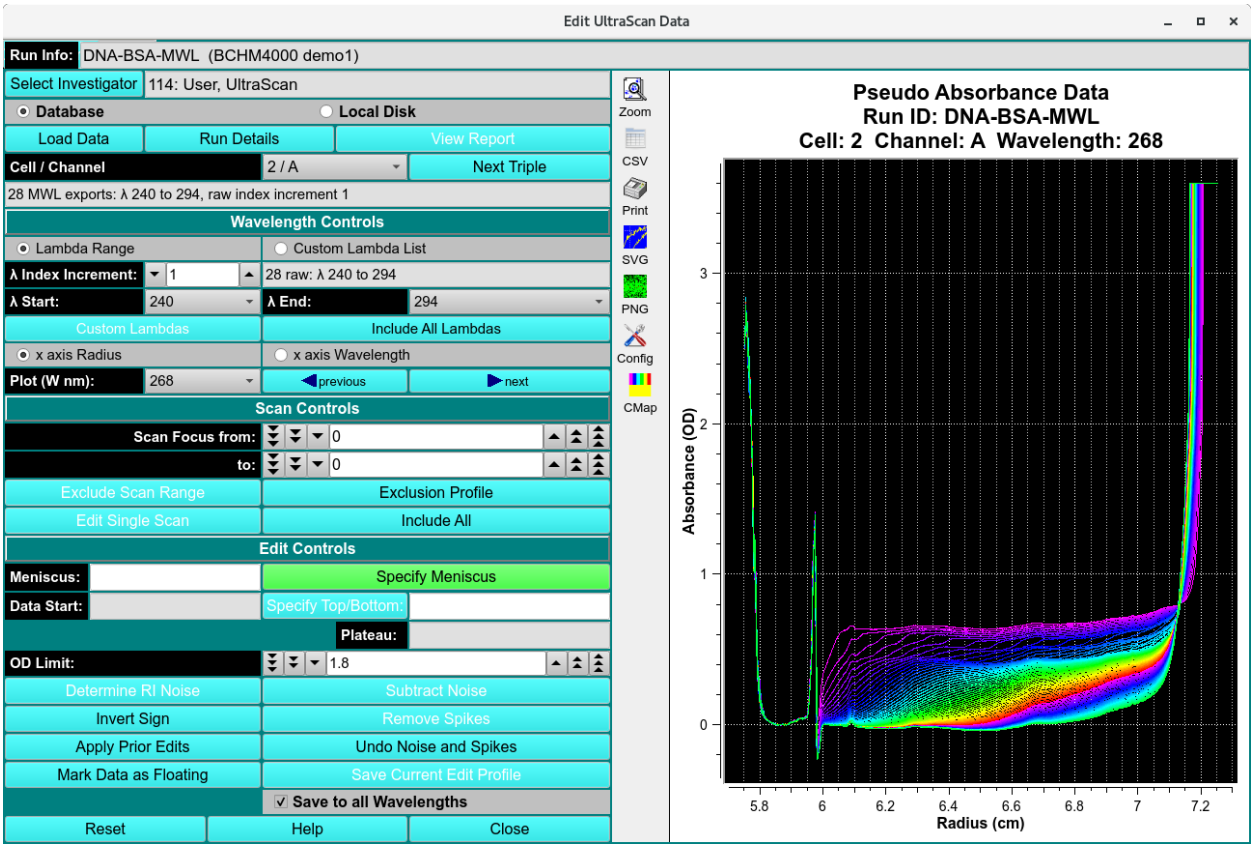
Note: When editing, the plot may be zoomed by selecting the Zoom button on the graph and dragging the mouse over the area desired. This operation may be repeated as needed. Each right mouse click will un-zoom one level. Clicking on the Zoom button again at any zoom level will restore the un-zoomed graph.

When zoomed, the graph may be panned by pressing the center mouse button (usually the scroll wheel) and dragging.

- **Step 1:** Load data from the database or a local data directory that contains the UltraScan-III data files previously converted from the Beckman raw data. Load choices are made in a **:ref:`Load Data Dialog <fe-data-loader>`_**.
- **Step 2:** Select the Cell / Channel / Wavelength Triple to be edited.
- **Step 3:** Specify the meniscus of the data by holding down the Control key and using the left mouse button. The meniscus value may be manually adjusted with the keyboard.
- **Step 4:** If the data were collected with the interference detector, specify the left and right edges of the air gap area of the data.
- **Step 5:** Specify the left and right edges of the data to be analyzed. Please note: Do not pick the left data edge too close to the meniscus. During meniscus fitting, the evaluated meniscus positions may reach inside of the data range and violate the boundary conditions of the finite element solution. This will cause the meniscus fit to fail.
- **Step 6:** Specify the location of the scan plateau. This is the radial position where most scans have a stable plateau, but the selected position should not reach into the back-diffusion region. The most appropriate point tends to be close to the right edge of the data range, but not so far to the right that it extends into the region where the concentration of the later scans curves upward at the bottom of the cell due to back-diffusion.
- **Step 7:** Make any other optional adjustments to the data that are necesary and save the edit profile. When saving, a pop-up message is presented asking for an edit ID. The default for this ID is the current date and time in the form of YYMMddhhmm (Year / Month / Day / Hour / Minute), but this default can be supplemented with a suffix of your own choice.
- **Step 8:** Repeat the above process for each Triple (Cell / Channel / Wavelength combination) in the data set.

Note: The above process may be reset to any point by pressing the appropriate button at the associated specification entry.

26.3 Edit Multi-Wavelength (MWL) Data



Where the data loaded in an editing session is comprised of multiple wavelengths (more than two), the main editor window is changed to the above form. Instead of selecting Triples to edit, this MWL edit procedure consists of selecting Cell / Channel pairs and the wavelengths within those doubles. Once editing in the normal way has been completed, the user may save the edit for either the specific Triple ("Save Current Edit Profile") or for all of the wavelengths of the current double ("Save to all Wavelengths").

Most of the edit controls for MWL are the same as for the standard *Data Editor Window*; and are not repeated here. The steps and control objects described herein are those specific to MWL processing.

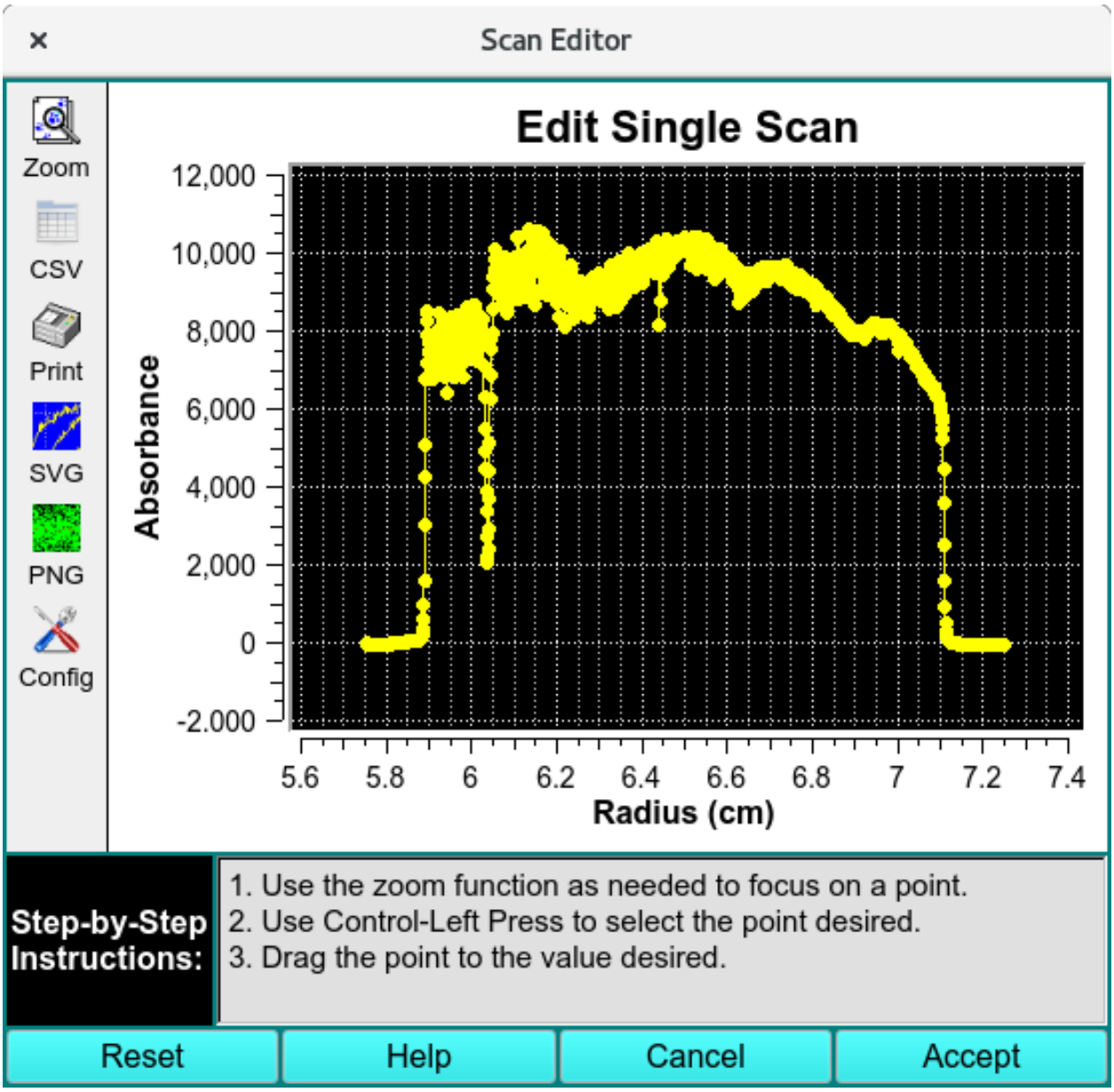
26.3.1 Functions specific to MWL:

Cell / Channel	This button removes all data and sets the program to a known starting point.
Lamdba Range	Display this help.
Close	Close the editing session.
Cell / Channel	Select the double for which to perform editing.
Lamdba Range	Select this radio button to indicate that wavelength selection will be by start/end/increment specification.
Custom Lambda List	Alternatively, select this radio button to indicate that wavelengths will be individually indicated in a list dialog.
λ Index Increment:	Select the wavelength range increment.
λ Start:	Select the wavelength range start value.
λ End:	Select the wavelength range end value.
Custom Lambdas	Click to bring up a list dialog in which individual output wavelengths can be enumerated.
Include All Lambdas	Reset the implied output wavelength range or list to the full set of loaded dataset values.
x axis Radius	Select this radio button to indicate that the plots of data should have an X axis of Radius (the default).
x axis Wavelength	Select this radio button to indicate that the plots of data should have an X axis of Wavelength; and, therefore, that each plot is for a selected Radius.
Plot (W nm):	Select the Wavelength (in NM) or (if the title of this button is "**Plot (R cm):**" when the "**x axis Wavelength**" radio button was selected) select the Radius (in CM) to plot and edit.
(left arrow) previous	Click to change the plot to the one for the previous Wavelength or Radius value.
(right arrow) next	Click to change the plot to the one for the next Wavelength or Radius value.
Save Current Edit Profile	Click this button (in the MWL case) to save the current edit for the current Cell / Channel / Wavelength only.
Save to all Wavelengths	Click this button to save the current edit for all wavelengths of the currently selected Cell / Channel.

26.3.2 Editing steps specific to MWL

- **Step 1:** Load data from the database or a local data directory. Select an entire runID so that MWL data is recognized as such.
- **Step 2:** Select the Cell / Channel to be edited; specify the Lambda (wavelength) range to be edited; and select a single wavelength for display and edit.
- **Step 3:** Specify editing parameters - such as Meniscus, Data Range, and Plateau - in the normal Control-Click way.
- **Step 4:** Save editing for either the specific Triple selected or to be applied to all wavelengths of a cell / channel double.

26.4 Edit Specific Scans

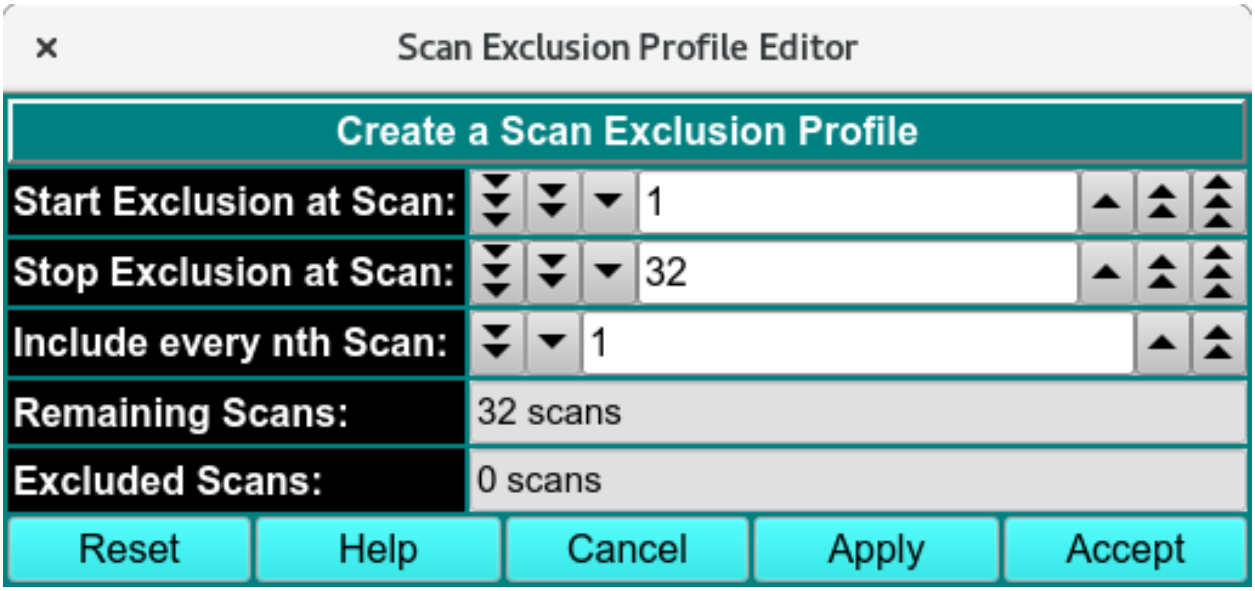


Edit Individual Scans

1. Use the zoom function as needed to focus on a point.
2. Use Control-Left Press to select the point desired.
3. Drag the point to the value desired.

26.5 Exclusion Profile

Exclude scan ranges and intervals with this dialog.



Exclude Scans

Triple== Fluorescence Data File Manager ==

Index

- *Process:*

This module allows the user to manage scan files created by the AOS Fluorescence Detector data acquisition software.

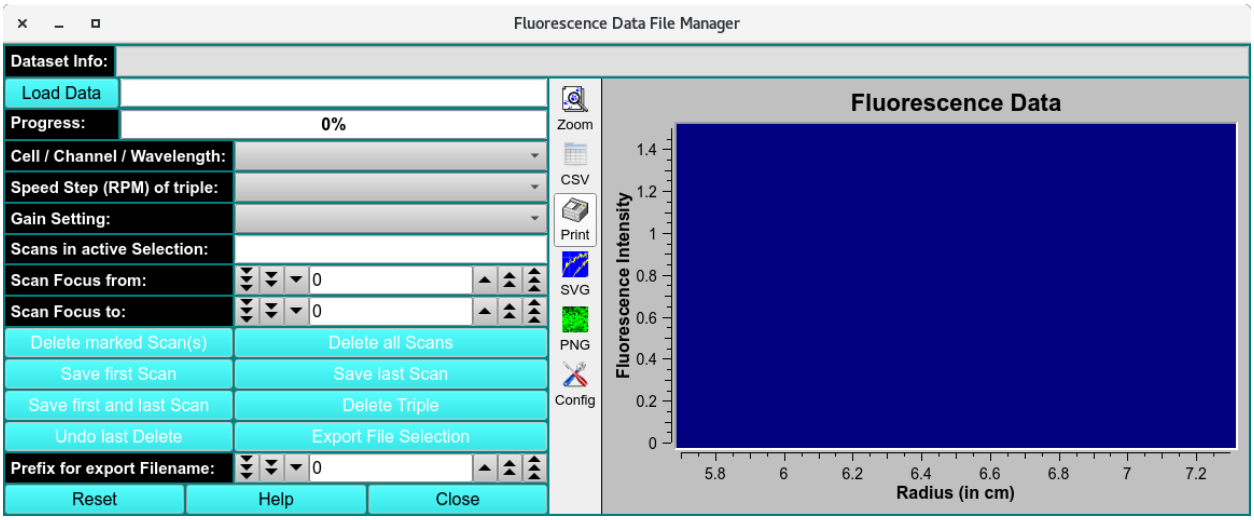
1. The AOS data acquisition software outputs scan files with a range of different naming conventions to accommodate different cells, channels, and importantly, different gain or voltage settings.
2. When experimental data are acquired with a singe gain setting, the output files are names A00001.FI1, A00001.FI2, … for channel A, and B00001.FI1, B00001.FI2, … for channel B. When multiple gain settings are used to scan the data, a prefix (1_A00001.FI1, 2_A00001.FI1, …) is added, incrementing the number of the prefix by one for each gain setting.

This procedure allows the user to distinguish between scans for different gain settings, but this requires renaming so that a run can be imported into the OpenAUC format. UltraScan-III recognizes only scan filenames without the prefix.

This module allows the user to select files with gainsets, speeds, and subsets of scans that are supposed to be combined into a logical run, or experiment.

Note: Depending on the experiment performed, it may be necessary to combine different gains from different speeds into a single run (for example, for equilibrium experiments, where different gains may need to be used to measure different gradients evolving at different speeds, where low speeds produce shallow gradients (requiring a higher gain setting) and higher speeds form steeper gradients that may require a lower gainset for optimal dynamic range). The program allows the user to export all selected Triples, speeds, gain settings and scans to a new directory, where the selected scans are saved with the standard file naming convention used for single gain settings.

The program can also be used to merge scan files from arbitrary many directories into a single directory to create a combined run directory suitable for a single data import action in UltraScan-III.



Fluorescence File Manager

The input to this module consists of fluorescence data only, scan files from other optical systems are ignored.

CHAPTER

TWENTYSEVEN

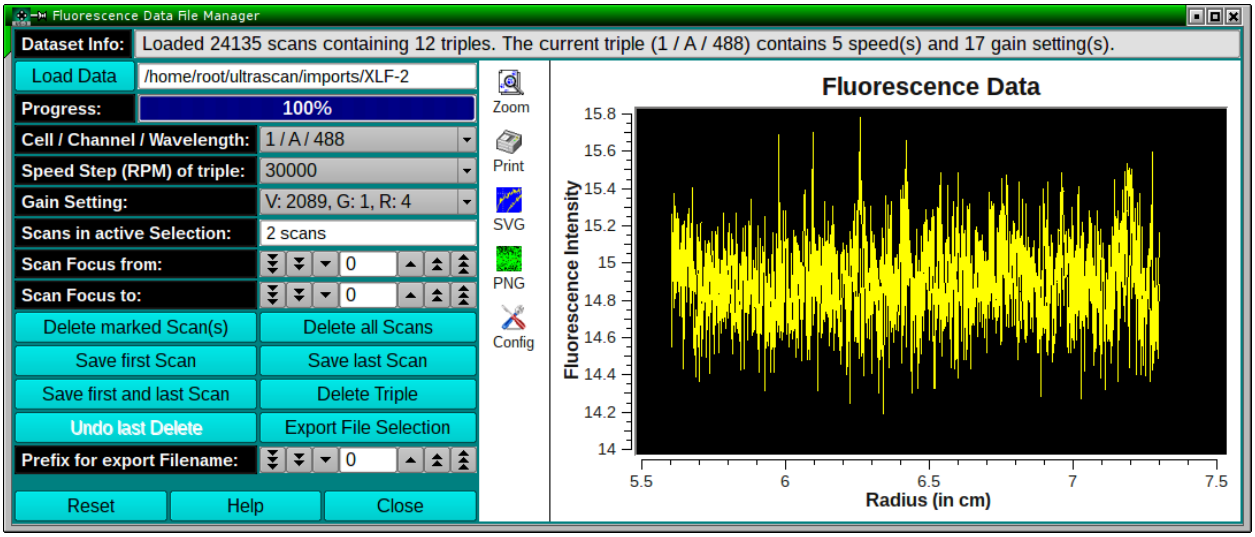
PROCESS:

1. **Load Data:** Load fluorescence data by clicking on **Load Data**.

A file manager dialog that allows user to navigate to the directory containing the desired fluorescence data will appear. You do not need to select any files at this stage, only the directory that contains all the desired scans needs to be selected.

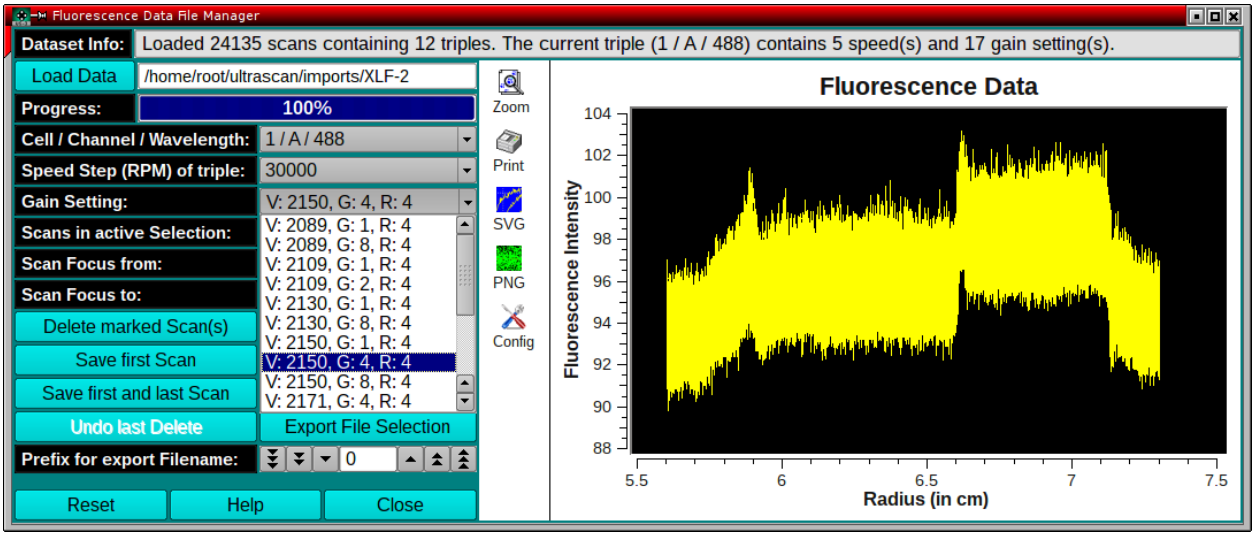
Note: Only one directory can be chosen at this stage.

Once loaded, the **Dataset Info** field will indicate a summary of the number scans in this directory, the number of Triples, speeds and gain settings for this directory. All scans for the first Triple, speed and gain setting will be plotted in the graph window:



Loaded Fluorescence File Manager

Once loaded, the listboxes will be populated with all Triples, speeds and gain settings. The gain settings will be listed if they differ by voltage, gain set or range:

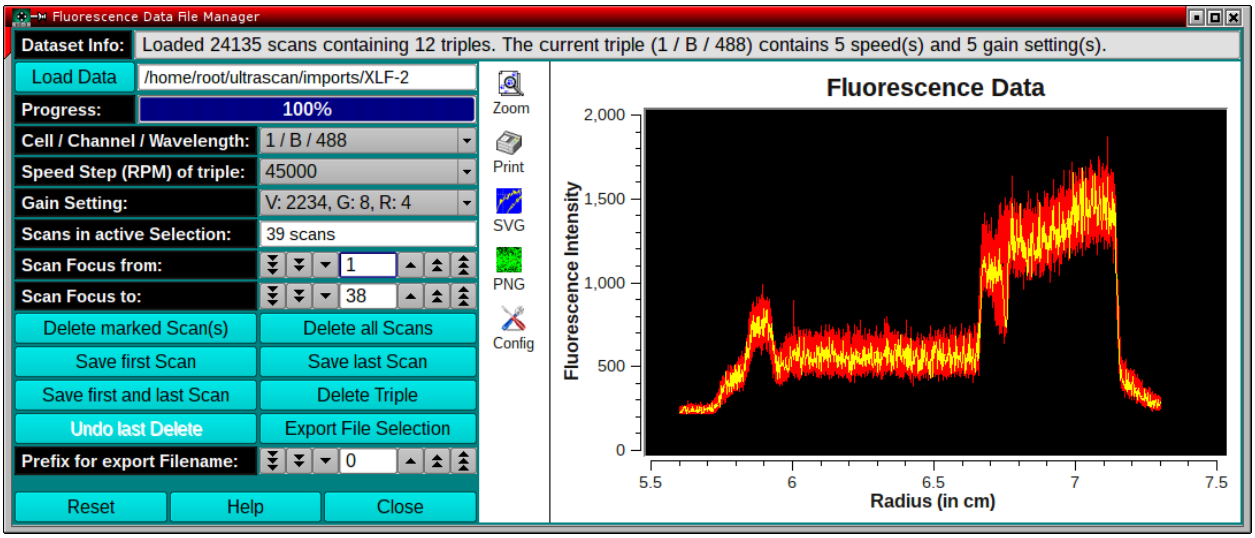


Parameter Details Listed

Tip: If every Triples, speeds or gain settings are listed, a quick review is possible by simply clicking with the mouse on any listbox item and scrolling through the list by holding down an arrow key button.

---

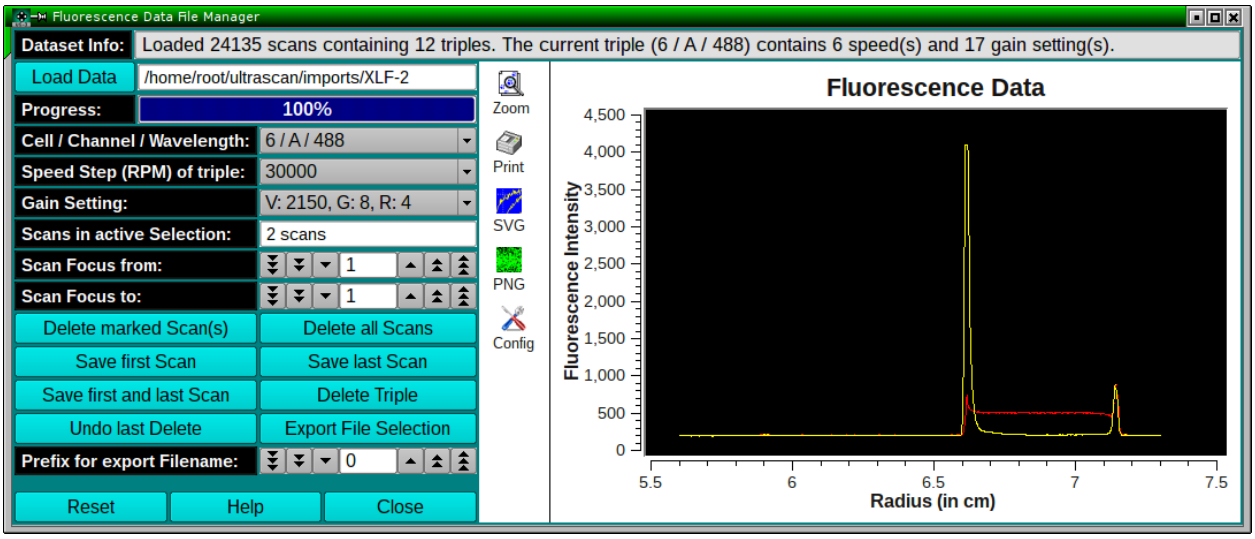
2. **File Exclusion:** There are multiple ways to exclude files before sorting all files into a single directory. For example, if the channel leaked, or the data are unusable for some other reason, **Delete Triple** will remove the entire Triple from the list. Alternatively, a scan file selection allows the user to save a subset of scans. Scans marked for exclusion will be marked in red, all others in yellow. To perform the exclusion, click on **Delete marked Scan(s)**:



Exclude Scans

Additional convenience functions are available to quickly delete a subset of scans. **Delete all Scans** will remove all scans for the active Triple, speed, and gain setting selection, but not other scans from this Triple. To save only the first scan of a selection, click on **Save first Scan**. The first scan may be needed for cases where the meniscus has to be determined visually. Similarly, **Save last Scan** will only save the last scan of the selection

shown on the screen. This is useful for picking the last scan in an equilibrium experiment, which is presumably closest to equilibrium. **Save first and last Scan** will save both:



Exclude Scans

3. **Exporting selected Files:** Once all scans that are not desired have been marked for deletion, the remaining scans can be reviewed by selecting the desired Triple, speed and gain setting, and then be exported by clicking on **Export File Selection**. A new target directory should be created where the file selection can be written with sorted filenames suitable for import into US3. All exported files will be written with standard filenames of the format A00001.FI1, A00002.FI1, …

4. **Merging Files from multiple directories:** To combine files from different directories (for example, different speeds from an equilibrium experiment that was collected with multiple methods), you can use the following procedure: First, load the data from the directory that contains the earliest experimental data. Process all files as needed using the scan exclusion routine shown above, keeping only the desired scans. Select a prefix for this experiment where the magnitude of the prefix number corresponds to the time of the experimental dataset. Files will later be sorted based on the filename, with files containing prefix **1_** sorted before files with prefix **2_**, and so forth. If you have more than 10 different files, you need to start with a prefix **10_** or larger, because **10_** is sorted before **2_**. Accumulate all runs into a single, newly created directory. Once merging all files into this directory, use this module once more to convert and merge all files with prefixes into a single new directory without prefixes. The second newly created directory will then contain the merged files suitable for import into UltraScan-III.

CHAPTER

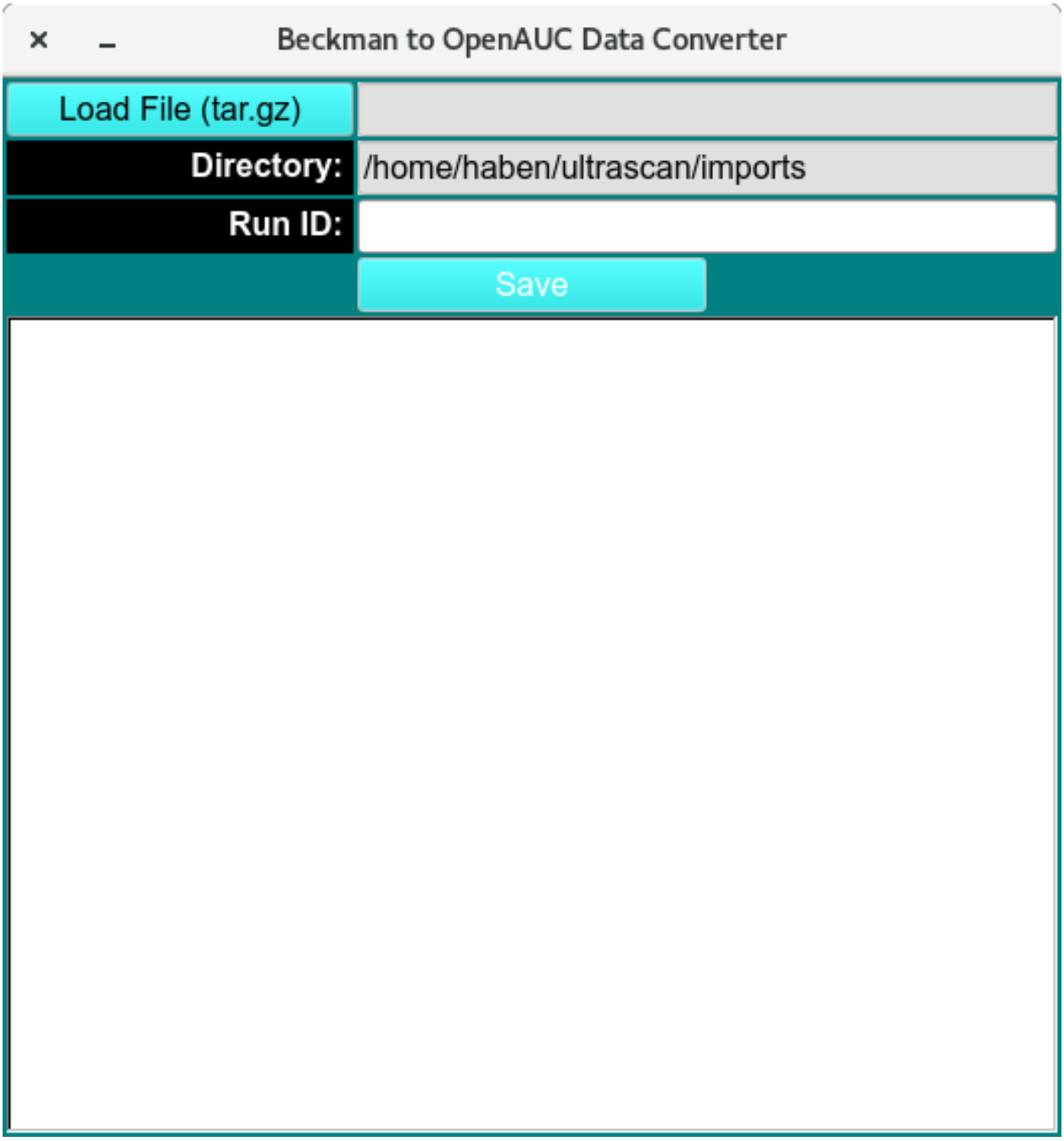
TWENTYEIGHT

BECKMAN TO OPENAUC DATA CONVERTER

Index

- *Process:*

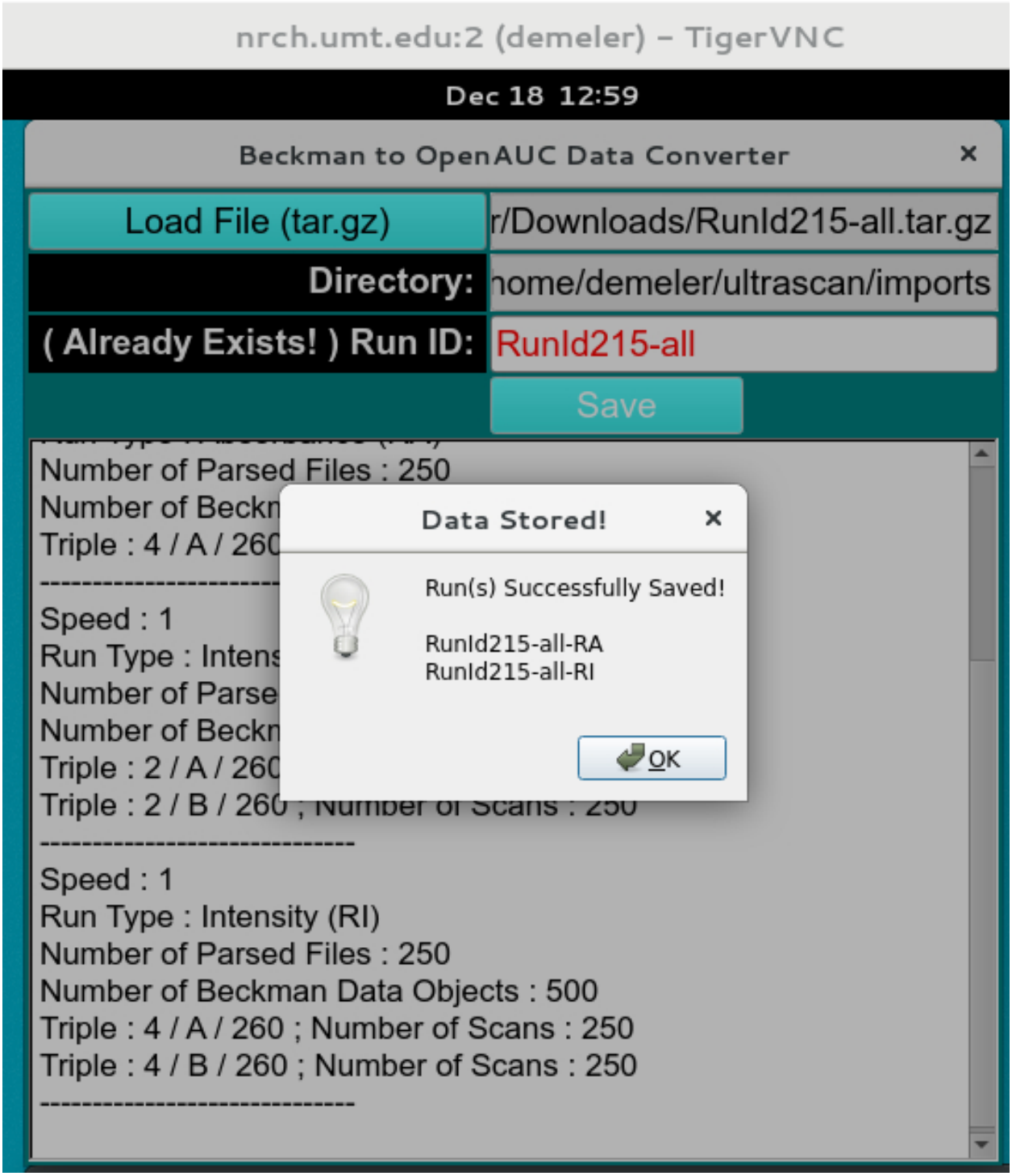
The Beckman to OpenAUC converter module converts legacy formatted (.tar.gz) files generated by the Beckman web application for import and conversion into OpenAUC formatted files.



Beckman (.tar.gz) to OpenAUC Data Converter

28.1 Process:

1. Click on "Load File (tar.gz) in the GUI dialog and navigate to the $HOME/Downloads/RunID*.tar.gz file.
2. Click on "Save"



Converted Beckman (.tar.gz) Data

Triple========== Export Data in Legacy Format:
==========

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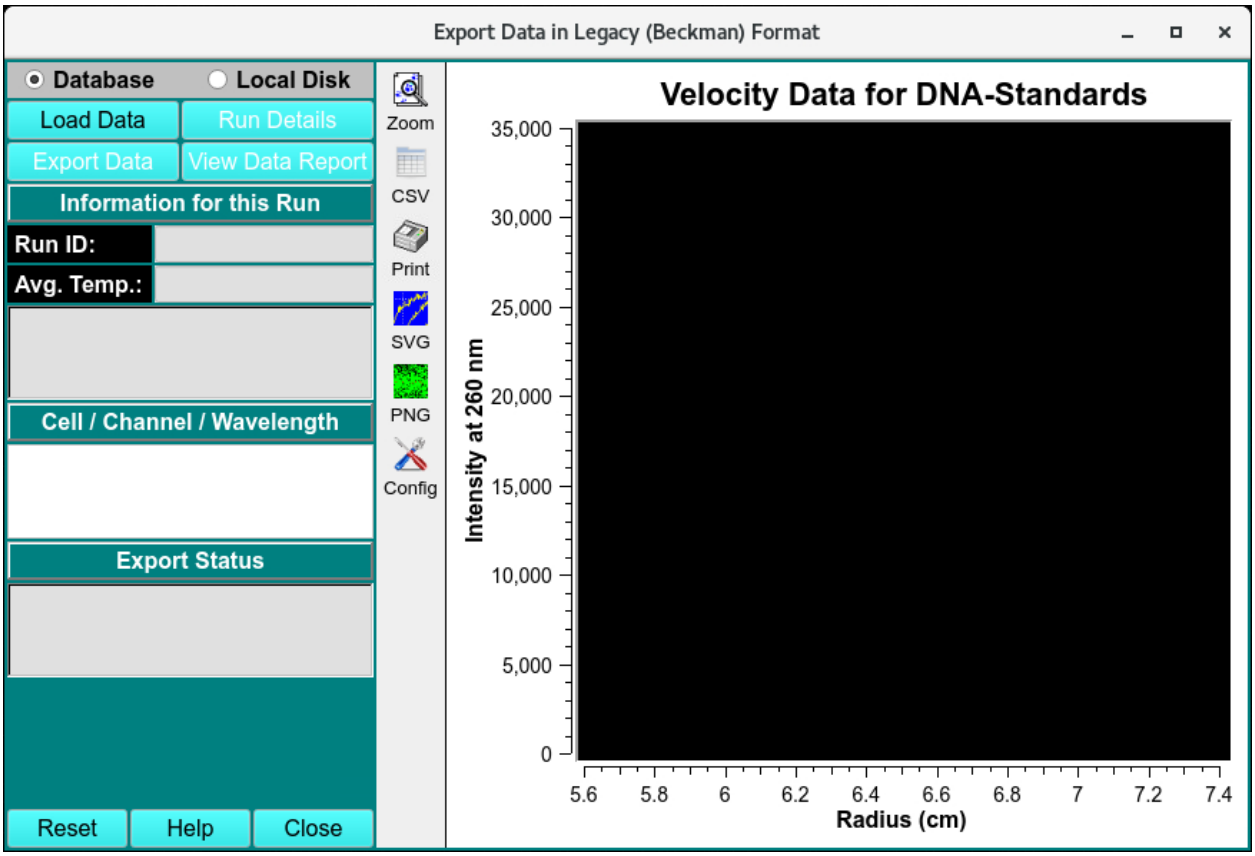
- *Beckman (tar.gz) to OpenAUC Data Converter:*
- *OpenAUC Converter Functions:*

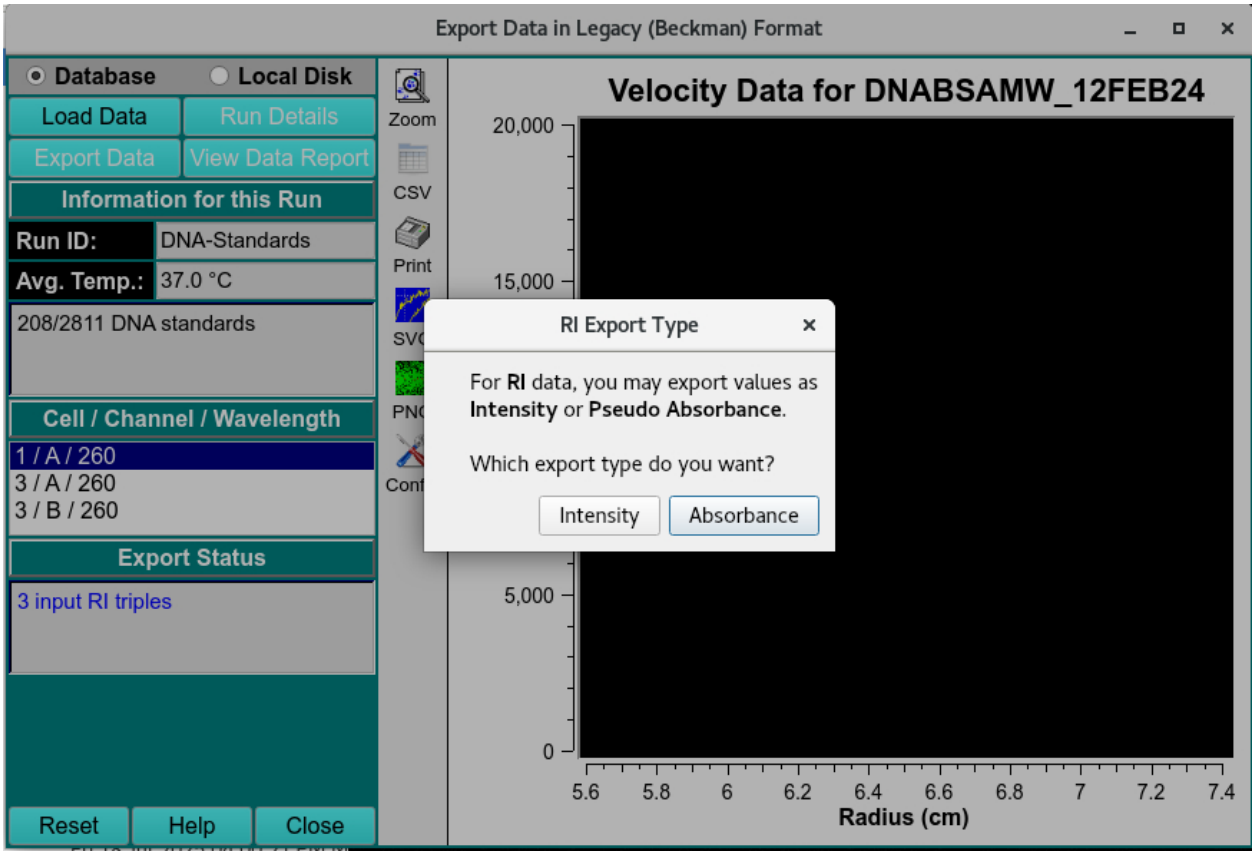
This module enables you to load a run with one or more triples in Ultrascan-III format (.auc) and export the data to files in legacy (Beckman) format.



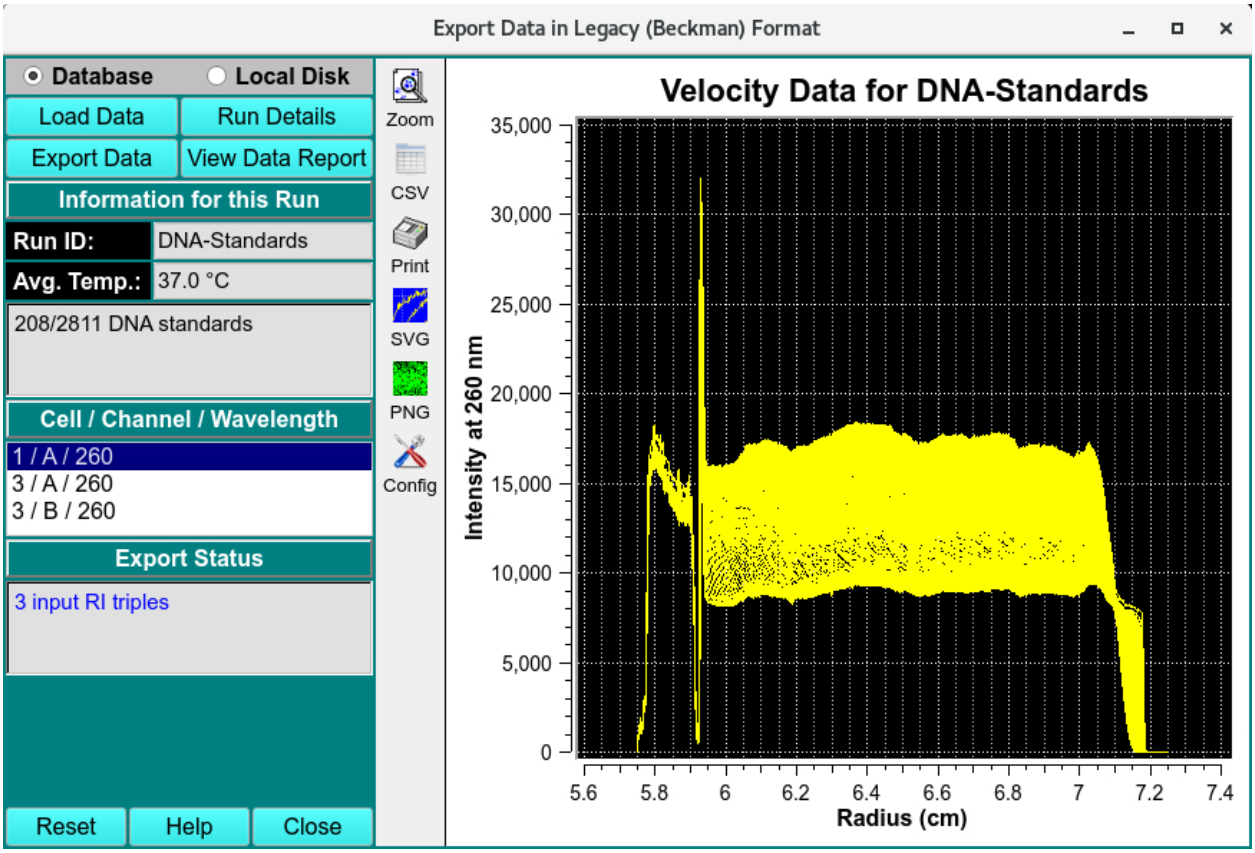
Export Data into Legacy (Beckman) Format

Once loaded by clicking **Load Data**, sedimentation velocity data that has been converted to pseudo-absorbance from intensity by the Import module can be exported as Intensity(RI) or as Pseudo-absorbance (RA) data.

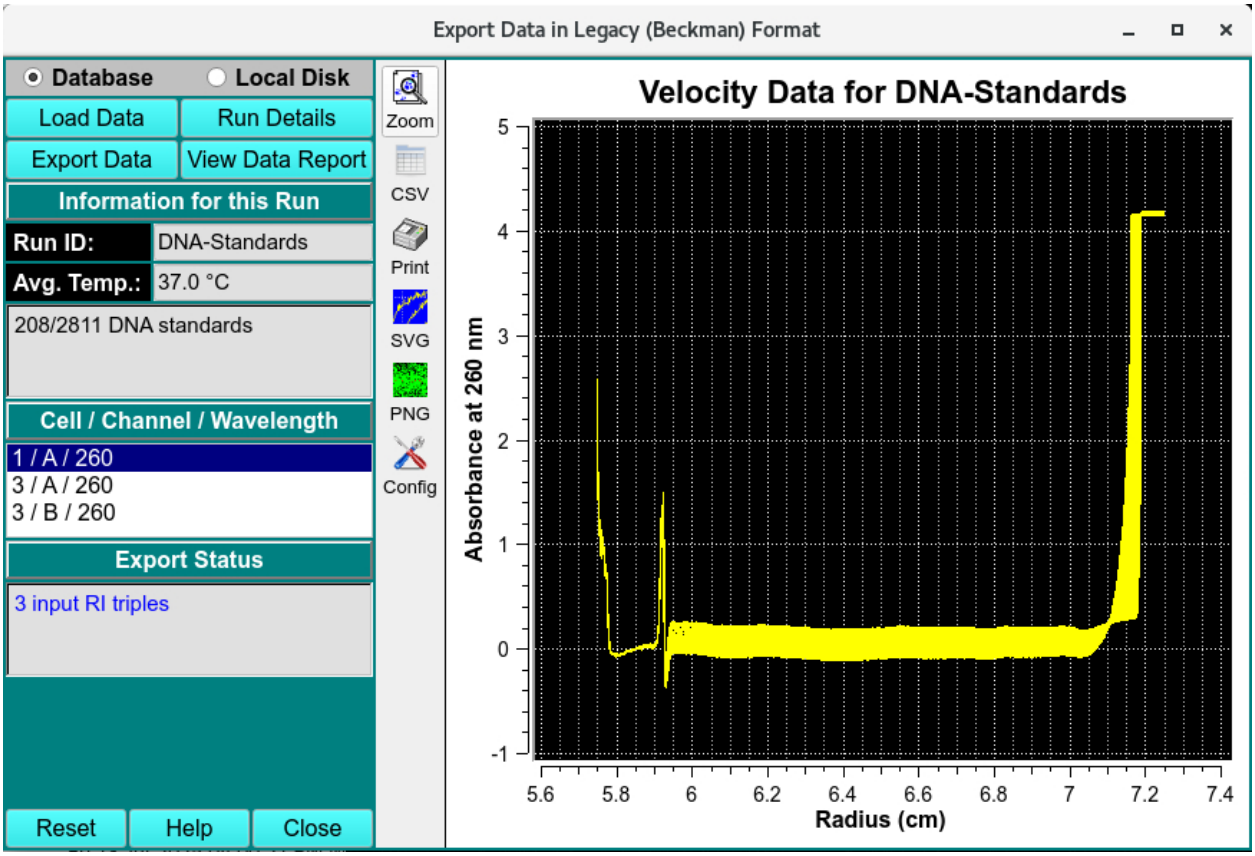
A popup dialog will appear to prompt the type the user wants to export



Data Type Prompt

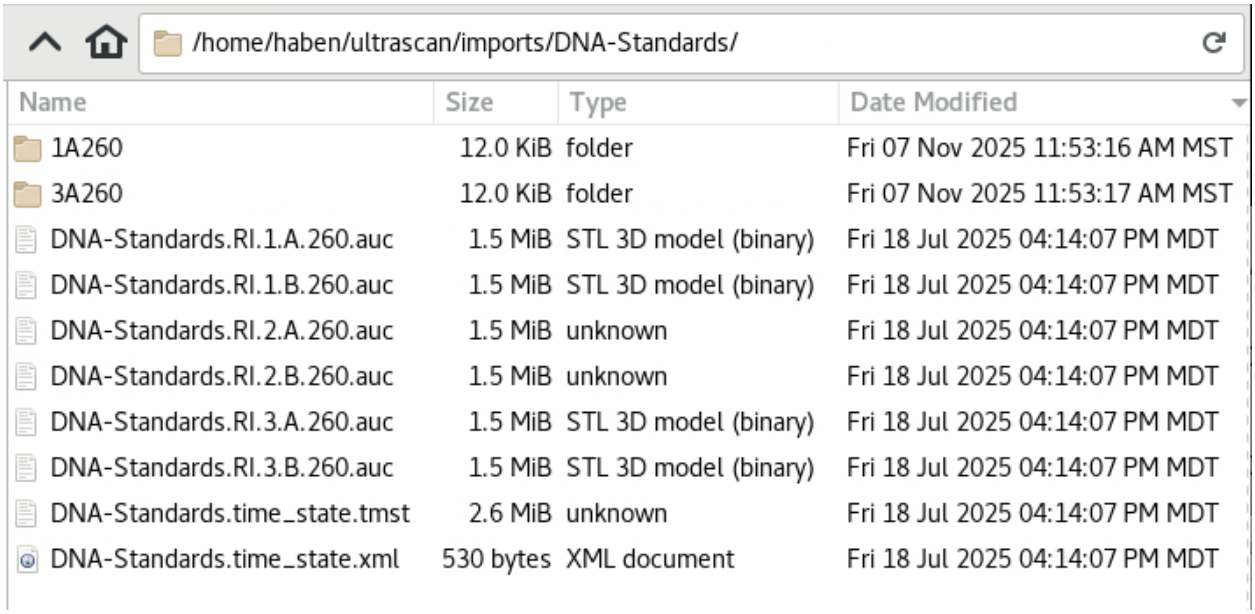


Intensity(.RI)

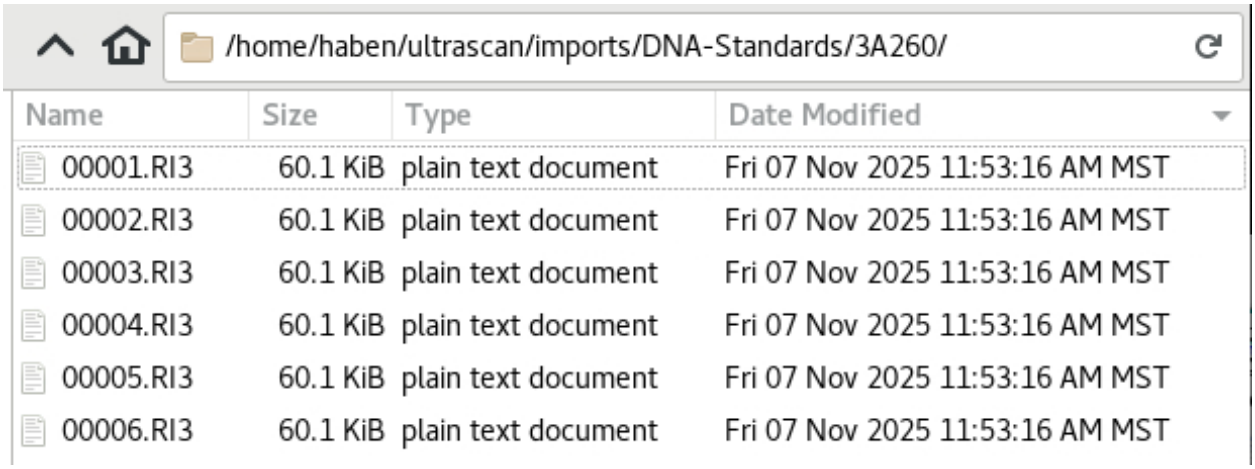


Pseudo-absorbance (.RA)

Note: Legacy Beckman and Ultrascan-III format are organized as in the table below, so when importing into Ultrascan-III, Beckman format files have to be compressed into .tar.gz files and converted into Ultrascan-III format using *Beckman to OpenAUC Data Converted*.



The .auc files are defined as:



The legacy (Beckman) files configured as:

CHAPTER

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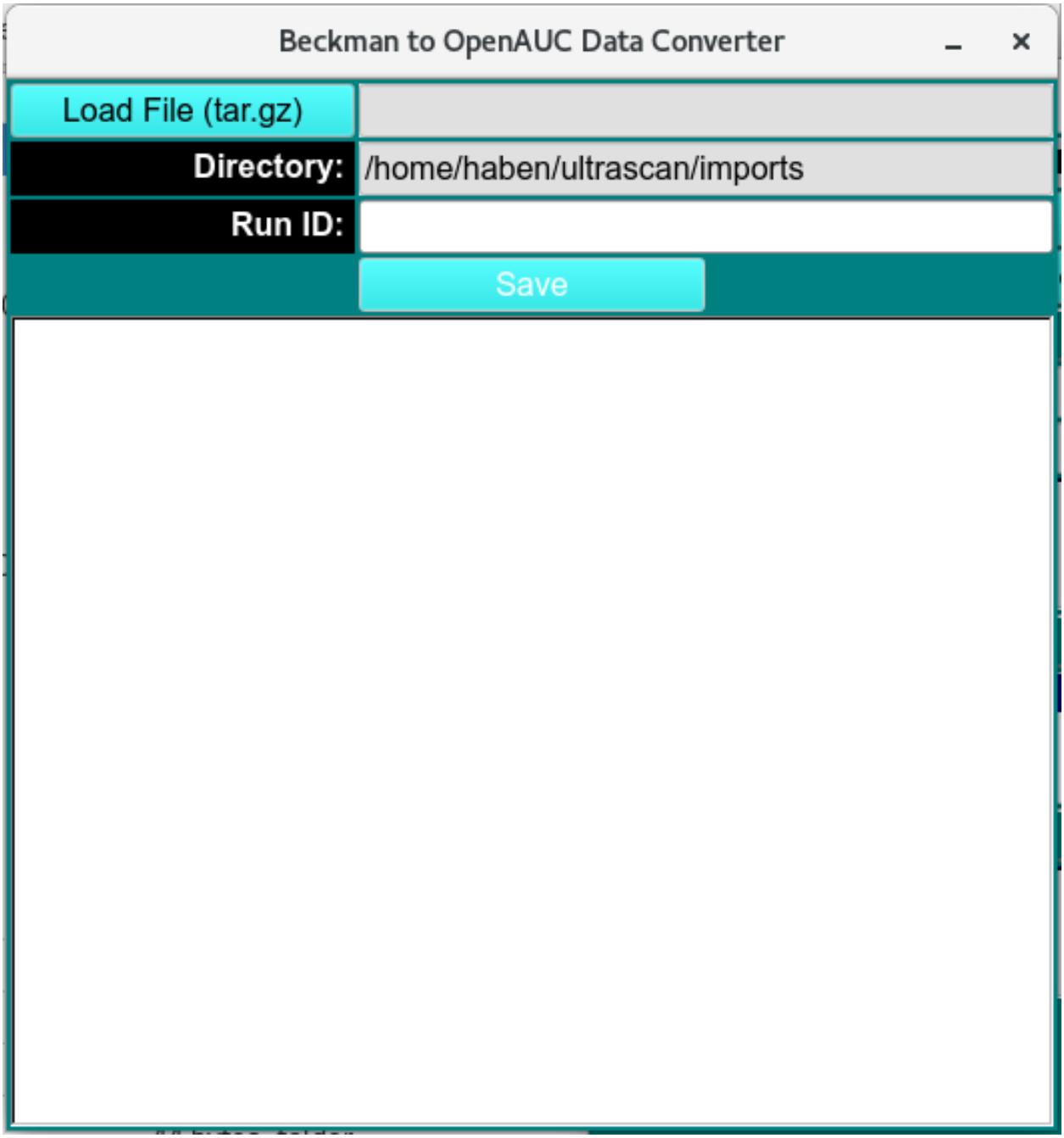
BECKMAN (TAR.GZ) TO OPENAUC DATA CONVERTER:

This conversion tool allows the user to convert tar.gz archives created by the Beckman-Coulter's web-based data acquisition application into the OpenAUC format for use by Ultrascan-III.



Beckman to OpenAUC Data Converter

CHAPTER

THIRTY

OPENAUC CONVERTER FUNCTIONS:

Database	Select to specify data input from the database.
Local Disk	Select to specify data input from local disk or the database.
Load Data	Click here and, in the resulting *Load Data Dialog*, select a run ID for the data set to load.
Run Details	Pop up a dialog showing run details.
Export Data	Click to export all Triples of the data to files in legacy (Beckman) format.
View Data Report	Generate a report file and view it in a dialog.
Run ID:	The Run identifier string is displayed for the loaded data sets.
Avg. Temp.:	The average temperature over all the scans of the currently selected Triple.
(experiment description)	A text string is displayed giving a fairly detailed description of the experiment.
Cell / Channel / Wavelength	One or more rows of raw data Triples. Click on a row to display its data in the plot window to the right.
Export Status	This text window displays continually updated summaries of computational activity and results.
(right side plot)	This plot shows the Experimental data for the currently selected Triple.
Reset	Reset all parameters to defaults.
Help	Display this and other documentation.
Close	Close the window and exit.

CHAPTER

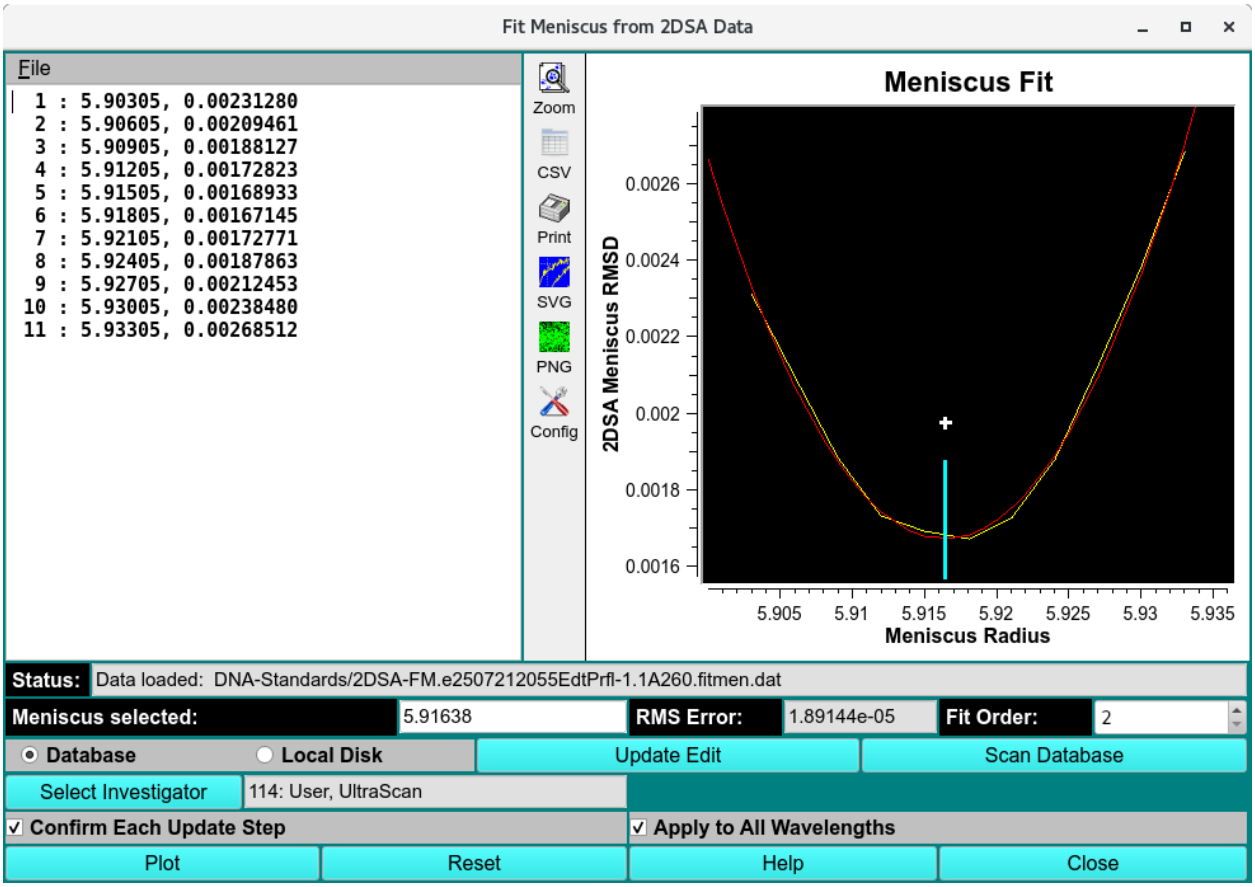
THIRTYONE

FIT MENISCUS

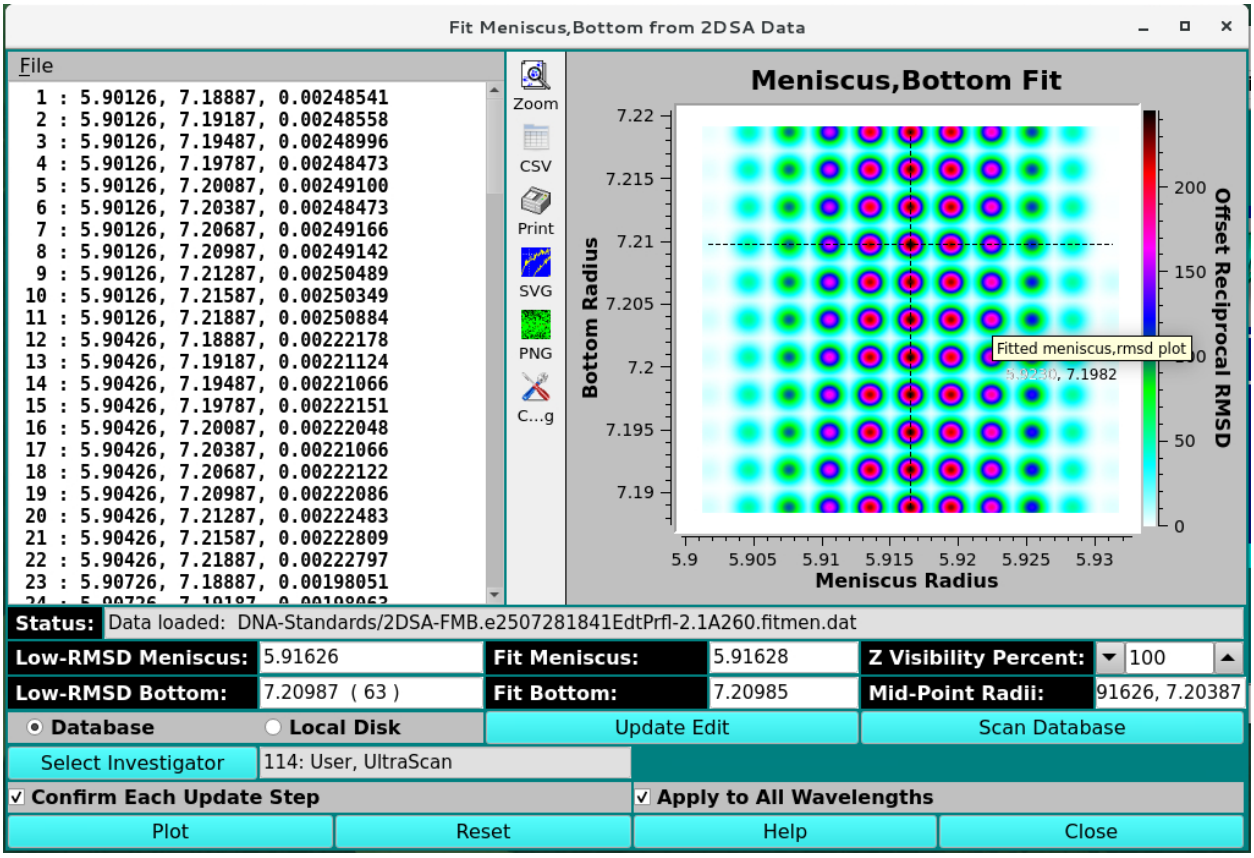
Index

- *Fit Meniscus Functions:*

This module enables you to modify an edited data set's meniscus value based on meniscus-fit models. A polynomial fit to a set of meniscus-RMSD points is plotted and its low point presented as the default new meniscus value. That value - or a manually modified one - can then replace the value in the associated Edited data set. The model with closest meniscus value in the set of fit-meniscus models that were obtained by a front-end or back-end meniscus fit 2DSA run can be preserved, while all other models in the set (and any associated noises) are removed.



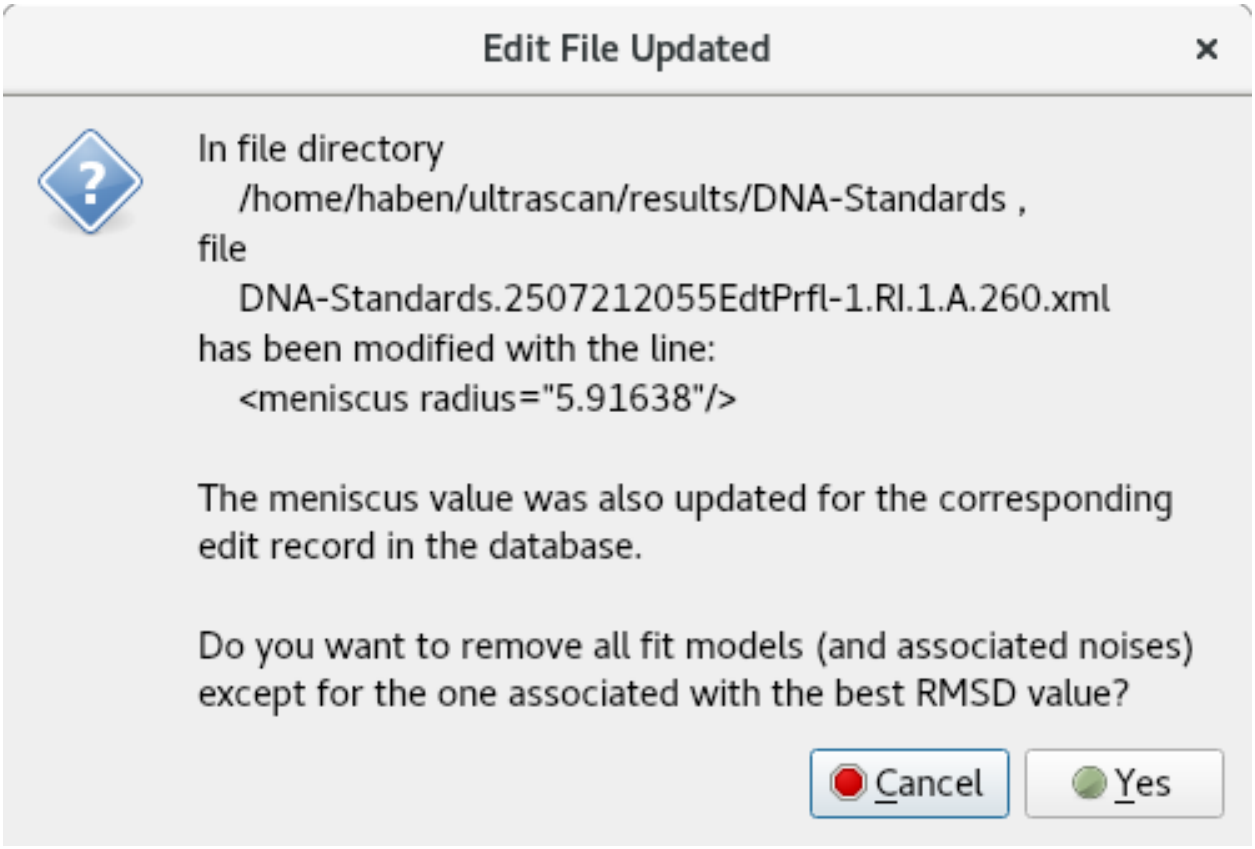
Meniscus Fitting



Meniscus and Bottom Fitting

31.1 Fit Meniscus Functions:

(Meniscus-RMSD points)	The main left-side text box displays a list of meniscus-RMSD values from a selected fit output by 2DSA. The fit to display is chosen by a *File->Load* selection in a FitMen Load Dialog that allows navigation to a ./results/directory and choice of a fitmen.dat file.
(right side plot)	This plot shows a yellow curve connecting all fitmen radius-RMSD points; a red fitted polynomial curve; and a vertical bar with annotation delineating the meniscus value at the low point of the polynomial curve.
Status:	A text box here displays the status of database scan or data load.
Fit Order	The counter here may be modified to select the order of the polynomial to fit to meniscus-radius points.
Meniscus selected:	The text box here is initially loaded with the meniscus value of the low point in the fitted curve.It may be edited to, for example, fit more closely to an actual entry in the meniscus-RMSD list. It's value at the time Update Edit is clicked determines the replacement value in an Edited data set and model and noises to be preserved after the update.
RMS Error:	Displays the RMS error of the curve fitted to meniscus, RMSD points.
Database	Select to include database model and noise records in the clean-up of non-selected models and noises after edit update; and to include the database Edited data in meniscus update.
Local Disk	Select to limit any model and noise removal and any menicus value Edited data update to local disk.
Update Edit	Click this button to update the associated Edited data with the currently selected meniscus value. After that update, a *Edited File Update Dialog* appears that also allows models and noises with meniscus values not the closest to the selected one to be removed.
Scan Database	Click this button to scan the database for any "fitmen" data sets that do not already have an associated local disk file. Any such files will be downloaded and their presence reported in the Status text box.
Plot	Click this button to re-plot the currently selected fitmen set. This is generally not necessary.



Edited File Update Dialog

Window Controls

Reset	Indicate that parameters are to be reset and the plots re-displayed based on original parameters.
Help	Display this detailed help.
Close	Close all windows and exit.

CHAPTER

THIRTYTWO

ULTRASCAN-III GOOD MANUFACTURING PRACTICES (GMP)

Triple================================ Automated GMP workflow setup ================================

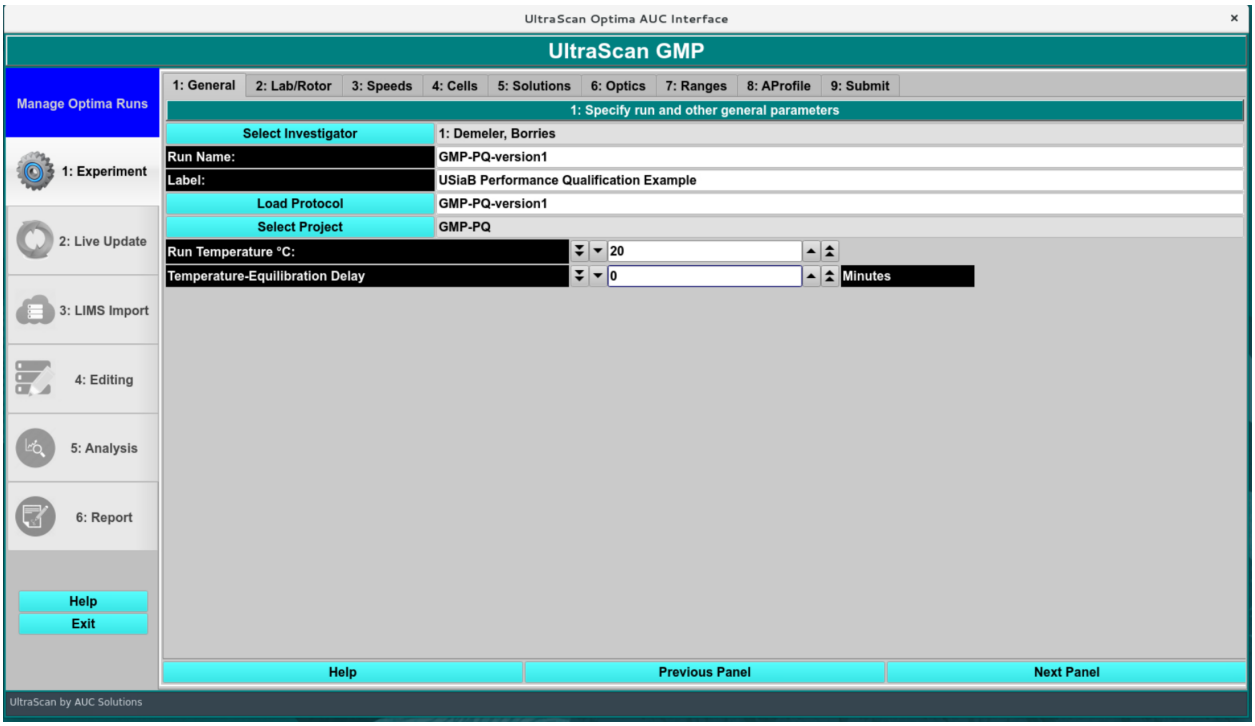
Index

- *Setting up an experiment in the GMP module:*
  - *Set up the Lab and Rotor information:*
  - *Speed Selection:*
  - *Cells:*
  - *Solutions:*
  - *Optics:*
  - *Ranges:*
  - *Analysis Profile/Report Settings:*
  - *Integration and Report settings:*
  - *2DSA Settings:*
  - *PCSA settings:*
  - *Submitting the run:*
- *LIMS Import:*
- *Edit Screen:*
- *Data Analysis:*
- *Data validation:*
- *Report:*

32.1 Setting up an experiment in the GMP module:

By default, the investigator should be set to the person logging into the system. This user requires at least user level 2 to proceed. Enter a unique run ID for the experiment (GMP-PQ-version1), a verbose label (60 chars), give the protocol a name and select a project. There is no need to populate the project, but create a *New Project*, rename the project and *Save*, select it from the list, and close the Project dialog.

The "Temperature-Equilibration Delay" can be set to zero minutes because no real instrument is involved. The temperature should be set to the same temperature that was used in the simulation settings.



GMP Modules Access

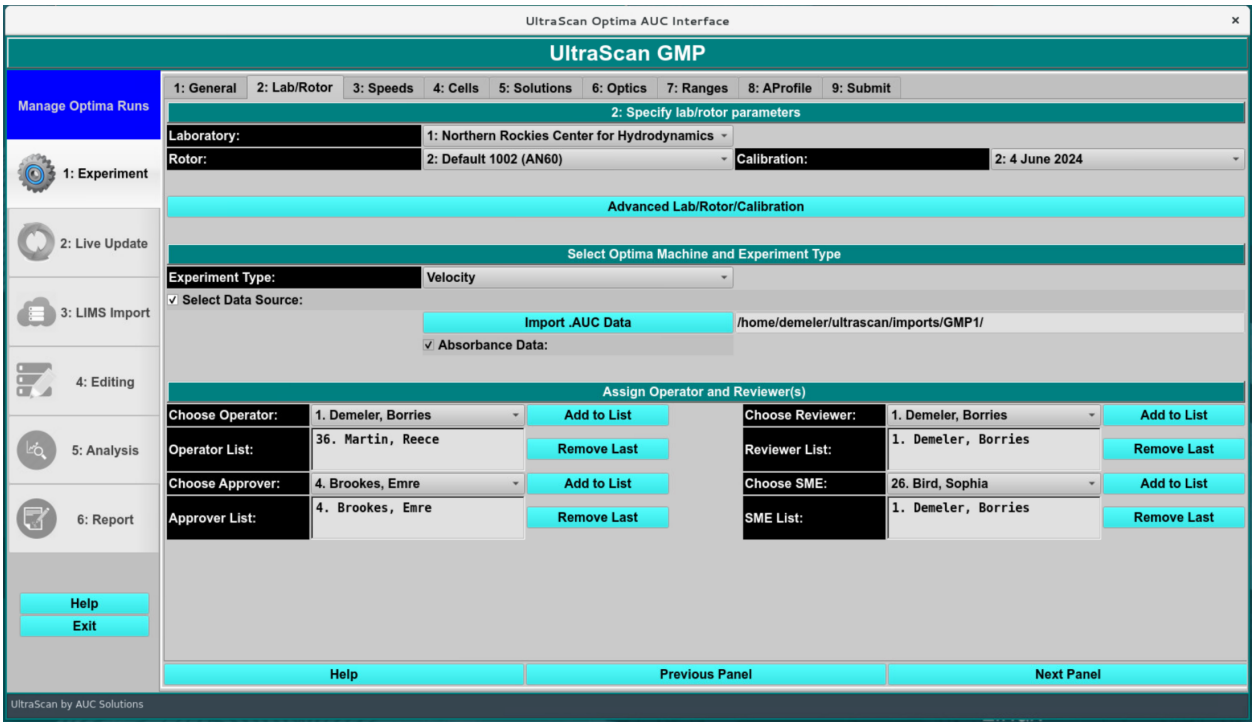
32.1.1 Set up the Lab and Rotor information:

In this screen, the laboratory, rotor, data source and GMP operator, reviewers, approver and SME information needs to be chosen. If you do not have any operators, reviewers or approvers defined, please see your IT administrator to assign these roles to the relevant users. Select one of each and click "Add to List" to activate their GMP role for this project. **Select the same rotor and calibration that was used to simulate the data.** The data source should be set to import AUC data from disk, and point to the directory containing your simulated dataset.

The rotor setting also determines if the stretching calculation is used. If the simulation rotor is used, no stretching will be calculated. Calibrated rotors will include a rotor stretch value that is used for the bottom, and also will be added to the meniscus position specified in the simulation of the simulation dataset. To calculate the theoretical offset due to rotor stretching, go to the *Utilities:Rotors* module and select the rotor chosen for the simulation, and calculate the stretch distance using the first and second order coefficients:

The rotor stretch value (in cm) is given by: (Rotor Stretch Coefficient 1)*(rotor speed) + (Rotor Stretch Coefficient 2) x (rotor speed)^2

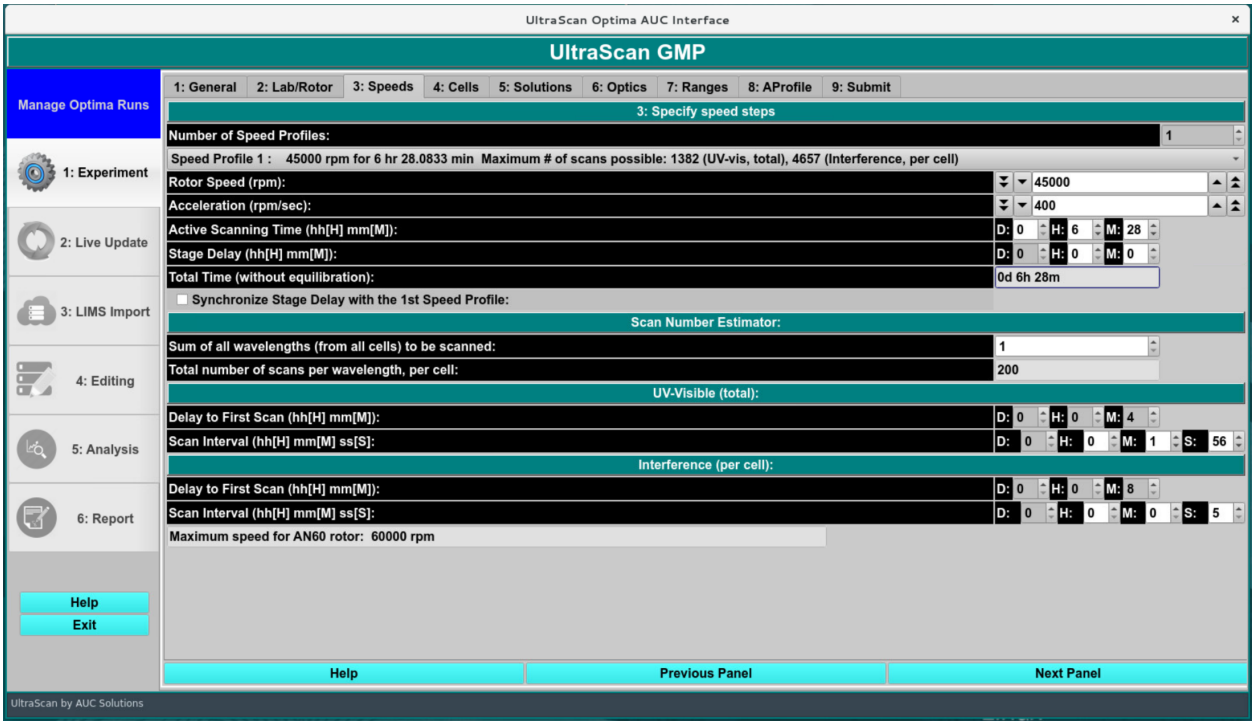
This value should be added to the simulated meniscus and should be recovered from the analysis as the meniscus position (available in the edit profile xml file).



Lab/Rotor

32.1.2 Speed Selection:

In this screen, the rotor speed, data collection interval between successive scans, data collection duration (active scanning time), and the total number of scans in the experiment are specified. For imported data, all fields in this screen are parsed from the input data and should not be changed.

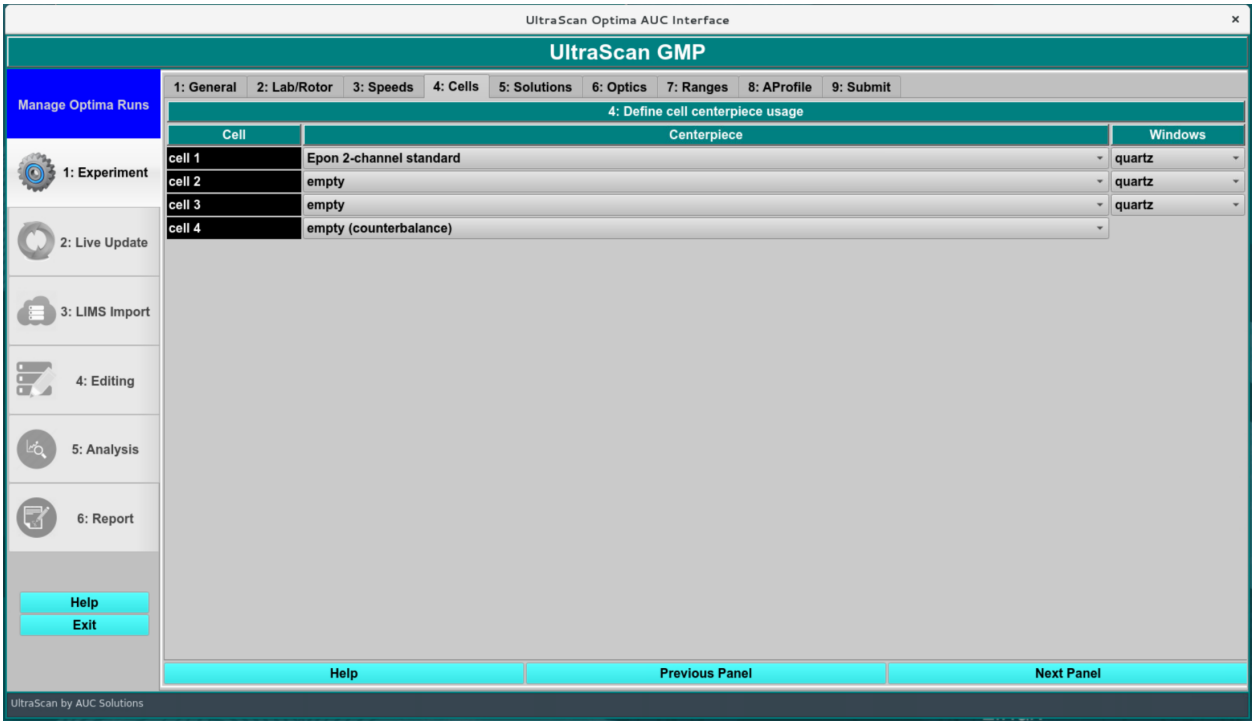


Speeds

32.1.3 Cells:

As in the previous screen, all settings in this screen are parsed from the imported data and **should not be changed, unless a different centerpiece was used. Make sure to use the same centerpiece as was used in the simulation program,** to reflect the appropriate centerpiece geometry in the simulation.

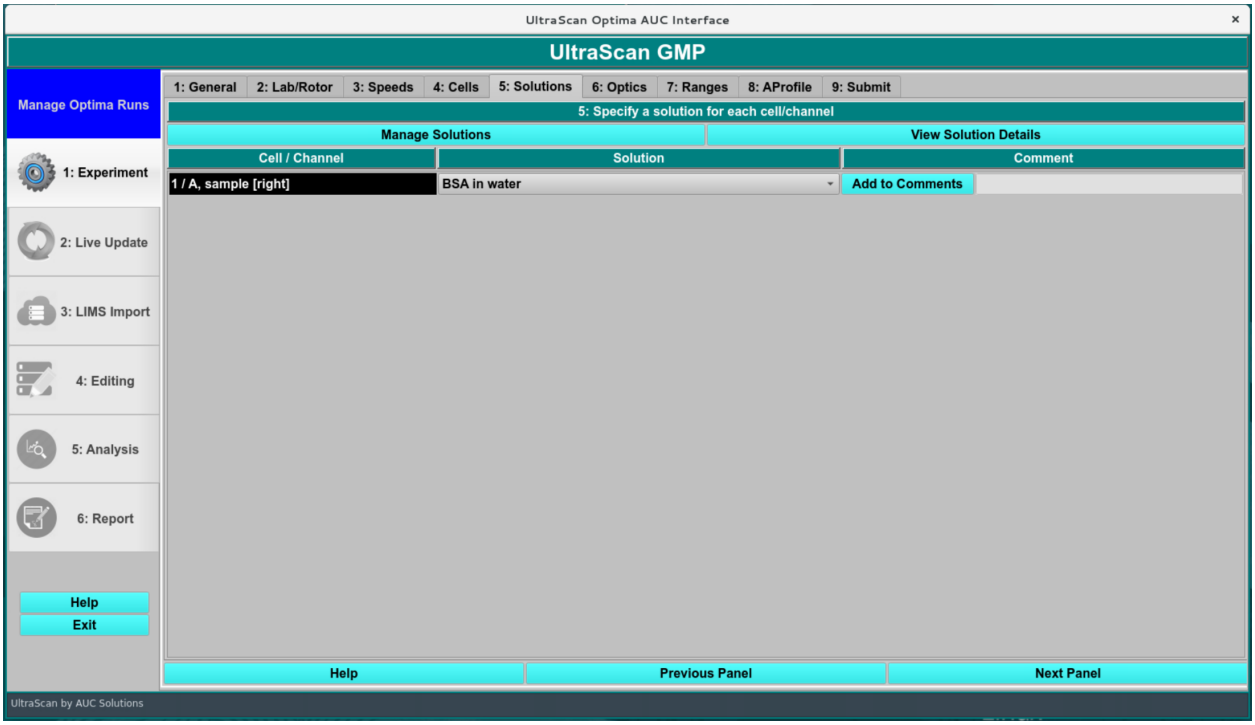
Note: Leave the cell position unchanged!



Cells

32.1.4 Solutions:

In this screen **the same solution that was used for the simulation must be selected.** Optionally, add some comments for the cell content description (left blank here).



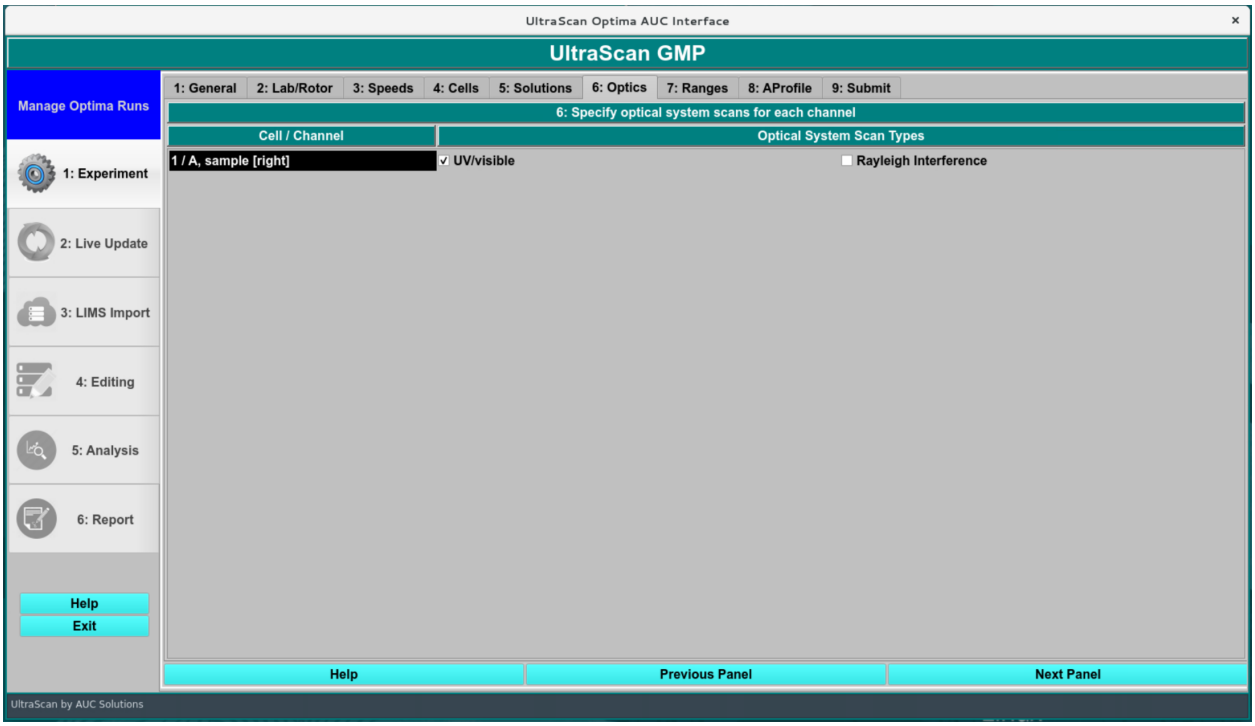
Solutions

The 'OK' button brings up the 'Live Update' window which instructs the user to start the run from the instrument panel

Alternatively, the user can press 'Return to Managing Optima Runs?' to return to the 'Manage Optima Runs' window.

32.1.5 Optics:

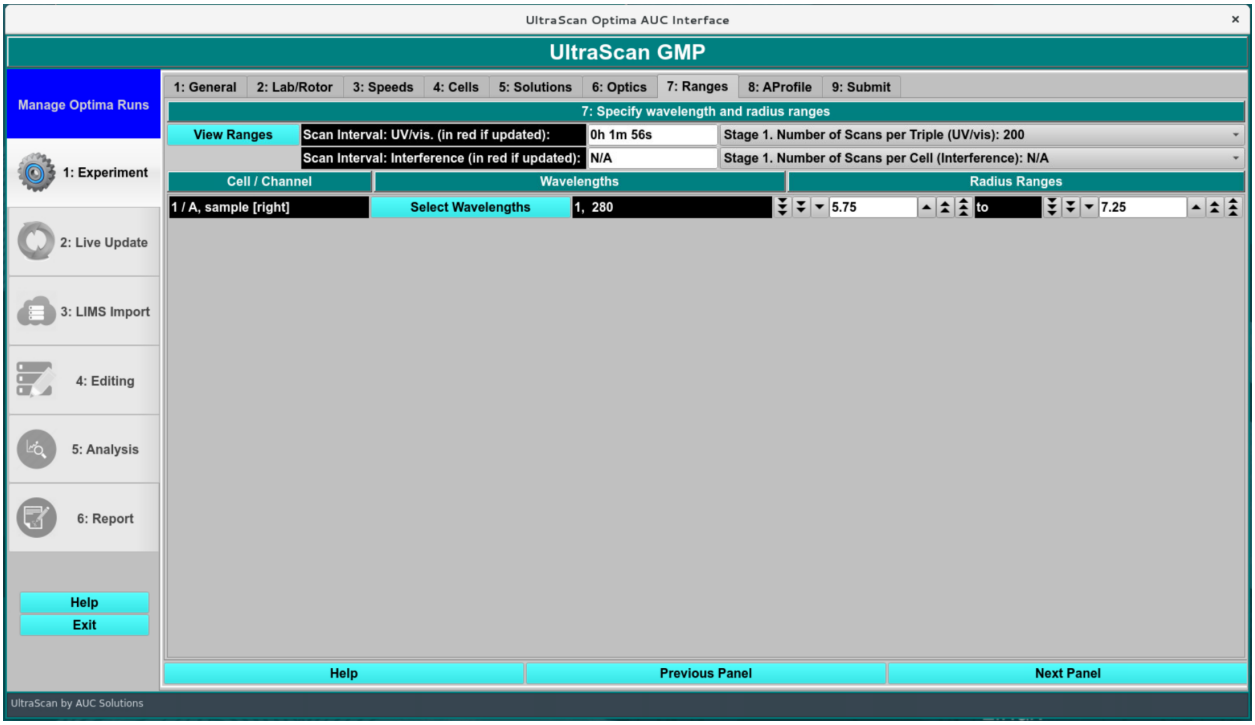
The optical system simulated is reflected here and should not be changed.



Optics

32.1.6 Ranges:

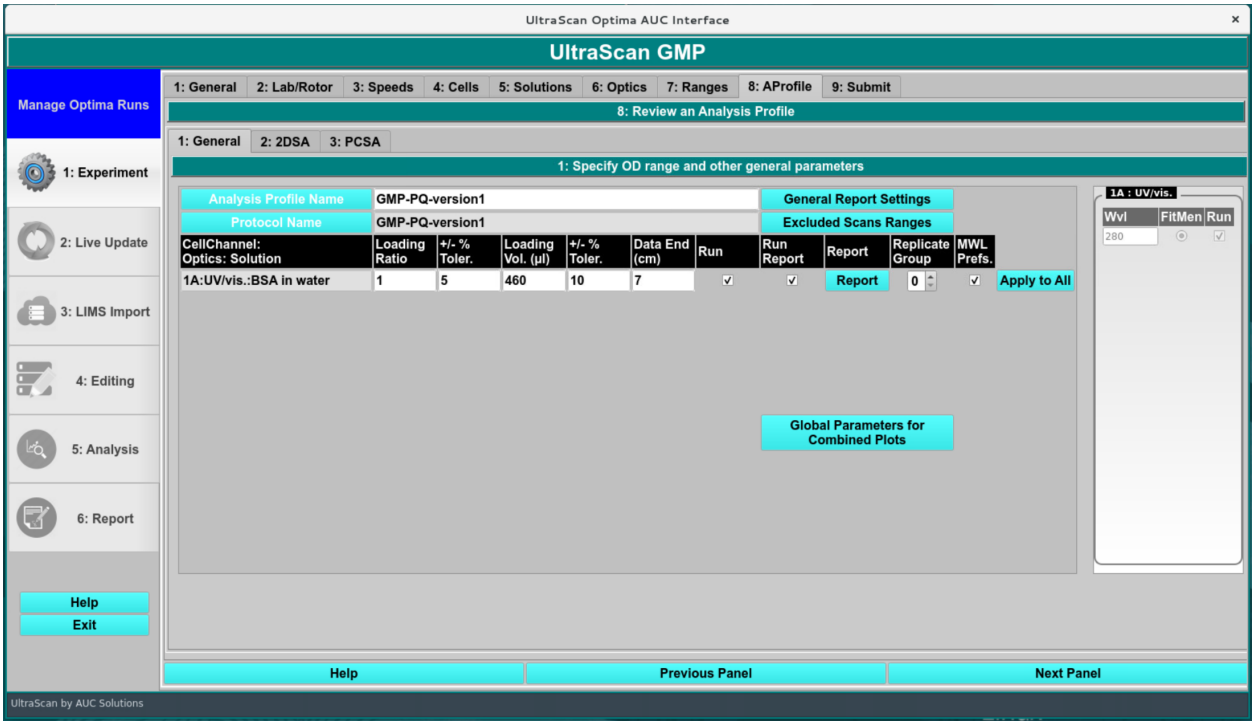
All ranges for wavelength and radial observation are indicated on this screen. The user doesn't have to change anything here because these details are parsed from the input data.



Ranges

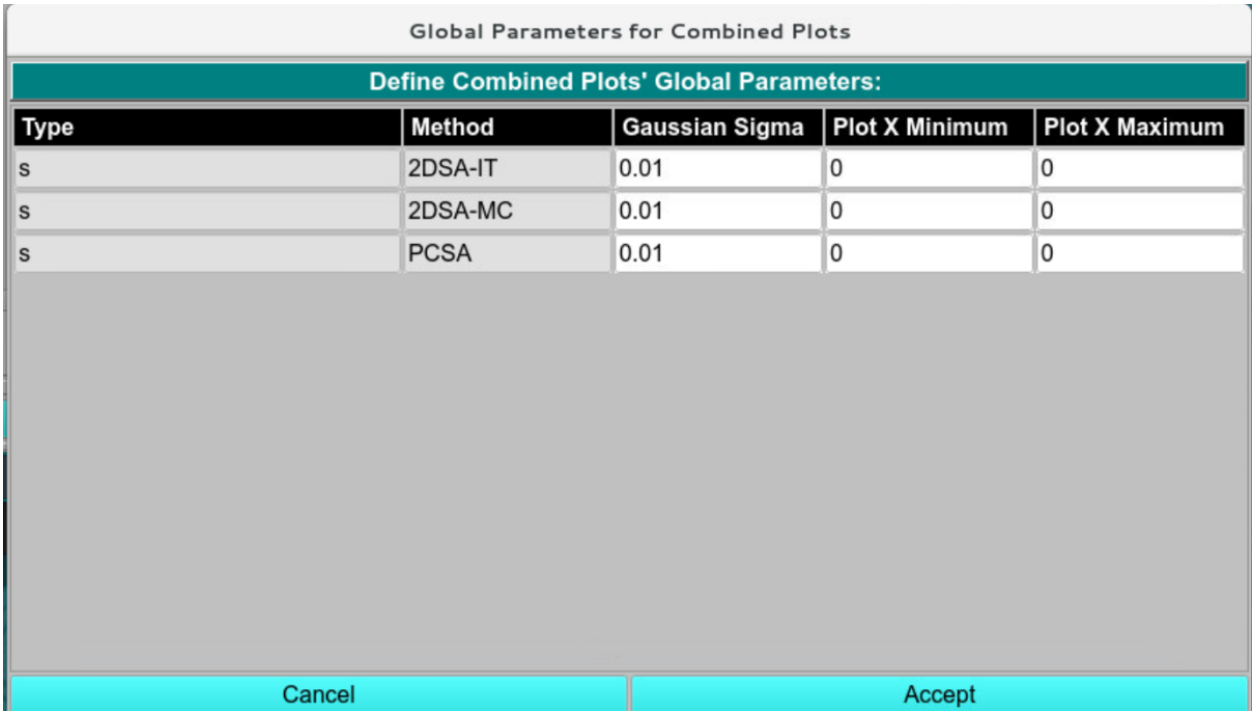
32.1.7 Analysis Profile/Report Settings:

In this screen the general report settings are entered, and integration limits are defined. The "Run" column needs to be checked for the dataset to be processed. A report will be generated if the "Run Report" is checked. The loading volume will be calculated based on the centerpiece geometry and the meniscus position by calculating the sector volume. This should match what was selected during the analysis. The data end is the bottom of the data range considered in the fitted data. This can be overwritten by using the manual edit function after the data import.



Analysis Profile

The global plot parameters can be edited for each analysis type individually. Manual x ranges can be chosen, as well as the regularization parameter for the plotting smoothness.

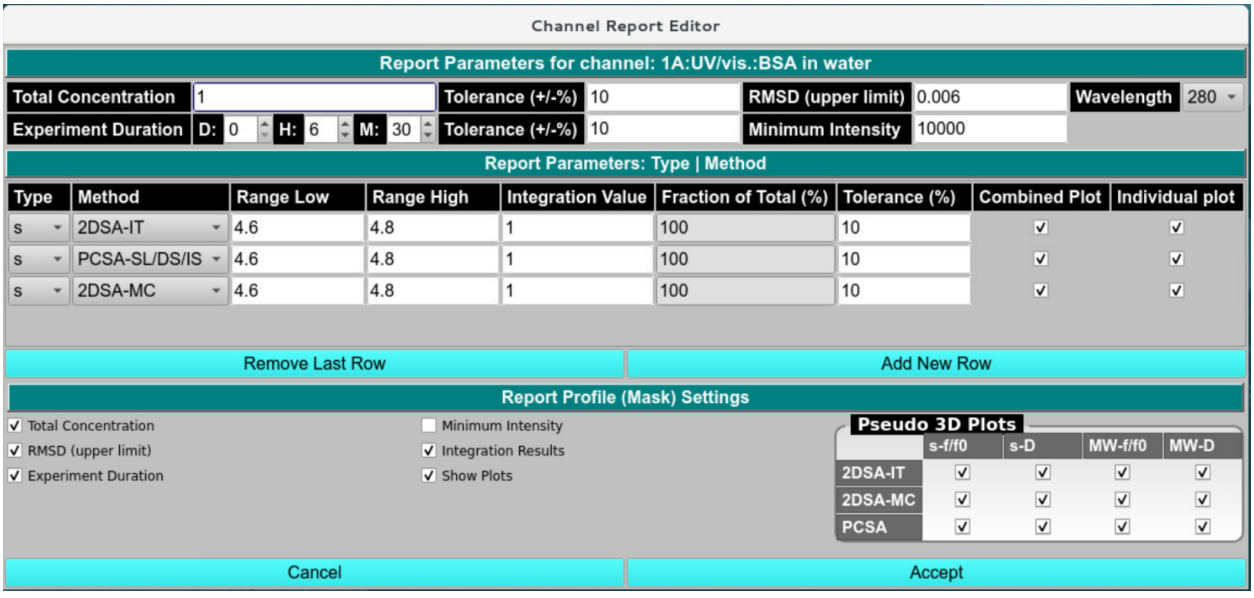


Plot Parameters

32.1.8 Integration and Report settings:

Clicking the *Report* button in the previous screen brings up the integration and report settings dialog. Here the following items need to be set:

- 1. Simulated Concentration
- 2. Maximum RMSD
- 3. Intensity minimum for this wavelength (only needed for actual experimental data acquired with UV/vis absorbance optics)
- 4. experimental duration of the experiment (should match the simulated time)
- 5. Integration ranges. - Integration reports can be provided for s, D, MW, and radius, as well as for 2DSA-IT, 2DSA-Monte Carlo, and PCSA models. The ranges should be matched to the simulated data. If no noise is simulated and the analysis grid matches the simulation grid, the fitted values should match exactly within the round-off limits of a single-precision floating data type (usually 6-7 significant digits). If noise is simulated, a range should be specified. Values that fall between the range limits will be averaged into a single weight-average value for reporting that can be compared to the target value. Accuracy of the reproduced values is a function of the simulated noise. The 2DSA-Monte Carlo analysis should provide the best weight average values. Statistical properties of the model returns from the Monte Carlo analysis can be checked with the Velocity:GA initialization module.
- 6. The report can provide differential and pseudo-3D contour plot representations for each combination listed. It is optional to provide these in the report. If multiple datasets are included in a report, overlay plots can be included as well (combined plots).



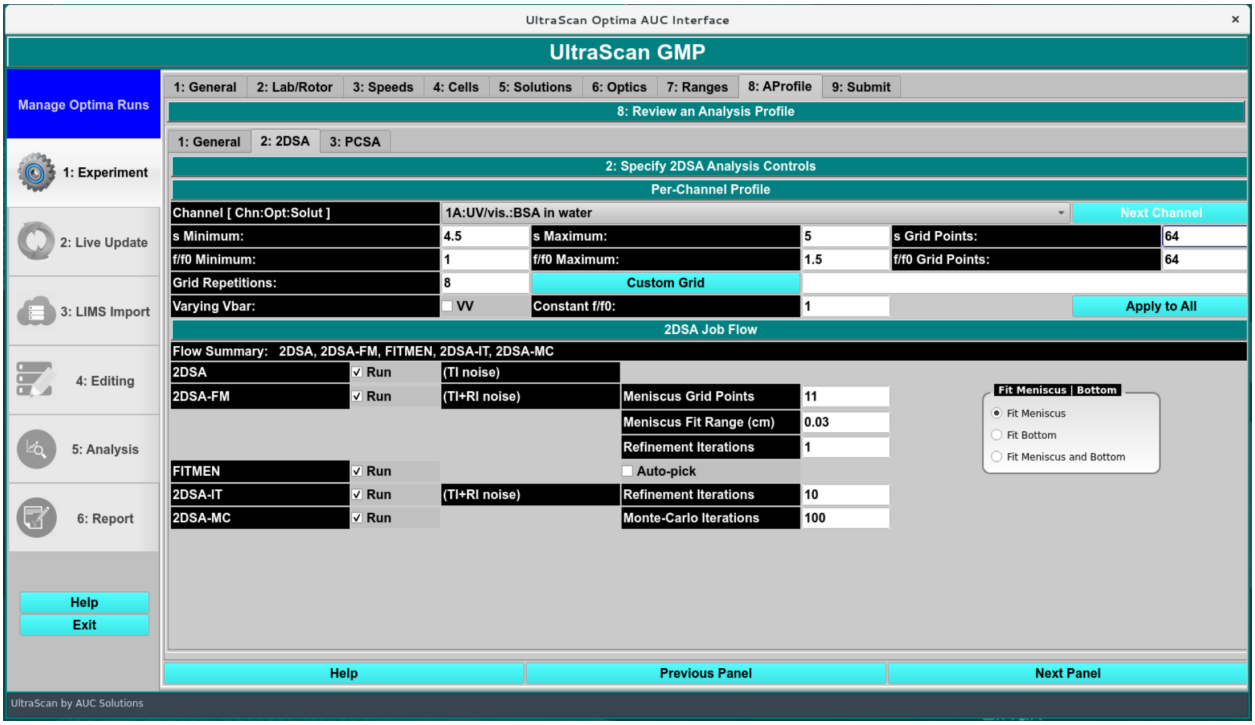
Integration and Report

32.1.9 2DSA Settings:

The 2DSA settings should closely match the simulation parameters for optimal resolution. Although larger grids can be set, the resolution needs to be increased by increasing the s or f/f0 grid point number to afford the same resolution. To compare resolutions in the grid, divide the maximum-minimum parameter range by the number of grid points to get the interval. The smaller the interval, the higher the chance the correct value used for the simulation is recovered. Using a custom grid that includes the appropriate value can provide an exact answer.

In this screen, specify the 2DSA workflow and analysis parameters, such as s-value and f/f0-value ranges, noise fitting, boundary condition fits, and Monte Carlo details. A manual or automatic meniscus/bottom position fit is available.

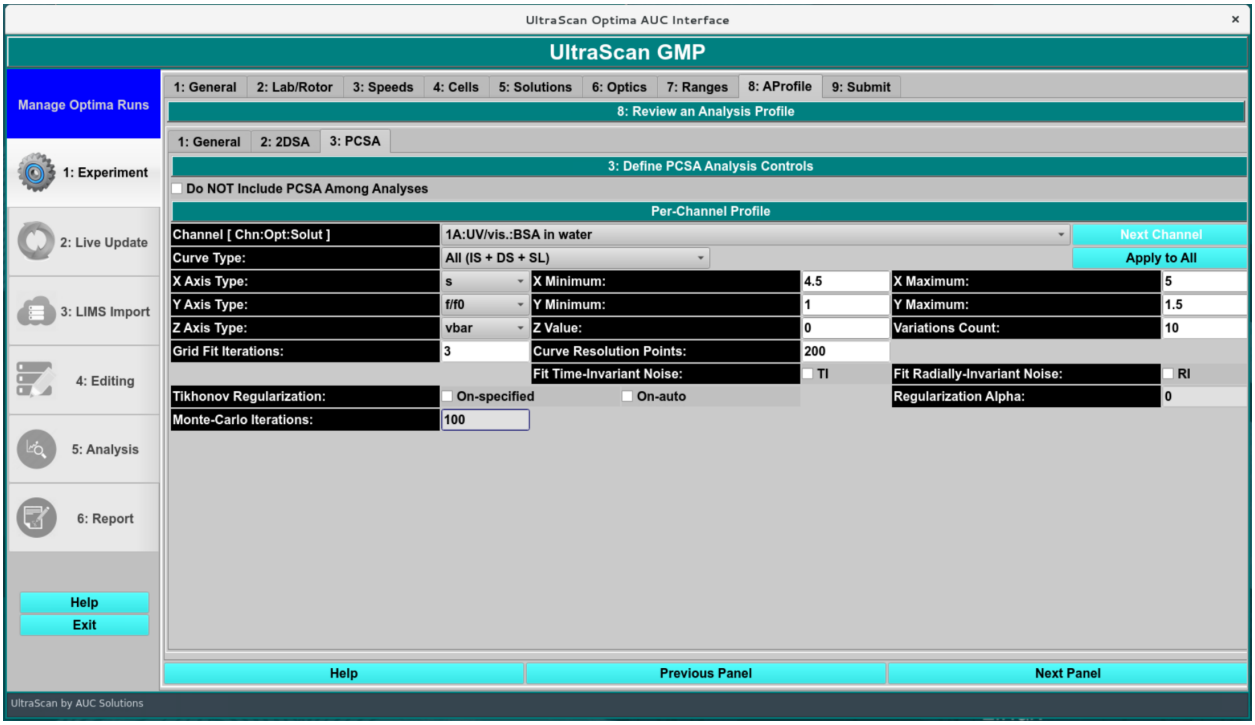
The recommended analysis flow chart is explained in this :doc:`Flowchart <../start_page>.



GMP 2DSA Analysis Profile

32.1.10 PCSA settings:

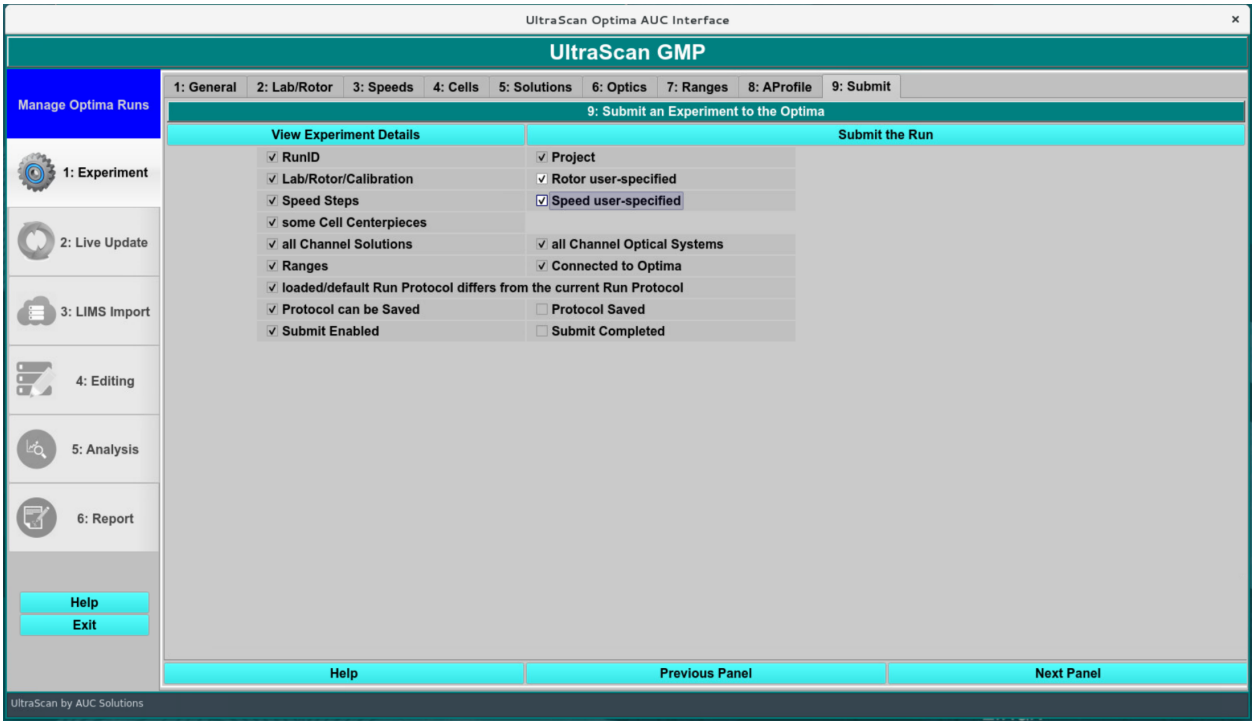
For PCSA analysis, select the appropriate fitting ranges (the same ranges as in the 2DSA analysis can be chosen), making sure the grid covers the target values. All other values should be left as default values.



GMP PCSA Analysis Profile

32.1.11 Submitting the run:

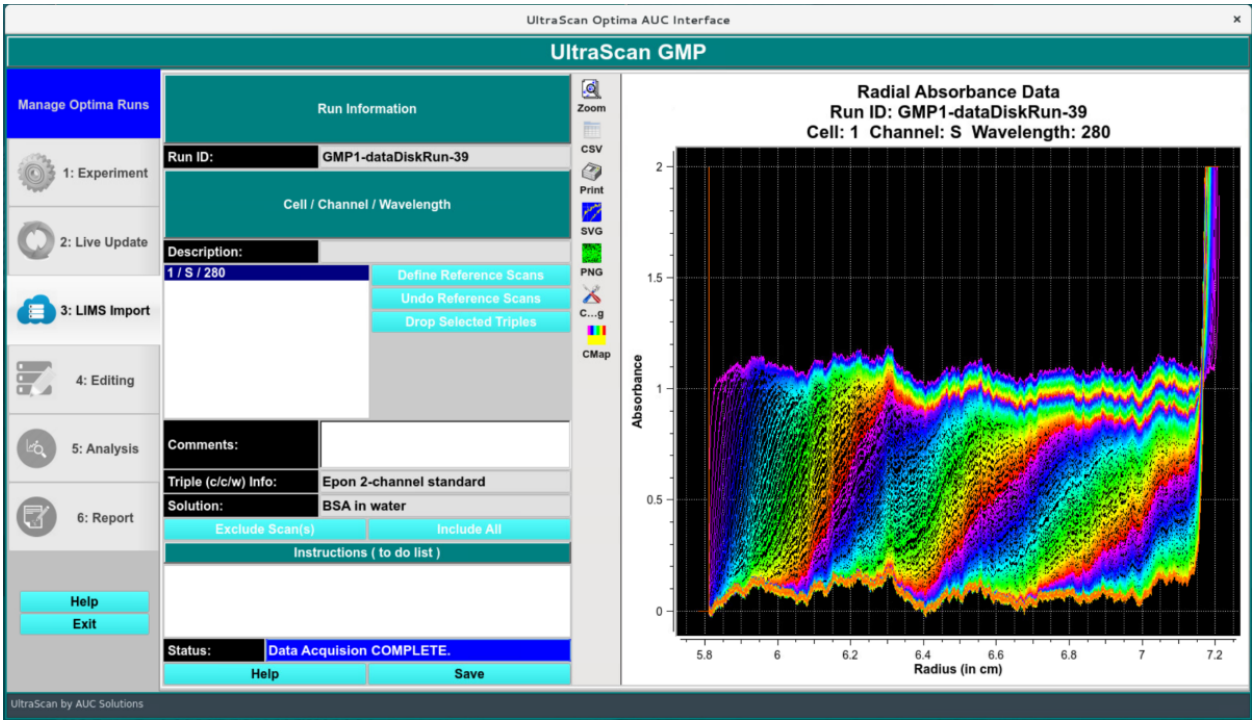
The submission screen performs a sanity check on the GMP analysis, making sure all essential parameters for the analysis have been set. A unique name for the protocol and for the runID must be chosen for each analysis by the user, although protocols can be reused as long as no changes have been made to the protocol. Submission of a run from disk prompts the data import screen to be shown as indicated on the next page.



Protocol Submission

32.2 LIMS Import:

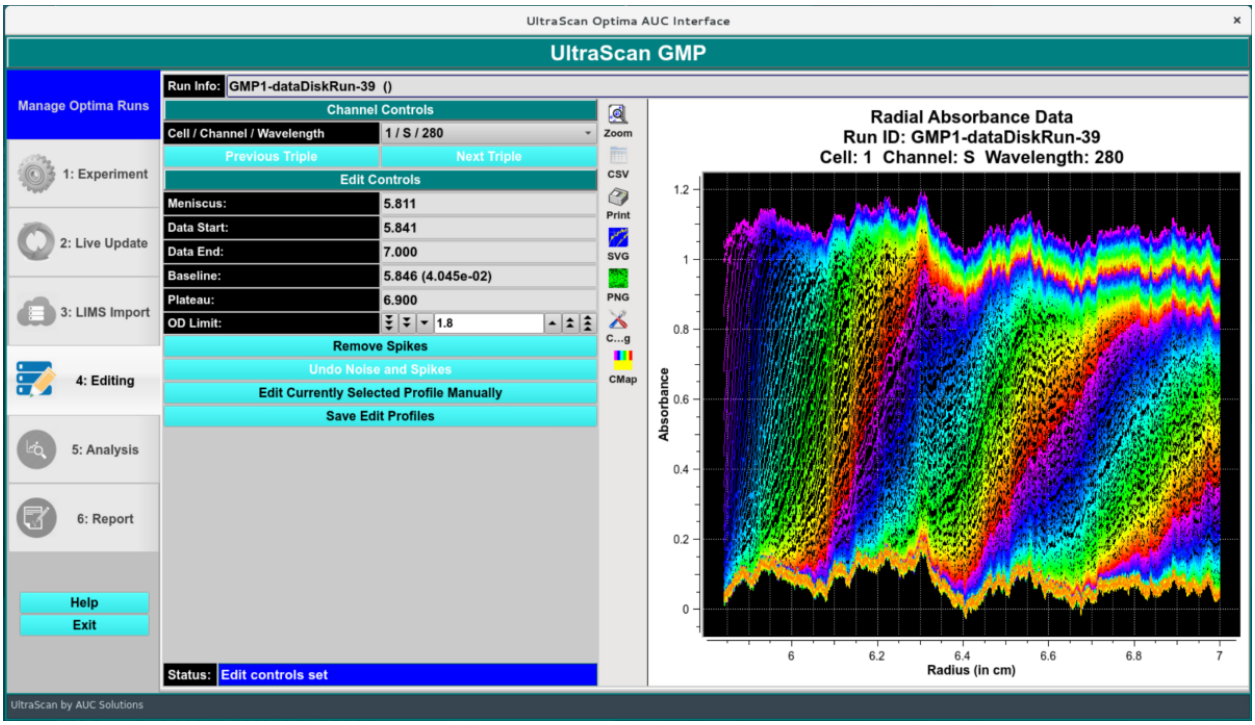
In the LIMS import screen, the user inspects the data and saves/imports it in the LIMS database for downstream analysis.



LIMS Import

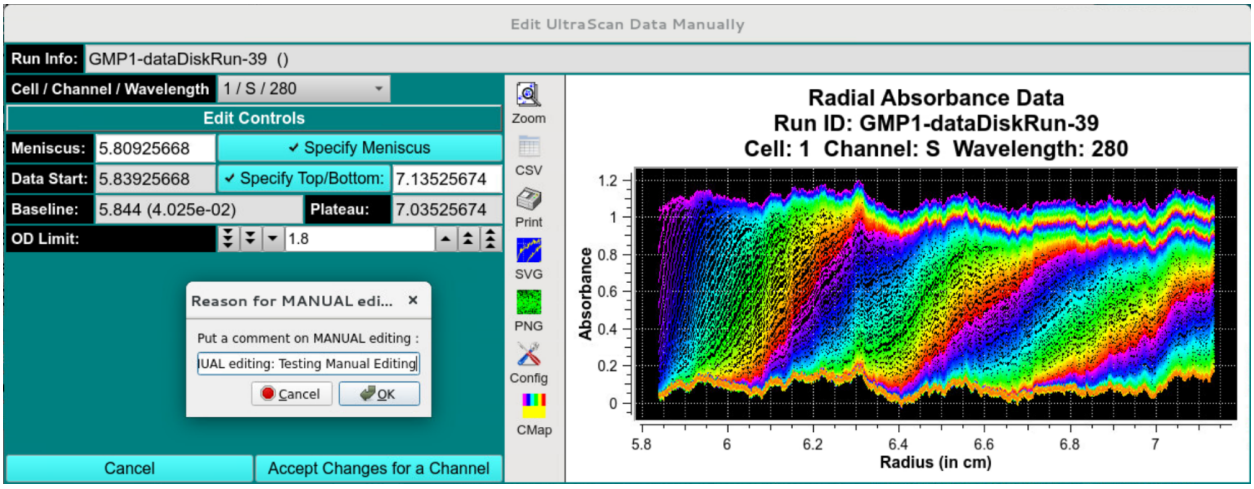
32.3 Edit Screen:

In the first step after import, the data are edited, typically in an automated fashion.



Edit Profile

Optionally, the manual editing option can be invoked, and new limits for the boundary conditions (meniscus position) and the data range can be specified. Any changes from the automated workflow must be documented with an electronic signature panel.

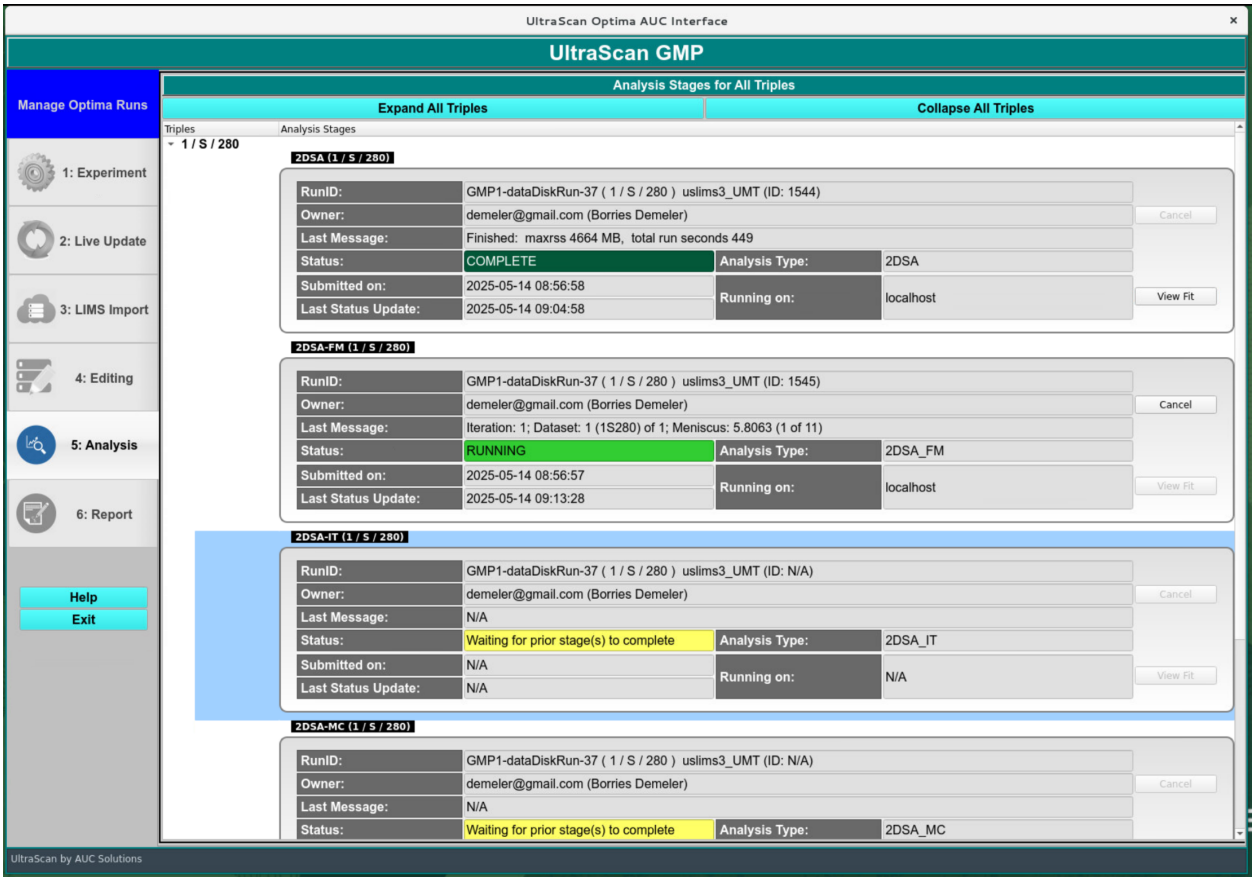


Manually Edited Profile

32.4 Data Analysis:

The data analysis should commence immediately after the editing step has been electronically signed. Individual analysis steps are listed in sequential order for each Triple in the analysis, and their status is indicated as "Waiting for prior stage(s) to complete", "QUEUED", "RUNNING", or "COMPLETE".

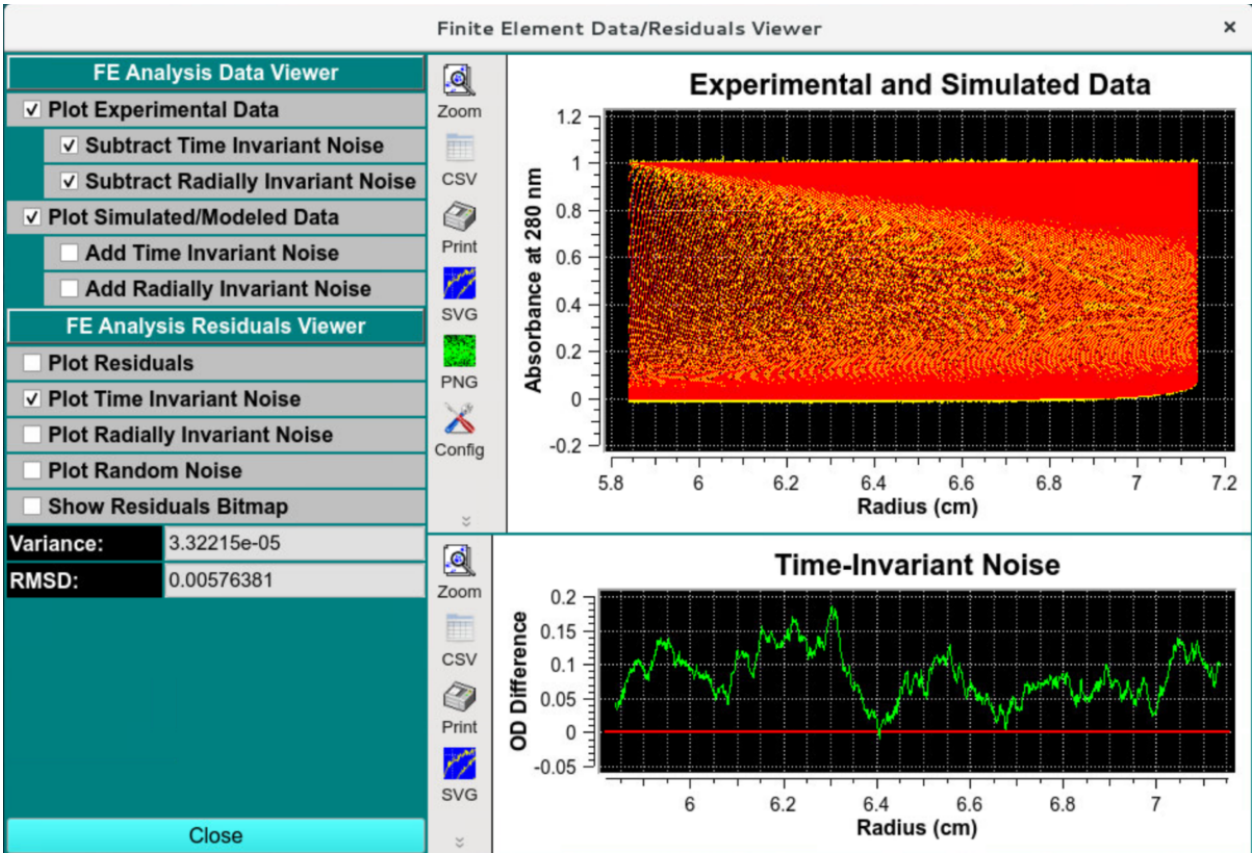
When all analysis steps are in "COMPLETE" stage, the GMP workflow will switch to the report generation and produce a GMP report. Review signatures from the operator, reviewers and approvers will be solicited via email, and added to the GMP report after signing. The electronic signature panel in the GMP menu will be used to review reports and provide the final signatures.



GMP Automated Analysis Stages

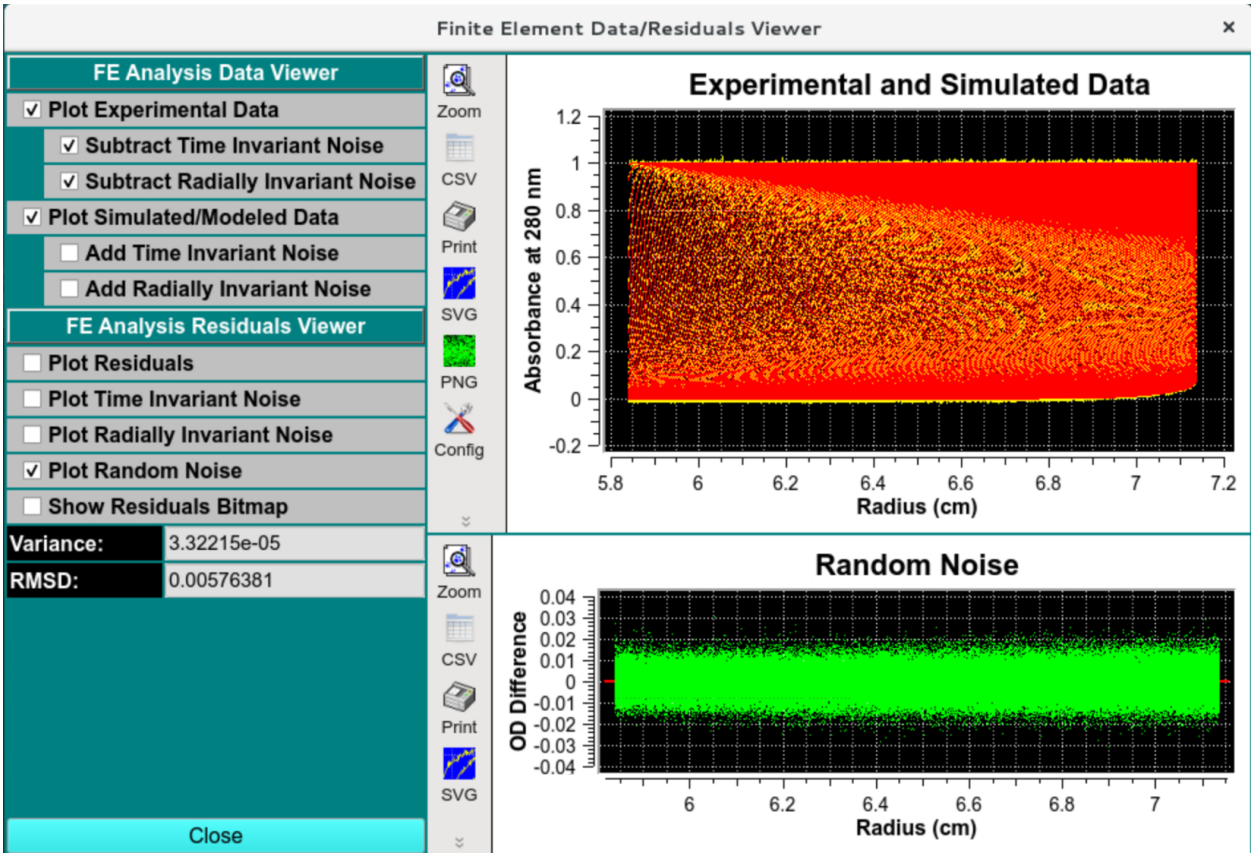
32.5 Data validation:

To validate that the software calculated the expected values, inspect the generated data by reviewing the experimental and simulated data overlays, inspect the magnitude of the RMSD, and the randomness of the residuals, and most importantly the integration results for concentration and sedimentation coefficient, or any other parameters specified in the report.



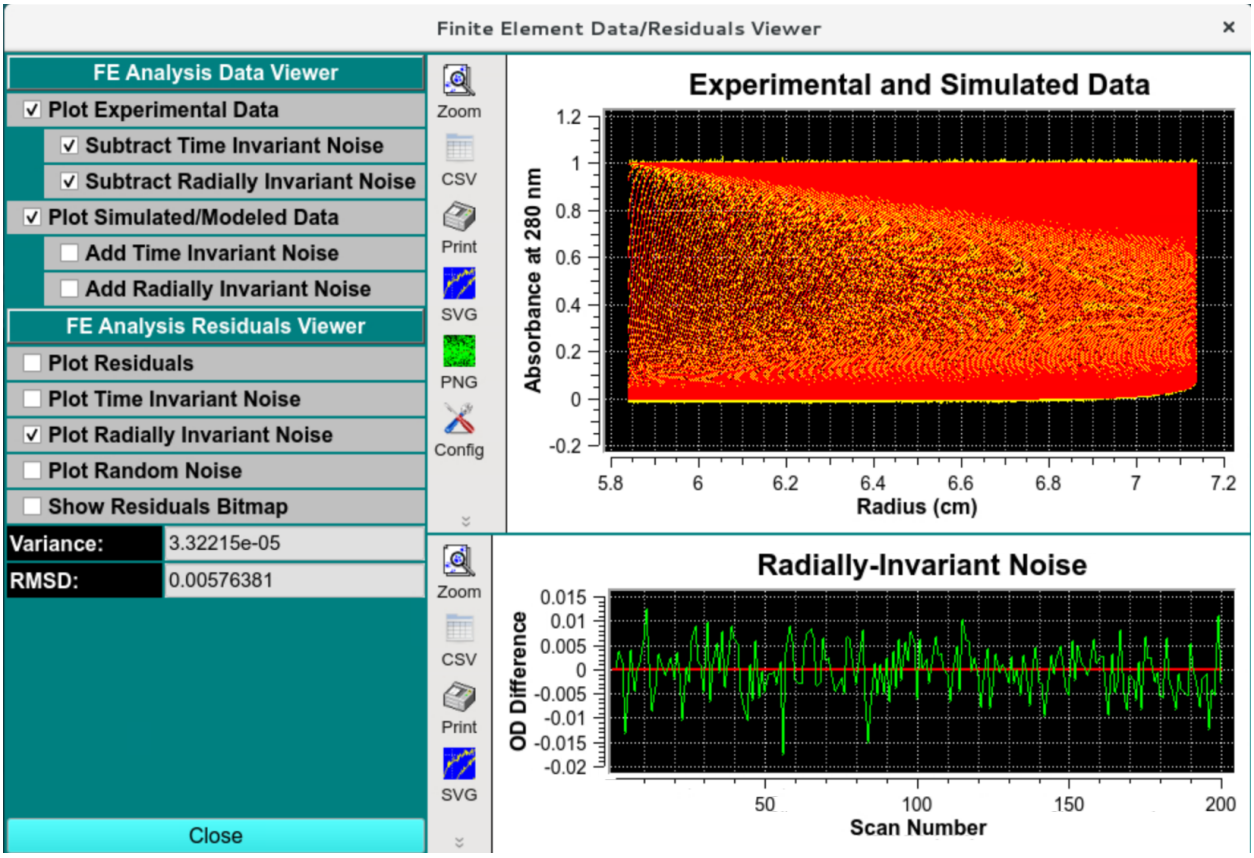
Data and Residual Viewer - Time Invariant Noise

Verify that the time invariant noise has been removed and the overlay appears to be smooth. The residuals should appear random, and the RMSD should be commensurate with the simulated stochastic noise amplitude:



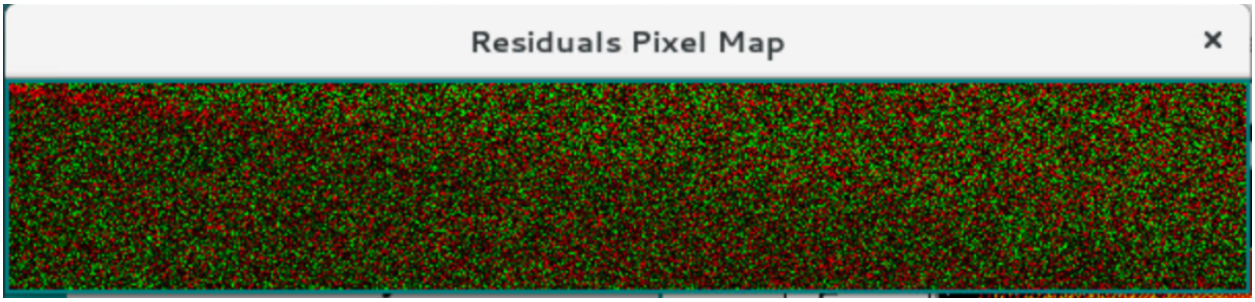
Data and Residual Viewer - Random Residuals

Radially invariant noise should look random and have the simulated amplitude:



Data and Residual Viewer - Radially Invariant Noise

The residual pixel map should show a mostly random pattern of a grainy red/green pattern without a visible diagonal pattern.



Pixel Bitmap

32.6 Report:

Quality attributes analyzed by the GMP analysis routine should faithfully reproduce the parameter values for the fitted parameters listed in model parameters, and should agree among multiple analysis types.

Distribution Information: View Model Distributions

Comparison Between Run/Simulation Results and Report Parameters:

Parameter:	Target Value:	Tolerance, %:	Measured Value:	PASSED ?
Total Concentration	1 (factor 1)	10%	1.0011e+00	YES
RMSD limit	0.006		0.00576374	YES
Loading Volume	460	10%	477.08	YES

Integration Results: Fraction of Total Concentration:

Type:	Range:	weight Avg.:	Integration from Model (target):	Fraction % from Model (target):	Tolerance, %:	PASSED ?
s	[4.6 - 4.8]	4.6913e-13	1.0011e+00 (1)	100.00% (100%)	10	YES

2DSA-IT

Comparison Between Run/Simulation Results and Report Parameters:

Parameter:	Target Value:	Tolerance, %:	Measured Value:	PASSED ?
Total Concentration	1 (factor 1)	10%	1.0010e+00	YES
RMSD limit	0.006		0.00576365	YES
Loading Volume	460	10%	477.08	YES

Integration Results: Fraction of Total Concentration:

Type:	Range:	weight Avg.:	Integration from Model (target):	Fraction % from Model (target):	Tolerance, %:	PASSED ?
s	[4.6 - 4.8]	4.6920e-13	9.8808e-01 (1)	98.71% (100%)	10	YES

2DSA-MC

Comparison Between Run/Simulation Results and Report Parameters:

Parameter:	Target Value:	Tolerance, %:	Measured Value:	PASSED ?
Total Concentration	1 (factor 1)	10%	1.0010e+00	YES
RMSD limit	0.006		0.00576381	YES
Loading Volume	460	10%	477.08	YES

Integration Results: Fraction of Total Concentration:

Type:	Range:	weight Avg.:	Integration from Model (target):	Fraction % from Model (target):	Tolerance, %:	PASSED ?
s	[4.6 - 4.8]	4.6919e-13	9.9638e-01 (1)	99.54% (100%)	10	YES

PCSA

Complete!!

32.7 GMP Authorized User Roles and Permissions

Index

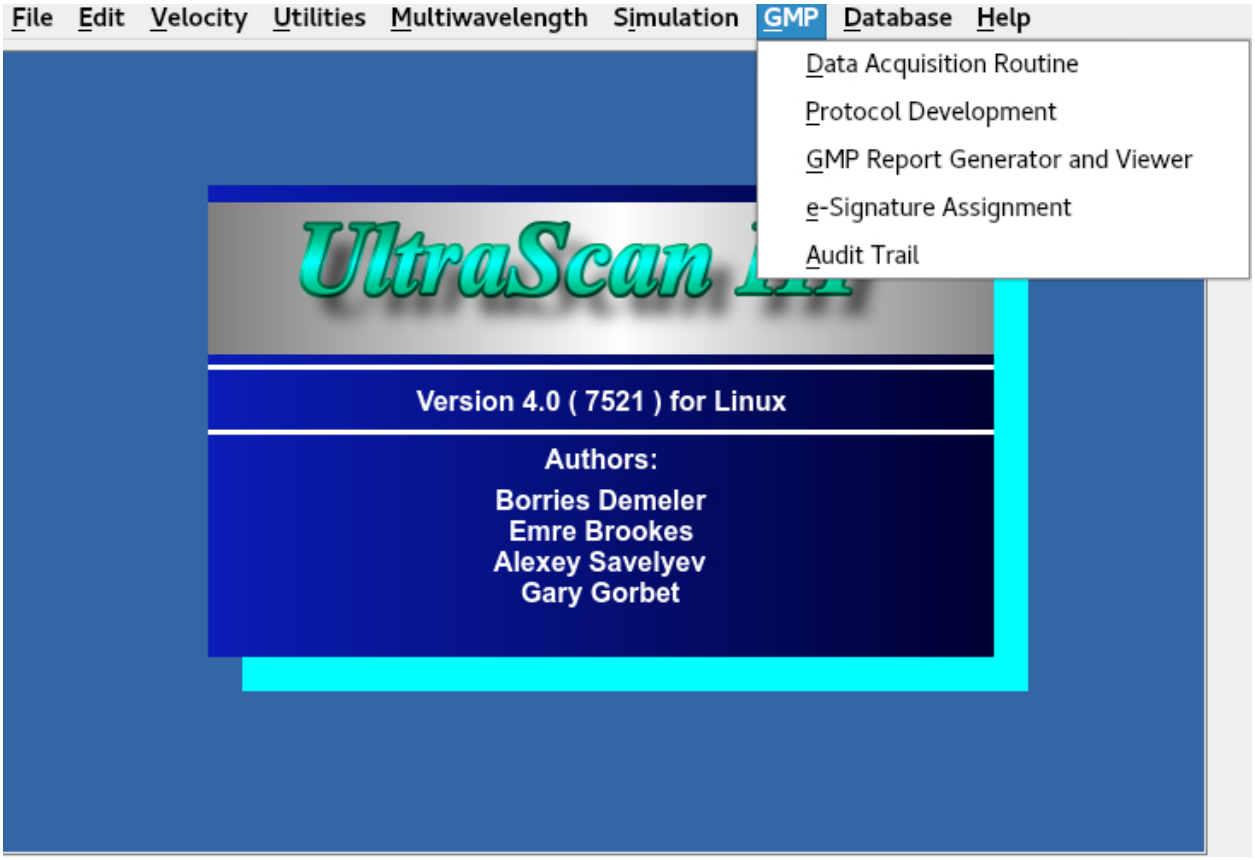
- *User Roles and Permissions:*

32.7.1 User Roles and Permissions:

Role	Responsibilities	Permissions	Restricted Actions
Operator	Conducting AUC experiments in accordance with standard operating procedures (SOPs), and ensuring raw data is properly collected	Setting up and conducting experiments	Modifying data after submission and approving or rejecting experiments.
Reviewer	Reviews experiment results for scientific accuracy and SOP compliance	Access completed experiment data, and view audit trails.	Conducting experiments, modifying data, and approving or rejecting experiments.
Approver	Verifies that the experiment conducted is compliant with SOPs, and that other compliance documentation is in order.	Access completed experiment data, and view audit trails.	Conducting experiments, modifying data, and validating scientific accuracy of experiments.
Subject Matter Expert (SME)	Provides expertise on scientific accuracy and experiment validity.	Access completed experiment data.	Conducting experiments, and modifying data.

Role Assignment: Roles are assigned from the lab-specific UltraScan3 LIMS portal by a system administrator (Level 0 user), whose role is restricted to assigning user levels and roles. Each role is mutually exclusive to ensure a clear separation of responsibilities and maintain GMP compliance.

Note: The GMP module can be accessed using the drop-down 'GMP' menu located at the top of the Ultrascan-III module



GMP Modules Access

32.8 Good Manufacturing Practices (GMP) Data Acquisition

32.8.1 Define Another Experiment

Experiment Sections:

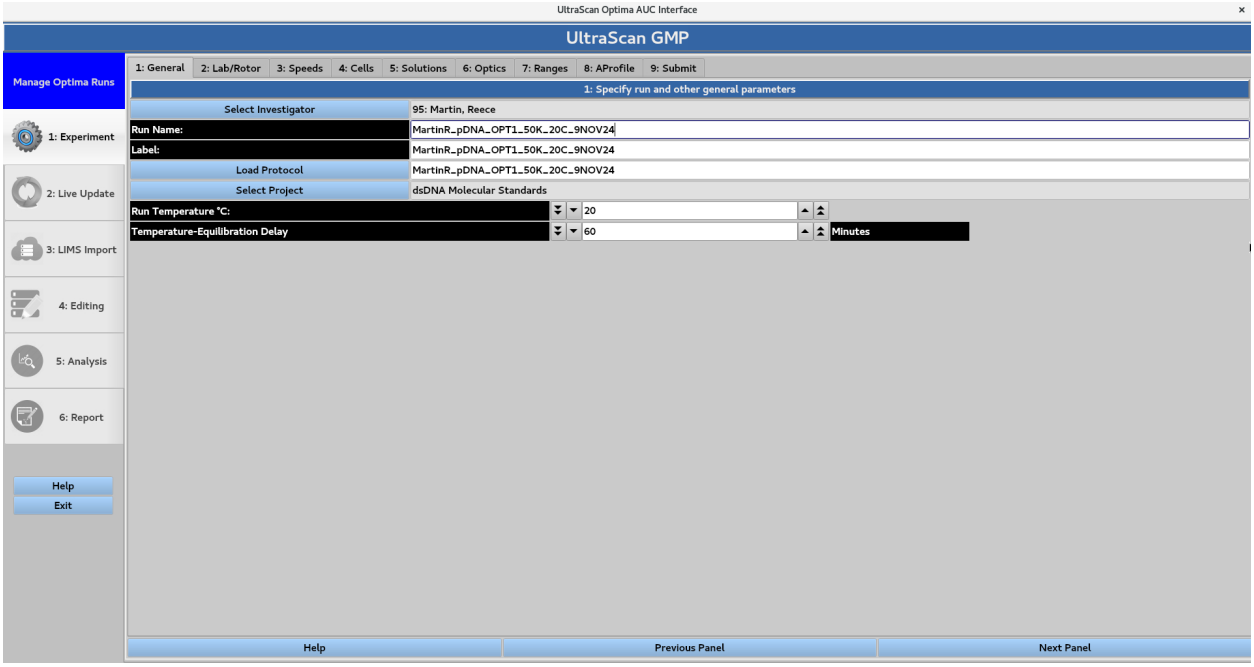
- *1. General*
  - *1.1 Select Investigator*
  - *1.2 Load Protocol*
  - *1.3. Select Project*
- *2. Lab/Rotor*
  - *2.1. Advanced Lab/Rotor/Calibration*
- *3. Speeds*
- *4. Cells*

- *5. Solutions*
  - *5.1. Manage Solutions*
  - *5.2. Solution details*
- *6. Optics*
- *7. Ranges*
  - *7.1. View Ranges*
  - *7.2. Select Wavelengths*
- *8. AProfile (Analysis Profile)*
  - *8.1. General*
  - *8.2. 2DSA Controls*
  - *8.3. PCSA Controls*
- *9. Submit*
  - *9.1. View Experiment details*
  - *9.2. Submit the Run/Save the protocol*

This module allows for the creation and submission of protocols and runs for experiments. Sections can be accessed in any order using the navigation bar at the top of the window or sequentially using the **Previous Panel** and **Next Panel** buttons at the bottom of the window.

1. General

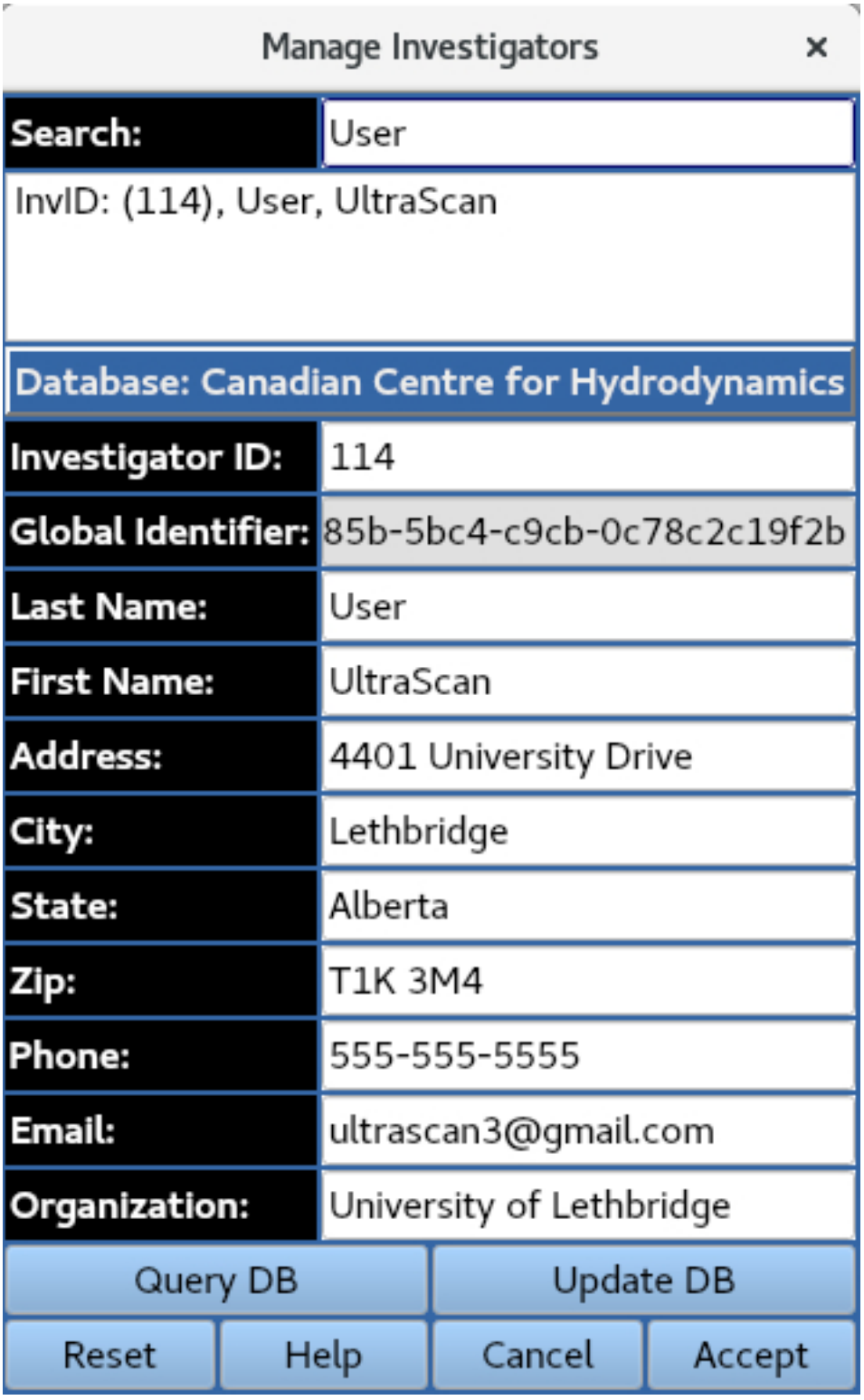
Assign a Run Name, Label, and Protocol Name, as well as the run temperature and temperature-equilibration delay length.



In this page, the investigator, the protocol and the project are selected

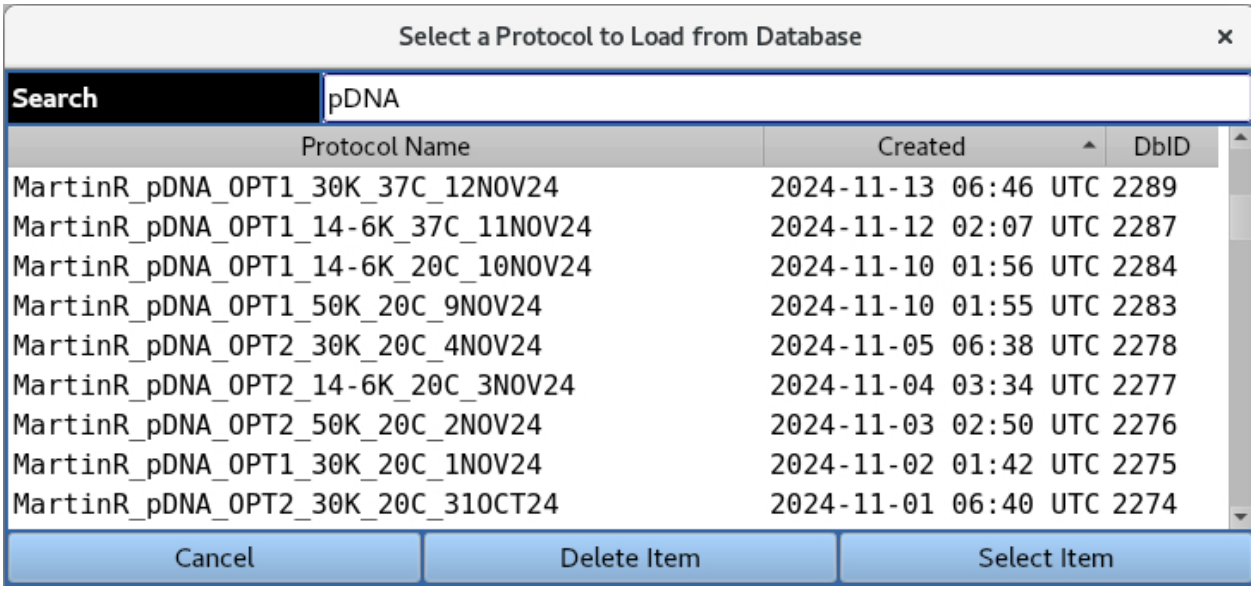
- *Select Investigator*
- *Load Protocol*
- *Select Project*

1.1 Select Investigator



1.2 Load Protocol

Load in the parameters from a pre-existing protocol in the database,

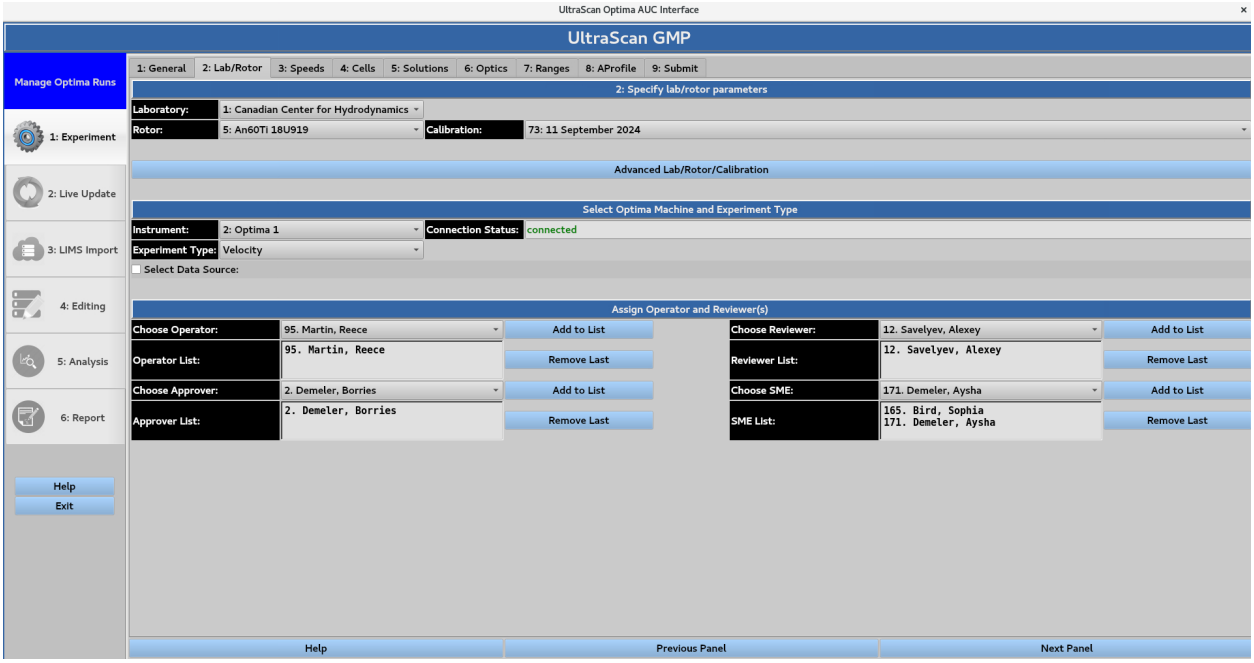


1.3. Select Project

Assign the run to a user-created Project in the :doc:`Project Management <../../us_project> module. Each section requires information on the goals, molecules being analyzed, and experimental design.

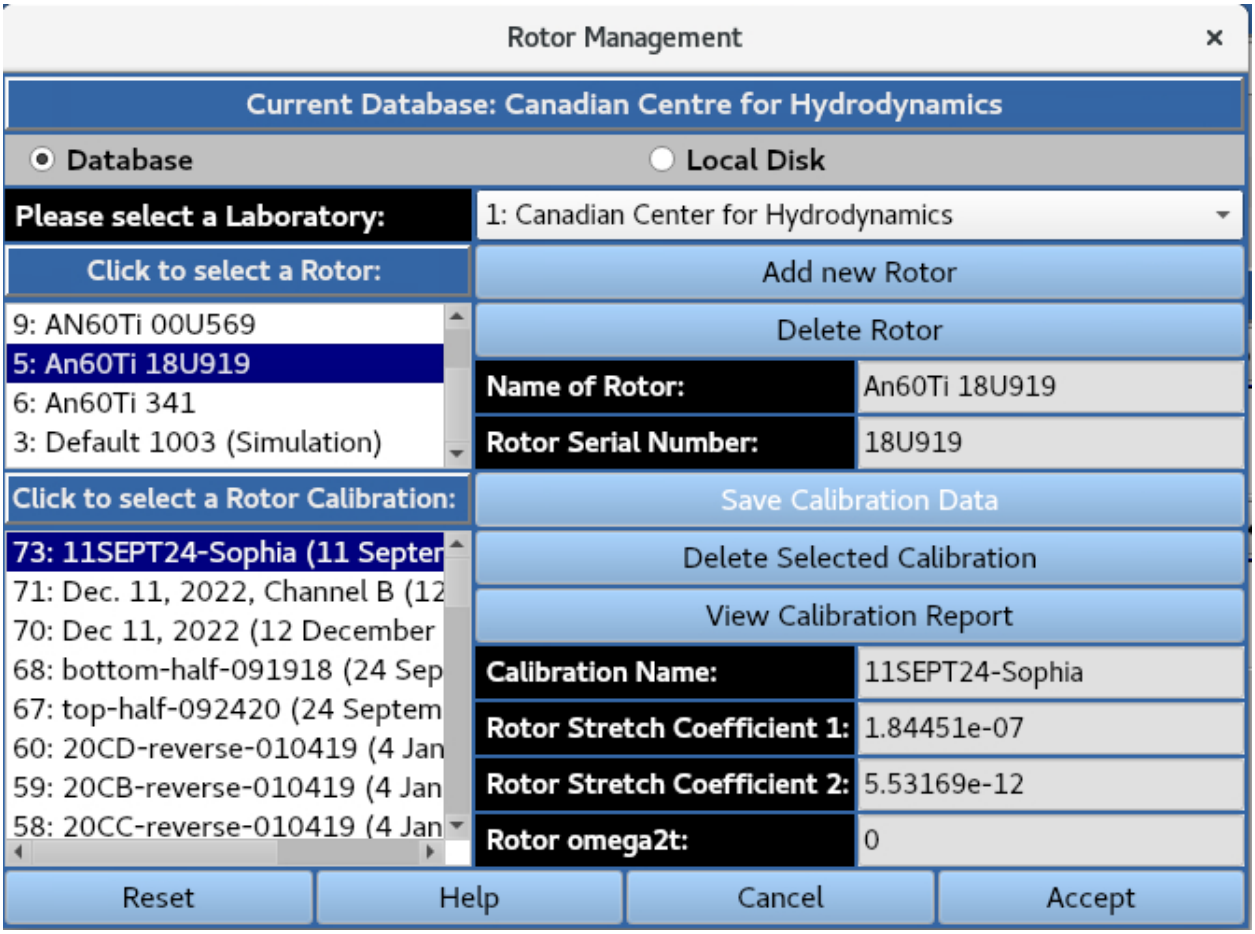
2. Lab/Rotor

Selection of the hardware and experiment type, along with the assignment of roles. The user selects the laboratory, rotor, rotor calibration profile, instrument, and experiment type using the drop-down menus. The roles of Operator, Reviewer, Approver, and Subject Matter Expert (SME) can then be assigned from the drop-down menus.



2.1. Advanced Lab/Rotor/Calibration

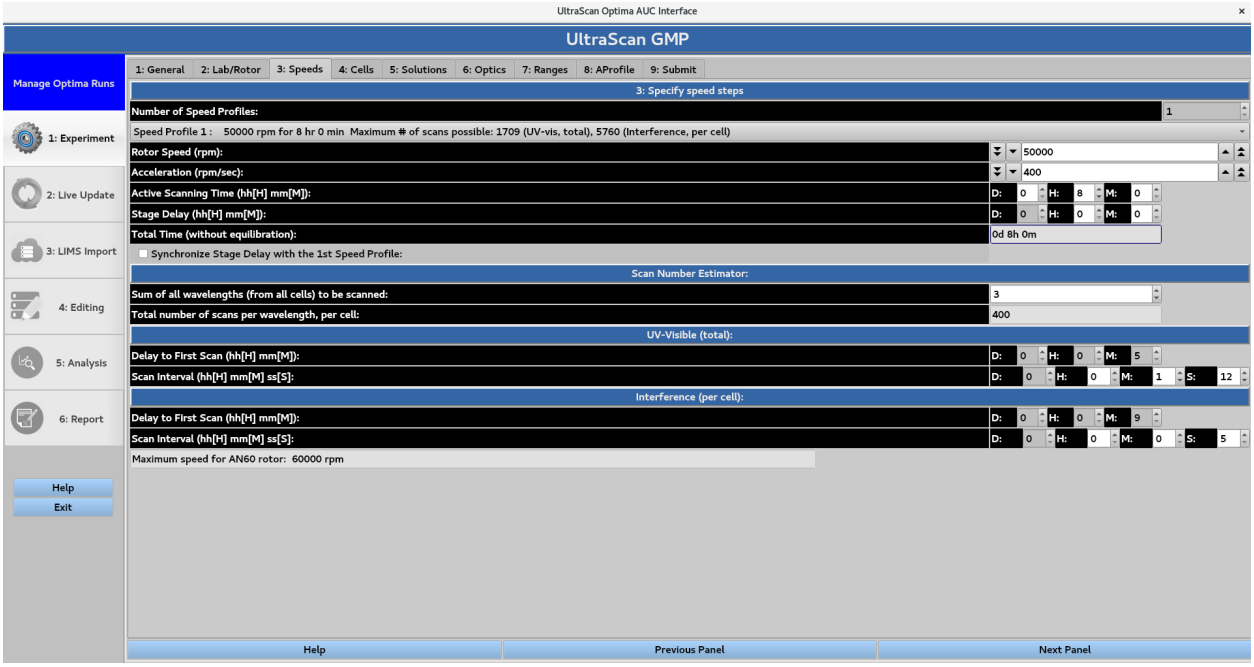
Management of rotors and associated rotor (stretch) calibration profiles is accessed by calling the :doc:`Rotor Management <../../rotor>.



3. Speeds

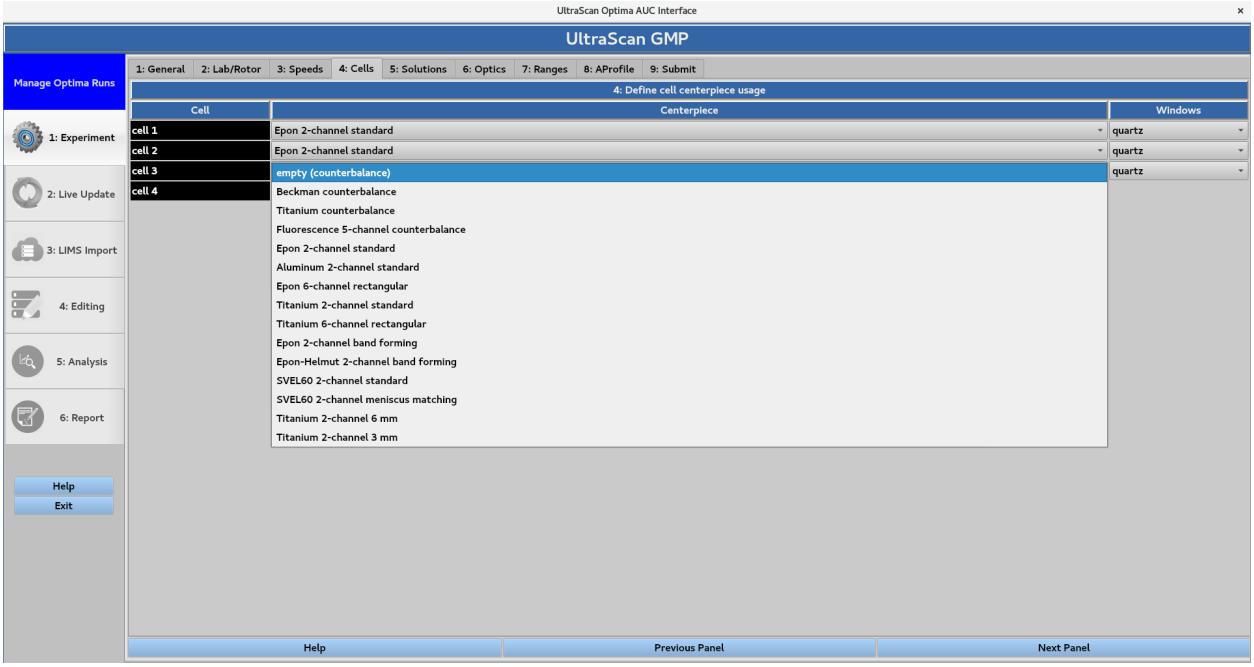
Set run parameters including rotor speed, experiment duration, and scanning in interval for each optical system.

- Under the **3: Specify speed steps** subheading, the number of speed profiles in the GMP module is restricted to 1.
- The **Rotor speed** in revolutions per minute (rpm) can be set by typing the numerical value, or by using the arrows.
- The **Acceleration** in revolutions per minute per second (rpm/sec) can also be set.
- The **Active Scanning Time** can be set in terms of days, hours and minutes, while the **Stage Delay** can be set in terms of hours and minutes.
- The sum of these values gives the **Total Time (without equilibration)**. Under **Scan Number Estimator**, the **Sum of all wavelengths (from all cells) to be scanned** gives the number of distinct wavelengths to be scanned.
- The **Total number of scans per wavelength per cell** is the number of scans for a single wavelength that will be collected for a cell (which produces data for channels A and B).
- The remaining options should typically stay unchanged.



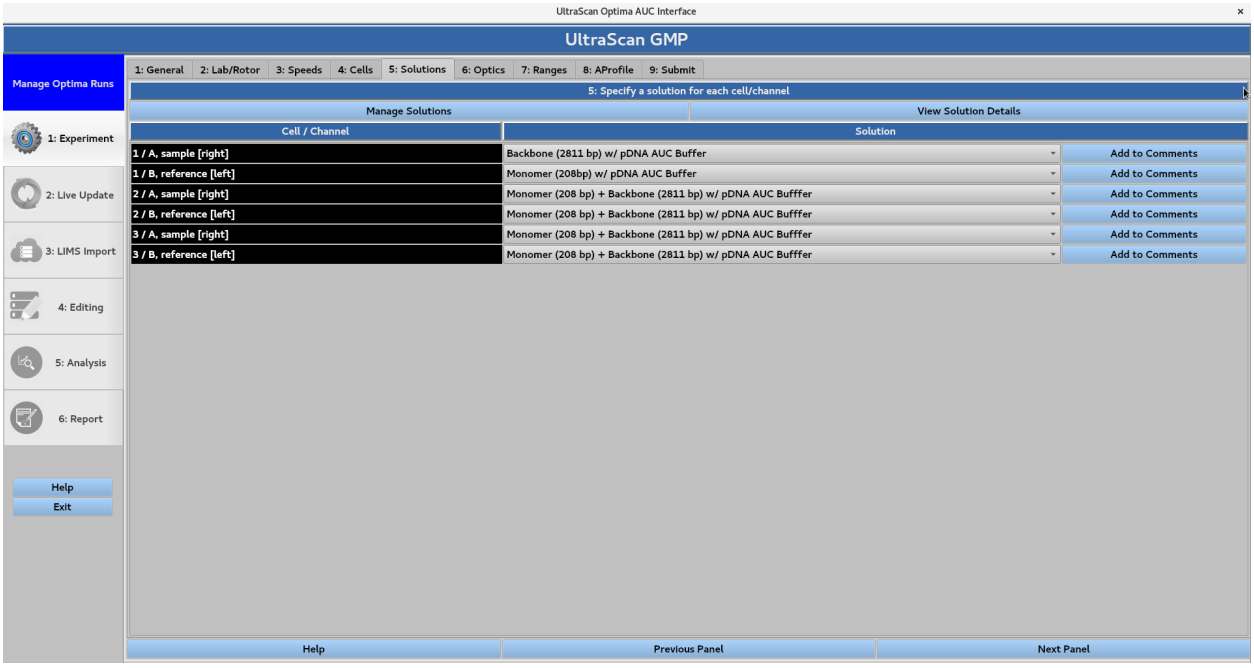
4. Cells

Assignment of centrepiece or counterbalance type, as well as window type for each cell. Correct centerpiece selection is important as the cell bottom position defines the boundary conditions used when fitting the data. Selecting **empty counterbalance**, **Beckman counterbalance**, **Titanium counterbalance**, and **Fluorescence 5-channel counterbalance** will prevent the instrument from collecting scans for the specified holes. The **Windows** selection (options are **quartz** or **sapphire**) does not affect the parameters of the experiment.



5. Solutions

Define the solutions comprised of analytes and buffers that will be measured in the experiment.



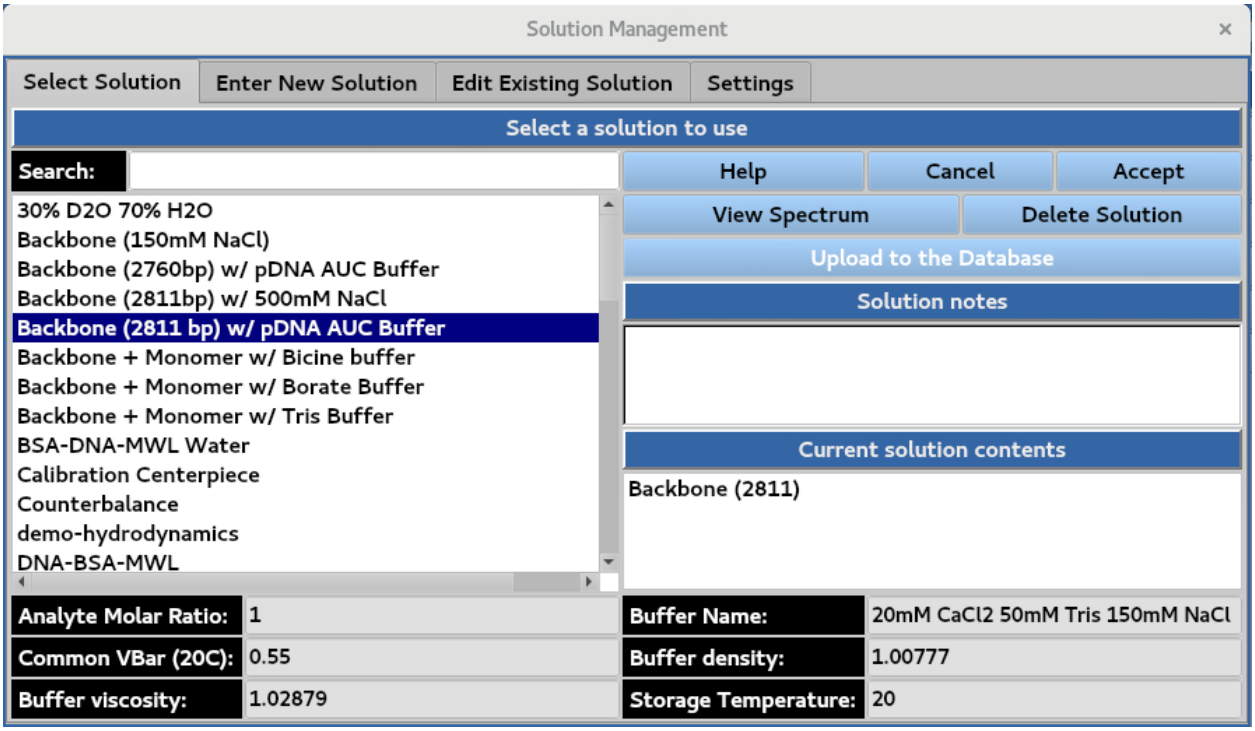
Pre-defined solutions can be assigned to each channel. **Add to comments** can be used to add notes pertaining to the solution in each channel.

In this page:

- *5.1 Manage Solutions*
- *5.2. Solution Details*

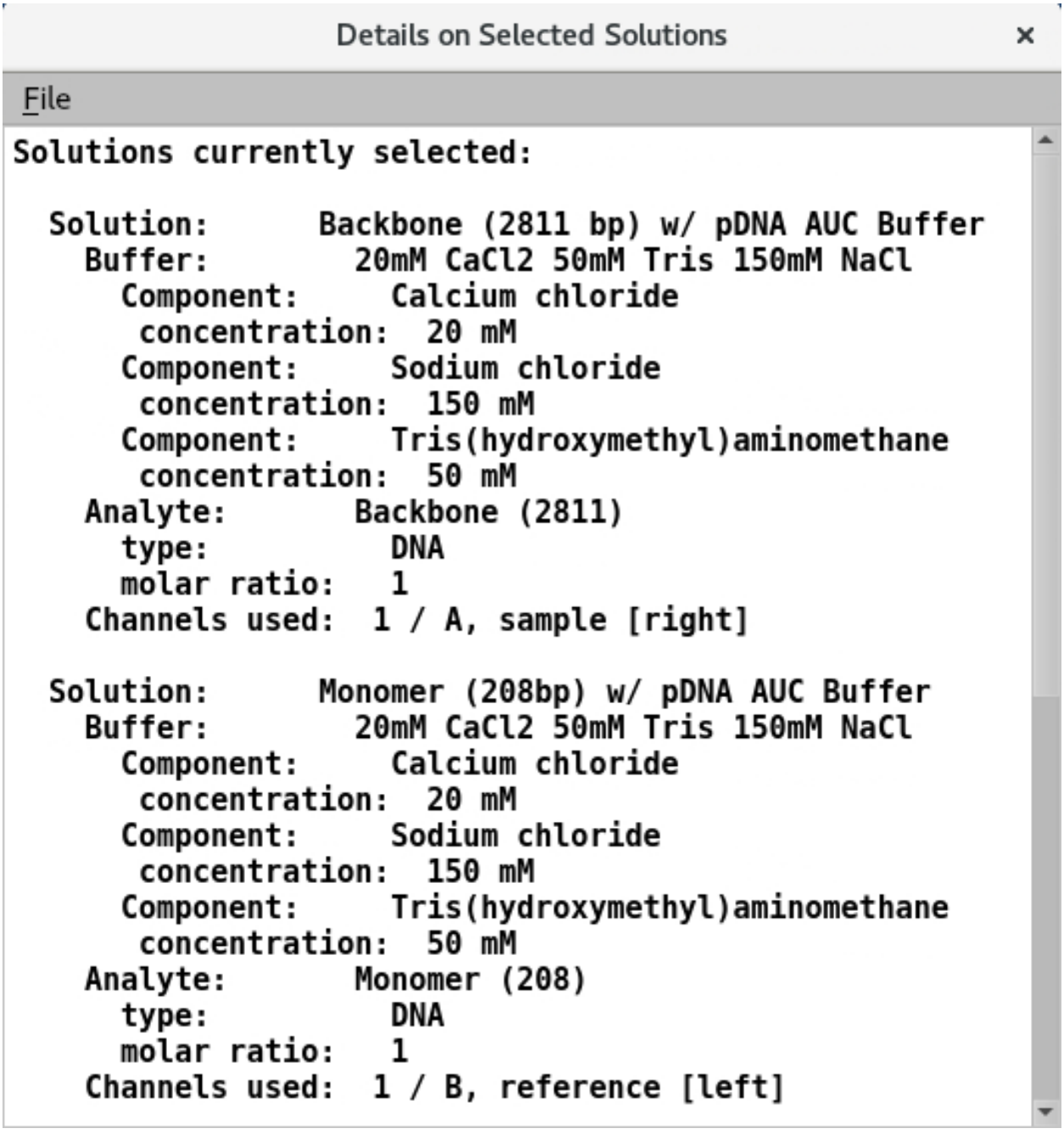
5.1. Manage Solutions

Creation and management of solutions. Existing solution properties can be viewed in the **Select Solution** window by selecting the solution.



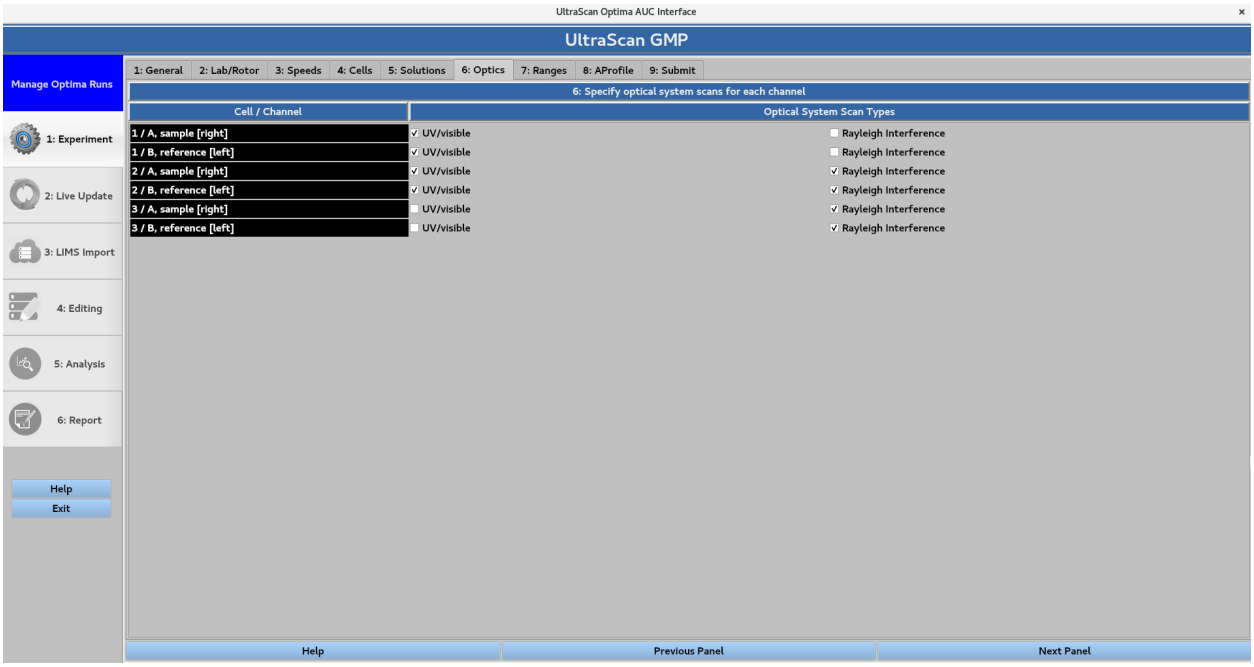
5.2. Solution details

Information on the analytes and buffer of the assigned solutions


Details on Selected Solutions
File
Solutions currently selected:
Solution: Backbone (2811 bp) w/ pDNA AUC Buffer
Buffer: 20mM CaCl2 50mM Tris 150mM NaCl
Component: Calcium chloride
concentration: 20 mM
Component: Sodium chloride
concentration: 150 mM
Component: Tris(hydroxymethyl)aminomethane
concentration: 50 mM
Analyte: Backbone (2811)
type: DNA
molar ratio: 1
Channels used: 1 / A, sample [right]
Solution: Monomer (208bp) w/ pDNA AUC Buffer
Buffer: 20mM CaCl2 50mM Tris 150mM NaCl
Component: Calcium chloride
concentration: 20 mM
Component: Sodium chloride
concentration: 150 mM
Component: Tris(hydroxymethyl)aminomethane
concentration: 50 mM
Analyte: Monomer (208)
type: DNA
molar ratio: 1
Channels used: 1 / B, reference [left]

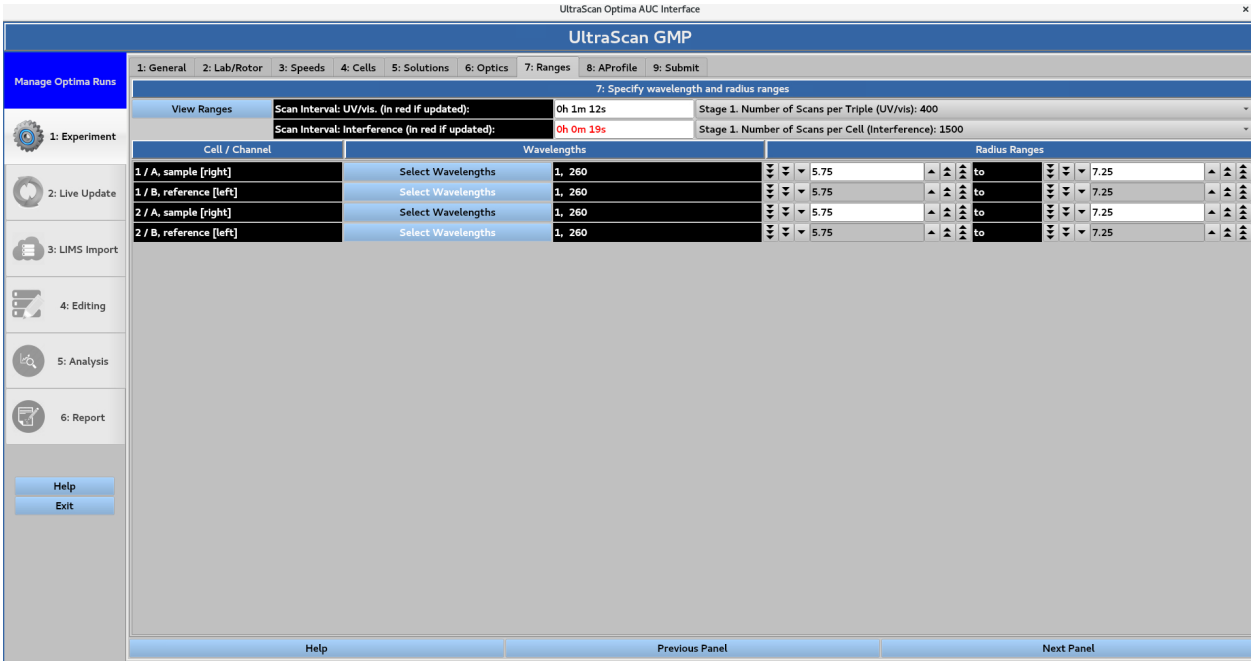
6. Optics

Define which optical system(s) will be used to scan each cell. The **Optical System Scan Type** options are **UV/visible** or **Rayleigh Interference**. Note that the instrument collects scans on a cell-by-cell basis, so the selections for a cell apply to both of its channels.



7. Ranges

Define the wavelength ranges for each cell to be scanned using UV/visible optics. When the wavelength ranges are defined for each cell, the **Scan interval** for the UV/visible and Rayleigh Interference optics appear highlighted in red if updated. This also changes the **Number of scans per Triple (UV/vis)** and the **Number of Scans per Cell (Interference)**. These changes are then reflected in the **3: Speeds** window.

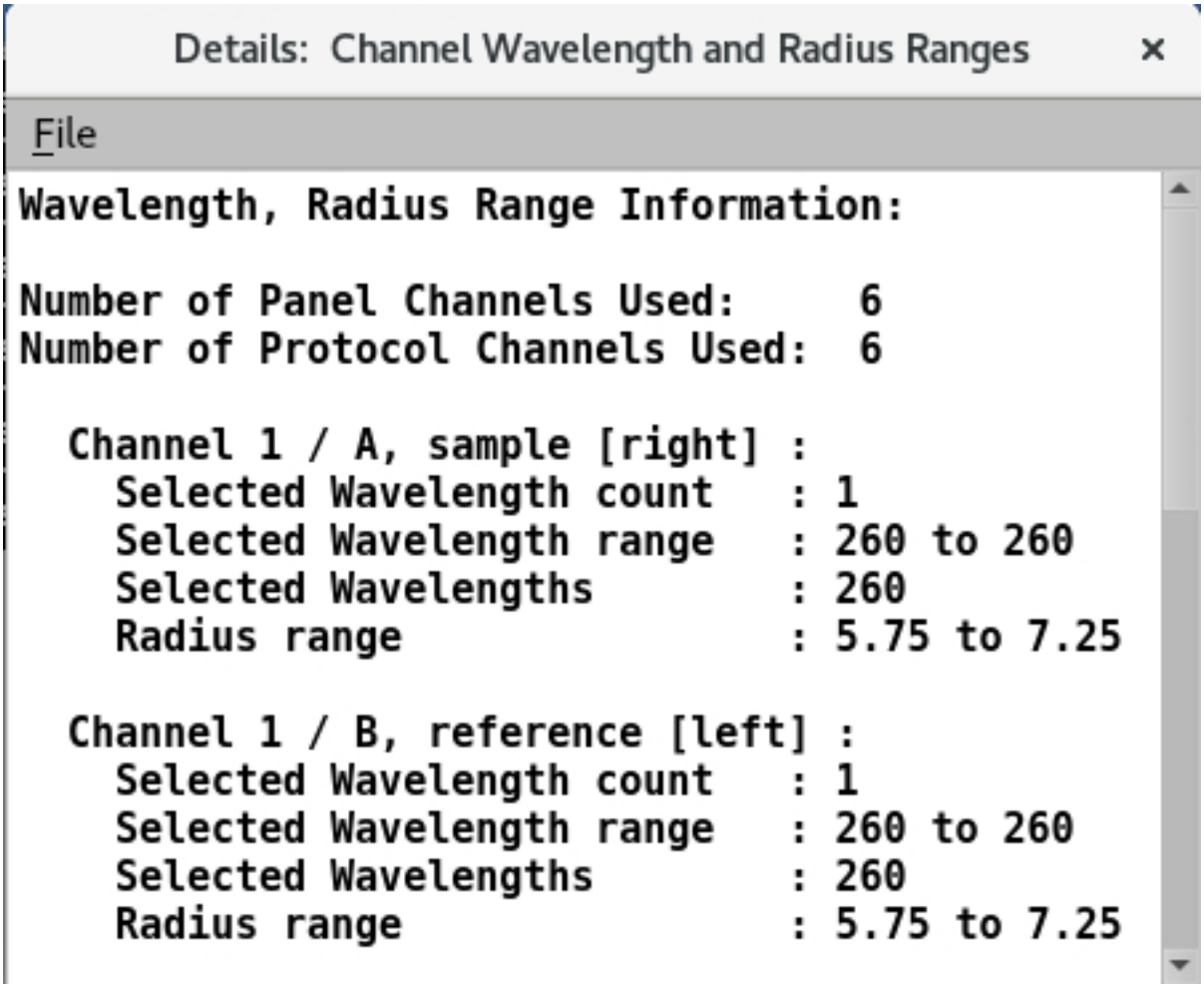


In this page:

- *7.1. View Ranges*
- *7.2. Select Wavelengths*

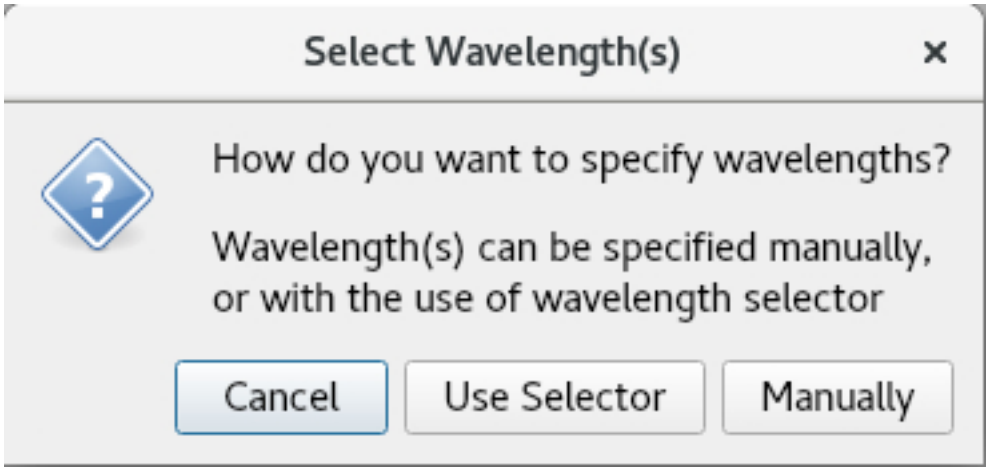
7.1. View Ranges

View the wavelength and radius ranges for each channel in a file

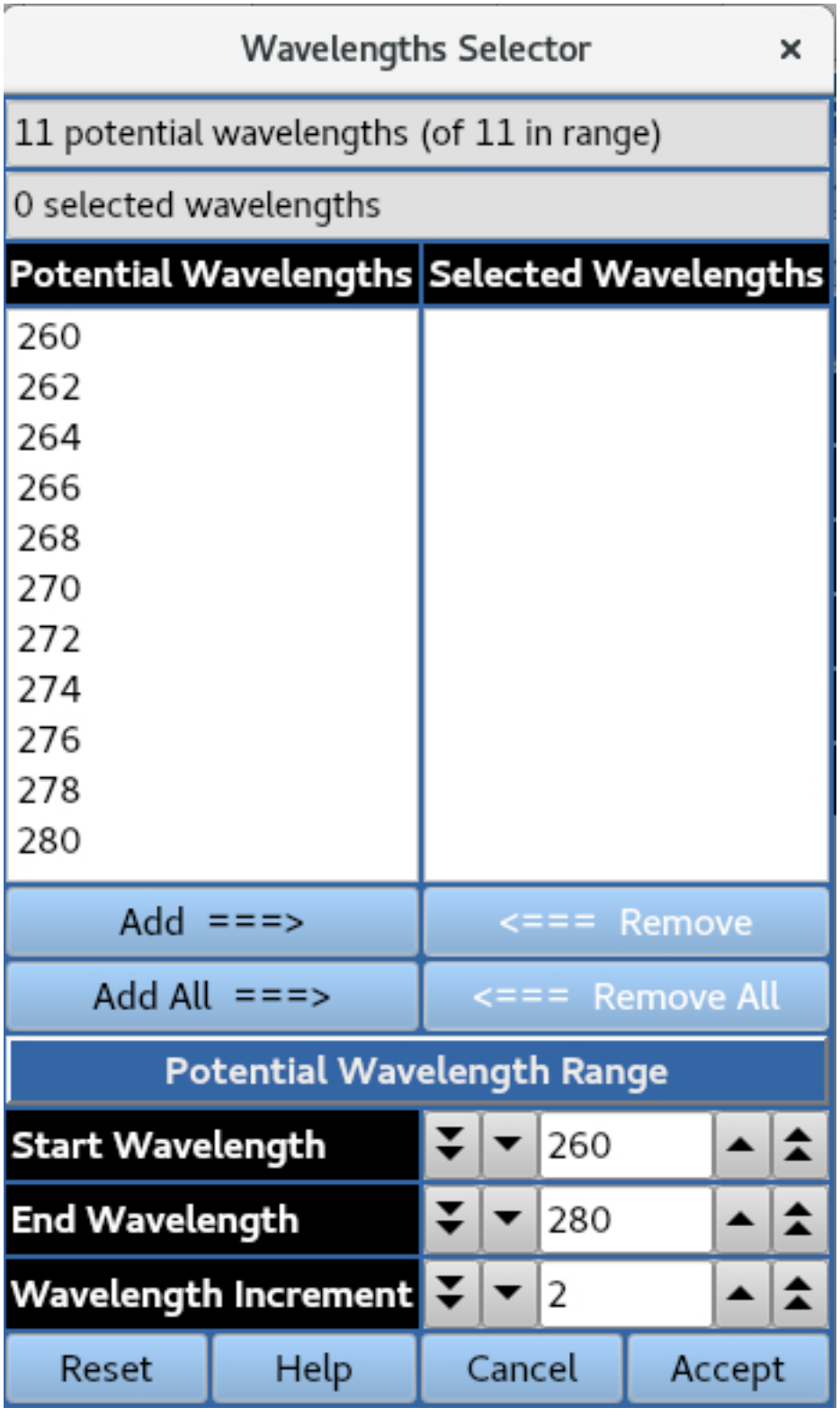


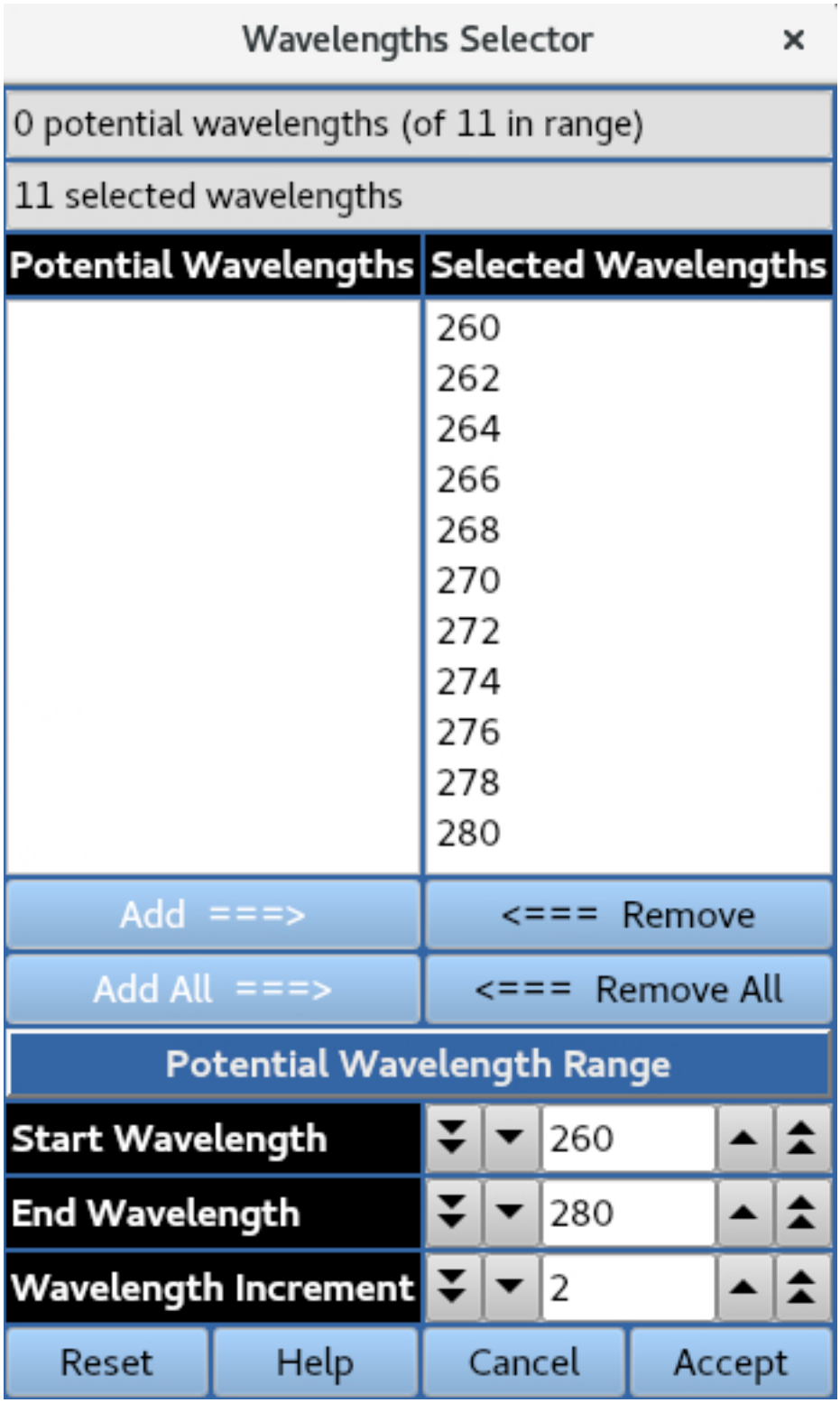
7.2. Select Wavelengths

Assign wavelengths to be scanned on a cell-by-cell basis

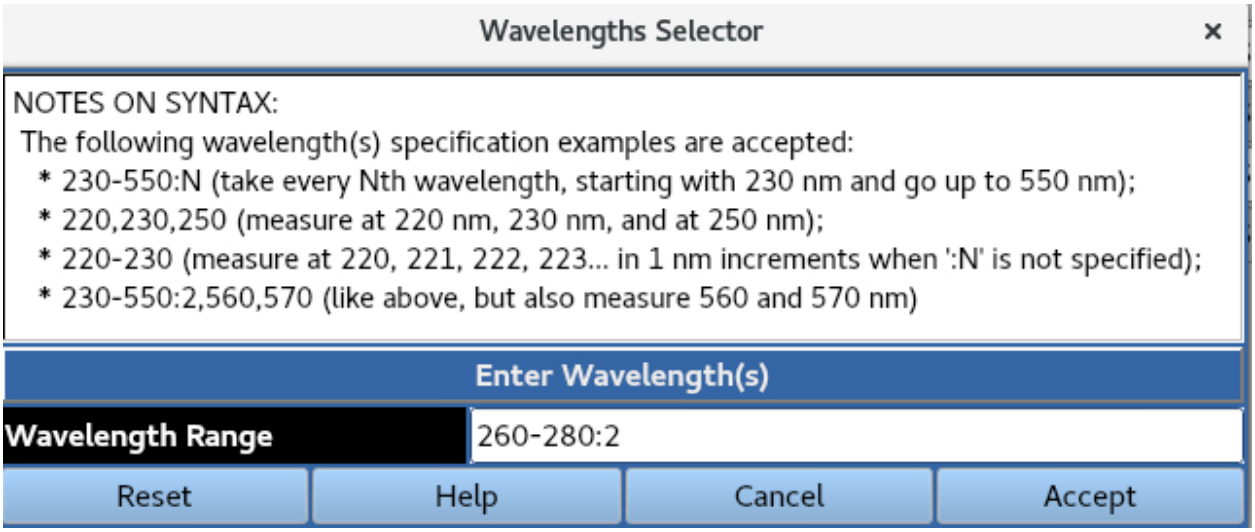


Use Selector allows for wavelengths at a specified increment across a defined range to be presented for selection.


Wavelengths Selector
11 potential wavelengths (of 11 in range)
0 selected wavelengths
Potential Wavelengths
Selected Wavelengths
260
262
264
266
268
270
272
274
276
278
280
Add ===>
<=== Remove
Add All ===>
<=== Remove All
Potential Wavelength Range
Start Wavelength
260
End Wavelength
280
Wavelength Increment
2
Reset
Help
Cancel
Accept



Manually uses the syntax listed below to define the wavelengths to be used



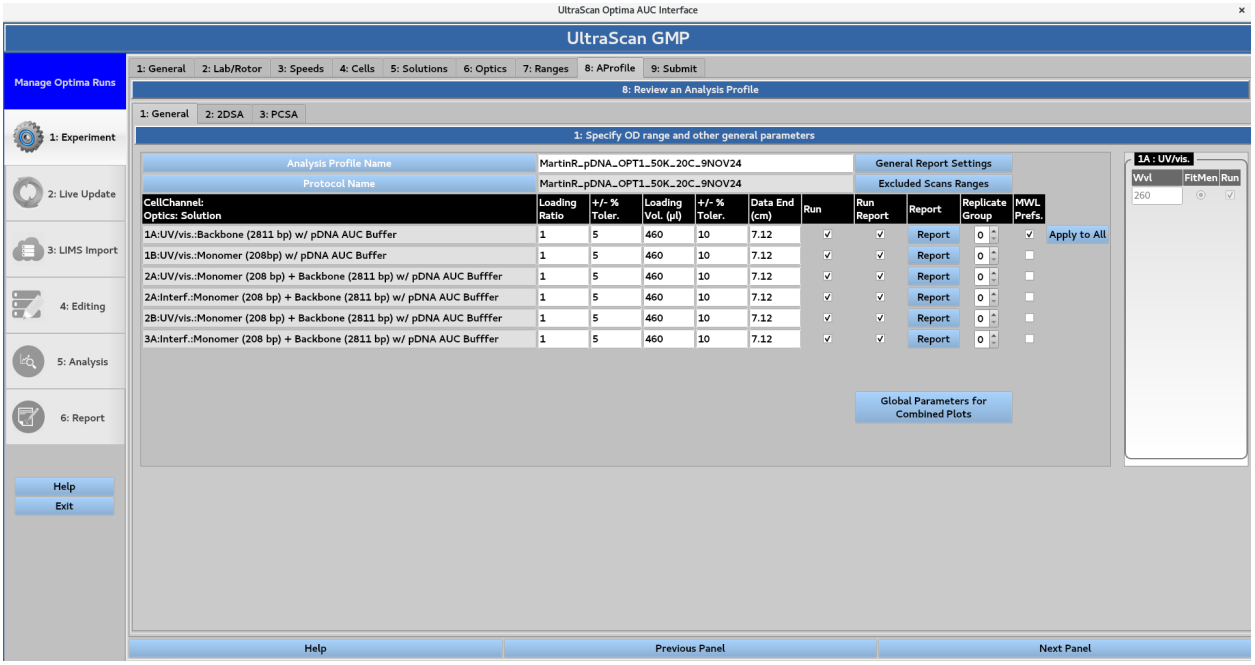
8. AProfile (Analysis Profile)

Define edit profiles, analysis parameters and report settings for the experiment.

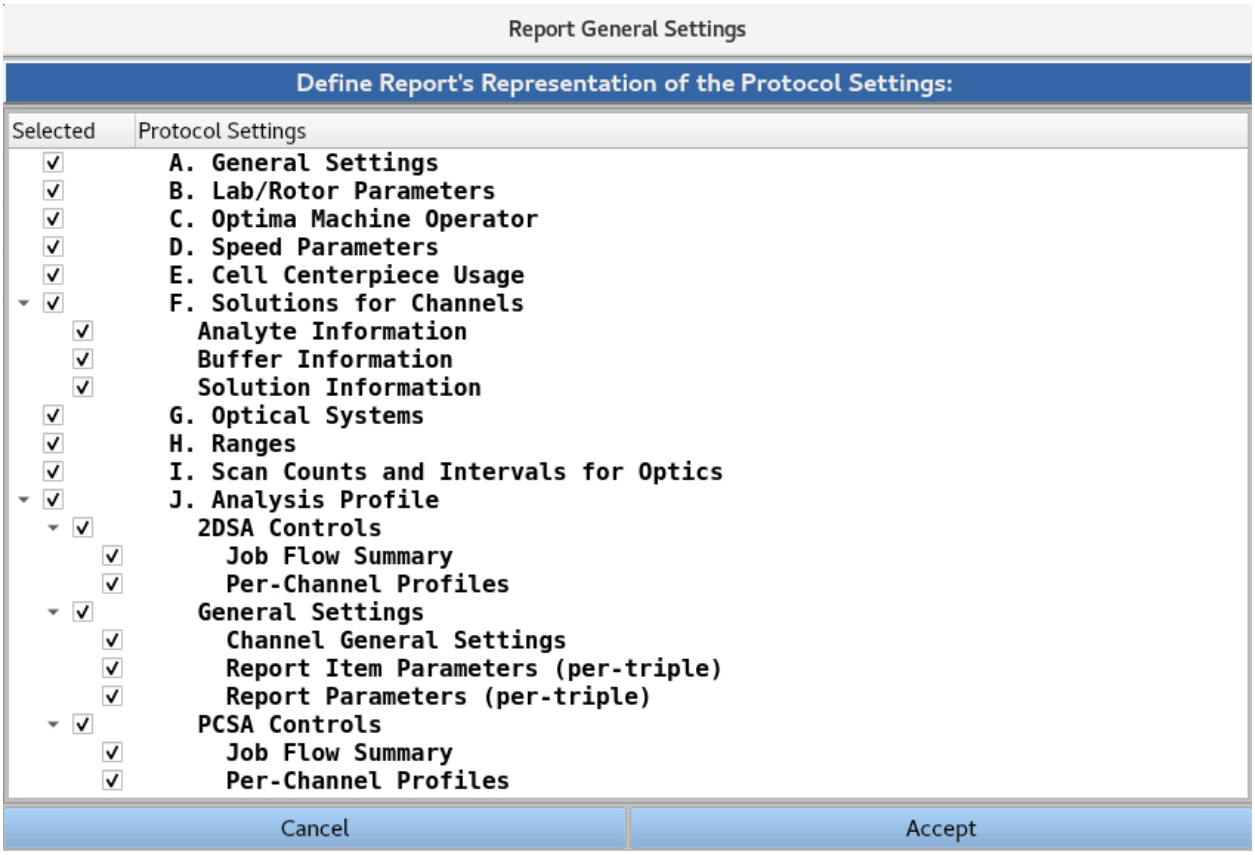
In this page:

- *8.1. General*
- *8.2. 2DSA*
- *8.3. PCSA*

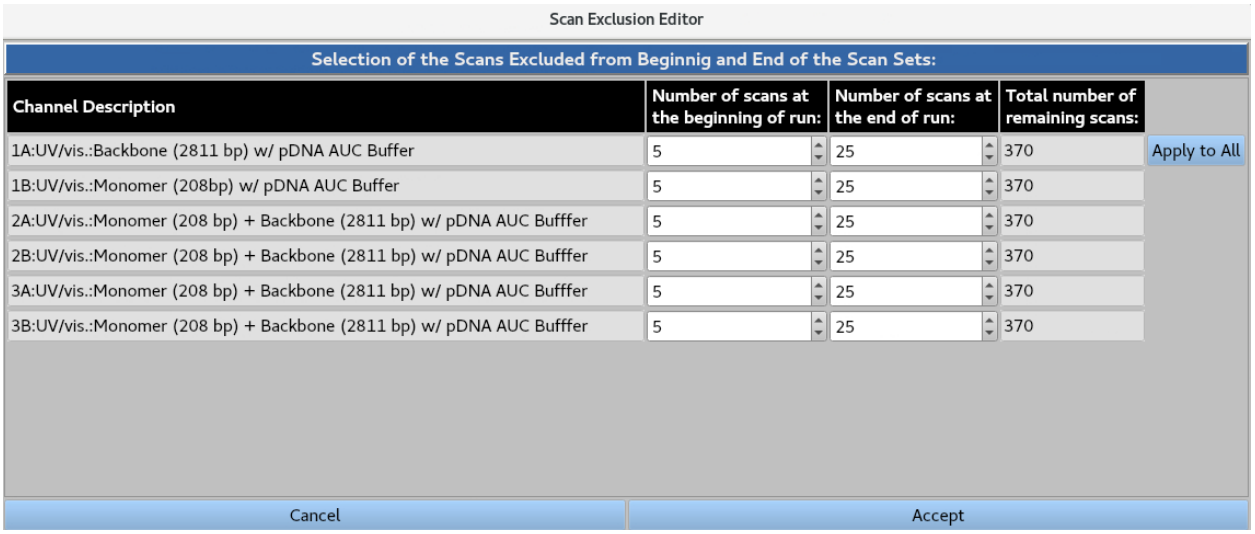
8.1. General



Define numerical values for **Loading Ratio**, the **Â± %Tolerance**, **Loading Volume (Î¼L)**, the **Â± %Tolerance**, and **Data End (cm)**. The user can then select checkboxes to specify whether the analysis should be **Run**, and included in the **Run Report**.

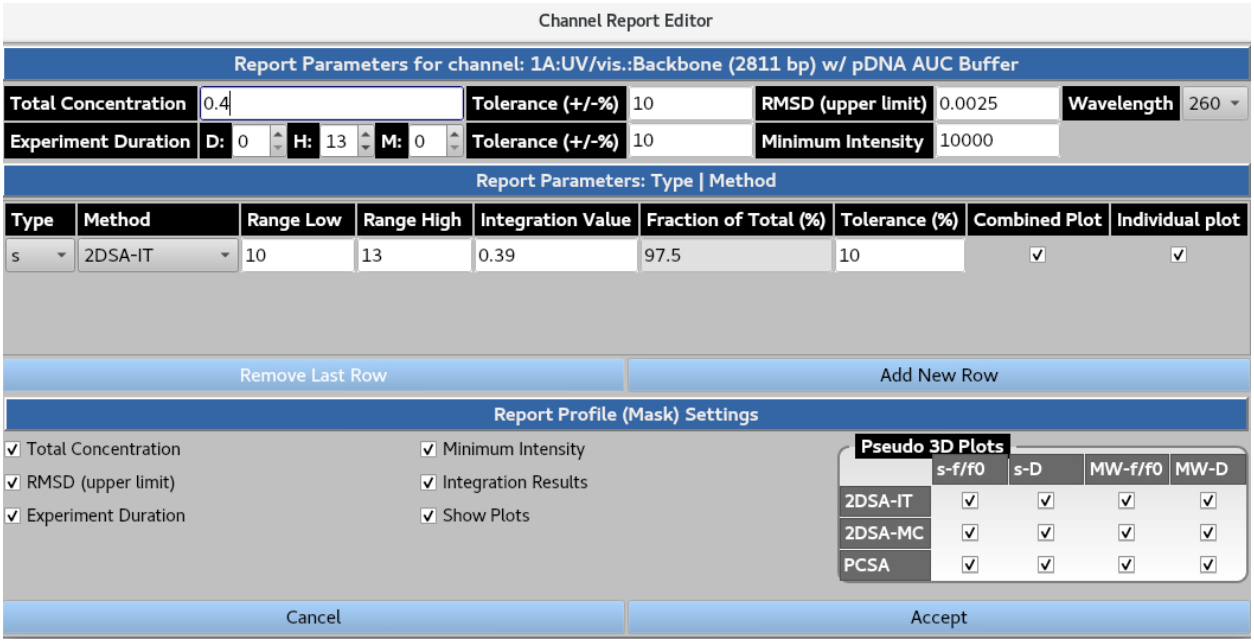


The **Report General Settings** windows allow the user to select/deselect which of the protocol settings (Sections 1-8) will be presented in the report. The {down arrow} button can be used to collapse or expand each subsection.

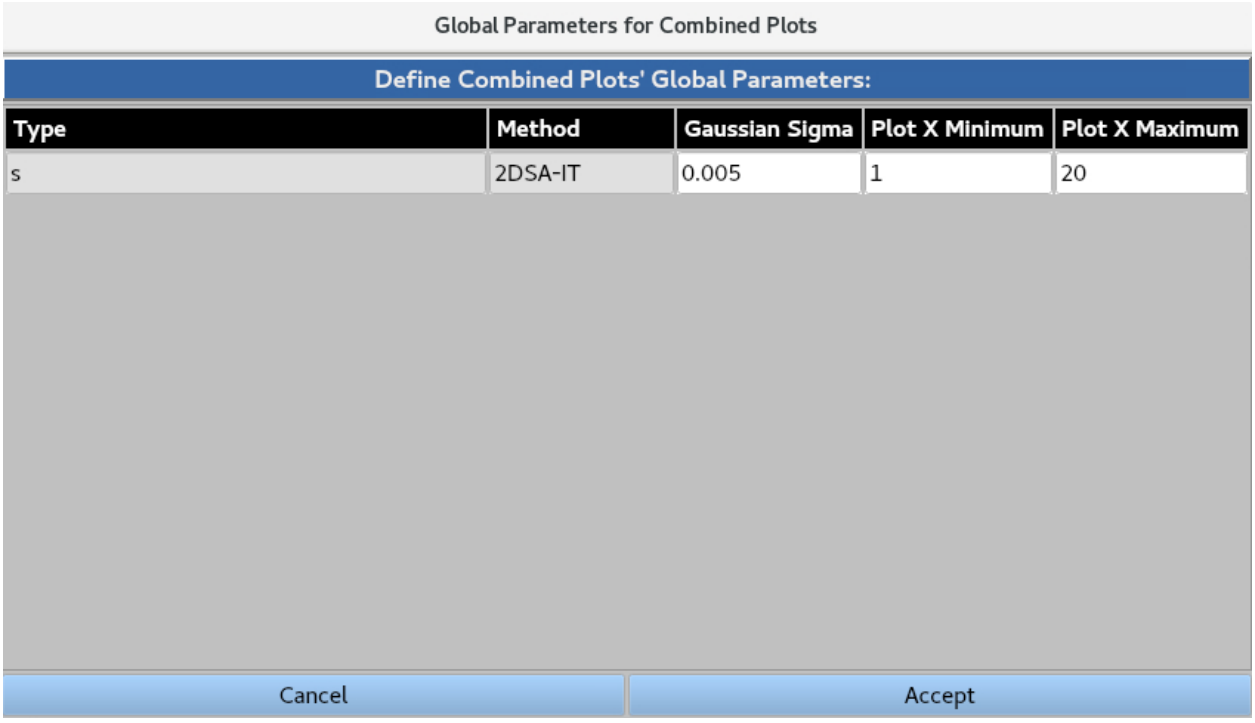


The **Scan Exclusion Editor** allows the user to numerically define the number scans to be removed from the

beginning and end of each channel, either by typing the value or using the arrows. The **Apply to All** button applies the values for the first channel to all channels.

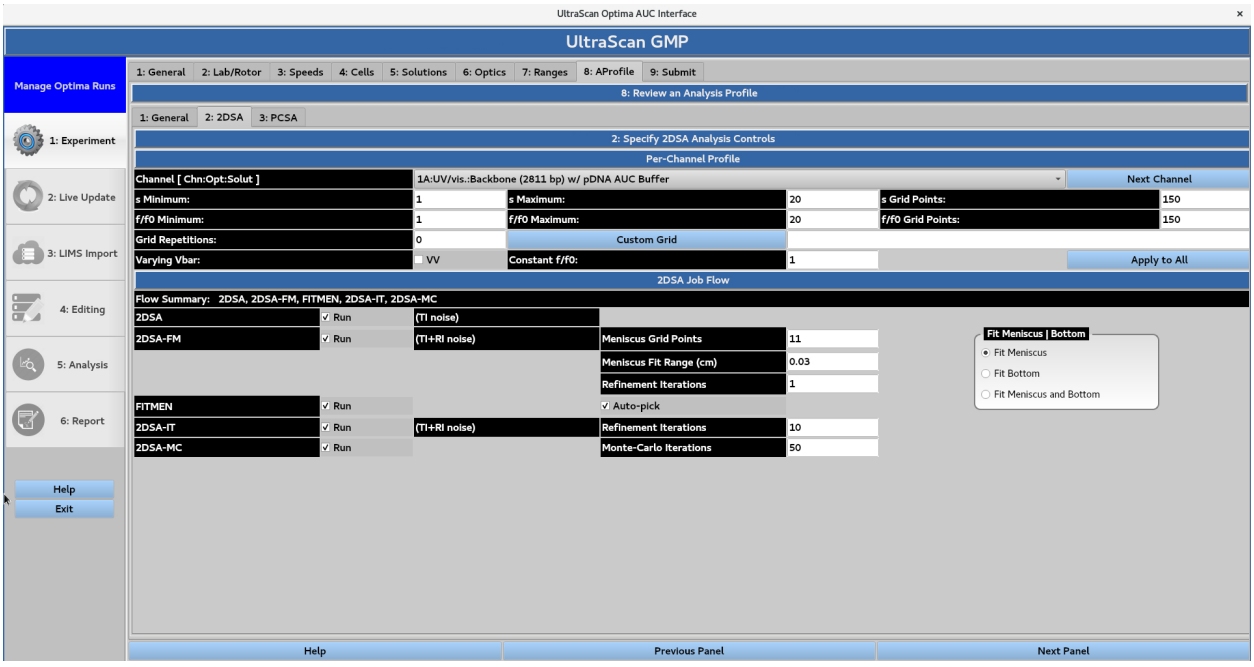


The **Report** button summons the channel-specific report settings window. The user defines the **Total Concentration**, the **Â± %Tolerance**, **RMSD (upper limit)**, the **Â± %Tolerance**, and the wavelength to extract these metrics from. The user then lists the expected **Experiment Duration**, the **Â± %Tolerance**, and **Minimum Intensity** at the selected wavelength from the xenon flash lamp that is deemed acceptable. In the **Report Profile (Mask) Settings** the user can select the following: **Total Concentration**, **Minimum Intensity**, **RMSD (upper limit)**, **Integration Results**, **Experiment Duration** and relevant plots (as well as the pseudo-3D plots for each parameter and analysis type) they wish to be included in the report.



Global Parameters for Combined Plots allows the user to define the plot minimum, plot maximum and Gaussian sigma for each plot type.

8.2. 2DSA Controls



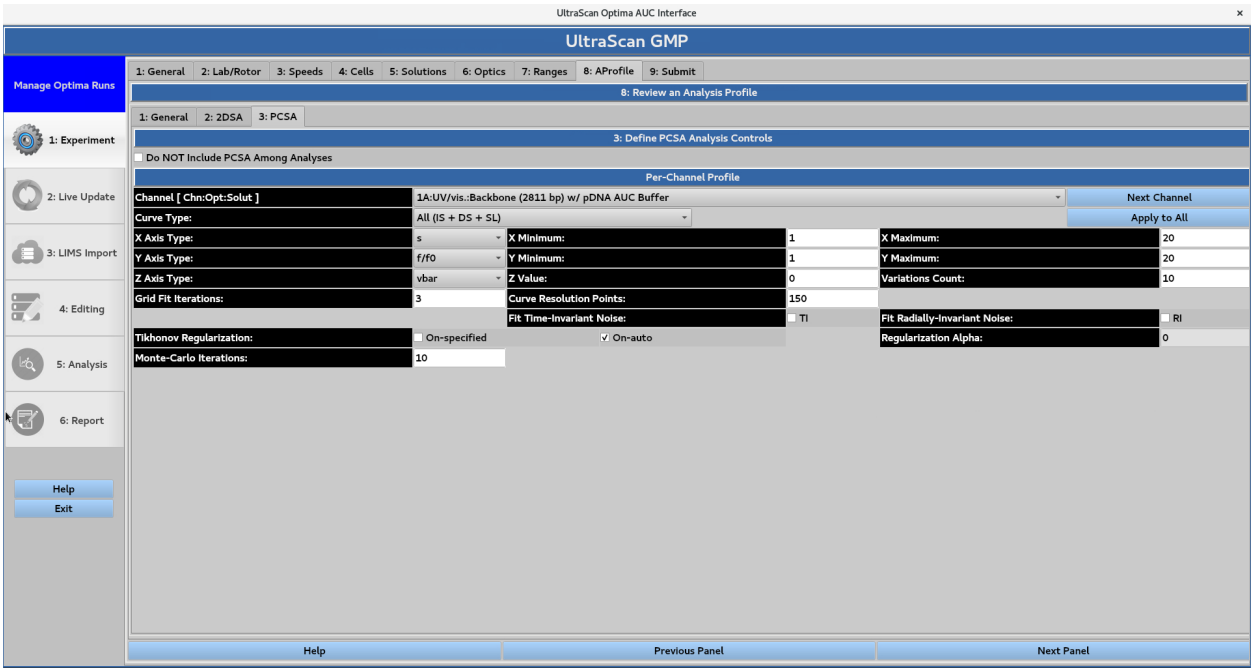
The user can define the 2-dimensional spectrum analysis (2DSA) controls for each channel. The grid size is defined by the minimum and maximum values for the sedimentation coefficient (s) and frictional ratio ($f/f0$),

while the grid resolution is defined by the number of points for each of the parameters. A 2DSA **Custom Grid** (CG) can also be used if the user requires higher resolution in certain regions of the data, and only needs lower resolution in other regions. The user can also select to vary the partial specific volume or **Vbar** parameter and hold the frictional ratio constant at a specified value. The **Apply to All** button can be used to apply these settings to all of the channels.

The **2DSA Job Flow** panel allows the user to control which analysis steps are performed, including:

- 2DSA
- 2DSA with meniscus fit (FM), with a user-defined number of grid points, range (cm), and number of refinement iterations. The user can also choose to fit **only the bottom of the channel** or **both the meniscus and the bottom.**
- FITMEN meniscus processing, with the option to **Auto-pick** the meniscus position
- 2DSA with iterative refinement (IT), with a user-defined number of iterations
- 2DSA with Monte Carlo iterations (MC), with a user-defined number of Monte Carlo iterations.

8.3. PCSA Controls



The user can select the option to not include the parametrically-constrained spectrum analysis (PCSA) in the submitted jobs. If the box is left unchecked, the user can define the PCSA controls for each channel.

The desired curve type is selected from the drop-down menu, and the user defines parameters for the x, y, and z axes.

- The x and y axes are defined by minimum and maximum values
- The z axis is assigned a value and a variation count

- Numerical values are then assigned for the grid fit iterations and curve resolution points

Optional settings include:

- Time- and radially-invariant noise fitting
- Tikhonov regularization, with either an auto-picked or manually-selected alpha
- Number of Monte Carlo iterations.

The **Apply to All** button can be used to apply these parameters and settings to all of the channels.

9. Submit

Save a protocol to the database or submit a run to an instrument.

In this page:

- *9.1. View Experiment Details*
- *9.2. Submit the Run/Save the Protocol*

9.1. View Experiment details

Summary of the information defined in sections 1-8

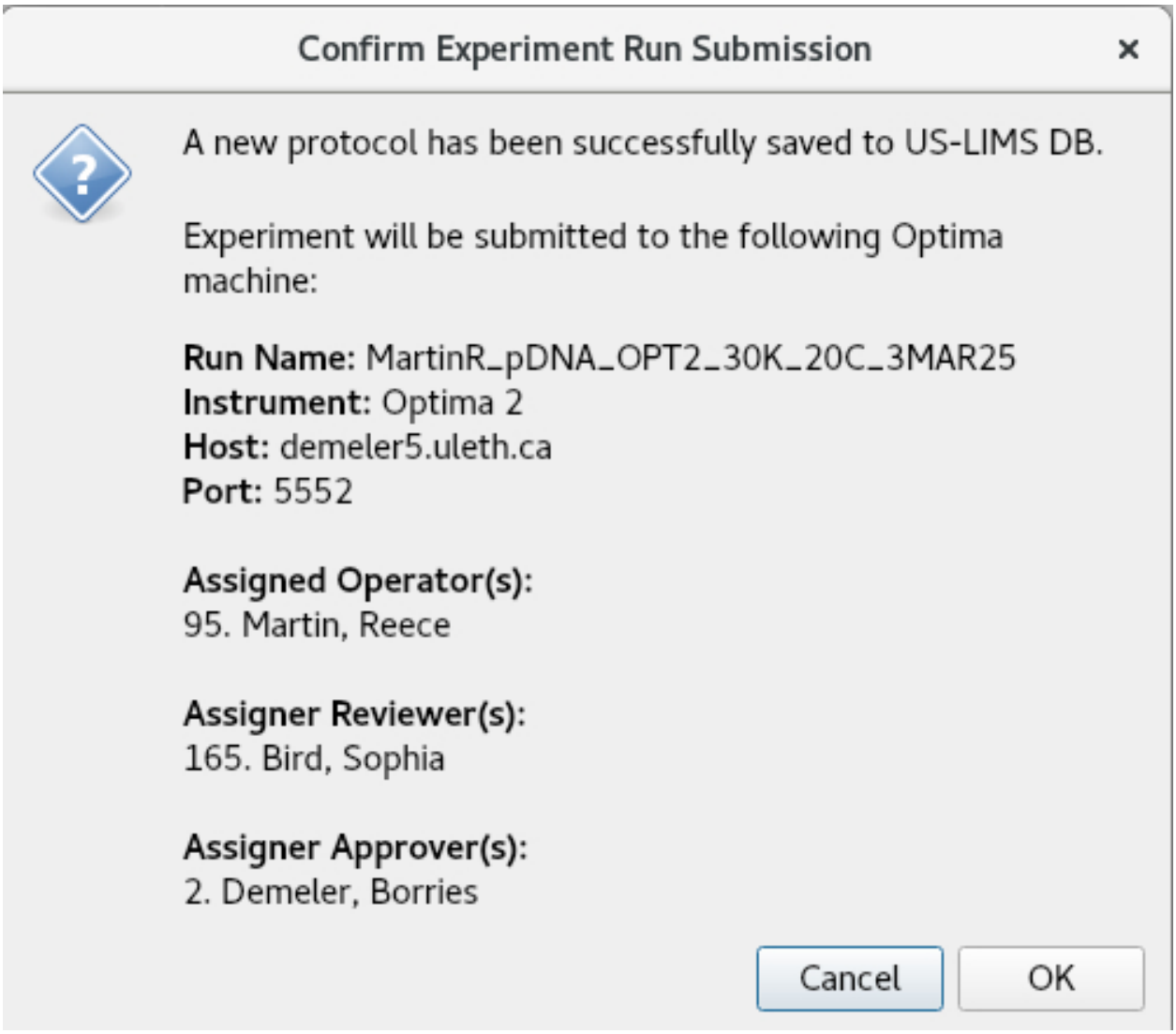
Details on Experiment Controls

File

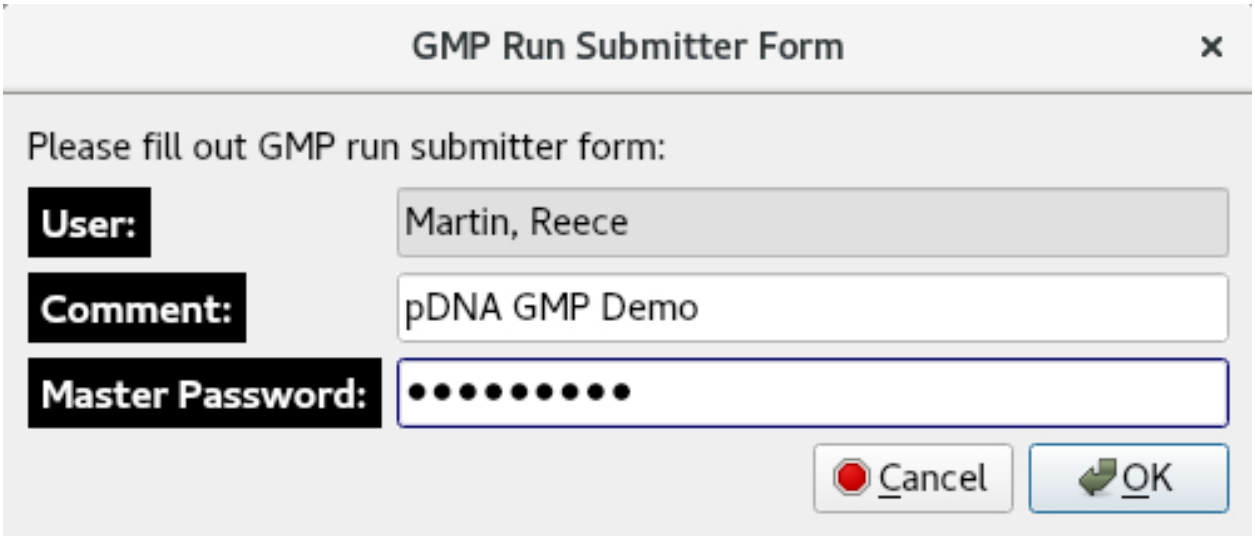
```
Experiment Control Information:

General
  ALL SPECIFIED:              no
  Investigator:               95: Martin, Reece
  DB / Disk:                  DB
  RunId:
  Project:                    dsDNA Molecular Standards

Rotor
  ALL SPECIFIED:              YES
  USER CHANGES:               YES
  Laboratory:                 Canadian Center for Hydrodynamics
  Rotor:                      An60Ti 18U919
  Calibration:                11 September 2024
  InstrumentID:               2
  OperatorID:                 95
  Experiment Type:            Velocity

Speeds
  ALL SPECIFIED:              YES
  USER CHANGES:               no
  Number Speed Steps:         1
    Step 1 :
      Speed:          50000 rpm
      Acceleration:   400 rpm/sec
      Duration:       8 h 0 m
      Delay (accel.):0 h 5 m
      Stage Delay:    0 h 0 m
      Scan Interval: 0 h 1 m 12 s

Cells
  ALL SPECIFIED:              YES
  Specified Centerpieces:     4
    cell 1 : Epon 2-channel standard  ( quartz )
    cell 2 : Epon 2-channel standard  ( quartz )
    cell 3 : Epon 2-channel standard  ( quartz )
    cell 4 : Beckman counterbalance
  Counterbalance:             Beckman counterbalance

Solutions
  ALL SPECIFIED:              YES
  Number Channels Used:       6 of 6
  Channel 1 / A, sample [right] :  protocol comments --
    Backbone (2811 bp) w/ pDNA AUC Buffer
                  :  run comments --
    Backbone (2811 bp) w/ pDNA AUC Buffer, plus more
  Channel 1 / B, reference [left] :  protocol comments --
```

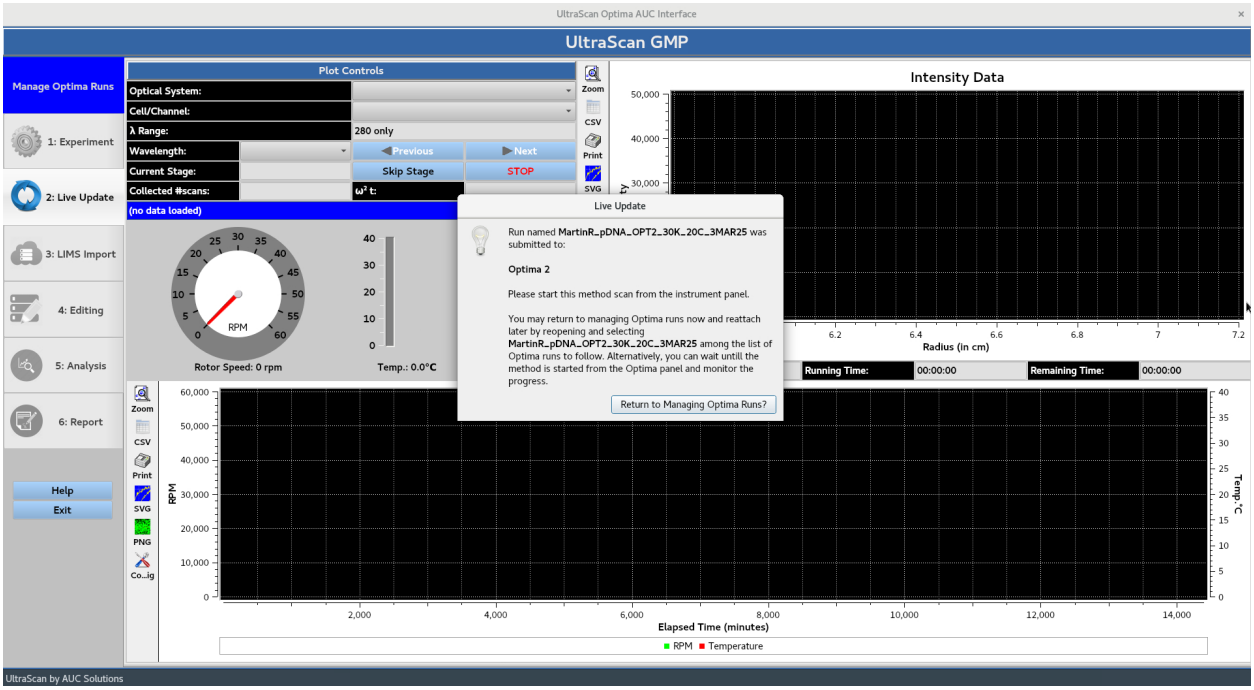
9.2. Submit the Run/Save the protocol

If the **Run Name** field in 1: General is left empty, the submission will be saved as a protocol instead of a run. All sections (1-8) must be completed before the button in the top-right corner becomes active and allows for run submission or protocol saving. Submission prompts the following window:



The **OK** button prompts the **GMP Run Submitter Form**, allowing the user to add a comment and prompting them for their master password for authentication.



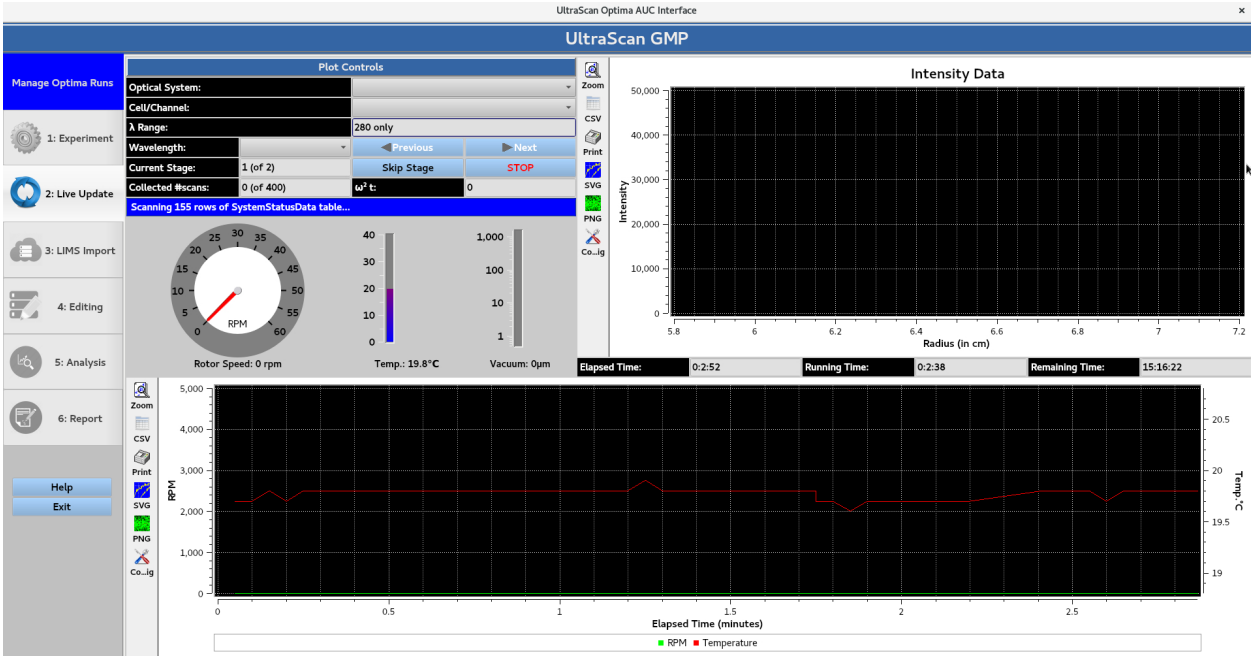


The **OK** button brings up the **Live Update** window which instructs the user to start the run from the instrument panel. Alternatively, the user can press **Return to Managing Optima Runs?** to return to the **Manage Optima Runs** window.

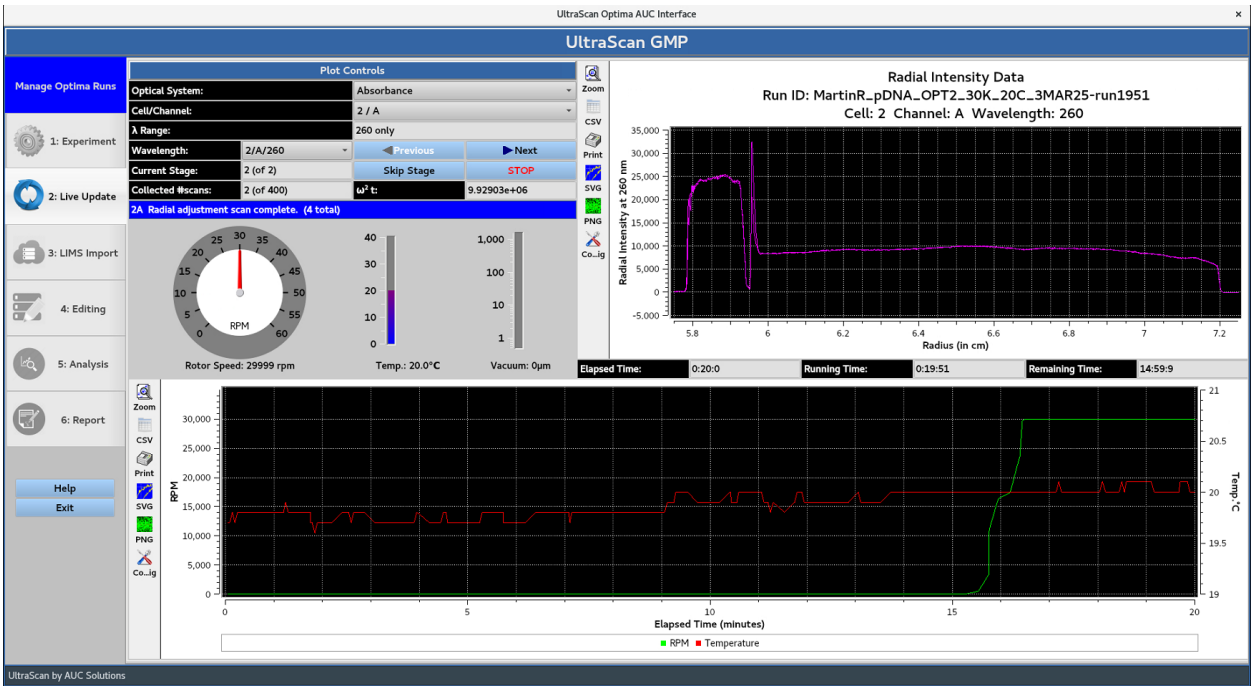
Triple========== GMP Data Acquisition ==========

Acquire, import, edit, and analyze data from the Optima AUC.

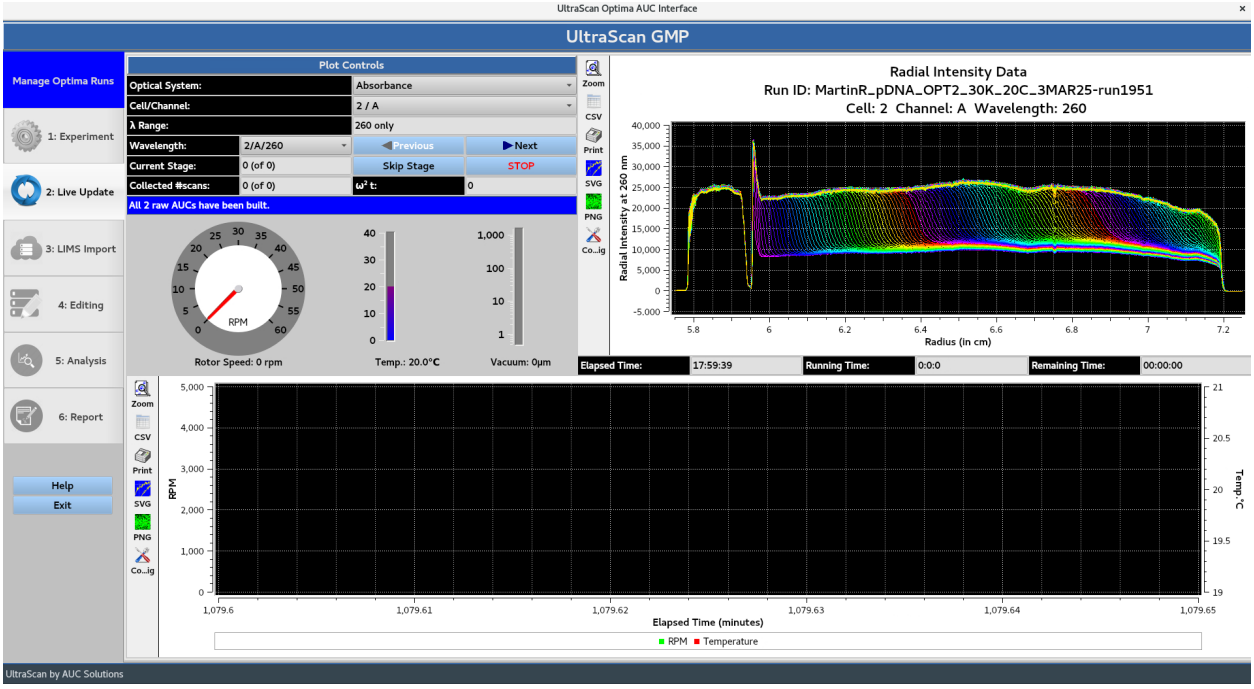
After a run is initiated from the instrument, the user is presented with the 'Data Acquisition Window'. Which displays the current temperature, vacuum level, as well as the elapsed, running, and remaining time for the experiment.



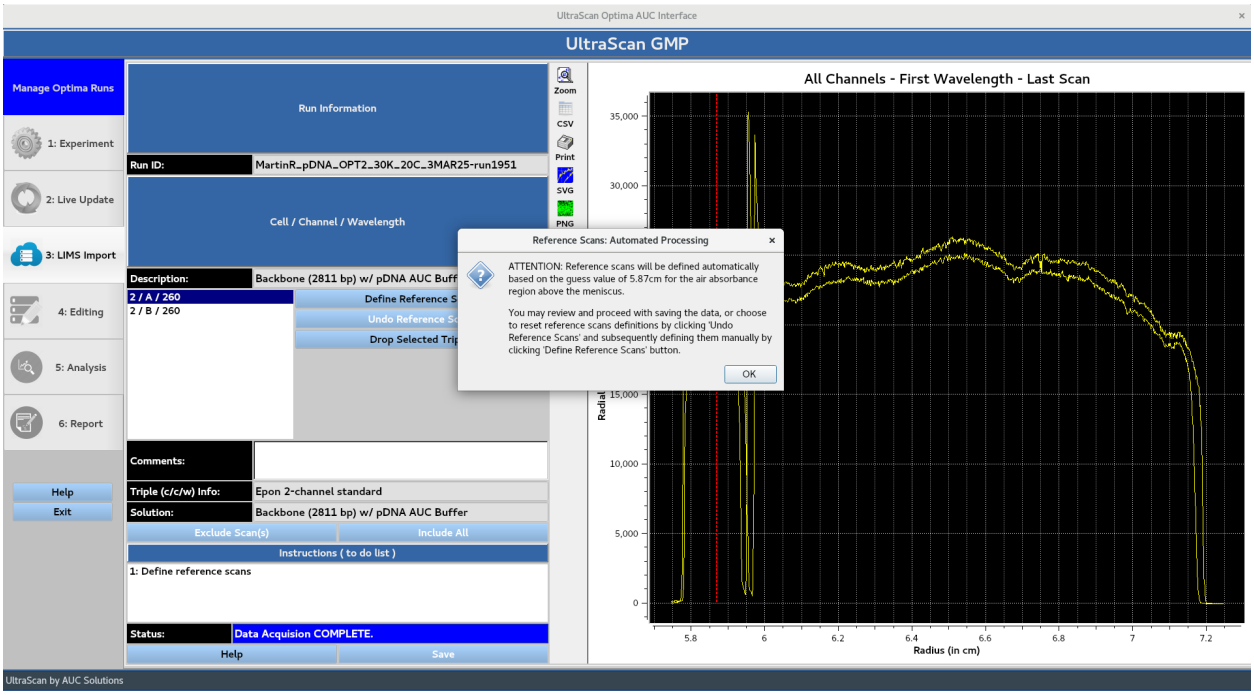
Once the machine reaches the desired rotor speed and begins acquiring scans, **Optical System** allows for the user to select the intensity/interference data. Each cell/channel/wavelength Triple can be viewed using the **Cell/Channel** and **Wavelength** drop down menus, which can be cycled through using the **Next** and **Previous** buttons. **Skip Stage** allows for the current stage to be skipped. While **STOP** allows for the run to be stopped, and prompts the :doc:`GMP e-signature <../gmp_signature> window for user authentication.

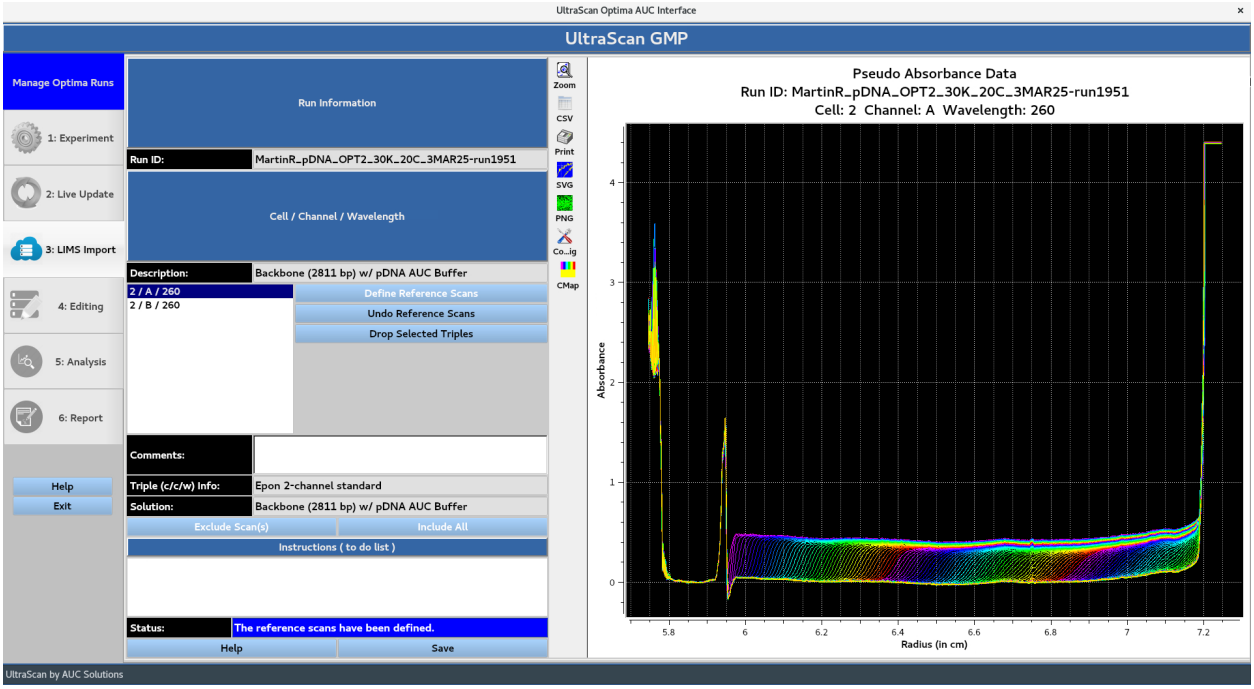


Once all of the scans defined in the experiment have been collected, the **Remaining Time** will be 00:00:00.



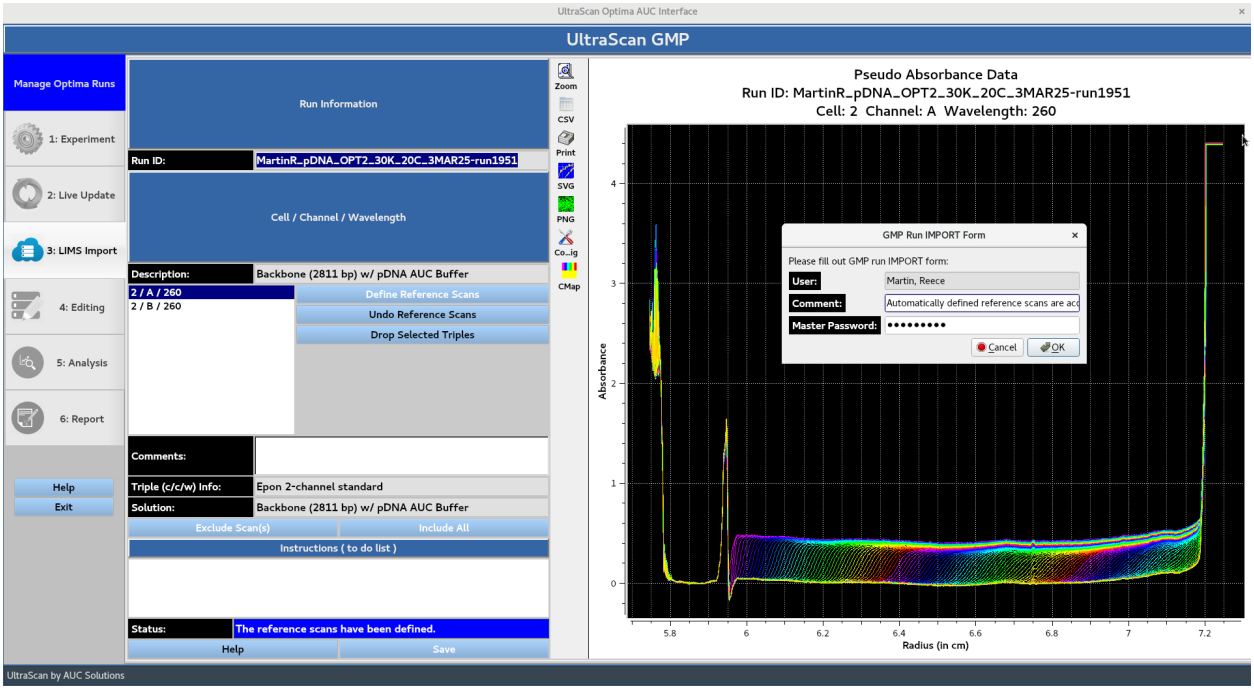
Next, the program moves to the 'IMPORT' stage, which displays the last scan of each Triple from the intensity data. The user is prompted with the **Reference Scans: Automated Processing** message:



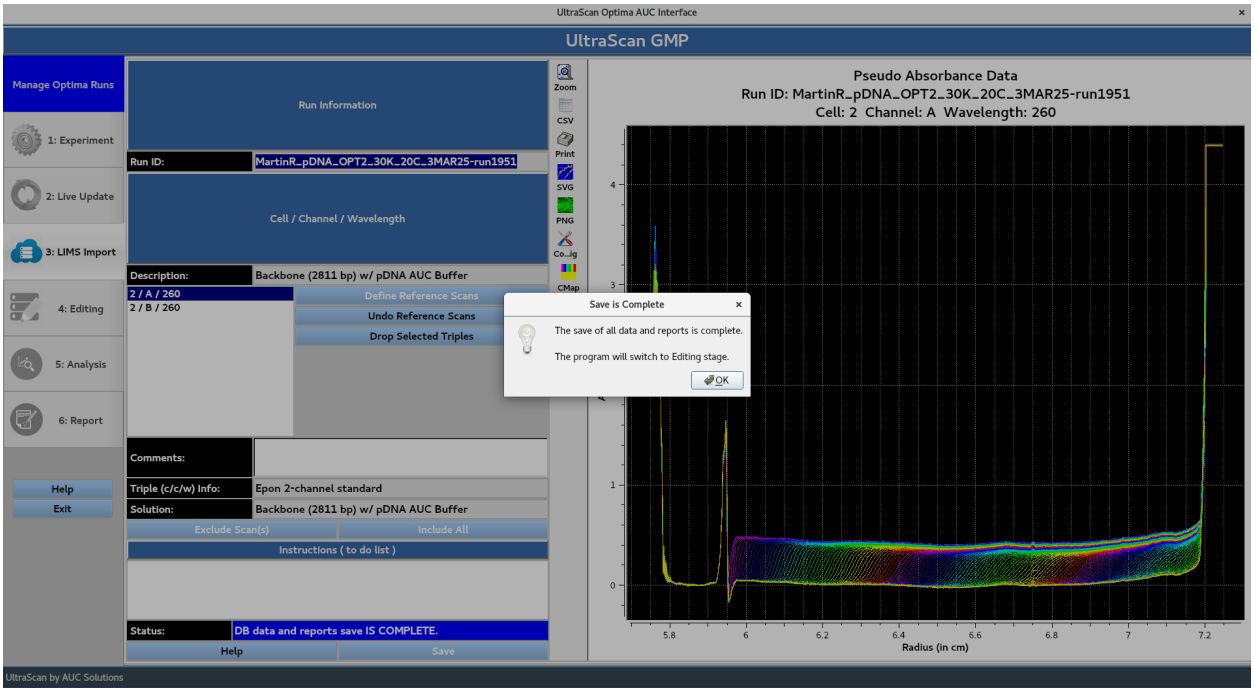


Clicking **OK** displays the converted absorbance data following the automated reference scan processing. If the user is not satisfied with the automated reference scans processing, the **Undo Reference Scans** button allows for them to be defined manually using the **Define Reference Scans** button. **Drop Selected Triples** can be used to drop unwanted Triples.

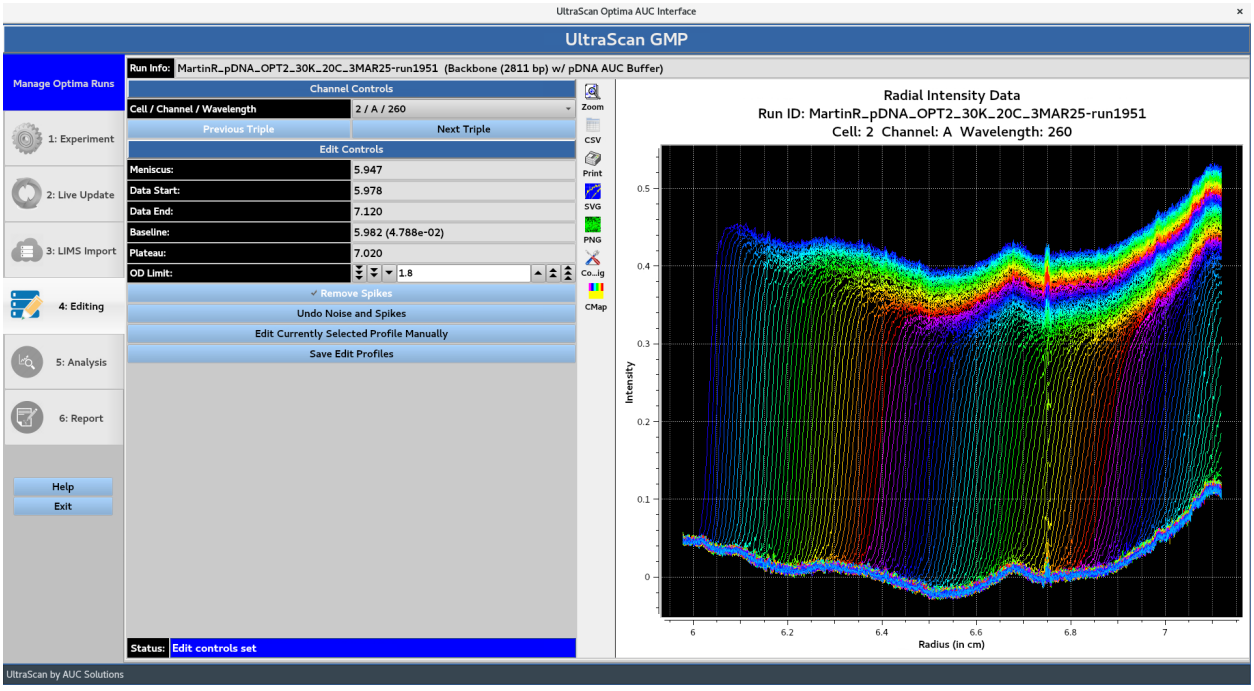
The 'Save' button prompts the 'GMP Run IMPORT Form', allowing the user to add a comment and prompting them for their master password for authentication.



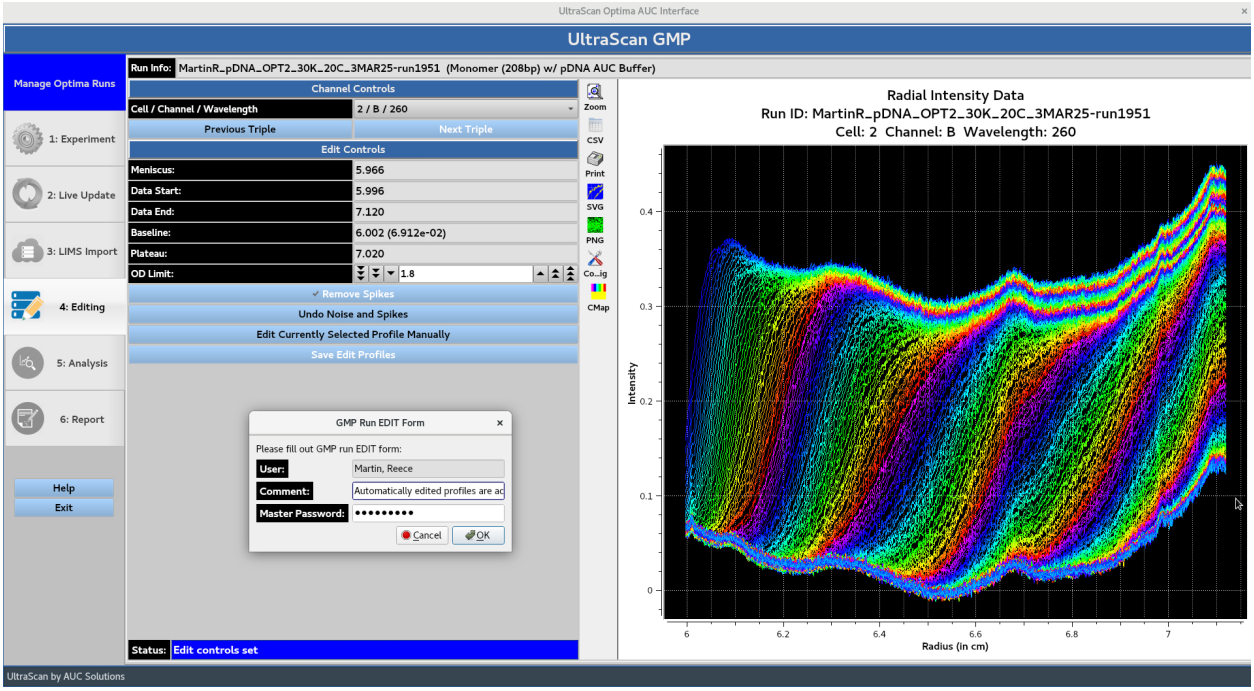
Once the data has been saved, the user is notified that the program is proceeding to the editing stage:



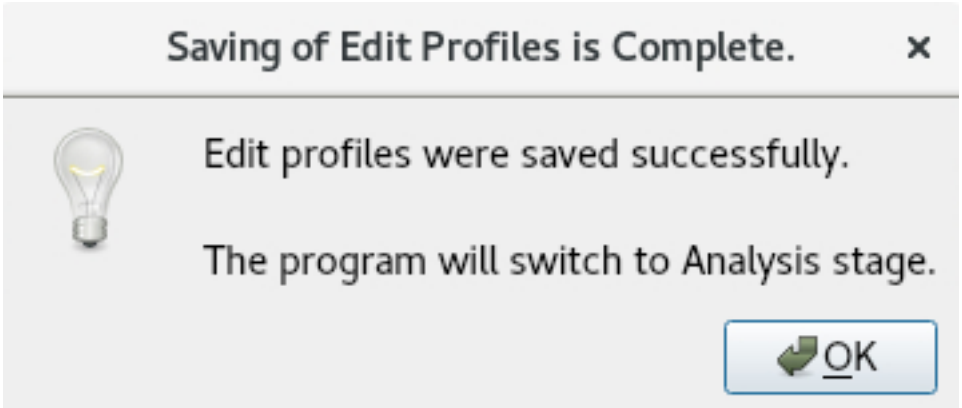
The editing window presents the user with the data edited in accordance with 'Data End' value and 'Excluded Scan Ranges' defined in the **Aprofile**. The 'Meniscus' position and 'Data Start' value are defined through automated processing. If the user is not satisfied with the automated processing, the **Edit Currently Selected Profile Manually** button allows for the 'Meniscus' position and 'Data End' values to be defined manually. **Remove Spikes** removes some sharp, irregular intensity spikes from data. Each cell/channel/wavelength Triple can be cycled through using the **Previous Triple** and **Next Triple** buttons.



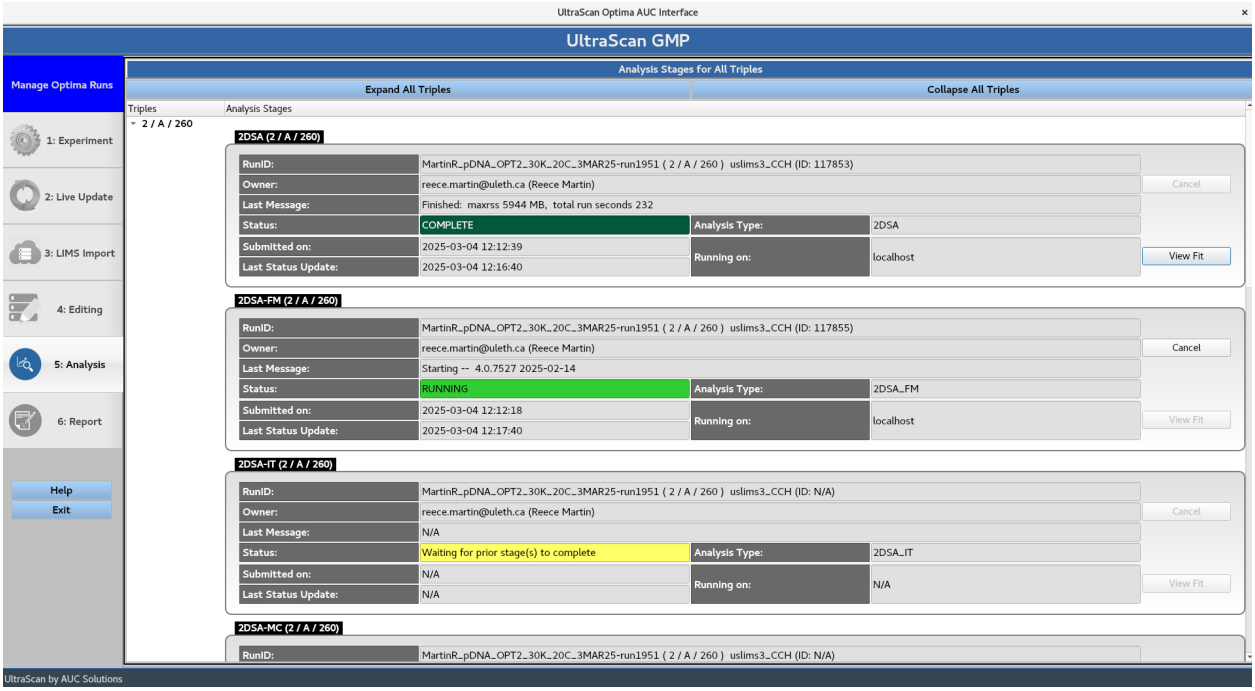
The **Save Edit Profiles** button prompts the **GMP Run EDIT Form**, allowing the user to add a comment and prompting them for their master password for authentication.



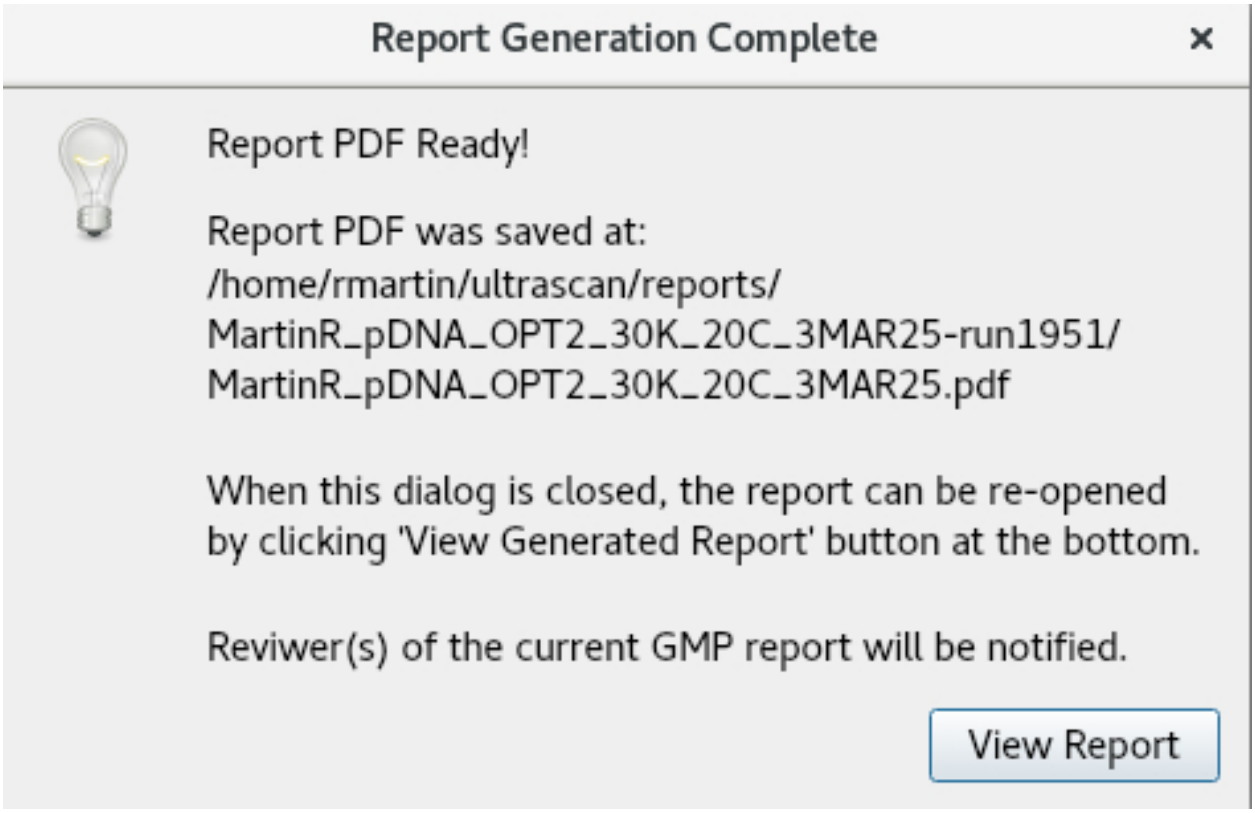
Once the edited profiles have been saved, the user is notified that the program is proceeding to the analysis stage:



The analysis window displays the current status of the submitted jobs producing the models specified in the **Aprofile**. **Cancel** can be used to stop ongoing jobs. **View Fit** simulates completed model overlays and displays them. The **Expand All Triples** and **Collapse All Triples** buttons control all Triples, while the {down arrow} button can be used to collapse or expand each individual Triple.



Once the analysis stage is complete, the models are simulated and the report is built. Which prompts the **Report Generation Complete** message box.



The **View Report** button opens the PDF file to be viewed:

Triple========== Protocol Development ==========

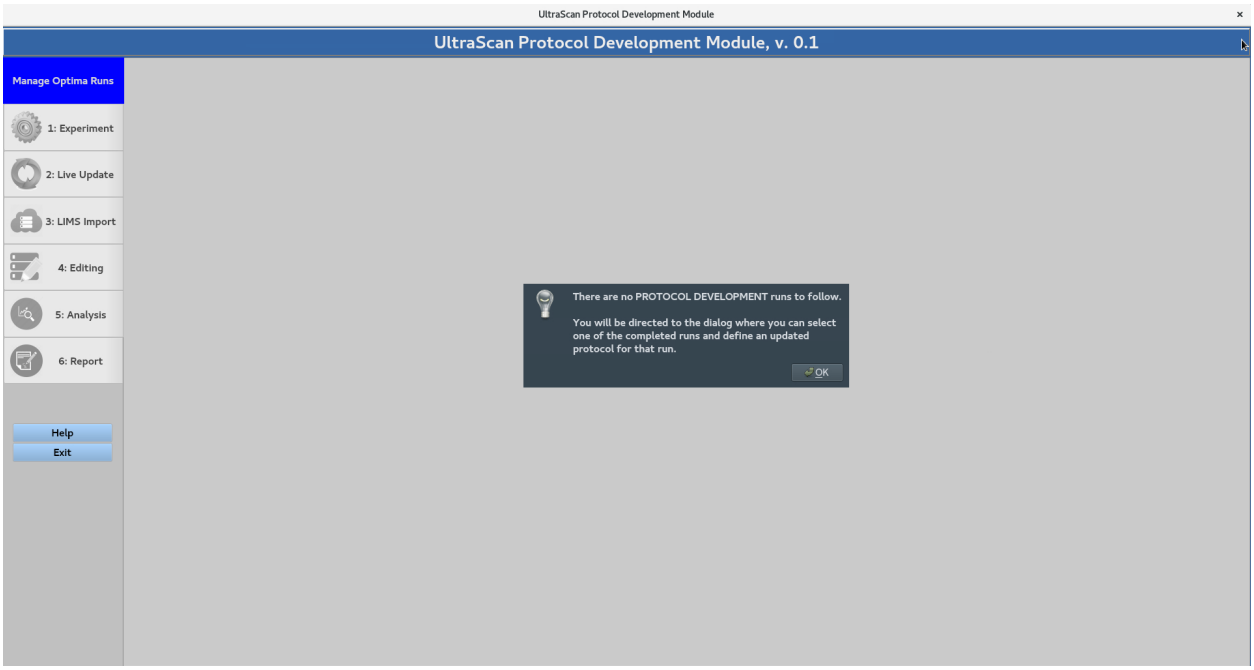
Index:

- *Manage Optima Runs:*
- *Select Completed Autoflow Run:*
- *1: General*
- *2: Lab/Rotor*
- *3: Speeds*
- *4: Cells*
- *5: Solutions*
  - *5b: Solution Details*
- *6: Optics*
- *7: Ranges*
  - *7a: View Ranges*
- *8: AProfile (Analysis Profile)*
  - *8.1: General*
- *8.2: 2DSA*
- *8.3: PCSA*

- *9: Submit*
  - *9a: View Experiment Details*
  - *9b: Submit the Run*

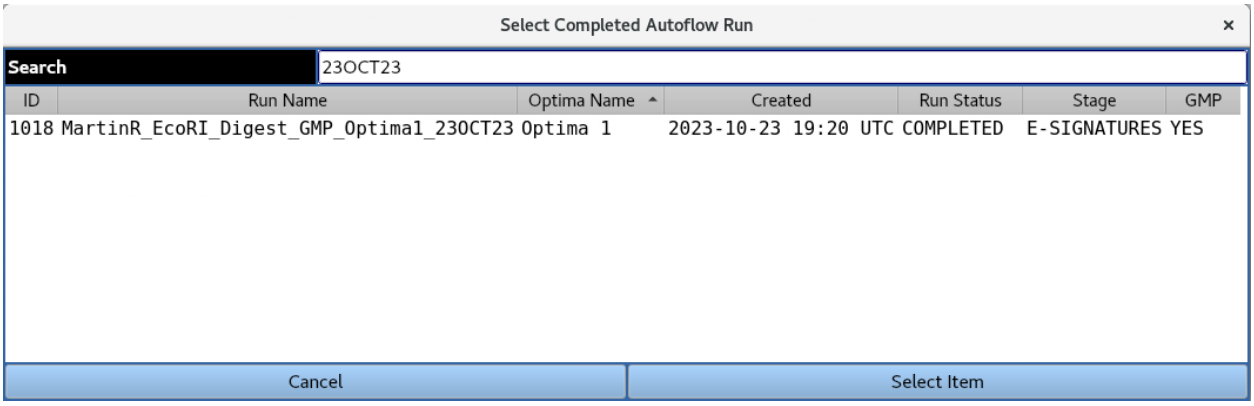
Allows for analysis profile optimization by re-attaching GMP runs at the **LIMS Import** stage with a modified **Aprofile**

32.9 Manage Optima Runs:



32.10 Select Completed Autoflow Run:

Provides a list of completed GMP runs in the database, as well as the current reporting stage (Report generation or collection of e-signatures).

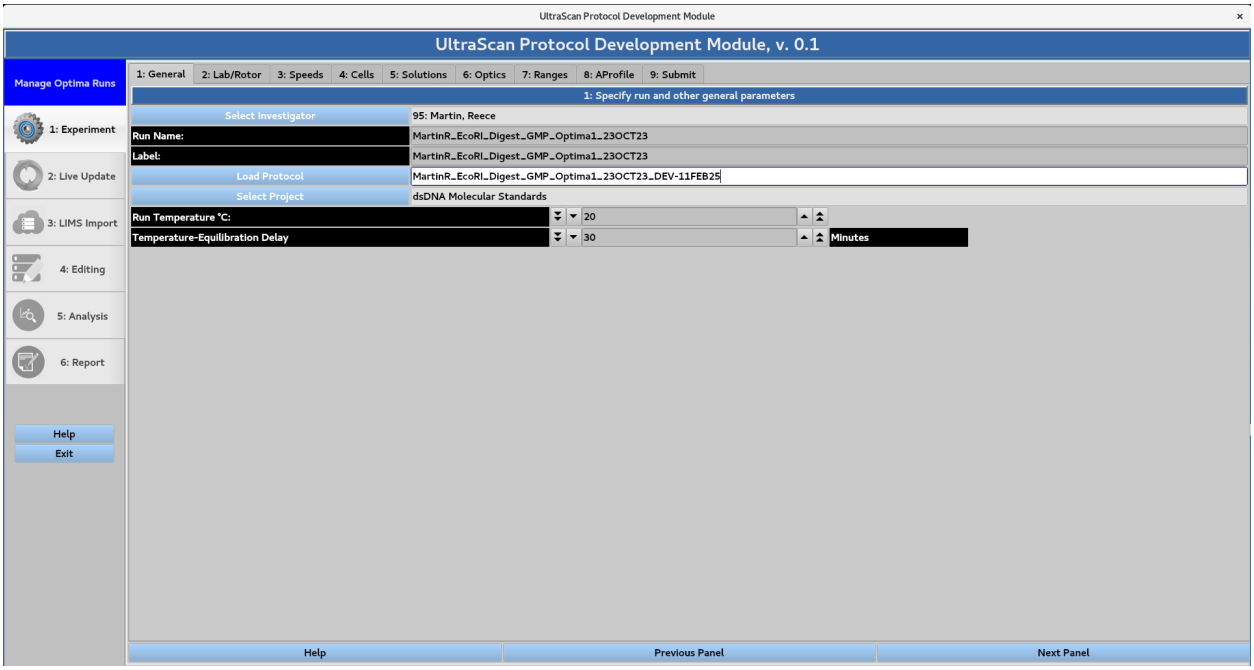


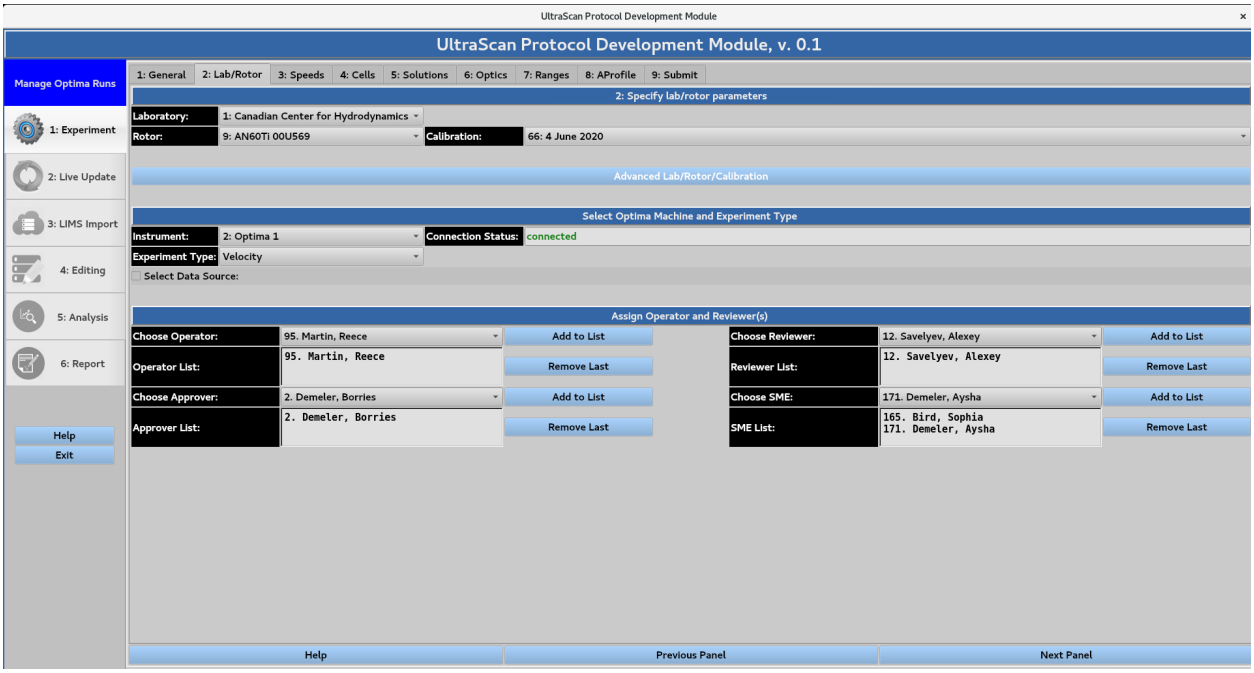
A GMP run can be loaded from the database by highlighting it and clicking the **Select Item** button. This opens a modified **Data Acquisition Routine**, where the user can assign a new protocol name. However, the run name and experimental parameters cannot be changed, as the run is being reimported for editing and analysis.

Sections 1-9 can be accessed in any order using the navigation bar at the top of the window or sequentially using the **Previous Panel** and **Next Panel** buttons at the bottom of the window.

32.11 1: General

The protocol name is the only information that can be changed, since the source GMP run (Run Name, Label, Run Temperature and Temperature-Equilibration Delay) remains unchanged.



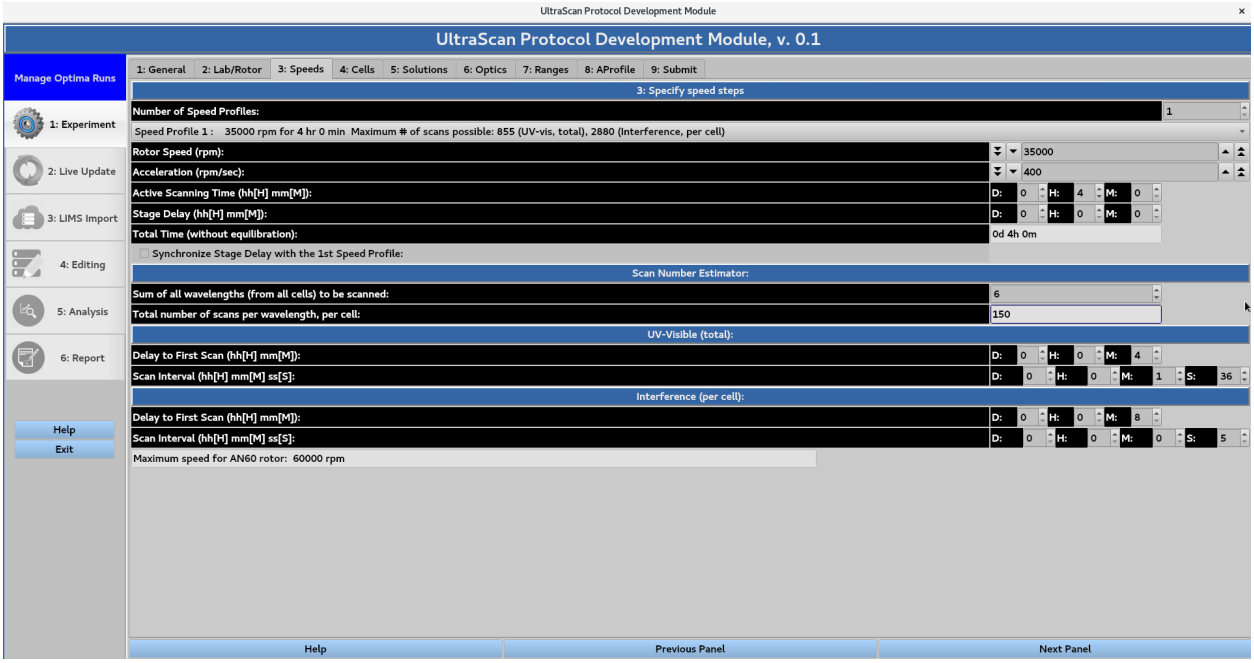


32.12 2: Lab/Rotor

The user cannot change the laboratory, rotor, rotor calibration profile, instrument, and experiment types. The roles of Operator, Reviewer, Approver, and Subject Matter Expert (SME) can be re-assigned from the drop-down menus. See here for more information: UltraScan GMP Help.

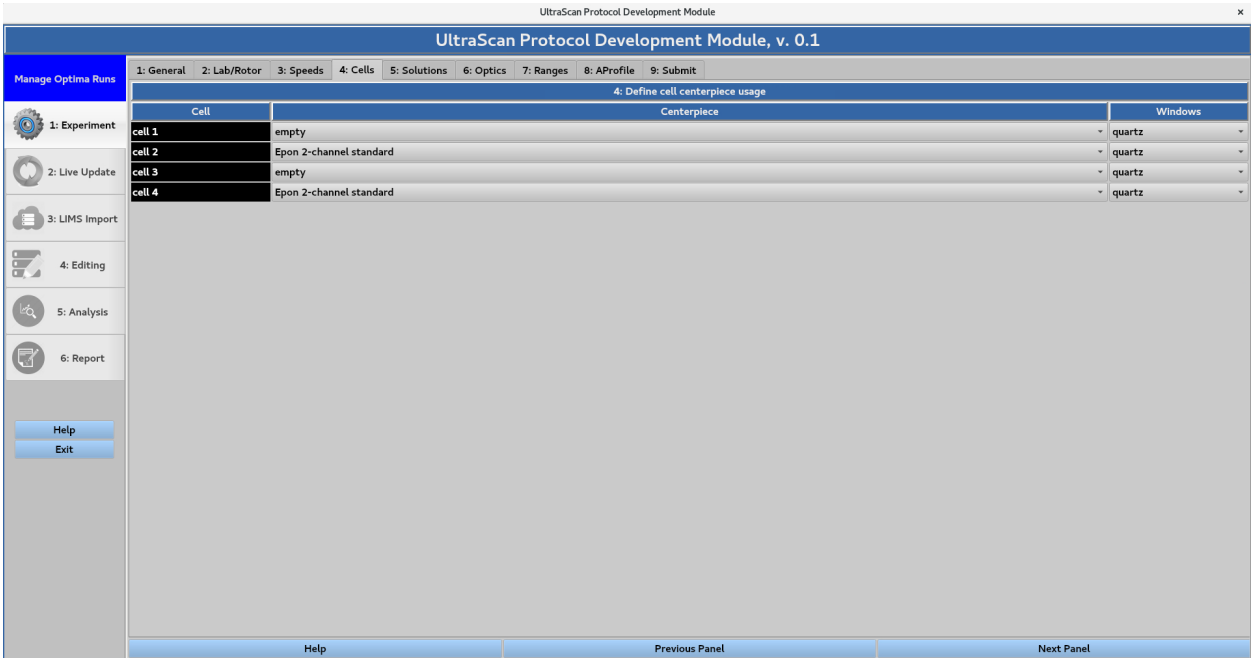
32.13 3: Speeds

None of the run parameters in this window can be modified.



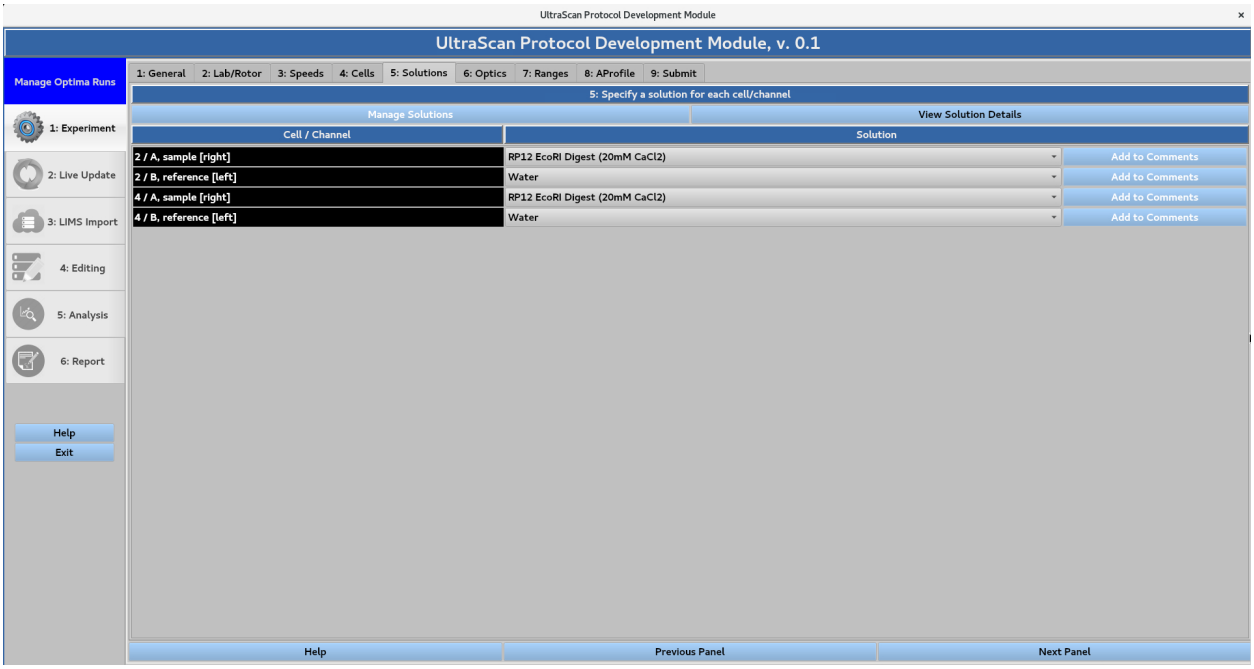
32.14 4: Cells

None of the run parameters in this window can be modified.

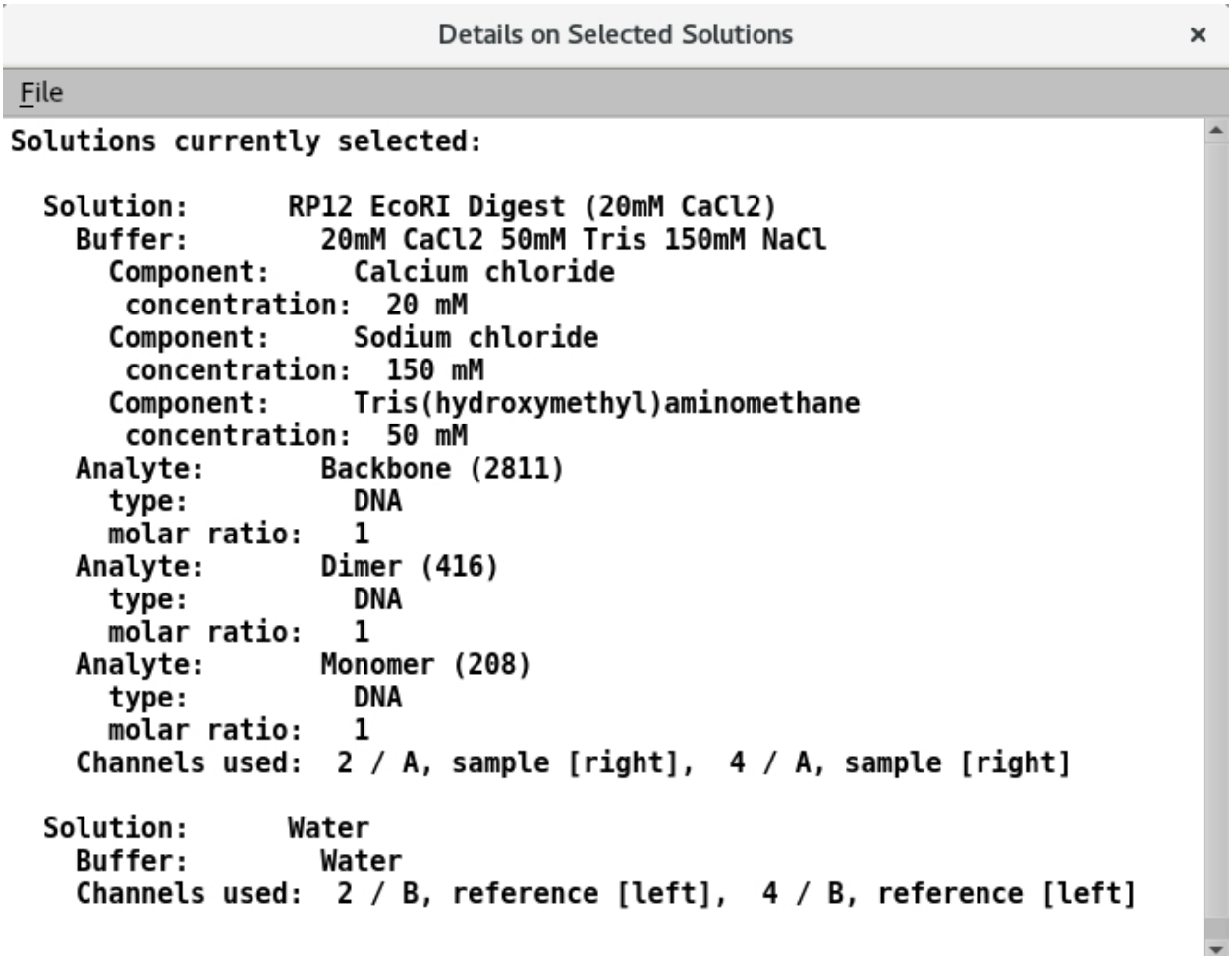


32.15 5: Solutions

None of the run parameters in this window can be modified.

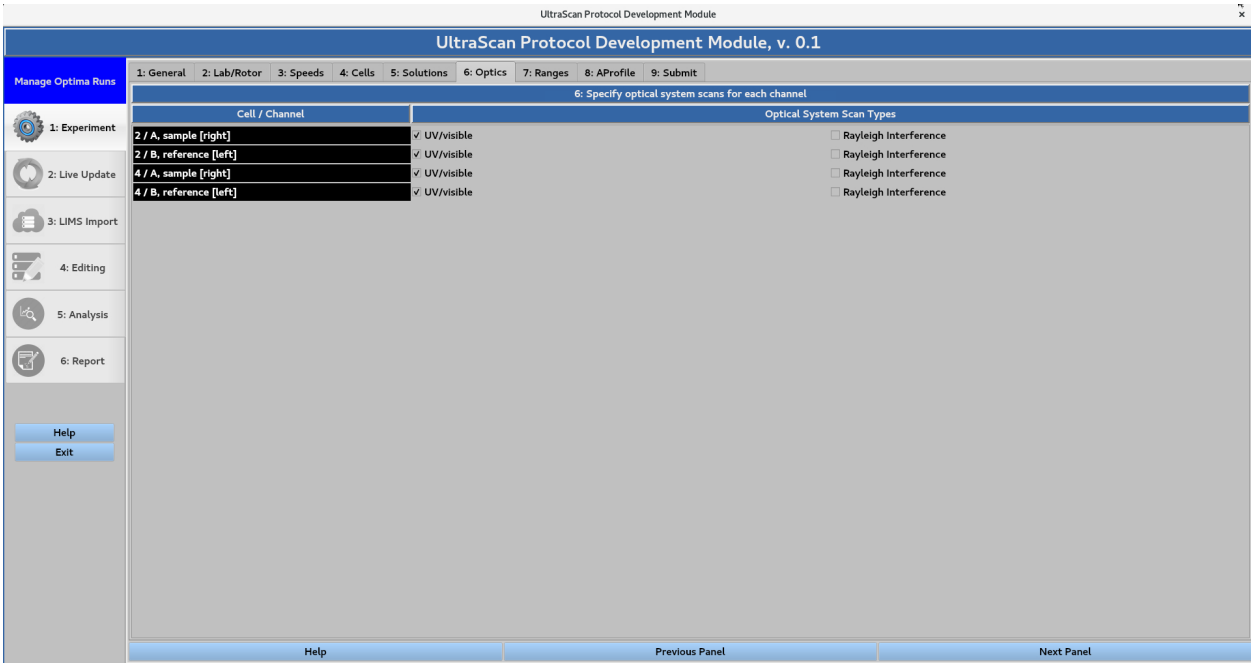


32.15.1 5b: Solution Details



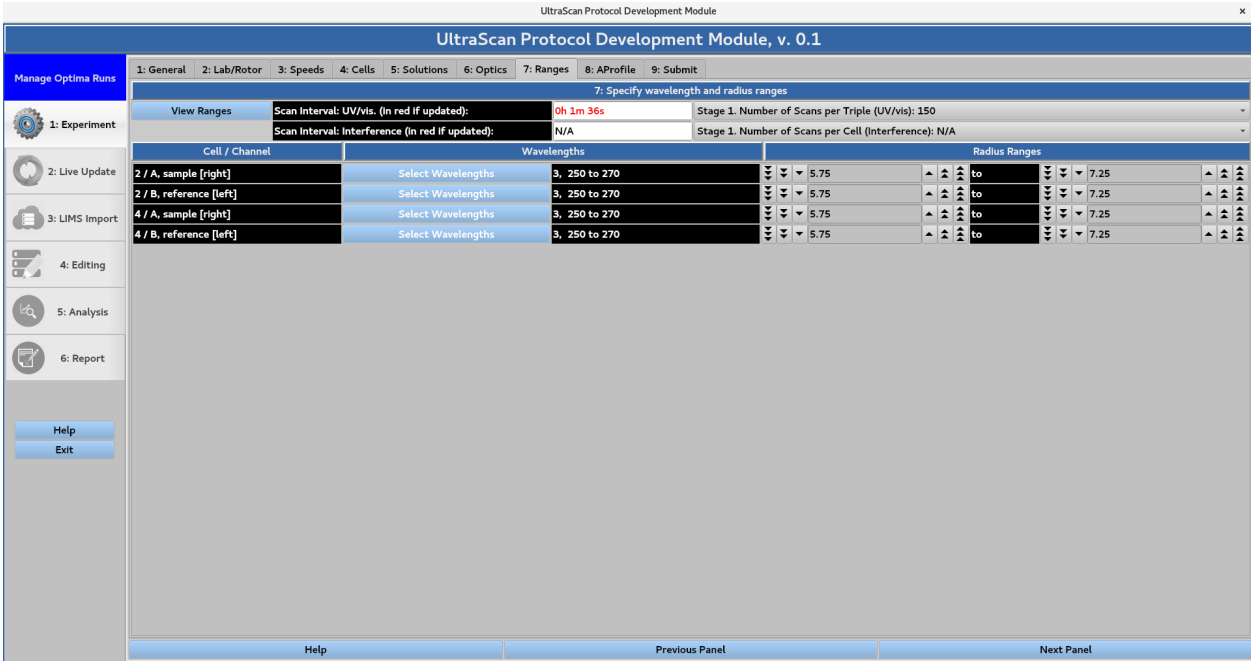
32.16 6: Optics

None of the run parameters in this window can be modified.

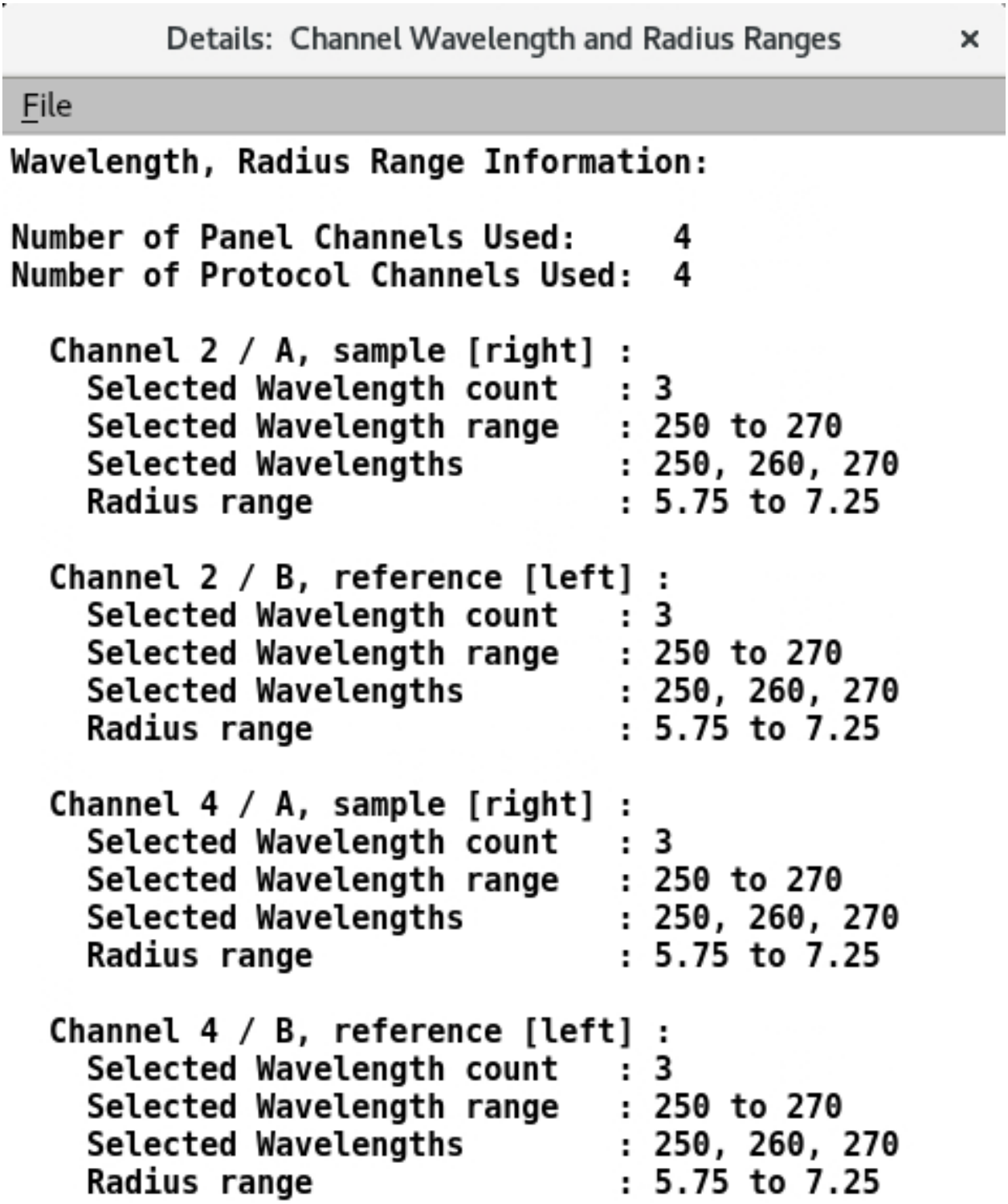


32.17 7: Ranges

None of the run parameters in this window can be modified.



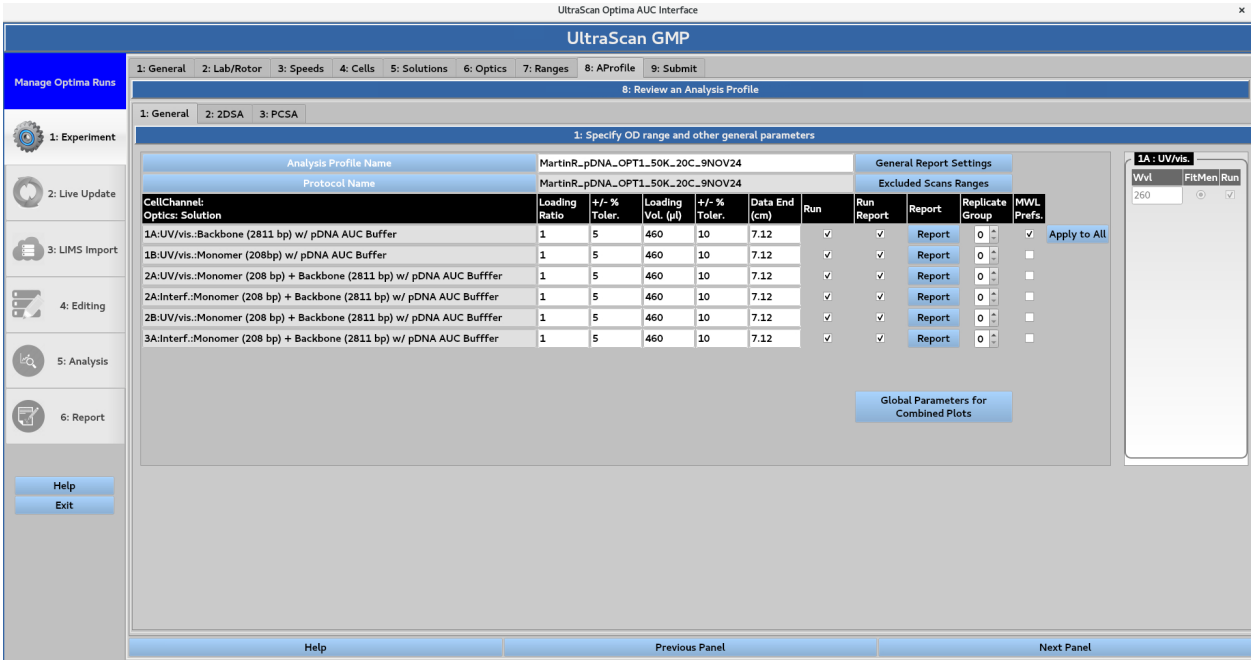
32.17.1 7a: View Ranges



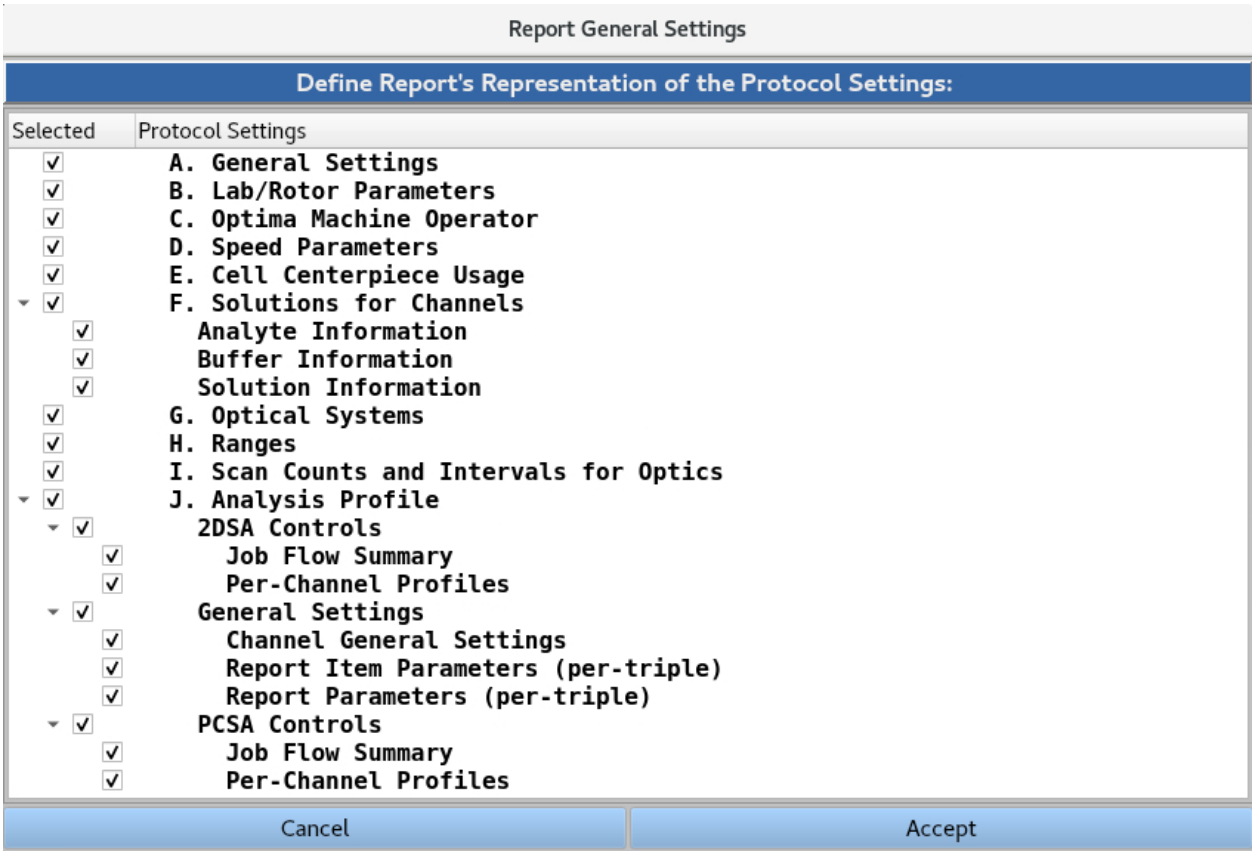
32.18 8: AProfile (Analysis Profile)

Define edit profiles, analysis parameters and report settings for the experiment.

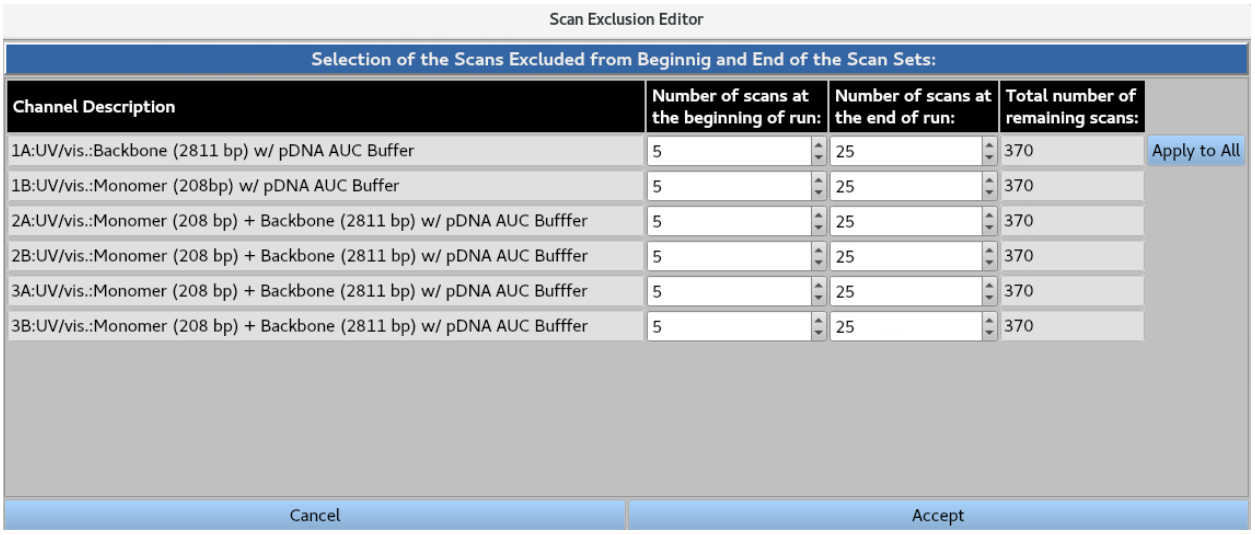
32.18.1 8.1: General



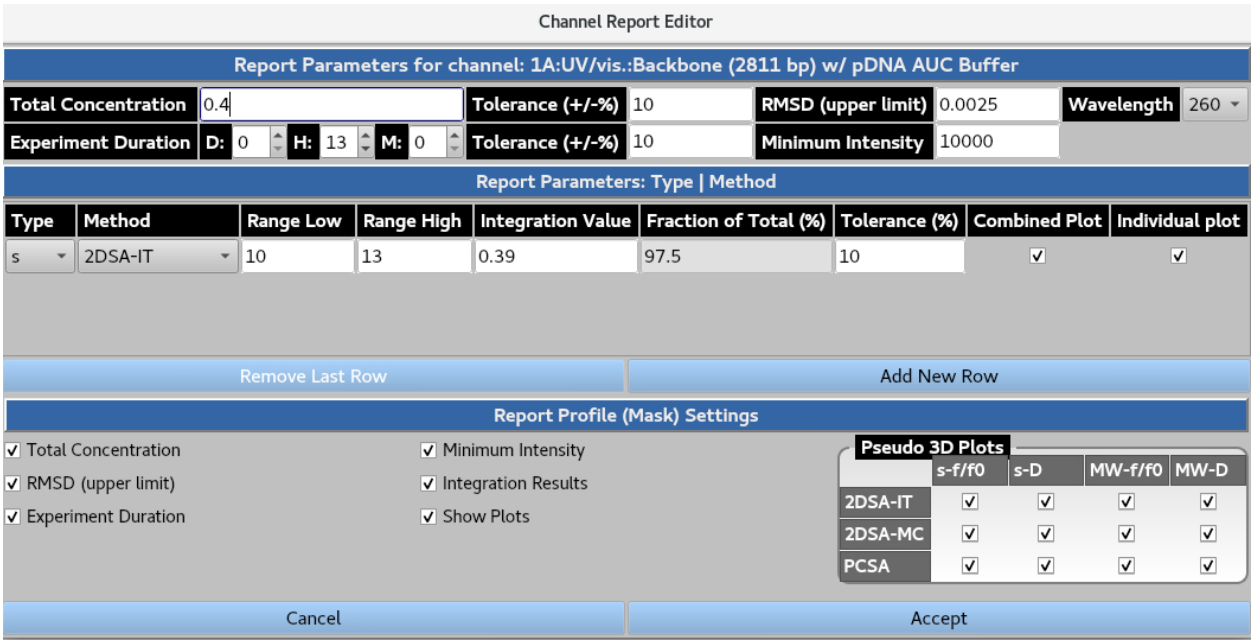
Define numerical values for ‘Loading Ratio’, the ‘Â± %Tolerance’, ‘Loading Volume (Î¼L)’, the ‘Â± %Tolerance’, and ‘Data End (cm)’. The user can then select checkboxes to specify whether the analysis should be ‘Run’, and included in the ‘Run Report’.



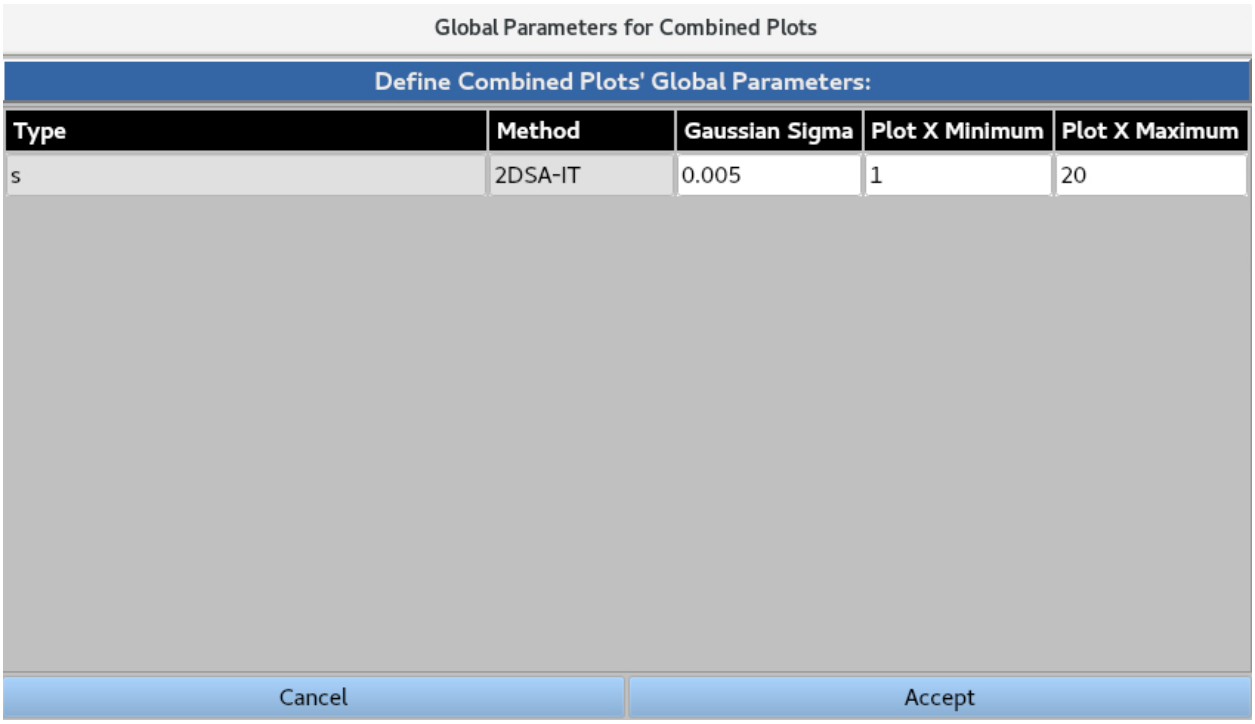
The **Report General Settings** windows allow the user to select/deselect which of the protocol settings (Sections 1-8) will be presented in the report. The {down arrow} button can be used to collapse or expand each subsection.



The **Scan Exclusion Editor** allows the user to numerically define the number scans to be removed from the beginning and end of each channel, either by typing the value or using the arrows. The **Apply to All** button applies the values for the first channel to all channels.



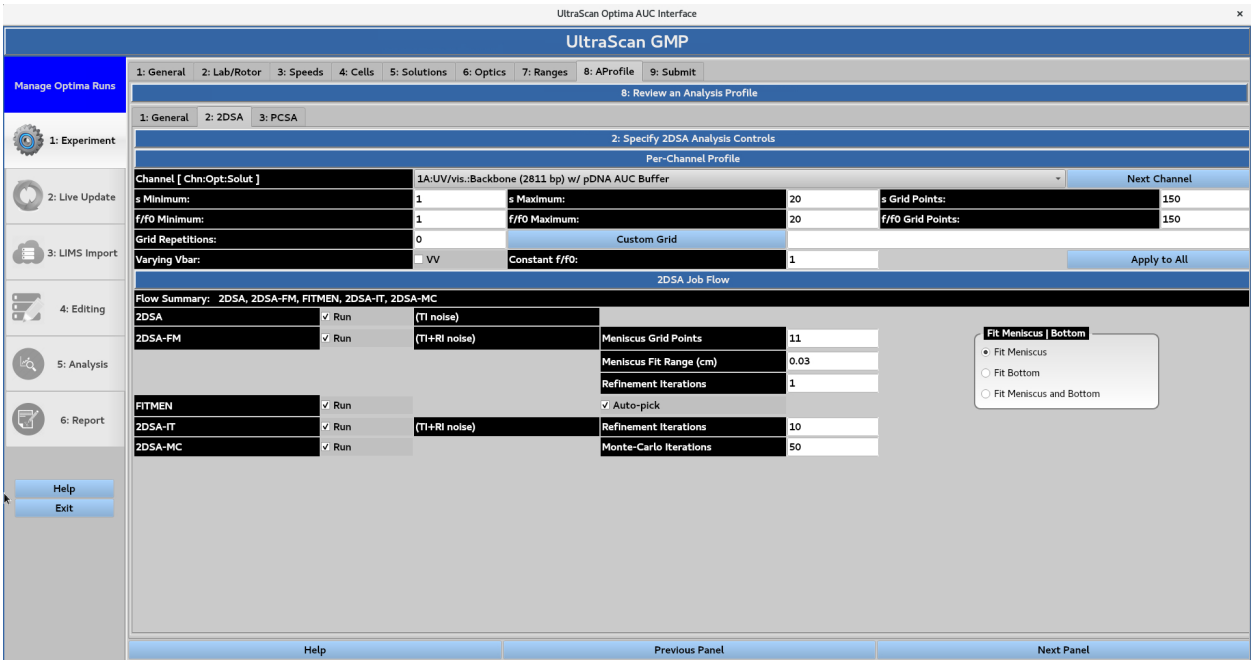
The **Report** button summons the channel-specific report settings window. The user defines the 'Total Concentration', the 'Â ± %Tolerance', 'RMSD (upper limit)', the 'Â± %Tolerance', and the wavelength to extract these metrics from. The user then lists the expected 'Experiment Duration', the 'Â± %Tolerance', and 'Minimum Intensity' at the selected wavelength from the xenon flash lamp that is deemed acceptable. In the 'Report Profile (Mask) Settings' the user can select the following: 'Total Concentration', 'Minimum Intensity', 'RMSD (upper limit)', 'Integration Results', 'Experiment Duration' and relevant plots (as well as the pseudo-3D plots for each parameter and analysis type) they wish to be included in the report.



Global Parameters for Combined Plots allows the user to define the plot minimum, plot maximum and

Gaussian sigma for each plot type.

32.19 8.2: 2DSA

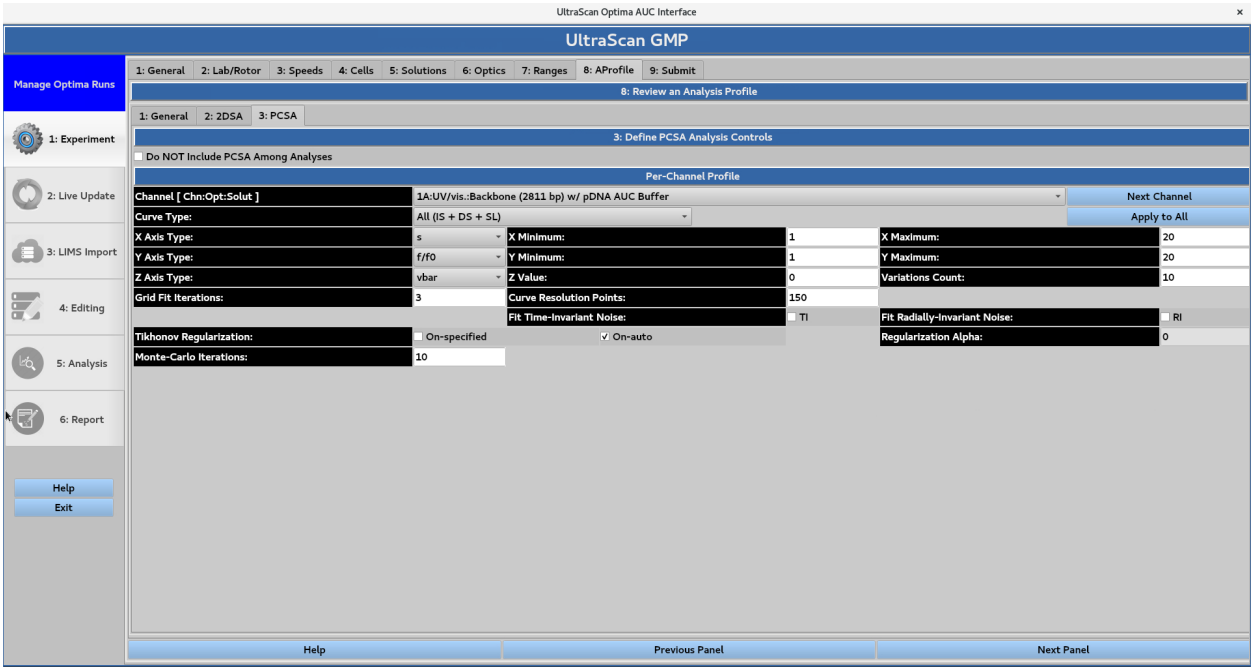


The user can define the 2-dimensional spectrum analysis (2DSA) controls for each channel. The grid size is defined by the minimum and maximum values for the sedimentation coefficient (s) and frictional ratio (f/f0), while the grid resolution is defined by the number of points for each of the parameters. A 2DSA **Custom Grid** (CG) can also be used if the user requires higher resolution in certain regions of the data, and only needs lower resolution in other regions. The user can also select to vary the partial specific volume or 'Vbar' parameter and hold the frictional ratio constant at a specified value. The **Apply to All** button can be used to apply these settings to all of the channels.

The '2DSA Job Flow' panel allows the user to control which analysis steps are performed, including:

- 2DSA
- 2DSA with meniscus fit (FM), with a user-defined number of grid points, range (cm), and number of refinement iterations. The user can also choose to fit **only the bottom of the channel** or **both the meniscus and the bottom.**
- FITMEN meniscus processing, with the option to 'Auto-pick' the meniscus position
- 2DSA with iterative refinement (IT), with a user-defined number of iterations
- 2DSA with Monte Carlo iterations (MC), with a user-defined number of Monte Carlo iterations.

32.20 8.3: PCSA



The user can select the option to not include the parametrically-constrained spectrum analysis (PCSA) in the submitted jobs. If the box is left unchecked, the user can define the PCSA controls for each channel.

The desired curve type is selected from the drop-down menu, and the user defines parameters for the x, y, and z axes.

- The x and y axes are defined by minimum and maximum values
- The z axis is assigned a value and a variation count
- Numerical values are then assigned for the grid fit iterations and curve resolution points

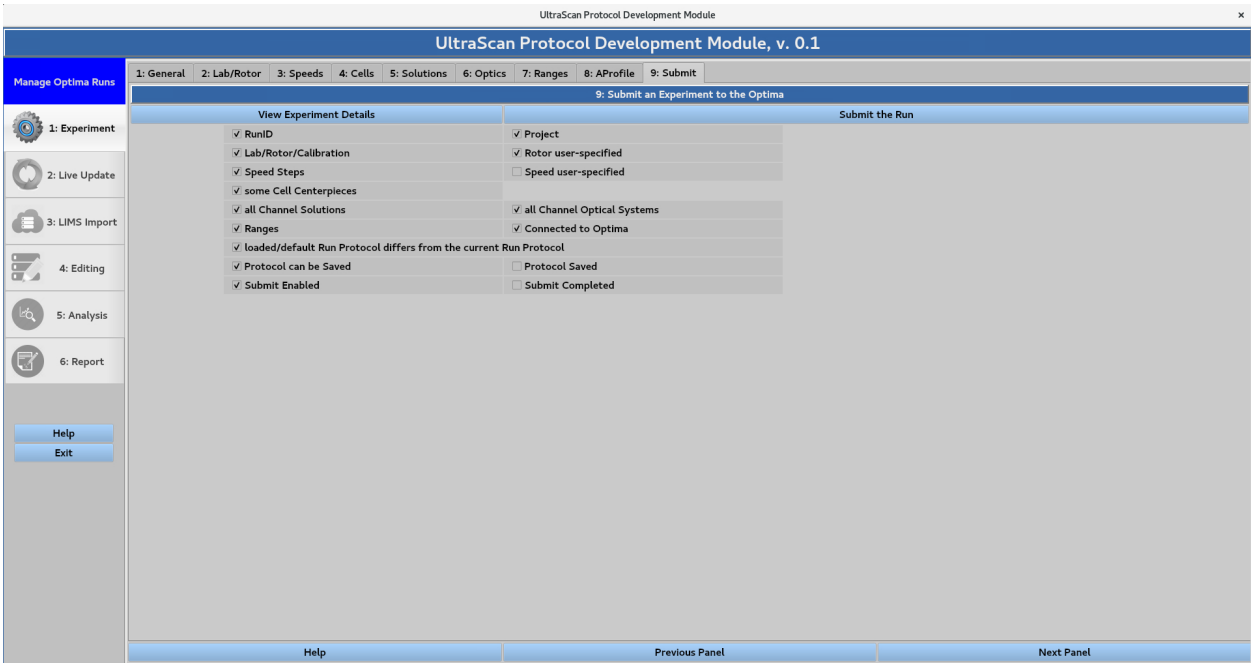
Optional settings include:

- Time- and radially-invariant noise fitting
- Tikhonov regularization, with either an auto-picked or manually-selected alpha
- Number of Monte Carlo iterations.

The 'Apply to All' button can be used to apply these parameters and settings to all of the channels.

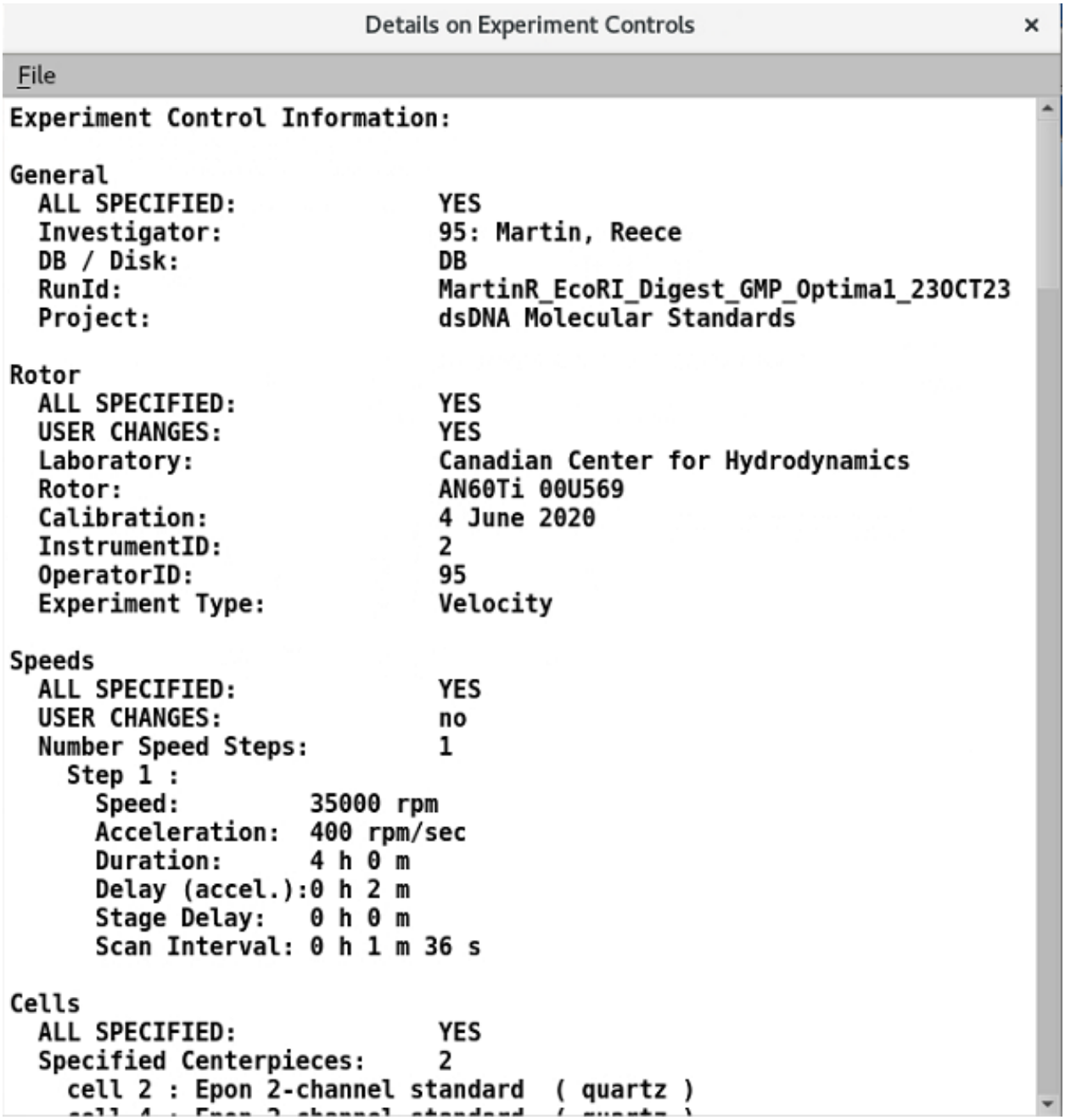
32.21 9: Submit

Submit the protocol development run for processing



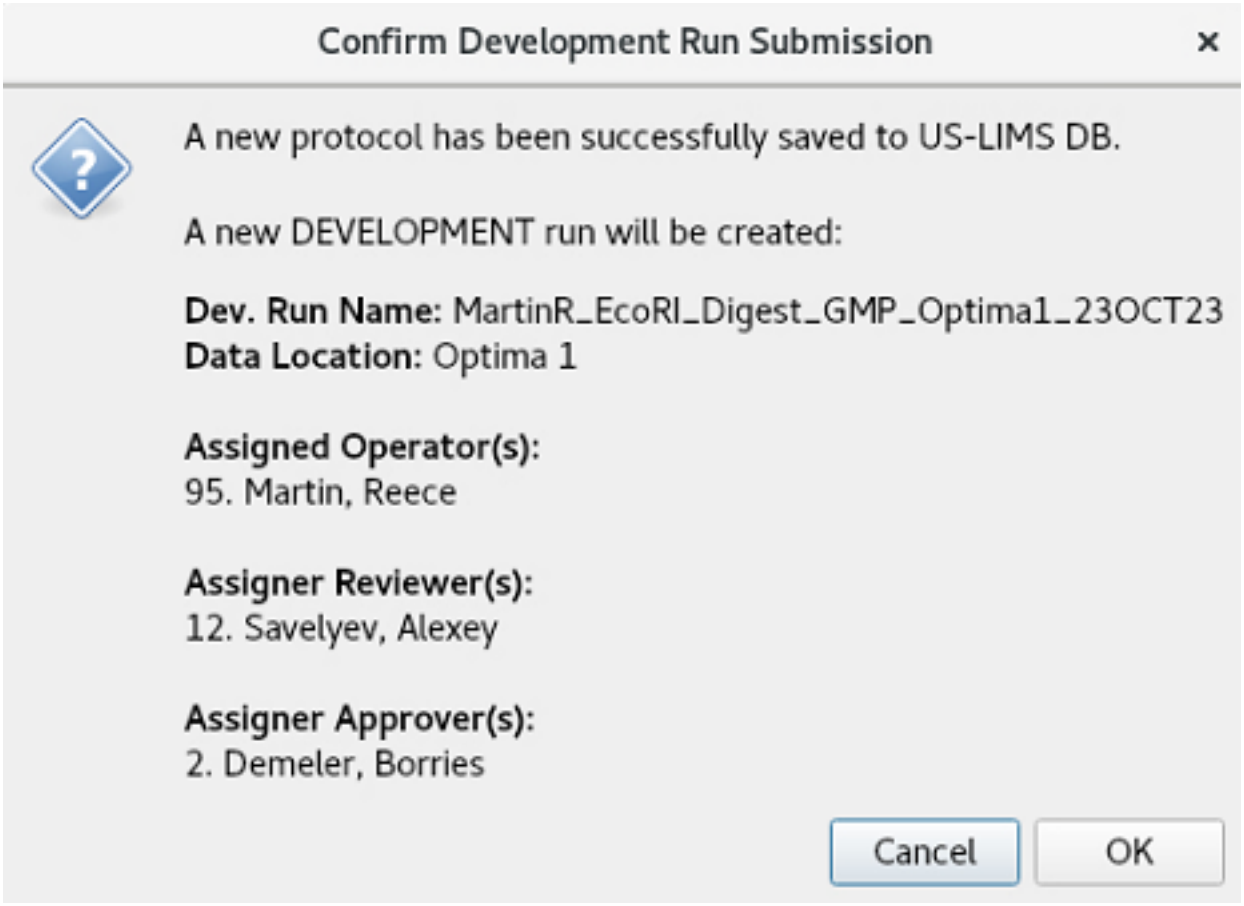
32.21.1 9a: View Experiment Details

Summary of the information defined in sections 1-8.

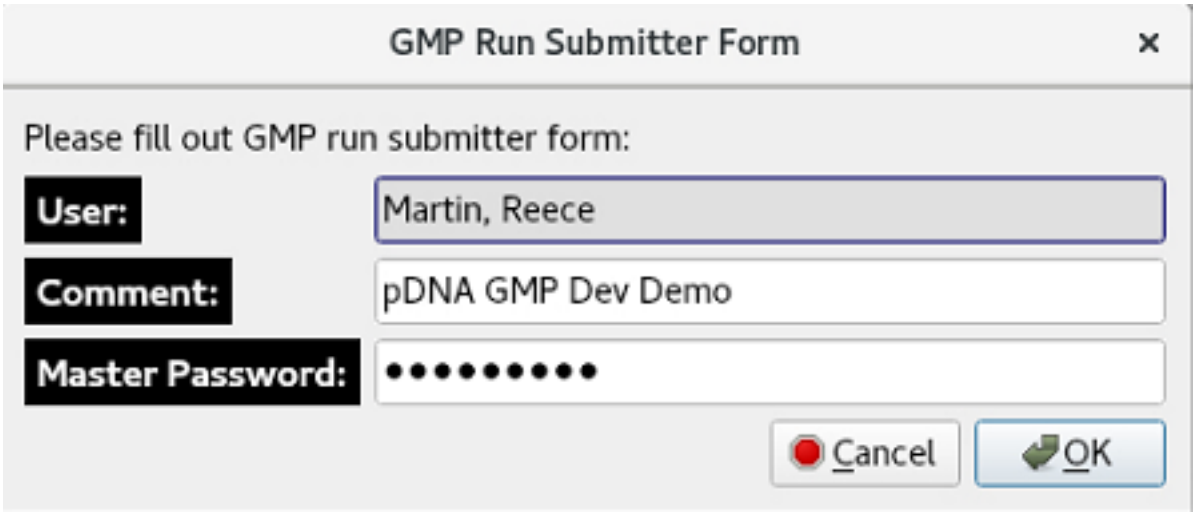

Details on Experiment Controls
File
Experiment Control Information:
General
ALL SPECIFIED: YES
Investigator: 95: Martin, Reece
DB / Disk: DB
RunId: MartinR_EcoRI_Digest_GMP_Optima1_23OCT23
Project: dsDNA Molecular Standards
Rotor
ALL SPECIFIED: YES
USER CHANGES: YES
Laboratory: Canadian Center for Hydrodynamics
Rotor: AN60Ti 00U569
Calibration: 4 June 2020
InstrumentID: 2
OperatorID: 95
Experiment Type: Velocity
Speeds
ALL SPECIFIED: YES
USER CHANGES: no
Number Speed Steps: 1
Step 1 :
Speed: 35000 rpm
Acceleration: 400 rpm/sec
Duration: 4 h 0 m
Delay (accel.):0 h 2 m
Stage Delay: 0 h 0 m
Scan Interval: 0 h 1 m 36 s
Cells
ALL SPECIFIED: YES
Specified Centerpieces: 2
cell 2 : Epon 2-channel standard (quartz)

32.21.2 9b: Submit the Run

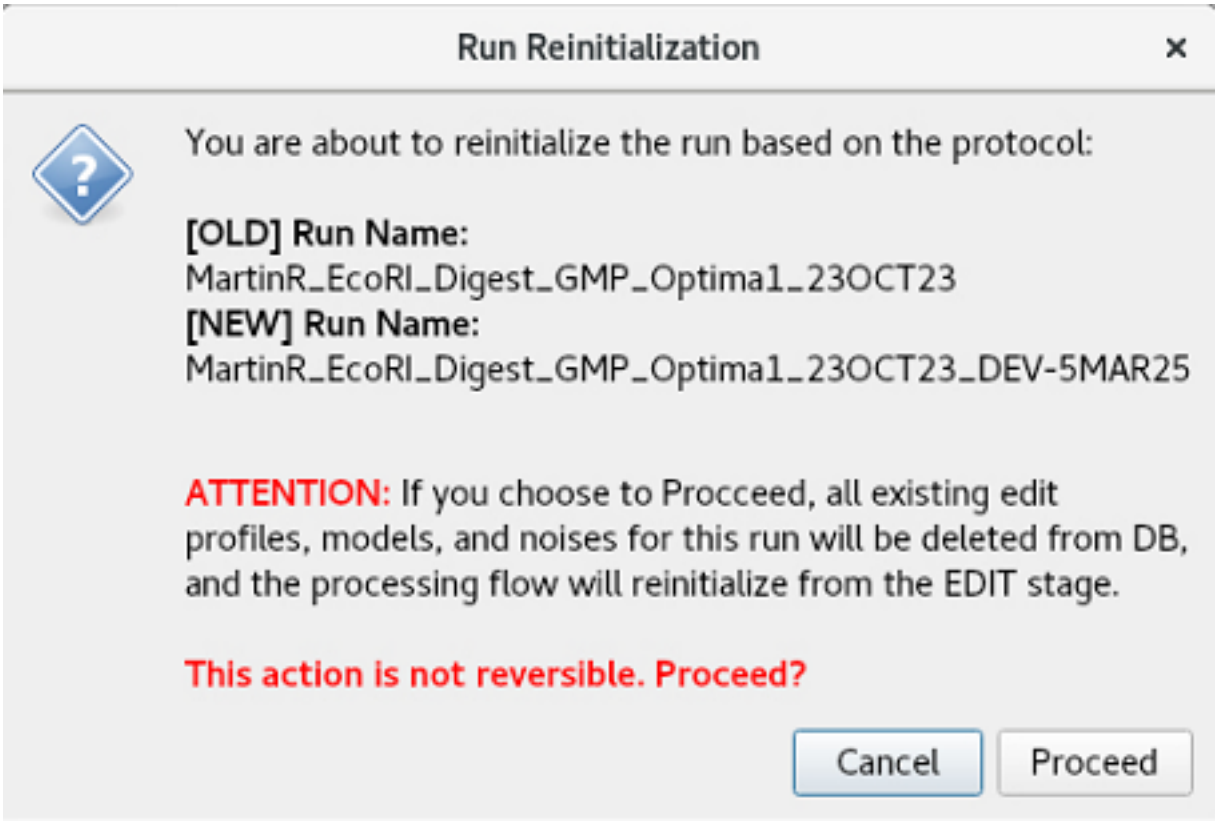
A new protocol name and GMP e-signer roles must be assigned before the 'Submit the Run' button becomes active and allows for submission. Submission prompts the following window:



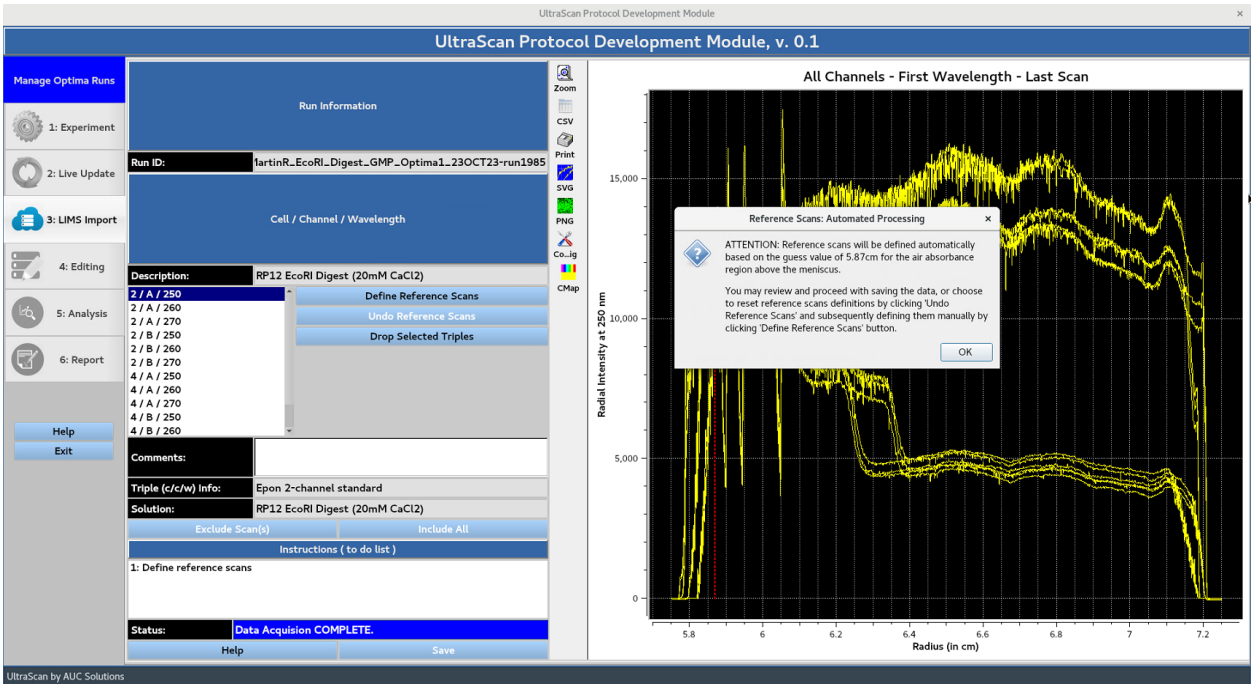
The 'OK' button prompts the 'GMP Run Submitter Form', allowing the user to add a comment and prompting them for their master password for authentication.



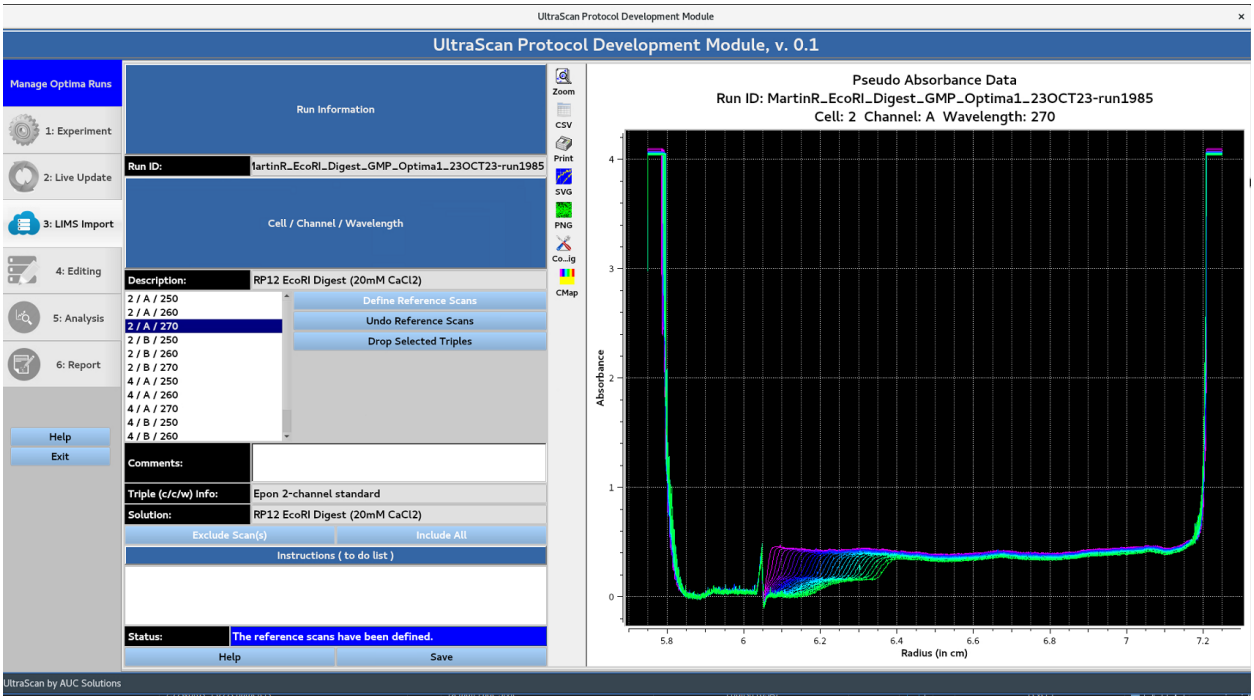
The 'OK' button brings up the 'Run Reinitialization' window, which informs the user that 'Proceed' will delete existing edit profiles, models, and noises, and reinitialize at the EDIT stage.



The program moves to the 'IMPORT' stage, which displays the last scan of each Triple from the intensity data. The user is prompted with the **Reference Scans: Automated Processing** message:

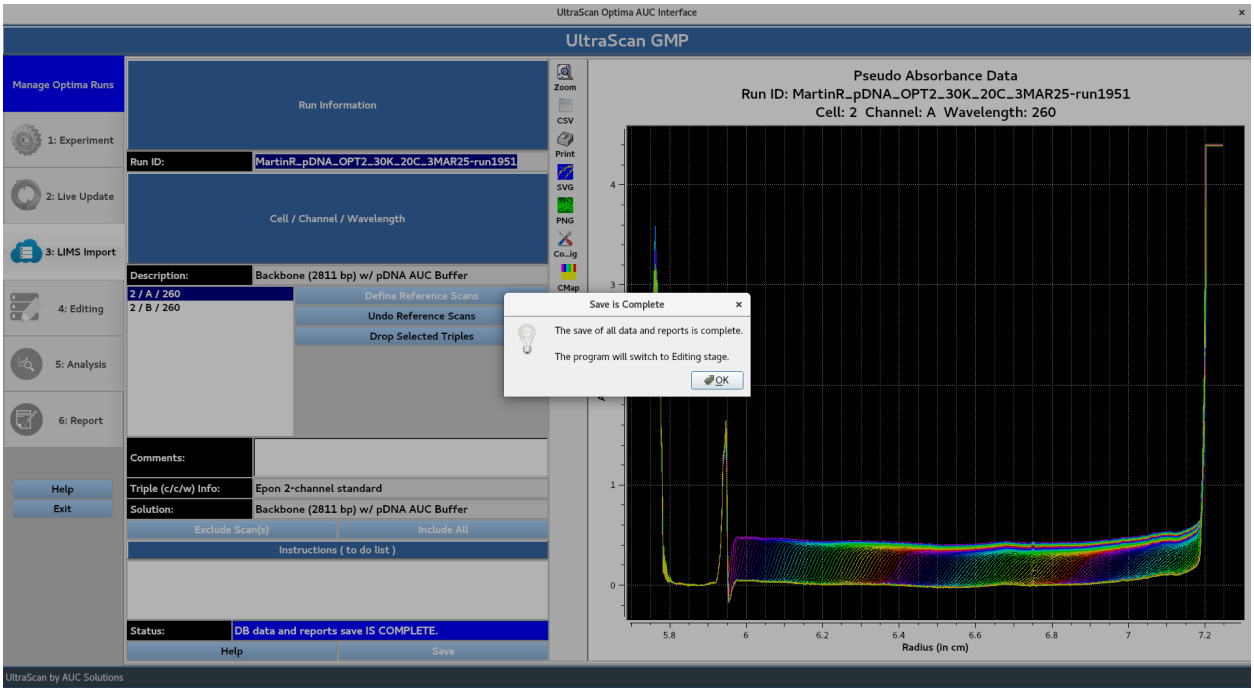


Clicking **OK** displays the converted absorbance data following the automated reference scan processing. If the user is not satisfied with the automated reference scans processing, the **Undo Reference Scans** button allows for them to be defined manually using the **Define Reference Scans** button. **Drop Selected Triples** can be used to drop unwanted Triples.

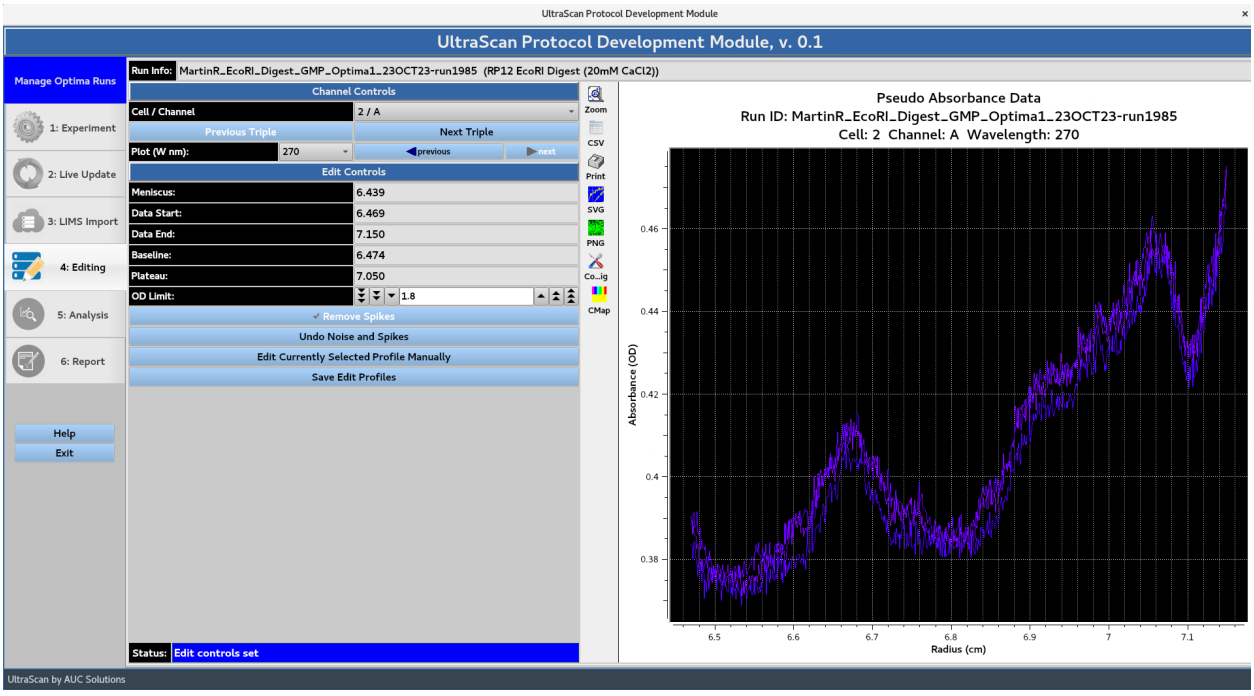


The **Save** button prompts the **GMP Run IMPORT Form**, allowing the user to add a comment and prompting them for their master password for authentication.

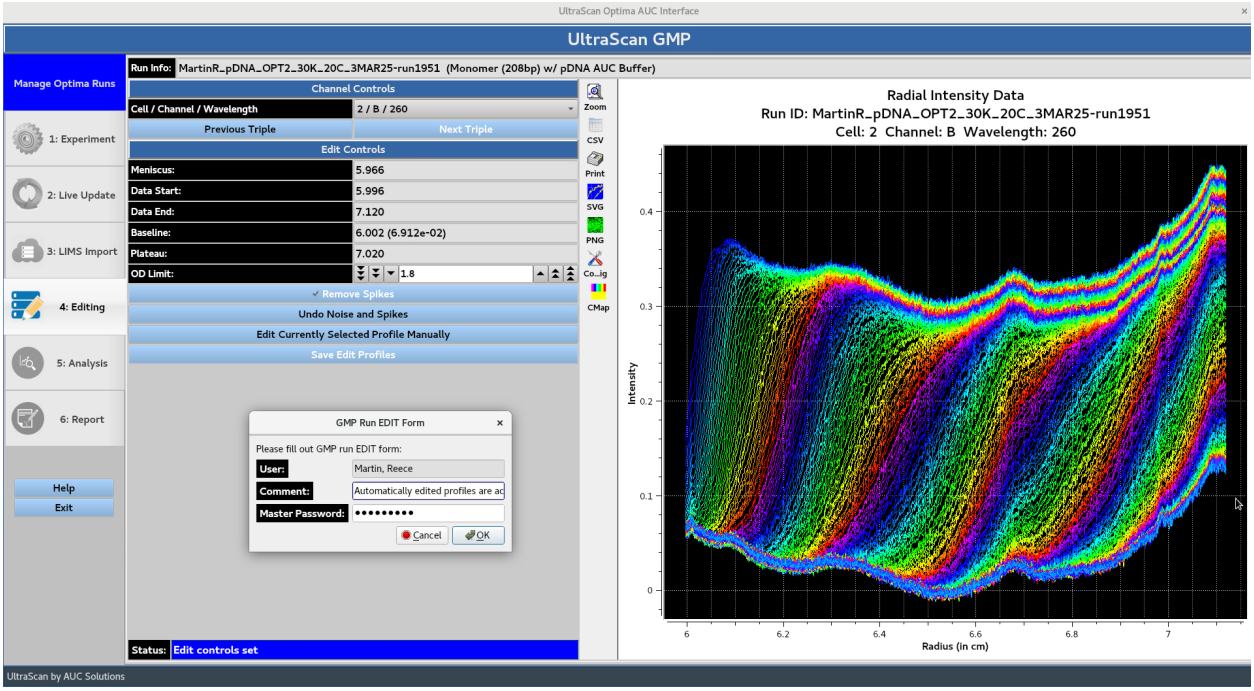
Once the data has been saved, the user is notified that the program is proceeding to the editing stage:



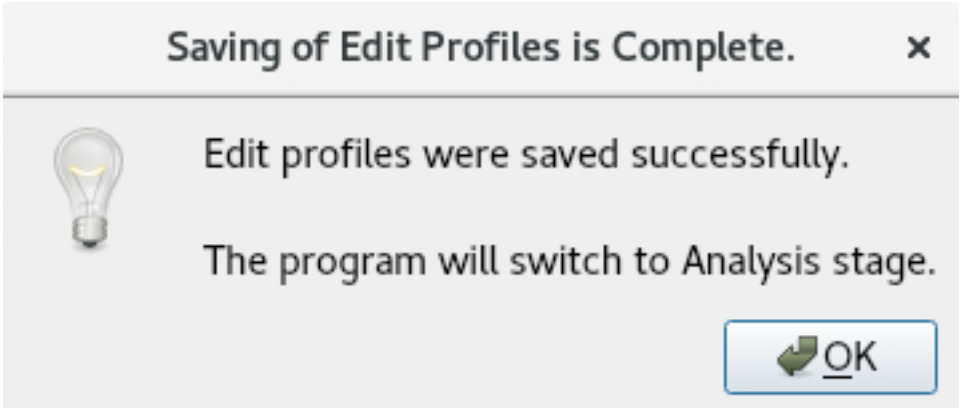
The editing window presents the user with the data edited in accordance with 'Data End' value and 'Excluded Scan Ranges' defined in the **Aprofile**. The 'Meniscus' position and 'Data Start' value are defined through automated processing. If the user is not satisfied with the automated processing, the **Edit Currently Selected Profile Manually** button allows for the 'Meniscus' position and 'Data End' values to be defined manually. **Remove Spikes** removes some sharp, irregular intensity spikes from data. Each cell/channel/wavelength Triple can be cycled through using the **Previous Triple** and **Next Triple** buttons.



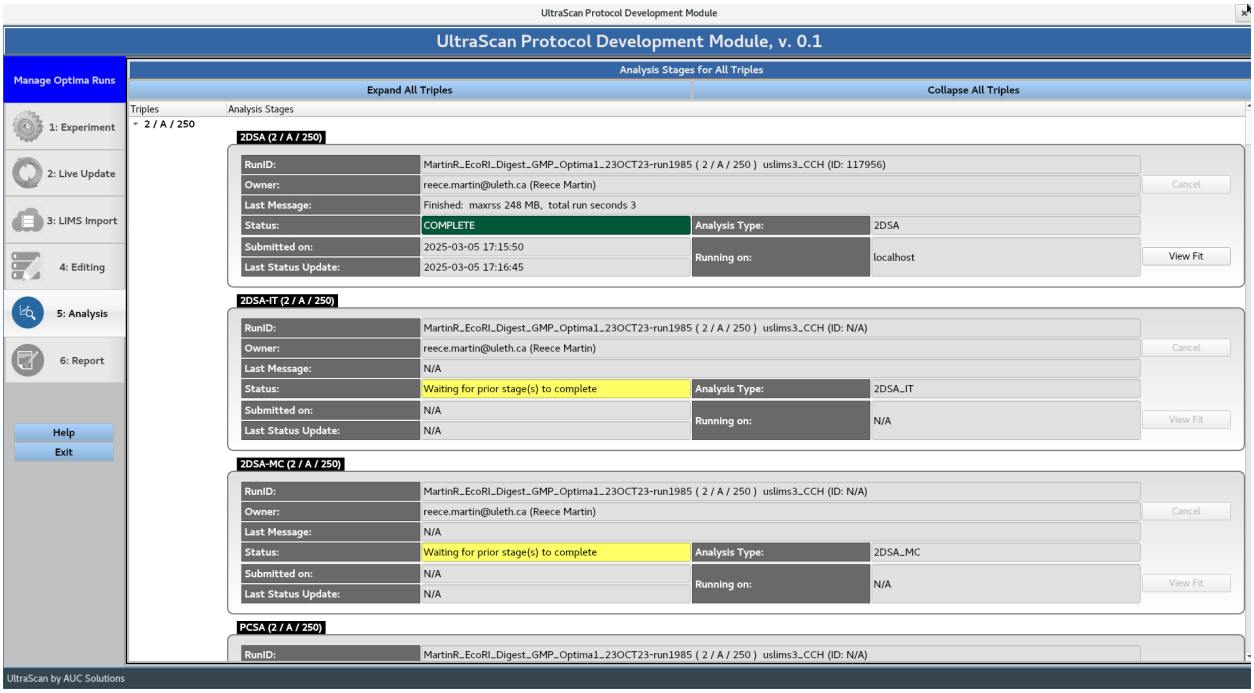
The **Save Edit Profiles** button prompts the **GMP Run EDIT Form**, allowing the user to add a comment and prompting them for their master password for authentication.



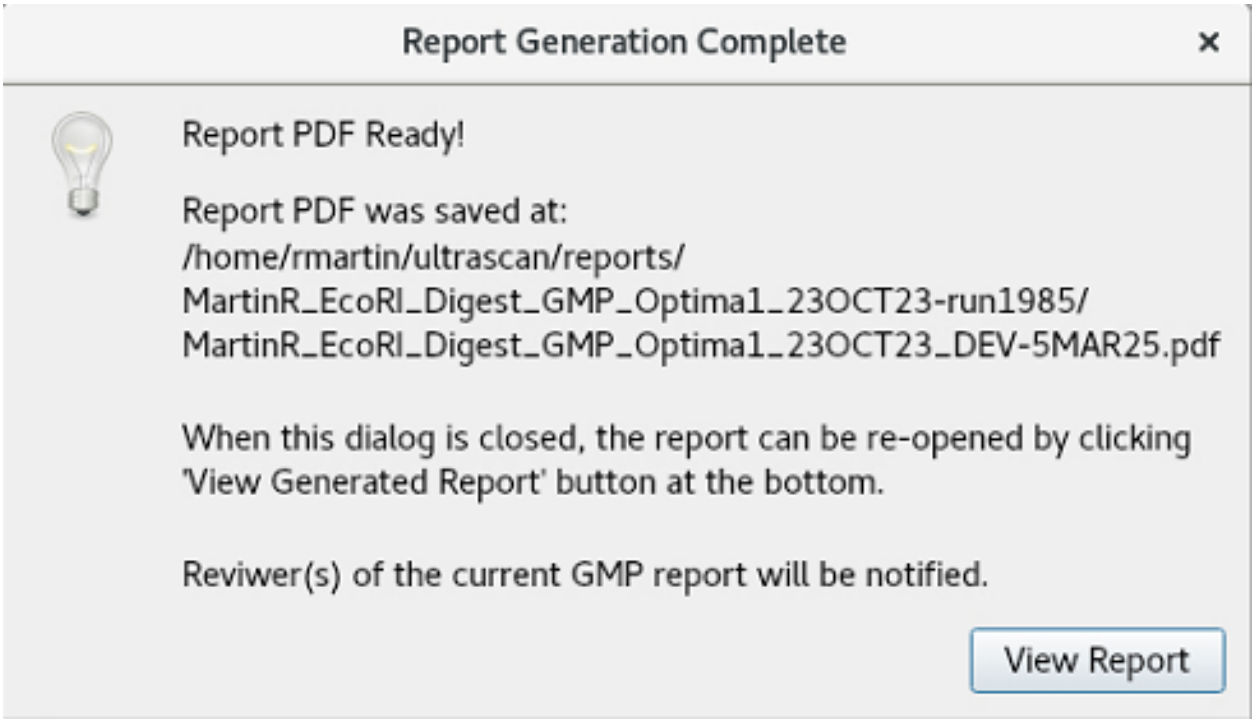
Once the edited profiles have been saved, the user is notified that the program is proceeding to the analysis stage:



The analysis window displays the current status of the submitted jobs producing the models specified in the **Aprofile**. **Cancel** can be used to stop ongoing jobs. **View Fit** simulates completed model overlays and displays them. The **Expand All Triples** and **Collapse All Triples** buttons control all Triples, while the {down arrow} button can be used to collapse or expand each individual Triple.



Once the analysis stage is complete, the models are simulated and the report is built. Which prompts the **Report Generation Complete** message box.



The **View Report** button opens the PDF file to be viewed:

32.22 GMP Report Generator and Viewer

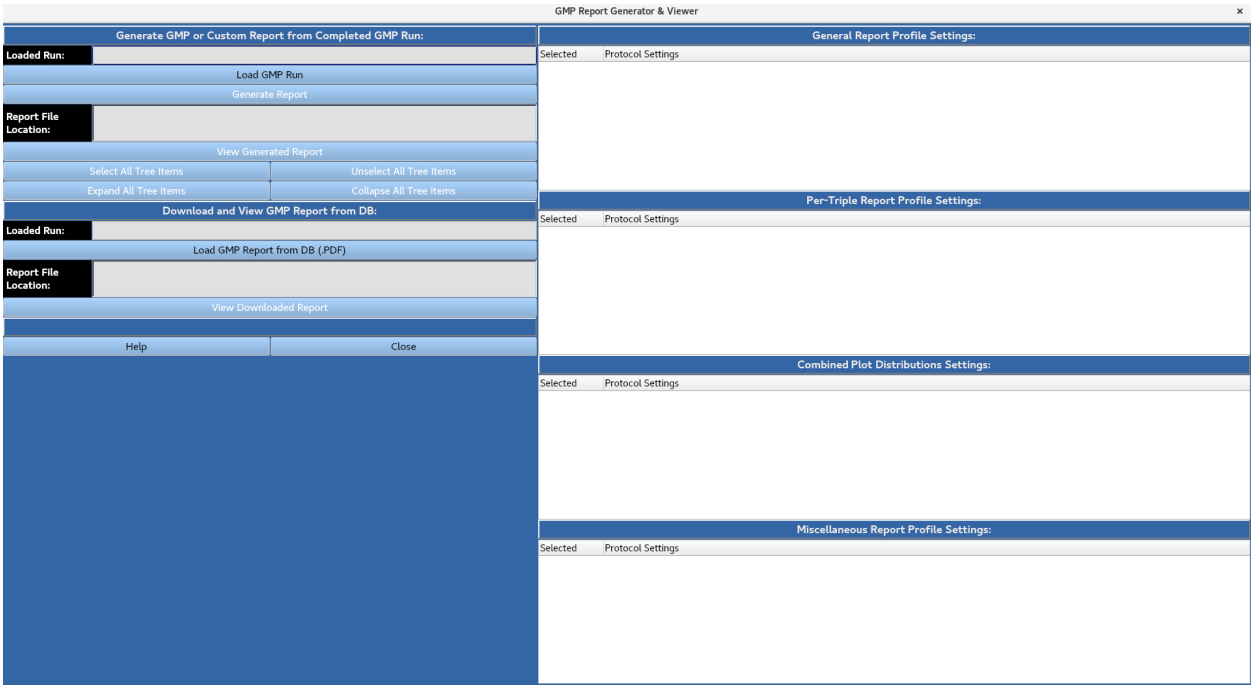
Index:

- *3a: Load GMP Run*
- *3b: Load GMP Report from DB (.PDF)*

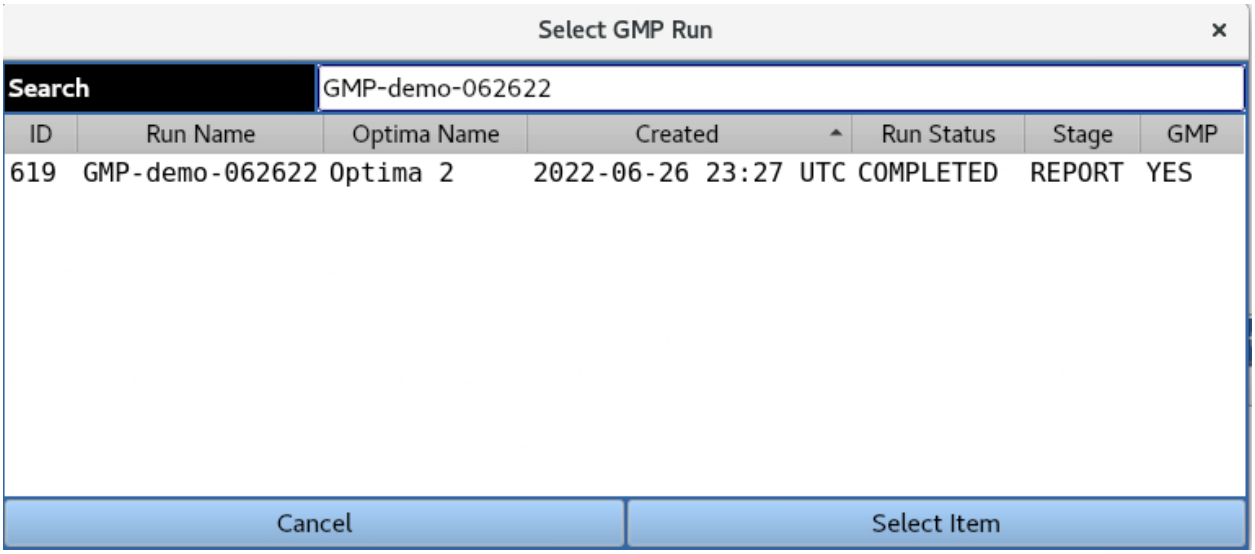
Allows for the generation/viewing of GMP run reports.

32.22.1 3a: Load GMP Run

Provides a list of completed GMP runs in the database, as well as the current reporting stage (Report generation or collection of e-signatures).

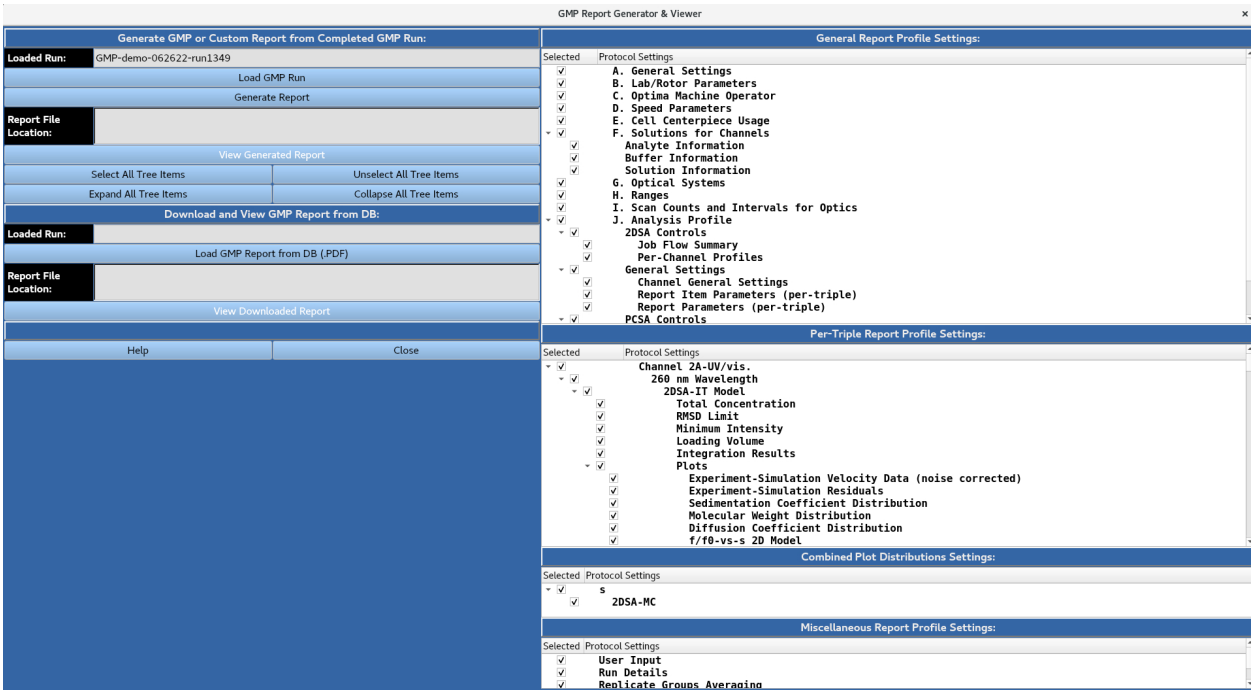


A GMP run can be loaded from the database by highlighting it and clicking the 'Select Item' button. The user is then presented with the below message indicating the report profile used was defined in the original GMP run submission, and that any changes to the report profile settings will result in the generation of a non-GMP report.



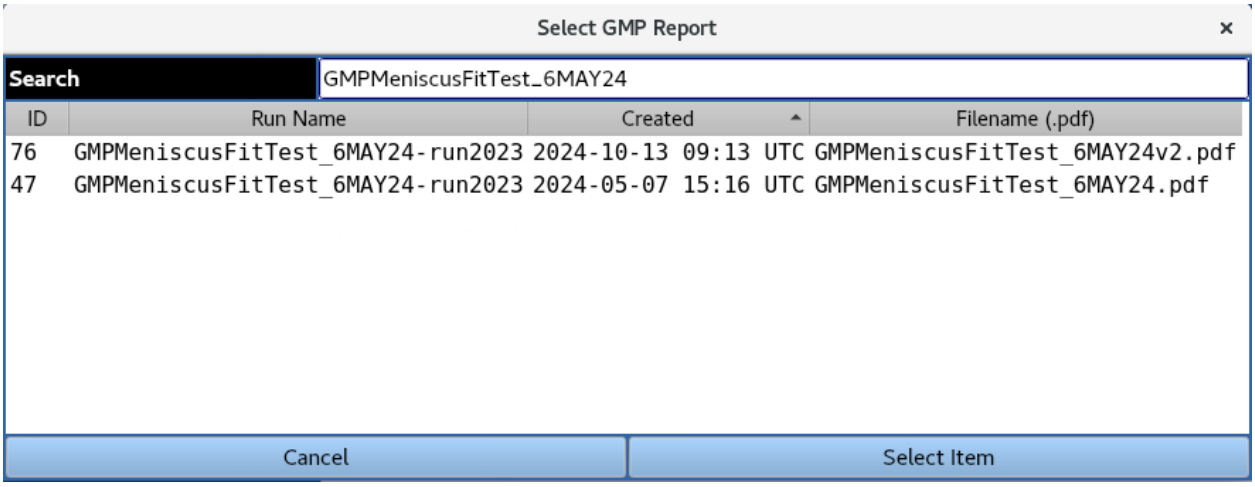
The user then has the option to expand/collapse and select/unselect all tree items for the report settings using the buttons, or manually expand/collapse and select/unselect individual report items using the check boxes in the portion of the window.

The user then has the option to expand/collapse and select/unselect all tree items for the report settings using the buttons, or manually expand/collapse and select/unselect individual report items using the check boxes in the portion of the window.

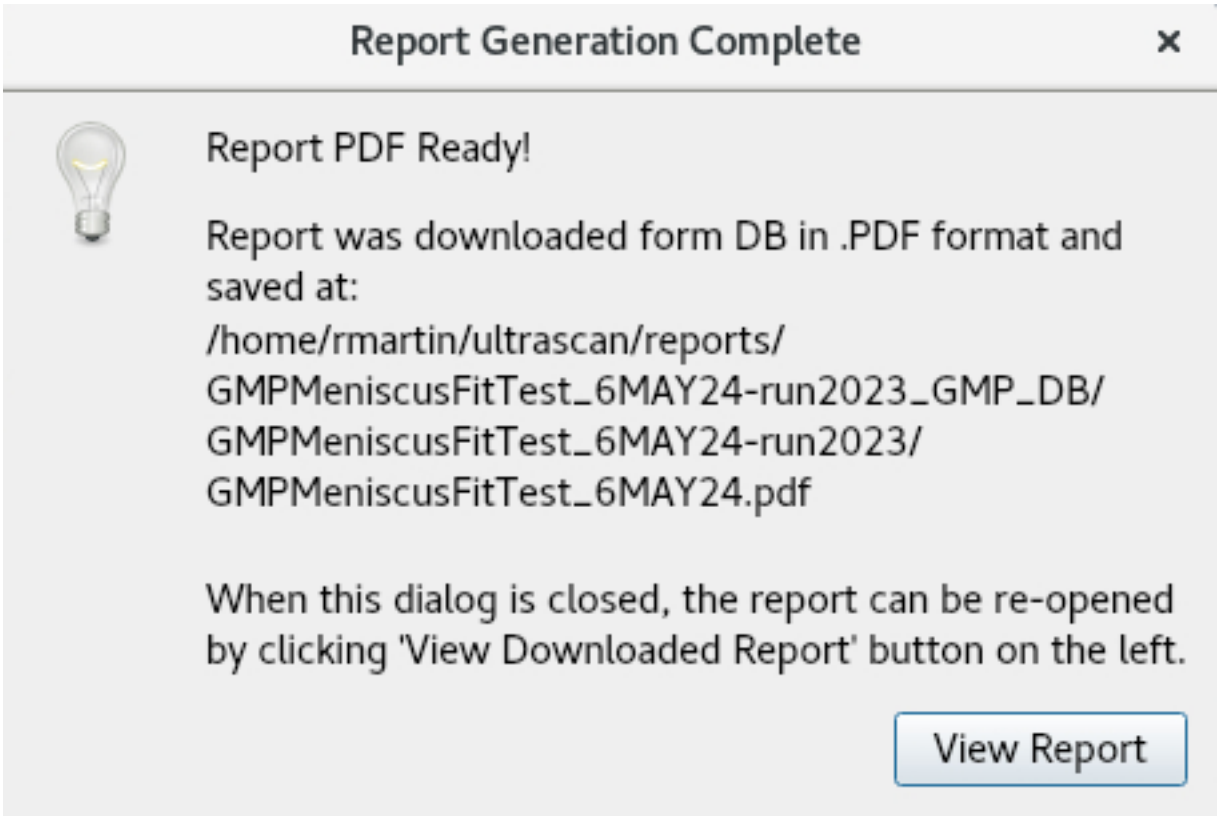


32.22.2 3b: Load GMP Report from DB (.PDF)

Allows for the viewing of a previously-generated GMP report stored in the database as a PDF file.

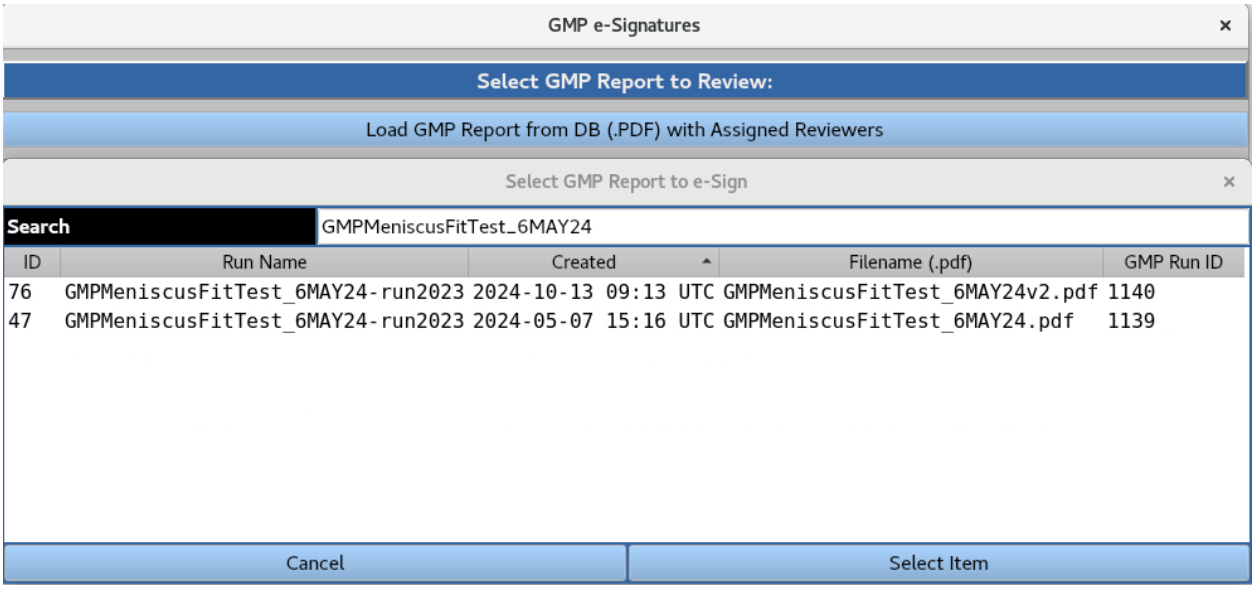


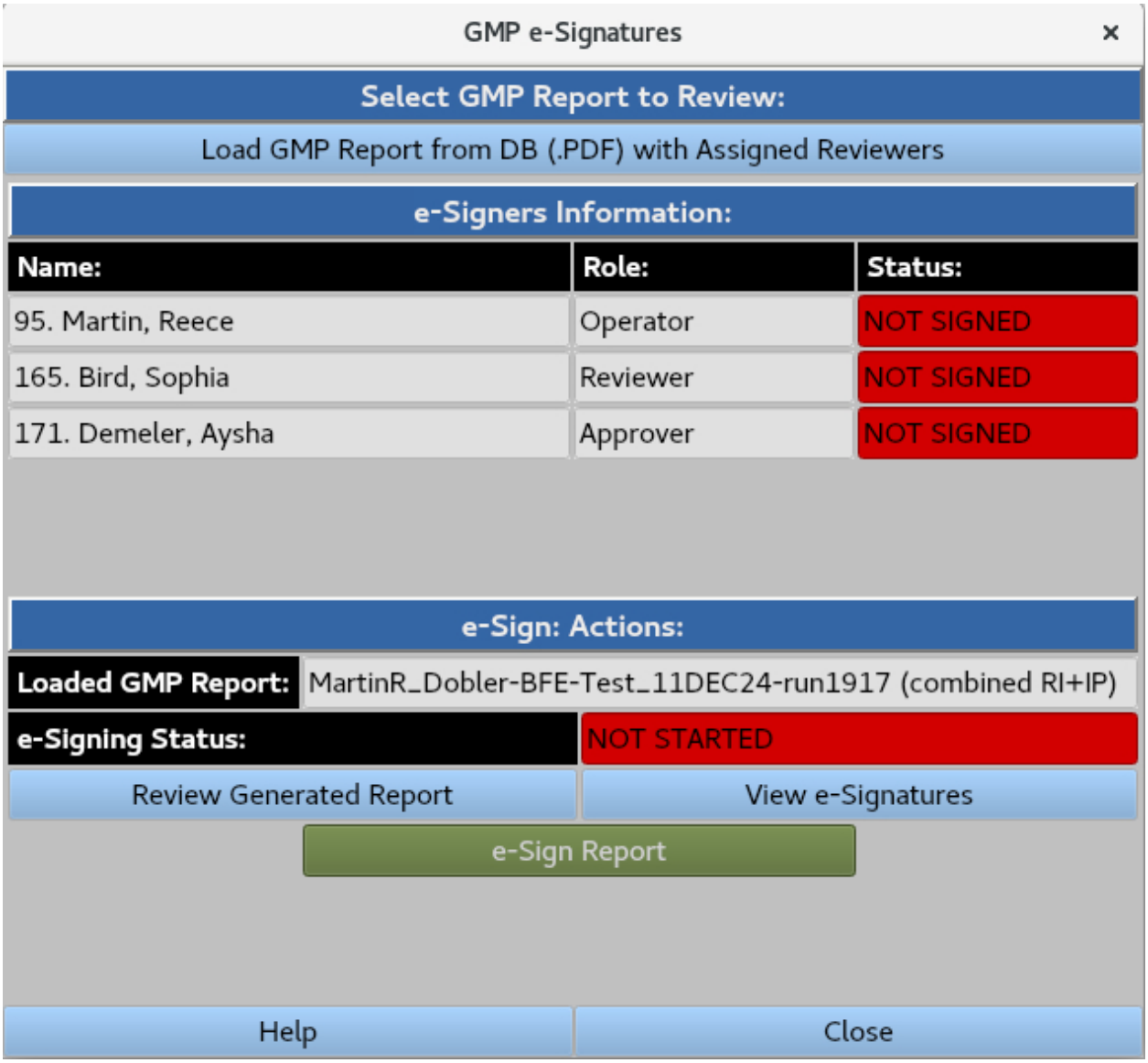
A GMP report PDF file can be loaded from the database by highlighting it a clicking the 'Select Item' button. The below message appears indicating the report was downloaded from the database and saved to the specified location. The **View Report** button opens the PDF file to be viewed.



32.23 GMP e-Signatures

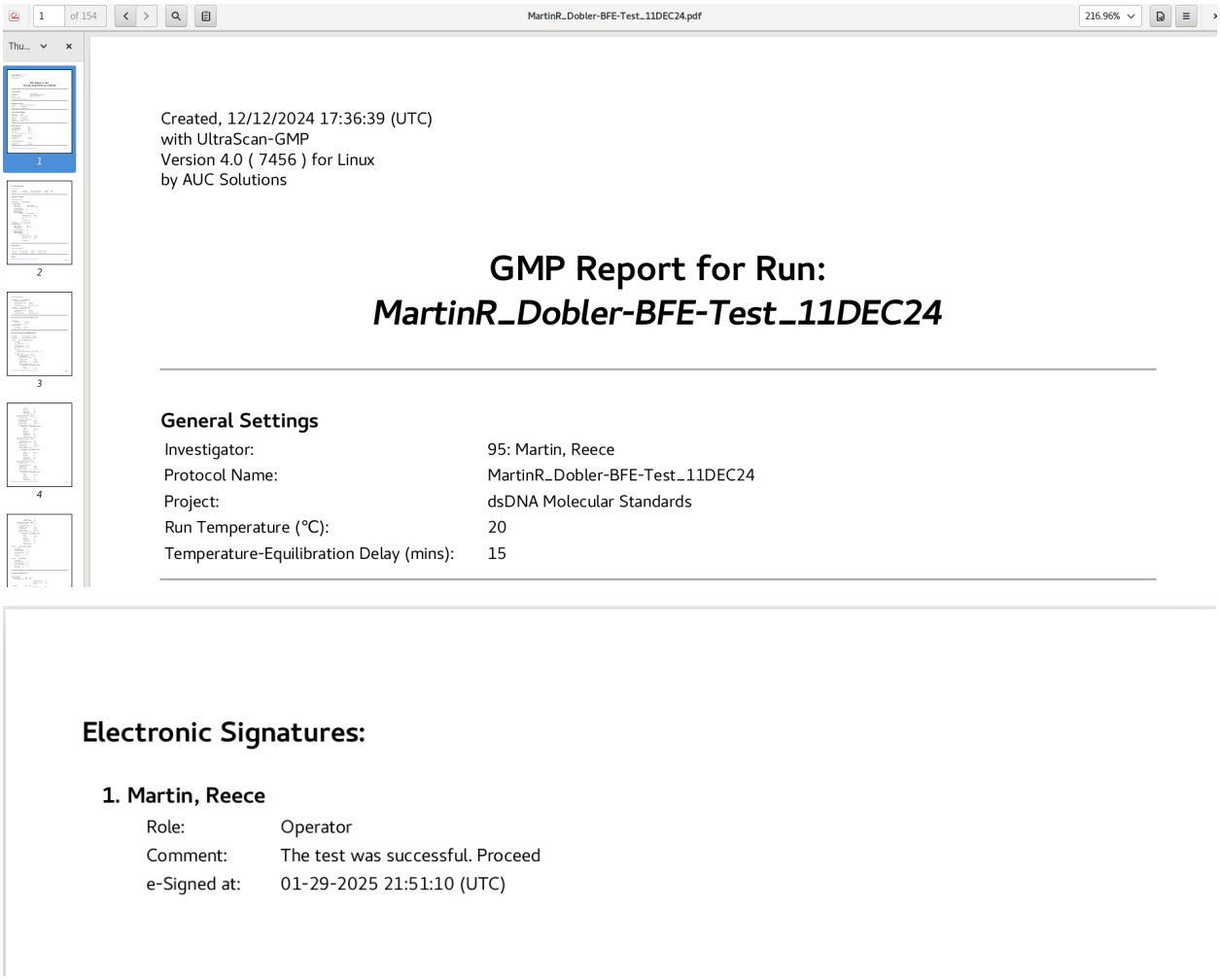
Enables users with assigned roles to load and electronically sign GMP report PDF files.



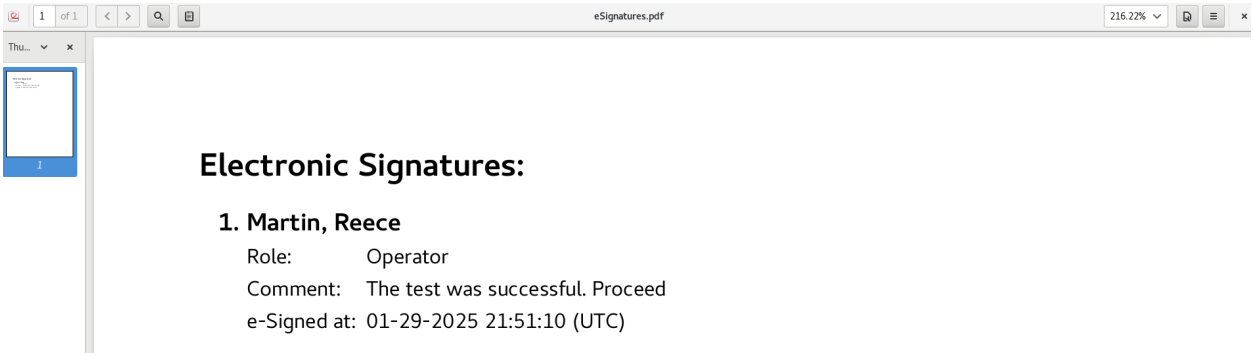


A GMP report PDF file can be loaded from the database by highlighting it a clicking the 'Select Item' button. The user is presented with the following interface:

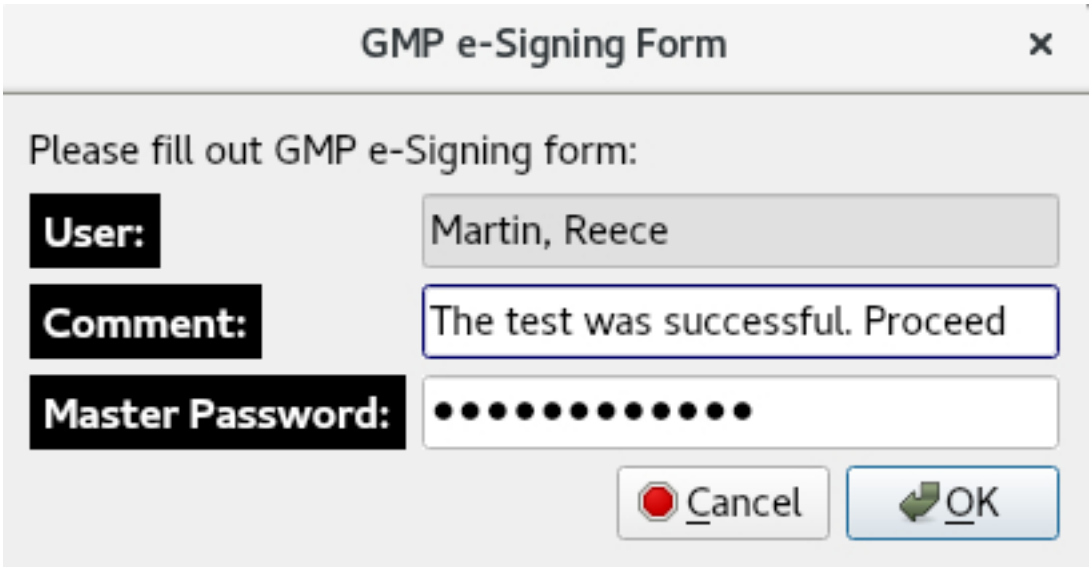
'Review Generated Report' opens the PDF file from the database, which includes the current e-signatures.



View e-signatures' opens a PDF file containing only the current e-signatures

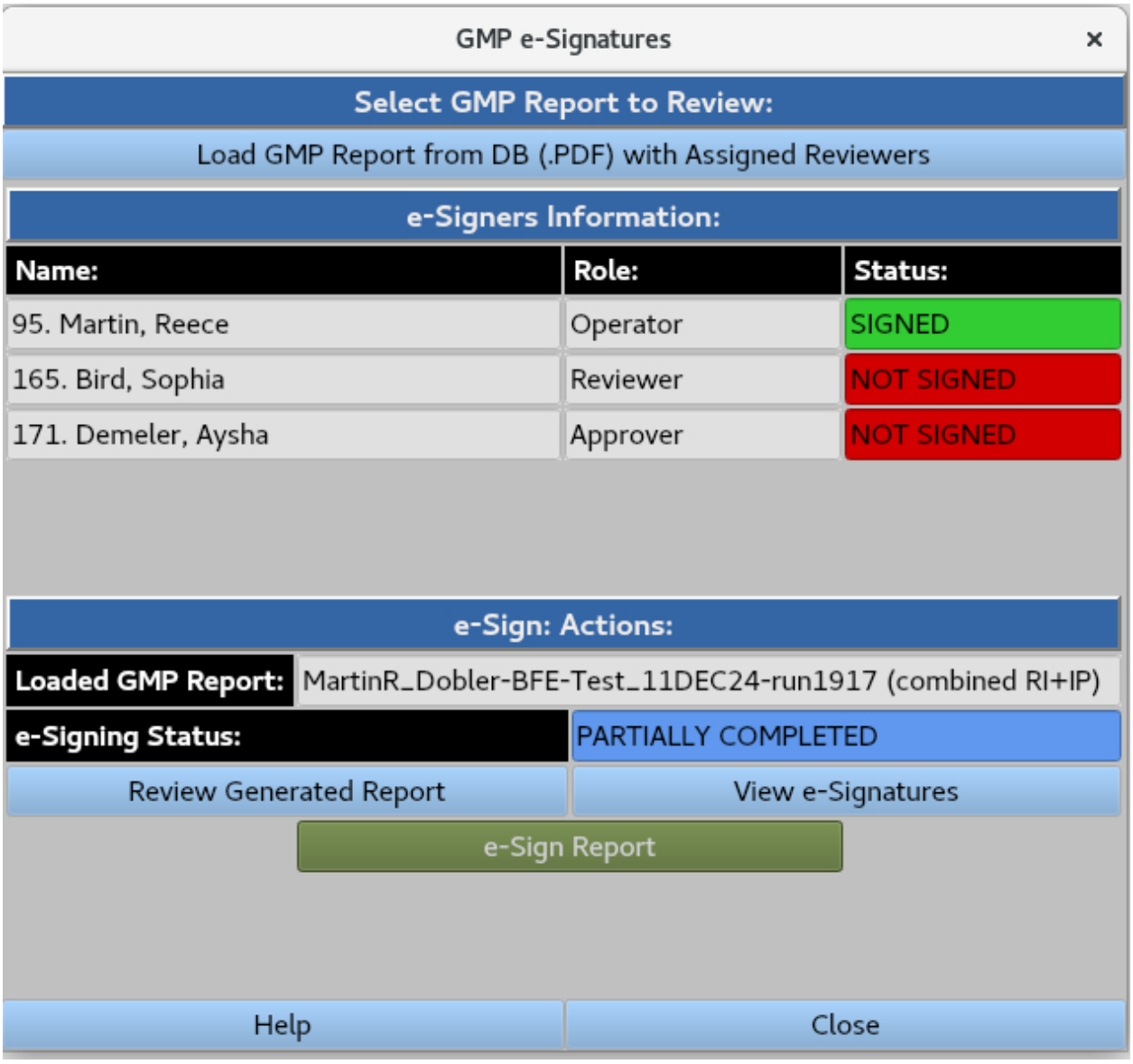


'e-Sign Report' prompts the 'GMP e-Signing Form', allowing the user to add feedback in the form of a comment, and requiring their master password for authentication.



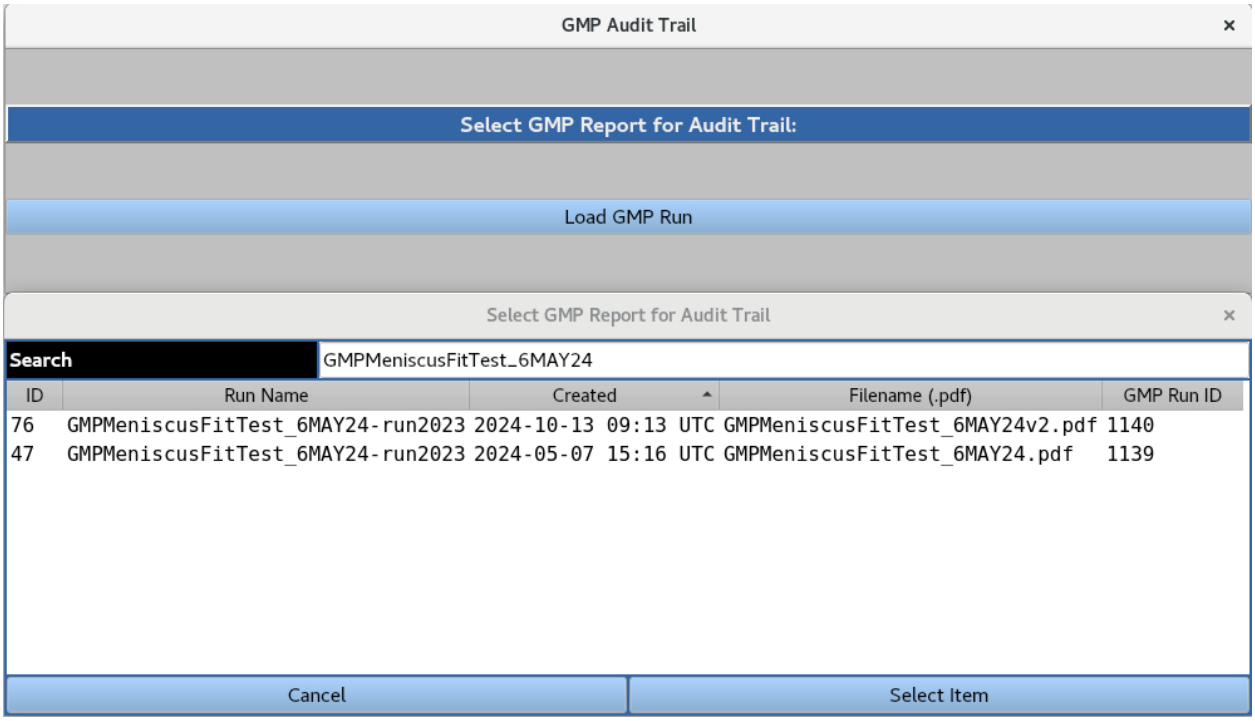
Completing this form prompts the 'Successful e-Signing' notification.

Returning to the GMP e-signature interface updates the e-signing status and prompts the next e-signer for electronic signature via email (email settings are configured by the System Administrator).

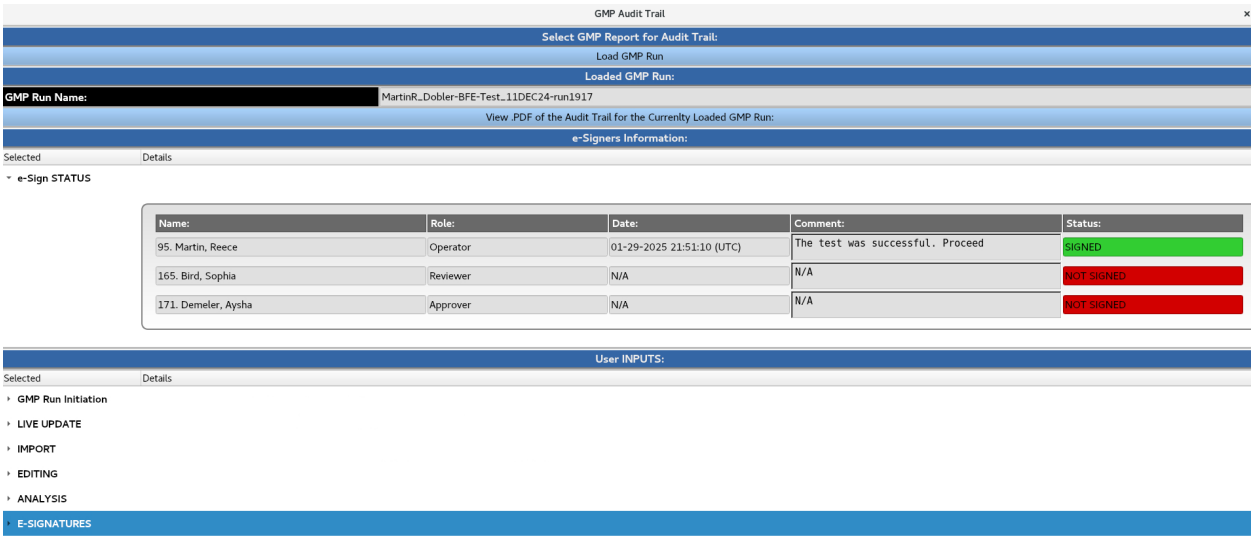


32.24 Audit Trail

Logs all user interactions and electronic signatures for a GMP report PDF file.



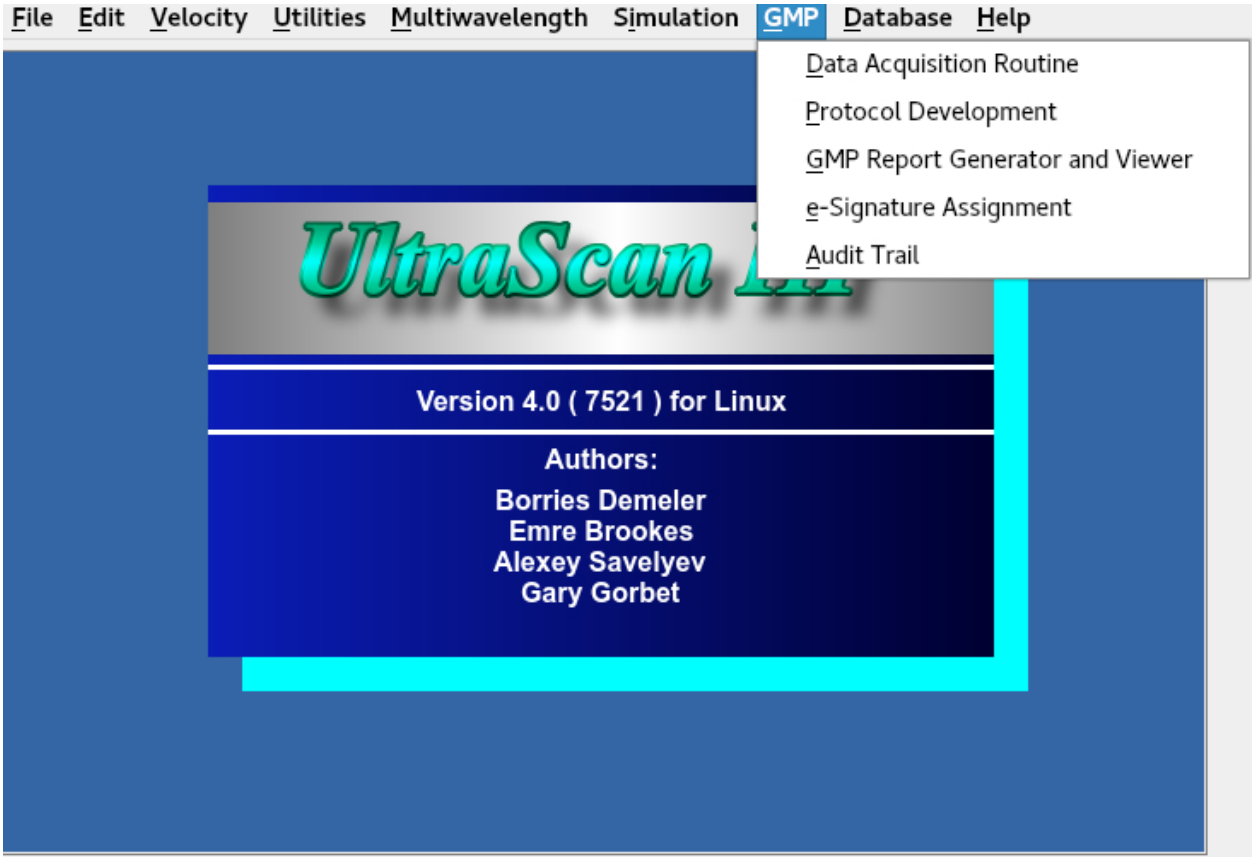
A GMP run can be loaded from the database by highlighting it and clicking the **Select Item** button. The user is then presented with the following interface:



e-Sign STATUS section records the user ID, name, role, date, time, comments, and signing status for the GMP run.

User INPUTS section logs all user interactions at various stages of the GMP run to ensure traceability. Each entry includes the user ID, date, time, and associated comments.

View .PDF of the Audit Trail of the Currently Loaded GMP Run loads the audit trail PDF file containing a record of all user interactions and e-signatures associated with the GMP run. This PDF file is a summary of the 'e-Sign STATUS' and 'User INPUTS' sections.



GMP Modules Access

CHAPTER

THIRTYTHREE

RESEARCH AND DEVELOPMENT AUC DATA ACQUISITION MODULE

Index

- *Data Acquisition*

33.1 General: Create Experiment

Index

- *Panel Tab Options:*
- *General Panel:*
  - *General Functions:*
- *Shared Panel Functions:*

33.1.1 Panel Tab Options:

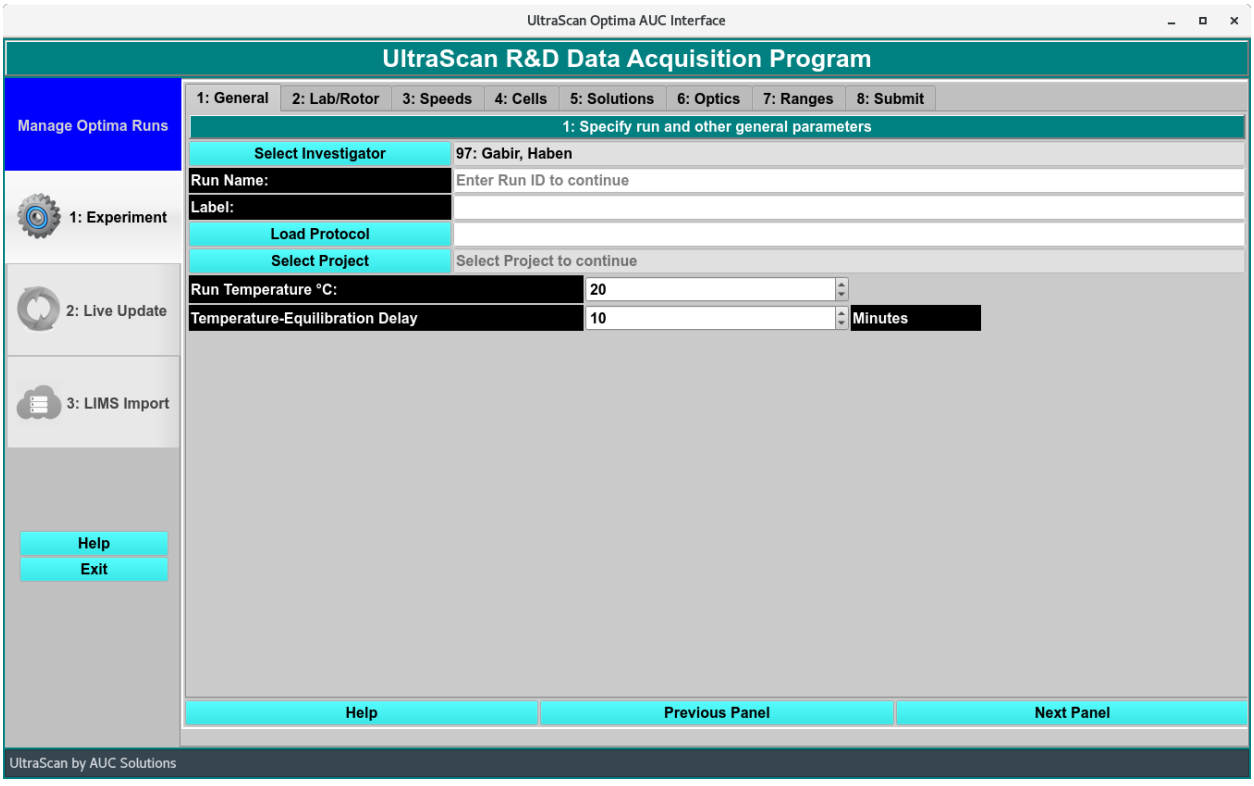
In each panel, tabs are visible at the top of the window to enable the user to move to another panel, to perform specific experiment subtasks.

1. *General*: A panel whose primary purpose is to specify the experiment run ID (a description string), select a parent project name, and possibly change data source (database or local disk) or database investigator name.
2. :doc:`Lab/Rotor <rotor>: A panel whose primary purpose is to select the Laboratory, Rotor, and Calibration values for the experiment.
3. :doc:`Speeds <speeds>: A panel whose primary primary purpose is to specify one or more speed steps. For each step, speeds and durations may be given.

4. :doc:`Cells <cells>: A panel whose primary purpose is to select the centerpieces (or counterbalance) for the cells, along with a quartz/sapphire windows selection.
5. :doc:`Solutions <solutions>: A panel whose primary purpose is to specify the solution to be used in each cell/channel.
6. :doc:`Optics <optics>: A panel whose primary purpose is to select the optical system used to scan.
7. :doc:`Range <range>: A panel whose primary purpose is to select the wavelength ranges and cell radius measured.
8. :doc:`Submit <submit>: A panel whose primary purpose is save and upload the protocol to the Optima/Instrument and database.

33.1.2 General Panel:

Using this panel, you can define general experiment parameters, such as Run ID and Project, along with data source (database or local disk) and investigator. As with all panels, a set of tabs allows you to navigate to other panels in order to perform specialized subtasks relating to experiment creation. Links to and summaries of the panels are given in the final section of this page.



General Definitions Panel

General Functions:

Run ID:	Enter a unique run identifier string that will identify the experiment to be run.
Select Project:	Click this button to open a full Project dialog in which a new project may be created or an existing one modified. Hitting the Accept button in that dialog will cause selection of a project for the experiment.
(project list):	Pulling down and releasing on an item in the existing projects list causes the created experiment to be associated with that project.
Select Investigator:	Click this button to open a full database Investigator dialog in which an investigator may be chosen.
(investigator text):	Any investigator chosen will be displayed in this text box. You may manually modify this text.
Database:	Select this radio button to indicate that data used in creating the experiment comes from the database.
Local:	Disk Select this radio button to indicate that data used in creating the experiment comes from the local disk.

33.1.3 Shared Panel Functions:

A panel status box and multiple buttons are shared by all panels.

Status:	A text box shows the status of completion of needed parameterization for all the panels. Only when all panels have a 'X' state is upload of controls possible.
Help:	Display the help manual page for the current panel.
Status:	Update Click this button to re-read elements of all panels to compose updated status summary text.
Previous Panel:	Convenience button to move to the panel previous to the current one.
Next Panel:	Convenience button to move to the panel that follows the current one.
Close:	Close the Experiment dialog. Note that any changes to database values or any upload of experiment controls is governed to elements within the panels of the Experiment dialog. This Close button merely closes the dialog, regardless of experiment definition state.

33.2 Lab/Rotor: Define Lab and Rotor Conditions

Index

- *Panel Tab Options:*
- *Lab/Rotor Panel:*

    - *Lab/Rotor Functions:*

- *Shared Panel Functions:*

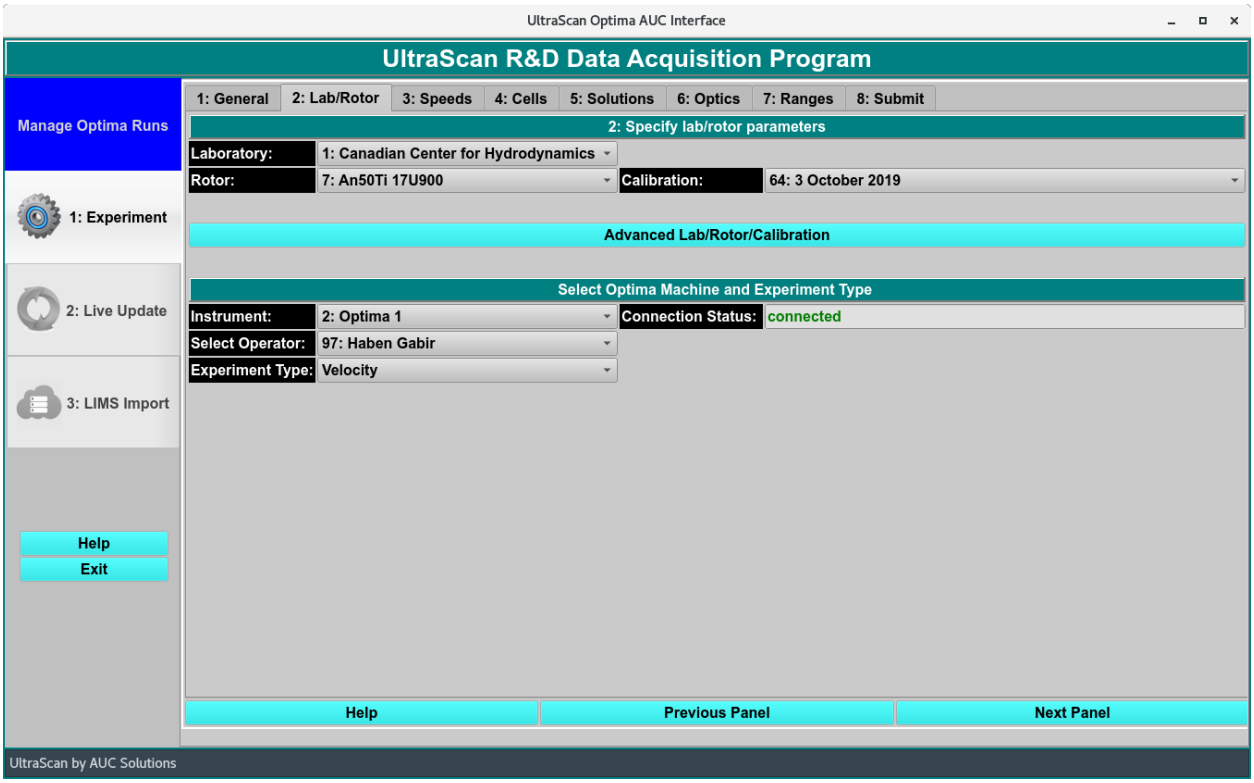
33.2.1 Panel Tab Options:

In each panel, tabs are visible at the top of the window to enable the user to move to another panel, to perform specific experiment subtasks.

1. :doc:`General <general>: A panel whose primary purpose is to specify the experiment run ID (a description string), select a parent project name, and possibly change data source (database or local disk) or database investigator name.
2. *Lab/Rotor*: A panel whose primary purpose is to select the Laboratory, Rotor, and Calibration values for the experiment.
3. :doc:`Speeds <speeds>: A panel whose primary primary purpose is to specify one or more speed steps. For each step, speeds and durations may be given.
4. :doc:`Cells <cells>: A panel whose primary purpose is to select the centerpieces (or counterbalance) for the cells, along with a quartz/sapphire windows selection.
5. :doc:`Solutions <solutions>: A panel whose primary purpose is to specify the solution to be used in each cell/channel.
6. :doc:`Optics <optics>: A panel whose primary purpose is to select the optical system used to scan.
7. :doc:`Range <range>: A panel whose primary purpose is to select the wavelength ranges and cell radius measured.
8. :doc:`Submit <submit>: A panel whose primary purpose is save and upload the protocol to the Optima/Instrument and database.

33.2.2 Lab/Rotor Panel:

Using this panel, you can specify the rotor to be used on a proposed experiment. Specifications include the laboratory, the rotor type, and the calibration. The :doc:`Rotor Management <../rotor> module help page has detailed explanations of these items.



Lab and Rotor Panel

Lab/Rotor Functions:

Laboratory:	Pull down a list and release on the laboratory to which the experiment belongs.
Rotor:	Pull down a list and release on the type of rotor to be used in the experiment.
Calibration:	Pull down a list and release on the calibration (in date string form) for use in the experiment.
Advanced Lab/Rotor/Calibration:	Click on this button to open a full Rotor dialog in which laboratory/rotor/calibration information may be created/selected. Selections in that dialog will be reflected in the other Experiment Rotor Panel GUI elements.

33.2.3 Shared Panel Functions:

A panel status box and multiple buttons are shared by all panels. The :doc:`General <general> tab help page has detailed explanations of these items.

33.3 Speeds: Define Speed and Duration Profiles

Index

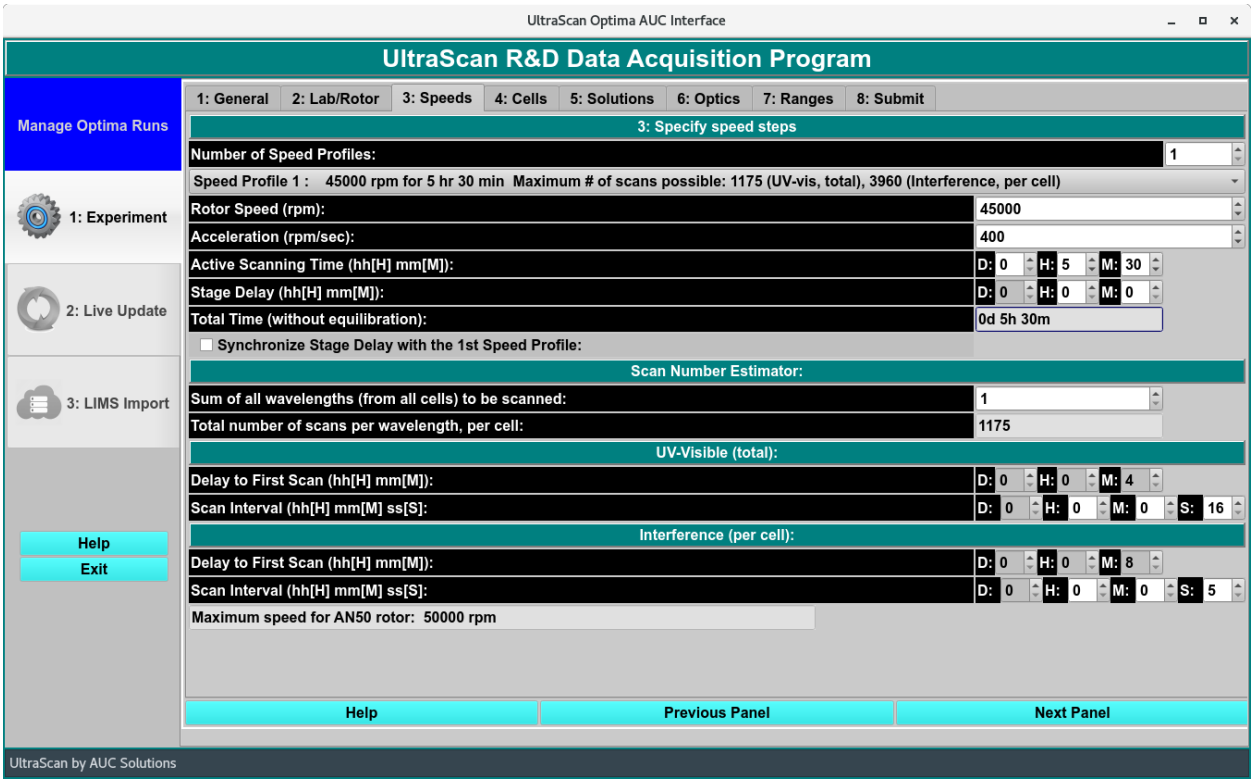
- *Panel Tab Options:*
- *Speeds Panel:*
  - *Speed Functions:*
- *Shared Panel Functions:*

33.3.1 Panel Tab Options:

In each panel, tabs are visible at the top of the window to enable the user to move to another panel, to perform specific experiment subtasks.

1. :doc:`General <general>: A panel whose primary purpose is to specify the experiment run ID (a description string), select a parent project name, and possibly change data source (database or local disk) or database investigator name.
2. :doc:`Lab/Rotor <rotor>: A panel whose primary purpose is to select the Laboratory, Rotor, and Calibration values for the experiment.
3. *Speeds*: A panel whose primary primary purpose is to specify one or more speed steps. For each step, speeds and durations may be given.
4. :doc:`Cells <cells>: A panel whose primary purpose is to select the centerpieces (or counterbalance) for the cells, along with a quartz/sapphire windows selection.
5. :doc:`Solutions <solutions>: A panel whose primary purpose is to specify the solution to be used in each cell/channel.
6. :doc:`Optics <optics>: A panel whose primary purpose is to select the optical system used to scan.
7. :doc:`Range <range>: A panel whose primary purpose is to select the wavelength ranges and cell radius measured.
8. :doc:`Submit <submit>: A panel whose primary purpose is save and upload the protocol to the Optima/Instrument and database.

33.3.2 Speeds Panel:



Speed Profile Panel

Speed Functions:

Number of Speed Profiles:	Select a counter value that specifies the number of constant rotor speeds to be used for the duration of the experiment. Most often, this will be "1".
(step description):	This pull-down menu selection sets the speed step for which the speed and duration to follow apply.
Rotor Speed (rpm):	Select a counter value to specify the rotor speed for the speed step.
Acceleration Profile (rpm_sec):	Select an acceleration value for the transition from zero speed or from the speed of the previous step.
Length of Experiment (Hours):	Specify the duration in hours of the portion of the experiment at the current rotor speed.
Length of Experiment (Minutes):	Specify the residual minutes of duration of the portion of the experiment at the current rotor speed. This specification may be a floating point value.
Time Delay for Scans (Hours):	Specify the duration in hours of the portion of the experiment spent in transitioning from zero speed or the previous speed to the current constant rotor speed.
Time Delay for Scans (Minutes):	Specify the residual minutes of duration of the portion of the experiment spent in transitioning from zero speed or the previous speed to the current constant rotor speed. This specification may be a floating point value.

33.3.3 Shared Panel Functions:

A panel status box and multiple buttons are shared by all panels. The :doc:`General <general> tab help page has detailed explanations of these items.

33.4 Cells: Define Cell Centerpieces and Windows

Index

- *Panel Tab Options:*
- *Cells Panel:*
  - *Cells Functions:*
- *Shared Panel Functions:*

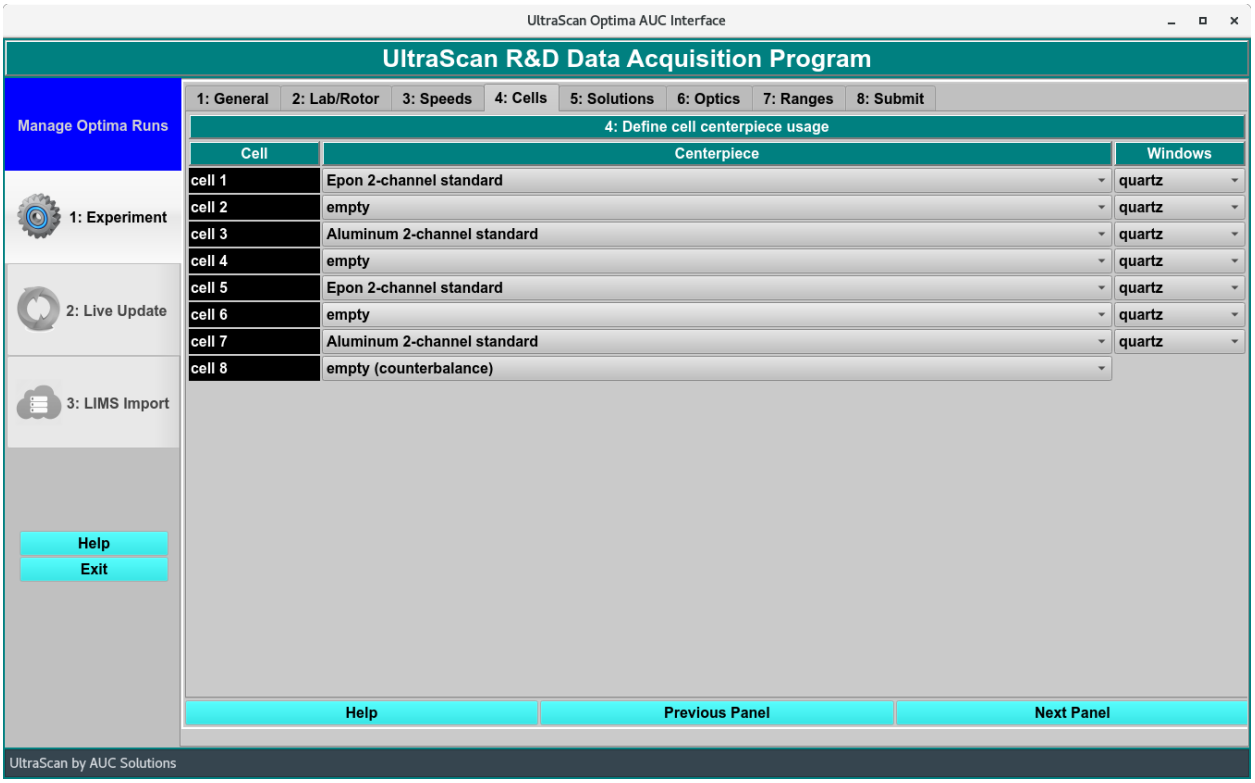
33.4.1 Panel Tab Options:

In each panel, tabs are visible at the top of the window to enable the user to move to another panel, to perform specific experiment subtasks.

1. *General*: A panel whose primary purpose is to specify the experiment run ID (a description string), select a parent project name, and possibly change data source (database or local disk) or database investigator name.
2. *Lab/Rotor*: A panel whose primary purpose is to select the Laboratory, Rotor, and Calibration values for the experiment.
3. *Speeds*: A panel whose primary primary purpose is to specify one or more speed steps. For each step, speeds and durations may be given.
4. *Cells*: A panel whose primary purpose is to select the centerpieces (or counterbalance) for the cells, along with a quartz/sapphire windows selection.
5. *Solutions*: A panel whose primary purpose is to specify the solution to be used in each cell/channel.
6. *Optics*: A panel whose primary purpose is to select the optical system used to scan.
7. *Range*: A panel whose primary purpose is to select the wavelength ranges and cell radius measured.
8. *Submit*: A panel whose primary purpose is save and upload the protocol to the Optima/Instrument and database.

33.4.2 Cells Panel:

Using this panel, you can define centerpiece and windows information for each cell of a proposed experiment. As with all panels, a set of tabs allows you to navigate to other panels in order to perform specialized subtasks relating to experiment creation. Links to and summaries of the panels are given in the final section of this page.



Cell Components Panel

Cells Functions:

Centerpiece:	For each cell, a centerpiece or counterbalance may be selected from a pull-down list.
Windows:	For each cell, **quartz** or **sapphire** windows type may be selected from a pull-down list.

33.4.3 Shared Panel Functions:

A panel status box and multiple buttons are shared by all panels. The *General* tab help page has detailed explanations of these items.

33.5 Solutions: Define the Solutions per Channel

Index

- *Panel Tab Options:*
- *Solutions Panel:*
- *Shared Panel Functions:*

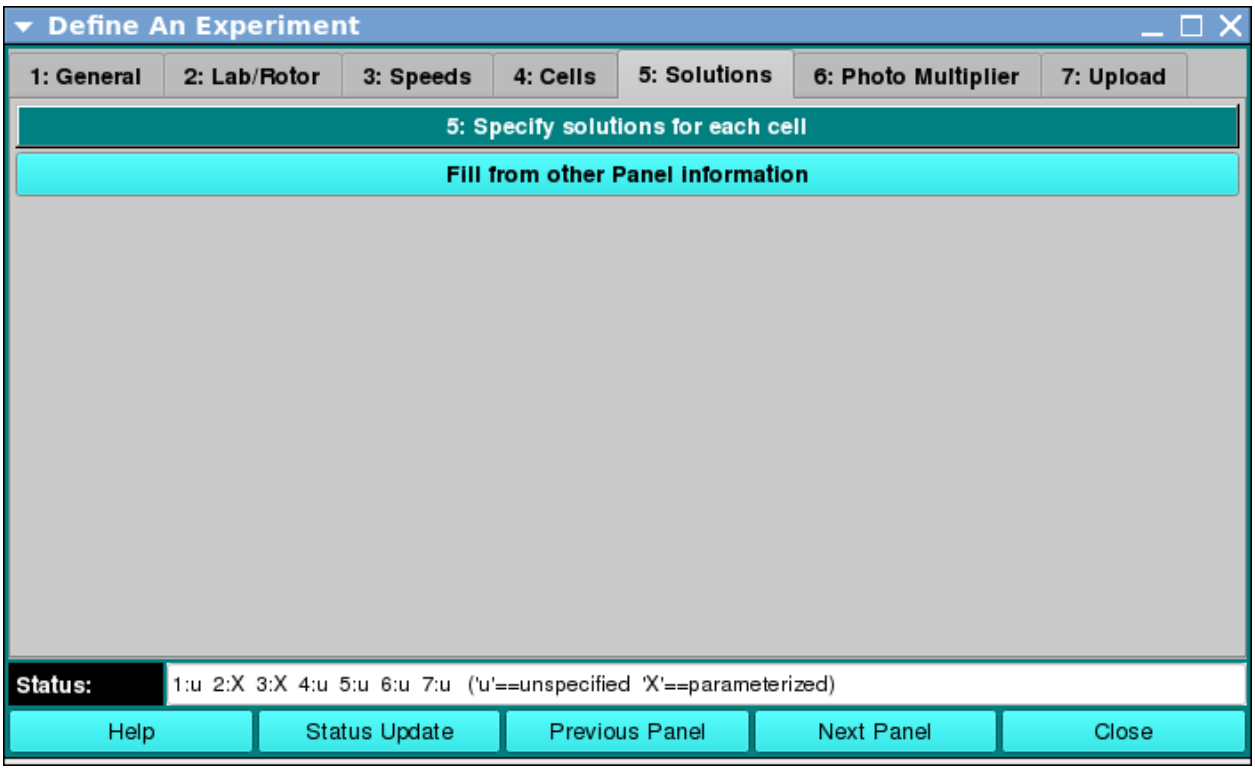
33.5.1 Panel Tab Options:

In each panel, tabs are visible at the top of the window to enable the user to move to another panel, to perform specific experiment subtasks.

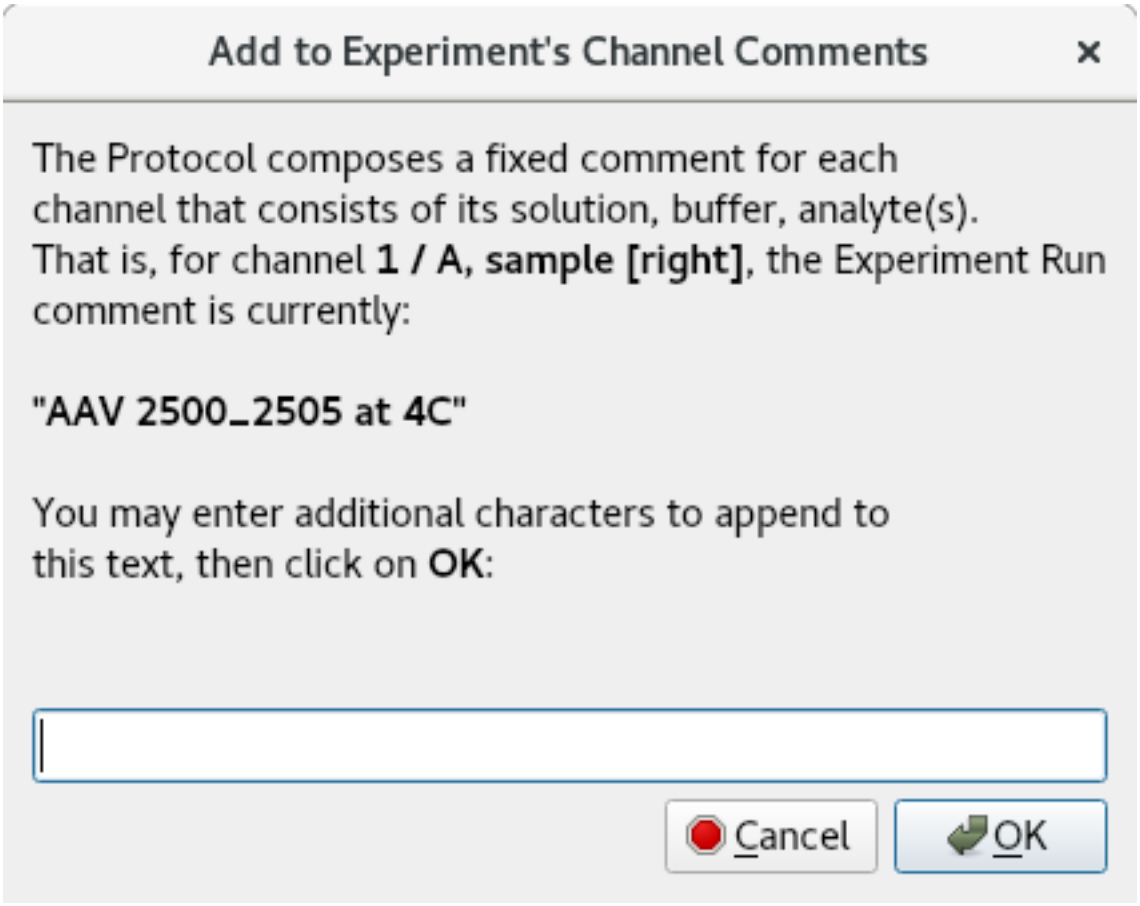
1. :doc:`General <general>: A panel whose primary purpose is to specify the experiment run ID (a description string), select a parent project name, and possibly change data source (database or local disk) or database investigator name.
2. :doc:`Lab/Rotor <rotor>: A panel whose primary purpose is to select the Laboratory, Rotor, and Calibration values for the experiment.
3. :doc:`Speeds <speeds>: A panel whose primary primary purpose is to specify one or more speed steps. For each step, speeds and durations may be given.
4. :doc:`Cells <cells>: A panel whose primary purpose is to select the centerpieces (or counterbalance) for the cells, along with a quartz/sapphire windows selection.
5. *Solutions*: A panel whose primary purpose is to specify the solution to be used in each cell/channel.
6. :doc:`Optics <optics>: A panel whose primary purpose is to select the optical system used to scan.
7. :doc:`Range <range>: A panel whose primary purpose is to select the wavelength ranges and cell radius measured.
8. :doc:`Submit <submit>: A panel whose primary purpose is save and upload the protocol to the Optima/Instrument and database.

33.5.2 Solutions Panel:

Using this panel, you can specify the solutions used in each channel. *Comments* can also be included and appended to the solution name. The :doc:`Solutions Management <../solutions/index> module help page has detailed explanations of these items.



Solutions Panel



Solution Name

33.5.3 Shared Panel Functions:

A panel status box and multiple buttons are shared by all panels. The :doc:`General <general> tab help page has detailed explanations of these items.

33.6 Optics: Define Optical System Scan Types

Index

- *Panel Tab Options:*
- *Optics Panel:*
- *Shared Panel Functions:*

33.6.1 Panel Tab Options:

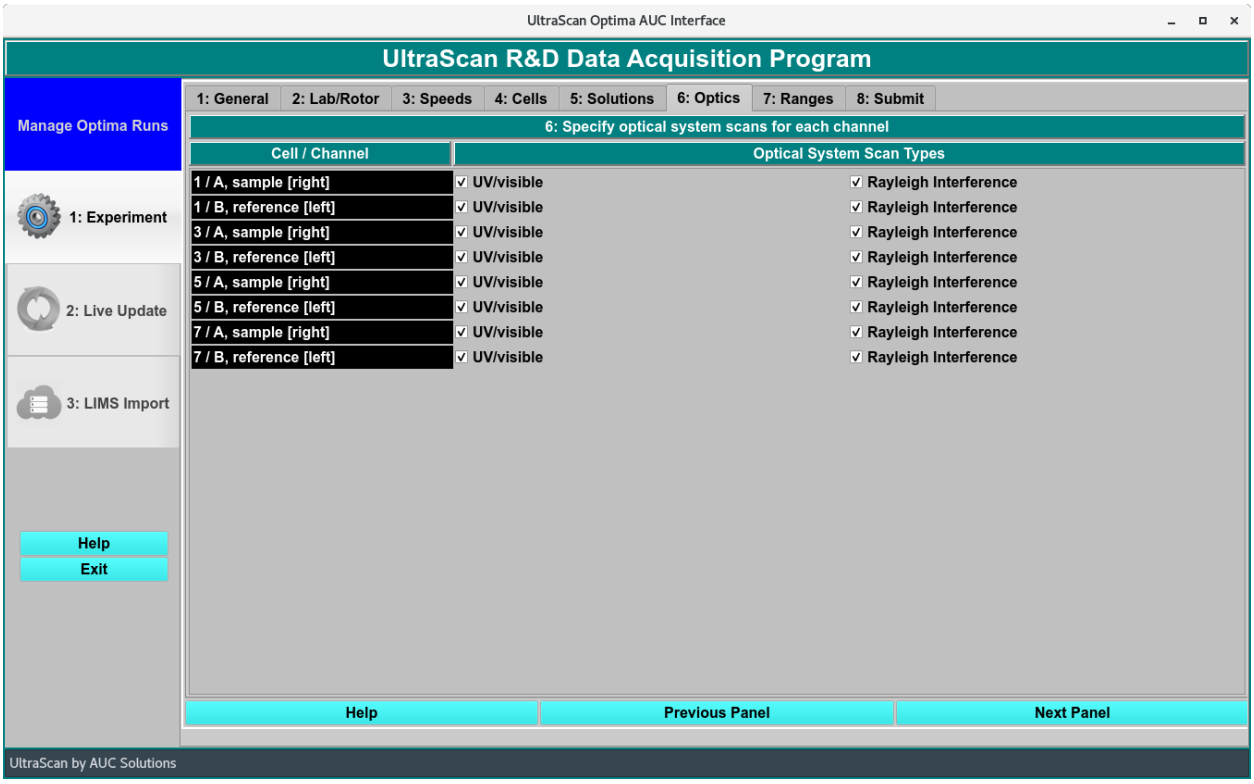
In each panel, tabs are visible at the top of the window to enable the user to move to another panel, to perform specific experiment subtasks.

1. *General*: A panel whose primary purpose is to specify the experiment run ID (a description string), select a parent project name, and possibly change data source (database or local disk) or database investigator name.
2. *Lab/Rotor*: A panel whose primary purpose is to select the Laboratory, Rotor, and Calibration values for the experiment.
3. *Speeds*: A panel whose primary primary purpose is to specify one or more speed steps. For each step, speeds and durations may be given.
4. *Cells*: A panel whose primary purpose is to select the centerpieces (or counterbalance) for the cells, along with a quartz/sapphire windows selection.
5. *Solutions*: A panel whose primary purpose is to specify the solution to be used in each cell/channel.
6. *Optics*: A panel whose primary purpose is to select the optical system used to scan.
7. *Range*: A panel whose primary purpose is to select the wavelength ranges and cell radius measured.
8. *Submit*: A panel whose primary purpose is save and upload the protocol to the Optima/Instrument and database.

33.6.2 Optics Panel:

Define which optical system(s) will be used to scan each cell. The 'Optical System Scan Type' options are **UV/visible** and **Rayleigh Interference**.

Note: The instrument collects scans on a cell-by-cell basis, so the selections for a cell apply to both of its channels.



Select Optical System Panel

33.6.3 Shared Panel Functions:

A panel status box and multiple buttons are shared by all panels. The *General* tab help page has detailed explanations of these items.

Triple==========
Ranges: Define Wavelengths and Scanning Range of Cell Radius
==========

Index

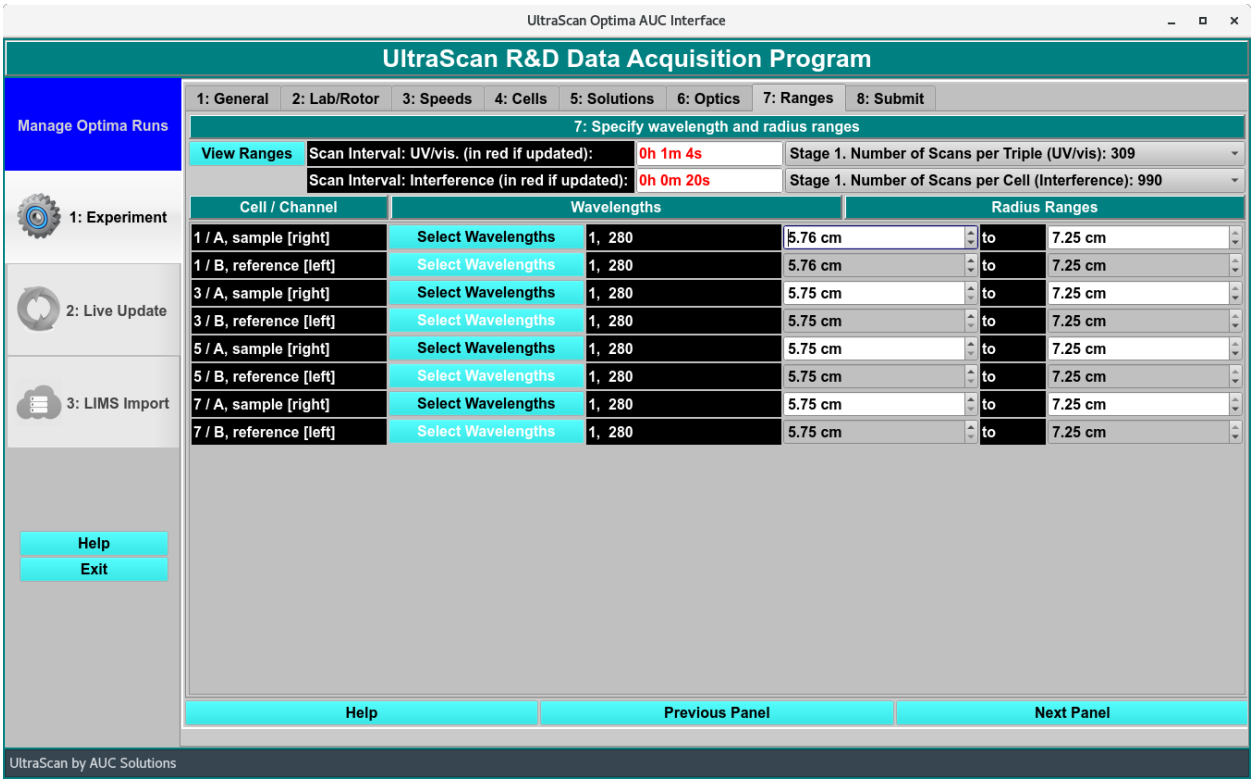
- *Panel Tab Options:*
- *Ranges Panel:*
  - *Range Functions:*
- *Shared Panel Functions:*

33.7 Panel Tab Options:

In each panel, tabs are visible at the top of the window to enable the user to move to another panel, to perform specific experiment subtasks.

1. *General*: A panel whose primary purpose is to specify the experiment run ID (a description string), select a parent project name, and possibly change data source (database or local disk) or database investigator name.
2. *Lab/Rotor*: A panel whose primary purpose is to select the Laboratory, Rotor, and Calibration values for the experiment.
3. *Speeds*: A panel whose primary primary purpose is to specify one or more speed steps. For each step, speeds and durations may be given.
4. *Cells*: A panel whose primary purpose is to select the centerpieces (or counterbalance) for the cells, along with a quartz/sapphire windows selection.
5. *Solutions*: A panel whose primary purpose is to specify the solution to be used in each cell/channel.
6. *Optics*: A panel whose primary purpose is to select the optical system used to scan.
7. *Range*: A panel whose primary purpose is to select the wavelength ranges and cell radius measured.
8. *Submit*: A panel whose primary purpose is save and upload the protocol to the Optima/Instrument and database.

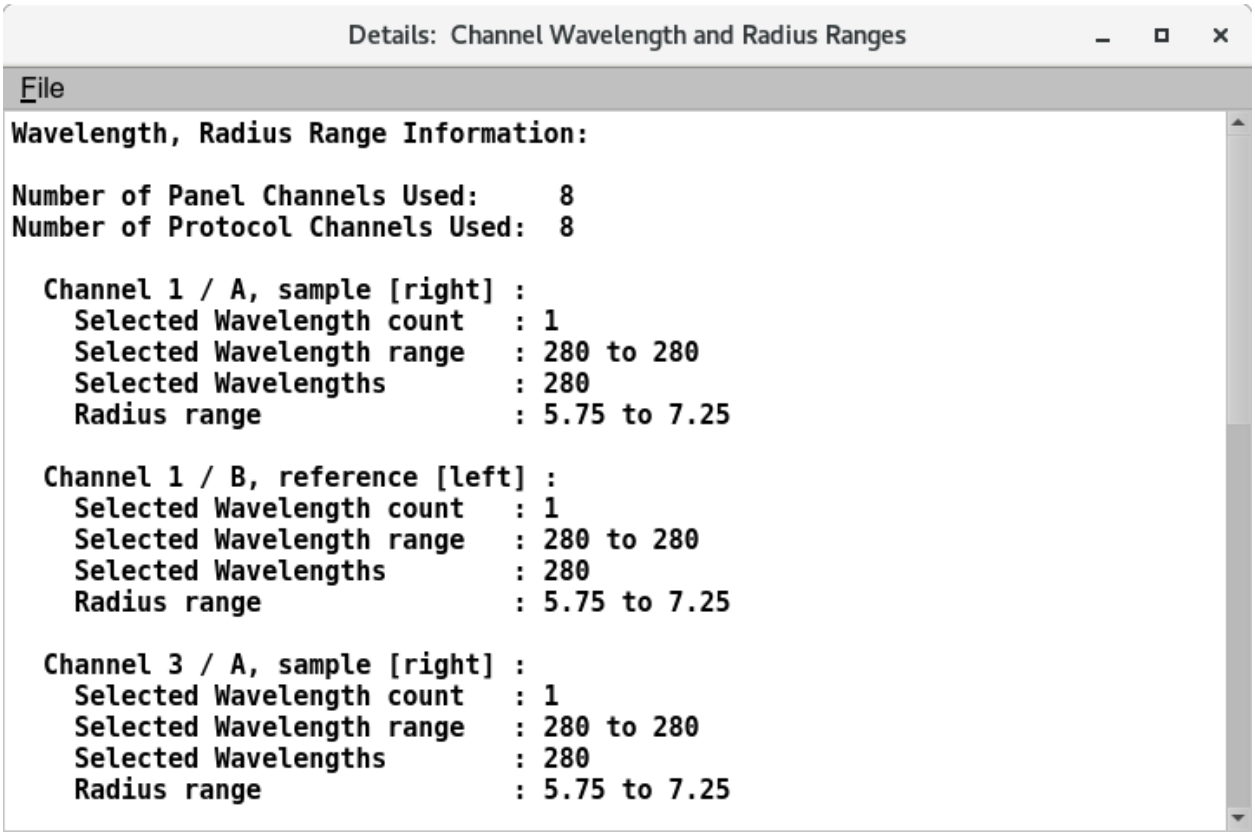
33.8 Ranges Panel:



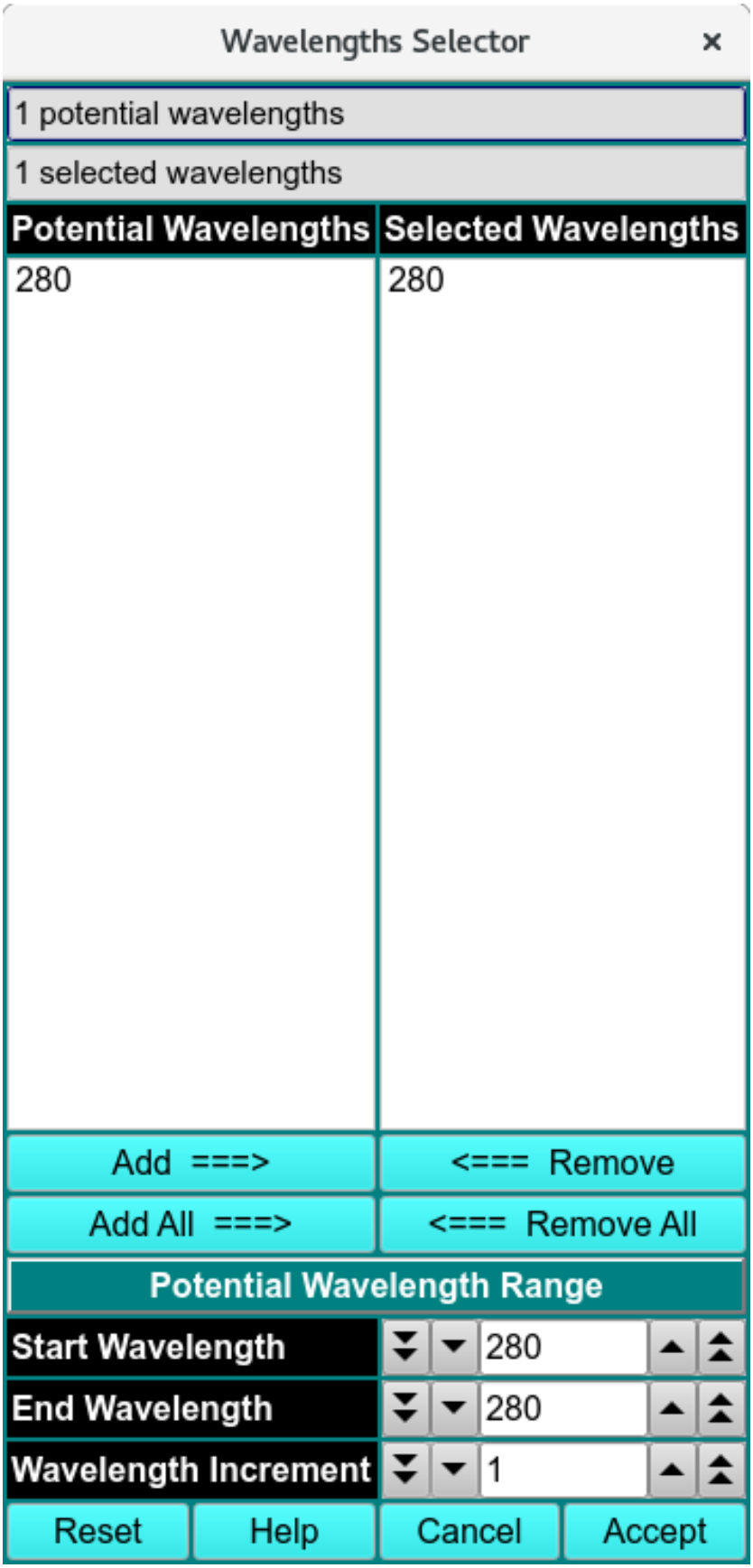
Wavelength and Radius Ranges Panel

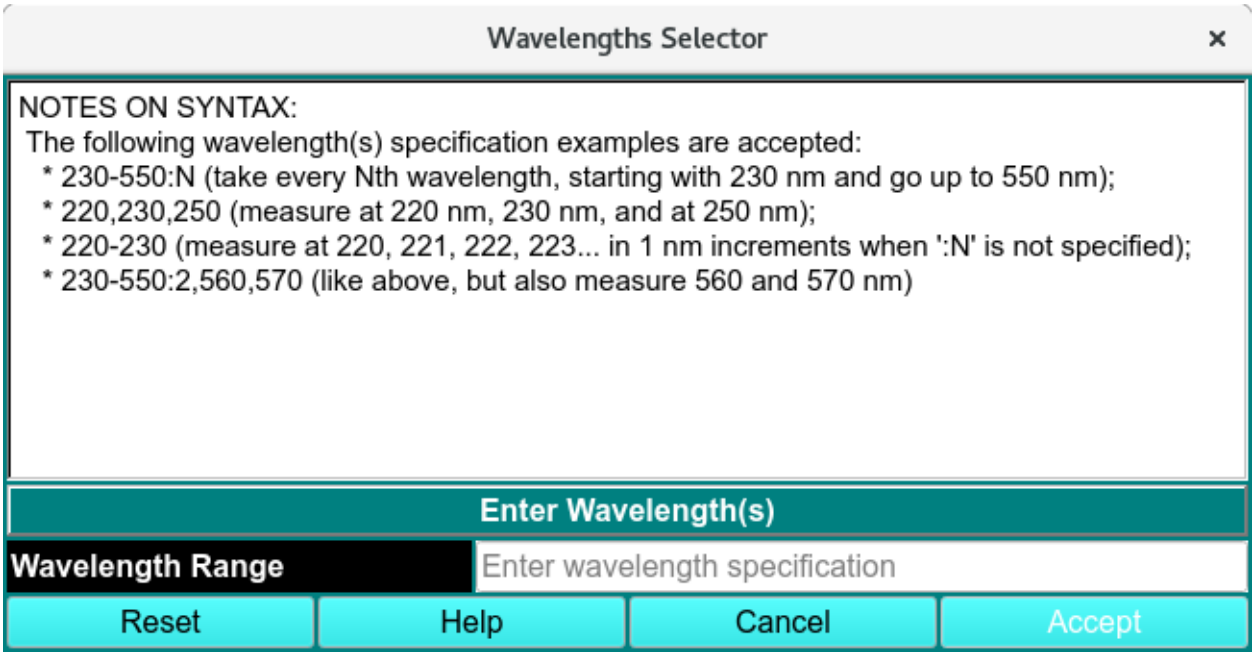
33.8.1 Range Functions:

View Ranges:	Click to see a *details.txt* file on of the cell channels, wavelength measured and radius range scanned.
Scan Interval: UV/vis.(in red if updated)	The interval time of each UV/vis scan.
Scan Interval: Interference(in red if updated)	The interval time of each interference scan.
Stage 1. Number of Scans per Triple (UV/vis):####	Number of totals UV/vis scans per Triple (cell/channel/wavelength).
Stage 1. Number of Scans per Triple (Interference):####	Number of totals Interference scans per Triple (cell/channel/wavelength).
Cell / Channel	The cell number and channel selected to be measured.
Wavelengths	The *wavelength ranges* can be picked by a selector dialog or manually dialog options.
Radius Range	For each cell, users can select the cell radius range to be measured from 5.75 cm to 7.25 cm. e



Details of Ranges Selected


Wavelengths Selector
1 potential wavelengths
1 selected wavelengths
Potential Wavelengths
Selected Wavelengths
280
280
Add ===>
<=== Remove
Add All ===>
<=== Remove All
Potential Wavelength Range
Start Wavelength
280
End Wavelength
280
Wavelength Increment
1
Reset
Help
Cancel
Accept



Wavelength Selection Options

33.9 Shared Panel Functions:

A panel status box and multiple buttons are shared by all panels. The *General* tab help page has detailed explanations of these items.

33.10 Submit: Submission Checklist and Save Protocol

Index

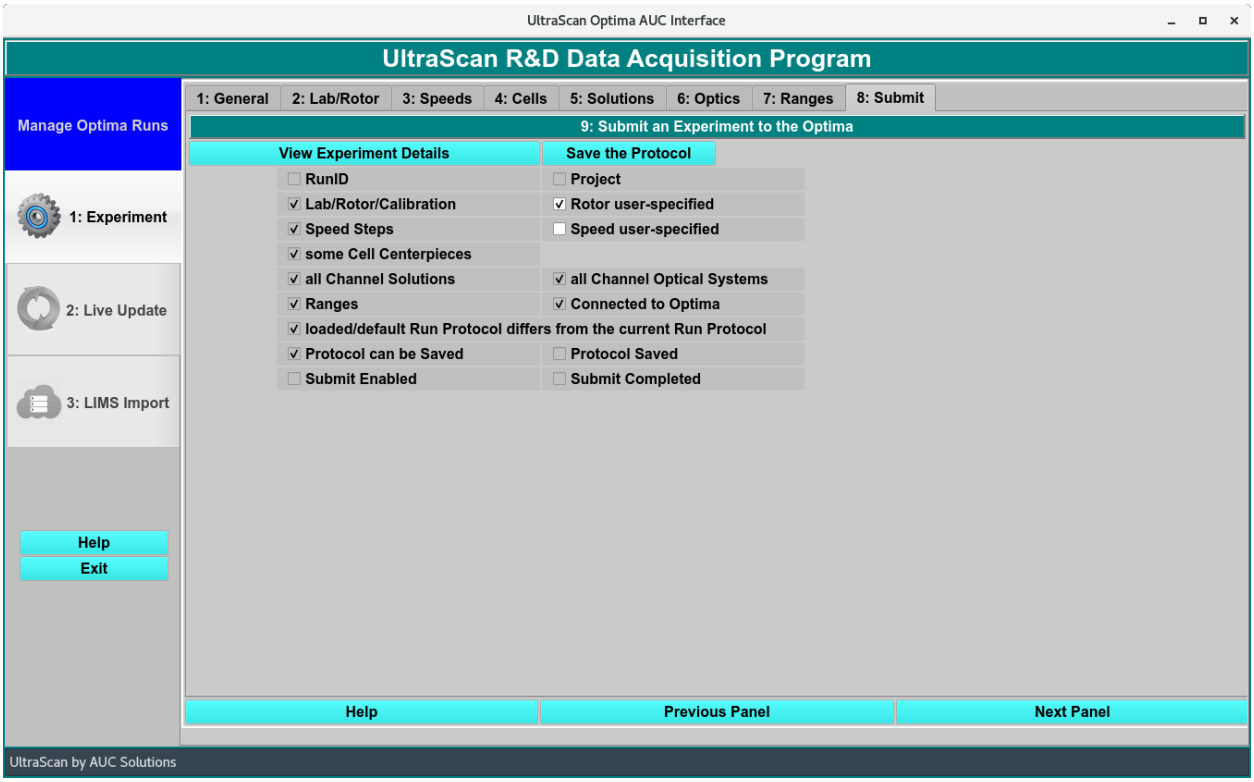
- *Panel Tab Options:*
- *Submit Panel:*
  - *Submit Functions:*
  - *Checklist to Save:*
- *Shared Panel Functions:*

33.10.1 Panel Tab Options:

In each panel, tabs are visible at the top of the window to enable the user to move to another panel, to perform specific experiment subtasks.

1. *General*: A panel whose primary purpose is to specify the experiment run ID (a description string), select a parent project name, and possibly change data source (database or local disk) or database investigator name.
2. *Lab/Rotor*: A panel whose primary purpose is to select the Laboratory, Rotor, and Calibration values for the experiment.
3. *Speeds*: A panel whose primary primary purpose is to specify one or more speed steps. For each step, speeds and durations may be given.
4. *Cells*: A panel whose primary purpose is to select the centerpieces (or counterbalance) for the cells, along with a quartz/sapphire windows selection.
5. *Solutions*: A panel whose primary purpose is to specify the solution to be used in each cell/channel.
6. *Optics*: A panel whose primary purpose is to select the optical system used to scan.
7. *Range*: A panel whose primary purpose is to select the wavelength ranges and cell radius measured.
8. *Submit*: A panel whose primary purpose is save and upload the protocol to the Optima/Instrument and database.

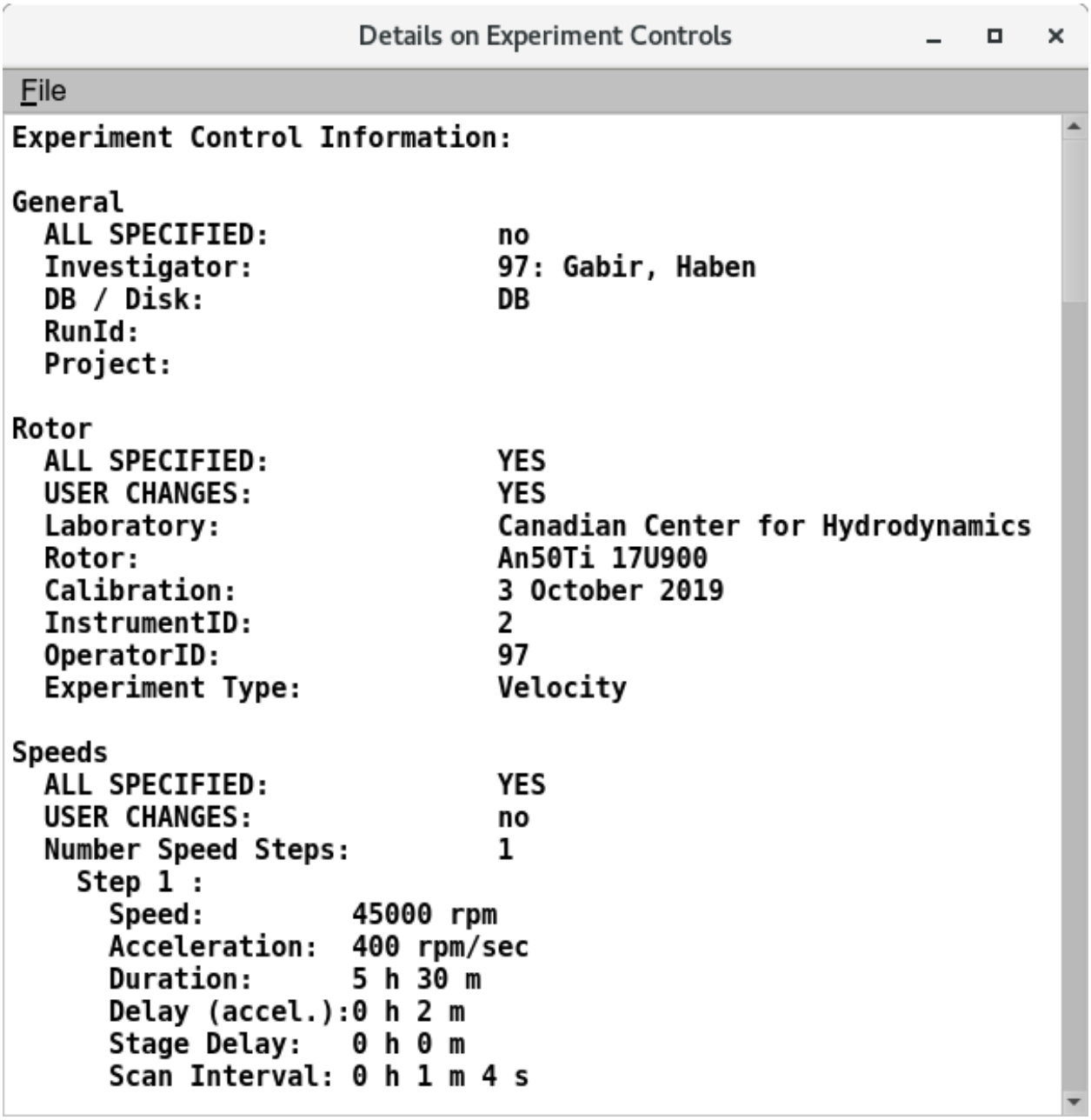
33.10.2 Submit Panel:



Submission Panel

Submit Functions:

View Experiment Details	Once all the Panels have been filled out, the details of the experiment can be viewed in a *Details* dialog.
Save the Protocol	Click to save the protocol once all the check ark boxed are checked.



Experimental Details in the Submission Panel

Checklist to Save:

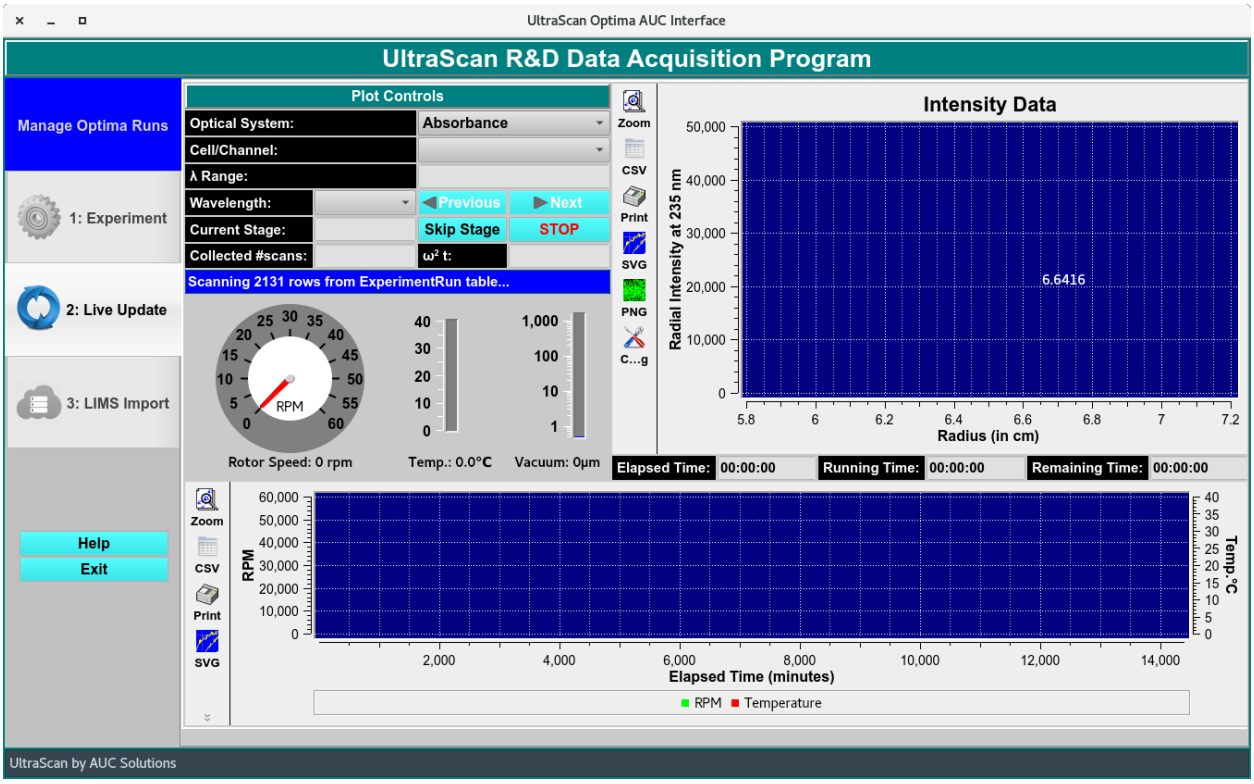
- RunID
- Project
- Lab/Rotor/Calibration
- Rotor user-specified
- Speed Steps
- Speed user-specified
- Some Cell centerpieces
- All Channel Solutions
- All Channel Optical Systems
- Ranges
- Connected to Optima
- loaded/default Run Protocol differs from the current Run Protocol
- Protocol can be Saved
- Protocol Saved
- Submit Enabled
- Submit Completed

33.10.3 Shared Panel Functions:

A panel status box and multiple buttons are shared by all panels. The *General* <*general*> tab help page has detailed explanations of these items.

>Triple========== 2: Live Update ==========

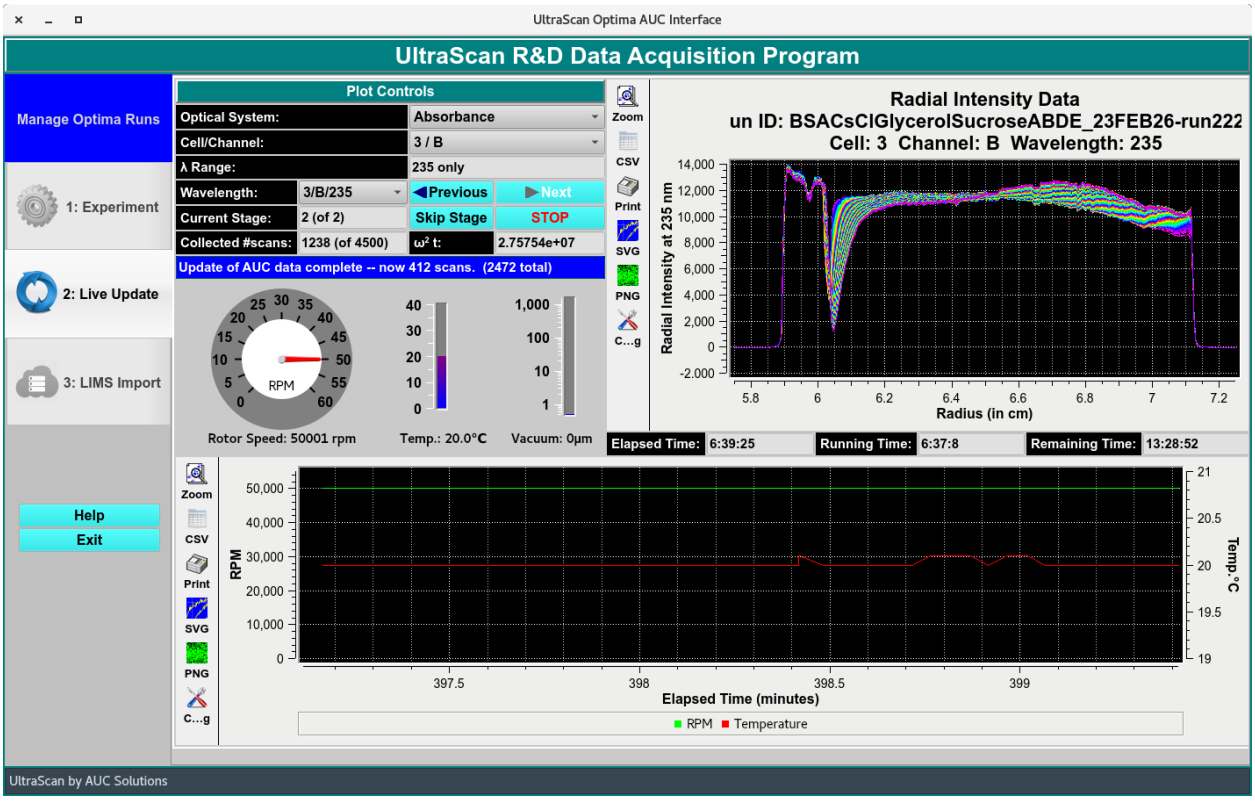
This live-update module allows users to view the on-going scan collections for protocols sent via the R&D Data Acquisition Module.



Live Update Window

1. Select the run with the RUNNING Optima Run status and in the LIVE_UPDATE stage in **Manage Optima Run** window. Click **Select Optima Run to Follow** to open the live update window.

Allow the window to update and query all scans.



Live Update Window

2. To navigate through the Triple by using the cell/channel pulldown menu, the wavelength pulldown menu which has all the Triples listed and the previous and next bottoms next to the the Wavelength list.

Note: The **Skip Stage** bottom allows user to skip any of the pre-defined stage (for example: temperature delay) and the **STOP** bottom allows user to stop run. Usually this is to stop run when a lack(s) is notices. It is not recommended to stop a run before the pre-defined run time has finished. Use the ASTFEM simulation module to estimate appropriate runtime for experiment).

---

3. **Collected scan:** - The number of scan collected in total.
4. **Blue text box** - updates status of collection stage.
5. **Plots**

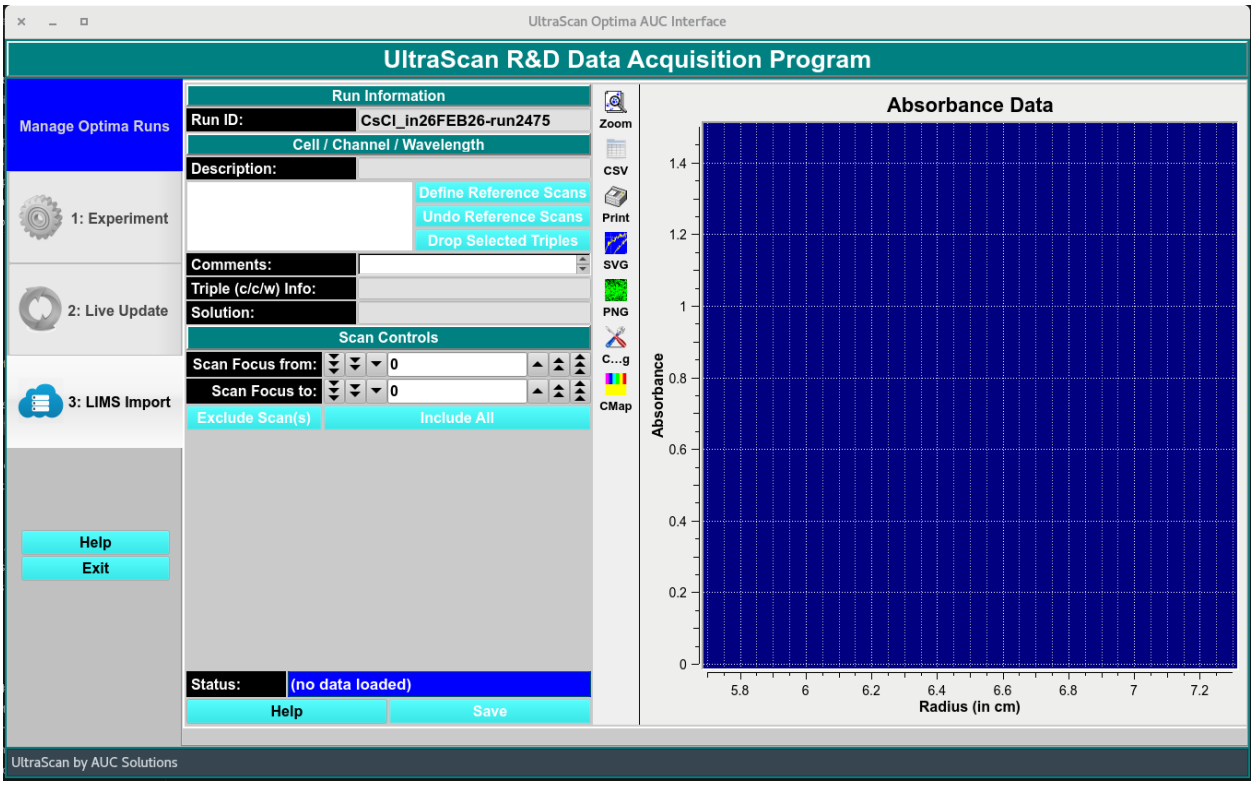
- Top Plot: Displays the scans collected.
- Bottom Plot: Displays the speed (rpm), time, temperature (°C) and ω^2T measured for the duration of run.

33.11 3: LIMS Import

Index

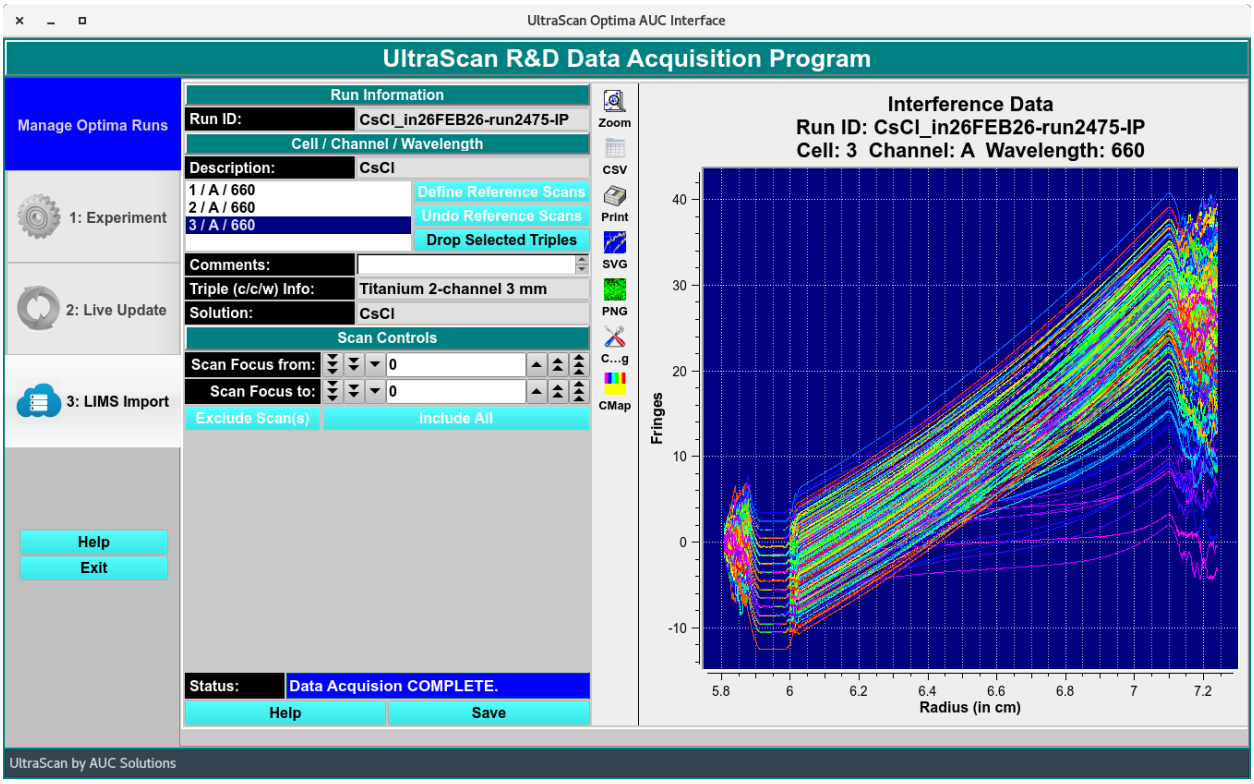
- *LIMS Import Functions:*

After a run is completed, the data is imported by first defining the reference scans, allowing users to drop any invalid Triples; or scans and the resulting edit profile is then imported into the database by clicking **Save**.

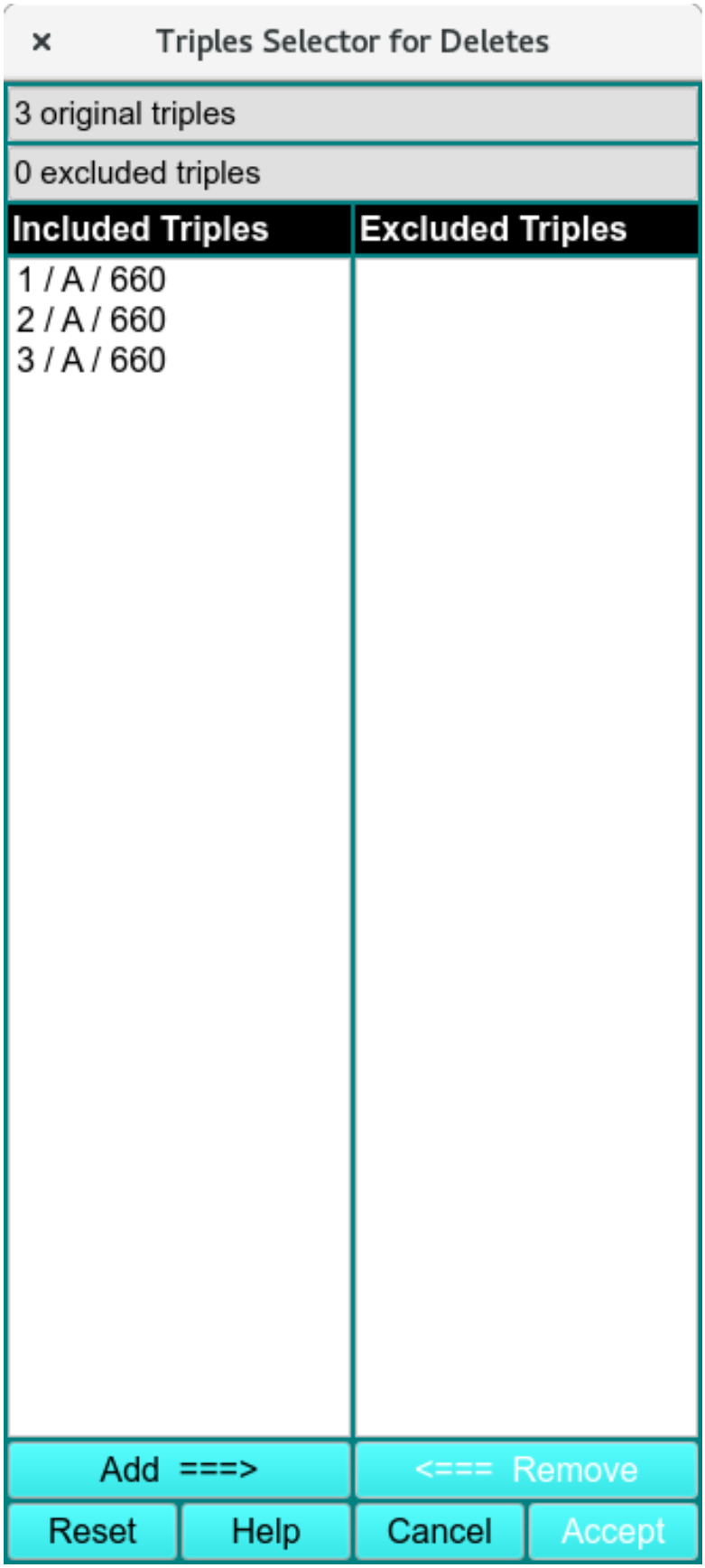


Main Import Window in the Data Acquisition Package

Once Run is completed, define reference scans, drop Triples, and the edit profile is imported to the database.



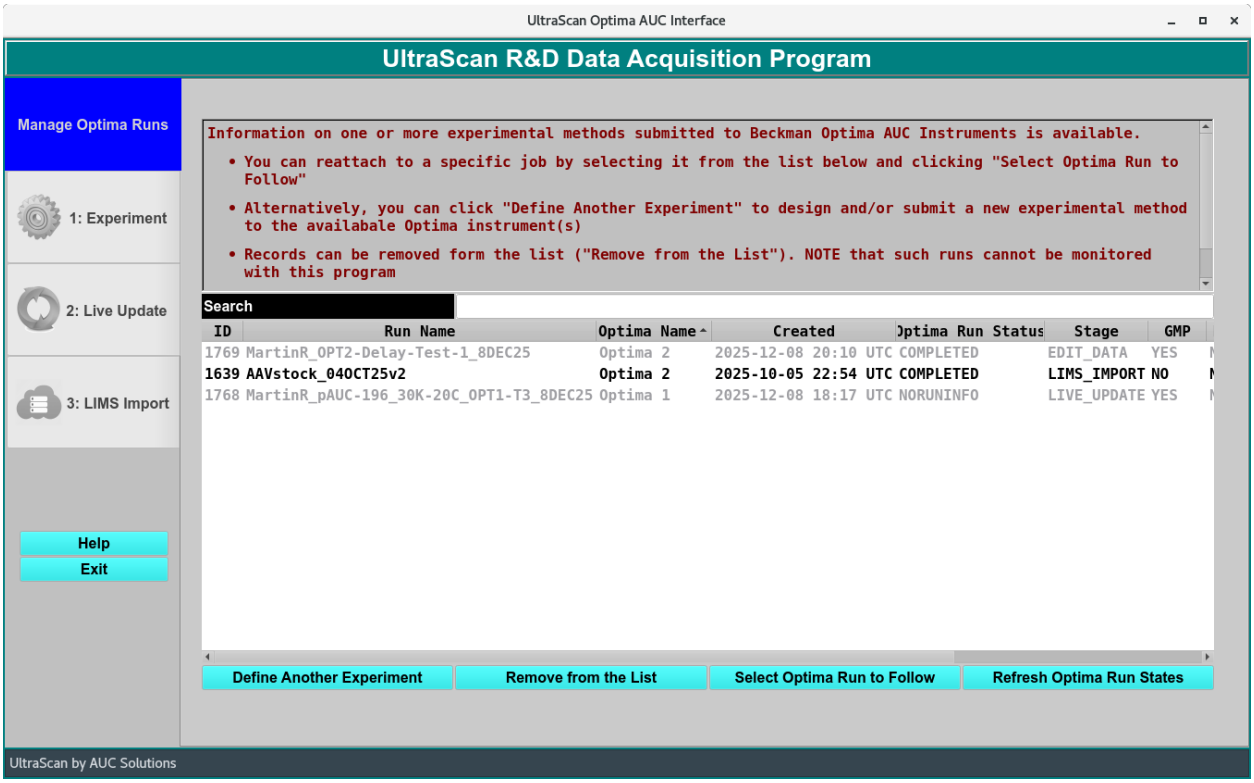
Imported data - Import Window


Triples Selector for Deletes
3 original triples
0 excluded triples
Included Triples
Excluded Triples
1 / A / 660
2 / A / 660
3 / A / 660
Add ===>
<=== Remove
Reset
Help
Cancel
Accept

33.11.1 LIMS Import Functions:

Run ID:	Run identification label selected by user.
Description:	Solution name
Define Reference Scans	Click bottom to accept the reference selection automatically generated.
Undo Reference Scans	Undo reference selection automatically made and make manual selection.
Drop Selected Triples	Delete selected Triple when *Triples Selector for Deletes* window appears.
Comments:	Add comments to sample solution label.
Triple (c/c/w) Info:	Cell Centerpiece.
Solution:	Solution added in the :doc:`Cells <cells> Experiment tab.
Scan Focus from:	Select Scan start range to exclude.
Scan Focus to:	Select Scan end range to exclude.
Exclude Scan(s)	Exclude selected scans.
Include All	Include all scans that were excluded.
Status:	Data and profile status update.
Help	Link to the help documentation of this Import Module.
Save	Save current profiles.

33.12 Data Acquisition



Submission Panel

CHAPTER

THIRTYFOUR

MULTI-WAVELENGTH DATA VIEWING, ANALYSIS AND REPORTING

34.1 Multi-Wavelength Raw Data Viewer

Index

- *MWL Viewer Process:*
- *MWL Viewer Functions:*

This module presents options for viewing multi-wavelength data for purposes of quality control. The user may, in this way, identify any problems in the experiment and get a sense of the wavelength, radial, and intensity ranges represented. Two types of input may be loaded for viewing.

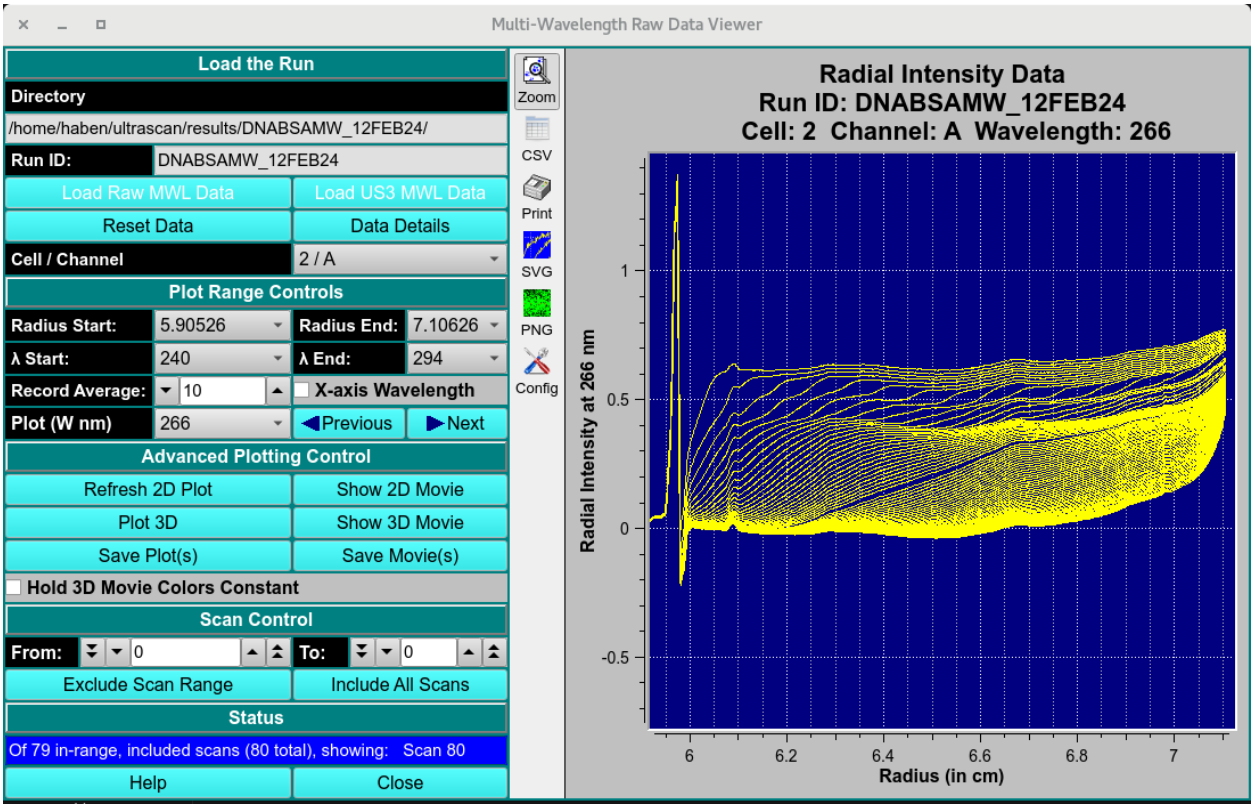
- **Raw MWL:** The unprocessed .mwrs files of a run.
- **US3 MWL:** The already imported US3 .auc files for an experiment.

34.1.1 MWL Viewer Process:

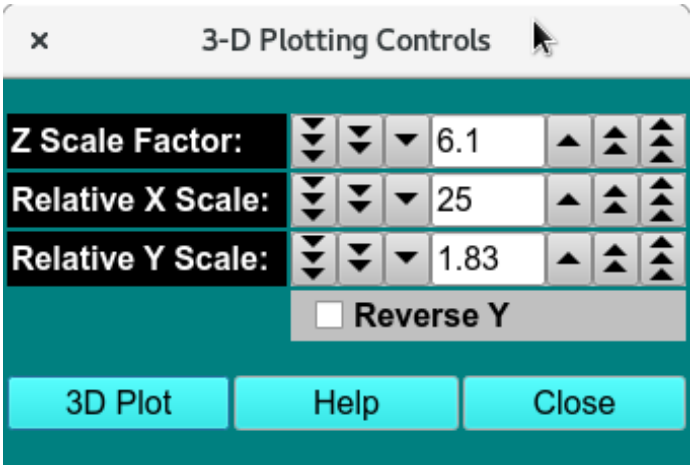
- **Load MWL Data:** Load multi-wavelength data to view by clicking on **Load Raw MWL Data** or **Load US3 MWL Data**, depending on the available or desired data type.
- **Define 2-D Plot and Ranges:** Set radius and lambda ranges; determine averaging count; choose the X-axis (radius or lambda); and possibly use the plot window Config option to refine the resulting 2-dimensional plot.
- **Show a 2-D Movie:** Once a 2-D plot and its ranges have been established, show a 2-D movie of all lambda/radius records to view the progression of data through record ranges.
- **Show and Refine 3-D Plot:** Click on **Plot 3D** to bring up a control window. Within that dialog, choose scales and bring up a 3-D plot window. Continue to refine scales and orientation until the 3-D plot is as desired.
- **Show a 3-D Movie:** Once a 3-D plot, with its scales and orientation, has been established, show a 3-D movie to see the change in data over time (scans). Note that scans may be excluded and a range of scans chosen to refine this movie.

- **Save Plots and Movies:** When plots and movies are all in a desired and informative state, you may save the plots and save frame files from the movies for processing by external imaging software.

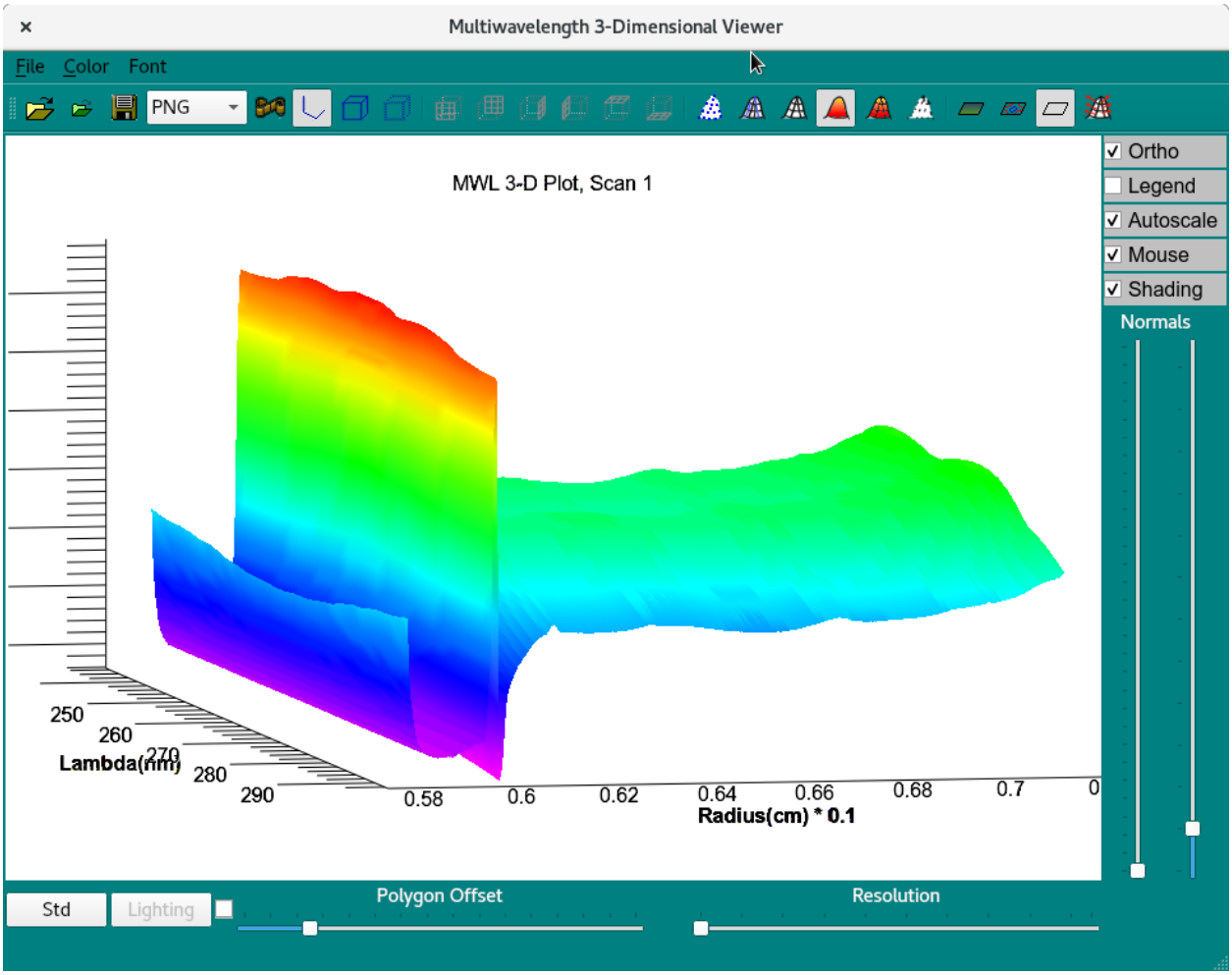
The main MWLR_Viewer window is shown here in the state it appears after loading US3 MWL data.



Main Window



3D-Display Control

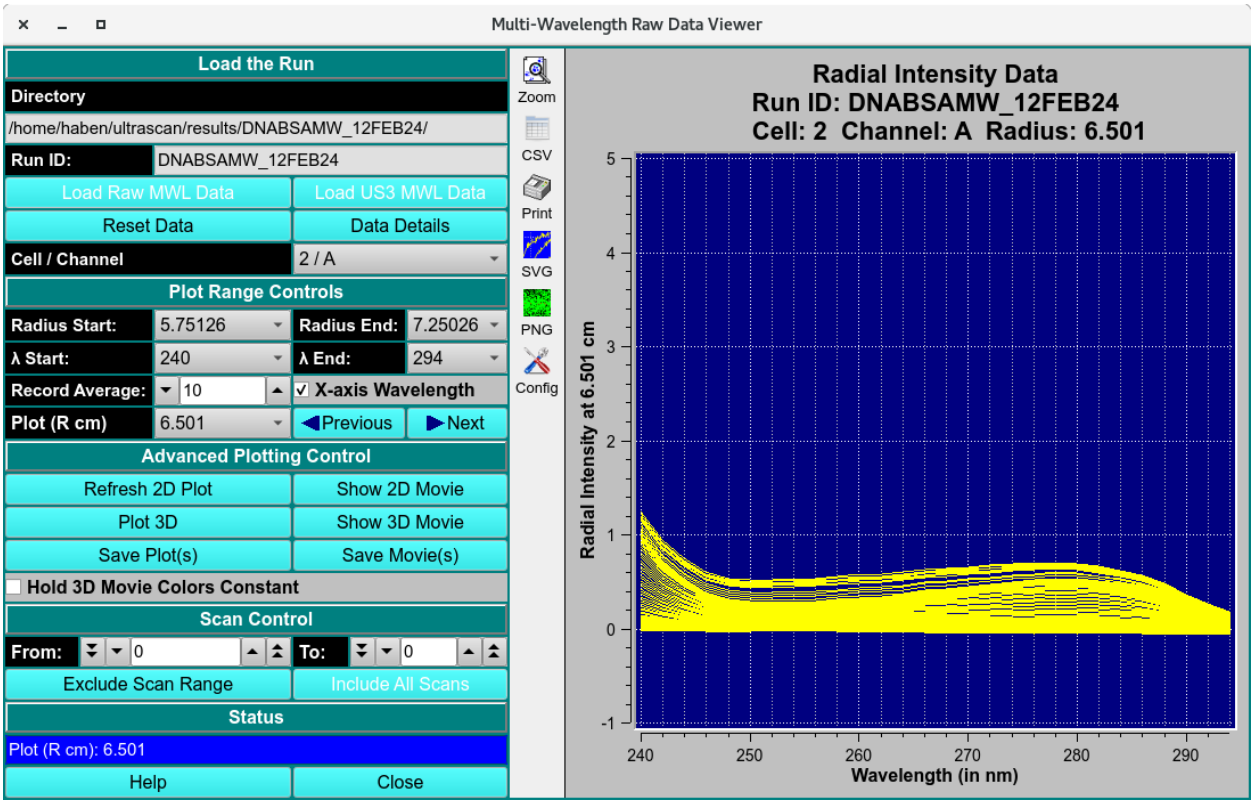

Multiwavelength 3-Dimensional Viewer
File Color Font
PNG
MWL 3-D Plot, Scan 1
Ortho
Legend
Autoscale
Mouse
Shading
Normals
250
260
270
280
290
Lambda(nm)
0.58
0.6
0.62
0.64
0.66
0.68
0.7
Radius(cm) * 0.1
Std
Lighting
Polygon Offset
Resolution

3D Viewer

34.1.2 MWL Viewer Functions:

Directory:	A text box to the right of this label shows the full path to currently loaded data.
Run ID:	A text box to the right of this label shows the currently loaded data's run identifier
Load Raw MWL Data:	Click here and, in the resulting **Load Data Dialog**, select a run from the **imports** directory that contains a set of **.mwrs** files to load.
Load US3 MWL Data:	Click here and, in the resulting **Load Data Dialog**, select a run from the **results** directory that contains a set of already imported **.auc** files to load.
Reset Data	Clear data and plots in preparation for loading a new data set.
Data Details	Click on this button to bring up a text dialog that shows details about the currently loaded data.
Cell / Channel	Select the Cell/Channel of data to view.
Radius Start:	Select the start radius in cm. from the list of available radius values.
Radius End:	Select the end radius in cm. from the list of available radius values.
λ Start:	Select the start lambda in nm. from the list of available wavelength values.
λ End:	Select the end lambda in nm. from the list of available wavelength values.
Record Average:	Select the number of wavelength records over which to compute an average for each record.
X-axis Wavelength	Check this box to change the plot to one in which the X-axis is wavelength and records are radius records.
Plot (W nm) or **Plot (R cm)**	Select the identifier of the lambda or radius record to plot.
Previous	Click to plot the previous lambda/radius record.
Next	Click to plot the next lambda/radius record.
Refresh 2D Plot	Click this button to replot the 2-D plot after any changes in controls or ranges.
Show 2D Movie	Click here to begin a 2-D movie over the lambda/radius record range.
Plot 3D	Click to open a **3D Plot Control Dialog**, from which a **3D Plot Window** may be opened or replotted. Note that if a 3D Plot Window is already opened and scan ranges have been changed, this button can be clicked to re-plot the 3D Plot for the new first scan.
Show 3D Movie	Click here to begin a 3-D movie over the scan range.
Save Plot(s)	Click this button to save the current 2-D and/or 3-D plot.
Save Movie(s)	Click here to re-play 2-D and/or 3-D movies and save the PNG files of each frame in the reports directory.
Hold 3D Movie Colors Constant	Check this box before beginning a 3-D movie in order to preserve the correspondence of colors in the currently selected color map to the maximum amplitude range. By default, colors are mapped to each frame's amplitude range.
From:	Select the first scan of a scan range focus.
To:	Select the last scan of a scan range focus. The **from** and **to** values may be used to exclude bad scan curves or to limit the range of a 3-D plot or movie.
Exclude Scan Range	Click on this button to exclude the currently selected range of scans.
Include All Scans	Click this button to revert to the full range of data scans.
Status	The text box below this banner shows the current status of the ongoing or just-completed process.
Help	Click to show this help documentation.
Close	Click to close the MWLR viewer

The default 2-D plot and movie frames is of wavelength records with radius as the X axis. You may change to radius records with lambda as the X axis by checking the X-axis Wavelength box. With this checked, the main window would be similar to the following.



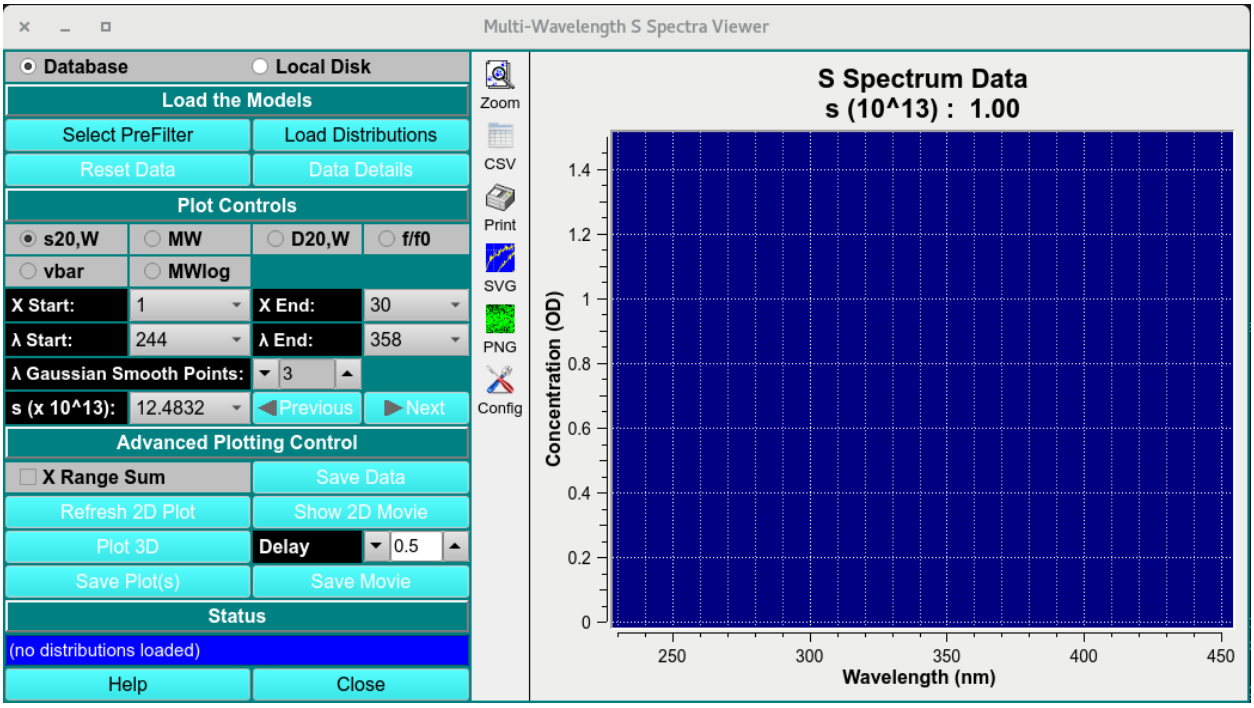
Alternate Record and X-axis

34.2 Multi-Wavelength S Spectra Viewer

Index

- *Process:*
- *MWL Spectra Viewer Functions:*
  - *MWL Spectra Viewer Plot controls*
  - *MWL Spectra Viewer Advanced Controls:*

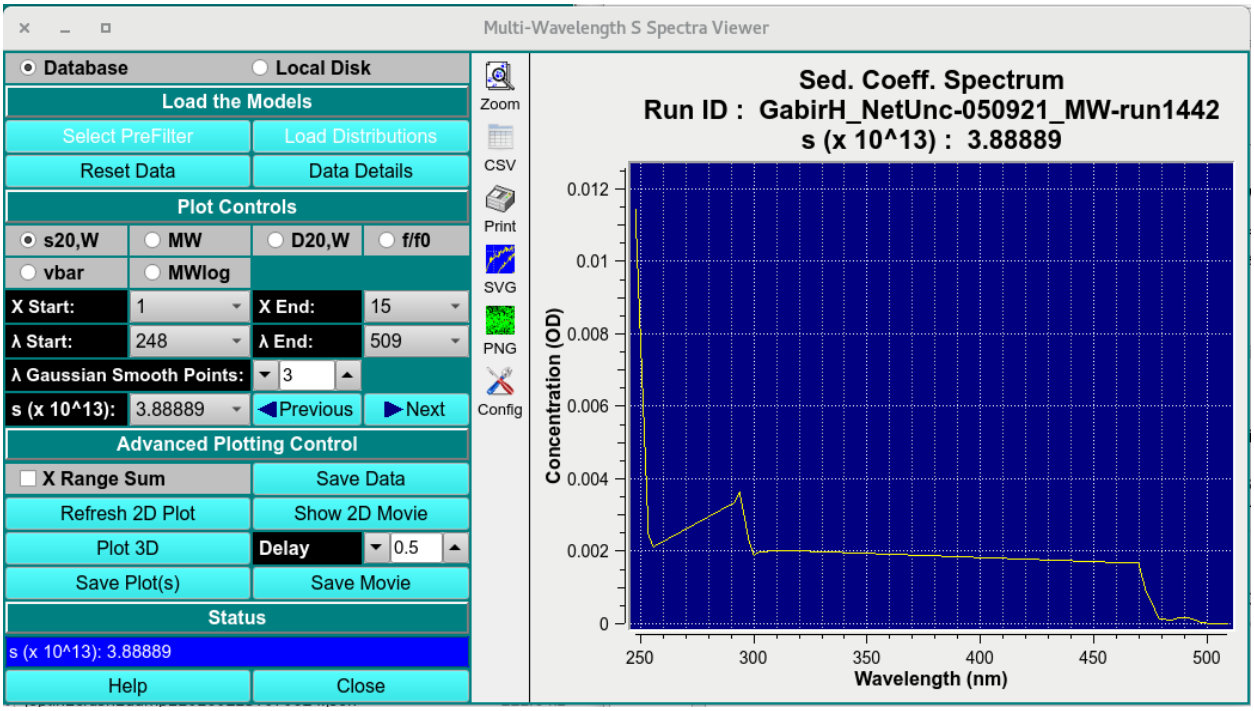
This module allows the user to view the multi-wavelength 2DSA models in a 2-D or 3-D plots.



Main Window

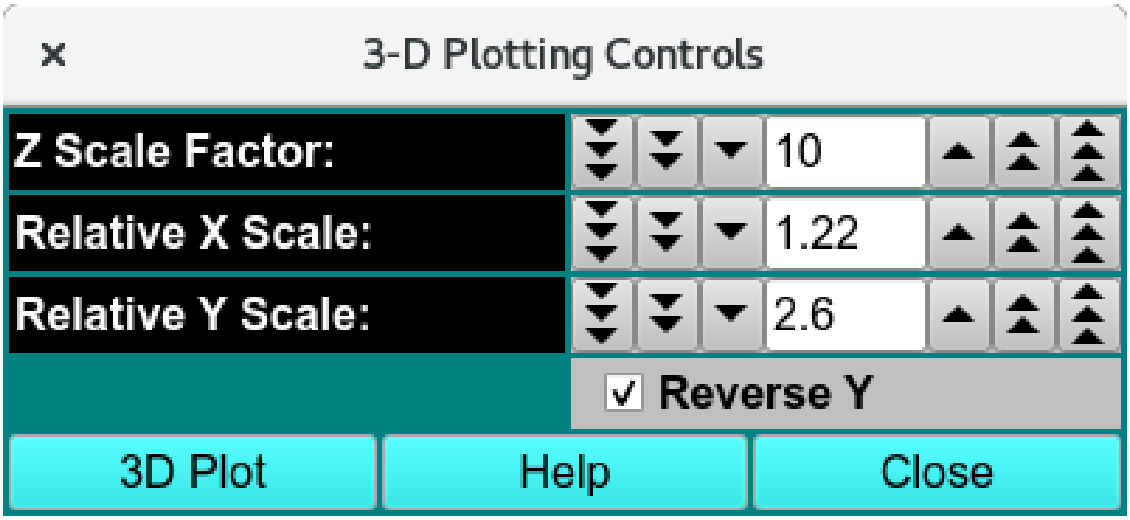
34.2.1 Process:

- **Load MWL 2DSA-IT:** Load multi-wavelength 2DSA-IT models to view by clicking on **Select Pre-Filter** then **Load Distribution** and select the 2DSA-IT models.

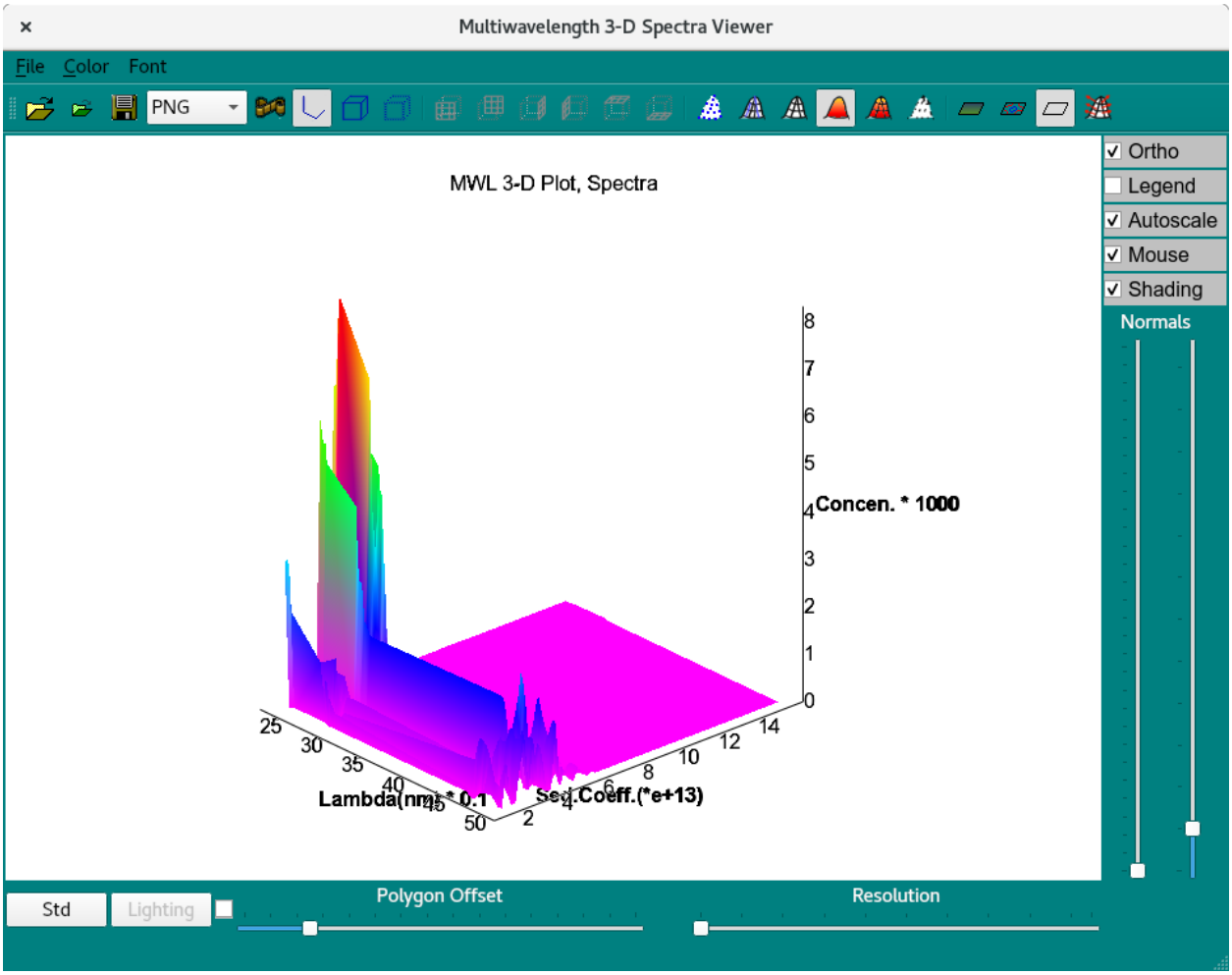


Loaded Window

1. **Plot Controls:** Set Z axis (S20,W, MW, D20,W, f/f0, vbar or MWlog) and range start (X start) and end (X End); set lambda start and end.
2. **Advancing Plotting Control** Once a 2-D plot and its ranges have been established, show a 2-D movie of all lambda/radius records to view the progression of data through record ranges.
3. Click on **Plot 3D** to bring up a control window. Within that dialog, choose scales and bring up a 3-D plot window. Continue to refine scales and orientation until the 3-D plot is as desired.
4. **Show a 3-D Movie:** Once a 3-D plot, with its scales and orientation, has been established, show a 3-D movie to see the change in data over time (scans). Note that scans may be excluded and a range of scans chosen to refine this movie.

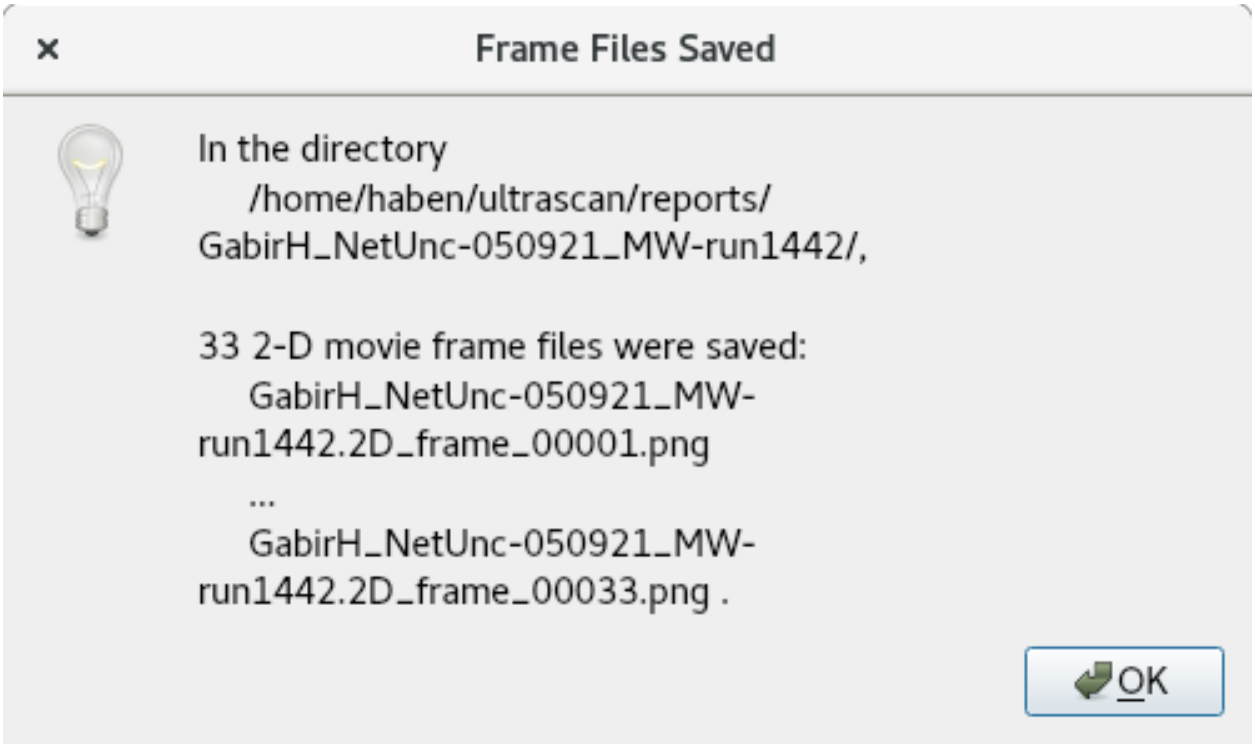


3D-Display Control



3D Viewer

- **Save Plots and Movies:** When plots and movies are all in a desired and informative state, you may save the plots and save frame files from the movies for processing by external imaging software.



save movie as png files

34.2.2 MWL Spectra Viewer Functions:

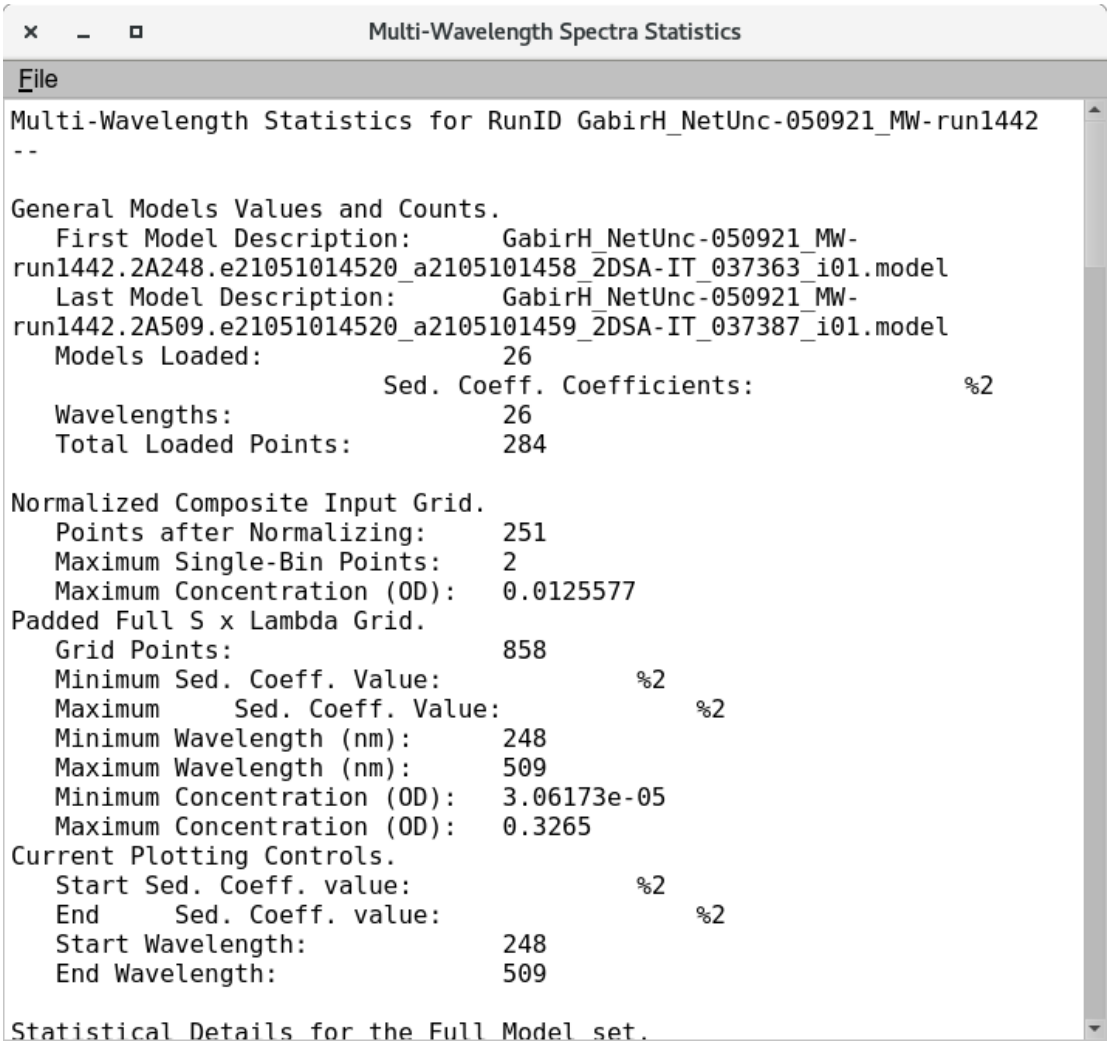
Select PreFilter	Load a run dataset as a Pre-filter using the "Select Run(s) as Models Pre-Filter (DB)".
Load Distribution	Select the distributions to plot on in the "../Load Distribution Model(s).
Reset Data	Reset loaded data.
Data Details	Open a detailed statistical:ref:*report <stats>* on selected dataset.

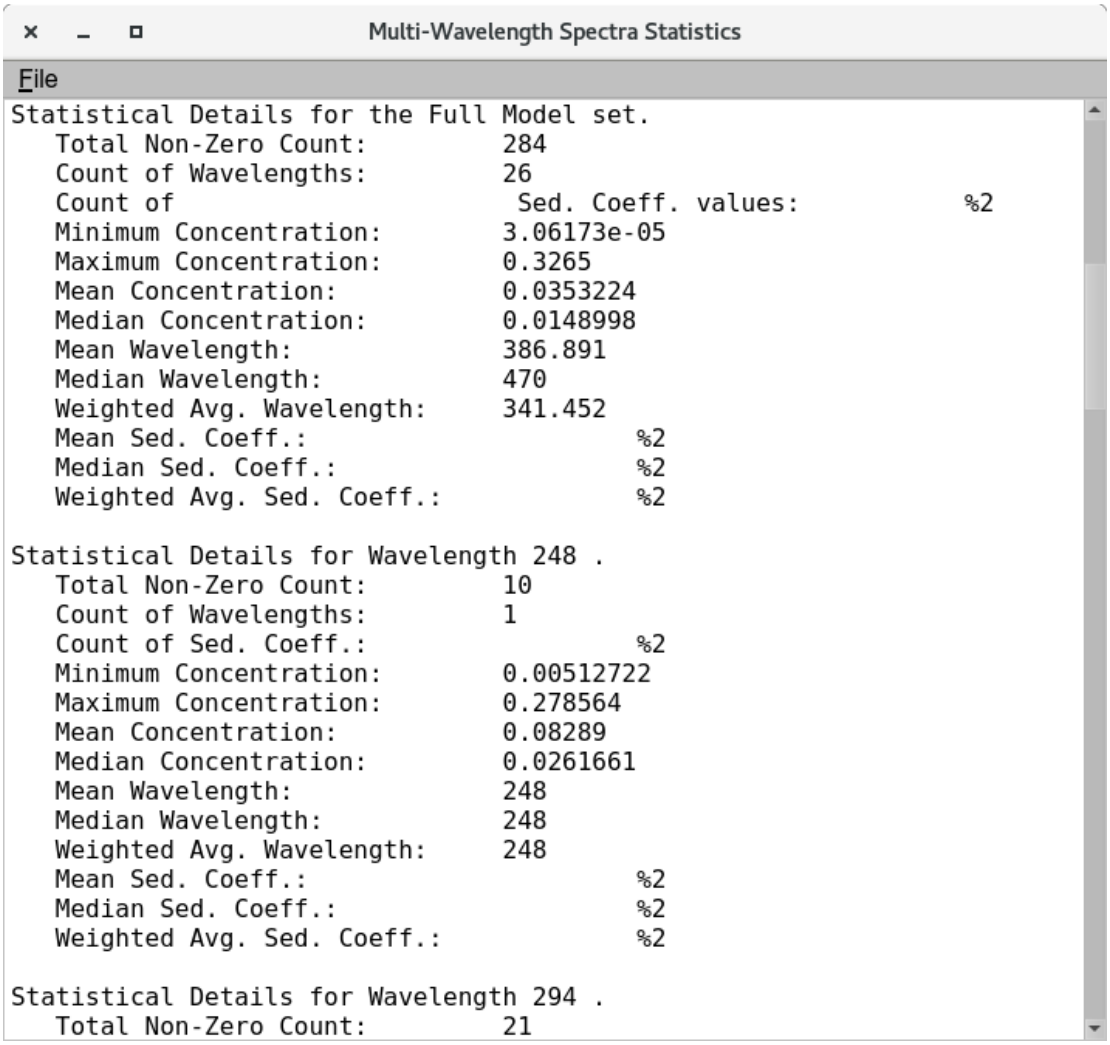
MWL Spectra Viewer Plot controls

X start and end	Start and end range of selected Z axis attribute.
λ start and end	Start and end range of wavelength measured.
λ gaussian smooth points:	Gaussian smoothing function applied to each spectra at any given Z value.
S (x10^13):	View the wavelength attribute distribution for any S point (Z-axis).

MWL Spectra Viewer Advanced Controls:

Save data	Save the 2-D plot data.
Refresh 2D plot	Refresh plot.
Show 2D movie	Show spectra of each S value.
Plot 3D	Plot the 3-D graph with X as wavelength, Z as user selected attribute, and Y as signal intensity.
Delay	Frame delay between frames.
Save plots	Save plots to $HOME/ultrascan/reports.
Save movie	Save movie to $HOME/ultrascan/reports.
Status box	Update status box.
Help	Help documentation to this module.
Close	close window.





Statistical Details of loaded multi-wavelength dataset

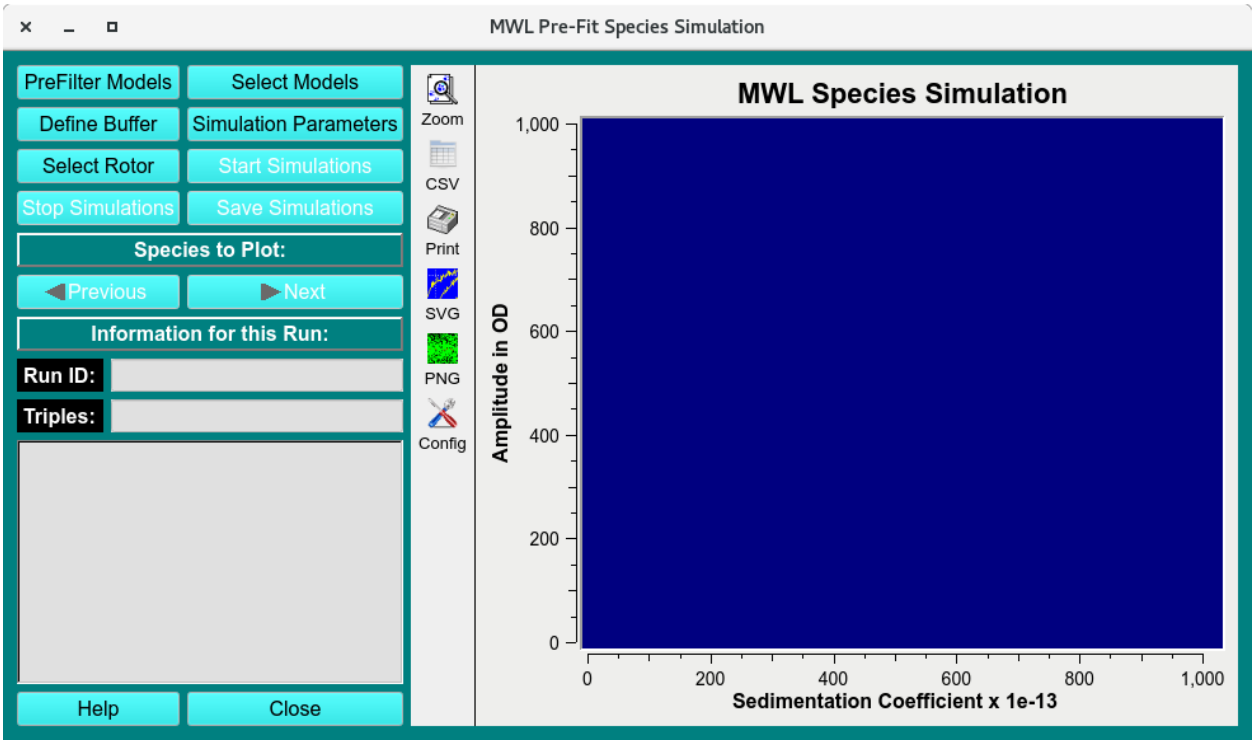
Triple========== MWL Pre-Fit Species Simulation ==========

Index

- *MWL Pre-Fit Species Process:*
- *MWL Pre-Fit Species Functions:*

Generate simulation based on mw- runs

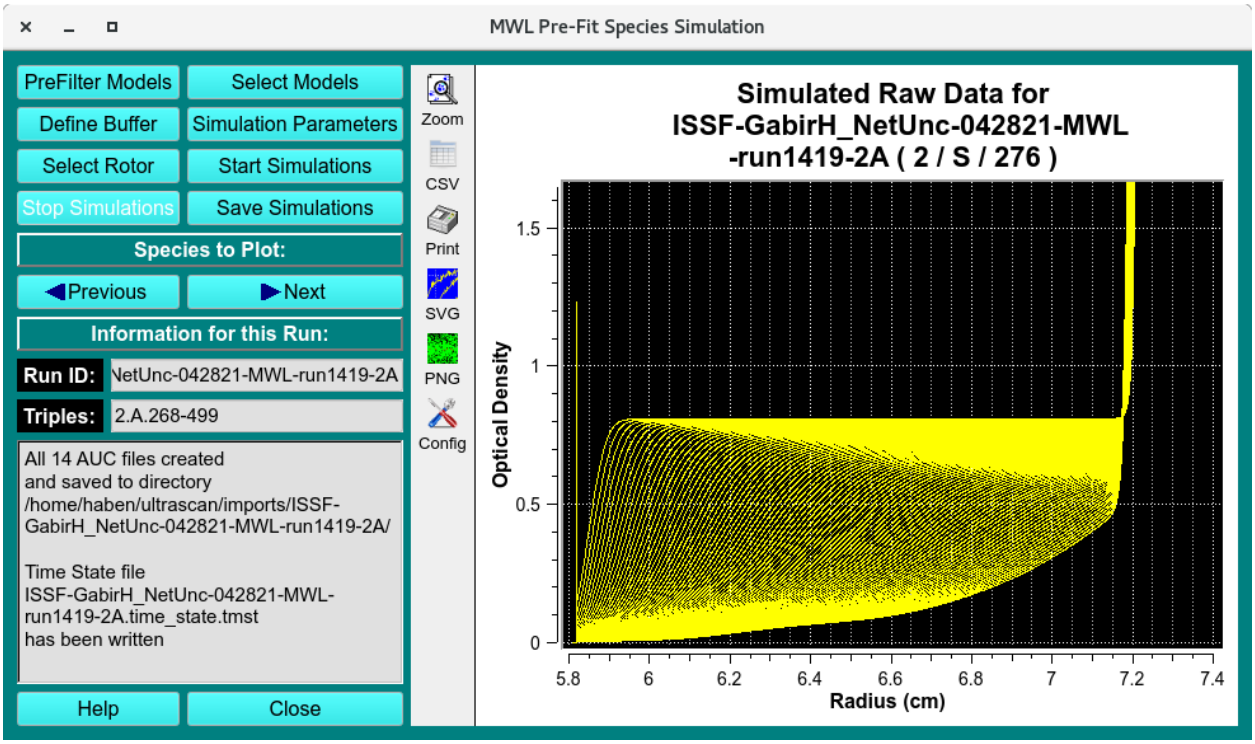
This module is used to simulate a multi-wavelength experiment using Finite Element (ASTFEM) or Finite Volume (ASTFVM) methods. Multi-Wavelength 2DSA-IT models are loaded from the database. Simulation parameters are specified, the same as the experimental parameters. A simulation is then calculated and displayed. The simulation may be saved as a synthetic version of raw experimental data with a **ISSF-** prefix added. This module generates simulations with synchronous time grid based on the 2DSA-IT models loaded.



MWL Pre-Fit Species Simulation main window

34.3 MWL Pre-Fit Species Process:

1. **PreFilter Models:** Select the pre-filter to find the distribution models.
2. **Select Model:** Load the 2DSA-IT models on which the simulation will be based.
3. **Define Buffer:** Optionally (and rarely), you may wish to specify buffer conditions. This may be necessary if your intention is to simulate and save an artificial version of a specific experimental data set. Normally, the buffer conditions are those of water at 20 degrees Centigrade.
4. **Simulation Parameters:** Secondly, open a dialog to specify parameters governing the ../simulation.
5. **Select Rotor:** Load the rotor and rotor calibration used in the pre-filter models and select the simulation rotor.
6. **Start Simulation:** Initiate calculation of the specified simulation. (If an error has occurred, users can stop the simulation by clicking **Stop Simulation**)
7. **Save Simulations** After simulation, users can save the resulting **ISSF** auc file to disk for later import.



Simulated Dataset

34.4 MWL Pre-Fit Species Functions:

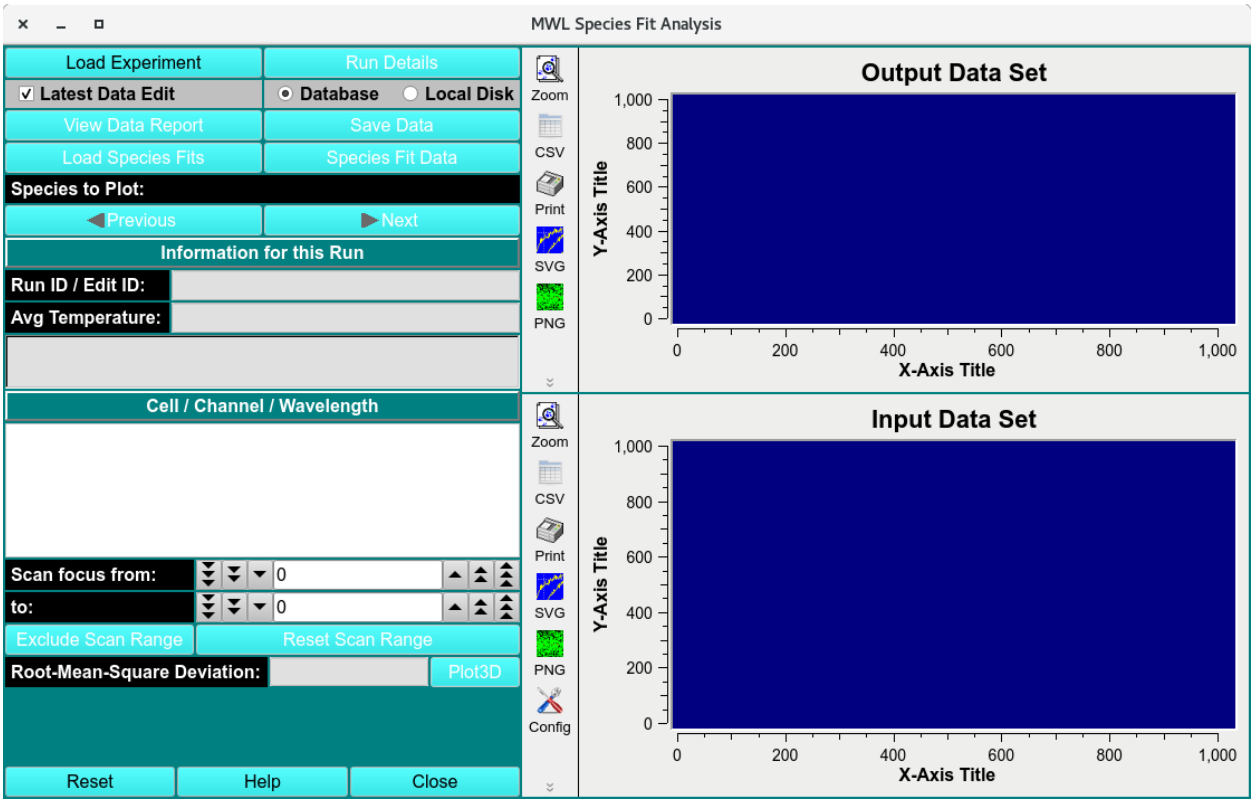
PreFilter Models	Load a run dataset as a Pre-filter using the "Select Run(s) as Models Pre-Filter (DB)".
Select Models	Select the distributions to base simulation on in the "**:doc:`Load Distribution Model(s) <../common_dialogs.html#load-distribution-model>`_**. Click and drag to select multiples model distributions. Click on **Select Models** first to select models across runs.
Define Buffer	Open the *Buffer Management* dialog to select the buffer the simulations will use for density and viscosity values.
Simulation Parameters	Click to set the *Simulation Parameters* profile.
Select Rotor	Click to open the *Rotor Management* dialog. Set the Rotor type and the associated calibration profiles for the simulation.
Start Simulations	Once all parameters have been entered, click to start simulation.
Stop Simulations	Click to stop simulation.
Save Simulations	Save simulation once generated. Simulations are saved in $HOME/ultrascan/imports/
Run ID:	The Run name automatically generated for simulations. Simulations will have the "ISS-" prefix.
Triple:	The Triple initially loaded using **Select Models**.
Help	The Help documentation of this "MWL Pre-Fit Species Simulation" Module.
Close	Close window.

34.5 MWL Species Fit Analysis

Index

- *MWL Species Fit Process:*
- *MWL Species Fit Functions:*

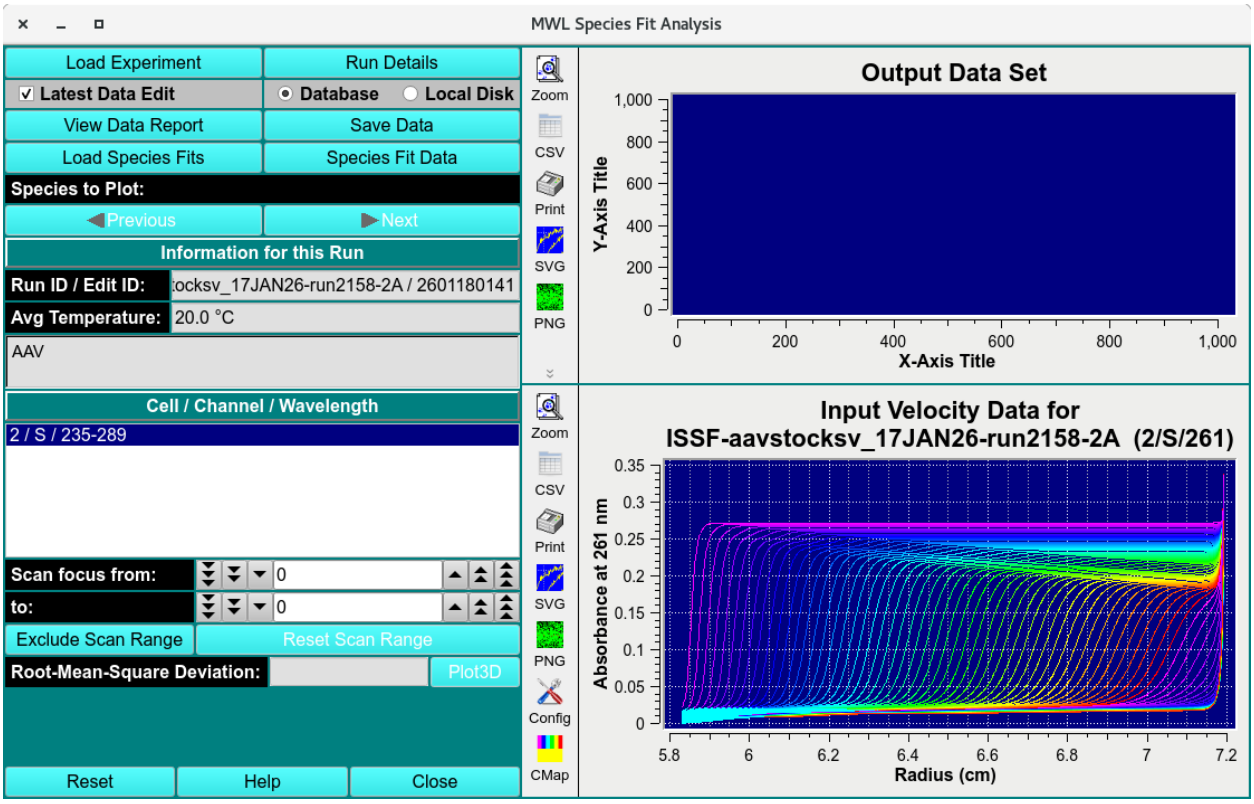
This module inputs the time-synchronized 2DSA-IT–based **ISSF** simulations and performs spectral deconvolution. For each spectral interval the measured absorbance is separated by the analytes' extinction-coefficient spectra into **SSF-ISSF-** datasets. The **SSF-ISSF-** outputs are solute-resolved, spectrum-separated datasets generated for every solute



Multi-Wavelength Species Fit Analysis Main Window

34.5.1 MWL Species Fit Process:

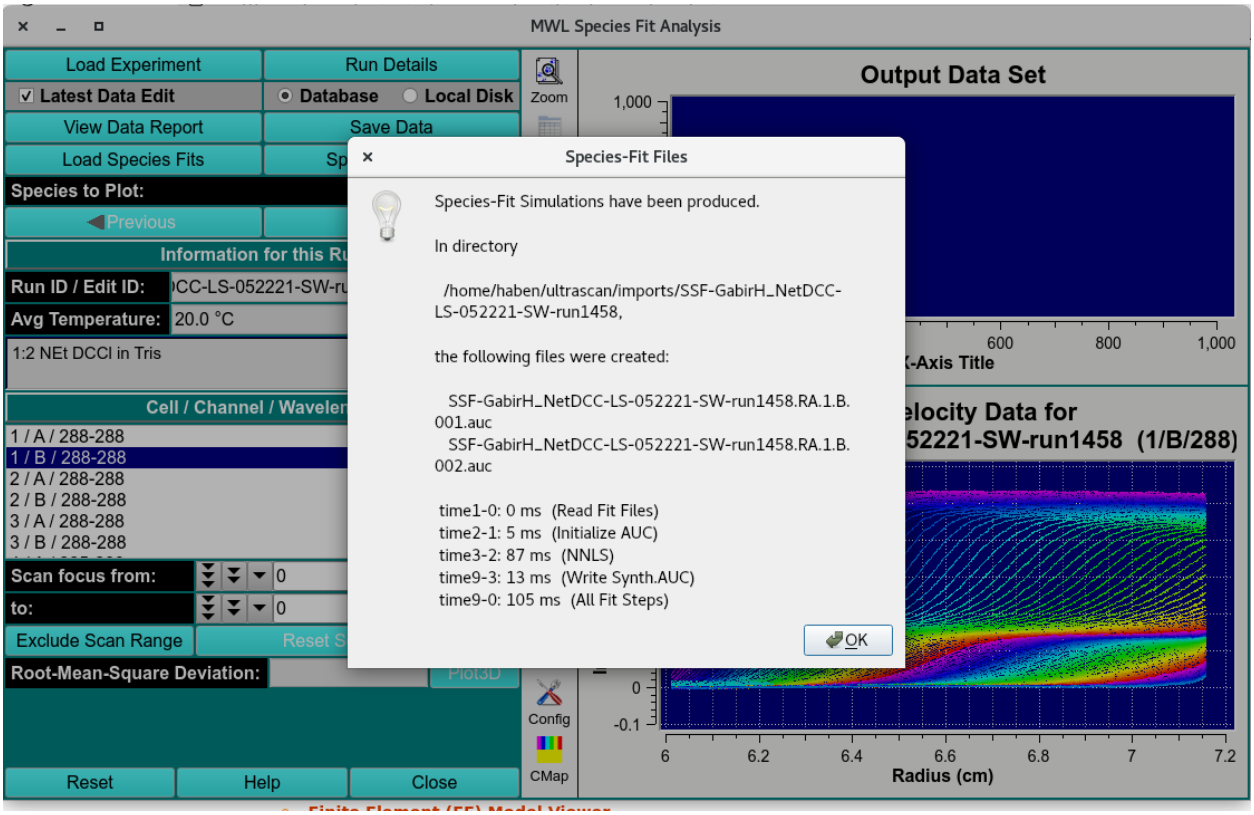
1. Load the experiment **ISSF-** file using a Load Run Data Dialog.



Loaded ISSF- Data

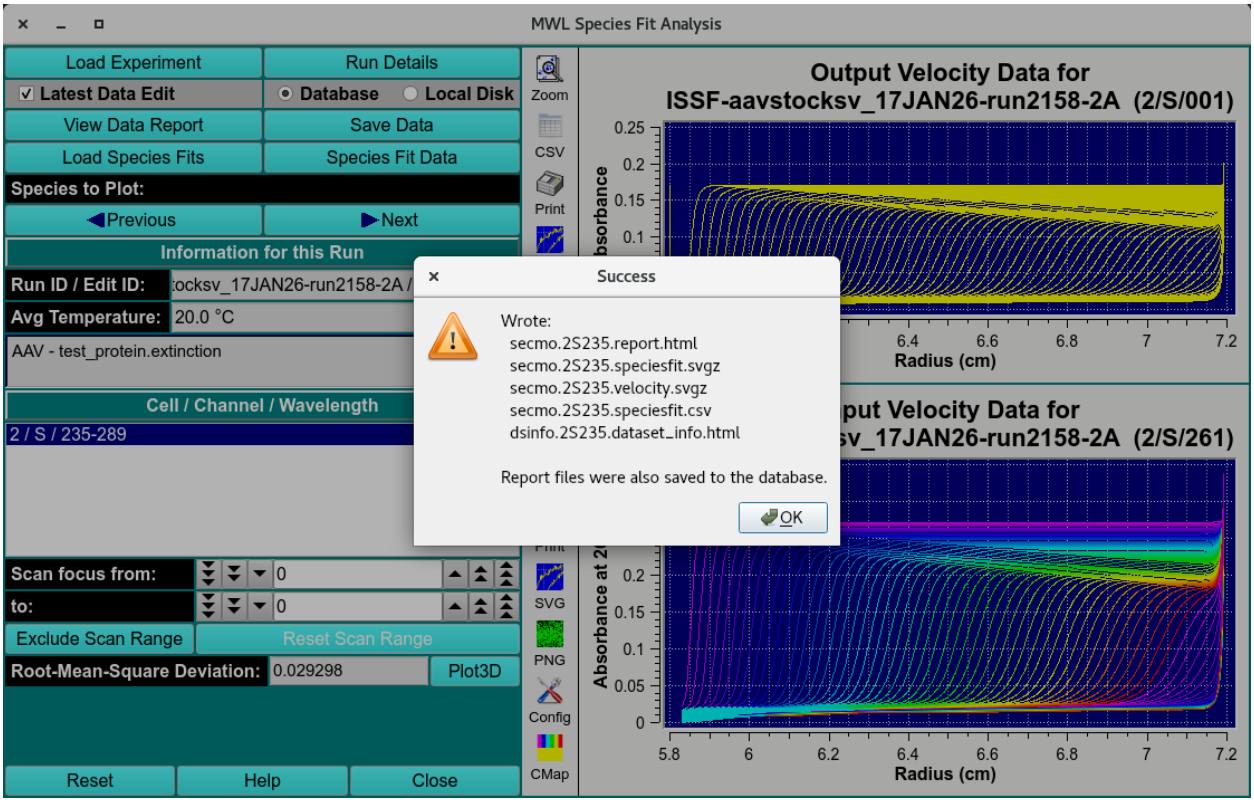
2. Load the extinction-coefficient spectra file as a csv. This file can be generated by measuring serial dilution spectra of analyte and fitting using :doc:`Spectrum Fitter <../us_extinction>.

3. Click **Species Fit Data** and a simulation **SSF-ISSF-** dataset for every analyte extinction-coefficient spectra inputted will be generated and saved in the $HOME/ultrascan/imports folder.



Saved Deconvoluted Data

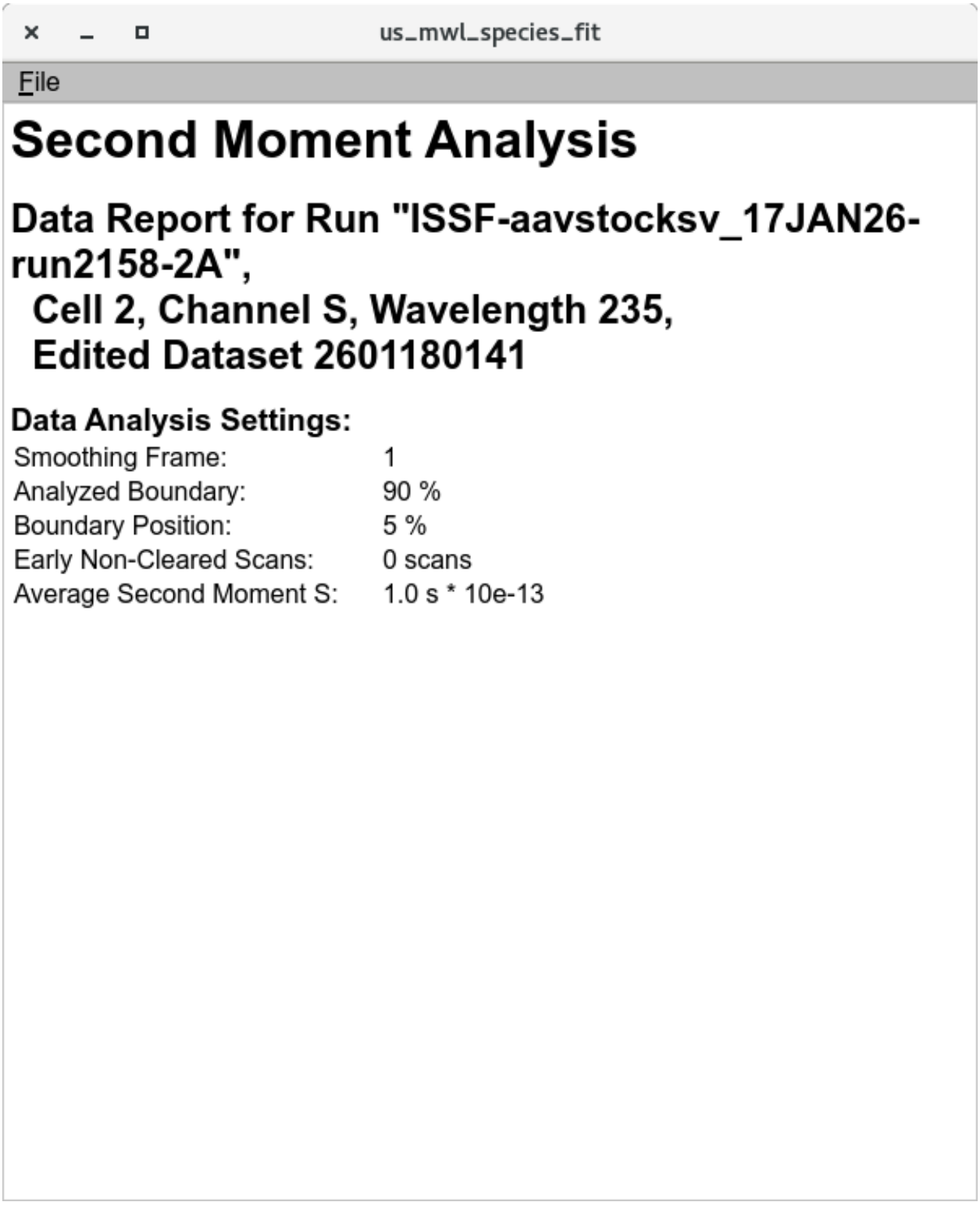
4. Click **Save Data** and the datasets and deconvolution quality report will be saved in the database.



Saved Deconvoluted Data and Report

34.5.2 MWL Species Fit Functions:

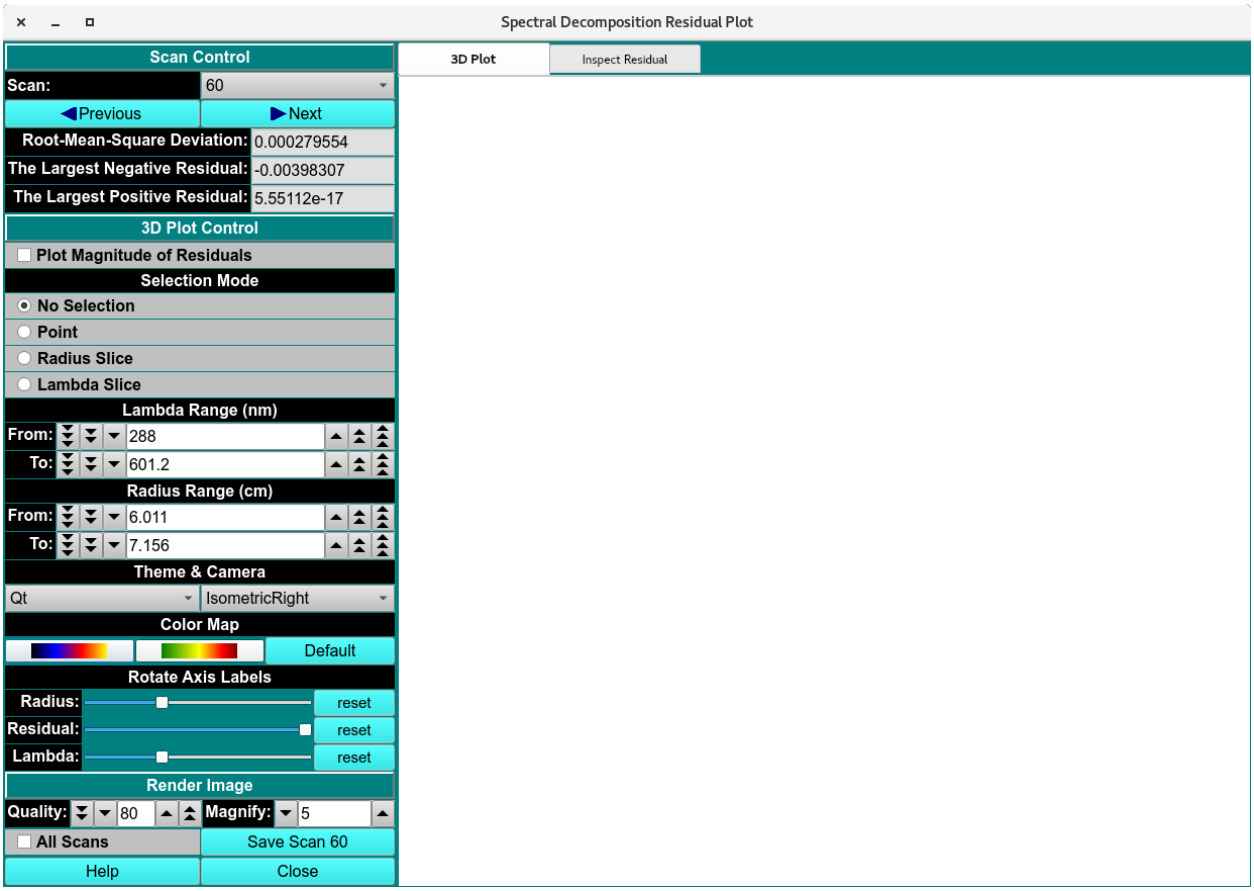
Load Experiment	Load the time-synchronized **ISSF-** simulations generated in :doc:`MWL Pre-Fit Species Simulation module <mwl_species_sim>.
Run Details	Bring up a :doc:`Run Details Dialog <../run_details> with a summary of data and run details.
Latest Data Editor	Select latest edit of all data displayed.
Database	Load or save data from the database.
Local Disk	Load or save data from the local disk.
View Data Report	View the *data report* file.
Save Data	Save the simulated data in $HOME/ultrascan/imports
Load Species Fits	Load the extinction-coefficient spectra files of 2 or more species from local file. Files will be saved in csv format from :doc:`Spectrum Fitter <../us_extinction>
Species Fit Data	Generate and save **SSF-ISSF-** simulation files.
Species to Plot:	Plot one of the at least 2 species deconvoluted from the original dataset.
Previous	Navigate to the previous datasets.
Next	Navigate to the next datasets.
Run ID/ Edit ID:	The run and edit identification labels.
Avg Temperature:	Average temperature of experiment.
Solution name:	Name of the original solution.
Cell / Channel / Wavelength	Triple identifier.
Scans focus	Select a scan range.
Exclude Scan range	Exclude selected scan range.
Reset Scan range	Reset all excluded scans.
Root-Mean-Square Deviation:	The Root-Mean-Square Deviation of the deconvolution.
Plot3D	RMSD, residuals and fitting plots of the *Deconvolution*.
Reset	Reset window.
Help	Open the MWL Species Fit Analysis help documentation.
Close	Close window.



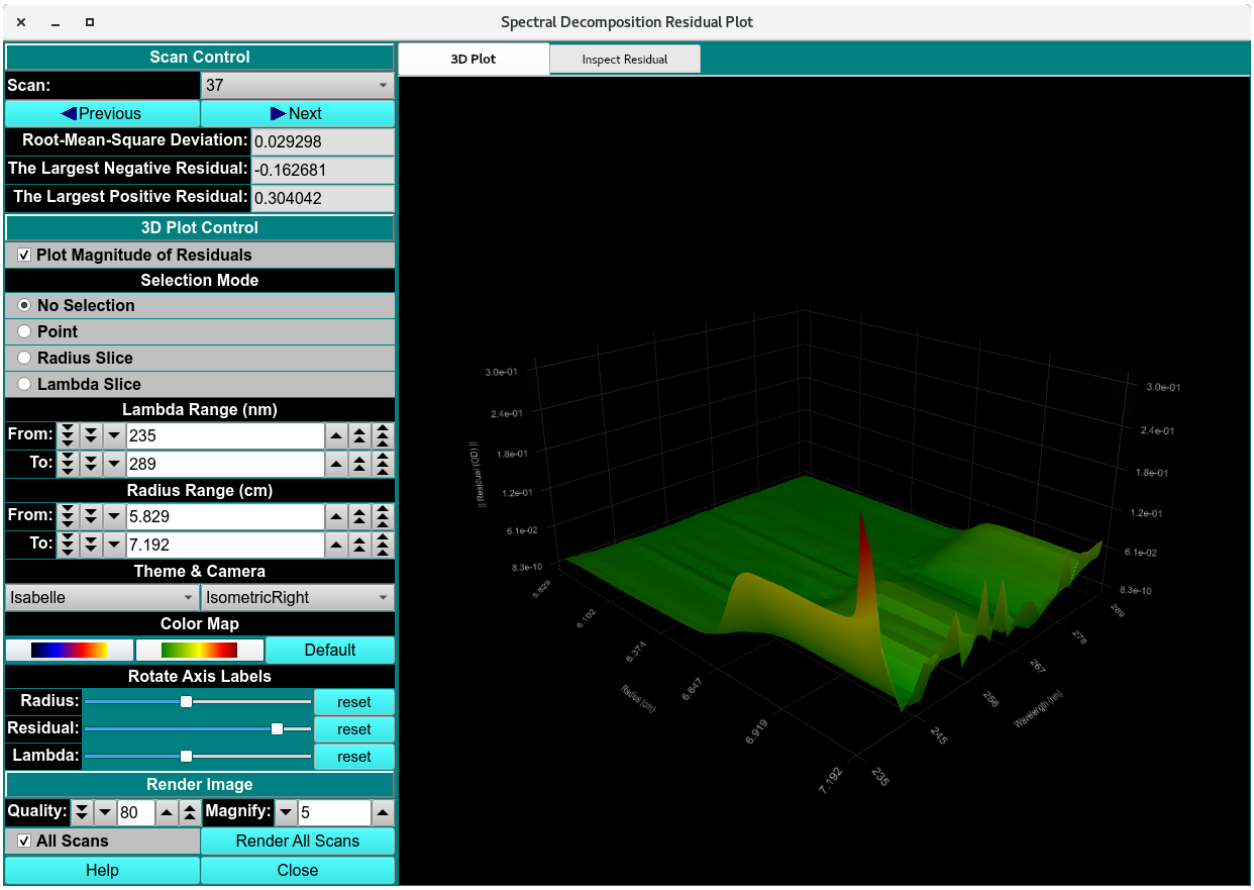
Data Report

Spectral Decomposition Residual Plot

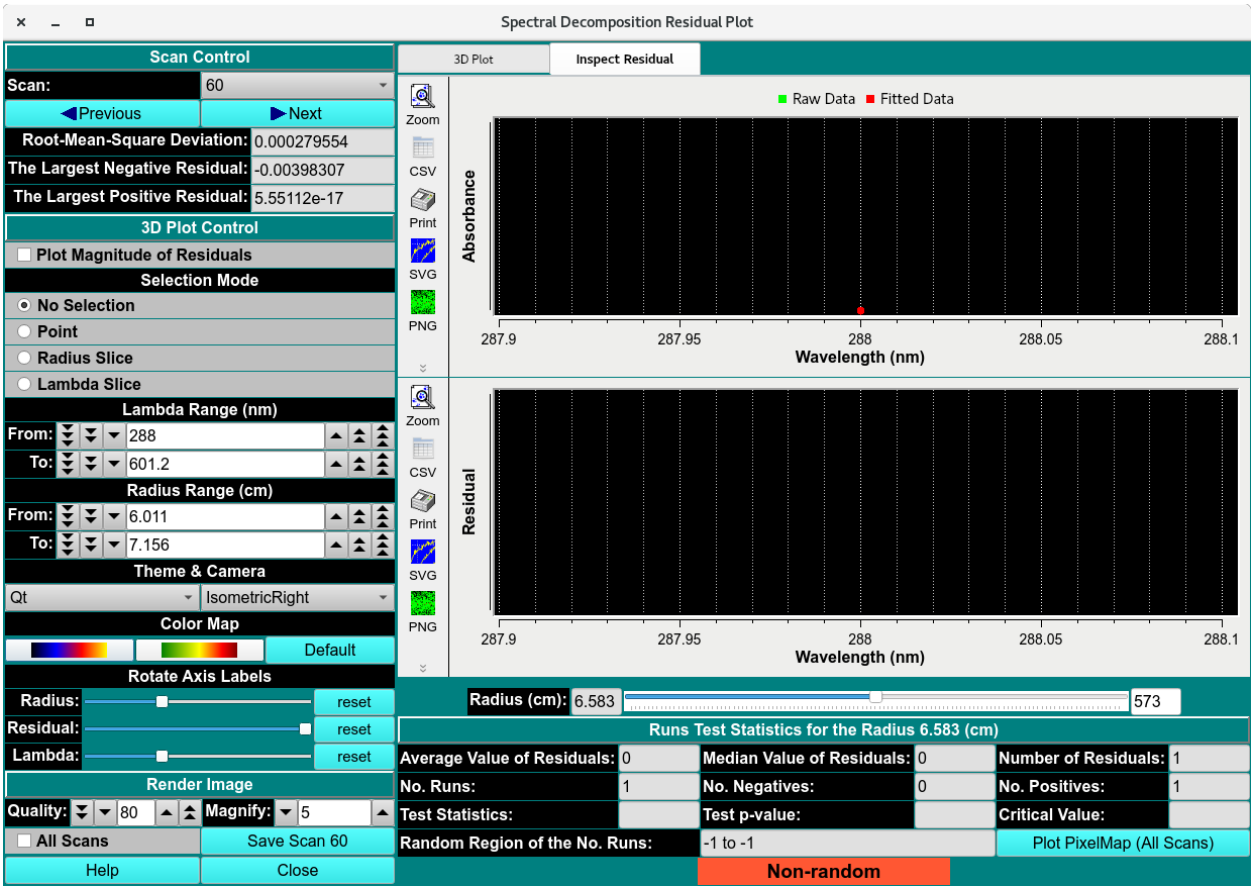
Assess the deconvolution RMSD and Residuals



Empty Spectral Decomposition Residual Plot



Spectral Decomposition Residual Plot



Empty Spectral Decomposition Residuals

Spectral Decomposition Residuals

Links to Associated Modules:

- :doc:`Spectrum Fitter <us_extinction> - Generate the extinction basis spectrum for each species in dataset.
- **:ref:`Analyte <analyte/edit-analyte-spectrum>`_** - Associate an extinction basis spectrum file to an analyte.
- :doc:`Flowchart to Multi-Wavelength Sedimentation Velocity Analysis <../start_page-mw>.

CHAPTER

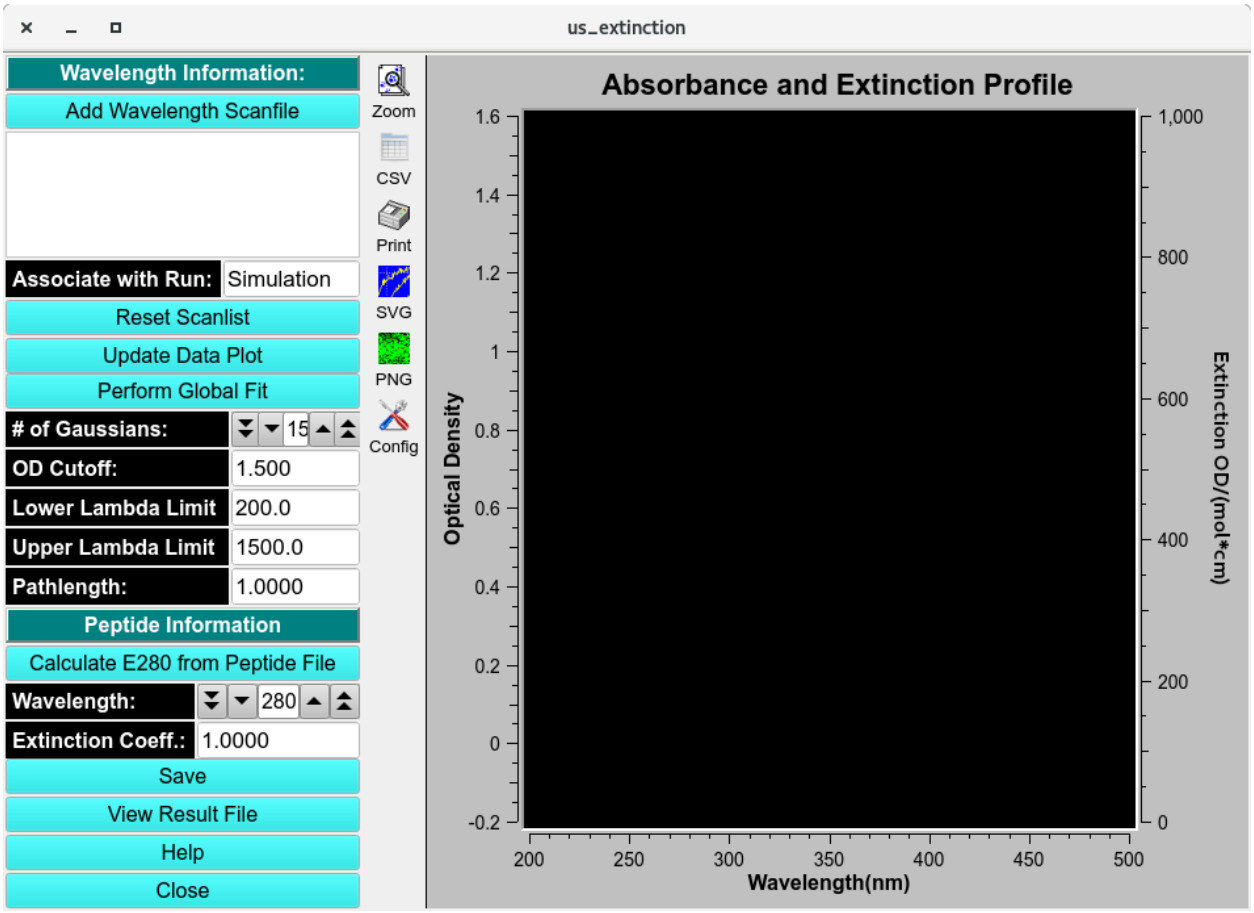
THIRTYFIVE

SPECTRUM FITTER

Index

- *Process:*

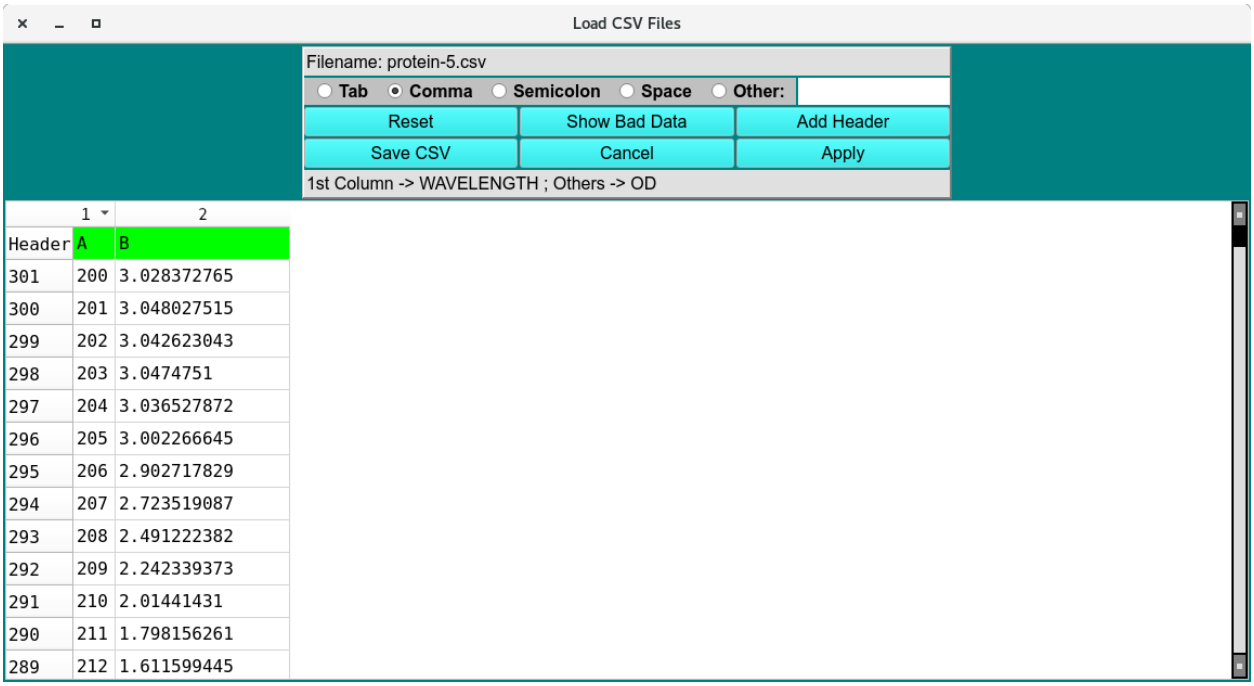
The Spectrum Fitter is a module for analyzing spectral scan files collected from component serial dilutions. It applies multiple global nonlinear least-squares (NLS) fitting algorithms to model dilution-dependent extinction curves and determine the solute's intrinsic extinction coefficient (intrinsic spectrum). Fitted results are exported as a .dat file for downstream analysis.



Spectrum Fitter

35.1 Process:

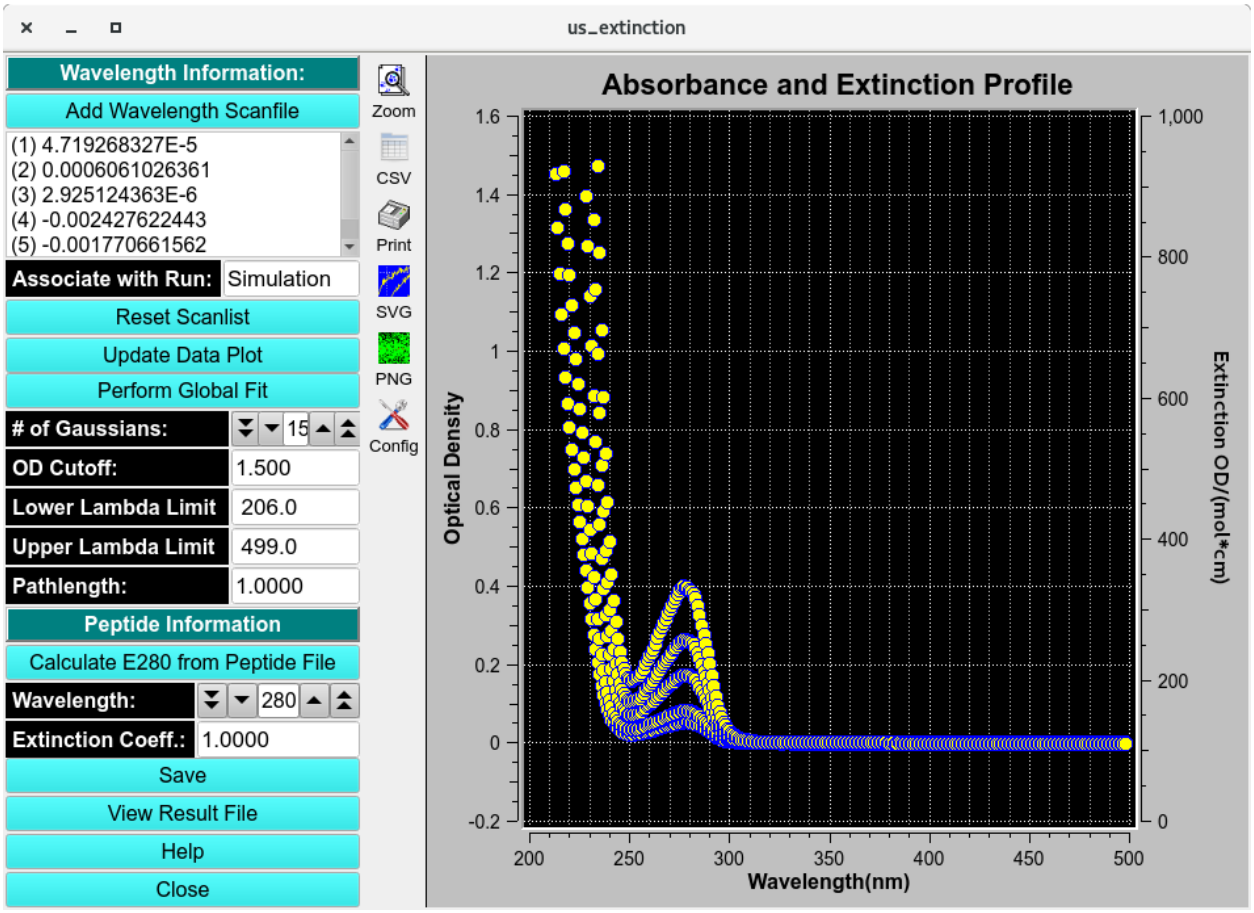
1. **Load Scanfiles:** Click **Add Wavelength Scanfile** to navigate to a file manager and load the serial dilution spectrum scan .csv files. A **Load CSV Files** window will pop-up to select the Delimiter-separated value, edit the header and inspect data.



	1	2
Header	A	B
301	200	3.028372765
300	201	3.048027515
299	202	3.042623043
298	203	3.0474751
297	204	3.036527872
296	205	3.002266645
295	206	2.902717829
294	207	2.723519087
293	208	2.491222382
292	209	2.242339373
291	210	2.01441431
290	211	1.798156261
289	212	1.611599445

Load CSV file

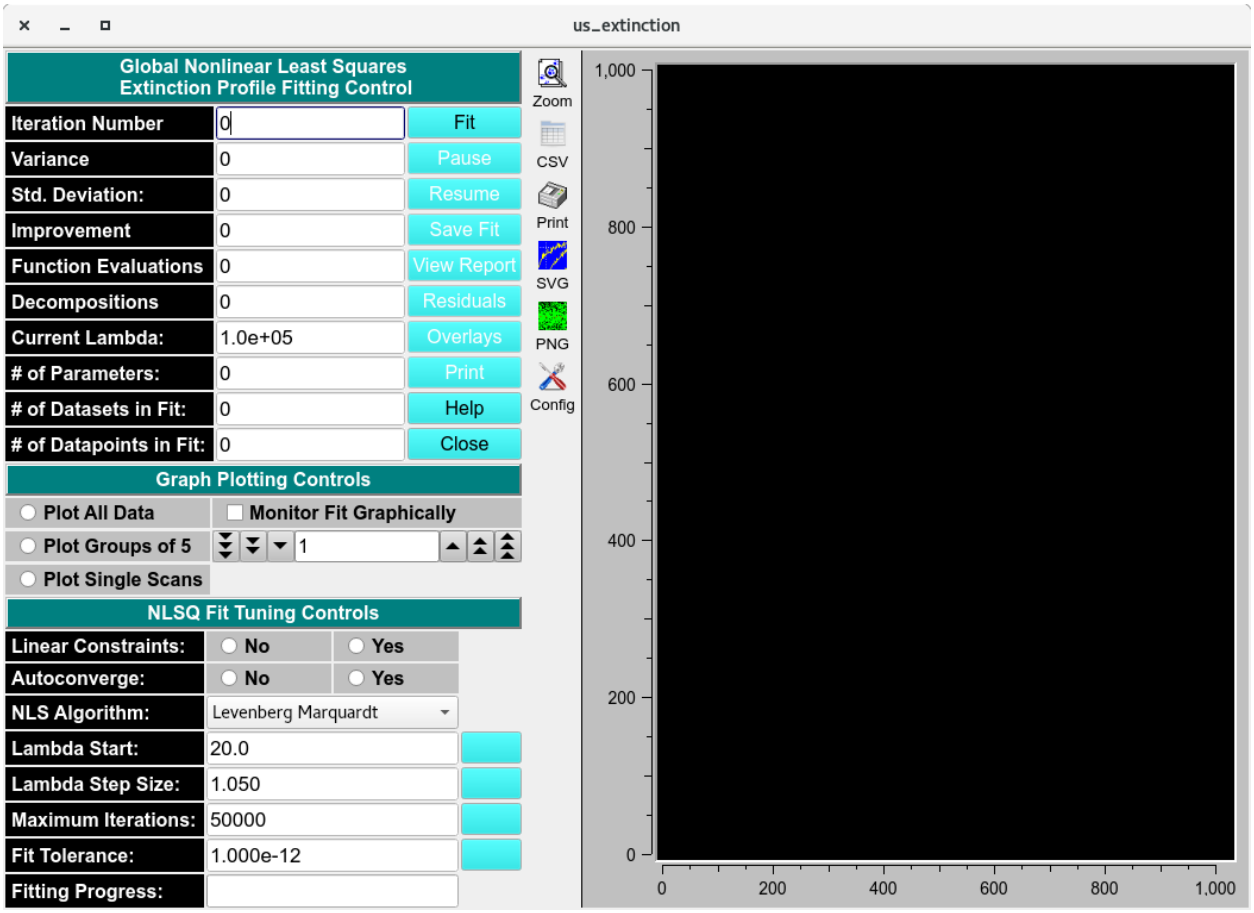
Click **Accept** and the data will be imported and displayed on the Absorbance and Extinction Profile plot. Repeat for each scanfile in the dilution.



Loaded Spectrum Fitter

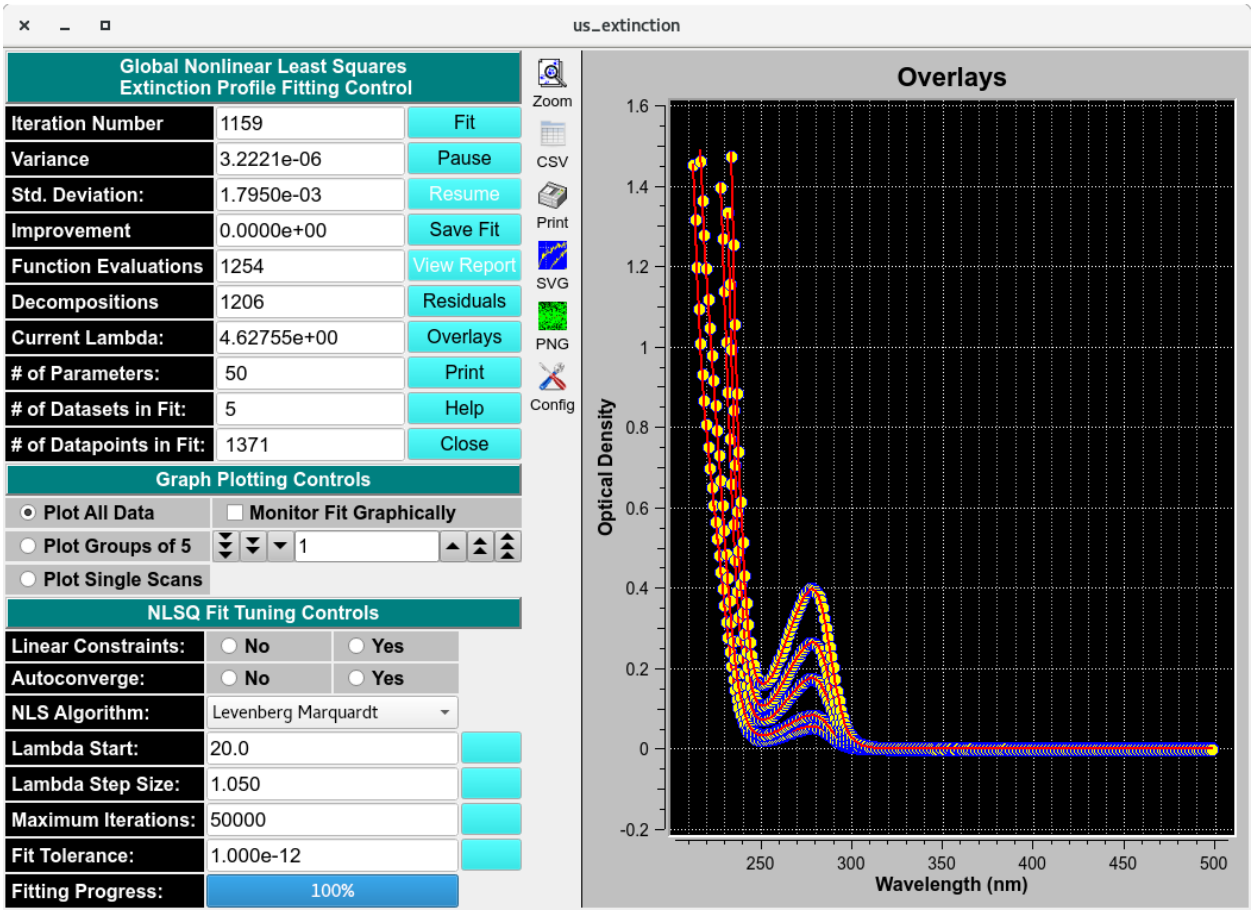
2. **Perform Global Fit:**

Click **Perform Global Fit** to navigate to the fitting controls for the Global Non-Linear Least Square (NLS) algorithms. For typical users the default settings are sufficient and users can **Fit** without any modifications.

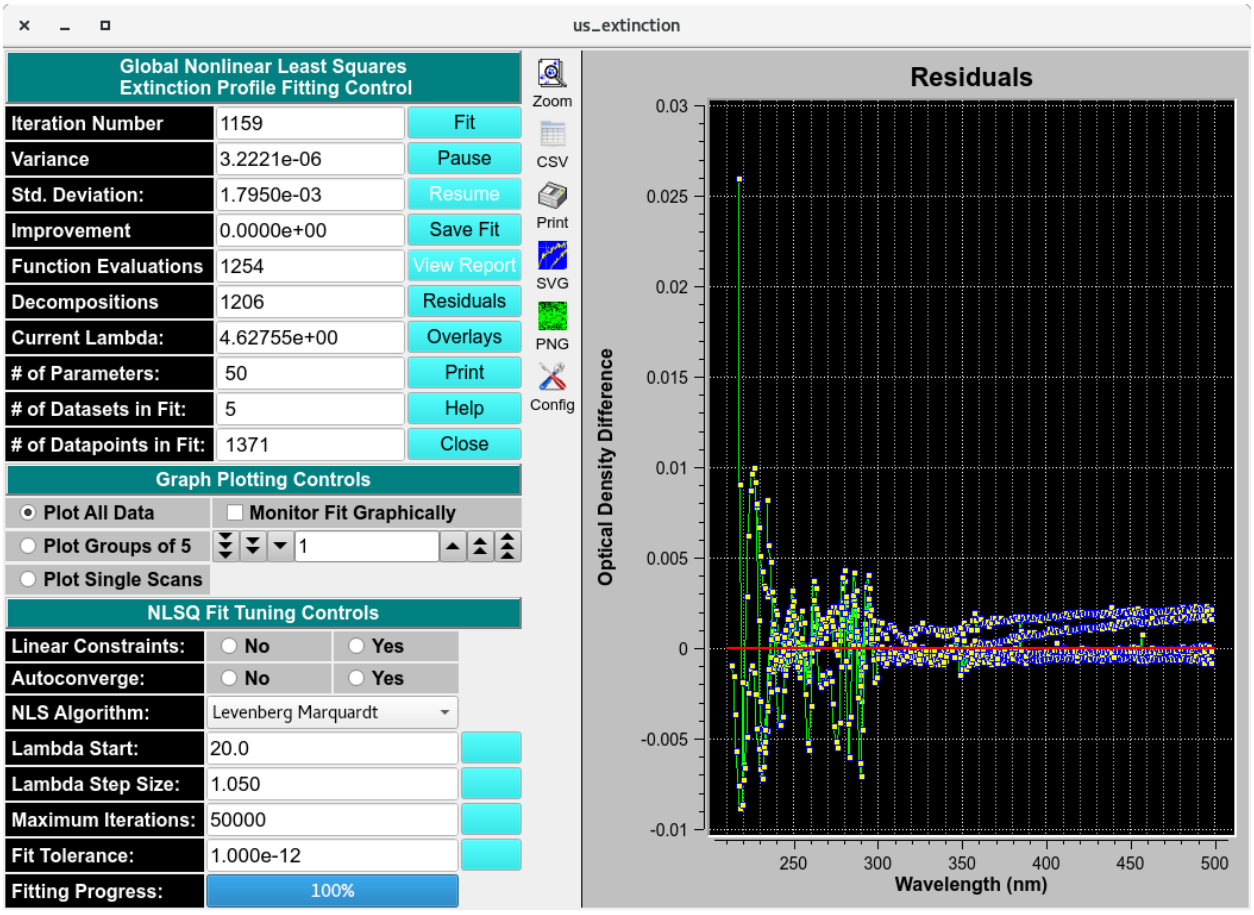


NLS Fitting Controls

3. **Global Fit Results:** Select **Residuals** or **Overlay** to display the results in the adjacent Overlay/Residual plot.



NLS Fitted Overlay



NLS Fitted Residuals

The results of the fit can be saved in the main us_extinction window. Enter a name and save the basis spectrum.

CHAPTER

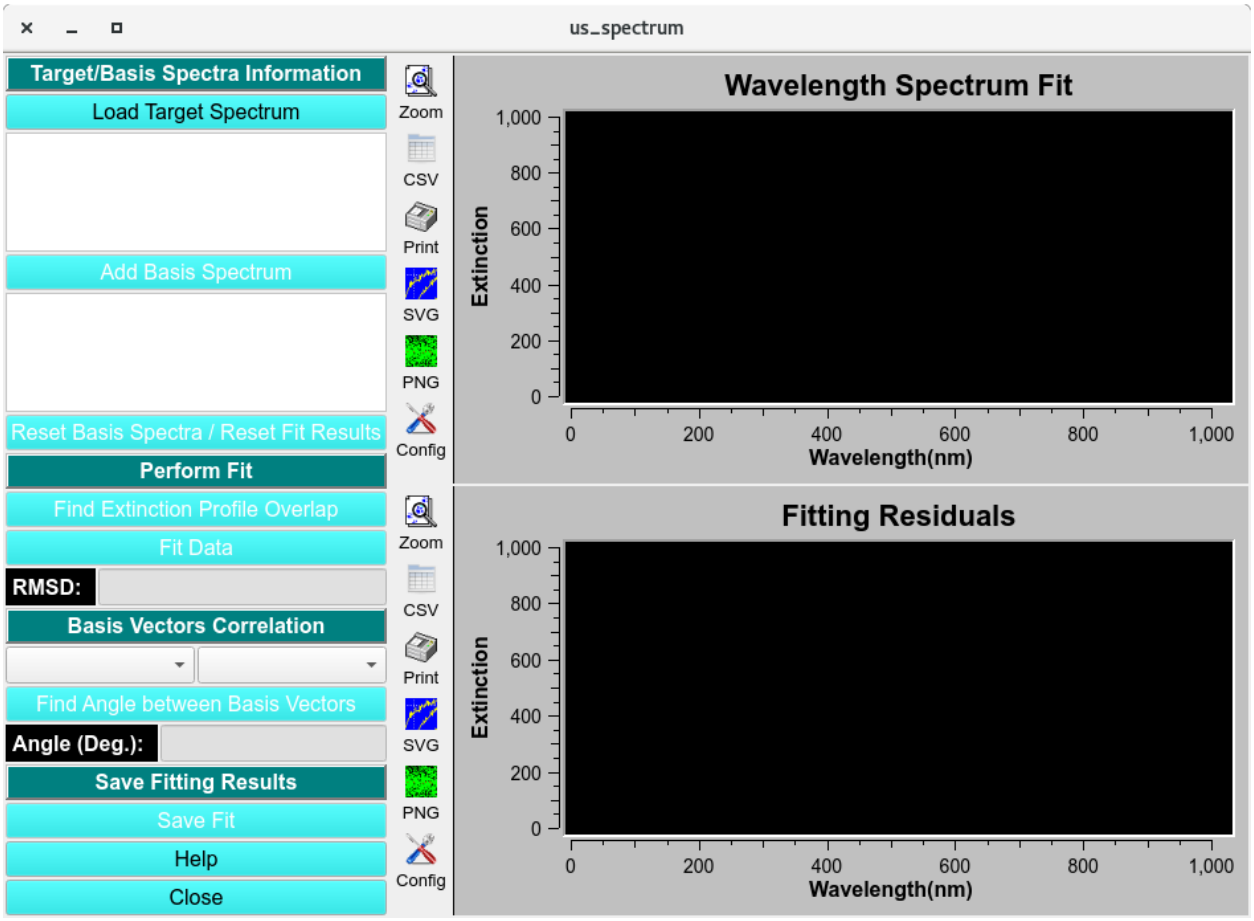
THIRTYSIX

SPECTRUM DECOMPOSITION

Index

- *Process:*

The Spectrum Decomposition module helps you interpret the components of a mixture spectra by fitting to the known intrinsic extinction profiles (basis vectors) of each component. Using CSV inputs, you can load a mixture spectrum and one or more basis spectra, fit the basis set to the mixture, and quantify the percent contribution of each component. The module also includes a vector-angle tool to compare spectra and measure the angle between two basis vectors.

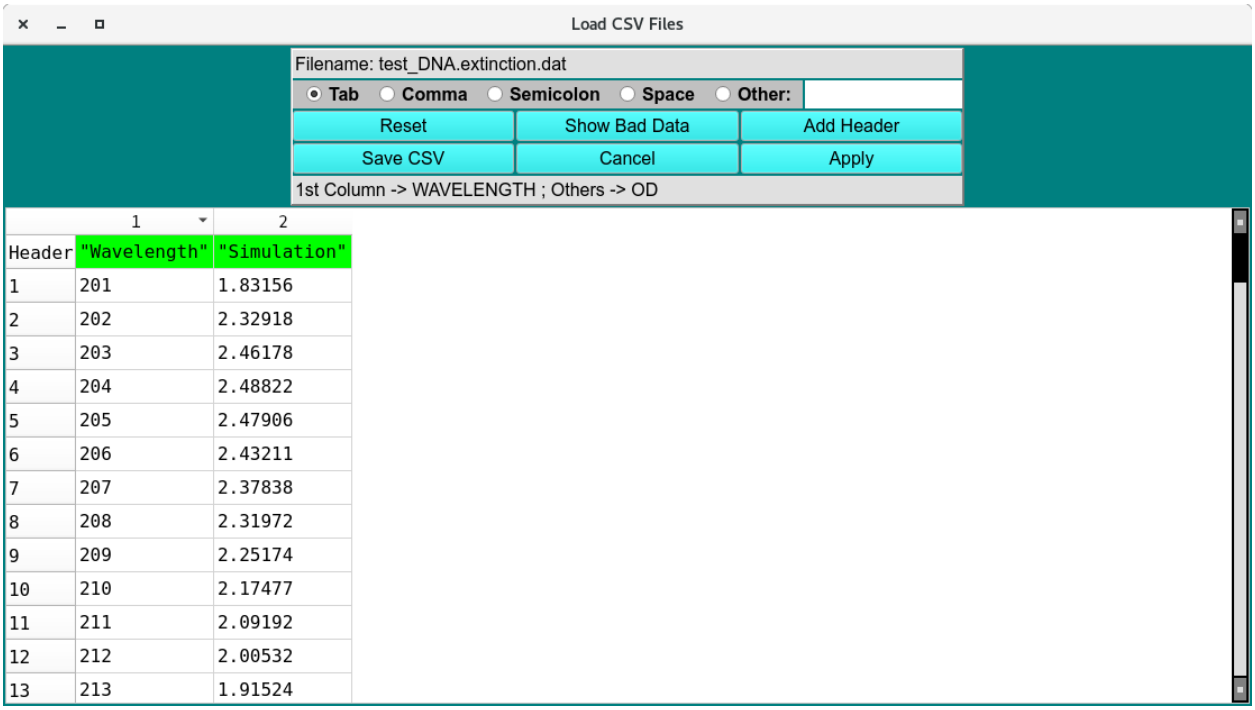


Spectrum Decomposition

36.1 Process:

1. **Load Target Spectrum:** and **Load Basis Spectrum**,

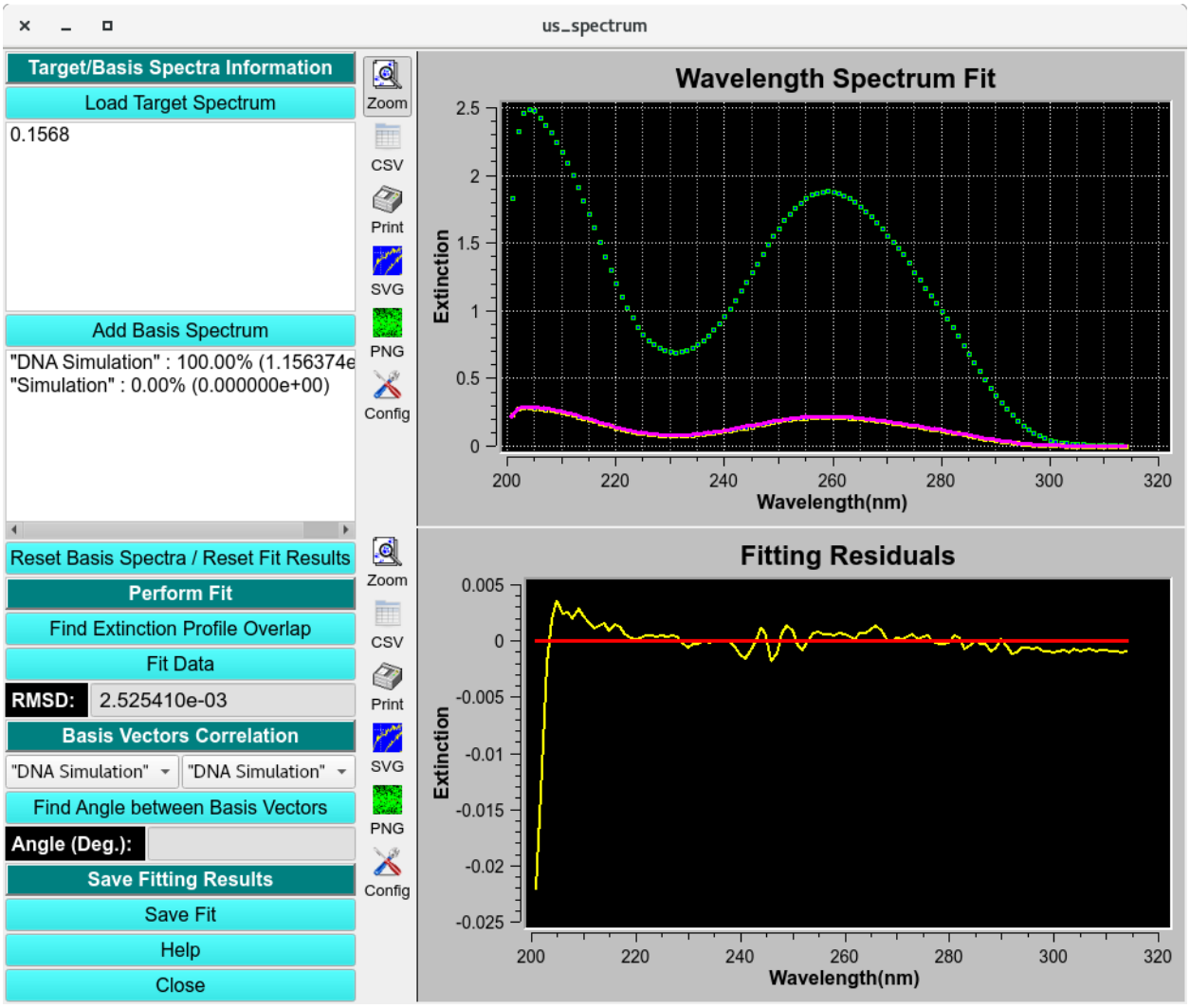
To load the target spectrum, click the **Load Target Spectrum** button and select the csv file from a file manager window. A **Load CSV Files** window will pop-up to select the Delimiter-separated value, edit the header and inspect data.



	1	2
Header	"Wavelength"	"Simulation"
1	201	1.83156
2	202	2.32918
3	203	2.46178
4	204	2.48822
5	205	2.47906
6	206	2.43211
7	207	2.37838
8	208	2.31972
9	209	2.25174
10	210	2.17477
11	211	2.09192
12	212	2.00532
13	213	1.91524

Load CSV File

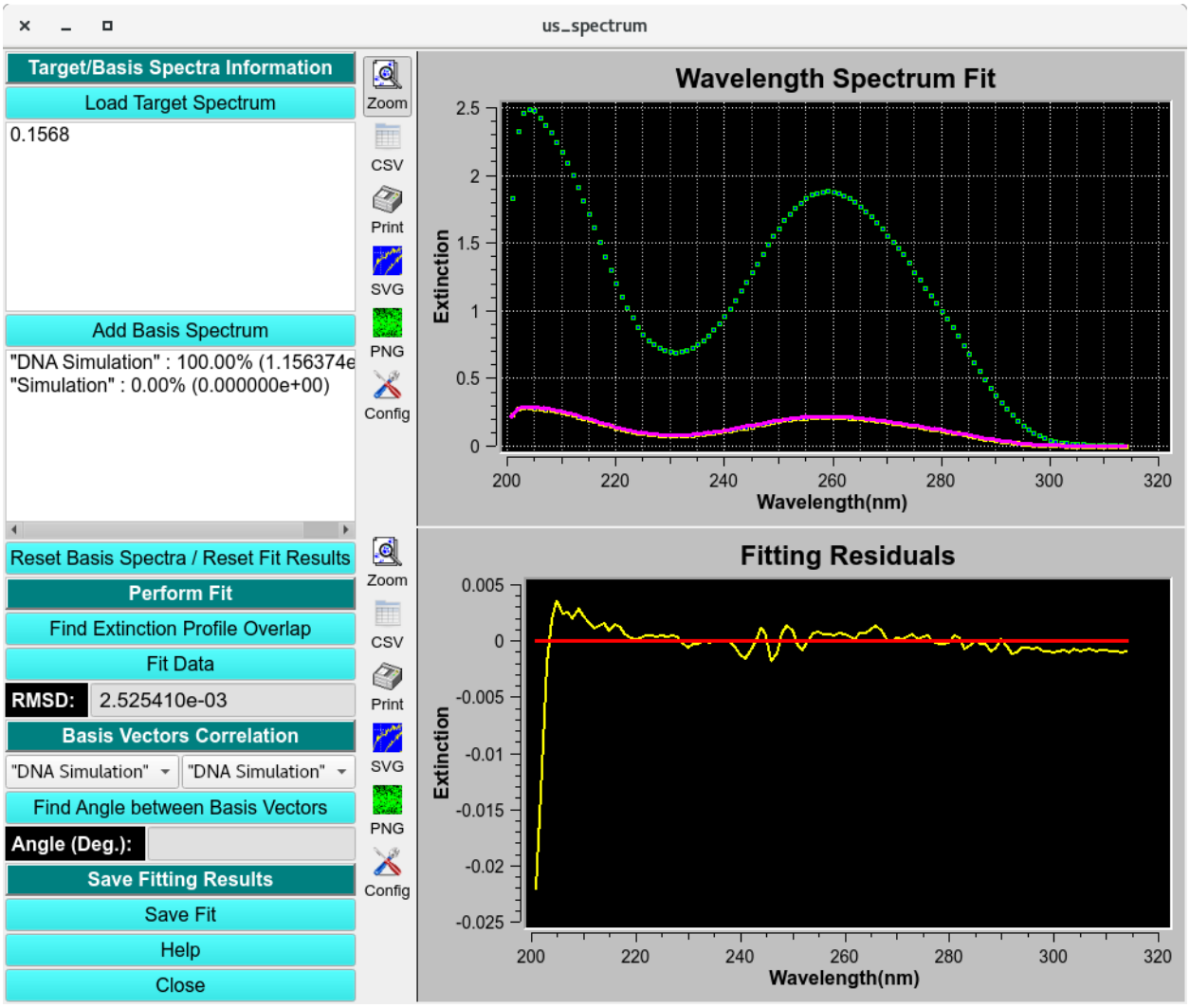
2. Click **Accept** and the data will be imported and displayed on the Absorbance and Extinction Profile plot. Repeat for each basis vector by clicking the **Add Basis Spectrum** button and load the intrinsic spectrum profile.



Loaded Spectrum Decomposition

3. **Find profile overlap** and **Fit Data**

Cut the dat by clicking **Find Extinction Profile Overlay** and trimming the data to the overlapping wavelength regions. After fitting the data and the solution curve and the residuals of the fit displayed it the **Wavelength Spectrum Fit** and **Residuals** plots.



Spectrum Decomposition

CHAPTER

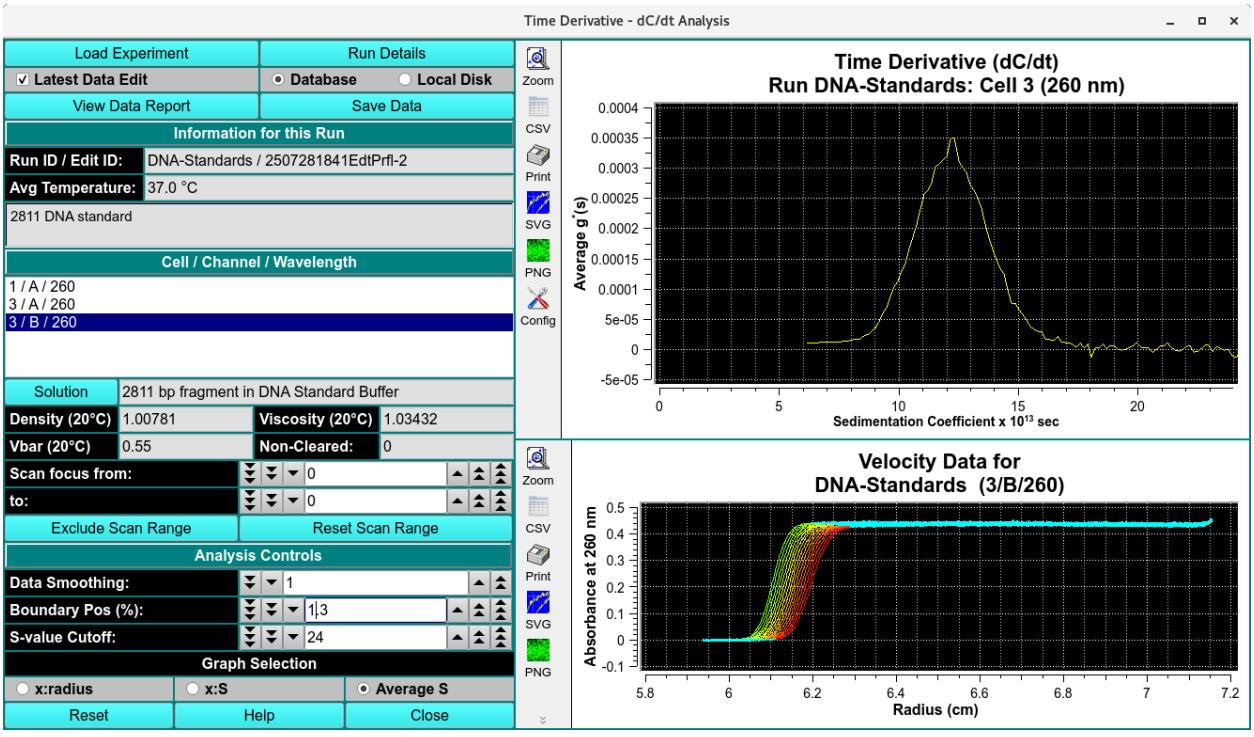
THIRTYSEVEN

TIME DERIVATIVE (DC/DT) DATA ANALYSIS

Index

- *(dC/dt) Functions:*

The DCDT or Time Derivative method will calculate g(S) differential sedimentation profiles by subtracting consecutive scan pairs from each other and mapping the resulting difference curves to the S domain to obtain dcdt curves. The advantage of this method is its ability to nicely subtract out time invariant noise, such as window scratches and dirt, as well as refractive index hetereogeneities in the windows. The latter is particularly important for low-concentration interference data. Unlike the *van Holde-Weischet method <vhw_enhanced>*, the Time Derivative method does not correct for diffusion. In order to obtain accurate results with this method, it is important that you use only a small scan range over which diffusion has not changed significantly.



37.1 (dC/dt) Functions:

Load Experiment	Click on this button and, in the resulting Load Data Dialog, select an edited data set to load.
Run Details	Bring up a *Details for Raw Data* dialog with a summary of data and run details.
Latest Data Edit	Uncheck to allow choosing an edit other than the latest one for the experimental data.
Database	Check to specify data input from the database.
Local Disk	Check to specify data input from local disk.
View Data Report	Create a results text file and display its contents in a text dialog.
Save Data	Create several data and report files based on input data and vHW parameters.

Information for this Run

Run ID / Edit ID:	The main run title of the data and an edit identifier are displayed.
Avg Temperature:	The average temperature of solute is displayed in Celsius.
(description)	The text box below the one for temperature shows a full data description string.

Cell / Channel / Wavelength

Cell / Channel / Wavelength	The text box below this label gives cell, channel and wavelength Triples available in this data set. Highlight the desired value.
Solution	Click this button to open a *Solution Management dialog* that allows changes to buffer and analyte characteristics of the data set.
Density (20°C)	Shows the density value for the loaded experiment. Click the Solution button to open a dialog in which density and other values may be changed.
Viscosity (20°C)	Shows the viscosity value for thes loaded experiment. Click the Solution button to open a dialog in which viscosity and other values may be changed.
Vbar (20°C)	Shows the vbar value for the loaded experiment. Click the Solution button to open a dialog in which vbar and other values may be changed.
Skipped	The count of experiment data scans skipped.
Scan focus from:	Choose the first of a range of scan numbers that may potentially be excluded from analysis.
To:	Choose the end of a range of scan numbers that may potentially be excluded from analysis. The From/To scan range is illustrated in both plots to the right.
Exclude Scan Range	If the From/To scan range selections are as desired, click on this button to exclude the indicated scans from analysis.
Reset Scan Range	Reset to the full range of scans

Analysis Controls

Data Smoothing:	Choose the number of points to use for any smoothing of raw input data.
Boundary Pos. (%):	Choose the percent of the plateau-baseline range that is to be added to the baseline to form the beginning of analysis span.
S-value Cutoff:	Choose the sedimentation coefficient value to form the maximum X value of the Time Derivative plot.

Graph Selection

x:radius	Select this radio button to choose a g(S) versus radius dC/dt plot.
x:S	Select this radio button to choose a g*(S) versus sedimentation coefficient dC/dt plot.
Average S	Select this radio button to choose an Average g*(S) versus sedimentation coefficient dC/dt plot.
(Time Derivative Plot)	The upper of the two right-side plots shows one of three types (x:radius, x:S, Average S) of Time Derivative (dC/dt) plots.
(Velocity Data Plot)	The lower of the right-side plots shows selected velocity data for which a time derivative calculation has been made.

Window Controls

Reset	Indicate that parameters are to be reset and the plots re-displayed based on original parameters.
Help	Display this detailed Time-Derivative help.
Close	Close all windows and exit.

CHAPTER

THIRTYEIGHT

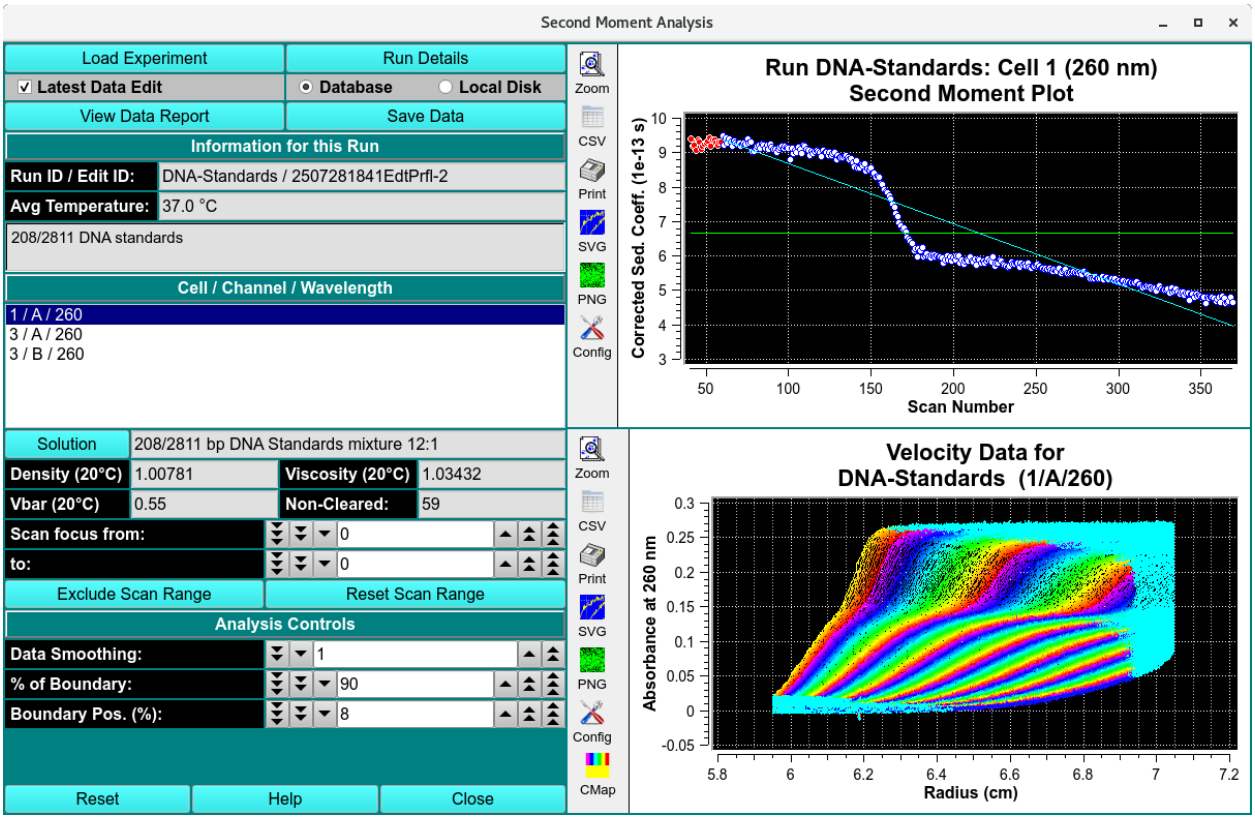
SECOND MOMENT DATA ANALYSIS

Index

- *Second Moment Functions:*

The second moment analysis will calculate weight-average sedimentation coefficients for each scan included in the analysis by finding the second moment point in the boundary. The second moment analysis can serve as a valuable diagnostics tool in identifying problems such as time-dependent degradation, aggregation and even concentration dependency.

Note: It is important to keep in mind that the second moment analysis is only valid for scans that have cleared the meniscus and still have a stable plateau. All other scans should be excluded from the analysis.


Second Moment Analysis
Load Experiment
Run Details
Latest Data Edit
Database
Local Disk
View Data Report
Save Data
Information for this Run
Run ID / Edit ID:
DNA-Standards / 2507281841EdtPrfl-2
Avg Temperature:
37.0 °C
208/2811 DNA standards
Cell / Channel / Wavelength
1 / A / 260
3 / A / 260
3 / B / 260
Zoom
CSV
Print
SVG
PNG
Config
Run DNA-Standards: Cell 1 (260 nm)
Second Moment Plot
Corrected Sed. Coeff. (1e-13 s)
Scan Number
Solution
208/2811 bp DNA Standards mixture 12:1
Density (20°C)
1.00781
Viscosity (20°C)
1.03432
Vbar (20°C)
0.55
Non-Cleared:
59
Scan focus from:
0
to:
0
Exclude Scan Range
Reset Scan Range
Analysis Controls
Data Smoothing:
1
% of Boundary:
90
Boundary Pos. (%):
8
Reset
Help
Close
CMap
Velocity Data for
DNA-Standards (1/A/260)
Absorbance at 260 nm
Radius (cm)

38.1 Second Moment Functions:

Load Experiment	Click on this button and, in the resulting Load Data Dialog, select an edited data set to load.
Run Details	Bring up a *Run Details Dialog* with a summary of data and run details.
Latest Data Edit	Uncheck to allow choosing an edit other than the latest one for the experimental data.
Database	Check to specify data input from the database.
Local Disk	Check to specify data input from local disk.
View Data Report	Create a results text file and display its contents in a text dialog.
Save Data	Create several data and report files based on input data and vHW parameters.
Run ID / Edit ID:	The main run title of the data and an edit identifier are displayed.
Avg Temperature:	The average temperature of solute is displayed in Celsius.
(description)	The text box below the one for temperature shows a full data description string.
Cell / Channel / Wavelength	The text box below this label gives cell, channel and wavelength Triples available in this data set. Highlight the desired value.
Solution	Click this button to open a *Solution Management dialog* that allows changes to buffer and analyte characteristics of the data set.
Density (20°C)	Shows the density value for the loaded experiment. Click the Solution button to open a dialog in which density and other values may be changed.
Viscosity (20°C)	Shows the viscosity value for these loaded experiment. Click the Solution button to open a dialog in which viscosity and other values may be changed.
Vbar (20°C)	Shows the vbar value for the loaded experiment. Click the Solution button to open a dialog in which vbar and other values may be changed.
Skipped	The count of experiment data scans skipped.
Scan focus from:	Choose the first of a range of scan numbers that may potentially be excluded from analysis.
To:	Choose the end of a range of scan numbers that may potentially be excluded from analysis. The From/To scan range is illustrated in both plots to the right.
Exclude Scan Range	If the From/To scan range selections are as desired, click on this button to exclude the indicated scans from analysis.
Reset Scan Range	Reset to the full range of scans.
Data Smoothing:	Choose the number of points to use for any smoothing of raw input data.
% of Boundary:	Choose the percentage of the range from concentration baseline to plateau that is to be used for analysis.
Boundary Pos. (%):	Choose the percent of the plateau-baseline range that is to be added to the baseline to form the beginning of analysis span.
(Second Moment Plot)	The upper of the two right-side plots shows the second moment plot of sedimentation coefficient for each scan.
(Velocity Data Plot)	The lower of the right-side plots shows selected velocity data for which a second moment calculation has been made.

Window Controls

Reset	Indicate that parameters are to be reset and the plots re-displayed based on original parameters.
Help	Display this detailed Second-Moment help.
Close	Close all windows and exit.

CHAPTER

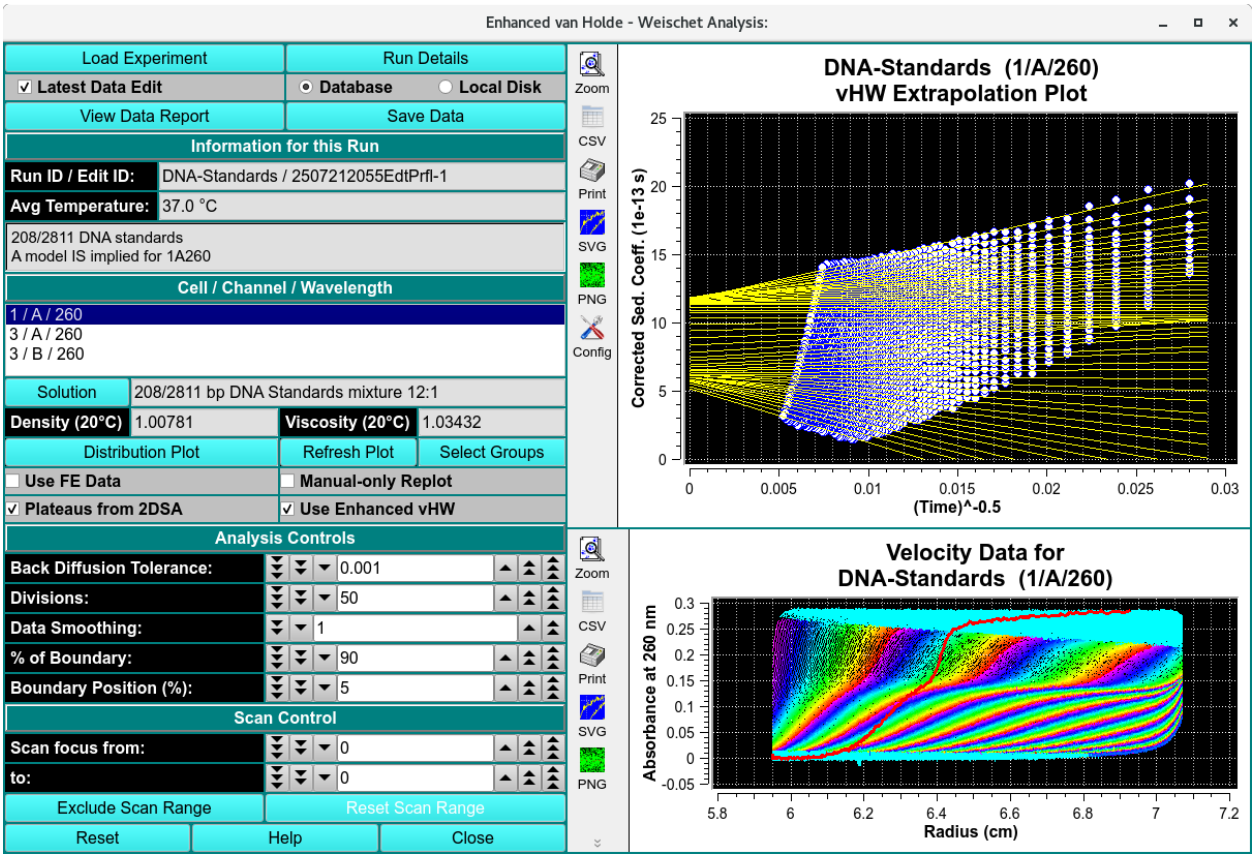
THIRTYNINE

VAN HOLDE-WEISCHET ENHANCED ANALYSIS

Index

- *vHW Enhanced Functions:*
- *van Holde-Weischet Analysis Steps*

The method of van Holde and Weischet (vHW) is a versatile, model-independent approach to determine sedimentation coefficient distribution by graphically analyzing sedimentation velocity experimental data. The vHW method's strength derives from its ability to differentiate boundary spreading due to diffusion from boundary spreading due to heterogeneity in the sedimentation coefficient. This program uses a number of enhanced techniques for vHW analysis.



39.1 vHW Enhanced Functions:

Load Experiment	Click on this button and, in the resulting **:doc:`Load Run Data Dialog <common_dialogs.html#data-loader>`_**, select an edited data set to load.
Run Details	Bring up a *Run Details Dialog* with a summary of data and run details.
Latest Data Edit	Uncheck to allow choosing an edit other than the latest one for the experimental data.
Database	Check to specify data input from the database.
Local Disk	Check to specify data input from local disk.
View Data Report	Create a results text file and display its contents in a text dialog.
Save Data	Create several data and report files based on input data and vHW parameters.
Run ID / Edit ID:	The main run title of the data and an edit identifier are displayed.
Avg Temperature:	The average temperature of solute is displayed in Celsius.
(description box)	The text box below the one for temperature shows a full data description string.

Cell / Channel / Wavelength The text box below this label gives cell, channel and wavelength Triples available in this data set. Highlight the desired value.

Solution	Click this button to open a Solution Management that allows changes to buffer and analyte characteristics of the data set.
Density (20°C)	Shows the density value for the loaded experiment. Click the Solution button to open a dialog in which density and other values may be changed.
Viscosity (20°C)	Shows the viscosity value for the loaded experiment. Click the Solution button to open a dialog in which viscosity and other values may be changed.
Distribution Plot	Click for a plot dialog that shows a *vHW Distribution Plot* illustrating sedimentation coefficient power related to separate species.
Select Groups	**Click on this button to begin a session of choosing vHW Extrapolation Plot division groups. As an initial pop-up dialog explains, you must click first above any group of division lines and then below it. Multiple groups may be chosen, with division line overlap between groups.** **Note:** The button label becomes **Clear Groups** after selection, allowing the set of selected groups to be discarded in favor of a new set of selections.
Refresh Plot	Click on this button at any time to refresh the right-side plots, based on current parameter settings.
Use FE Data	**add definition**
Manual-only Replot	Check this box to suppress automatic re-plotting with each parameter change. This may be desirable when you plan to change several settings, in order to avoid a replot with each change. With this box checked, you may force a plot with the Refresh Plot button.
Manual-only Replot	Check this box to suppress automatic re-plotting with each parameter change. This may be desirable when you plan to change several settings, in order to avoid a replot with each change. With this box checked, you may force a plot with the Refresh Plot button.
Plateaus from 2DSA	Check this box to have plateau values calculated from a model associated with any loaded noise values. With this box unchecked, each scan's plateau value is determined by a series of fits to initially designated plateaus.
Use Enhanced vWH	**add definition**

Analysis Controls

Back Diffusion Tolerance:	Select a tolerance value to fine tune exclusion of points affected by back diffusion, as indicated by the red line on the velocity data.
Divisions:	Choose the number of divisions (yellow lines in the vWH Extrapolation Plot) to use in vHW analysis.
Data Smoothing:	Choose the number of points to use for any smoothing of raw input data.
% of Boundary:	Choose the percentage of the range from concentration baseline to plateau that is to be used for analysis.
Boundary Position (%):	Choose the percent of the plateau-baseline range that is to be added to the baseline to form the beginning of analysis span. For typical analysis 5% is recommended.

Scan Control

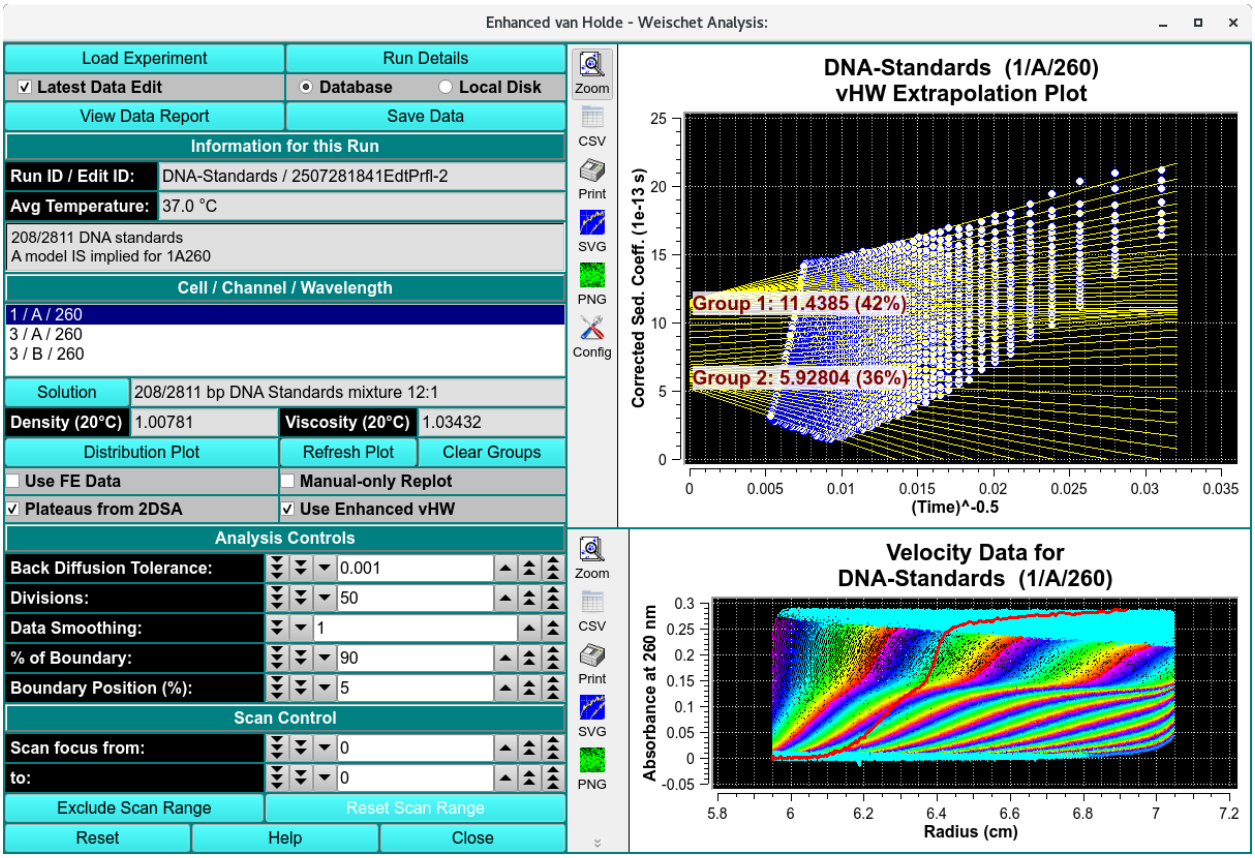
Scan Focus from:	Choose the first of a range of scan numbers that may potentially be excluded from analysis.
Scan Focus to:	Choose the end of a range of scan numbers that may potentially be excluded from analysis. The From/To scan range is illustrated in both plots to the right.
Exclude Scan Range	Exclude the scans indicated by the two preceding scan bounding counters.
Reset Scan Range	Resetting the scan range to include all excluded scans.

Window Controls

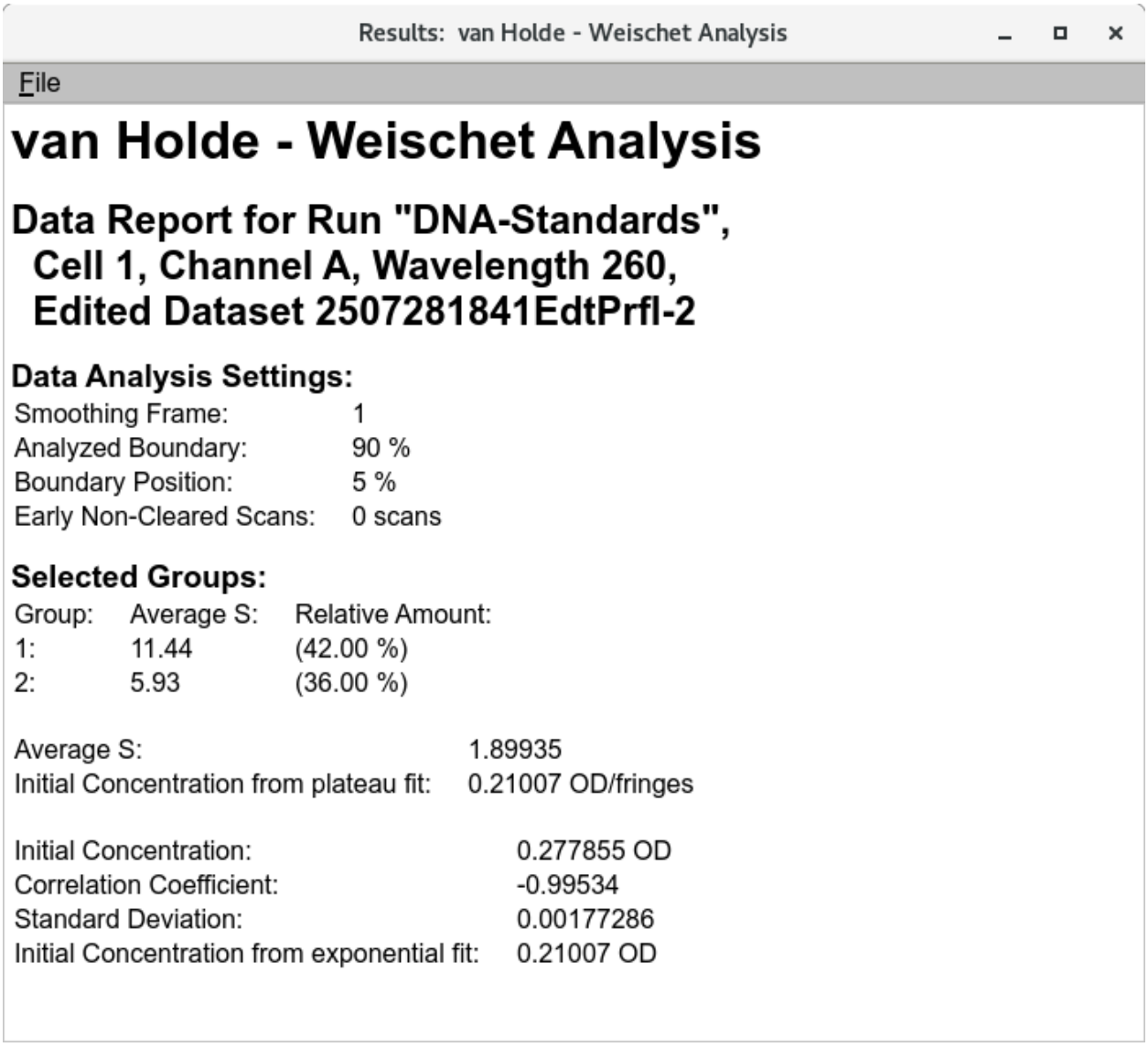
Reset	Indicate that window is reset and the plots are initiated.
Help	Display this detailed van Holde-Weischet Analysis help.
Close	Close all windows and exit.

39.2 van Holde-Weischet Analysis Steps

- **Step 1:** First, load a velocity dataset. Click on **Load Experiment** to select an edited velocity data set from the database or from local disk.
- **Step 2:** Secondly, select a model from database or disk. Simply click on a triple in the Cell / Channel / Wavelength. Enhanced van Holde-Weischet analysis is automatically applied according to the Analysis Controls
- **Step 3:** If necessary, adjust the Back Diffusion Tolerance, Divisions, % of Boundary and Boundary Position (%).
- **Step 4:** After analysis is complete, the resulting distribution and histogram are available for display in the Distribution Plot.
- **Step 5:** To choose the vHW Extrapolation Plot division groups, select **Select Group** and follow the prompt instructions. The vHW Extrapolation Plot will display the selected group.



vHW Extrapolation Plot division groups



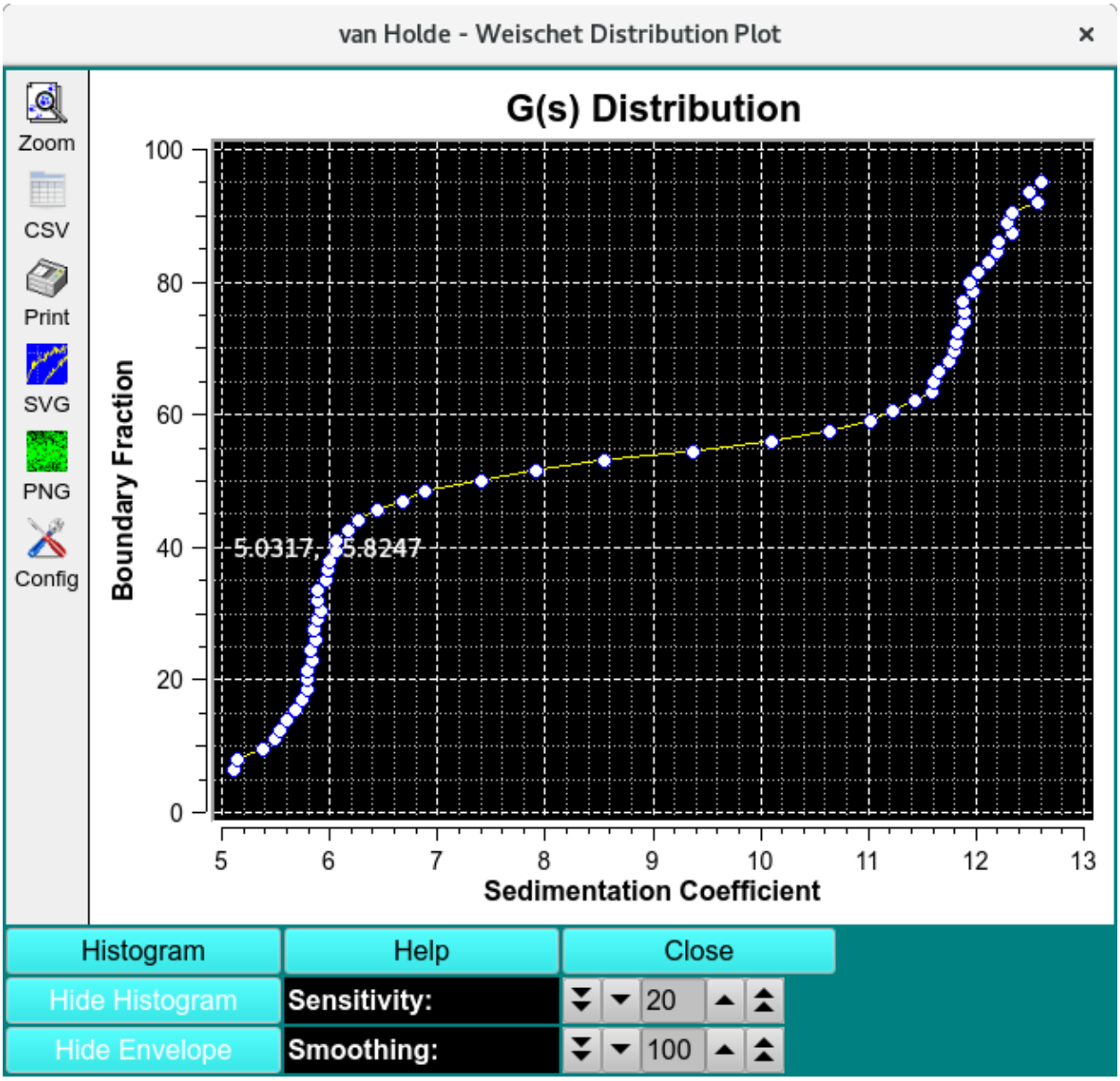
vHW Extrapolation Plot division groups

CHAPTER

FORTY

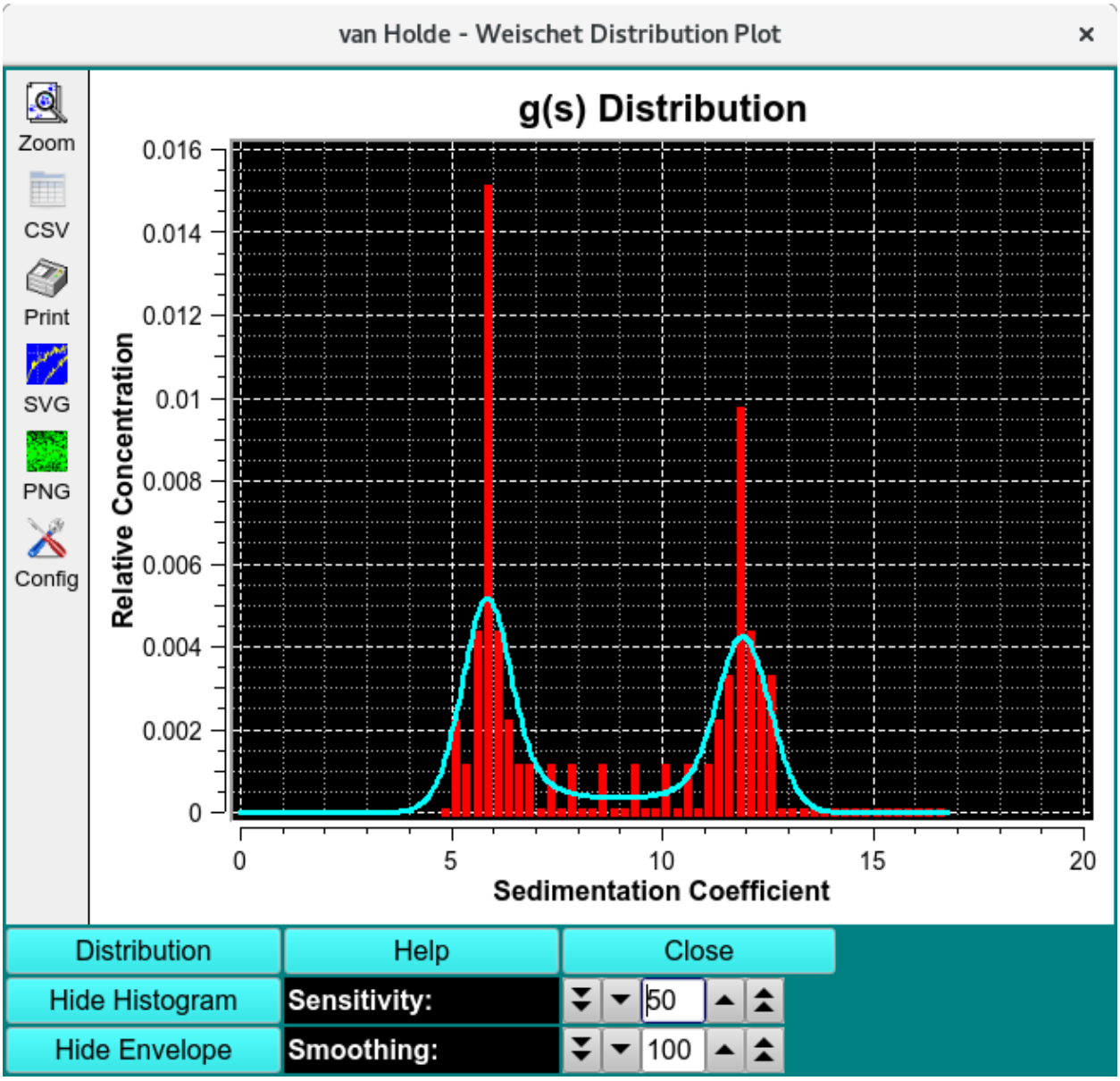
ULTRASCAN VHW DISTRIBUTION PLOT

In the van Holde-Weischet (vHW) Enhanced program, there is a Distribution Plot option that brings up a dialog in which the vHW analysis data is displayed as a distribution plot or a histogram plot. In either case, a clustering of data around several sedimentation coefficient points indicates the presence of multiple species in the solution.


van Holde - Weischet Distribution Plot
Zoom
CSV
Print
SVG
PNG
Config
G(s) Distribution
Boundary Fraction
100
80
60
40
20
0
5.0317, 25.8247
5
6
7
8
9
10
11
12
13
Sedimentation Coefficient
Histogram
Help
Close
Hide Histogram
Sensitivity:
20
Hide Envelope
Smoothing:
100

40.1 vWh Dis. Functions:

Histogram	Click on this button to change the plot from Distribution to Histogram.
Distribution	Click on this button to change the plot from Histogram to Distribution. This is the alternate label to the same button as above.
Hide Histogram	Select hiding the histogram, showing only an envelope. The alternate label for this button is **Show Histogram**
Hide Envelope	Select hiding the envelope, showing only a histogram. The alternate label for this button is **Show Envelope**
Sensitivity	Change the sensitivity factor up or down to primarily affect the nature of the histogram.
Smoothing	Change the smoothing factor up or down to primarily affect the nature of the envelope.
Help:	Display detailed vHW Distribution/Histogram Plot help.


van Holde - Weischet Distribution Plot
Zoom
CSV
Print
SVG
PNG
Config
g(s) Distribution
Relative Concentration
0.016
0.014
0.012
0.01
0.008
0.006
0.004
0.002
0
0
5
10
15
20
Sedimentation Coefficient
Distribution
Help
Close
Hide Histogram
Sensitivity:
50
Hide Envelope
Smoothing:
100

CHAPTER

FORTYONE

2-DIMENSIONAL SPECTRUM ANALYSIS

This module enables you to perform 2-dimensional spectrum analysis on a chosen experimental data set. This module fits varying pairs of s and f/f0 with constant Vbar or varying pairs of s and Vbar with constant f/f0 Upon completion of an analysis fit, plots available include: experiment; simulation; overlaid experiment and simulation; residuals; time-invariant noise; radially-invariant noise; 3-d model. Final outputs may include one or more models and noises.

41.1 2-Dimensional Spectrum Analysis Module

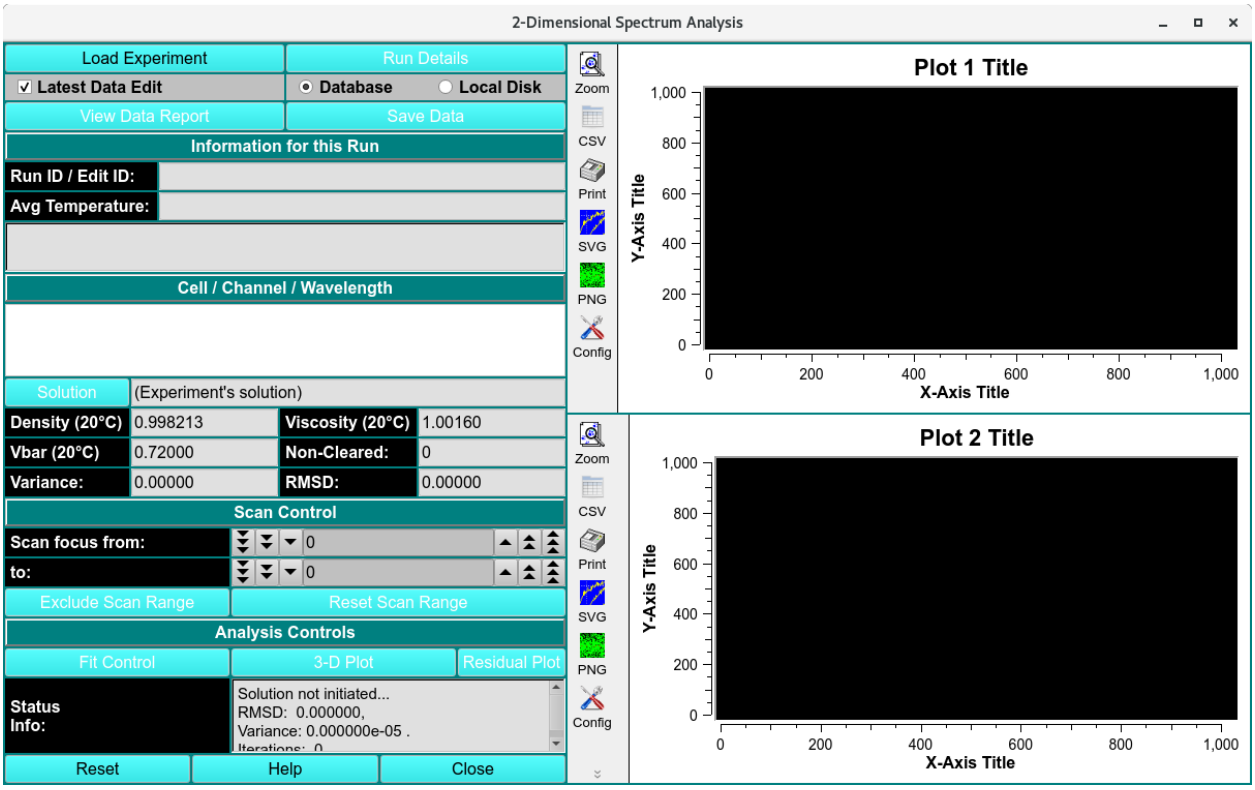
Index

- *2DSA Process:*
- *2DSA Main Window Functions:*

The 2DSA method is used for composition analysis of sedimentation velocity experiments. It can generate sedimentation coefficient, diffusion coefficient, frictional coefficient, f/f_0 ratio, and molecular weight distributions. The distributions can be plotted as 3-dimensional plots (2 parameters from the above list against each other), with the third dimension representing the concentration of the solute found in the composition analysis. The set of all such final calculated solutes form a model which are used to generate a simulation via Lamm equations. The simulation is plotted overlaying a plot of experimental data.

Every 2DSA pass can be repeated for a specified number of refinement iterations. These iterations, in turn, can be repeated for a specified number of meniscus points or Monte Carlo iterations.

Each refinement iteration proceeds over a defined grid of s and f/f_0 values. That grid is divided into subgrids as defined by a number of grid refinements in each direction.

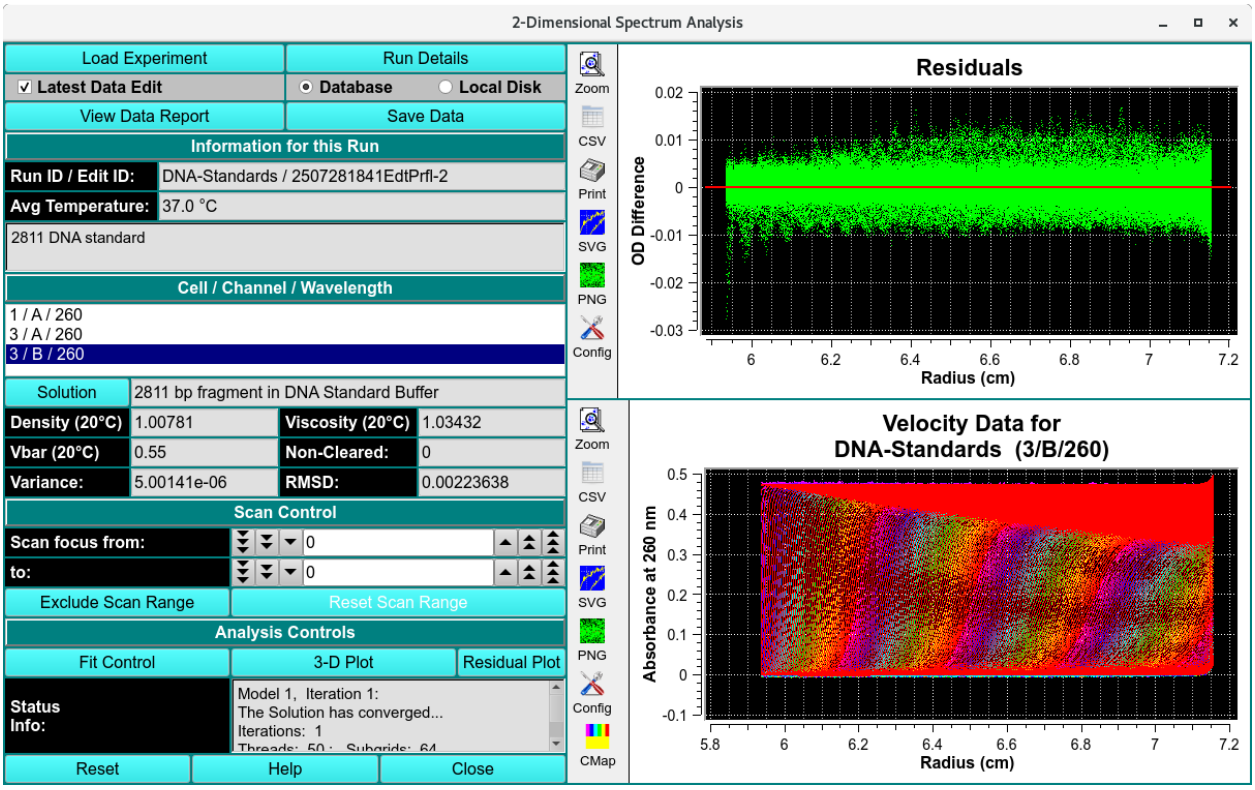


2-Dimensional Spectrum Analysis

41.1.1 2DSA Process:

- **Step 1:** First, load experimental velocity data. Click on Load Data to select an edited velocity data set from the database or from local disk.
- **Step 2:** Secondly, open an analysis control window by clicking on *Fit Control*. Within that dialog, define the grids and iterations that comprise the analysis.
- **Step 3:** Next, after having specified analysis parameters, begin the fit analysis by clicking **Start Fit**.
- **Step 4:** Display and Save Results: After simulation, a variety of options are available for displaying simulation results, residuals, and distributions. Report text files and graphics plot files can also be generated.

Once an analysis is complete, the main window will appear similar to this:



Simulated 2DSA Model

41.1.2 2DSA Main Window Functions:

Load Experiment	Click here and, in the resulting *Load Run Data Dialog* <../fe-data-loader>, select an edited data set to load.
Run Details	Pop up a dialog showing *Run Details Dialog*.
Latest Data Edit	Uncheck to allow choosing an edit other than the latest one for the raw experimental data.
Database	Select to specify data input from the database.
Local Disk	Select to specify data input from local disk.
View Data Report	Generate a report file and view it in a dialog. See the details of the report in *2DSA Process*.
Save Data	Save models and noises, as well as report and plot images files.
Run ID / Edit ID:	The Run identifier string and the Edit identifier (generally a time string) are displayed for loaded edit.
Avg Temperature:	The average temperature over all the scans of the loaded data.
(experiment description)	A text string is displayed giving a fairly detailed description of the experiment.
Cell / Channel / Wavelength	One or more rows of data edit Triples. If more than one, click on a row to select it as the data of interest.
Solution	Click this button to open a *Solution Management dialog* that allows changes to buffer and analyte characteristics of the data set.
Density (20°C)	Shows the density value for the loaded experiment. Click the Solution button to open a dialog in which density and other values may be changed.
Viscosity (20°C)	Shows the viscosity value for these loaded experiment. Click the Solution button to open a dialog in which viscosity and other values may be changed.
Vbar (20°C)	Shows the vbar value for the loaded experiment. Click the Solution button to open a dialog in which vbar and other values may be changed.
Skipped	The count of experiment data scans skipped.
Variance:	Variance value (square of RMSD) for residuals.
RMSD:	Root-Mean-Square-of-Differences for residuals.
Scan focus from:	Select a low scan of the range of values for exclusion from analysis.
to:	Select a high scan of the range of values for exclusion from analysis.
Exclude Scan Range	Initiate exclusion of the scans given in the above controls.
Reset Scan Range	Reset to the full range of scans.
Fit Control	Open a dialog to set analysis parameters and start a fit run. For details on the results of clicking this button, see :doc:`2DSA Analysis Control <2dsa_analys>.
3-D Plot	After a 2DSA model is fitted, open a control window for a :doc:`3-Dimensional plot <../3d_plot> of the final computed model.
Residual Plot	After a 2DSA model is fitted, open a plot dialog for a far more detailed set of result plots. See Finite Element Viewer/Residual Plot Dialog for further details.
Status Info:	This text window displays continually updated summaries of computational activity and results.
(upper right side plot)	Upon analysis completion, this plot is of the Residuals (Experimental minus Simulation minus any Noise).
(lower right side plot)	Upon analysis completion, this plot is of an overlay of the Experimental and Simulation data.

Window Controls

Reset	Indicate that window is reset and the plots are initiated.
Help	Display this detailed 2-Dimensional Spectrum Analysis Functions help.
Close	Close all windows and exit.

41.2 2-Dimensional Spectrum Analysis Process (2DSA)

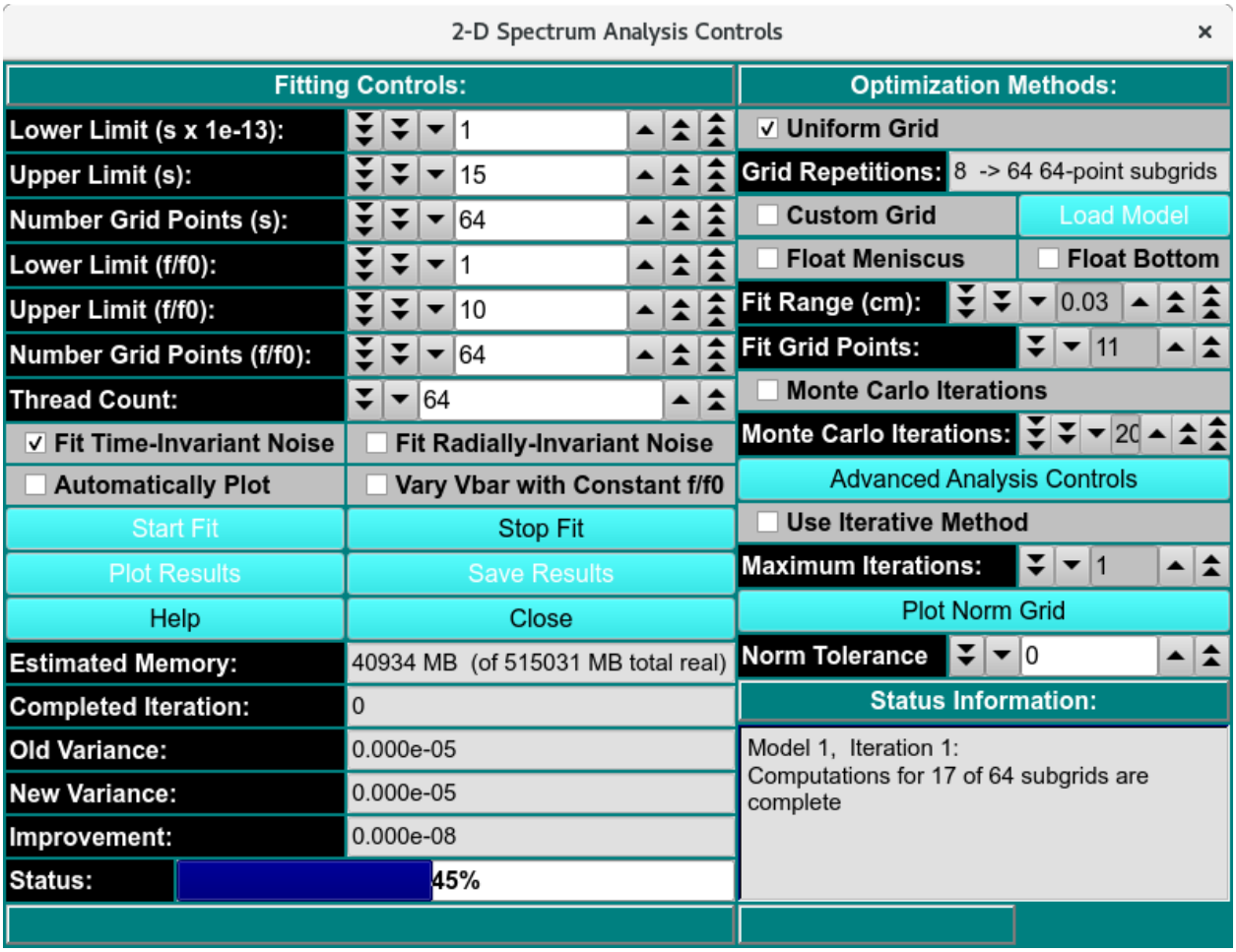
Index

- *Fitting Analysis Process:*
- *2DSA Analysis Control Functions:*
- *Fit Controls for Varying Vbar 2DSA:*
  - *Functions Specific to Varying Vbar*
- *Residual Plot Dialog*

The parameters of this dialog define and control an analysis run to find the set of solutes that best fits experimental data.

Each refinement iteration proceeds over a defined grid of s and f/f0 values. That grid is divided into subgrids as defined by a number of grid refinements in each direction. The base fit analysis pass thus defined may be repeated a specified number of times, with the second and subsequent refinement iterations adding calculated solutes from the previous iteration to subgrid solutes in each subgrid analysis.

A full set of analysis iterations may be repeated for either a range of meniscus points or a number of monte carlo iterations.



2-Dimensional Spectrum Analysis Controls

41.2.1 Fitting Analysis Process:

- **Step 1:** *Define the Grid and Grid points:* With *Fitting Controls*, define the overall solute grid by giving the lower and upper limits of s and f/f0, and the number of grid points for s and f/f0. The default of 64 grid points works well for typical sedimentation velocity datasets. Divide that total grid into subgrids using Grid Refinements.
- **Step 2:** *Set Refinement Iterations:* Secondly, set a value for Maximum Iterations of refinement passes.
- **Step 3:** *Set any Meniscus/Monte Carlo:* Thirdly, if desired, set parameters defining a Meniscus scan or a set of Monte Carlo iterations.
- **Step 4:** *Set Threads:* Next, after control values are set, define a number of threads that is appropriate to the complexity of the run and the number of processors or cores available on your machine. Typically, 8-16 threads work best for single-wavelength datasets.
- **Step 5:** *Start the Fit:* Begin the fit analysis by clicking "Start Fit".
- **Step 6:** *Display and Save Results:* After simulation, a variety of options are available for displaying simulation results, residuals, and distributions. Report text files and graphics plot files can also be generated.

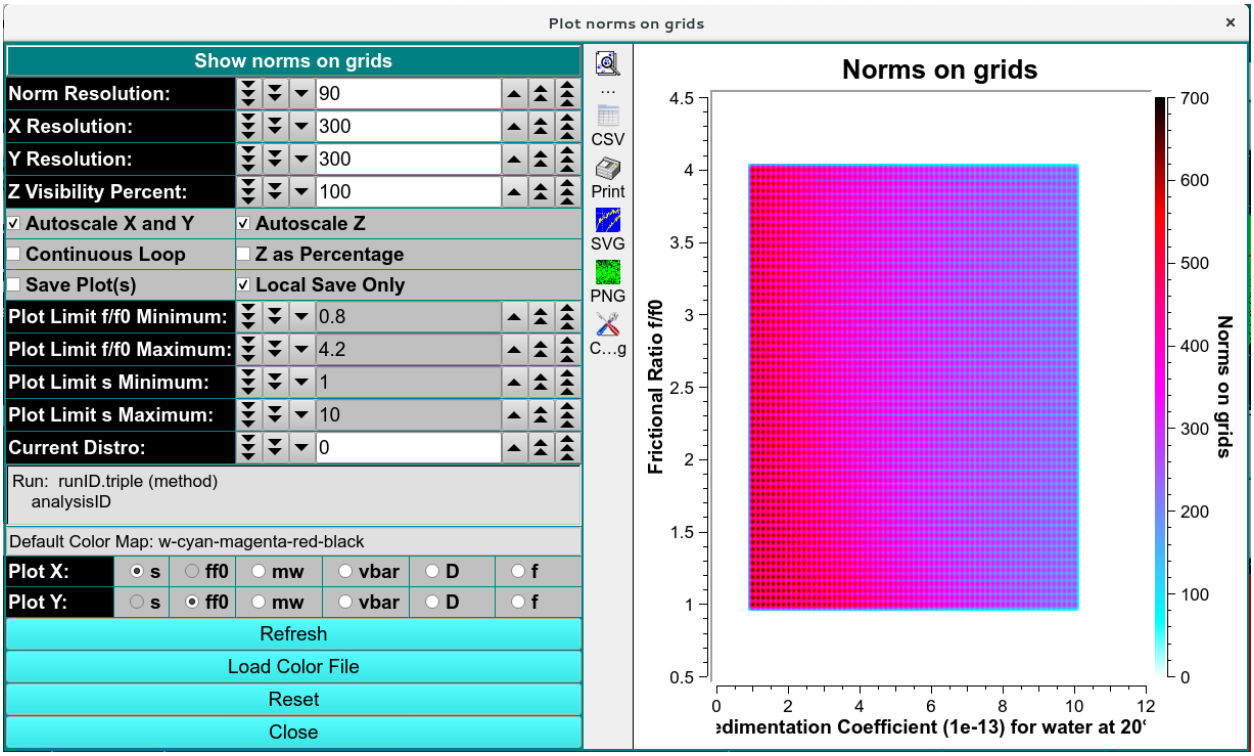
41.2.2 2DSA Analysis Control Functions:

Fitting Controls

Lower Limit (s x 1e-13):	Set a lower limit of sedimentation coefficient values to scan.
Upper Limit (s):	Set an upper limit of sedimentation coefficient values to scan.
Number Grid Points (s)	Set the total grid count of sedimentation coefficient points.
Lower Limit (f/f0):	Set a lower limit of frictional ratio values to scan.
Upper Limit (f/f0):	Set an upper limit of frictional ratio values to scan.
Number Grid Points (f/f0)	Set the total grid count of frictional ratio points.
Thread Count:	Specify by counter the number of threads to use for computations. This value is the total number of worker threads used at one time. The master thread generally has little work to do during computations, so the value may be set to your machine's total processors or cores, 8-16 threads work best for a typical single-wavelength datasets.
Fit Time-Invariant Noise	Check this box if you want to calculate time-invariant noise.
Fit Radially-Invariant Noise	Check this box if you want to calculate radially-invariant noise.
Automatically Plot	Check this box if you want plot dialogs to automatically open at the completion of all calculations.
Vary Vbar with Constant f/f0	Check this box if you want to vary vbar while holding f/f0 constant. With this box checked, the control window replaces the lower and upper limits of f/f0 with controls for the upper and lower limits of vbar.
Start Fit	Click to begin the fit analysis.
Stop Fit	If something seems wrong with the progress of analysis or if you realize you have parameterized incorrectly, click this button to abort the fit run.
Plot Results	Open 3-D and *Residual Plot Dialog* to display final results.
Save Results:	Save final model(s) and any noises generated. Also output report and plot image files.
Estimated Memory:	Text showing a memory use estimate based on chosen parameters.
Completed Iteration:	Display of the last completed refinement iteration number.
Old Variance:	The variance value for the previous iteration.
New Variance:	The variance value for the last completed iteration.
Improvement:	The difference between the variance value from the last iteration and the one preceding it.
Status:	A progress bar showing activity progress within each iteration pass.

Optimization Methods

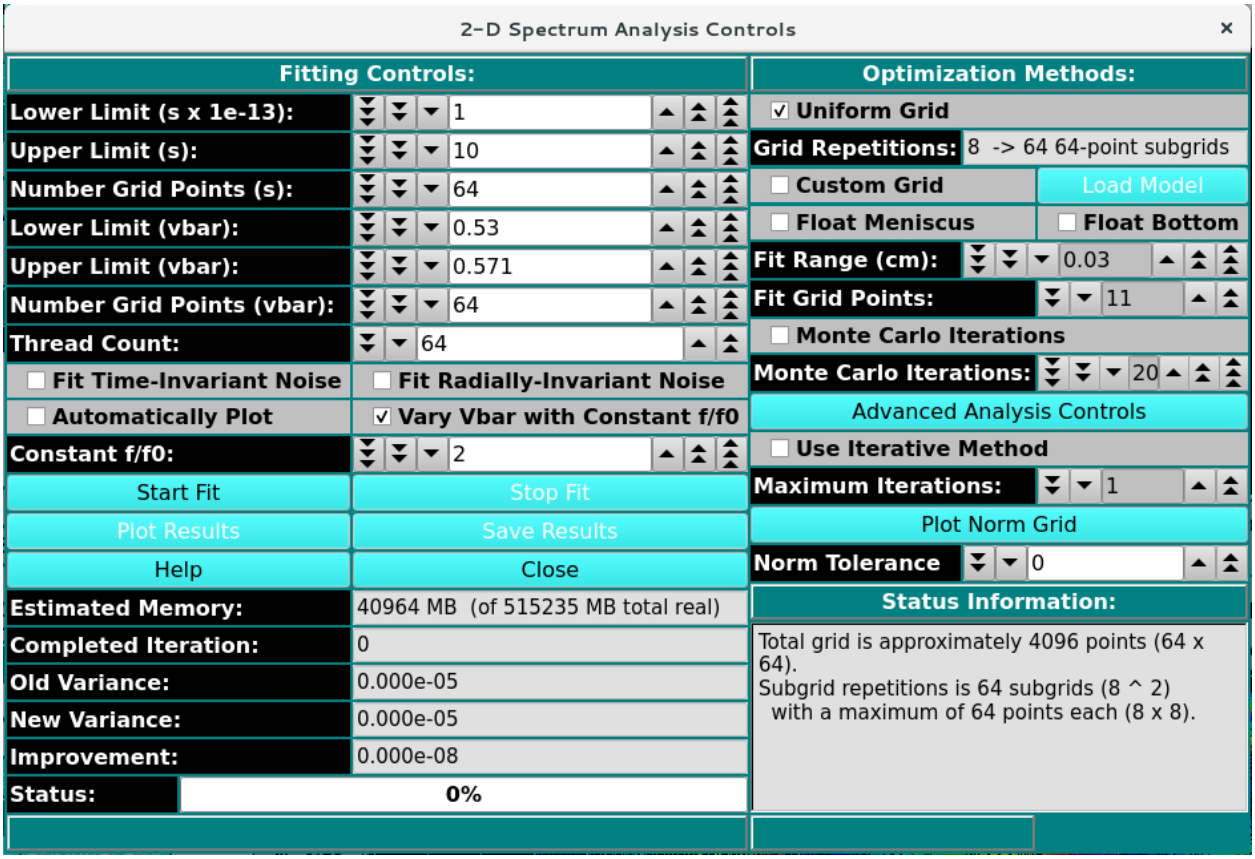
Uniform Grid	Check this box if Uniform Grid is your preferred optimization method. This is currently the only choice.
Grid Refinements:	The number of refinements (subgrid divisions) in each dimension (s and f/f0). The square of this number is the number of subgrids. The Number Grid Points given for each dimension, divided by Grid Refinements, is the approximate number of subgrid points in that dimension.
Custom Grid	Check this box to load a custom grid as the preferred optimization method. A Load Distribution dialog to load the custom grid generated by the *Custom Grid Editor (CG)*.
Float Meniscus:	Check this box if you wish to wrap the refinement iterations in outer iterations of meniscus scans. Note that this option means that Monte Carlo (below) may not be chosen.
Float Bottom:	Check this box to fit the Bottom position.
Fit Range (cm):	Select the total meniscus/bottom or both value ranges, centered around the original edited data's value, for which to perform iterations.
Grid Points:	Select the total number of meniscus/bottom points (iterations) to sample.
Monte Carlo Iterations:	Check this box if you wish to wrap the refinement iterations in an outer set of monte carlo iterations. The second and subsequent iterations use as "experiment" data input the previous iteration's simulated data with gaussian determined random variations. Note that as mentioned above, this choice and Meniscus are mutually exclusive. Monte Carlo iterations do not allow noise calculations.
Monte Carlo Iterations:	Select a number of Monte Carlo iterations to perform. A separate model is produced from each each iteration.
Advanced Analysis Controls	Click on this button to open the *2DSA Advanced Controls* dialog with control parameters not of interest to the typical user.
Use Iterative Method:	Check this box if you want to refine analysis fits with multiple refinement iterations.
Maximum Iterations:	Select the maximum number of refinement iterations. This number may not be reached if subsequent iterations achieve the same set of computed solutes or if their variances differ by a very small amount.
Plot Norm Grid:	Click to visualize the *normalized grid plot* of the 2DSA s–f/f0 grid.
Norm Tolerance:	Select the normalization tolerance of the grid.
Status Information:	The text box here is continually updated with summaries of analysis activity and iteration results.



Normalized Grid Plot of User-specified grid

41.2.3 Fit Controls for Varying Vbar 2DSA:

This variation of the 2DSA Analysis Control presents controls for varying vbar while holding frictional ratio constant.

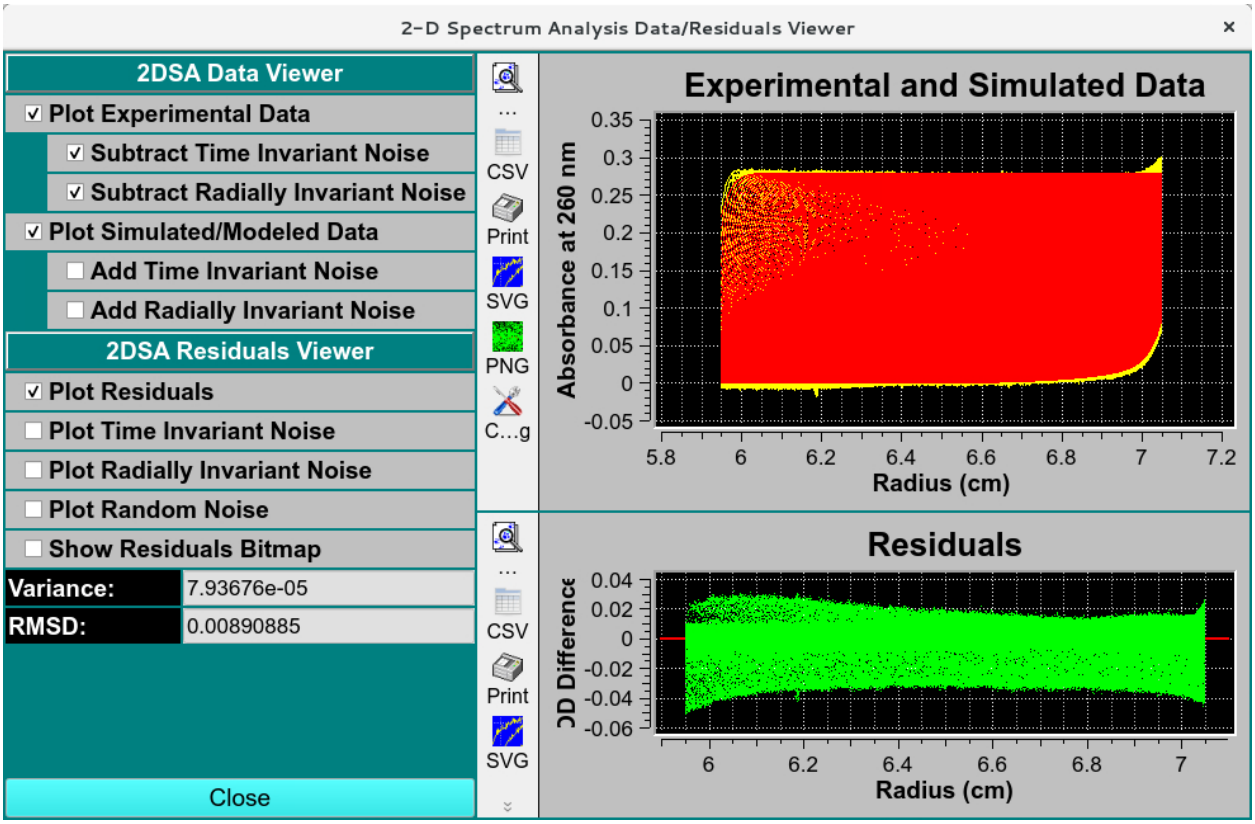


2DSA Fit Controls with Varying VBar

Functions Specific to Varying Vbar

Lower Limit (vbar):	Set a lower limit of vbar values to scan. The default is a little below the current experiment vbar value.
Upper Limit (vbar):	Set an upper limit of vbar values to scan. The default is a little above the current experiment vbar value.
Number Grid Points (vbar):	Set the total grid count of vbar points.
Constant f/f0:	Specify by counter the constant frictional ratio to use while varying the vbar values.

41.2.4 Residual Plot Dialog



2-D Spectrum Analysis Data / Residuals Viewer

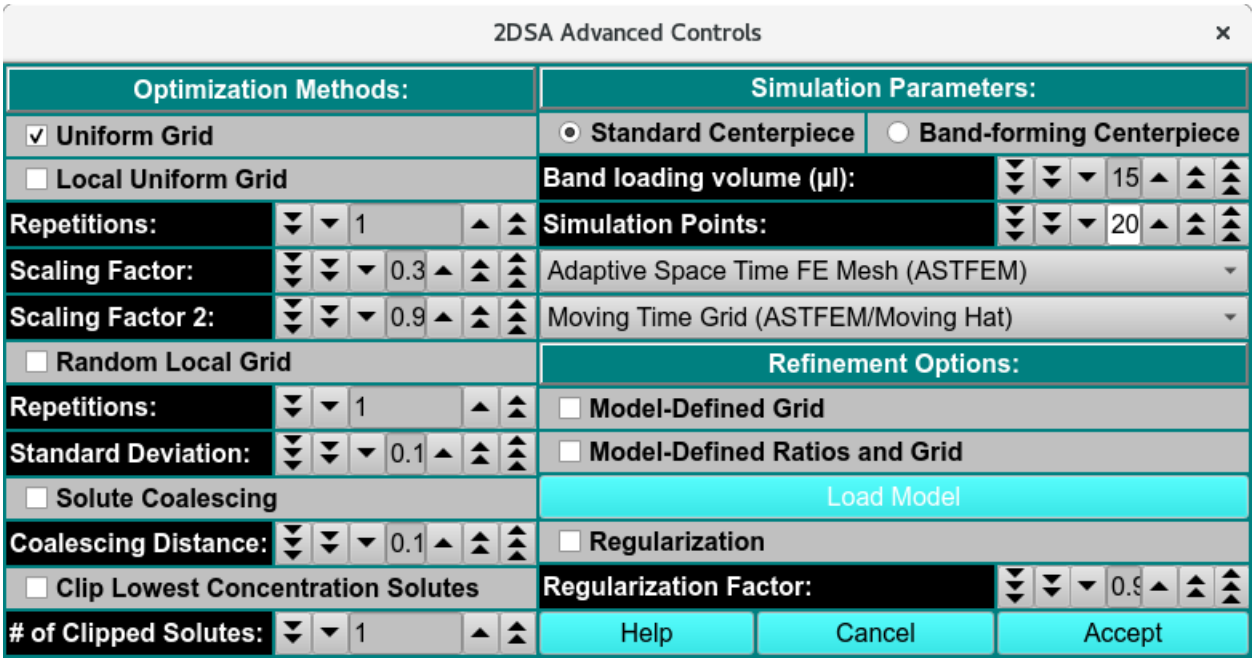
Detailed information on the results of a 2DS analysis is available at *Simulated Finite Element Model Viewer*

41.3 2DSA Advanced Analysis Control

Index

- *2DSA Adv. Functions:*

This dialog provides for definition of an analysis run with parameters not normally of interest to the typical user including use of band-forming centerpieces for Band sedimentation experiments. The advanced analysis choices presented in this dialog fall into three categories.



2DSA Advanced Controls

41.3.1 2DSA Adv. Functions:

Optimization Methods: Parameters in this section allow for methods beyond the default Uniform Grid method.

Uniform Grid	Use a uniform grid of the simulation parameter pairs.
Solute Coalescing	Select to merge solutes together.
Coalescing Distance:	Specify the distance between solutes to merge.
Clip Lowest Concentration Solutes	Select to clip low concentration solutes.
# of Clipping Solutes:	Specify the number of solutes to clip.

Simulation Parameters: Parameters here allow for changes in the base Lamm equation simulation computation method.

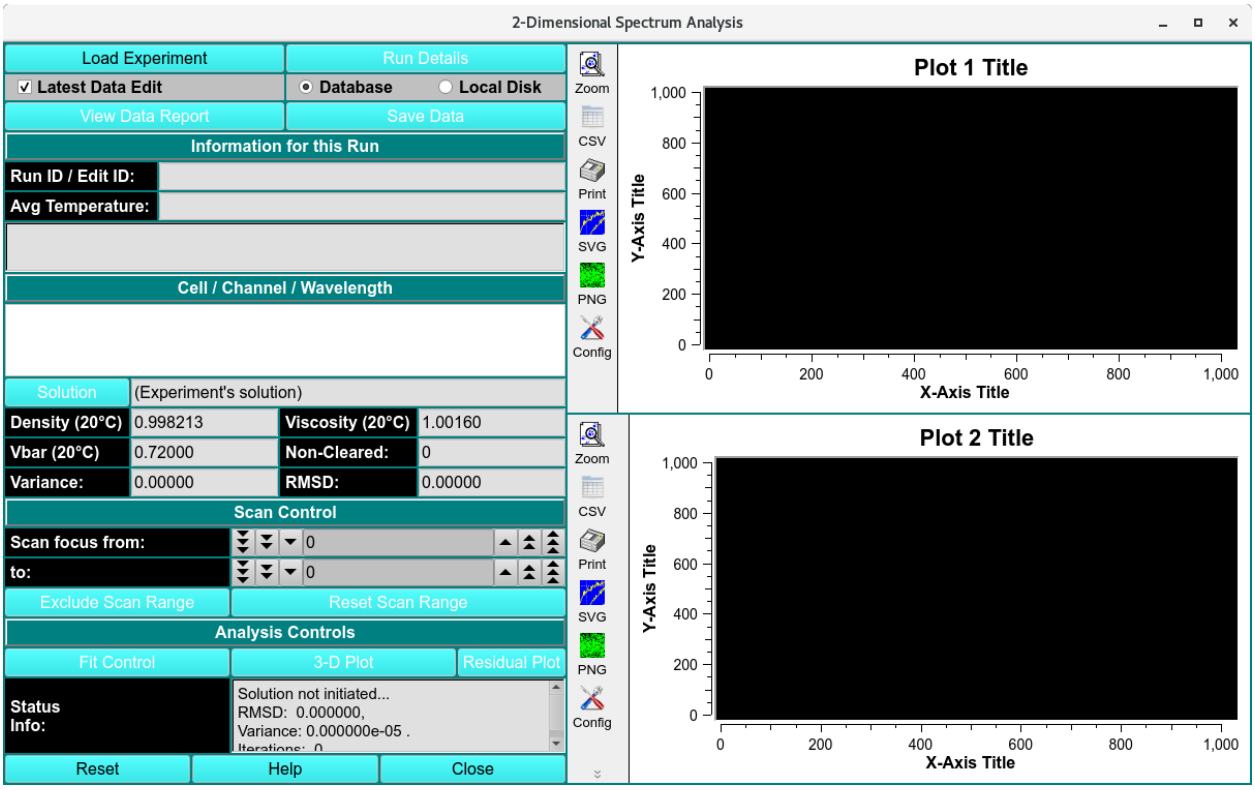
Standard Centerpiece	Check to specify a standard, non-band-forming standard 2-channel centerpiece
Band-forming Centerpiece	Check to specify a band-forming centerpiece if running a band forming sedimentation velocity experiment.
Band-loading volume (µL)	Specify the volume added to the reservoir in the band-forming centerpiece in microliters (µL).
Simulation Points:	Specify the number of discretized radial positions used when numerically integrating the Lamm equation. For typical analyses, 200 simulation points provide sufficient resolution. Increasing this value improves the precision of the simulation but also increases computational time.
(mesh type)	Select from one of several mesh types, including Adaptive Space Time (ASTFEM), Claverie, Moving Hat, user File, or Adaptive Space Volume (ASVFEM).
(grid type):	Select the grid type to use in the simulation: MOVING, FIXED.

Refinement Options: These parameters are also found in the standard analysis control dialog and the Regularization options.

Load Model	Load the model if Model-Defined grid or Model-Defined Ratios and Grid are selected.
Regularization	Apply regularization to model.
Regularization Factor:	Level of regularization to apply.

41.4 2DSA Overall Process

- **Step 1:** First, load experimental velocity data. Click on **:doc:`Load Data <../common_dialogs.html#data-loader>`_** to select an edited velocity data set from the database or from local disk.
- **Step 2:** Secondly, open an analysis control window by clicking on *Fit Control*. Within that dialog, define the grids and iterations that comprise the analysis.
- **Step 3:** Next, after having specified analysis parameters, begin the fit analysis by clicking "Start Fit".
- **Step 4:** Display and Save Results: After simulation, a variety of options are available for displaying simulation results, residuals, and distributions. Report text files and graphics plot files can also be generated.



2-Dimensional Spectrum Analysis

CHAPTER

FORTYTWO

PARAMETRICALLY CONSTRAINED SPECTRUM ANALYSIS

This module enables users to perform parametrically constrained spectrum analysis on a chosen experimental data set. The PCSA module fits sedimentation velocity data using linear or sigmoidal constraints to model the relationship between the s and f/f0 of heterogenous macromolecule mixtures. This module is ideal for interacting macromolecules and growing polymer chain systems. Upon completion of an analysis fit, plots available include: model lines; experiment; simulation; overlaid experiment and simulation; residuals; time-invariant noise; radially-invariant noise; 3-d model. Final outputs may include a model and computed noises.

fe-data-loader>`========== Parametrically Constrained Spectrum Analysis ==========

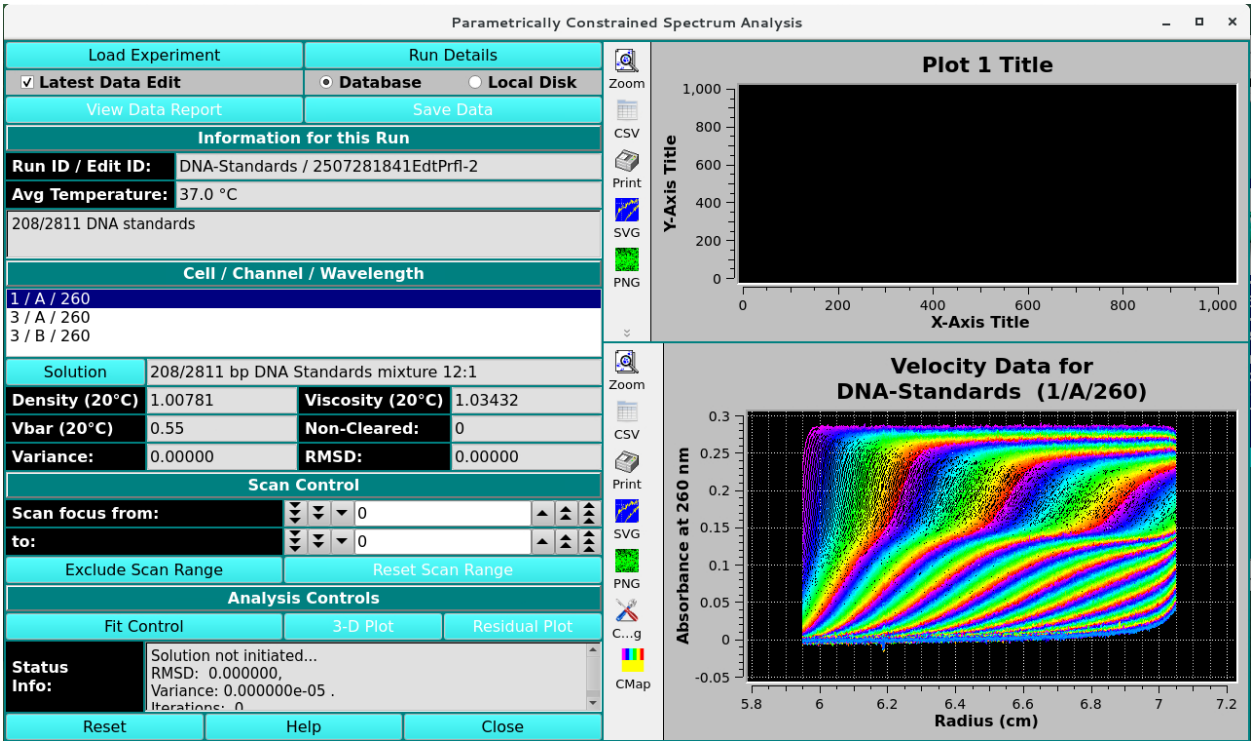
Index

- *PCSA Process:*
- *PCSA Functions:*

The PCSA method is used for **composition analysis** of sedimentation velocity experiments. It can generate *sedimentation coefficient*, *diffusion coefficient*, *frictional coefficient*, *f/f0 ratio*, and *molecular weight distributions*. The distributions can be plotted as *3-dimensional plots* <*3d_plot*> (2 parameters from the above list against each other), with the third dimension representing the concentration of the solute found in the composition analysis. The set of all such final calculated solutes form a model which is used to generate a simulation via Lamm equations. The simulation is plotted overlaying a plot of experimental data.

The PCSA pass proceeds for a set of models each of which consists of the solute points along a curve in s,f/f0 space. The model whose RMSD of the resulting residuals (simulation-experimental difference) is the lowest forms the starting point for a second phase which uses Levenberg-Marquardt to refine the model to a final output model. The set of initial curves is specified by a s and f/f0 ranges and a direct or implied number of variations in f/f0 end-points or sigmoid par1, par2 values. The type of curve used may be any of the following.

- **Straight Line**
- **Increasing Sigmoid**
- **Decreasing Sigmoid**
- **Horizontal Line [C(s)]**



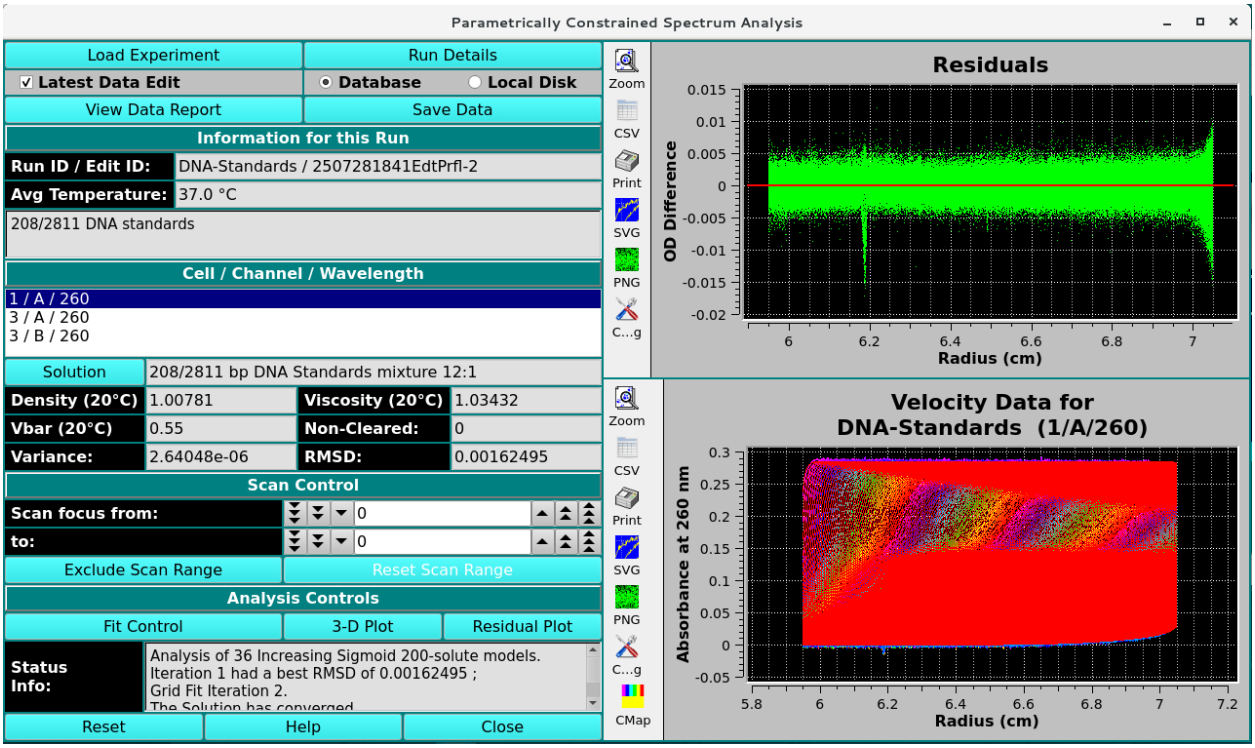
PCSA Simulation Window

42.1 PCSA Process:

- **Step 1:** *Load Experimental Data:* First, load experimental velocity data. Click on "Load Data" to select an edited velocity data set from the database or from local disk.
- **Step 2:** *Define the Analysis Fit:* Secondly, open an *Analysis control window* by clicking on "Fit Control". Within that dialog, define the ranges and counts that comprise the analysis.
- **Step 3:** *Start the Fit:* Next, after having specified analysis parameters, begin the fit analysis by clicking "Start Fit".
- **Step 4:** *Display and Save Results:* After simulation, a variety of options are available for displaying simulation results, residuals, and distributions. Report text files and graphics plot files can also be generated.

Upon completion of an analysis fit, plots available include: model lines; experiment; simulation; overlaid experiment and simulation; residuals; time-invariant noise; radially-invariant noise; 3-d model. Final outputs may include a model and computed noises. A detailed explanation of the results are found in *Parametrically Constrained Spectrum Analysis Results <pcsa_results>*

The main PCSA window is shown here in the state it appears after the completion of a fit run.



Fitted PCSA Simulation

42.2 PCSA Functions:

Load Experiment	Click here and, in the resulting **:doc:`Load Run Data Dialog <common_dialogs.html#data-loader>`_**, select an edited data set to load.
Run Details	Pop up a dialog showing Run Details Dialog.
Latest Data Edit	Uncheck to allow choosing an edit other than the latest one for the raw experimental data.
Database	Select to specify data input from the database.
Local Disk	Select to specify data input from local disk.
View Data Report	Generate a report file and view it in a dialog. See the details of the report in **:doc:`PCSA Process Results <pcsa_results.html#pcsa-report>`_**.
Save Data	Save models and noises, as well as report and plot images files.
Run ID / Edit ID:	The Run identifier string and the Edit identifier (generally a time string) are displayed for loaded edit.
Avg Temperature:	The average temperature over all the scans of the loaded data.
(experiment description)	A text string is displayed giving a fairly detailed description of the experiment.
Cell / Channel / Wavelength	One or more rows of data edit Triples. If more than one, click on a row to select it as the data of interest.
Solution	Click this button to open a Solution Management dialog that allows changes to buffer and analyte characteristics of the data set.
Density (20°C)	Shows the density value for the loaded experiment. Click the Solution button to open a dialog in which density and other values may be changed.
Viscosity (20°C)	Shows the viscosity value for these loaded experiment. Click the Solution button to open a dialog in which viscosity and other values may be changed.
Vbar (20°C)	Shows the vbar value for the loaded experiment. Click the Solution button to open a dialog in which vbar and other values may be changed.
Skipped	The count of experiment data scans skipped.
Variance:	Variance value (square of RMSD) for residuals.
RMSD:	Root-Mean-Square-of-Differences for residuals.
Scan focus from:	Select a low scan of the range of values for exclusion from analysis.
to:	Select a high scan of the range of values for exclusion from analysis.
Exclude Scan Range	Initiate exclusion of the scans given in the above controls.
Reset Scan Range	Reset to the full range of scans.
Fit Control	Open a dialog to set analysis parameters and start a fit run. For details on the results of clicking this button, see *PCSA Analysis Control*.
3-D Plot	After a PCSA model is fitted, open a control window for a 3-Dimensional plot of the final computed model.
Residual Plot	After a PCSA model is fitted, open a plot dialog for a far more detailed set of result plots. See **:doc:`PDSA Residual Plot Dialog <pcsa_results.html#pcsa-data-residuals-viewer>`_** for further details.
Status Info:	This text window displays continually updated summaries of computational activity and results.
Status Info	This text window displays continually updated summaries of computational activity and results.
(upper right side plot)	Upon analysis completion, this plot is of the Residuals (Experimental minus Simulation minus any Noise).
(lower right side plot)	Upon analysis completion, this plot is of an overlay of the Experimental and Simulation data.

Window Controls

Reset	Indicate that window is reset and the plots are initiated.
Help	Display this detailed Parametrically Constrained Spectrum Analysis help.
Close	Close all windows and exit.

42.3 Parametrically Constrained Spectrum Analysis Fitting Controls

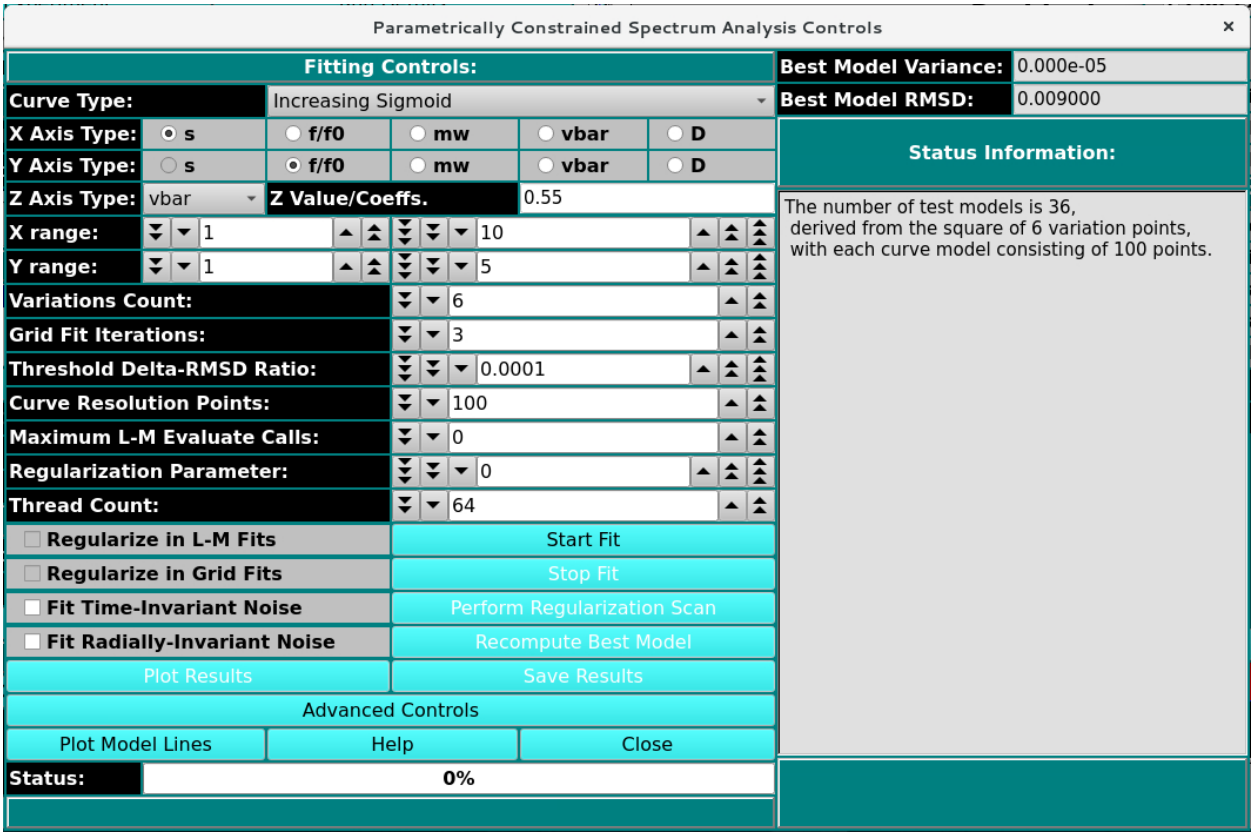
Index

- *PCSA Fitting Process:*
- *Fitting Controls Functions:*

The parameters of this dialog define and control an analysis run to find the set of solutes that best fits experimental data. Each analysis run proceeds over a defined set of curves in s and f/f0 space. The single analysis pass produces the model whose associated simulation differs the least from the experimental data, as determined by its RMSD value. Time-invariant and/or radially-invariant noise(s) may also be computed. The curves in a set cover a specified s and f/f0 range and vary according to a specified increment or count. Each model has a specified number of input solute points along its curve.

After an initial pass in which all the specified model curves are evaluated, the results are sorted by fitness (least RMSD). The best model then becomes the starting point for a second pass that utilizes Levenberg-Marquardt to refine the model. The result of that second pass is inserted as the final best model. For a straight line type, a specified variation count is given. Lines cover the full s range, with end points along the f/f0 range varying the specified number of times. Each implied starting point connects with each of the possible end points. So the square of the implied number of f/f0 end points is the number of models analyzed. If the type is Horizontal Line, the f/f0 value is the same for start and end points, so the number of models is the variations count itself (not its square).

For sigmoid types, a variation count is directly specified. The curves cover the full s and f/f0 ranges, with "par1" and "par2" values each varying the specified number of times, yielding a number of test models equal to the square of that count. The par1 value changes logarithmically from 0.001 to 0.5; and par2 changes linearly from 0.0 to 1.0.

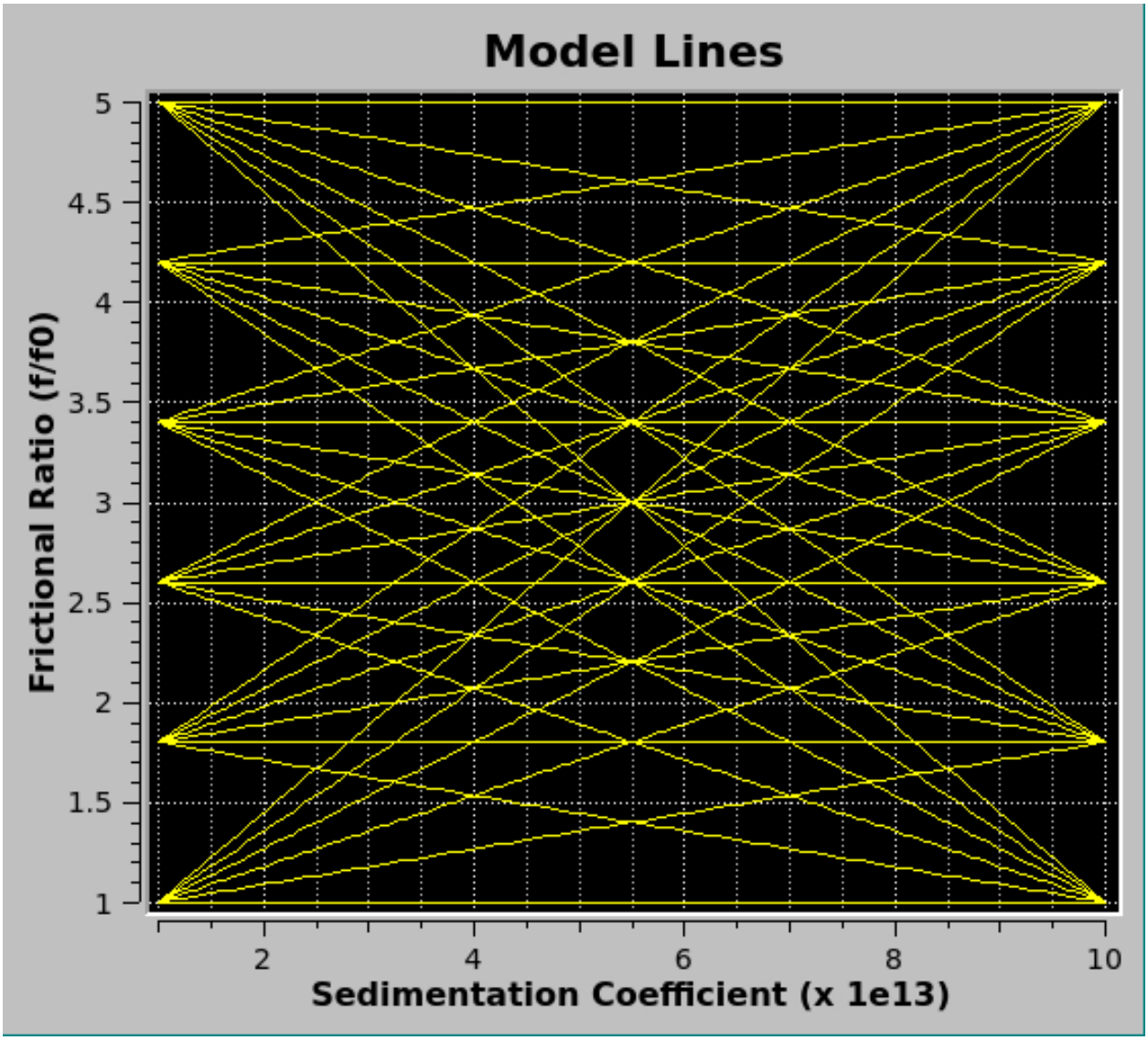


PCSA Fitting Controls

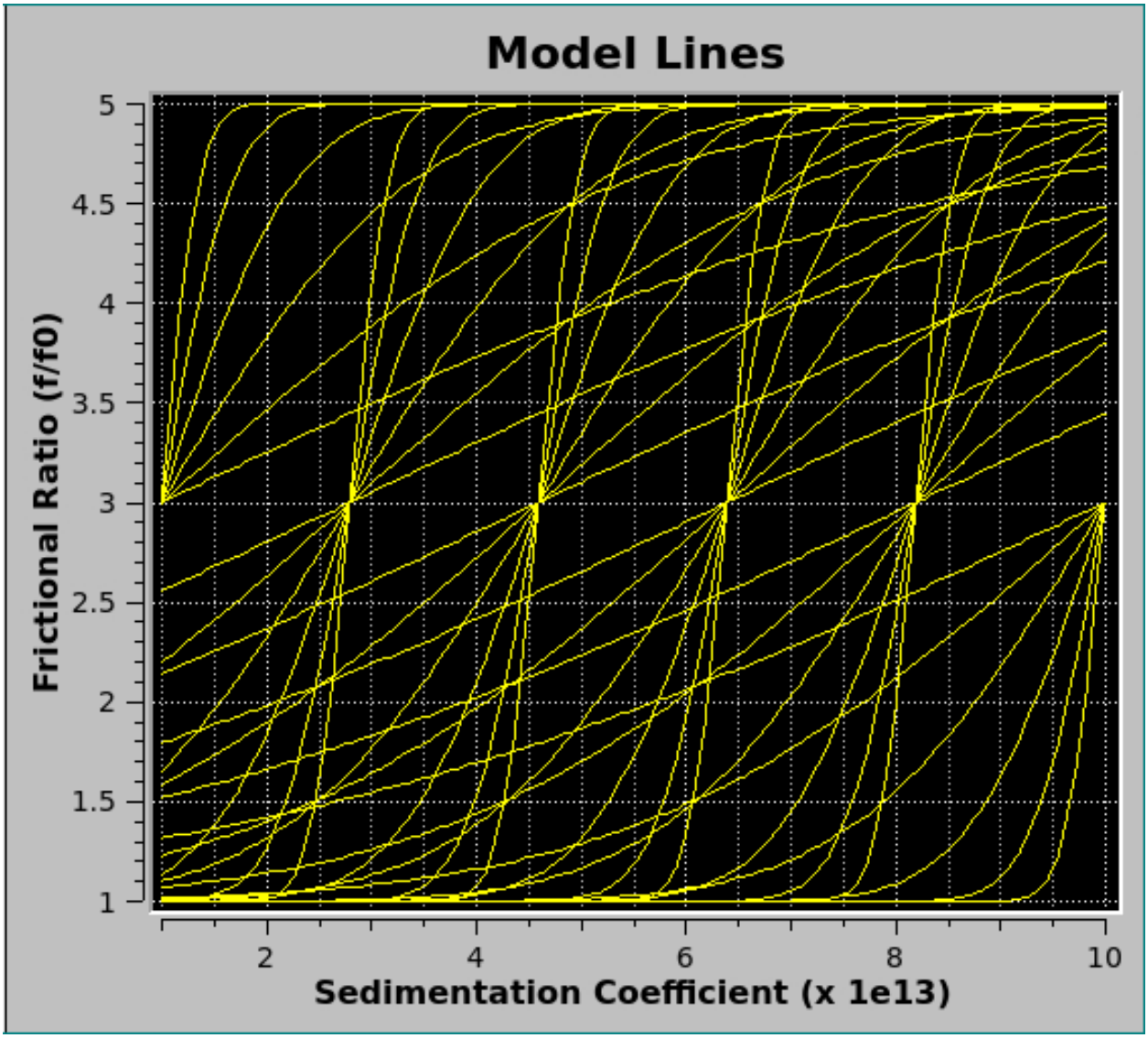
42.3.1 PCSA Fitting Process:

- **Set the Curve Type:** First, specify the type of curve you expect the species to follow. Choices are:

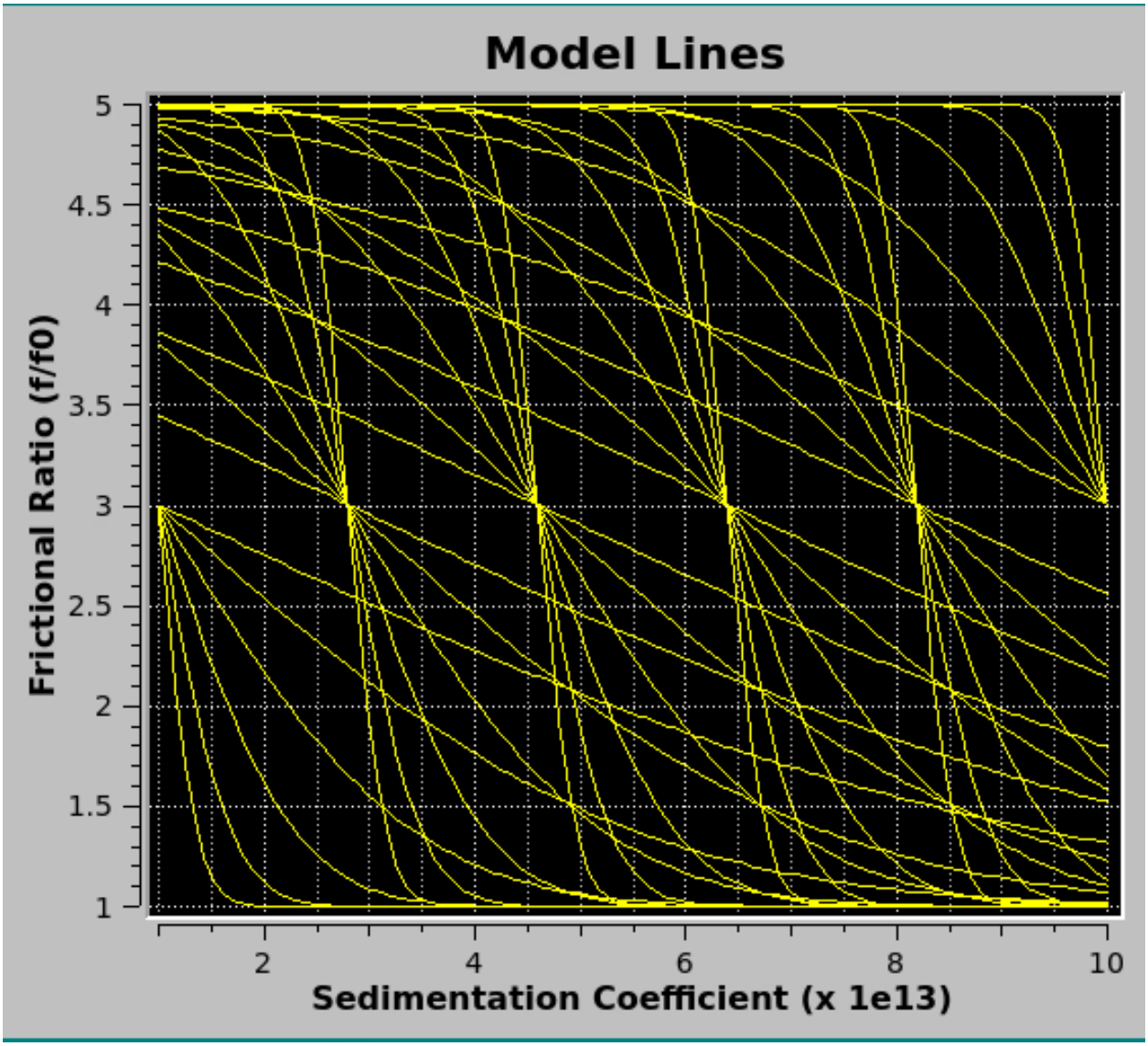
Straight Line	Increasing Sigmoid	Decreasing Sigmoid	Horizontal Line [C(s)]	Second-Order Power Law



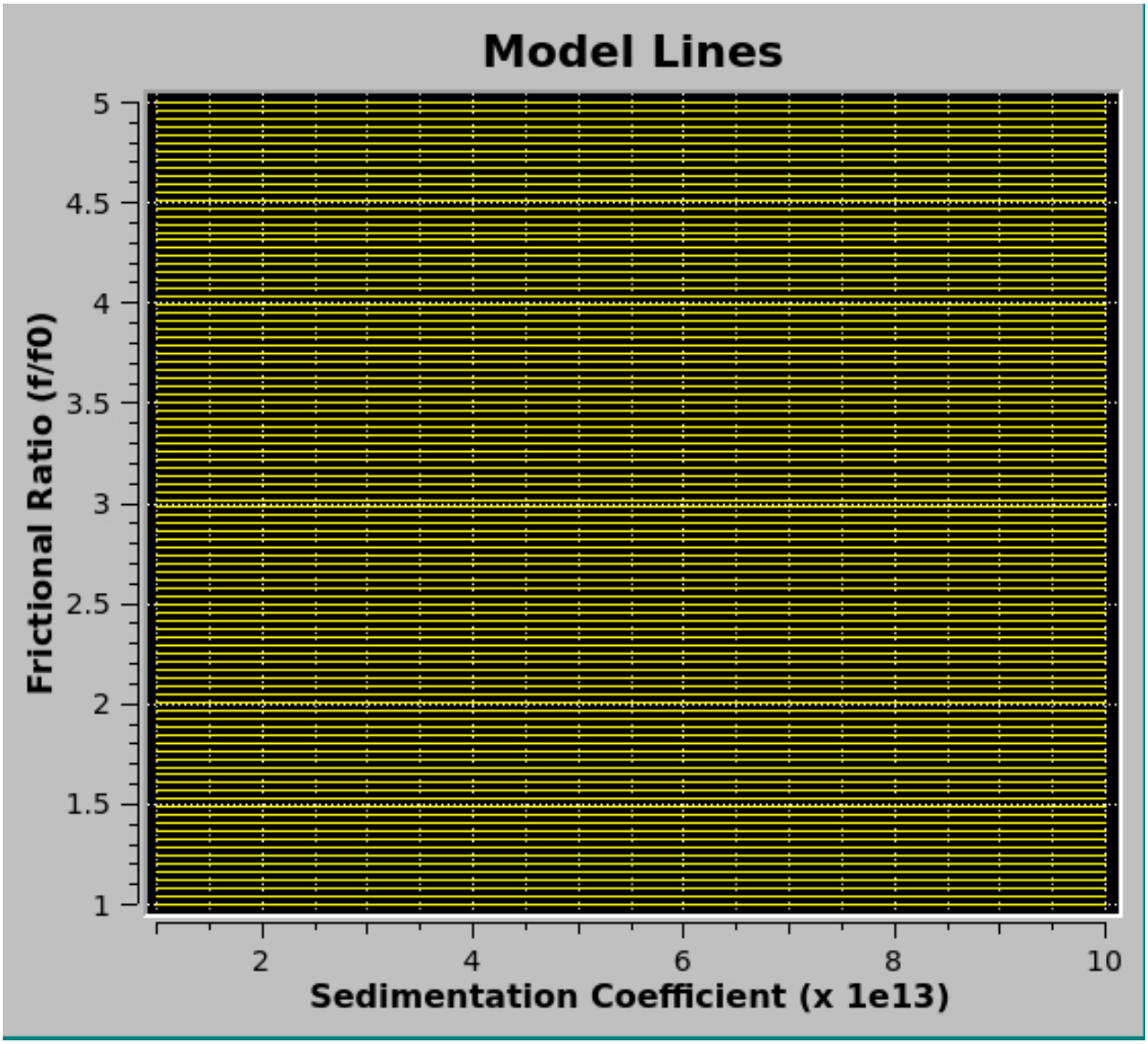
Straight Line



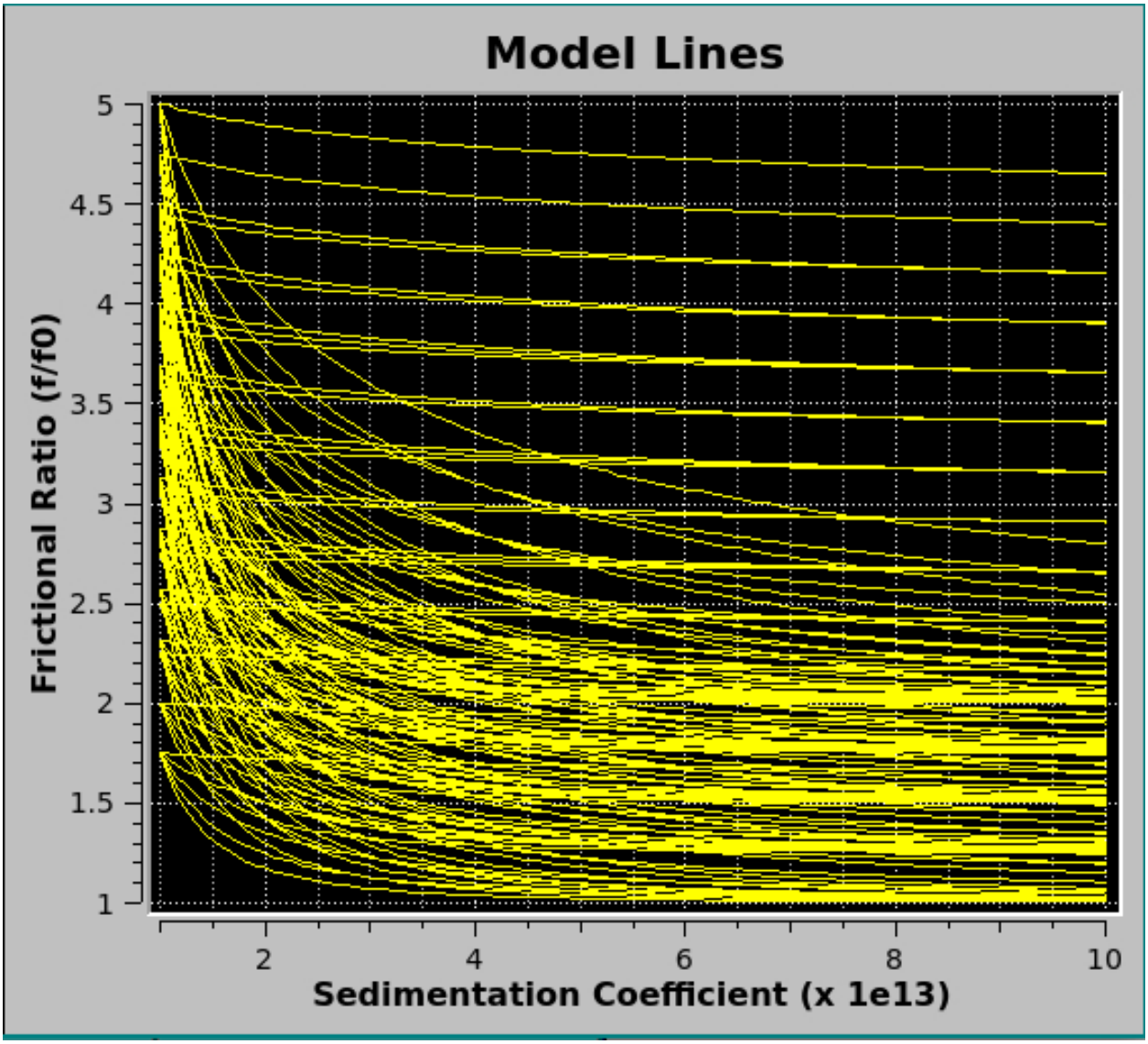
Increasing Sigmoid



Decreasing Sigmoid



Horizontal Line (Cs)

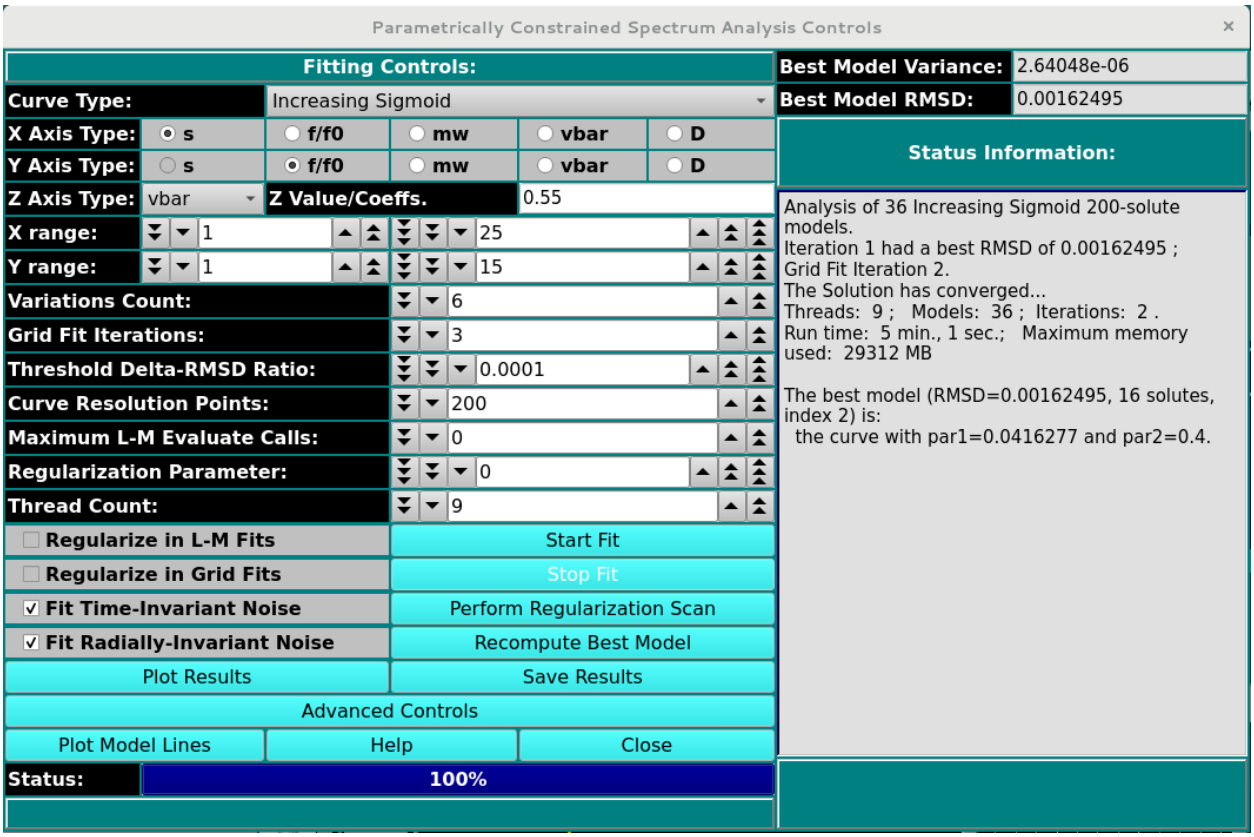


Power Law

Example lines of PCSA Fitting Curves

- **Define the Ranges:** Next, define the s and f/f0 ranges; and directly (sigmoids) or indirectly (lines) give the number of variations in each dimension.
- **Specify Resolution Points:** Specify how many solute points exist on each curve.
- **Set Threads:** Next, after control values are set, define a number of threads that is appropriate to the complexity of the run and the number of processors or cores available on your machine.
- **Start the Fit:** Begin the fit analysis by clicking "Start Fit".
- **Display and Save Results:** After simulation, a variety of options are available for displaying simulation results, residuals, and distributions. Report text files and graphics plot files can also be generated.
- **Regularize:** If you wish to perform Tikhonov Regularization, you can set a Regularization Parameter (Alpha) or determine a good parameter value by clicking "Perform Regularization Scan". When you

have a reasonable Alpha, click on Regularize Current Model.



Updated PCSA Fitting Controls after Simulation completed

Control dialog parameters differ slightly in interpretation, based on whether the curve type is a line or a sigmoid.

- If straight line is specified, the curve variation count gives the number of start and end point f/f0 variations.
- If horizontal line is specified, curve variations gives the total models, since start and end points always have the same f/f0 value.
- If one of the sigmoid types is specified, the specified number of variations is in both the par1 and par2 values of the sigmoid function:

$$Y = Y_{start} + Y_{diff} \times 0.5 \times erf\left(\frac{(x - par\,2)}{\sqrt{(2.0 \times par\,1)}}\right) + 0.5$$

Note: Notes on Fit Stages and Regularization:

- **During all model fitting runs, at least two fit stages occur:**

  - **Fixed Stage:** The models following fixed curves that were defined are evaluated and results ordered by best fitness (least RMSD).
  - **LM Stage:** The best model from the previous stage forms the starting point for a Levenberg-Marquardt fit sequence that results in a refined model being found. This refined model gets inserted at the top of the ordered models list.

- **If you wish to apply Tikhonov regularization in order to smooth concentration peaks in the final model, up to two additional stages may occur:**
  - **Scan Stage:** Starting with the best model from the LM stage, a range of Regularization Parameter (Alpha) values is tested in a separate dialog to derive an optimal Alpha.
  - **Final Stage:** The result of the LM stage is used as a starting point for one final model computation, this time using a given Alpha.
- **Where regularization is to be employed, then, the two most common run sequences are:**
  - **Fixed+LM+Scan+Final** – where the optimal Alpha is unknown and needs to be determined, then applied.
  - **Fixed+LM+Final** – where the optimal Alpha is already known or was determined in a previous sequence. This sequence may be employed after a …Scan… sequence that used moderate Variations Count and Curve Resolution Points for speed. Those Count and Points values may now be increased for accuracy and this Fixed+LM+Final sequence performed.
- **Note that clicking the Start Scan button really initiates a Fixed+LM+Scan sequence.**
  - When it completes, clicking on the Final Fit button results in the Final stage being performed.
- **Note that where regularization is not to be employed (Alpha=0), the only run sequences that will be used is:**
  - Fixed+LM – where the final LM model is the desired overall final model.

A detailed explanation of the PCSA specific results can be found in :doc:`Parametrically Constrained Spectrum Analysis Results <pcsa_results>

42.3.2 Fitting Controls Functions:

Curve Type:	Select from the list of types; either a line or one of the sigmoid types.
Lower Limit (s x 1e-13):	Set a lower limit of sedimentation coefficient values to scan.
Upper Limit (s):	Set an upper limit of sedimentation coefficient values to scan.
Lower Limit (f/f0):	Set a lower limit of frictional ratio values to scan.
Upper Limit (f/f0):	Set an upper limit of frictional ratio values to scan.
Variations Count:	Specify the number of curve variations in each direction to form the models. Usually, the square of this value is the number of models to analyze. The number of variations is applied to each of "par1" and "par2" in the sigmoid computations. In the Horizontal Line case, this value is the number of models.
Grid Fit Iterations:	Specify the number of iterations of fixed grid fits to perform. Each iteration after the first starts with a range of par1 and par2 values or f/f0 values that is refined from the best models of the previous iteration.
Threshold Delta-RMSD Ratio:	Specify the ratio of the delta in RMSD to previous iteration RMSD that is the threshold below which the iterative scan is taken to have converged.
Curve Resolution Points:	Specify the number of points to generate for each test line. This is the number of solute points that each model will have.
Maximum L-M Evaluate Calls:	Specify the maximum number of Levenberg-Marquardt evaluate calls to allow. The default value of 100 is almost always sufficient for L-M to converge. This value is used to insure that a run-away condition will not occur. Additionally, a value of "0" may be specified to suppress L-M and use the final fixed grid best model as the overall best model.
Thread Count:	Specify by counter the number of threads to use for computations. This value is the total number of worker threads used at one time. The master thread generally has little work to do during computations, so the value may be set to your machine's total processors or cores, 8-16 threads work best for a typical single-wavelength datasets.
Start Fit	Click to begin the fit analysis. The analysis will proceed through fixed curve analysis followed by Levenberg-Marquardt refinement, resulting in either a final model or the base model from which regularization may proceed.
Stop Fit	If something seems wrong with the progress of analysis or if you realize you have parameterized incorrectly, click this button to abort the fit run.
Perform Regularization Scan	Once fit analysis is complete, you may optionally click here to bring up a :doc:`Regularization Parameter Scan <pcsa_rpscan> dialog that sets the best value for Regularization Parameter.
Regularize Current Model	Any time you set the Regularization Parameter, either manually or through scan, you may then compute a final model, using that parameter, by clicking here.

Note: If the parameter is set to zero, the **Regularize Current Model** button relabeled to "Unregularize Current Model".

Regularize in L-M Fits	This box should normally remain unchecked, but you may check it in order to test regularization during the Levenberg-Marquardt stage of model fitting.
Regularize in Fixed Fits	This box should normally remain unchecked, but you may check it in order to test regularization during both the fixed-curves and L-M stages of model fitting.
Fit Time-Invariant Noise	Check this box if you want to calculate time-invariant noise.
Fit Radially-Invariant Noise	Check this box if you want to calculate radially-invariant noise.
Plot Results	Open 3-D and Residual plot dialogs to display final results.
Save Results:	Save final model(s) and any noises generated. Also output PCSA report and plot image files.
Advanced Controls	Click on this button to open the advanced controls dialog
Plot Model Lines	Click on this button to open a dialog showing the Model Lines specified (such as *Sigmoid Models*).

Note: Note that the dialog is automatically opened upon analysis completion. But this button can be used to re-display completed model lines if the dialog has been closed by the user. The dialogs after analysis include color-coded illustrations of best and near-best models: *Fitted Models*.

Status:	A progress bar showing activity progress within each iteration pass.
Best Model Variance:	The minimum variance value for models thus far computed is continually updated.
Best Model RMSD:	Simularly, the minimum RMSD value for models thus far computed is continually updated.

Note: Note that at completion of all model computations, this box and the one above will hold values for the final best model, which might not be a minimum (when regularizing, for example).

Status Information:	The text box here is continually updated with summaries of analysis activity and results.

Window Controls

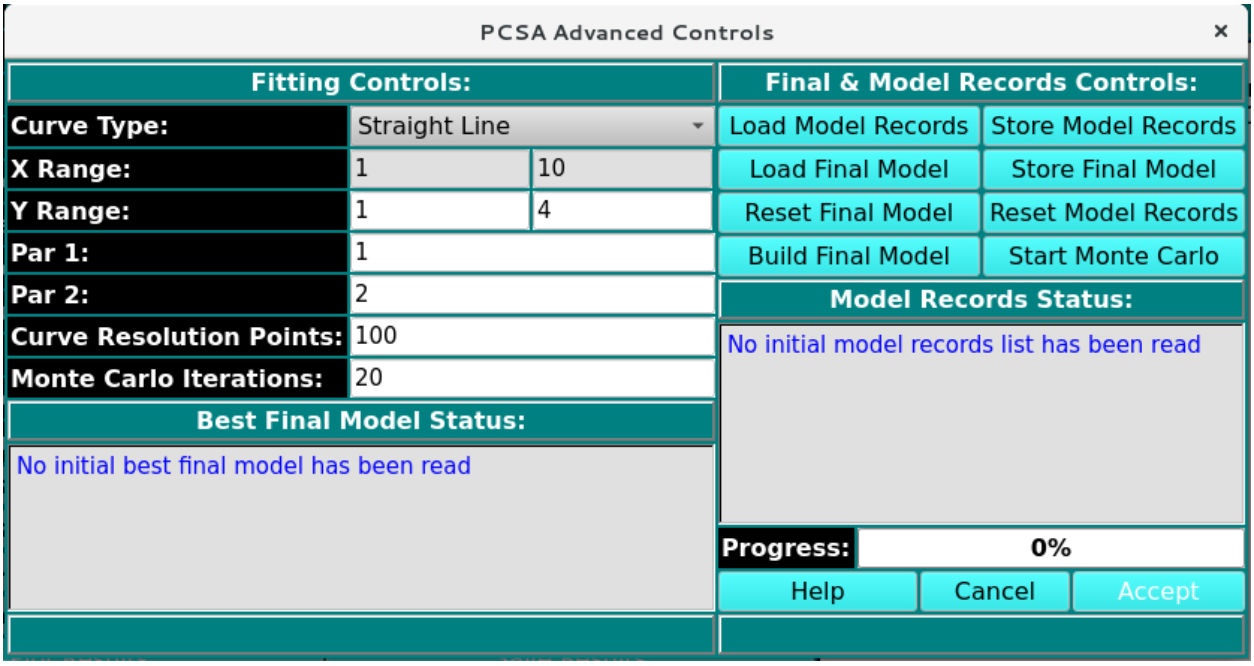
Reset	Indicate that window is reset and the plots are initiated.
Help	Display this detailed PCSA Fitting Controls help.
Close	Close all windows and exit.

42.4 PCSA Advanced Controls

Index

- *Advance Controls Functions:*

This dialog allows user manually adjust the Y range, par1/par2, the Curve resolution points and perform monte carlo analysis on the Best Fitted Model (BFM).



PCSA Advanced Controls

42.4.1 Advance Controls Functions:

Fitting Controls

Curve Type:	The module take the specifies curve type selected in the *PCSA Fit Controls* Window.
X Range	The module take the X parameter range specified in the *PCSA Fit Controls* Window.
Y Range	Adjust the Y parameter range
Par1	Adjust parameter 1 of the selected curve.
Par2	Adjust parameter 2 of the selected curve.
Curve Resolution Points	Adjust the number of points used to simulate along the curve selected.
Monte Carlo Iterations:	Set the number of monte carlo interations to perform
(Best Final Model Status)	Update the status of the window.

Final and Model Records Controls

Load Model Records	Load all the model records of the simulated data.
Load Final Model	load the model with the lowest RMSD.
Reset Final Model	Reset the loaded final model
Build Final Model	Re-simulated the final model withthe lowest RMSD
Store Model Records	Save all model records of the latest simulated data
Store Final Model	Save model with the lowest RMSD of the latest simulated data
Reset Model Records	Reset the loaded model record.
Start Monte Carlo	Start the
(Model Record Status)	
Progress	
Help	Display this detailed help.
Cancel	Close the dialog window and exit.
Accept	Close the dialog window, communicate the current Alpha value to the caller, and exit.

42.5 Regularization Parameter Scan and Plot for PCSA

Index

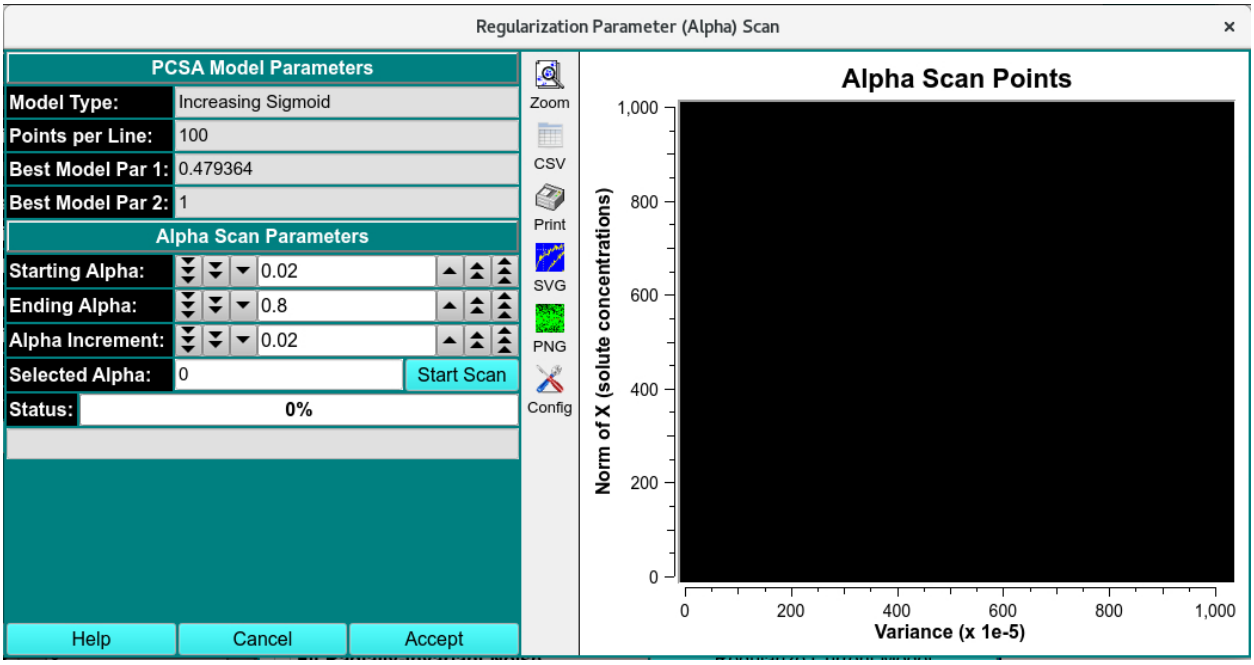
- *Sample Regularization Parameter Scan Dialogs:*
- *Regularization Parameter Functions:*

This dialog provides the means to perform a scan of possible regularization parameter (alpha) values. Regularization is done once the Best Fit Model (BFM) has been generated. The solution can include false-positive contributions from the stochastic noise in the data. Tikhonov regularization (TR) minimizes the contributions of the false positive components by adding a regularization parameter (alpha) - a number between 0.0 and 1.0 - where 0 has no regularization and 1 has maximum regulariation at the expense of goodness of fit. The final selected Alpha value is passed back to the Analysis Control dialog of PCSA. A full Parametrically Constrained pass can then be performed for the constant Alpha value.

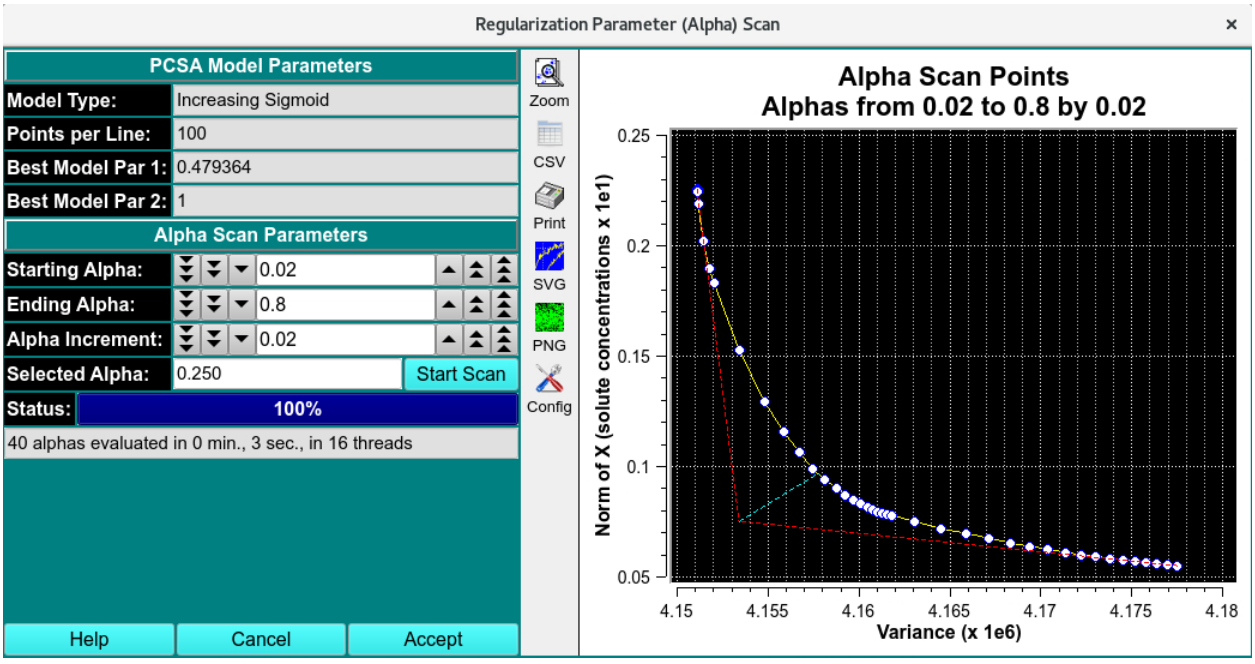
The dialog begins with the final best model from an initial PCSA scan. The user chooses a range and increment of Alpha values and initiates a scan that computes models, each of which will have an associated variance and x-norm-squared value. The plot produced tracks a line through each of those points. The “elbow” of the curve is analytically determined and graphically represented. The Alpha of the elbow point may be accepted as is. Alternatively, the user may select a different point by holding down the Ctrl key and the left mouse button while positioning the resulting vertical line at the desired curve point. At mouse button release, the Alpha is displayed beneath the vertical line and in the Selected Alpha text box. Clicking the Accept button will close the dialog and communicate the Alpha in the text box to the calling PCSA Analysis Control dialog. Note that the text box Alpha may be manually changed to refine the accepted value.

42.5.1 Sample Regularization Parameter Scan Dialogs:

The dialog images below show the plot results after a scan of Alpha values has completed. The yellow line connecting white dots tracks the variance, x-norm points associated with each Alpha. The red dashed lines are straight lines fitted to the first and last 5 points of the curve. The cyan line extends from the intersection of the two red lines to the point on the curve nearest to it. That nearest point is taken as the "elbow" of the curve; and its associated Alpha is the automatically determined best value. In the second image below, the user uses the Ctrl key and left mouse button to select an overriding elbow point and its associated Alpha value.



Before Regularization



After Regularization

Note:

1. The text box Alpha may be manually changed to refine the accepted value.
2. Model line point and variation values may be selected for speed in preparation for the Alpha scan. Once the Accept button is clicked here and focus is returned to the Analysis Control dialog, larger point and variation values may be used, along with the selected Alpha, in order to compute a more accurate final model.

42.5.2 Regularization Parameter Functions:

Model Type:	Documentation of the type of curve that defines models.
Points per Line:	Documentation of the number of solute points in input and scan models.
Best Model Par 1:	Documentation of the par1 value of the line that defines the input solutes of the model.
Best Model Par 2:	Documentation of the par2 value of the line that defines the input solutes of the model.
Starting Alpha:	A counter by which the user can set the starting Alpha of the scan to perform.
Ending Alpha:	A counter by which the user can set the ending Alpha of the scan to perform.
Alpha Increment:	A counter by which the user can set the increment between test Alphas in the scan to perform.
Selected Alpha:	A text box that is filled with the analytically determined Alpha or an Alpha associated with the curve point interactively set by the user using Ctrl-Left-Button mouse selection. This text box value can also be manually changed.
Start Scan	Click on this button once the scan Alpha range has been set, in order to begin a scan.
Accept	Close the dialog window, communicate the current Alpha value to the caller, and exit.

Window Controls

Help	Display this detailed help.
Cancel	Close the dialog window and exit.

42.6 Model Lines Plot for PCSA

Index

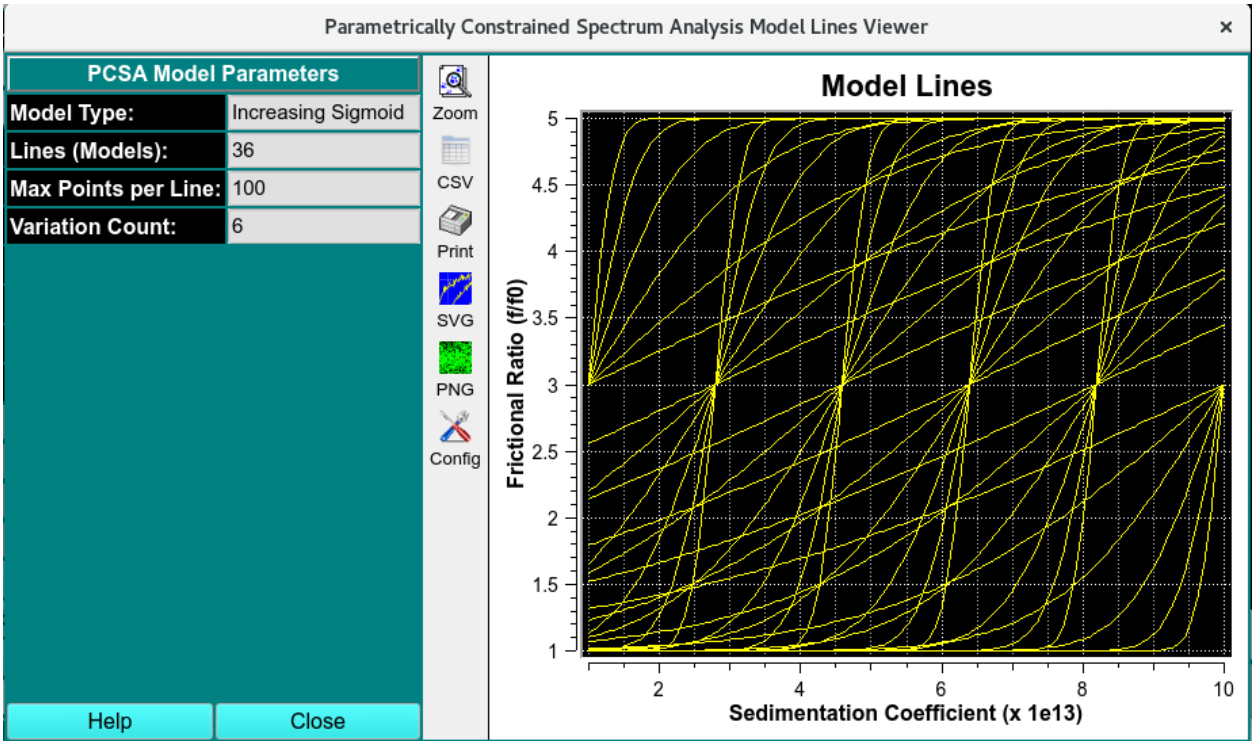
- *Dialog Sections:*
- *Sample Model Lines Dialogs:*
- *Model Line Functions:*

This dialog displays the lines in s x f/f0 space that define models evaluated by Parametrically Constrained Spectrum Analysis. The lines from the best models are illustrated with color coding. Worst-model lines are shown with dark gray lines or may be removed from the plot. A selected set of the best model lines contain white dots showing computed solute points, with size proportional to relative concentration.

The appearance of the plot may be governed by setting the cutoff points in terms of line counts for (1) the number of "elite" (best-model, lowest-RMSD) colored lines; (2) the number of the best models for which to show solute points; and (3) the number of overall visible lines. It is also possible to choose an alternative color map. The default color map ranges through the colors of the rainbow from red (best) to magenta (worst of the elite).

42.6.1 Dialog Sections:

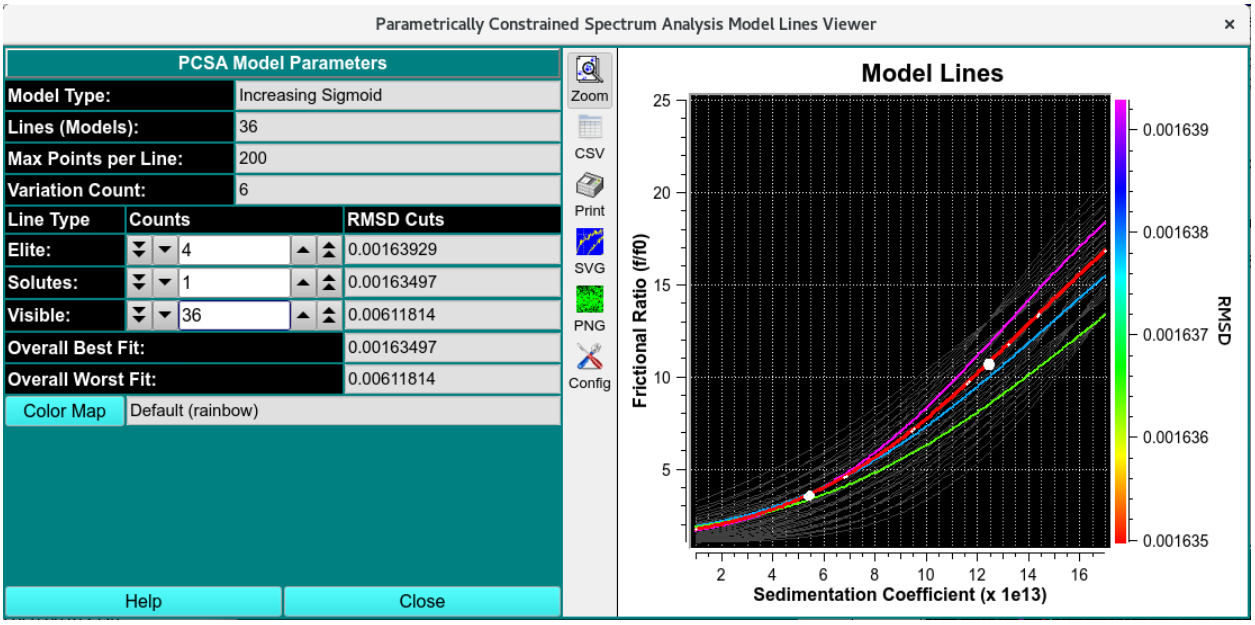
- **Model Documentation:** A set of read-only text boxes that document the type of analysis that was performed.
- **Display Controls:** A set of line counters and their accompanying read-only RMSD equivalents for specifying the cutoff points of elite, solute-containing, and visible lines. There are also read-only text boxes showing the best and worst overall RMSD values from the fits.
- **Color Map:** A button for choosing a file containing an alternative to the default color map.
- **General Control:** For example, the Close button.
- **Model Lines Plot:** Plotted lines and axes including the color gradient.



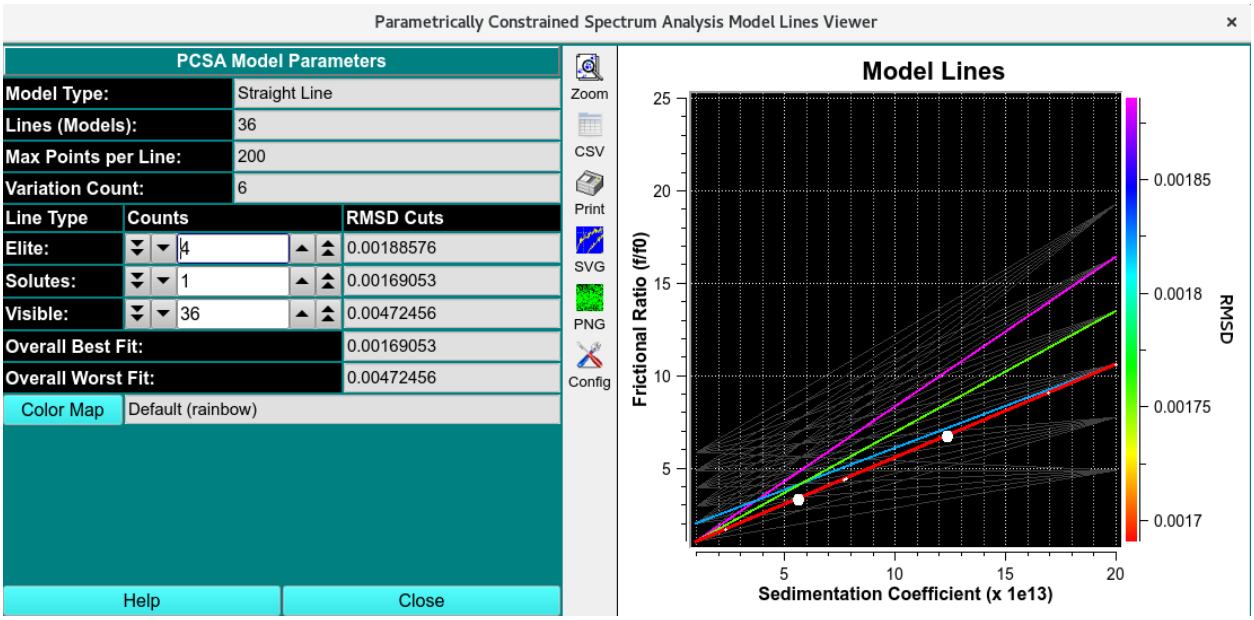
PCSA Model Lines Initiation Grid

42.6.2 Sample Model Lines Dialogs:

The plot dialog differs slightly based on whether the curve type is a line or a sigmoid. Below are images for sigmoids and straight line.



Simulated PCSA Sigmoidal Model Lines



Simulated PCSA Straight Model Lines

42.6.3 Model Line Functions:

Model Type:	Documentation of the type of curve that defines models.
Lines (Models):	Documentation of the number of model lines that were created.
Points per Line:	Documentation of the number of solute points in input models.
f/f0 Increment:	The frictional ratio increment between the set of start and end point variations for straight-line models.
Variation Count:	The number of variations in each of par1 and par2 in the computation of sigmoid model curves.
Elite:	To the right of this label is a counter for setting the number of colored best-model lines to display. To the right of the counter is the RMSD-cutoff equivalent.
Solutes:	To the right of this label is a counter for setting the number of best-model lines for which solute points are to be displayed. To the right of the counter is the RMSD-cutoff equivalent.
Visible:	To the right of this label is a counter for setting the number of visible lines to show. Normally, this should range from the Elite count to the number of total lines. To the right of the counter is the RMSD-cutoff equivalent.
Overall Best Fit:	The read-only text box to the right of this label documents the lowest overall RMSD value encountered.
Overall Worst Fit:	The read-only text box to the right of this label documents the highest overall RMSD value encountered.
Color Map:	This button opens a Load Color Map dialog in which an alternative color map file may be selected. To the right of the button is a text box giving the name for the current selection. A default UltraScan installation will contain a number of color map files. You can construct your own variation using the Color Gradient Generator. At the opening of the plot dialog, a default rainbow colors map is selected.
(Model Lines):	The plot on the right of the dialog window displays model lines as controlled by the user. The right side axis shows the color gradient annotated with corresponding RMSD values.
Help	Display this and other documentation.
Close	Close all windows and exit.

Note: that RMSD values increase from bottom to top, as is the convention with numeric annotation. But this means that the color corresponding to the best model is shown on the bottom.

Note: that the images and controls documented above are for model lines plots shown after a fit computation. There are also variations of this dialog showing pure single-color lines before any fit. There are 4 basic variations of this dialog depending on whether it is displayed before or after a fit and on whether the curve type is straight line or sigmoid.

triple========== Parametrically Constrained

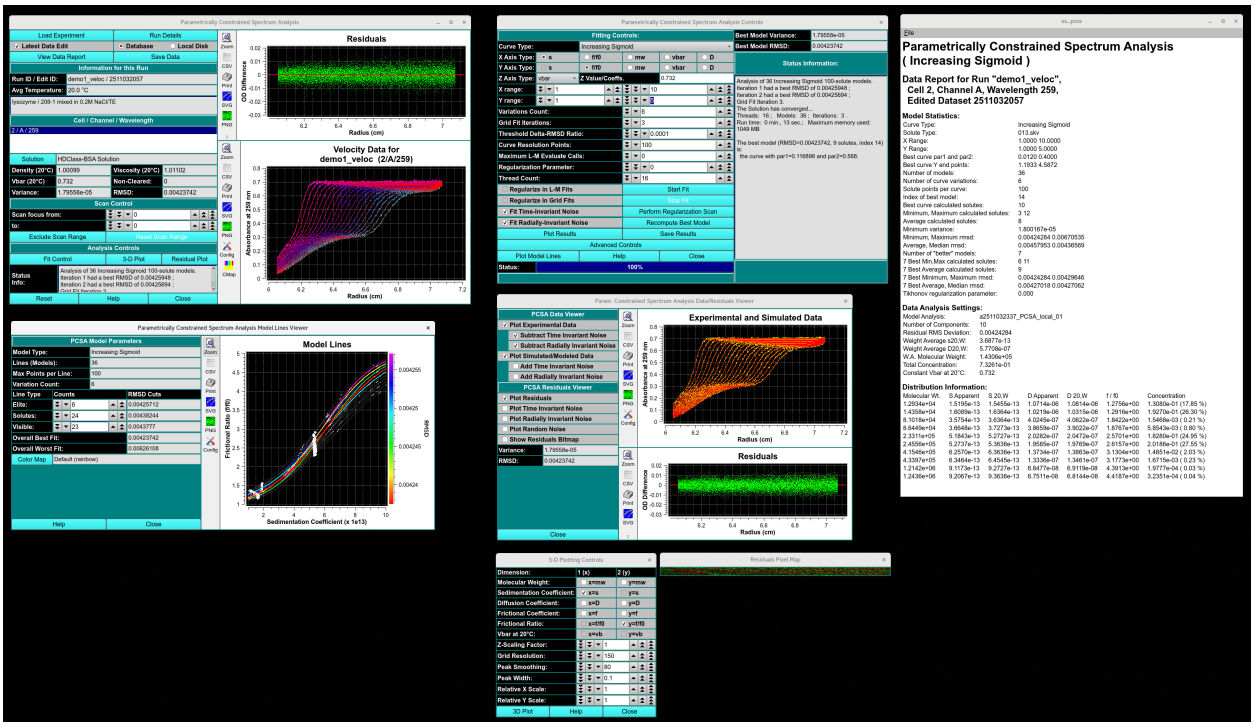
Spectrum Analysis Results ==========

Index

- *PCSA Data Residuals Viewer*
- *Residual Bit Map*
- *PCSA Report*
  - *Distribution Information Units:*

Upon completion of an analysis fit, windows and plots generated include:

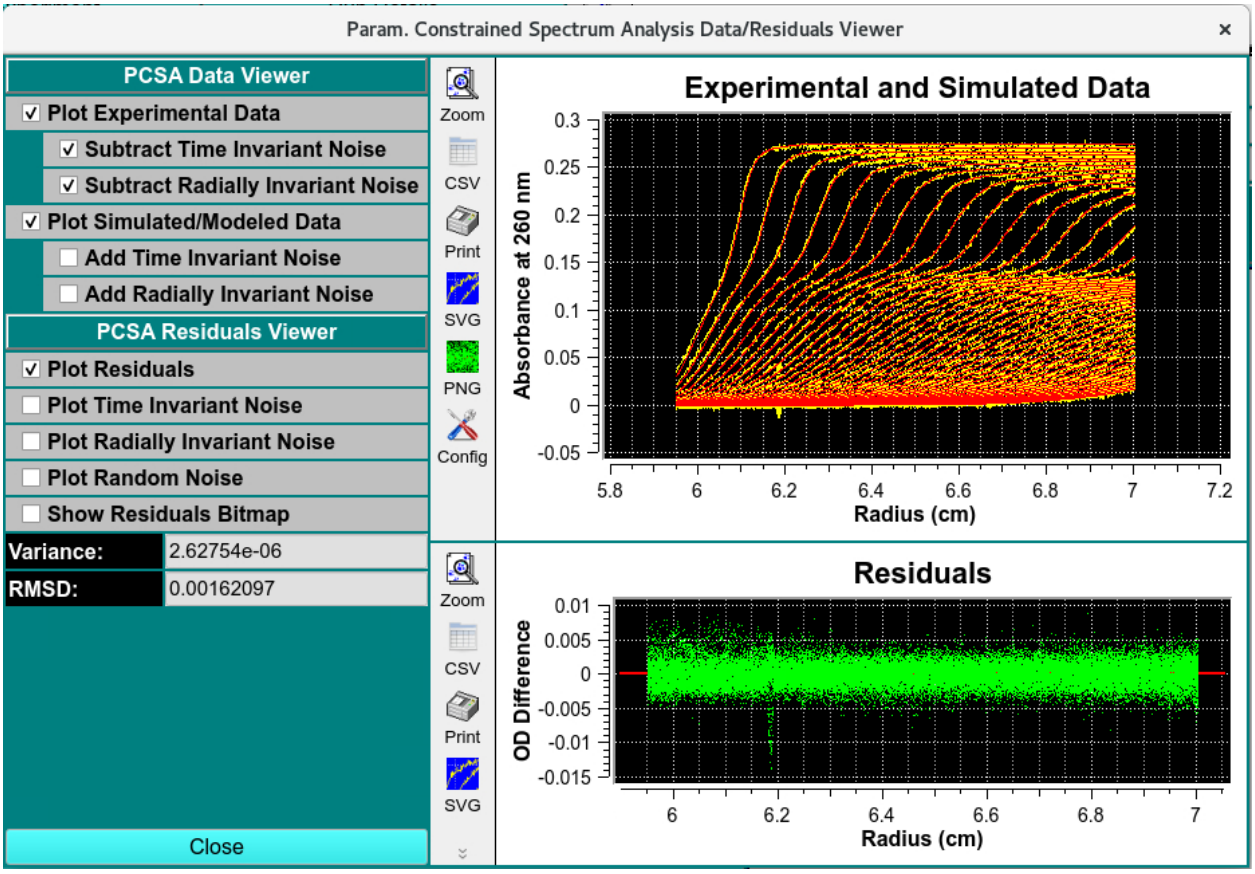
- *PCSA Fitting Controls* - Once the PCSA module has simulated the model with, the Status Information on the window updates to include the best model after the iterative simulations are done. the par1 and par2 for the curves updates.
- *PCSA model lines viewer* - simulated model lines;
- *PCSA data/residuals viewer* - Visual comparison of simulated and experimental data. Graphical display of time and radially invariant noises subtracted;
- *PCSA Report* - Generate a Simulated model Report file and view it in a dialog;
- *Bit Map residuals* Bit map of every scan as a pixel line and the positive or negative difference of the model to the data are highlighted as red (negative) and green(positive) pixels. an Excellent residual map will have a random distribution of red and green pixels;
- *3-Dimensional Plot Controls* - 3D plots of the simulated data



Simulated PCSA Results

42.7 PCSA Data Residuals Viewer

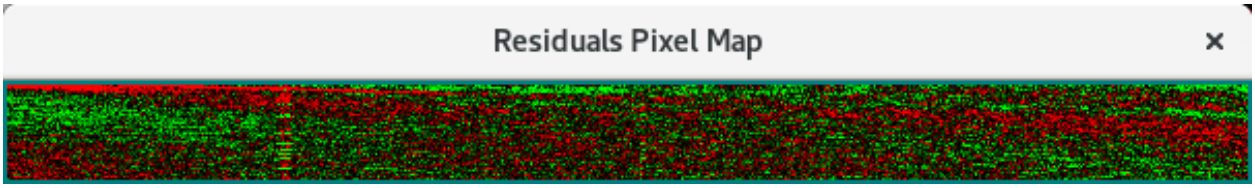
The simulation creates a data set with the same ranges as the edit experimental data set. The actual values for scan readings vectors are synthetically produced, as illustrated by the plot below.



PCSA Data and Residual Viewer

42.8 Residual Bit Map

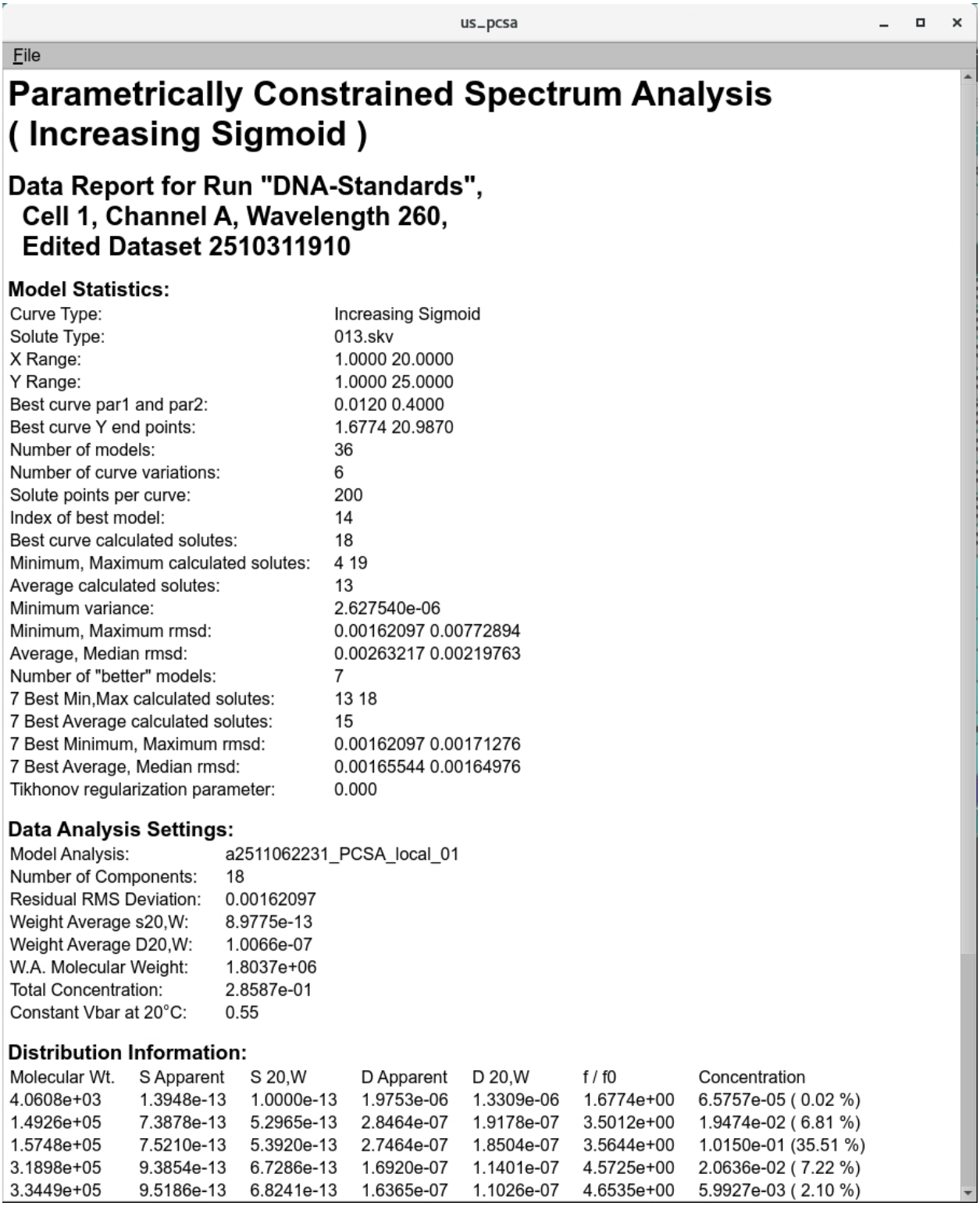
Experimental-Simulation residuals are plotted in another way in a bit map. This small window represents each residual #Scans x #Readings point as a color: green where simulation is greater than experimental; red where experimental is greater. A random distribution of colors throughout the bit map is indication of a good model fit.



Pixel Bit Map

42.9 PCSA Report

The "Save Data" button in the PCSA Controls produces a set of report files. One of these is displayable via the "View Report" button, which produces a dialog that shows the contents of a report. A dialog sample follows.



PCSA Report

Detailed 2DSA Report

- **Heading 1:** Analysis type

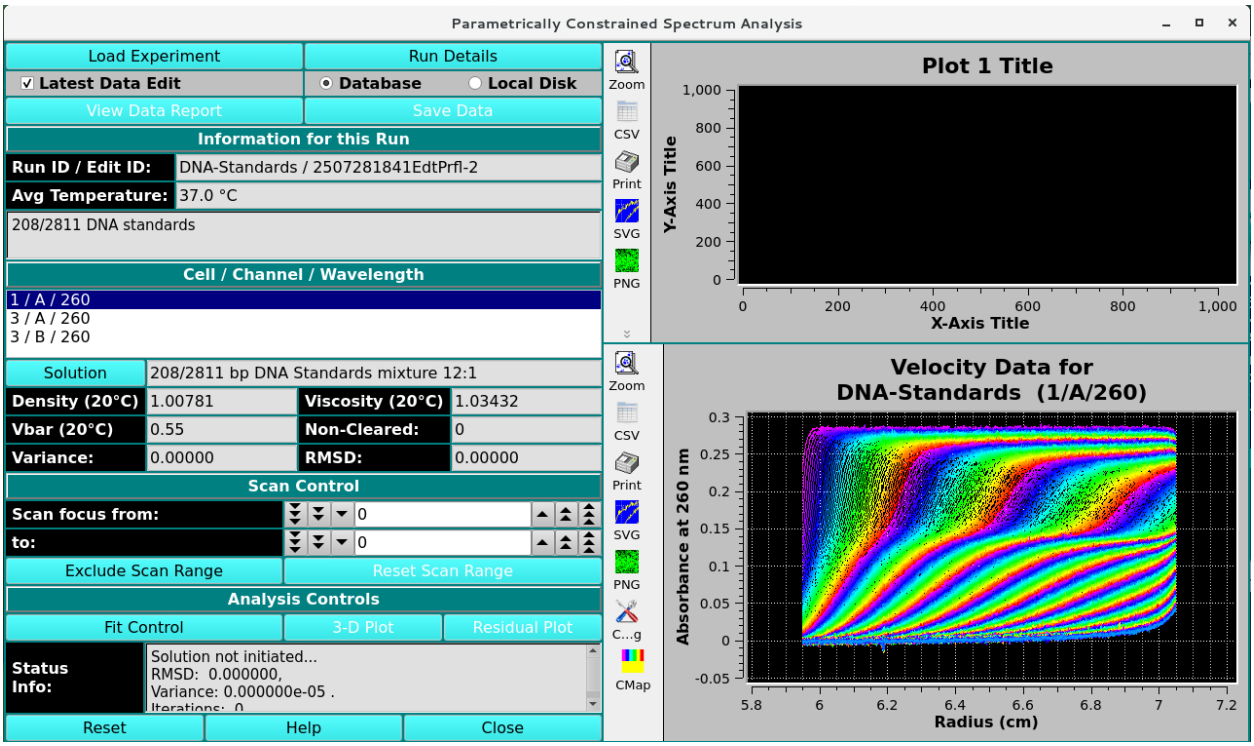
- **Heading 2:** This heading indicates the "dataset named in the Import module", the cell number, channel, and wavelength triple, the edit profile processed in the Edit data module,
- **Model Statistics:** The settings entered to generate the models and the. The model line used by the analysis, number_the type of analysis completed_the database request number_number of models in the analysis cohort
- **Data Analysis Settings:** The name of the model is determined by the analysis number_the type of analysis completed_the database request number_number of models in the analysis cohort
- **Number of components:** Number of components with unique s, and D values.
- **Residual RMS Deviation:** The RMSD determined by re-simulating the model in the finite element viewer module.
- **Model-reported RMSD:** The RMSD of the simulated model determined by message passing interface (MPI) or the 2DSA GUI.
- **Weight Averaged s20,W, D20,W, Molecular weight, and f/f0:** describe the weight averaged (using % of the total concentration) to calculate the s20,W, D20,W, Molecular weight, and f/f0 coefficients of each component at 20ºC and in water.
- **Total concentration:** The baseline and noise subtracted optical density at the measured wavelength of the absorbing components as a function of radius.
- **Constant vbar20:** The analyte vbar at 20ºC.

Note: Difference between the Residual RMSD and model-reported RMSD should be within 3 significant figures. Greater then 3 significant figures indicates an error in analysis.

42.9.1 Distribution Information Units:

- **Molec. Wt.:** Dalton
- **S Apparent:** seconds
- **S 20,W:** Seconds
- **D Apparent:** cm^2/s
- **D 20,W:** cm^2/s
- **f/f0:** unitless
- **Concentration:** optical density (OD) at measured wavelength, (% contribution)

42.10 PCSA Advanced Controls - load record



Parametrically Constrained Spectrum Analysis window

42.11 Reference:

Gorbet G., T. Devlin, B. Hernandez Uribe, A. K. Demeler, Z. Lindsey, S. Ganji, S. Breton, L. Weise-Cross, E.M. Lafer, E.H. Brookes, B. Demeler. A parametrically constrained optimization method for fitting sedimentation velocity experiments. Biophys. J. (2014) vol 106, 1741-50.

Also Available Here

CHAPTER

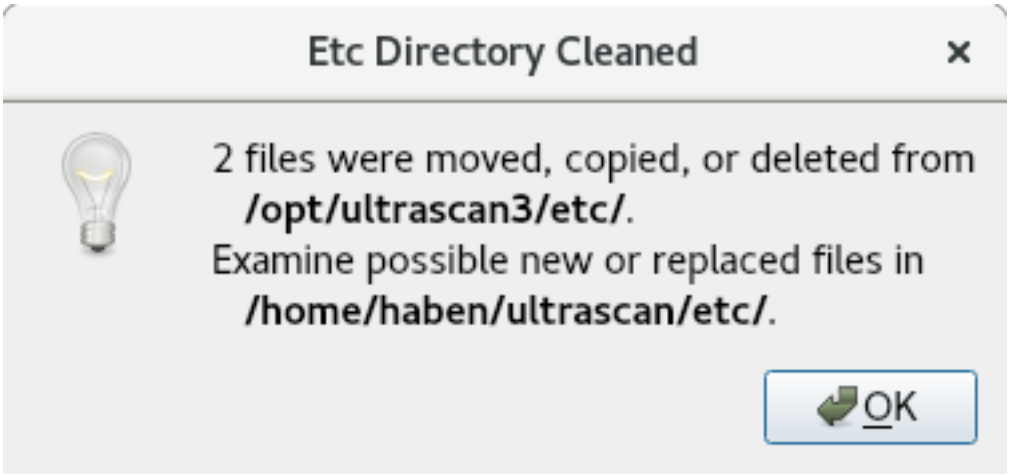
FORTYTHREE

INITIALIZE GENETIC ALGORITHM

Index

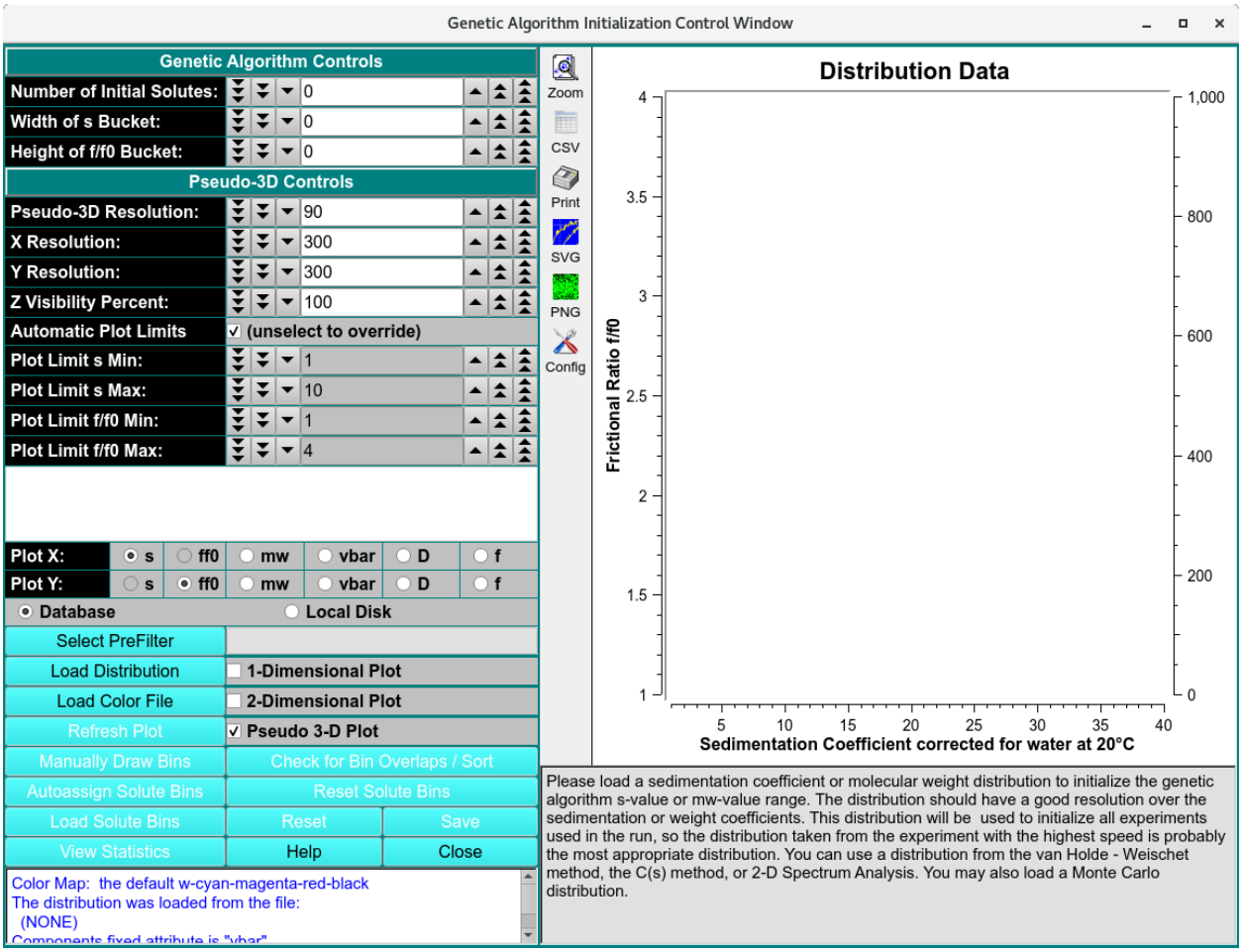
- *GA Functions:*
- *Statistics File Output*

When the **Initialize Genetic Algorithm module** is opened from *Velocity*, a **Etc Dorectory Cleaned** dialog will open. Press **OK** to continue to the Genetic Algorithm Initiation Window.



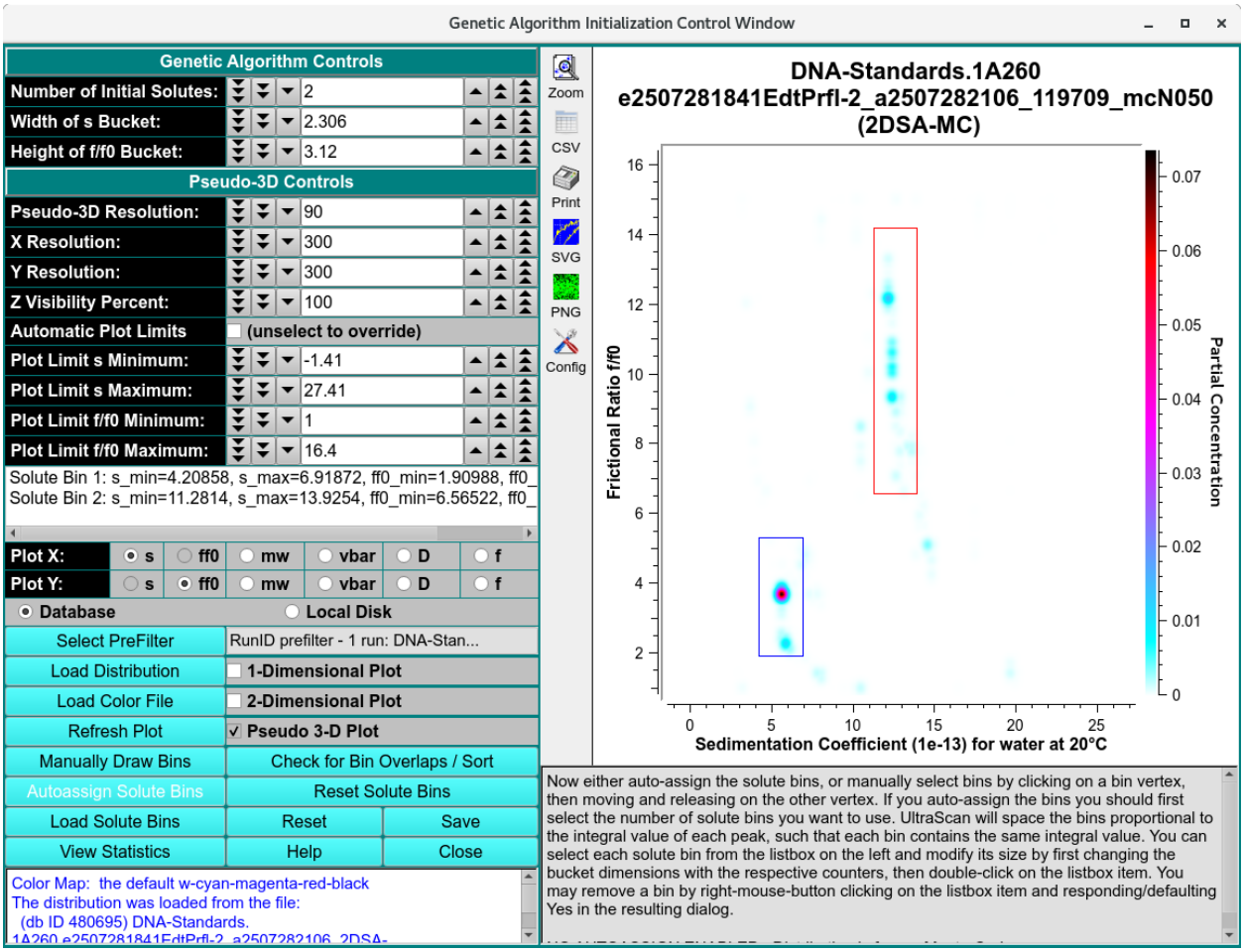
Etc Directory Cleaned

Press **OK** to continue to the Genetic Algorithm Initiation Window.



Initialize Genetic Algorithm Window

Solute distribution data can be used to generate data for use with Genetic Algorithm analysis programs. The output data results from choosing bins (or buckets) around solute points. These buckets can be modified for better results. Once they are as desired, data for Genetic Algorithm analysis can be generated and output. The input data may come in one of two flavors, each of which is treated differently. The data is either 2DSA type or Monte Carlo. The 2DSA is relatively sparse and generally requires fashioning buckets containing a single point. For a Monte Carlo distribution, buckets should contain multiple points and be constructed around clusters.



Genetic Algorithm populated with Monte Carlo Model

43.1 GA Functions:

Genetic Algorithm Controls

Number of Initial Solutes:	Choose the number of initial solute buckets when using the Autoassign Solute Bins option. If 0 or greater than the actual number of input solute points, all solute points will be binned. Otherwise, this number limits initial bins to those derived from the specified number of highest-concentration solute points.
Width of s Bucket:	Set the horizontal width of buckets for autoassigning or for resizing via list box double-click. Note that this may alternately read "Width of mw Bucket:".
Height of f/f0 Bucket:	Set the vertical height of buckets for autoassigning or for resizing via list box double-click. Note that "bucket" and "bin" are used interchangeably herein.

Pseudo-3D Controls

Pseudo-3D Resolution:	Choose near-100 flattening parameter for Gaussian distribution of frequency points.
X Resolution:	Choose the number of pixels to represent the full X (Sedimentation Coefficient or Molecular Weight) data range.
Y Resolution:	Choose the number of pixels to represent the full Y (Frictional Ratio) data range.
Z Visibility Percent:	Choose the percent of the Z (Frequency) range to add below the minimum Z in order to affect display of faint values.
Automatic Plot Limits	Select to automatically set X and Y plot range limits or unselect to allow explicitly setting these values.
Plot Limit s Min:	Choose the lower plot limit for sedimentation (or other X).
Plot Limit s Max:	Choose the upper plot limit for sedimentation (or other X).
Plot Limit f/f0 Min:	Choose the lower plot limit for f/f:sub:*0* (or other Y).
Plot Limit f/f0 Max:	Choose upper lower plot limit for f/f:sub:*0* (or other Y).
(list box)	This space is populated with lines giving the dimensions of bins generated manually or automatically, and displayed in the plot to its right. Clicking of a line causes the associated bin plot borders to be highlighted. Double-clicking a line results in the associated bin dimension being modified according to the bucket width and height values in the counter above. A right-mouse-button click of a line signals removal of the bin, if you default or click Yes in the resulting dialog.
Plot X:	Select one of the radio buttons to the right of this label to choose the X dimension of 3-D plots.
Plot Y:	Select one of the radio buttons to the right of this label to choose the Y dimension of 3-D plots.
Database	Select to specify data input from the database.
Local Disk	Select to specify data input from local disk.

Data Input and Outout Controls

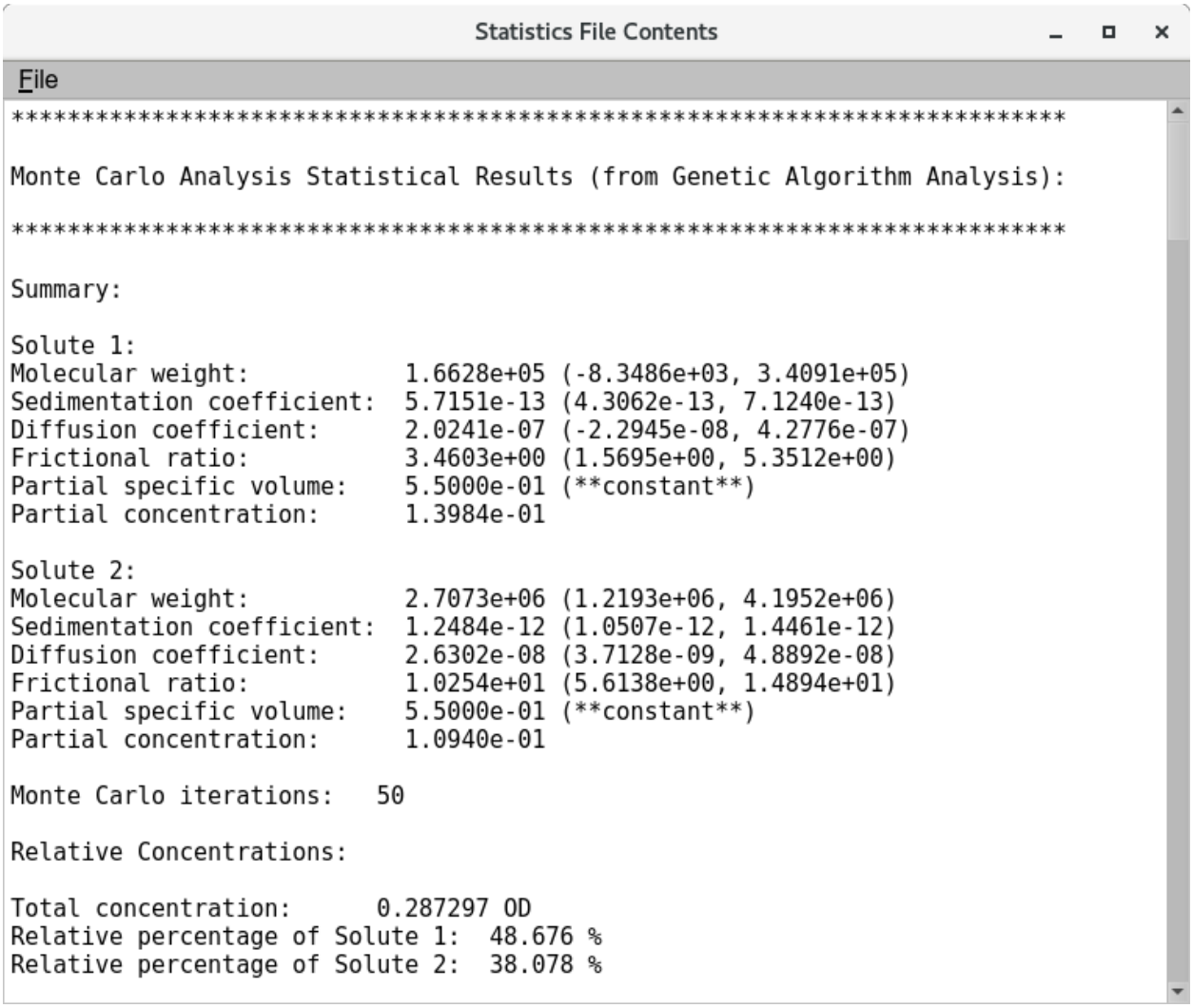
Select PreFilter(s)	Open Select Pre-Filter to select Run IDs on which to pre-filter lists of models for loading. This can significantly reduce model loading time, particularly with large model counts in the database. The text box to the right of this button will display a summary of runs selected as Pre-filters.
Load Distribution	Load model distribution data as specified through a Model Loader dialog.
Load Color File	Load a color map (.xml) for Z dimension display as specified through *Load Color File dialog*.
1-Dimensional Plot	Select to plot in one dimension, displaying the selected X-axis variable on the horizontal axis and the partial concentration on the Y-axis as a bar plot.
2-Dimensional Plot	Select to plot in 2 dimension, excluding the Z dimension.
Pseudo 3-D Plot	Select to plot in pseudo three-dimension, using a loaded or default color map to represent the "Z" dimension.
Refresh Plot	Re-draw the plot. This is sometimes necessary when one or more control parameters are changed.
Manually Draw Bins	Specify that you are about to manually draw bins in a 2-D or 3-D plot. Each bin is drawn by clicking and holding on one bin vertex, moving the mouse and showing a rubber band rectangle, then releasing when the diagonal vertex point is reached.
Check for Bin Overlaps / Sort	Click this button to check that no drawn bins overlap and to sort bins so they are ordered by the X,Y upper-left vertex.
Autoassign Solute Bins	Automatically draw bins around solute points, using bucket width and height values specified. If a number of initial solutes value is first specified, only that number of the bins with the highest concentration values will be initially drawn. Where bins overlap, additional rectangles will be drawn using the portion of the second rectangle that is not overlapping.
Reset Solute Bins	Erase bins from the plot and internal data base, so that a new set may be generated.
View Statistics:	View a text dialog that shows detailed statistics on drawn bin points.

Window Controls

Save:	Output a file with Genetic Algorithm data based on generated bins. If a Monte Carlo distribution was loaded, also generate a text file with detailed analysis.
Reset	Reset parameters to their default settings.
Help	Display this detailed Genetic Algorithm Initialize help.
Close	Close all windows and exit.

43.2 Statistics File Output

Once the bins for Genetic Algorithm analysis are selected, detailed statistical analysis can be performed on the Monte Carlo models. Click on 'View Statistics' to generate the detailed analysis.



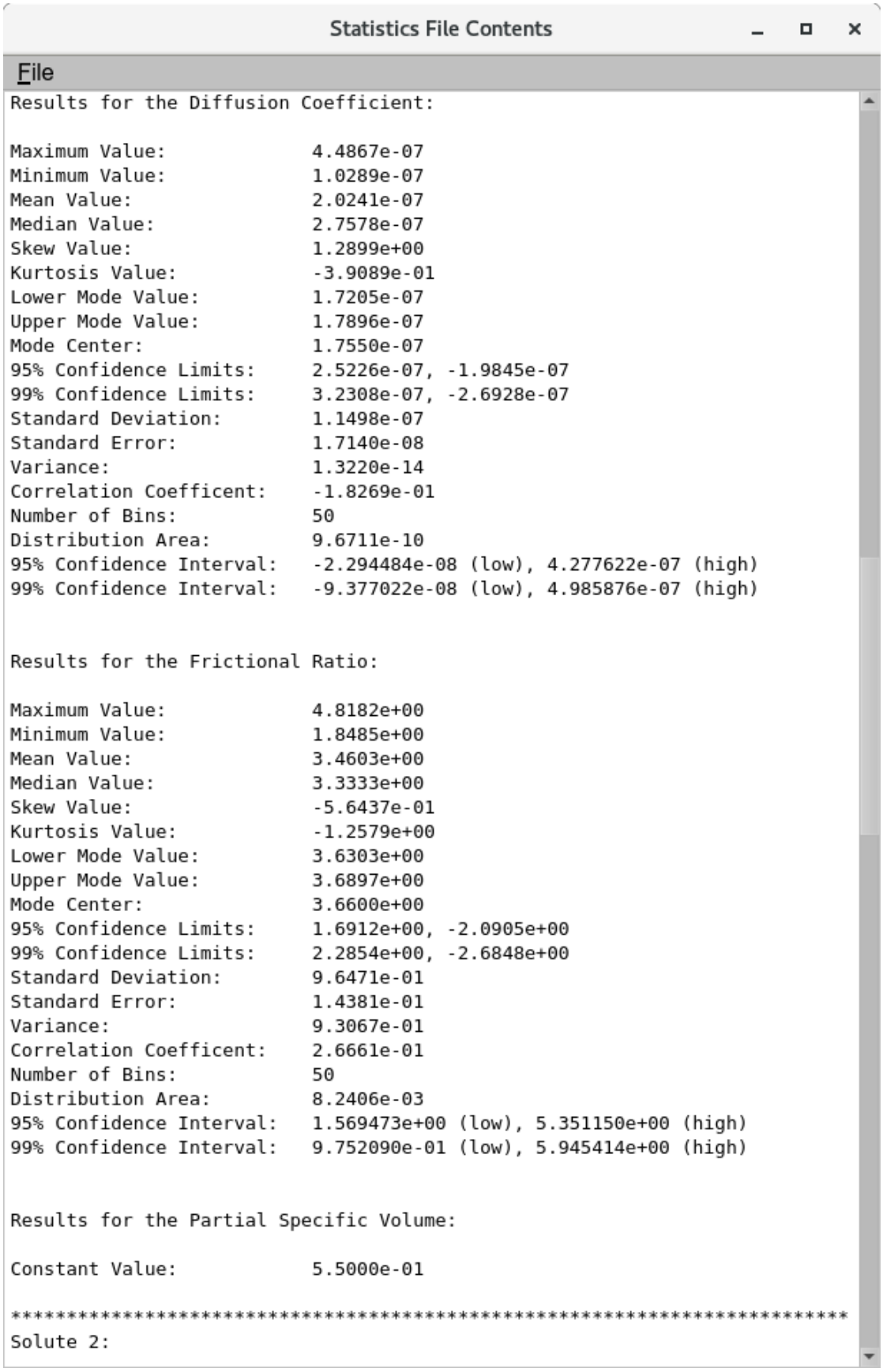
Summary Statistics File Content

For each selected bin, the GA statistical analysis includes the mean, along with the lower and upper 95% confidence intervals for the molecular weight, sedimentation coefficient, diffusion coefficient, and frictional ratio. The partial specific volume and partial concentration are treated as constant values. For a more detailed analysis, refer to the Detailed Results section of the report.

Statistics File Contents

File

```
Detailed Results:

********************************************************************************
Solute 1:

Results for the Molecular Weight:

Maximum Value:              3.7088e+05
Minimum Value:              7.4208e+04
Mean Value:                 1.6628e+05
Median Value:               2.2254e+05
Skew Value:                 1.0379e+00
Kurtosis Value:             -4.5244e-01
Lower Mode Value:           1.7508e+05
Upper Mode Value:           1.8101e+05
Mode Center:                1.7804e+05
95% Confidence Limits:      1.6287e+05, -1.8639e+05
99% Confidence Limits:      2.1776e+05, -2.4127e+05
Standard Deviation:         8.9098e+04
Standard Error:             1.3282e+04
Variance:                   7.9384e+09
Correlation Coefficent:     5.5536e-01
Number of Bins:             50
Distribution Area:          8.2323e+02
95% Confidence Interval:    -8.348571e+03 (low), 3.409140e+05 (high)
99% Confidence Interval:    -6.323269e+04 (low), 3.957981e+05 (high)


Results for the Sedimentation Coefficient:

Maximum Value:              7.0606e-13
Minimum Value:              5.1212e-13
Mean Value:                 5.7151e-13
Median Value:               6.0909e-13
Skew Value:                 1.5459e+00
Kurtosis Value:             -3.7679e-01
Lower Mode Value:           5.5867e-13
Upper Mode Value:           5.6255e-13
Mode Center:                5.6061e-13
95% Confidence Limits:      1.5179e-13, -1.2999e-13
99% Confidence Limits:      1.9607e-13, -1.7427e-13
Standard Deviation:         7.1883e-14
Standard Error:             1.0716e-14
Variance:                   5.1672e-27
Correlation Coefficent:     9.6685e-01
Number of Bins:             50
Distribution Area:          5.3242e-16
95% Confidence Interval:    4.306179e-13 (low), 7.124004e-13 (high)
99% Confidence Interval:    3.863378e-13 (low), 7.566805e-13 (high)


Results for the Diffusion Coefficient:
```


Statistics File Contents
File
Results for the Diffusion Coefficient:
Maximum Value: 4.4867e-07
Minimum Value: 1.0289e-07
Mean Value: 2.0241e-07
Median Value: 2.7578e-07
Skew Value: 1.2899e+00
Kurtosis Value: -3.9089e-01
Lower Mode Value: 1.7205e-07
Upper Mode Value: 1.7896e-07
Mode Center: 1.7550e-07
95% Confidence Limits: 2.5226e-07, -1.9845e-07
99% Confidence Limits: 3.2308e-07, -2.6928e-07
Standard Deviation: 1.1498e-07
Standard Error: 1.7140e-08
Variance: 1.3220e-14
Correlation Coefficent: -1.8269e-01
Number of Bins: 50
Distribution Area: 9.6711e-10
95% Confidence Interval: -2.294484e-08 (low), 4.277622e-07 (high)
99% Confidence Interval: -9.377022e-08 (low), 4.985876e-07 (high)
Results for the Frictional Ratio:
Maximum Value: 4.8182e+00
Minimum Value: 1.8485e+00
Mean Value: 3.4603e+00
Median Value: 3.3333e+00
Skew Value: -5.6437e-01
Kurtosis Value: -1.2579e+00
Lower Mode Value: 3.6303e+00
Upper Mode Value: 3.6897e+00
Mode Center: 3.6600e+00
95% Confidence Limits: 1.6912e+00, -2.0905e+00
99% Confidence Limits: 2.2854e+00, -2.6848e+00
Standard Deviation: 9.6471e-01
Standard Error: 1.4381e-01
Variance: 9.3067e-01
Correlation Coefficent: 2.6661e-01
Number of Bins: 50
Distribution Area: 8.2406e-03
95% Confidence Interval: 1.569473e+00 (low), 5.351150e+00 (high)
99% Confidence Interval: 9.752090e-01 (low), 5.945414e+00 (high)
Results for the Partial Specific Volume:
Constant Value: 5.5000e-01
****
Solute 2:

Detailed Statistics Results of GA Initialization Analysis

CHAPTER

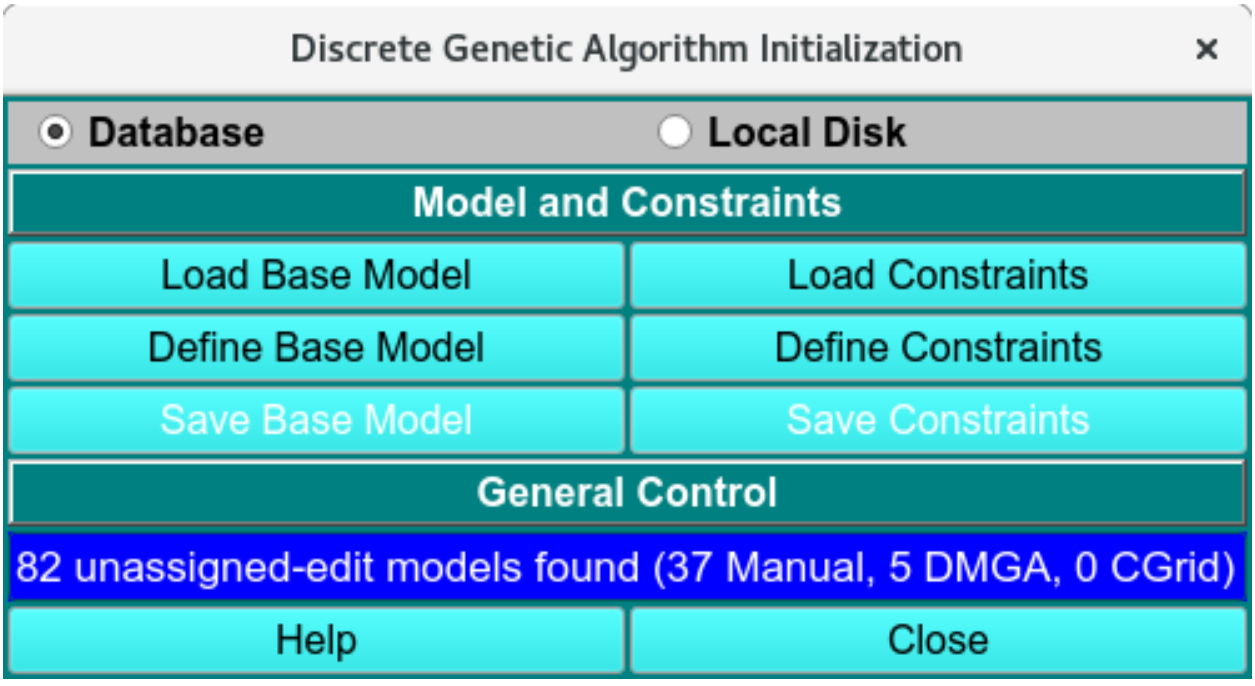
FORTYFOUR

DISCRETE MODEL GENETIC ALGORITHM

Index

- *DMGA Process:*

The Discrete Model Genetic Algorithm (DMGA) module employs user-defined models to fit sedimentation velocity data to parameters of those models. While there is no restriction on the type of model that can be built to fit a particular system, the method is most often used for fitting reversibly associating systems, such as oligomerization or self-association, or hetero-associating systems. The method uses genetic algorithms to optimize non-linear fitting problems using stochastic approaches. The user has to define a model and the *constraints* for the parameters of the model to initialize the DMGA routine. The fit is performed on the **LIMS** using the **Discrete Model GA Control**.



Discrete Genetic Algorithm Initialization

44.1 DMGA Process:

1. If an existing model is available, **Load Base Model** and use the *Load Distribution Model* dialog to select the desired model. If a model needs to be created first, use **Define Base Model** to open the *Model Editor window* and select **Save Base Model**.

2. If constraints for all parameters have been defined previously, **Load Constraints** will allow the user to select a predefined constrained from the *Load Distribution Model* dialog. If a set of constraints needs to be created first, use **Define Constraints** to open the *Discrete Model GA Constraints Editor <dmga_init_constr>*, enter the parameter constraints for the **Components** and any **Associations (reactions)**, and **Accept** the changes. At this point, **Save Constraints** in the main window will allow you to save the DMGA initialization to the LIMS database. By default, the constraints model will be named with the current timestamp, but selecting **Edit** allows you to use a custom name in the database.

3. To start the analysis, select the same constraints model in the **Setup Discrete Model GA Control** on the **LIMS** page.

CHAPTER

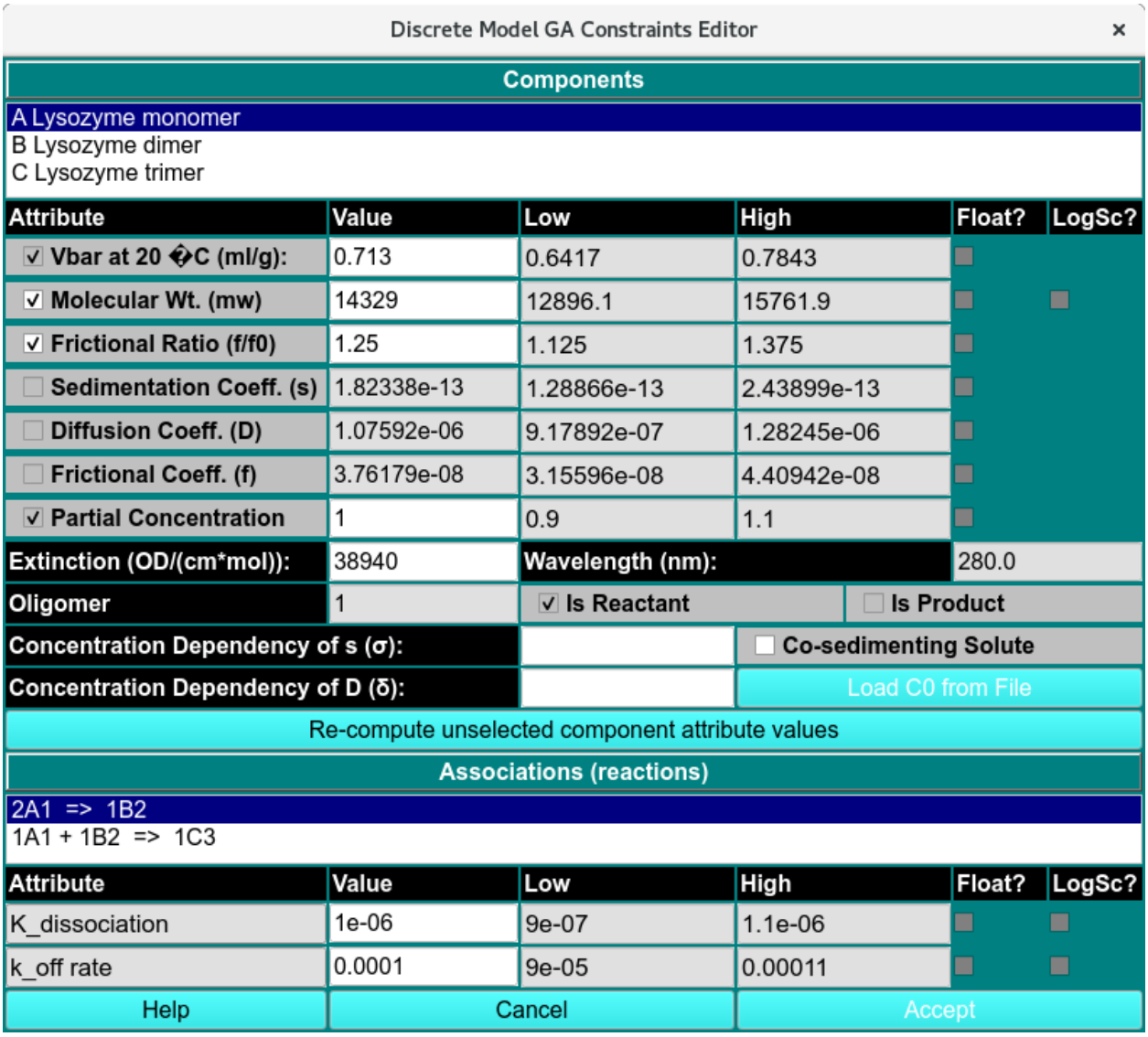
FORTYFIVE

DISCRETE MODEL GENETIC ALGORITHM CONSTRAINTS

Index

- *Component Parameters:*
- *DMGA Constraint Functions:*
  - *Component Parameters:*
  - *Association Parameters:*
  - *Window Controls:*

The Discrete Model GA Constraints Editor is a sub-module of the *DMGA Initialization* module that allows the user to define constraints for each parameter of the selected model, and to determine which parameters are held fixed and which are floated.



DMGA Contraints

45.1 Component Parameters:

For each component defined in the model, the following parameters can be defined:

- **Vbar at 20 °C (mL/g):**
- **Molecular Wt.(MW)**
- **Frictional ratio (f/f0)**
- **Sedimentation Coeff. (s)**
- **Diffusion Coeff. (D)**
- **Friction coeff.(f)**

- **Partial Concentration**

Multiple parameter combinations can be selected for fitting, but hydrodynamic constraints restrict the total number of parameters that can be fitted simultaneously. Select three out of the 6 possible parameters to define each component by selecting the checkbox in the front of the parameter name. Parameter values highlighted in white are writeable and can be changed in the white field. Greyed-out values are read-only and can be recomputed with the **Re-compute unselected component parameter values**. Any parameters selected for fitting need to be floated by selecting the **Float?** checkbox. Floated parameters require a **Low** and **High** limit between which the parameter is allowed to float. Select parameters can be set to a logarithmic scale by checking the **LogSc?** checkbox.

45.2 DMGA Constraint Functions:

45.2.1 Component Parameters:

Components	Displays the list of analytes included in the model.
Vbar at 20 °C (mL/g):	Displays the vbar value for the selected analyte. If this parameter is selected for fitting, enter the desired **Low** and **High** Vbar limits.
Molecular Wt. (MW)	Displays the molecular weight (MW) for the selected analyte. If this parameter is selected for fitting, enter the desired **Low** and **High** MW limits.
Frictional ratio (f/f0)	Displays the frictional ratio (f/f0) for the selected analyte. If this parameter is selected for fitting, enter the desired **Low** and **High** f/f0 limits.
Sedimentation Coeff. (s)	Displays the sedimentation coefficient (s) for the selected analyte. If this parameter is selected for fitting, enter the desired **Low** and **High** s limits.
Diffusion Coeff. (D)	Displays the diffusion coefficient (D) for the selected analyte. If this parameter is selected for fitting, enter the desired **Low** and **High** D limits.
Frictional coeff. (f)	Displays the frictional coefficient (f) for the selected analyte. If this parameter is selected for fitting, enter the desired **Low** and **High** f limits.
Partial Concentration	Displays the partial concentration for the selected analyte. This parameter is always fitted automatically using either the default or user-defined **Low** and **High** partial concentration limits.
Extinction (OD/(cm*mol)).	Displays the extinction value for the selected analyte at select wavelength (nm).
Wavelength (nm):	The wavelength value.
Oligomer	The oligomer number assigned to the selected analyte.
is (Reactant or Product)	Indicates whether the selected analyte is defined as a reactant or a product in the model.
Concentration Dependency of s (σ)	If the component has concentration dependent non-ideality of sedimentation, the sigma value (also called ks) can have a non-zero value.
Concentration Dependency of D (δ)	If the component has concentration dependent non-ideality of diffusion, the delta value can have a non-zero value.
Re-Compute unselected component attribute values	Recalculates the values of unselected component attributes based on the currently selected fitting parameters.

45.2.2 Association Parameters:

Association (reactions)	The association reactions defined for the model.
K_Dissociation	The dissociation constant (**K_Dissociation**) for the selected reaction.
K_off rate	The dissociation rate constant (**K_off**) for the selected reaction.

45.2.3 Window Controls:

Help	Open the DMGA constraints editor documentation.
Cancel	Reset and close window .
Accept	Accept the user-defined parameters.

CHAPTER

FORTYSIX

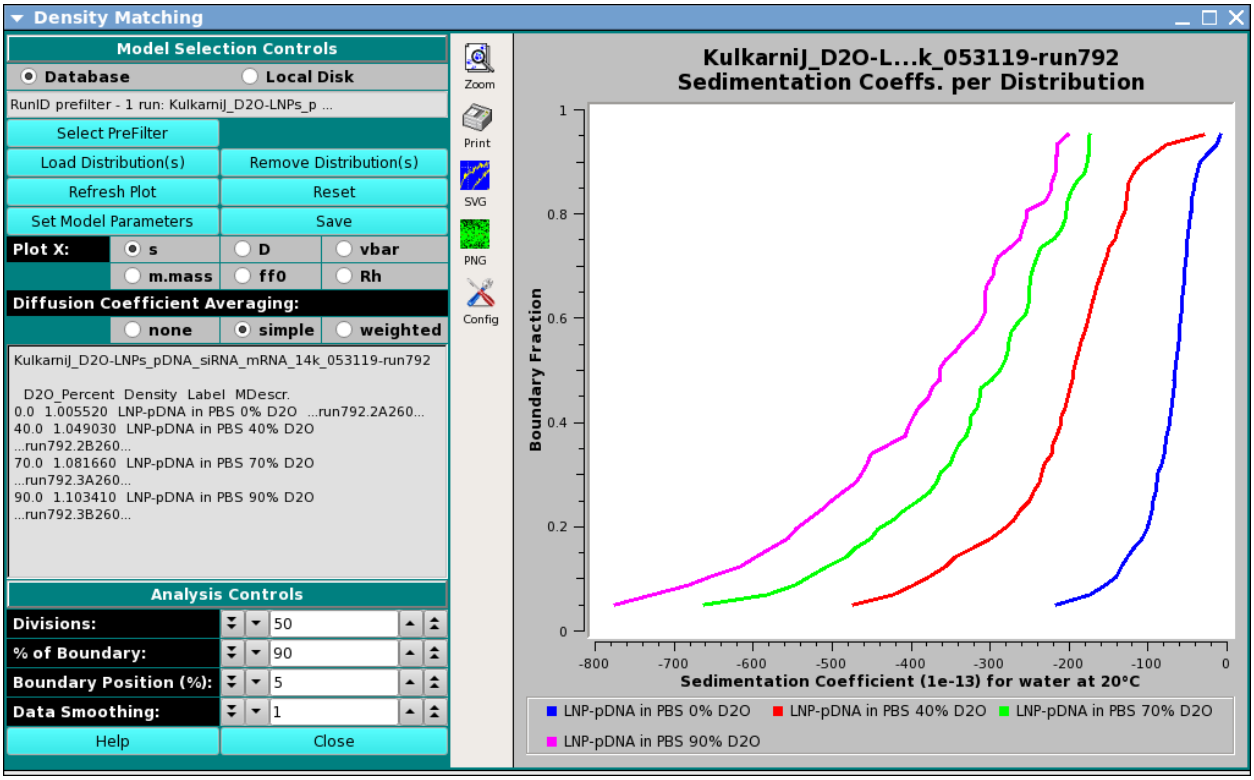
DENSITY MATCHING

Index

- *DM Functions:*
- *Density Matching Model Parameters:*
  - *DM. Model Para. Functions*

Density Matching (DM) experiments are used to determine the partial specific volume of colloidal particles by repeating the velocity experiment in a range of buffers with a different density, but identical chemical properties. Here it is important that the ionic strength and pH stay constant. This is most easily accomplished by using D2O (heavy water) which has mostly the same chemical properties as light water. The change in buoyancy incurred when changing the density of the solvent causes a change in sedimentation speed, which can be measured. The change is proportional to the density of the analyte. Density matching experiments exploit this difference.

It is recommended that users of this technique measure sedimentation velocity experiments in 4-6 different D2O concentrations, including one buffer that does not have any D2O in it (just light water). Finite element models from these experiments Function are imported into this program to elucidate the partial specific volume distribution for the entire solution mixture.



Density Matching Window

46.1 DM Functions:

Model Selection Controls

Database	Check this radio button to indicate that distribution data to be selected is in the database.
Local Disk	Check this radio button to indicate that distribution data to be selected is on local disk file systems.
(prefilter text):	Brief information about PreFilter selection(s) made.
Select PreFilter:	Open a *Models Pre-Filter dialog* to select Run IDs on which to pre-filter lists of models for loading.
Load Distribution(s):	Load model distribution data to use in density matching, as specified through a *Model Loader dialog*.
Remove Distribution(s):	Open a *Remove Models dialog* to pare down the list of selected models.
Refresh Plot:	Force a re-plot of the currently selected plot type.
Set Model Parameters:	Open a Model Parameters dialog to associate D20_Percent, Density, Label values with each model.
Save:	Generate CSV spreadsheet files and PNG plot files for all plot types and save them locally to a runID subdirectory of the */ultrascan/reports directory*.
Plot X:	Select the plot to currently be displayed by selecting its X axis: • **s** Sedimentation coefficient. • **D** Diffusion coefficient. • **vbar** Partial specific volume. • **m.mass** Molar Mass. • **f/f0** Frictional ratio. • **Rh** Hydrodynamic radius.
Diffusion Coefficient Averaging:	Select the means of averaging diffusion coefficients at each boundary fraction, across models: • **none** No averaging (maintain individual model D values). • **simple** Simple averaging. • **weighted** Weighted averaging, using concentrations.
(Distribution Information text box)	Information about Run and Model Distributions currently being used.
Divisions:)	Set the number of boundary fraction divisions to use.
% of Boundary::)	Set the extent in boundary percent over which plot values are computed and plotted.
Boundary Position (%)::)	Set the starting boundary percent.
Data Smoothing::)	Set the number of smoothing points to use in curves displayed.

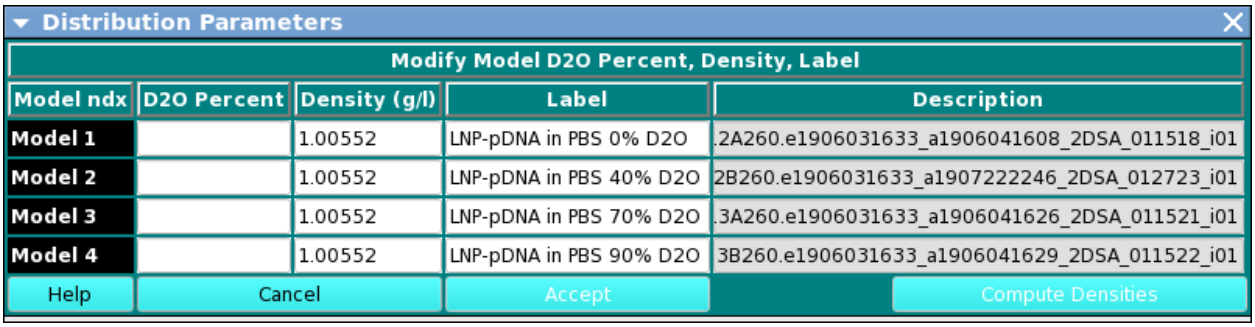
Window Controls

Reset Plot	Clear PreFilter and Distribution selection data.
Help	Display this detailed Density Matching help.
Close	Close all windows and exit.

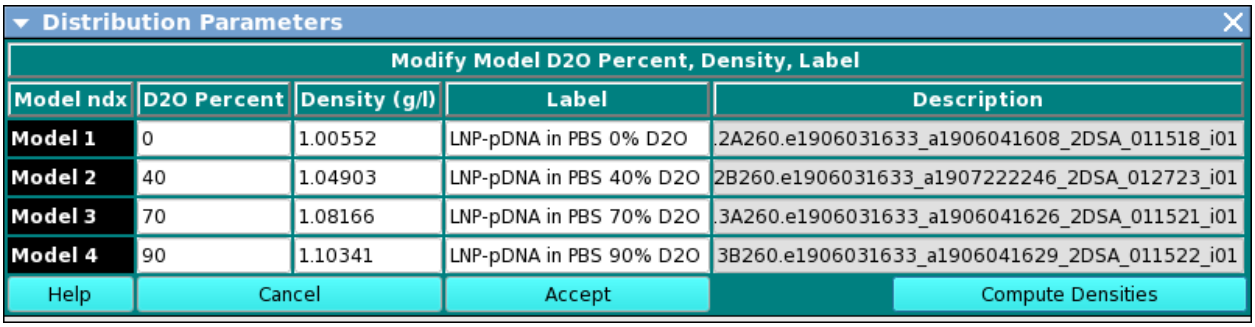
46.2 Density Matching Model Parameters:

An option in UltraScan-III US_Density_Match, this dialog allows specifying parameters associated with each model distribution. The dialog presents a row for each model selected in the calling Density Matching main window. In each model row, the user may add or change the value for (1) D2O (heavy water) percent; (2) density; or (3) model label. After setting all percent values, the Compute Densities button causes computation of the densities for non-zero-percent models, based on the zero-percent model density value. Once all model values are set, the Accept button closes the dialog and returns the values associated with each model.

Upon entry, the dialog appears as follows.



After completion of parameter setting, the dialog will appear similar to this:



46.2.1 DM. Model Para. Functions

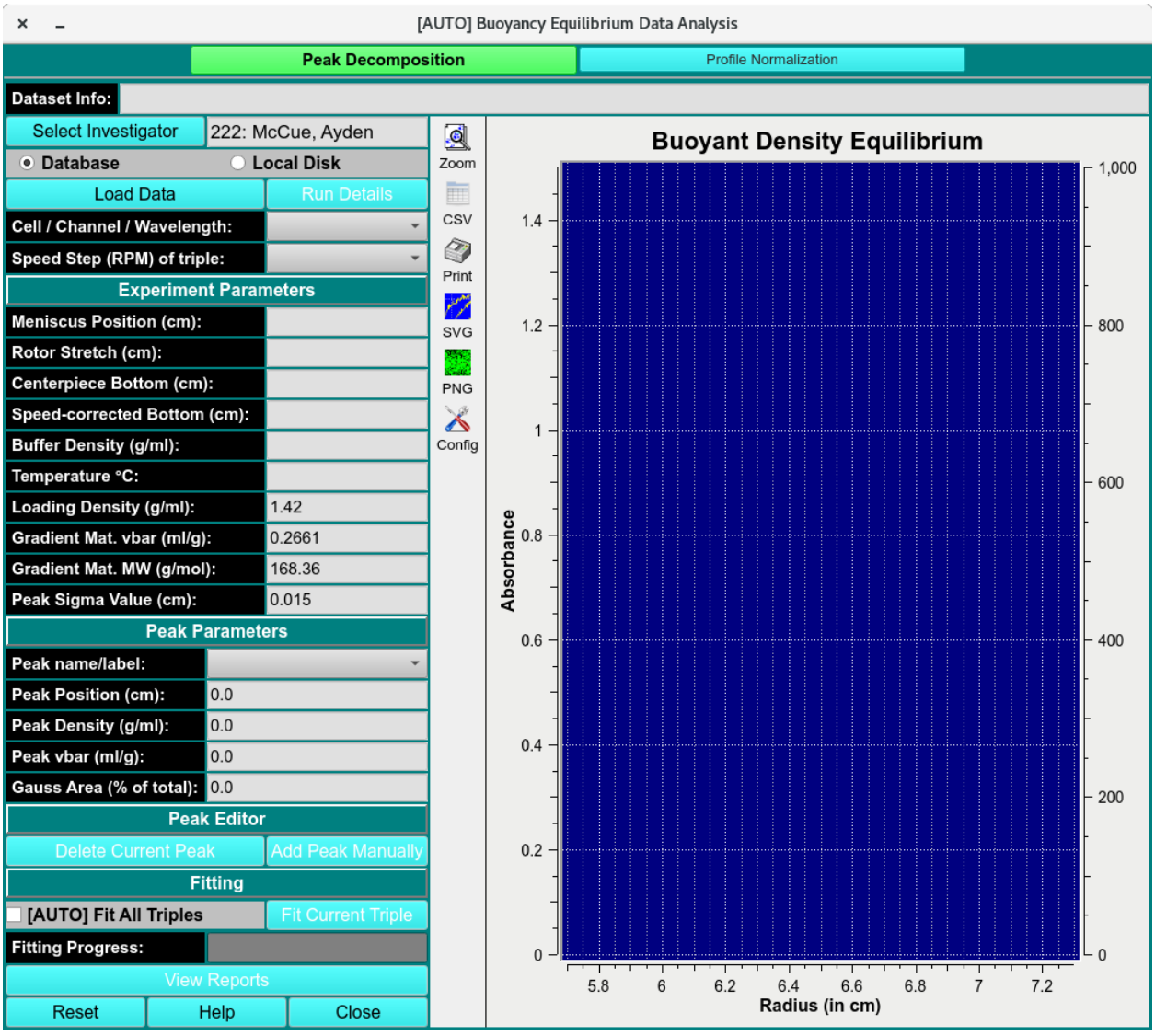
(D2O Percent column):	Text boxes allow setting D2O percent values for each model.
(Density (g/l) column)	Text boxes allow setting density values in grams per liter for each model. Initial values are computed and often need not be manually changed by the user.
(Label column):	Text boxes allow setting labels for each model that are used in the Legend of each plot.
(Description column):	Read-only text boxes display the description text associated with each model.
Help:	Display this documentation.
Cancel:	Exit this dialog without returning parameter values to the caller.
Accept:	Exit this dialog and return parameter values to the caller.
Compute Densities:	Compute densities values for all models with non-zero D2O percent values .

Triple========== Buoyancy Equilibrium Data Analysis ==========

Index

- *Peak Decomposition*
  - *Peak Decom. Functions:*
    - *Experimental Parameters*
    - *Peak Parameters*
    - *Peak Editor*
    - *Peak Fitting*
- *Profile Normalization*
  - *Profile Norma. Functions:*
  - *Plots*

Peak selection

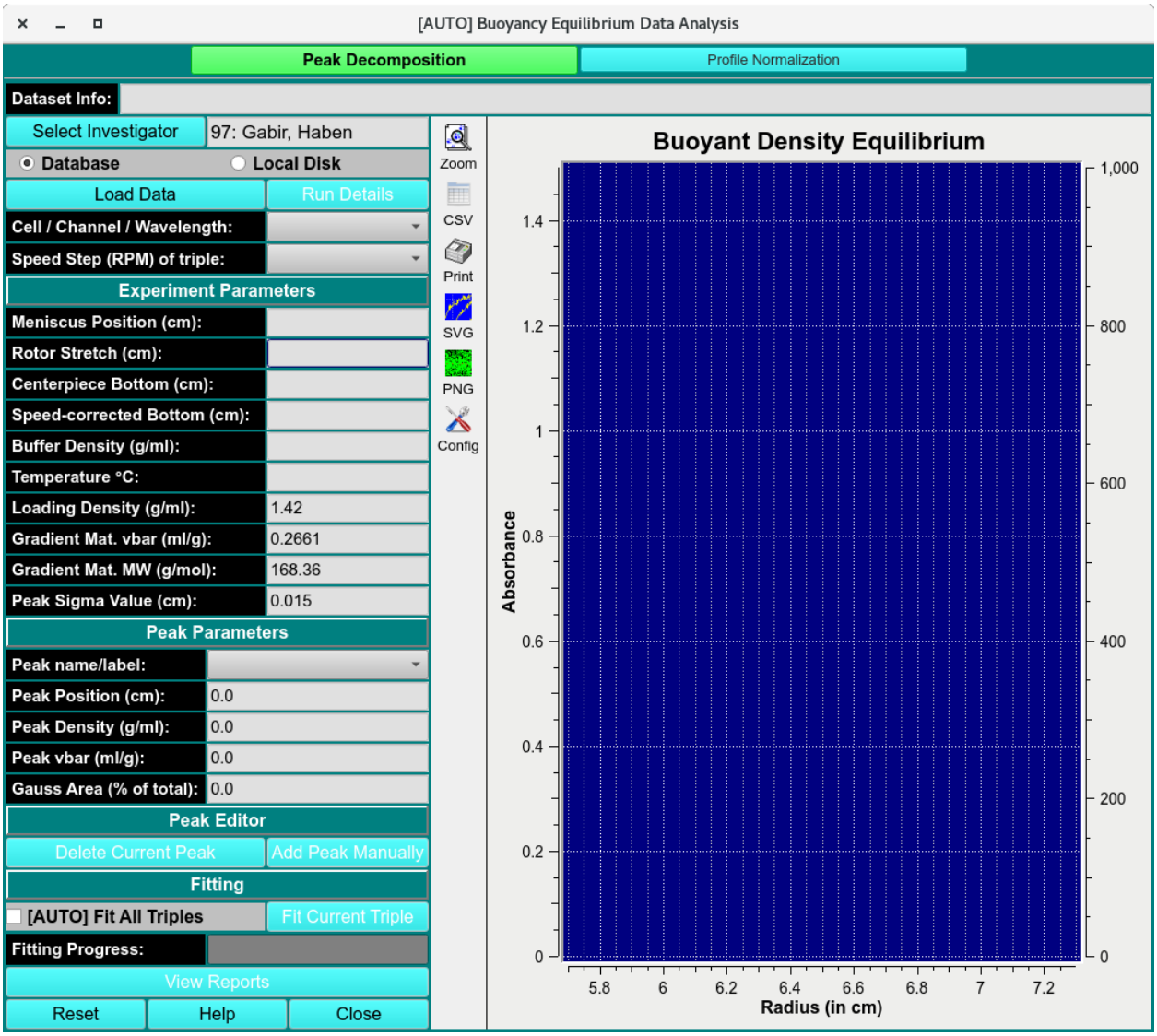


ABDE Analysis Main Window

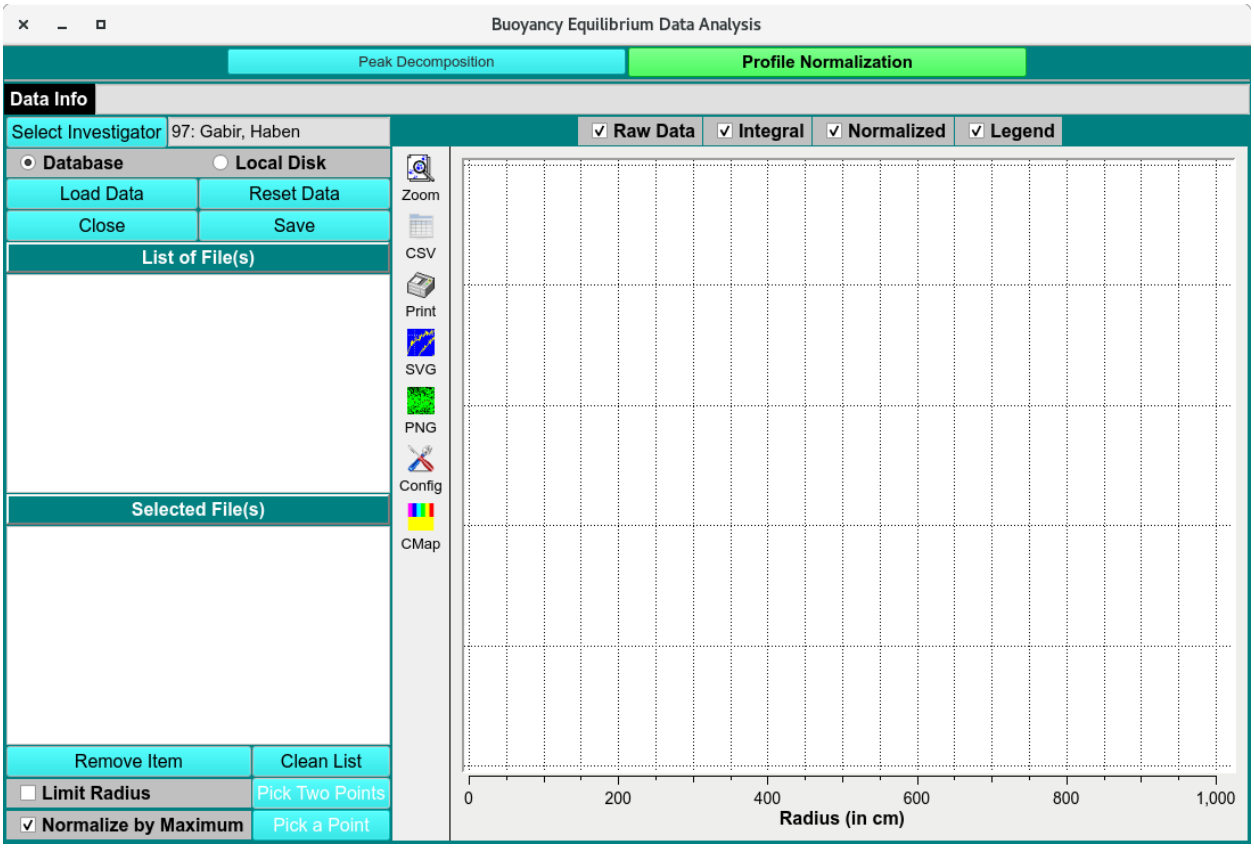
CHAPTER

FORTYSEVEN

PEAK DECOMPOSITION



Spectrum Fitter



Spectrum Fitter

47.1 Peak Decom. Functions:

Dataset Info:
Database
Local Disk
Load Data
Run Details
Cell /Channel/ Wavelength:
Speed Step (RPM) of Triple:

47.1.1 Experimental Parameters

Meniscus Position (cm):
Rotor Stretch (cm):
Centerpiece Bottom (cm):
Speed-corrected Bottom (cm):
Buffer Density (g/ml):
Temperature °C:
Loading Density (g/ml):
Gradient Mat. vbar (ml/g):
Gradient Mat. MW (g/mol):
Peak Sigma Value (cm):

47.1.2 Peak Parameters

Peak name/label
Peak Position (cm):
Peak Density (g/ml):
Peak vbar (ml/g):
Gauss area (1% of total):

47.1.3 Peak Editor

Delete Current Peak
Add Peak Manually

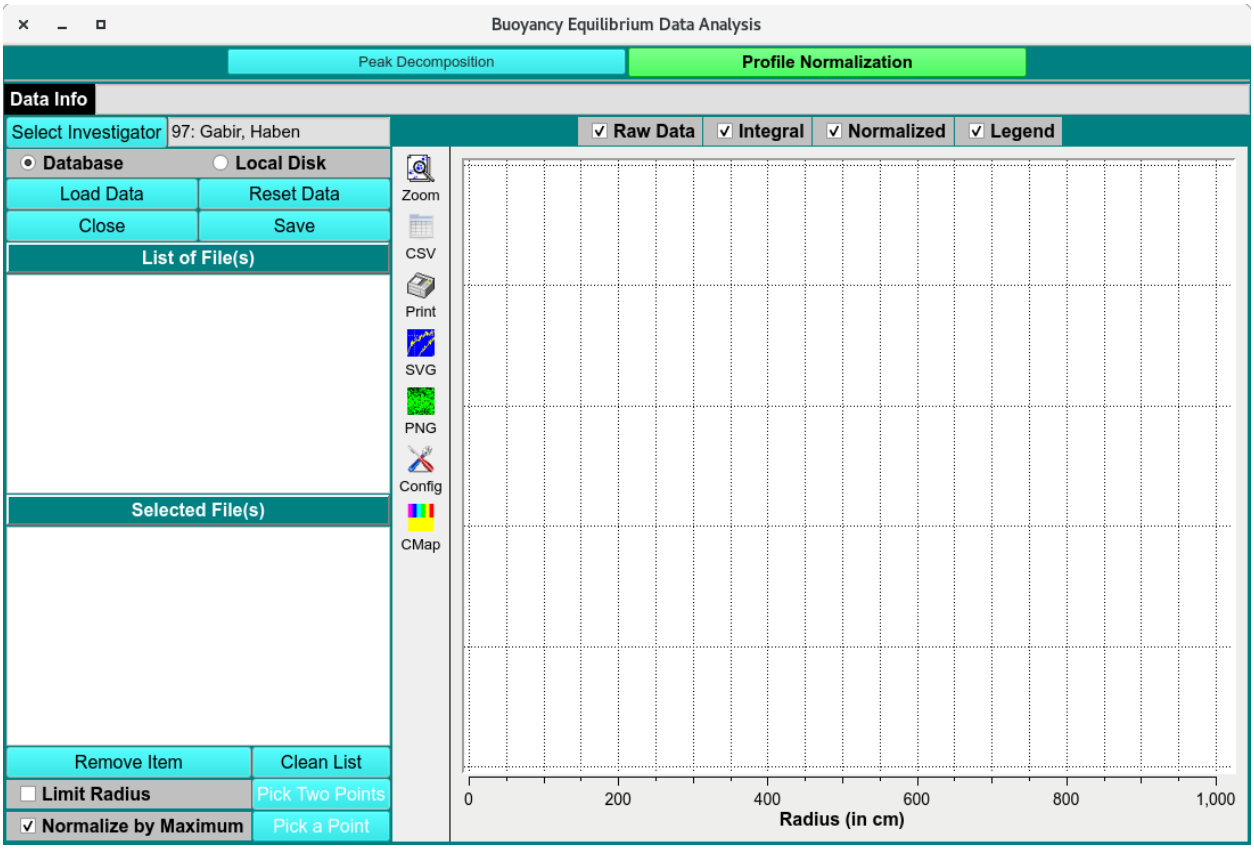
47.1.4 Peak Fitting

Fit All Triples
Fit Current Triples
Fitting Progress:
View Reports
Reset
Help
Close

CHAPTER

FORTYEIGHT

PROFILE NORMALIZATION



Spectrum Fitter

48.1 Profile Norma. Functions:

Load Data
Reset Data
Close
Save
List of File(s)
Selected File(s)
Remove Item
Clean List
Limit radius
Pick Two Points
Normalize by Maximum
Pick a Points

48.2 Plots

Raw Data
Integral
Normalized
Legend

CHAPTER

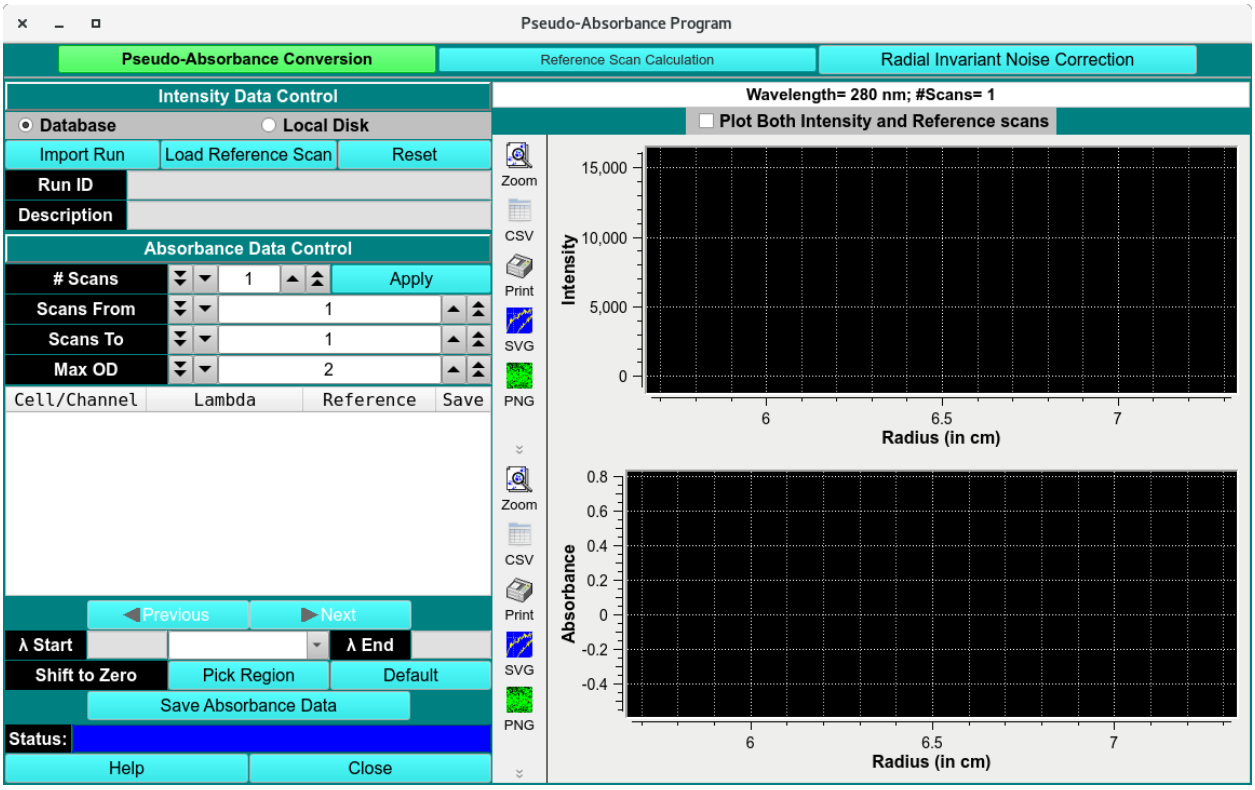
FORTYNINE

PSEUDO-ABSORBANCE PROGRAM

Index

- *Process:*
- *Pseudo-Absorbance Conversion*
  - *Pseudo-Absorbance Functions:*

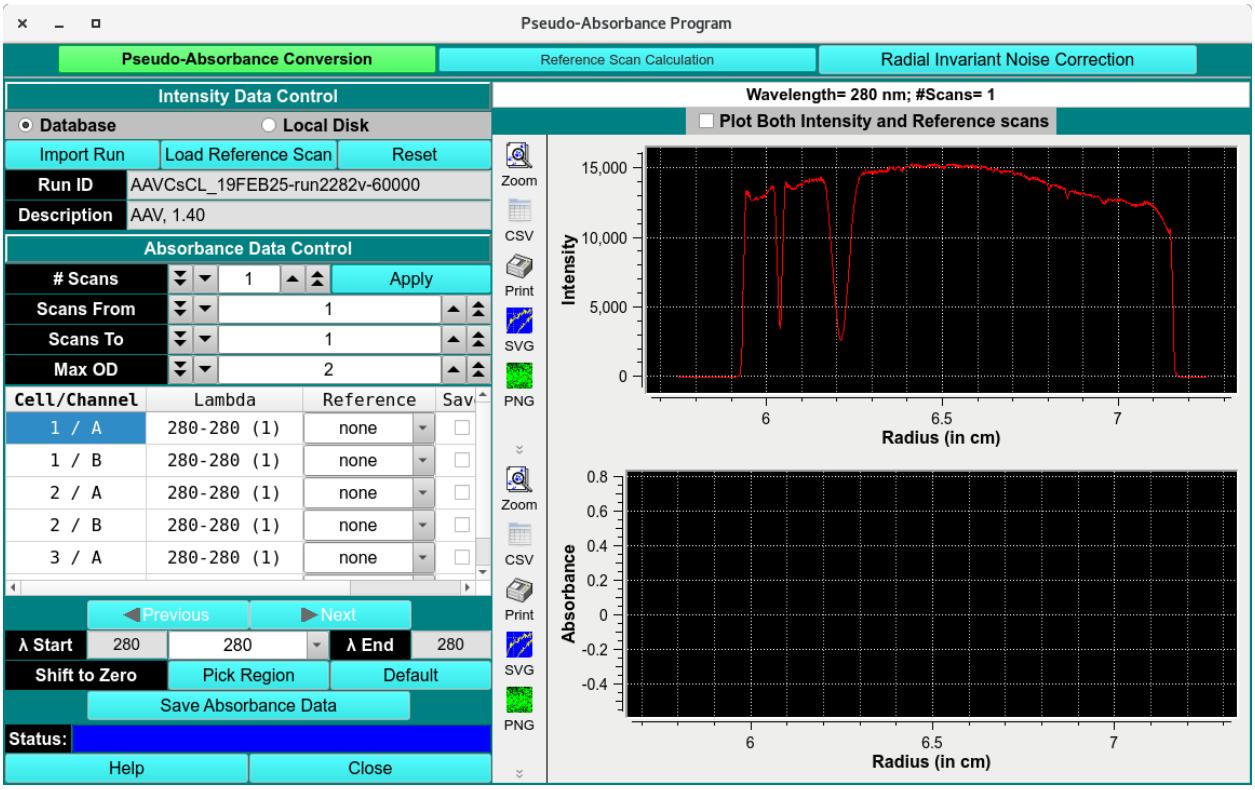
The Pseudo-Absorbance Program is the first processing step in the characterization and quantification of Analytical Buoyancy Density Equilibrium (ABDE) experiments used to determine AAV capsid loading. In ABDE, solutes are separated according to their buoyant density. This module initializes the data by converting raw intensity scans into pseudo-absorbance data through reference subtraction and baseline correction. Reference scans are defined and subtracted from the sample scans, and the resulting corrected data are saved with the prefix **PSAB-**. Reference scans used by this module must be acquired from the Reference channel (B) of the same centerpiece.



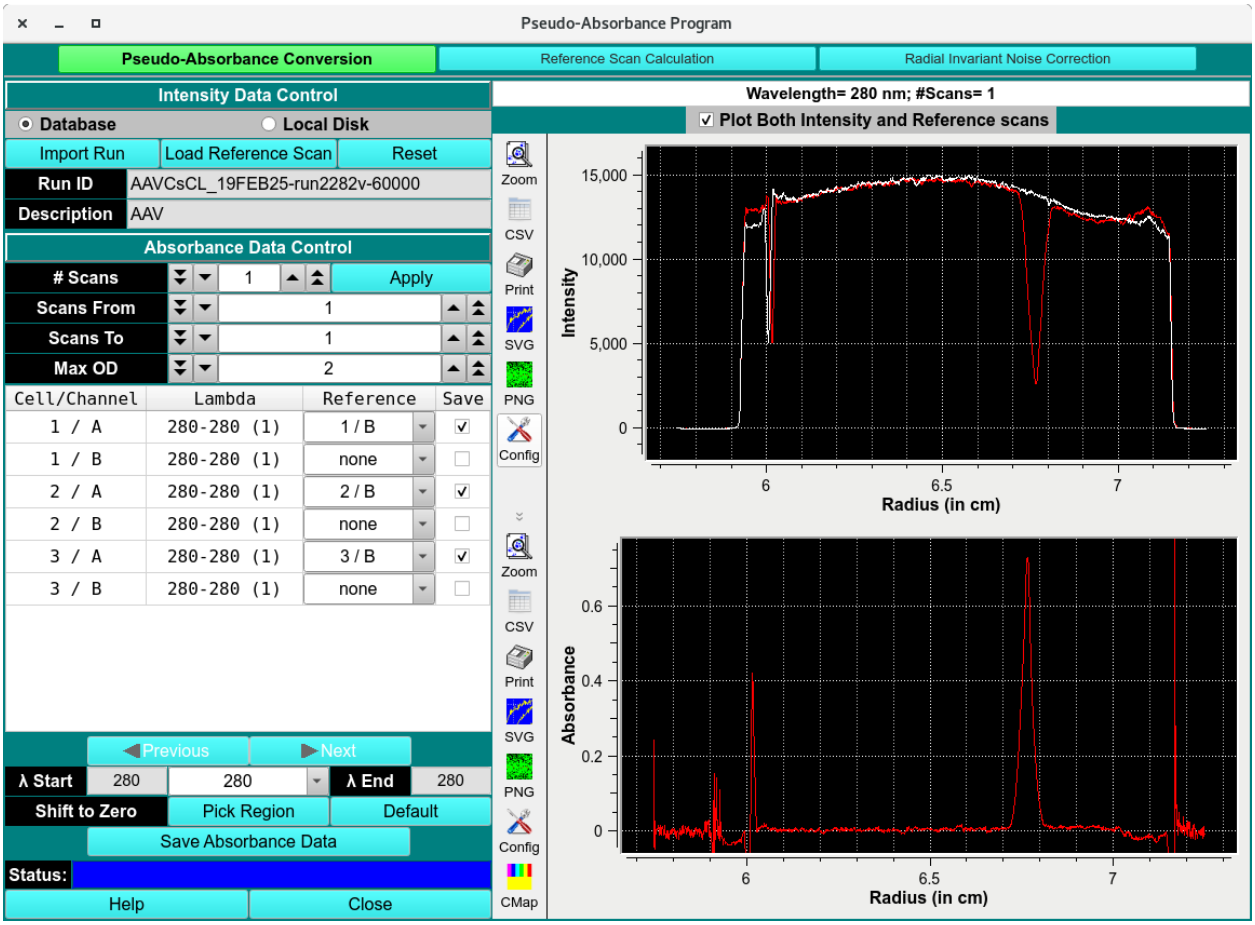
ABDE

49.1 Process:

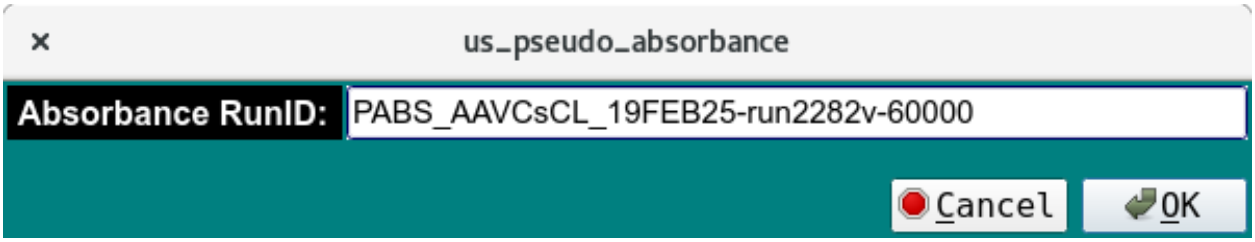
1. Click **Import Run** and load ABDE dataset using a Load Run Data Dialog.



2. Assign the reference channel in the Reference pulldown menu. The reference will automatically be subtracted. Select which Triple to save by clicking the **Save** checkbox.



3. Ensure the subtraction is saved by **apply**. Edit name and save **PABS-** auc file in $HOME/ultrascan3/imports.



49.2 Pseudo-Absorbance Conversion

49.2.1 Pseudo-Absorbance Functions:

Database	Select to specify data input from the database.
Local Disk	Select to specify data input from local disk or the database.
Import Run	Click here import the dataset using *Load Data Dialog*, select a run ID for the data set to load.
Load Reference Scan	Load a reference scan from a different dataset if no Reference Scan was included.
Reset	Reset all parameters to defaults.
Run ID	the Run identifier.
Description	The name of the solution and user inputted comments
# Scan	The number of the scan.
Scan From	Select Scan from
Scans To	Select Scan To
Max OD	The OD cutoff of the absorbance.
Cell/Channel	Cell and channel column
Lambda	The wavelength(s) samples were measured by.
Reference	Define the reference scans in this column.
Save	Select which Cell/Channel to save.
Previous	Navigate to the previous wavelength.
Next	Navigate to the next wavelength.
Lambda Start	Start range of the wavelengths detected.
Lambda End	End range of the wavelengths detected.
Shift to Zero	Select region at baseline and shift whole scan to baseline.
Pick Region	Define the region to baseline.
Default	Use default parameters.
Save Absorbance Data	Save the absorbance data.
Status	Status bar.
Plot Both Intensity and Reference Scans	checkbox to include both sample and reference scans.
Intensity Plot	Plot of Intensity data, before subtraction.
Absorbance Plot	Plot of absorbance data, after subtraction.
Help	Open help documentation of this module.
Close	Close window.

CHAPTER

FIFTY

CUSTOM GRID EDITOR (CG)

Index

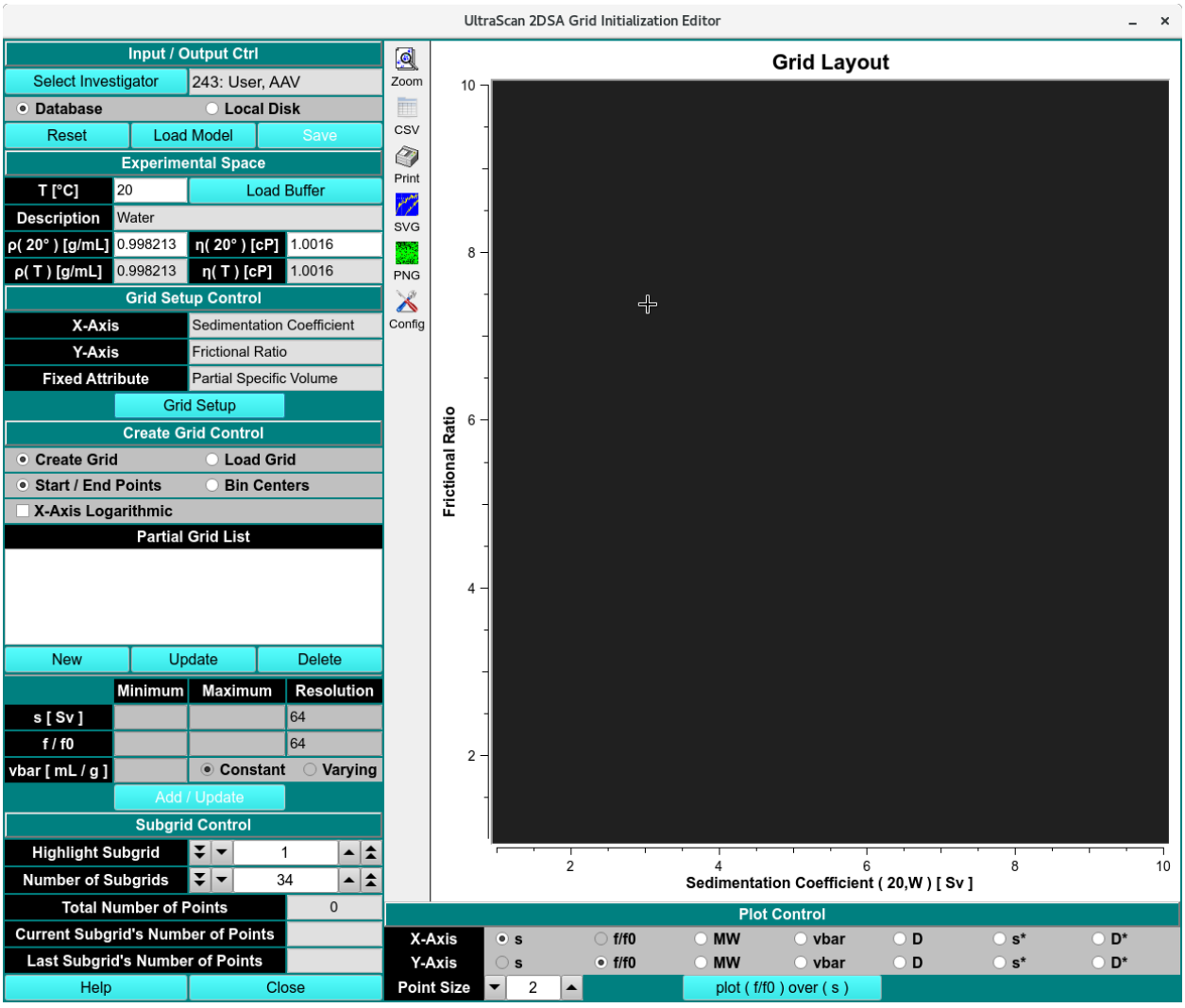
- *Processes:*
  - *Create a Grid*
  - *View or Edit an Existing Grid*
- *Custom Grid Functions:*
  - *Input/Output Controls*
  - *Experimental Space*
  - *Grid Setup Control*
- *Create Setup Control*
  - *Grid Spacing:*
  - *Subgrid Control*
  - *Plot Control*
  - *Set Z-Value Function:*

This module enables you to create and edit custom 2-dimensional initialization grids for the 2-dimensional spectrum analysis.

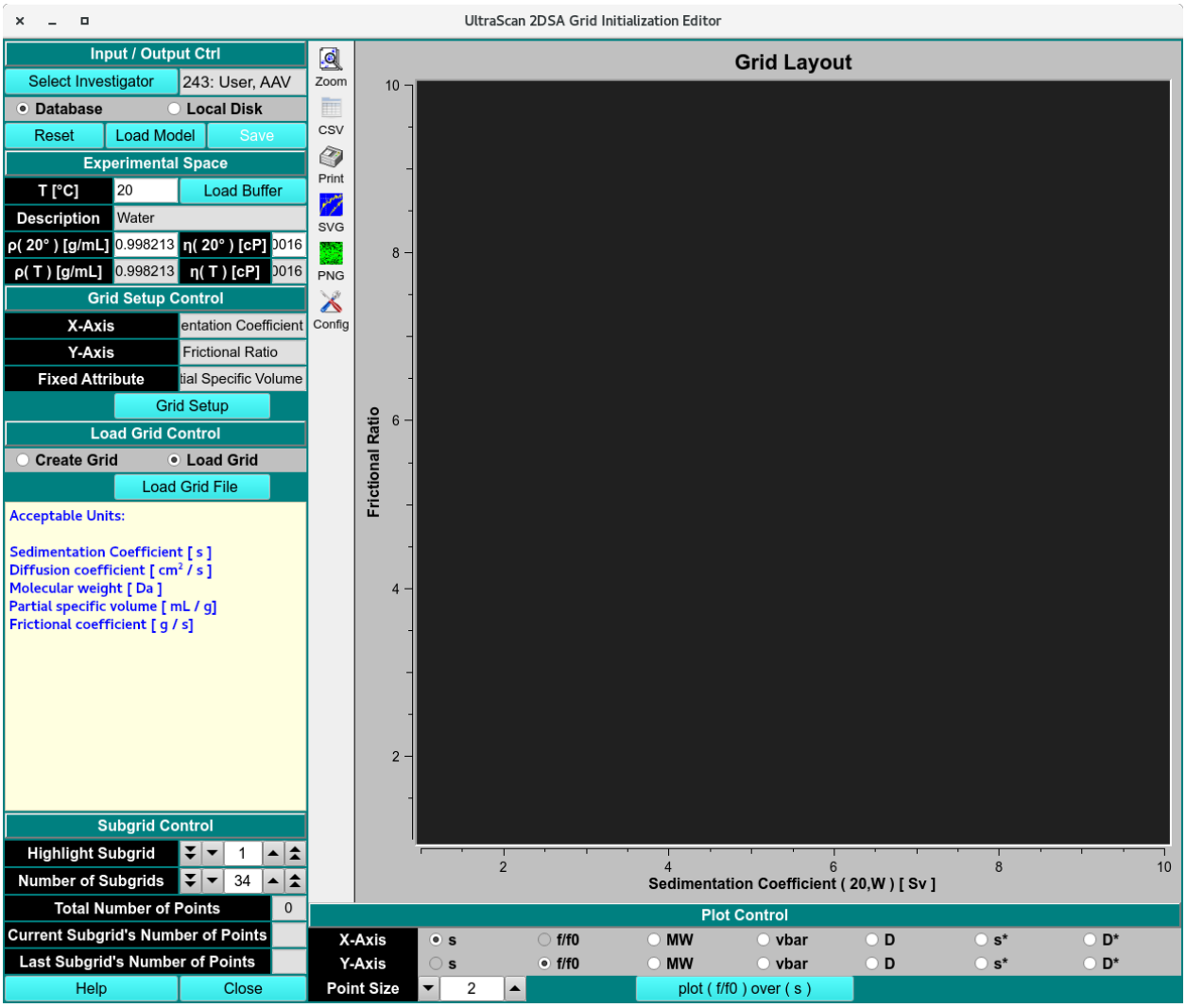
- Custom grids can be defined by varying two if the attributes listed and fixing one (**sedimentation coefficient**, **frictional ratio**, **molecular weight**, **vbar** and **diffusion coefficient**).
- It is also possible to define multiple grid regions and combine them into a single grid which defines multiple solute regions.
- The custom grid can be re-displayed in the other domains left unchanged.

Note: It is also possible to define grids that contain both sedimenting and floating species, in which case the grid needs to be defined for a fixed frictional ratio, but permits variations in vbar to account for the difference in buoyancy while maintaining positive molecular weights.

- All combined partial grids are saved as a special model structure that can be stored in the database or on disk to be submitted with either the desktop or supercomputing version of the 2-dimensional spectrum analysis.



Custom Grid Editor - Create Model



Custom Grid Editor - Load Model

50.1 Processes:

50.1.1 Create a Grid

1. Select the investigator to save the model to or from.
2. Click *Load Buffer* to define the experimental parameters of the dataset by loading the buffer used. Once selected the density and viscosity of the buffer will populate the Description, Density (ρ) and Viscosity (η) at 20°C and the selected temperature.
3. Under the Grid Setup Controls, select the two variable attributes and one fixed attribute.
4. Select *Create Grid* radio button and prerequisite the grid by picking *Start/End Points* or *bin centers* to start the grid points at the edge of bin or center.

Note: For unevenly distributed samples over a large s range, the x attribute can be spaced logarithmically clicking *X-Axis Logarithmic*. This is typically done for samples with aggregates.

5. To add a partial grid to the list, click *New* and the X and Y attribute minimum, maximum and resolution boxes will be available to edit.
6. Select *Constant* for the Fixed Attribute or *Varying* if prior information is available. Varying the fixed Attribute will create a 3-D box or line, for best results, contact Ultrascan-III Development Team
7. Repeat steps 5-7 for every partial grids needed.
8. Once the grid is created, the *Current Subgrid's Number of Points* must be ≤ 150. Increase or decrease *Number of Subgrids* to adjust *Current Subgrid's Number of Points* to the right value and *a diffusely spaced subgrid* (displayed as Yellow dots).

Note: Each subgrid should be small enough to fit into the computer memory of the analysis program. The more subgrids are defined, the more grids will need to be calculated. If sufficient cores are available, this could speed up the analysis, but it will also lead to increasingly inefficient use of the available cores as there will be CPU-idling in the later stages of the analysis when the outputs of multiple grids are merged.

9. Once the grid has been created, click *Save* and a *Save Grid Model* dialog will appear to edit name and save.

50.1.2 View or Edit an Existing Grid

Note: Users cannot change the grid setup for existing grids.

1. Select the investigator to load the model from.
2. Click *Load Model* to call Load Distribution Model dialog and Accept. The selected model grid will populate the Grid plot.
3. From the Partial Grid List, select the grid to edit and click *Update* or *Delete*. This will to enable the grayed out area.
4. Adjust the values of the X and Y attributes and click *Add/Update*.
5. Once the grid has been edited, click *Save* and a *Save Grid Model* dialog will appear to edit name and save.

50.2 Custom Grid Functions:

50.2.1 Input/Output Controls

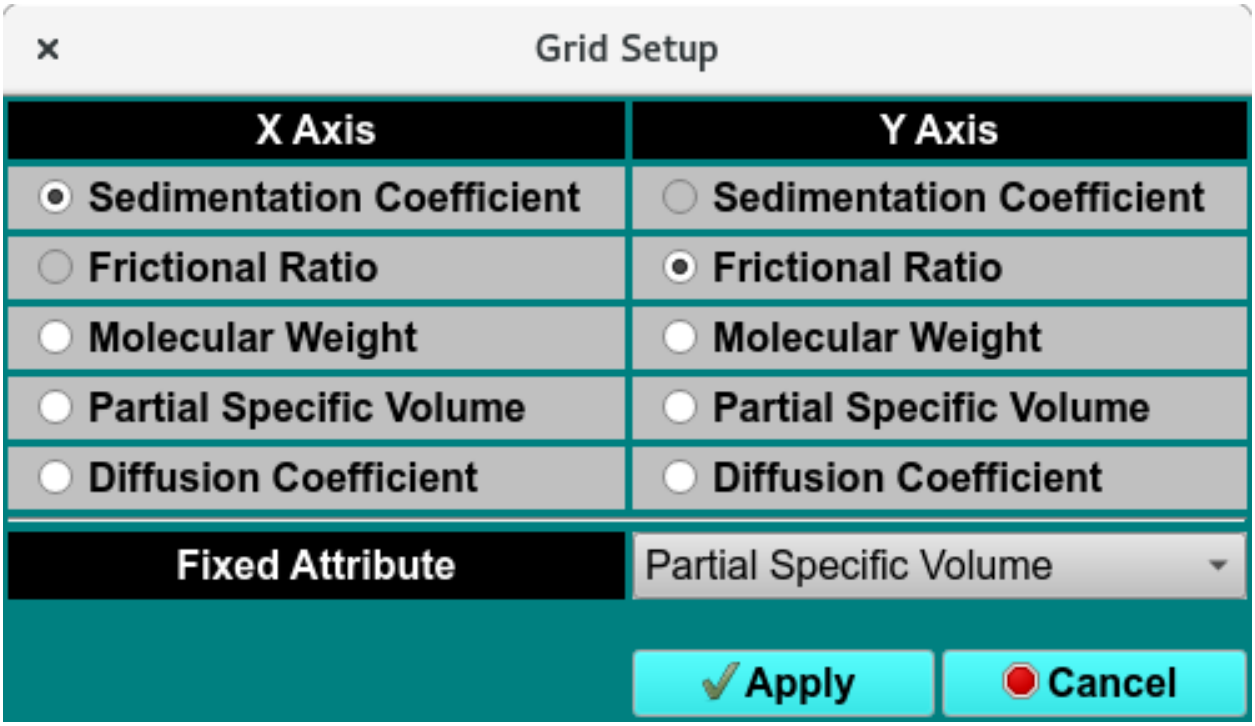
Select Investigator	Click here to retrieve the investigator management widget.
Database/Local Disk	Select the data target. If Database is selected, the model will be stored in the database. This is required if submitting a supercomputing job for the custom grid. Local Disk will only save the grid to the local disk for use in the desktop version of the 2-dimensional spectrum analysis.

50.2.2 Experimental Space

T [C]	The temperature selected by the user in the Buffer Management module. Manually setting the temperature will readout **arbitrary Value!!!** error.
Load Buffer	Load the buffer using the Buffer Management module.
Description	Read out the buffer selected or error if temperature is not set in the buffer management module.
p (20°C) [g/mL]	The density value is set by default to those of water at 20°C.
n (20°C) [cP]	The viscosity value are set by default to those of water at 20°C.
p (T°C) [g/mL]	The density value set by user.
n (T°C) [cP]	The viscosity value set by user.

Note: The density and viscosity values are set by default to those of water at 20°C. This means that all hydrodynamics variables are adjusted to standard conditions, which is what the analysis routines expect on input.For academic purposes, it is possible to change these values to see what the resulting grids look like. Under normal circumstances these values should never be adjusted.

50.2.3 Grid Setup Control



Custom Grid Editor - Grid Setup Dialog

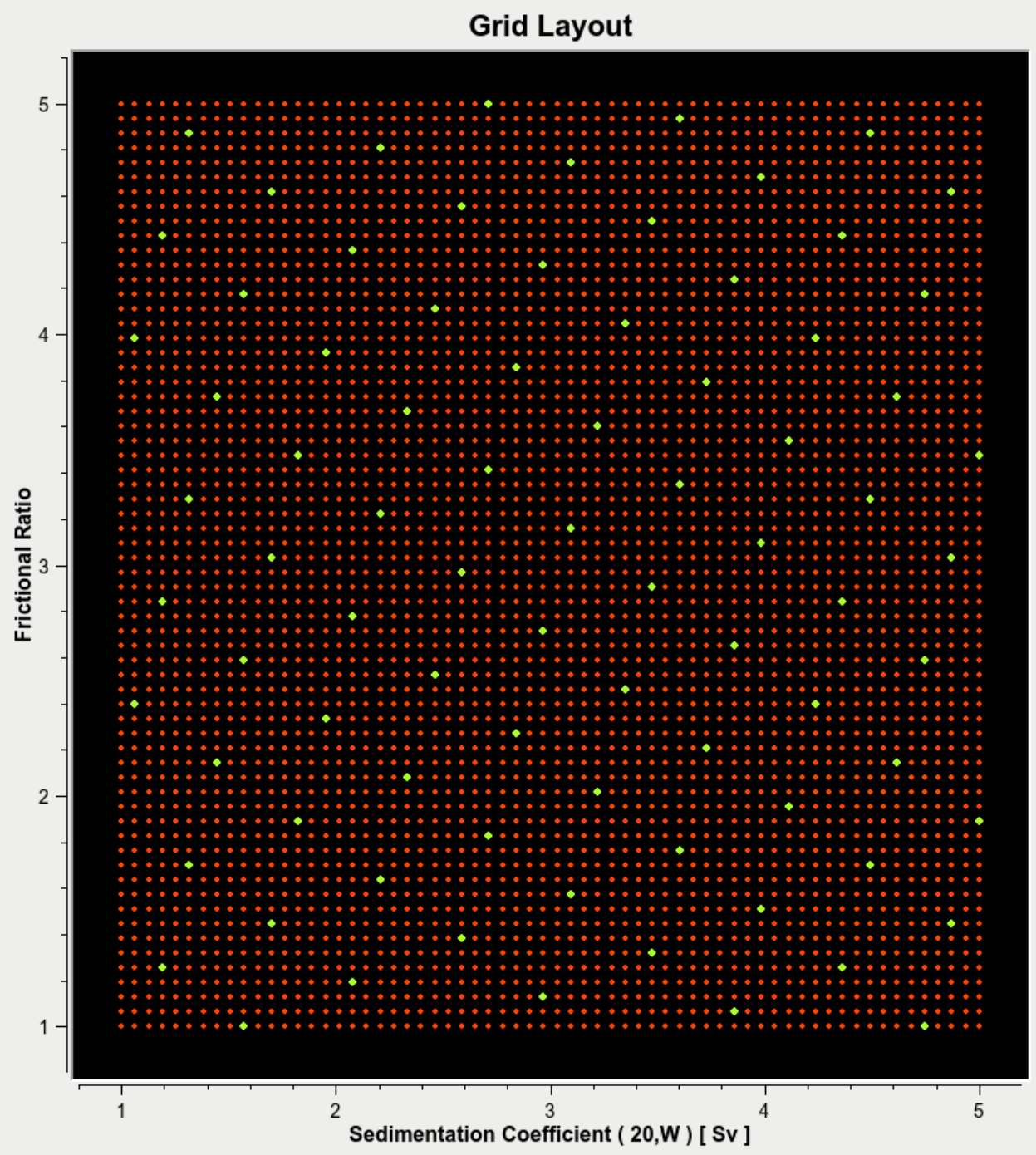
X-Axis	Displays the selected X-Axis attribute.
Y-Axis	Displays the selected Y-Axis attribute.
Fixed Attribute	Displays the selected fixed attribute.
Grid Setup	Click to choose the X-Axis, Y-Axis and fixed attributes.

50.3 Create Setup Control

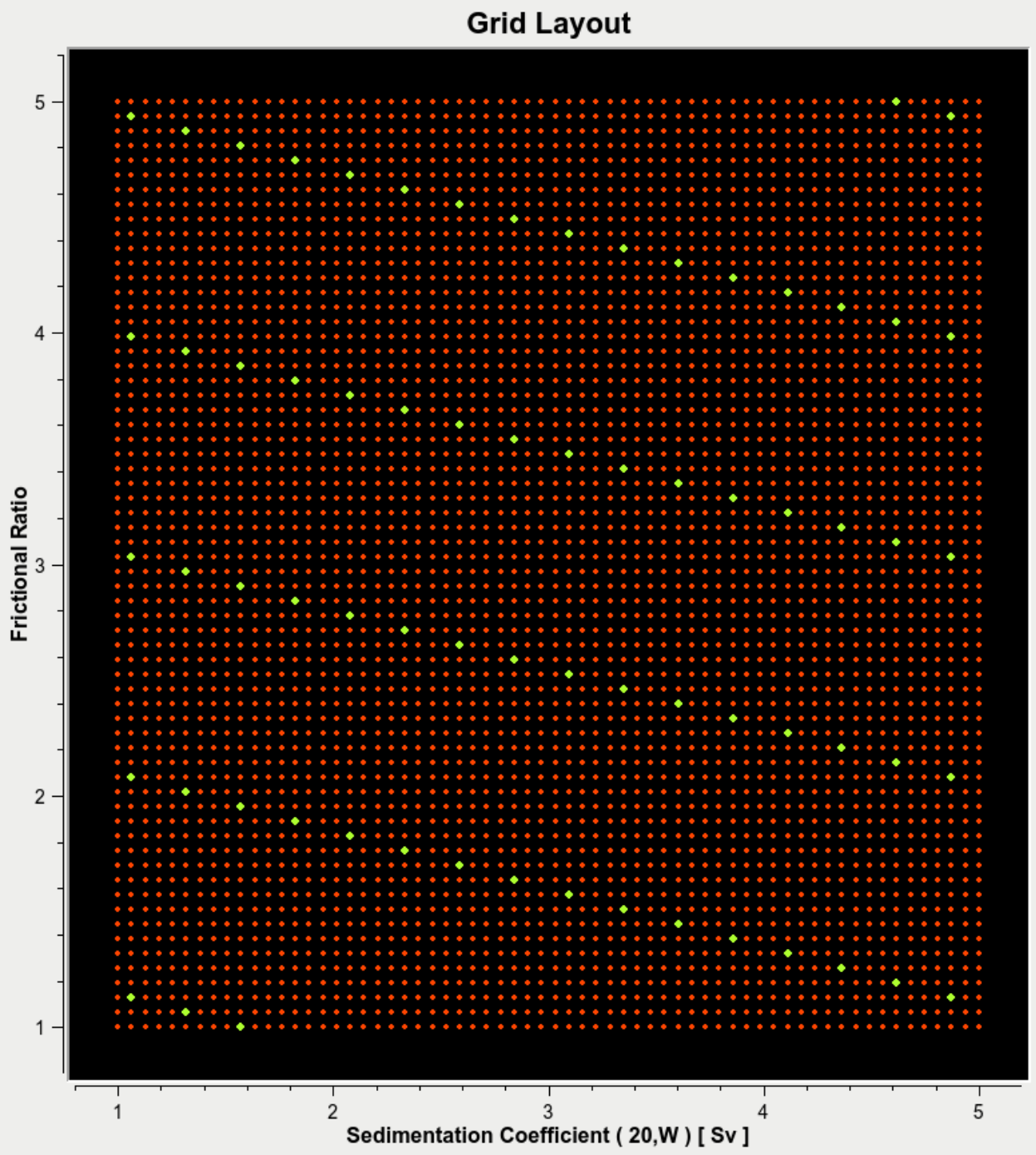
Create Grid	Radio button to call all the functions required to create a grid as show *above*.
Load Grid	Radio button to call all the functions required to load a pre-existing grid as show *above*.
Start / End Points	Select to start and end the grid points on the integer of the X and Y Axis attributes.
Bin Centers	Select to start and end the grid points on the middle of the X and Y Axis attribute integers.
Partial Grid List	List of the grids made.
New	Create new subgrid.
Update	Update and edit existing subgrid.
Delete	Delete existing subgrid.
Minimum and Maximum controls:	use these controls to adjust the minimum and maximum of the grid dimension. The labels will change according to the choice for X and Y dimensions.
Resolution	These controls allow you to set the grid resolution for any partial grid in both the X and Y dimensions. The label on these controls will change according to the selected X and Y dimensions, i.e., s-value or Mol. Weight Resolution:. The grid resolution should be high enough to cover the resolution contained in the experimental data. Excessive resolution will significantly increase the computational cost, without any appreciable benefit in goodness of fit. A too coarse resolution will not produce a good fit. If you see obvious step functions in the model that do not trace the underlying data well then you likely are not using sufficient resolution in your grid.
Constant	Keep the fixed attribute constant.
Varying	Vary the fixed attribute in the *Set Z-Value Function*
Add/Update	Apply and merge the grid parameters to total grid model.

50.3.1 Grid Spacing:

The parameter are required to be spaced evenly, and should not exclude major regions of the search space. Below are examples of Good and Bad grid spacing.



Good Spacing


Grid Layout
Frictional Ratio
1
2
3
4
5
Sedimentation Coefficient (20,W) [Sv]

Bad Spacing

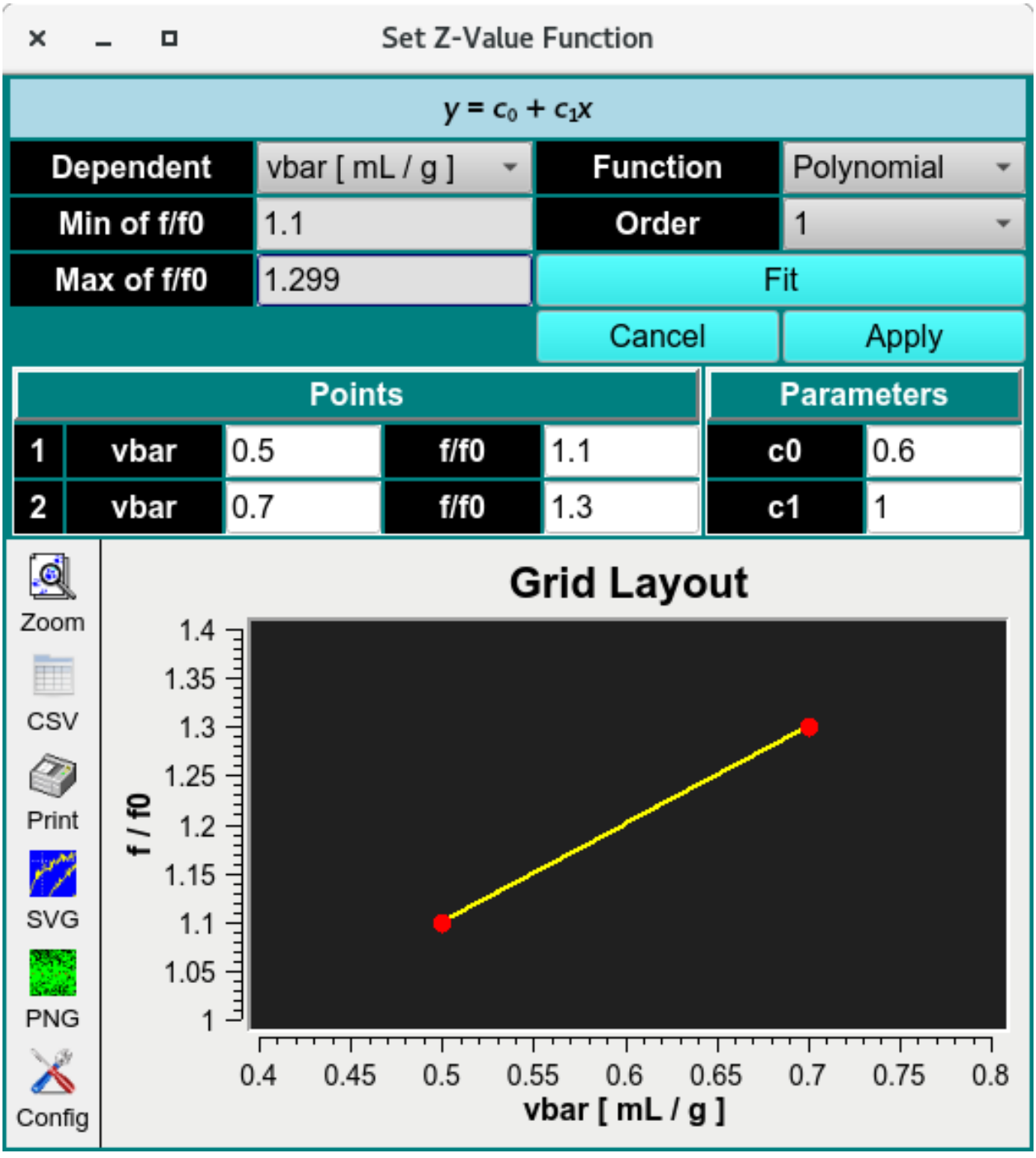
50.3.2 Subgrid Control

Highlight Subgrid	Highlight one of the total subgrids.
Number of Subgrids	The total number of subgrids. The number can be increased or decreased depending on computational availability or need, See .
Total Number of Points	Number of subgrids X Number of points per subgrid
Current Subgrid's Number of Points	The number of points in current subgrid.
Last Subgrid's Number of Points	The number of points in the last subgrid.
Help	Software documentation for the custom grid module.
Close	Close the window.

50.3.3 Plot Control

X-Axis and Y-Axis	these controls are used to switch between views of the defined grid. A rectangular grid may be defined in the sedimentation domain, which looks like this when displayed with the molecular weight view.
Point Size	Size of the points.
plot ( ) over ( )	Revert to the rectangular grid.

50.3.4 Set Z-Value Function:



Set Z-Value Function

The **Set Z-Value Function** (advanced control) in the Custom Grid module lets users vary the third variable using predefined *Z-Value Dependency Function* equations. These equations constrain the third attribute as a function of one of the two variable attributes, using prior information to define the relationship.

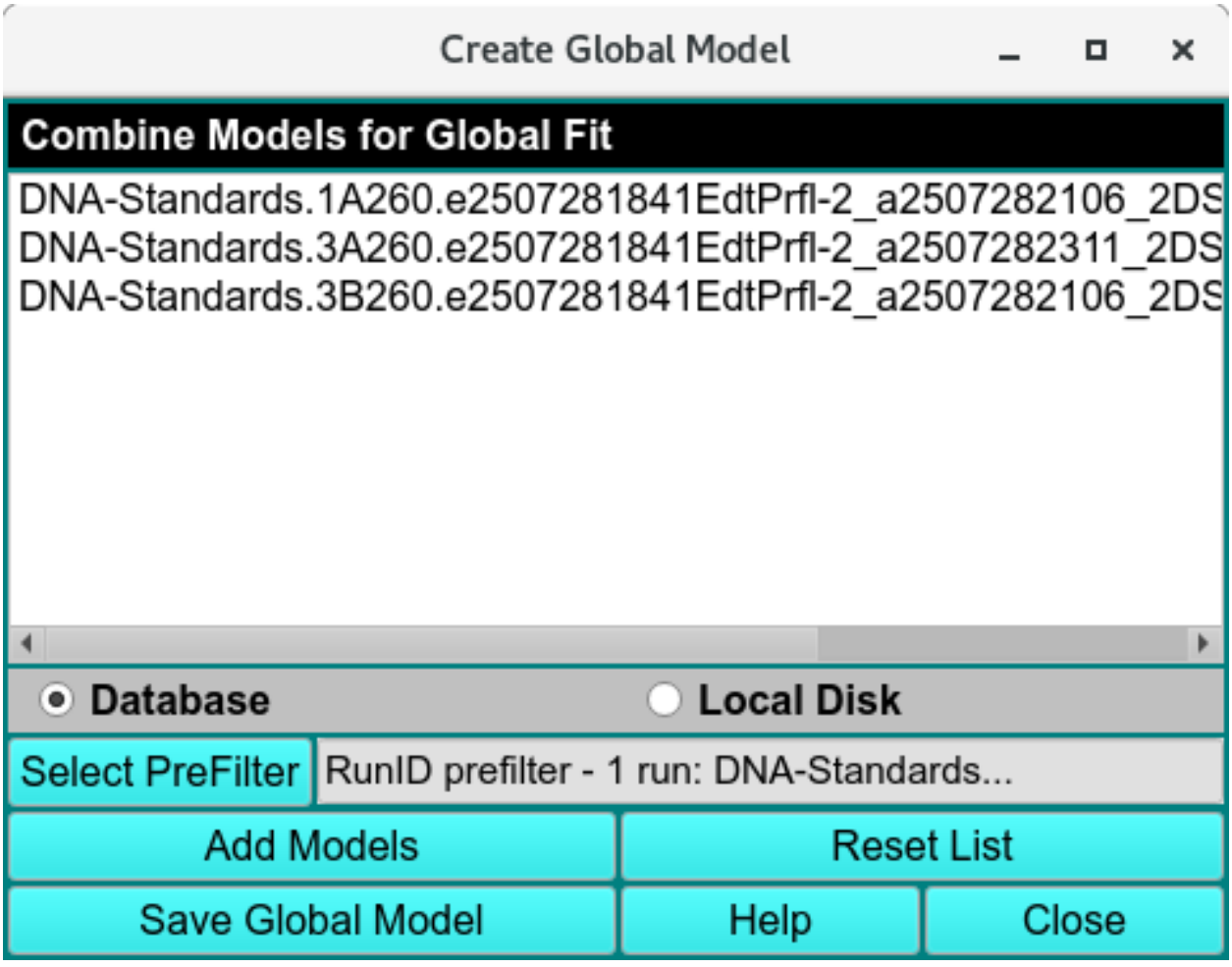
Dependent	The variable attribute the third attribute will be constrained to.
Min of (Fixed)	The minimum value of the Dependent.
Max of (Fixed)	The maximum value of the Dependent.
Function	Select between polynomial or exponential functions.
Order	Select the order of the polynomial functions (1-5).
Fit	Fit the function and function parameters enters.
Cancel	Cancel Fit.
Apply	Apply the fitted function to the main window.
Grid Points	Enter the interval of the x and y attributes
Grid Parameters	Enter the partial concentration of the x and y attributes
Grid Layout	Plot the function and parameters selected.

CHAPTER

FIFTYONE

CREATE GLOBAL MODEL

This utility combines selected models into a single global model that can be used in global fits. The most common use of its output is to form an input to the Initialize Genetic Algorithm program in which a global set of buckets may be created for use in a global fit Genetic Algorithm run.



Create Global Model

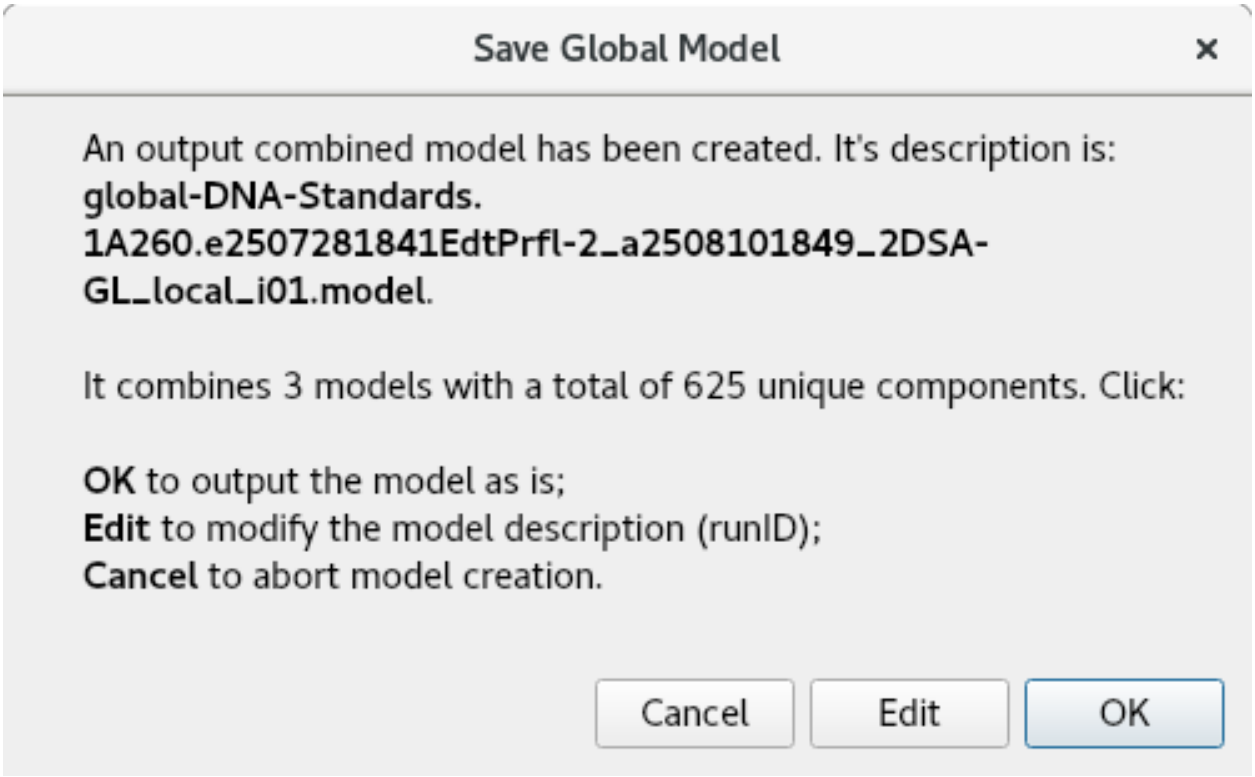
Window Controls

(models list)	Look here to see the list of component models so far selected.
Database	Select to specify model input from the database.
Local Disk	Select to specify model input from local disk.
Select PreFilter:	Open a *Models Pre-Filter dialog* to select Run IDs on which to pre-filter lists of models for loading.
Add Models	Open a *Model Loader dialog* to select model(s) to load into the list of component models to the global model.
Reset List	Clear the list of input models in order to start over in the model selection process.
Save Global Model	Open a dialog (see below) to save the output combined global model.

Window Controls

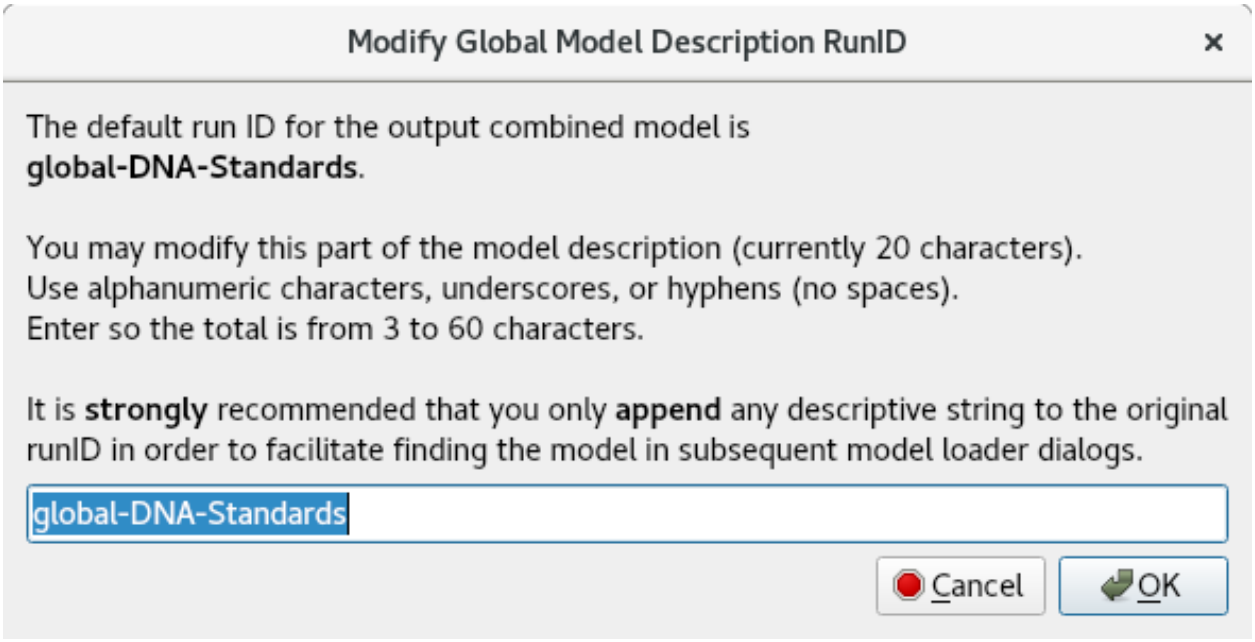
Help	Display this detailed Create Global Model help.
Close	Close all windows and exit.

When the Save Global Model button is clicked, a dialog reports on the combined global model to be created and allows the user options in saving it.



Save a Global Model

If the user wishes to modify the RunID part of the output model description, a click on the **Edit** button brings up a dialog for entering new ID text.



Edit Global Model Name

CHAPTER

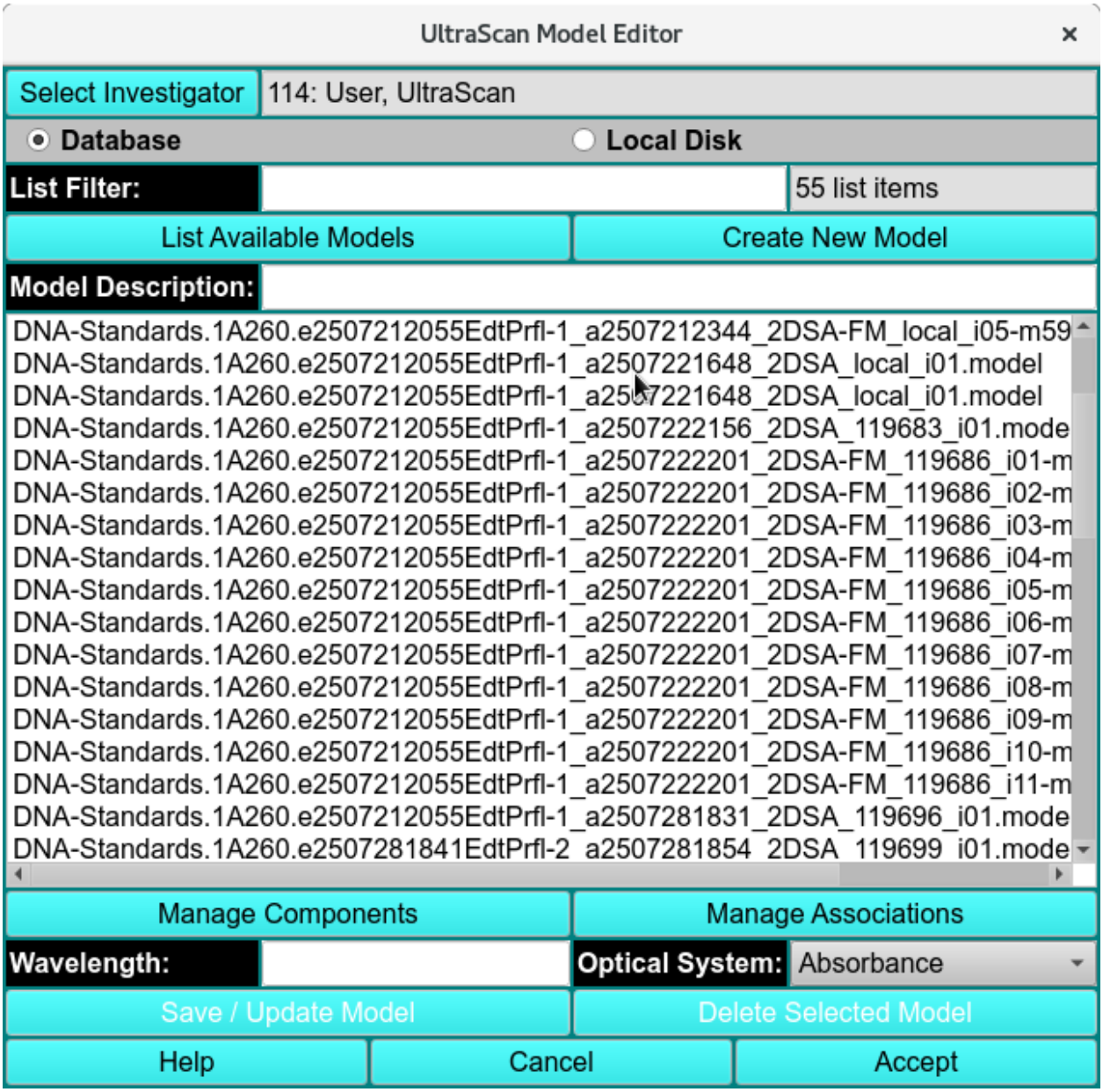
FIFTYTWO

ULTRASCAN MODEL EDITOR

Index

- *Model Editor Functions:*

Using this window, you choose a model to load and can create or edit a model.



Model Editor

52.1 Model Editor Functions:

Select Investigator	This button brings up an *Investigator Dialog* that allows selecting the current investigator for whom to list models.
Use Database	Check to select loading or editing the model from the database.
Use Local Disk	Check to select loading or editing the model from the hard disk.
List Available Models	This button initiates population of the list of models from the designated source (DB or disk).
Create New Model	This button designates that the model to be specified and used by the caller is new. An entry with the name **New Model** will be added to the list. It should be changed in the Model Description text box to an appropriate name.
Model Description:	The text box here echoes the name of the model currently selected in the list. The text box entry can be edited to change the model name; of use, particularly, when a new model is being created.
Manage Components	This button brings up a *Model Components Dialog* in which the model components and their properties can be specified.
Manage Associations	This button brings up a *Model Associations Dialog* in which reversible association chemical equations can be defined.
Wavelength:	Specify the model wavelength value in this text box.
Optical System:	Select the model optical system: Absorbance, Interference, or Fluorescence.
Global Identifier:	Read-only global identifier of the model.
Save / Update Model	Save the new or edited model to the database or local disk.
Delete Selected Model	Delete the selected model from the database or local disk.
Accept	Close the dialog and return the model selection to the caller.
Help	Show this documentation.
Cancel	Close the dialog and do not return the model selection to the caller.

CHAPTER

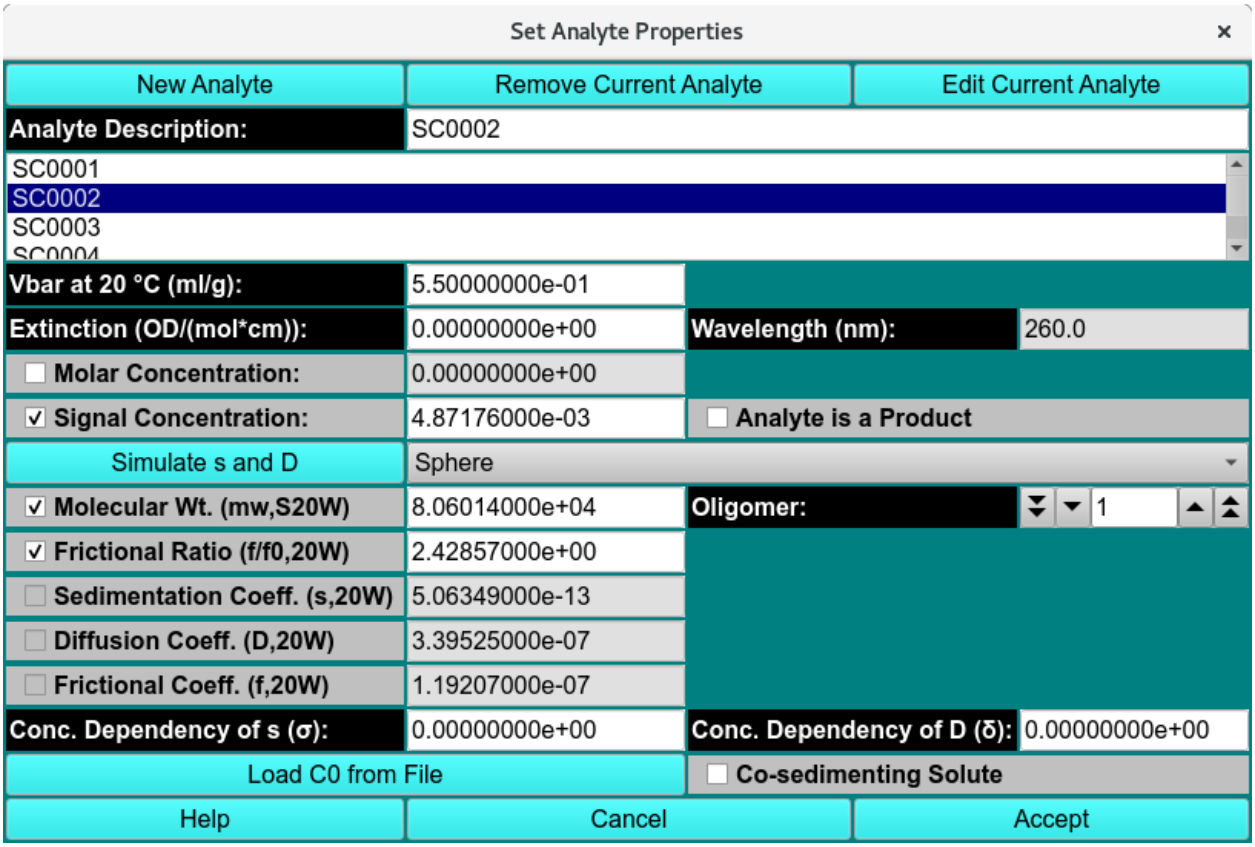
FIFTYTHREE

MODEL COMPONENTS

Index

- *Model Comp. Functions:*

In this dialog, you can set the properties of components of a model.



Analyte Properties

53.1 Model Comp. Functions:

New Analyte	Click to create a new analyte and add it to the list of model components, using an *Analyte Management dialog*.
Remove Current Analyte	Click to remove from the list of model components the analyte currently selected.
Edit Current Analyte	Click to replace a model components with a new or ready-made analyte and add it to the list of model components, using an *Analyte Management dialog*.
Analyte Description:	Enter in the text box the name to be given a newly created analyte or the new name to be given to a currently selected analyte.
(Analyte List)	Click on the name of an analyte to select it for editing of its properties.
Global Identifier	Read-only global identifier of the analyte.
Vbar at 20 °C (ml/g):	Enter the vbar value at 20 degrees centigrade.
Extinction (optical units)/Wavelength:	Enter the extinction value for the read-only-displayed wavelength.
Molar Concentration:	Read-only molar concentration calculated from Extinction and Signal Concentration.
Analyte Signal Concentration:	Enter the analyte's signal concentration value.
Simulate s and D	Bring up a *Shape dialog* that uses axial ratio to model s, D, and f from MW for 4 basic shapes.
(Shape)	Select molecule shape: Sphere, Prolate Ellipsoid, Oblate Ellipsoid, or Rod. The selection here should be made before any use of the Shape dialog.
(Parameters/Coefficients)	Exactly 2 of the 5 component parameters to follow should be checked and their values possibly modified. If a different pair than currently checked is desired, you must first uncheck one of the boxes to enable selection of an alternate. Values for unchecked parameters are calculated from the two selected.
Molecular Wt. (mw) / Oligomer:	Check Molecular Weight to enable setting a value for it in the text box. For a polygomer component, set the Oligomer value greater than 1.
Frictional Ratio (f/f0) (20W)	Select frictional ratio and set a value for it.
Sedimentation Coeff. (s) (20W)	Select sedimentation coefficient and set a value for it.
Diffusion Coeff. (D) (20W)	Select diffusion coefficient and set a value for it.
Frictional Coeff. (f) (20W)	Select frictional coefficient and set a value for it.
Analyte is a Product	click if the analyte is a product.
Oligomer	Set the oligomeric status.
Conc. Dependency of s (σ):	Set a non-zero value to specify concentration dependency and its sedimentation coefficient sigma factor.
Conc. Dependency of D (δ):	Set a non-zero value to specify concentration dependency and its diffusion coefficient delta factor.
Load C0 from File	Click this button to begin an input file dialog in which you can specify a file with first-scan concentration values.
Co-sedimenting Solute	Check this box to specify that the current component is a co-sedimenting solute.
Cancel	Close the dialog and do not return component selections to the caller.
Accept	Close the dialog and return component selections to the caller.

CHAPTER

FIFTYFOUR

MODEL ASSOCIATIONS

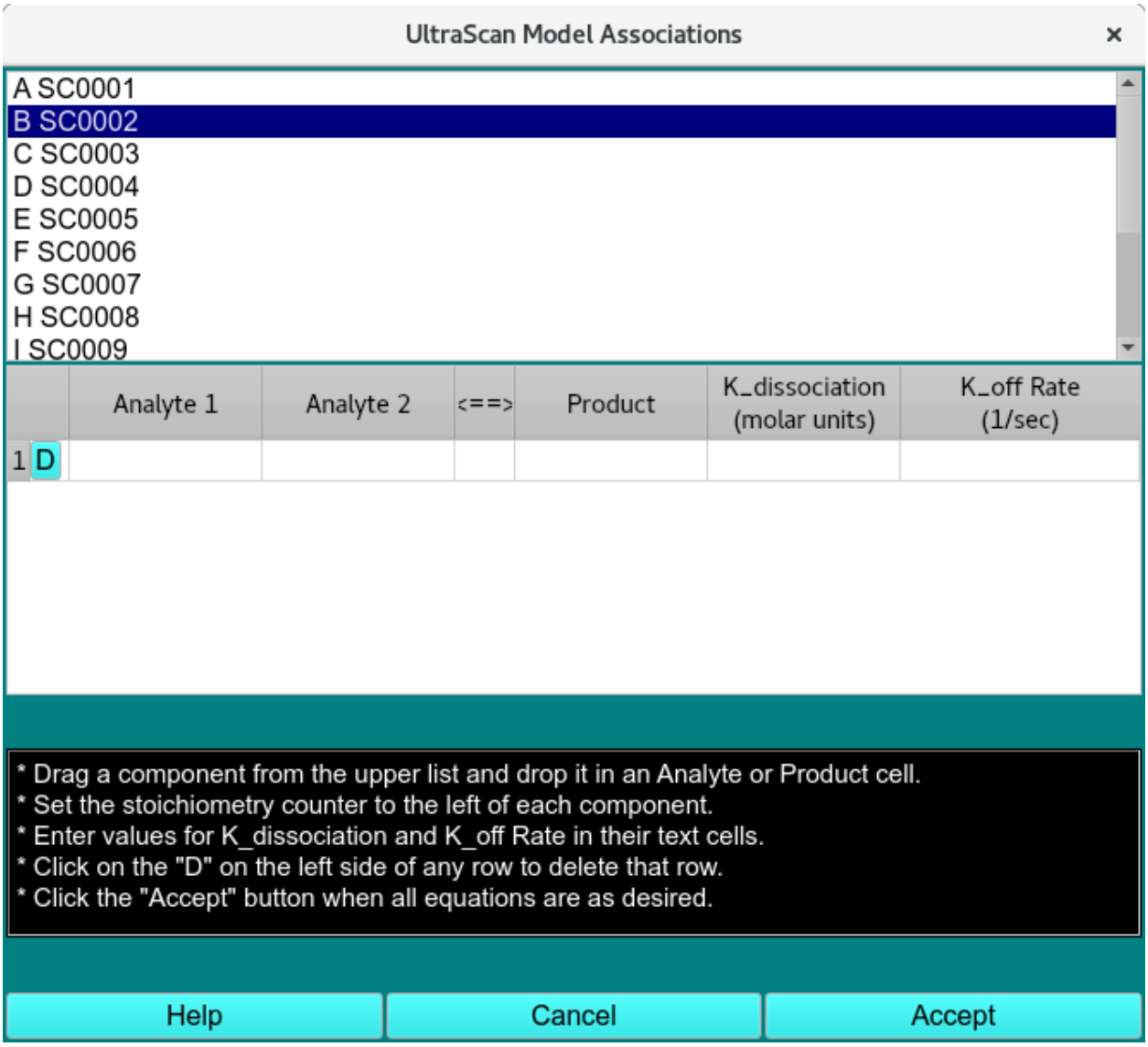
Index

- *Model Assoc. Functions:*

In this dialog, you can define reversible associations of model components. Associations are constructed in terms of chemical equations with one or two analytes on one side and a product on the other.

Each association row is built by dragging a model component in the upper list to an Analyte or Product cell. As each analyte is dropped, its code letter (A, B, C, …) appears in the cell with a subscript that represents the oligomer number of the molecule. At least one Analyte and one Product cell must be thus populated. The implied chemical equation must be balanced by setting the stoichiometry counter in each cell such that stoichiometry times oligomer value(s) on the left equals the product of stoichiometry and oligomer on the right. K_association and K_off Rate values should be entered for each row.

Once all association rows have been validly built, click the Accept button to set the reversible association parameters for the model. If a row is wrong and cannot be easily corrected, click the “D” button on the left side of the row to delete that row so that it can be rebuilt.



Create Associations

54.1 Model Assoc. Functions:

(analyte list)	The upper list of model components is populated from the model passed by the calling object. Members of this list can be dragged to the table below to construct association equations.
(chemical equation associations table)	A table with one or more rows is constructed to represent the reversible associations present in the model.
D	These buttons in the first table column allow you to delete the associated row.
Analyte 1	Each row for an association must have this column populated by dragging and dropping a list component.
Analyte 2	Each row for an association may optionally have this column populated by dragging and dropping a list component.
Product	Each row for an association must have this column populated by dragging and dropping a list component. Like the Analyte cell(s), a component will be represented by its code letter and a subscript showing its oligomer number. The stoichiometry counter should be set so stoichiometry-times-oligomer balances on the sides of the equation.
K_association	Enter the K_association value in molar units for each association row.
K_off Rate	Enter the K_off Rate value in reciprocal seconds for each association row.
Help	Show this documentation.
Cancel	Close the dialog and do not return association specifications to the caller.
Accept	Close the dialog and return association specifications to the caller.

Note: If you click on the **Accept** button and any equation does not balance, a dialog informing you of the imbalance will pop up. You then have the opportunity to correct the equation and **Accept** again or to click on the **Cancel** button.

CHAPTER

FIFTYFIVE

SIMULATION MODULE

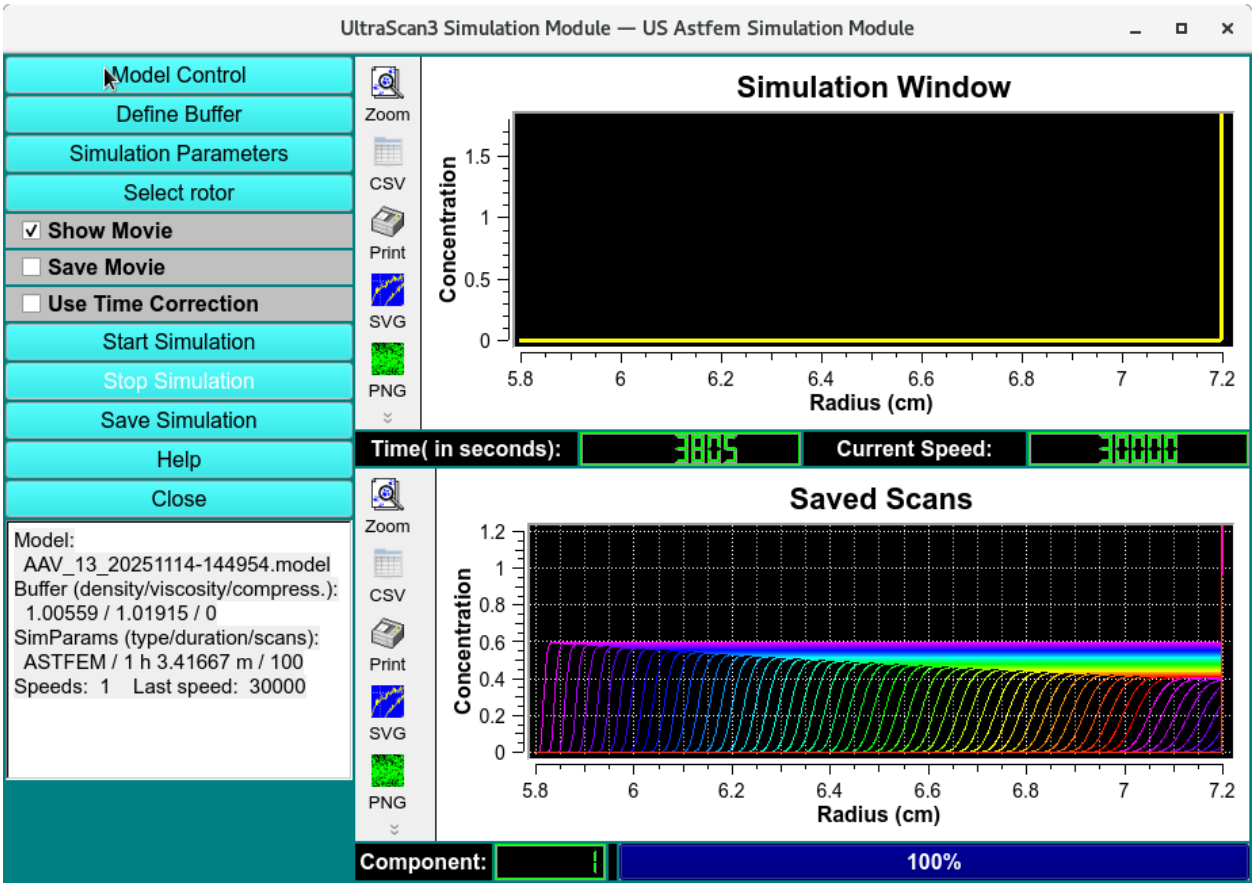
Index

- *Simulation Process:*
- *Simulation Functions:*
- *Clip*

This module is used to simulate an experiment using Finite Element (ASTFEM) or Finite Volume (ASTFVM) methods. A model is loaded from the database or from local disk. Simulation parameters are specified. A simulation is then calculated and displayed. The simulation may be saved as a synthetic version of raw experimental data. If model and simulation parameters indicate ideal conditions, the default method is Adaptive Space Time Finite Element Method (ASTFEM). For non-ideal cases - such as concentration dependence, co-sedimenting, or compressibility, the default is Adaptive Space Time Finite Volume Method (ASTFVM).

55.1 Simulation Process:

1. **Model Control:** First, load a model on which the simulation will be based.
2. **Simulation Parameters:** Secondly, open a dialog to specify parameters governing the simulation.
3. **Define Buffer:** Optionally (and rarely), you may wish to specify buffer conditions. This may be necessary if your intention is to simulate and save an artificial version of a specific experimental data set. Normally, the buffer conditions are those of water at 20 degrees Centigrade.
4. **Start Simulation:** Initiate calculation of the specified simulation.
5. **Display and/or Save Results:** After simulation, a variety of options are available for displaying simulation results and for optionally saving them to disk.



Simulation Main Window

55.2 Simulation Functions:

Model Control	Load a model from database or local disk using a *Model Editor* dialog.
Define Buffer	Normally not used, this button allows you to specify buffer conditions through a *Buffer Management* dialog.
Simulation Parameters	The majority of the parameters that govern the output simulation grid, duration, method, and so on are given in the result of clicking this button: a *Simulation Parameters* dialog.
Select Rotor	Load a rotor and its calibration data by opening the :doc:`Rotor Management <rotor> dialog.
Show Movie	Have intermediate scan curves displayed in the upper right plot window during simulation calculations by checking this box. This function allows the user to select a run time which allows the analyte sediments fully.
Use Time Correction	Specify with a check here that you want time corrections to be used in the course of simulation calculations.
Start Simulation	Click this button to begin simulation calculations.
Stop Simulation	If the simulation seems to be taking an inordinate amount of time and you suspect incorrect setting of parameters as the cause, you may click here to stop calculations so that you can review and modify parameters.
Save Simulation	Clicking this button leads to a standard output directory dialog in which you may specify a ./results folder on local disk into which to save a simulated raw experiment data set. This data set may be desirable for testing purposes, in which case it will need further editing through *Edit <us_edit>*.
Help	Display this documentation.
Close	Exit this dialog.
(status summary)	A text box displays a brief summary of model, buffer, and simulation parameters chosen thus far.
(Simulation Window)	This plot window shows intermediate scan curves as they are generated, when "Show Movie" has been checked.
(Saved Scans)	This plot window displays the final result of simulation calculations.
(progress counters and bars)	Several counters and bars illustrate ongoing progress during simulation calculation.

55.3 Clip

CHAPTER

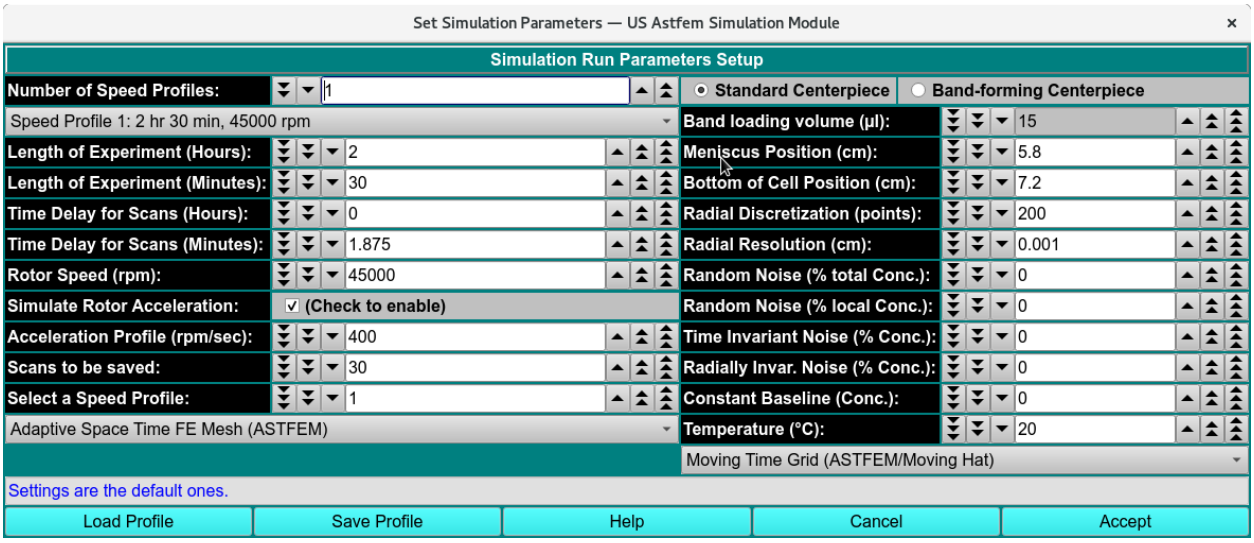
FIFTYSIX

SIMULATION PARAMETERS

Index

- *Simulation Param. Functions:*

An option in UltraScan-III Simulation Module, this dialog allows specifying parameters governing a simulation. The dialog allows multiple speed profiles in the course of the simulation. Items along the left side of the dialog refer to the currently selected speed step. Those on the right are for the general properties of the simulation, including radial ranges and simulation method.



Simulation Parameters

56.1 Simulation Param. Functions:

Number of Speed Profiles:	Specify the total number of speed profiles to define.
(Speed Profile…)	Select the speed profile for which parameters to be specified below apply.
Length of Experiment (Hours):	Current speed step duration in hours.
Length of Experiment (Minutes):	Current speed step duration in residual minutes (beyond duration given in hours).
Time Delay for Scans (Hours):	Current speed step time delay in hours.
Time Delay for Scans (Minutes):	Current speed step time delay in residual minutes (beyond delay given in hours).
Rotor Speed (rpm):	Current speed step rotor speed in revolutions per minute.
Simulate Rotor Acceleration:	Check the box to the right to enable simulation of gradual acceleration of the rotor initially or between steps with differing rotor speeds.
Acceleration Profile (rpm/sec):	If rotor acceleration is to be simulated, the value here gives the acceleration as a change in revolutions per minute for each second.
Scans to be saved:	The number of scans to be simulated and saved for this speed step.
Select a Speed Profile	Selecting a number here is an alternate way (to the "Speed Profile…" drop-down) to specify the speed step for which parameters above apply.
Standard Centerpiece	Check to specify a standard, non-band-forming centerpiece.
Band-forming Centerpiece	Check to specify a band-forming centerpiece.
Band-loading volume (ml):	Specify any band-loading volume in milliliters.
Meniscus Position (cm):	Specify the radial position of the meniscus.
Bottom of Cell Position (cm):	Specify the radial position of the simulation data bottom.
Radial Discretization (points):	Specify the number points for which concentrations are calculated over the radial range.
Radial Resolution (cm):	Specify the resolution of the radial range, the increment in centimeters between radius points.
Random Noise (% Conc.):	Specify the percent concentration of random noise to calculate and add to the simulation.
Time Invariant Noise (% Conc.):	Specify the percent concentration of time-invariant noise to calculate and add to the simulation.
Radially Invar. Noise (% Conc.):	Specify the percent concentration of radially-invariant noise to calculate and add to the simulation.
Temperature (°C):	Specify the temperature in degrees centigrade to assign to scans.
(mesh type):	Select the mesh or method type to use in the simulation: ASTFEM, CLAVERIE, MOVING-HAT, USER, ASTFVM.
(grid type):	Select the grid type to use in the simulation: MOVING, FIXED.
Load Profile	Open a dialog to load a previously saved simulation parameters profile in "sp-.xml" name form.
Save Profile	Open a dialog to save the current simulation parameters to a profile file in "sp-.xml" name form.
Help	Display this documentation.
Cancel	Exit this dialog without returning parameter values to the caller.
Accept	Exit this dialog and return parameter values to the caller.

CHAPTER

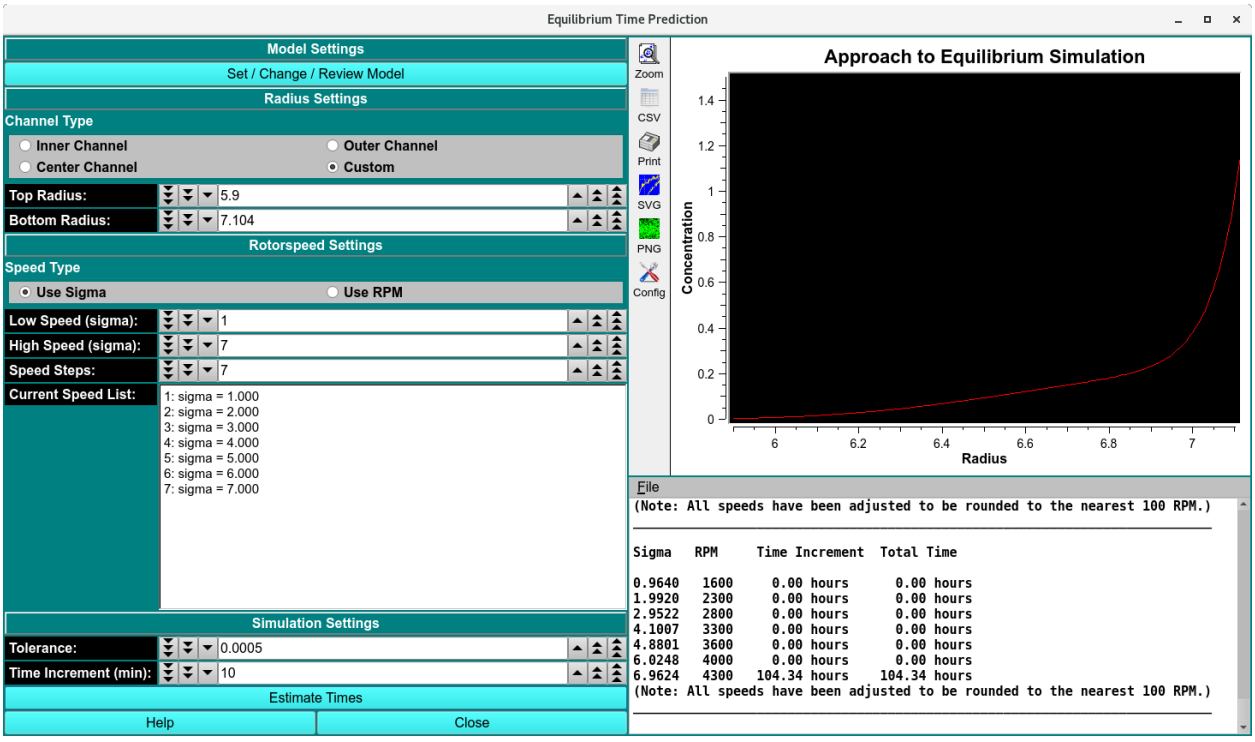
FIFTYSEVEN

ESTIMATING EQUILIBRIUM TIME

Index

- *How to use this program:*
- *Est. Equilibrium Time Functions:*

Note: Why a sedimentation velocity experiment is superior to equilibrium experiments. Link to Documents!!



Whenever you are planning or designing an equilibrium experiment, the question comes up of how long it takes to reach equilibrium at a certain speed. The time it takes depends on several factors:

- The model of the solution will govern the experiment simulation. Use the Model Editor to set the following parameters:

  - The diffusion coefficient. The smaller the diffusion coefficient, the longer it will take to reach equilibrium, since the sample will diffuse slower. Asymmetric and random coil molecules will take longer to equilibrate than globular molecules with the same molecular weight.
  - The sedimentation coefficient. The larger the sedimentation coefficient, the faster the molecule will equilibrate, provided the diffusion coefficient stays constant, which is rarely the case. As molecular weight and sedimentation coefficient increase, the diffusion coefficient will generally decrease, which has the opposite effect.
  - The molecular weight.
- The position inside the rotor. The further to the outside of the rotor the sample is positioned, the faster the sample will equilibrate, since the centrifugal force is larger at the outside.
- The column length. The longer the column (the more volume is loaded), the longer it will take to equilibrate.
- Finally the rotor speed. The larger the rotor speed, the faster the sample will approach equilibrium.
- It also matters if the system is at chemical equilibrium, and if that equilibrium can be established rapidly enough (on the time order of the experiment). If a monomer-dimer equilibrium experiment takes a long time to equilibrate (i.e., the equilibrium kinetics are not diffusion controlled), then it may take substantially longer than predicted by this program. THis program will only predict the experiment of a diffusion controlled kinetics environment.

57.1 How to use this program:

- When designing an equilibrium experiment, you should plan to acquire equilibria at multiple speeds (4-5 speeds, ranging from **sigma=1** to **sigma=5**). Using the finite element simulator programmed in this module, you can predict the time it takes to reach equilibrium for all practical purposes, i.e., the change over time is less than a certain tolerance value. You can then pre-program the speeds into the XLA and are not required to check for equilibrium every time before you change to the next speed.
- Note: Sigma is a measure of the curvature of the exponential. If it is too small, the curvature is shallow and there is not enough information. If it is too steep, most of the concentration points are going to be near zero, drowned out by bad s/n ratios. The happy medium is between 1.
- When simulating these times, you should simulate a molecule that is sedimenting slightly slower than the actual molecule, perhaps by simulating a system with a larger frictional ratio or larger axial ratio, and perhaps a non-globular model rather than a spherical model. While the equilibrium times may be longer than needed, at least they won't be too short. In this module you can change the parameters that influence the time it takes to reach equilibrium.

57.2 Est. Equilibrium Time Functions:

Set / Change / Review Model	Use this pushbutton to bring up the Model Editor.
Radius Settings	Chose a pre-dfined top and bottom radius settings with the Channel Type radio buttons. Use the Custom button to manually define the radius settings.
Rotor speed Settings	Select the appropriate radio button to enter the speeds in terms of sigma (reduced molecular weight term: **(M * ω^2 * (1 - vbar * ρ))/(2 * R * T)** or by RPM. • Low Speed: Select the first speed to be equilibrated • High Speed:Select the last speed to be equilibrated • Speed Steps: The total number of speeds to be simulated, including the high and low speed. These will be linearly spaced between the high and low speed. • Current Speed List: All selected speeds will be listed here. This field cannot be edited directly.

Note: All speeds will be adjusted to the nearest 100 rpm, so they can be programmed in the XLA, which only allows 100 rpm steps.

Simulation Settings

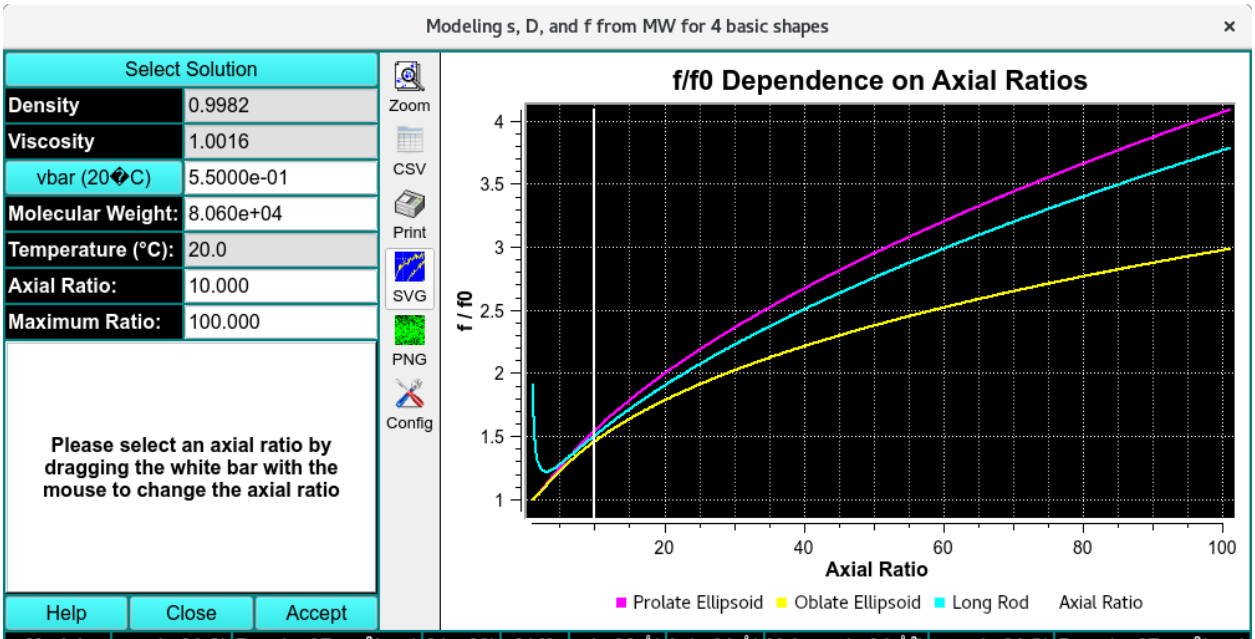
Tolerance:	This value is the tolerance setting that determines if a scan has reached equilibrium. Successive scans are subtracted from each other and the root mean square of the differences are calculated. If the difference is less or equal to this tolerance value, the program will consider the speed equilibrated and automatically proceed to the next speed. For a 3 mm column with 0.001 cm resolution the accuracy of the computer allows about $1.0x10^{-5}$ for the smallest practical tolerance (noise-free data).
Time Increment (min):	This is the time between successive scans. The larger the time difference, the coarser is the time determination, but the more reliable is the time estimation. A good default setting is 15 minutes, to be used with a 5 x 10^{-4} tolerance setting.
Estimate Times Pushbutton	This function runs the simulation and populates the graph and the text window. The text can be saved with the File dropdown list above the text window.

CHAPTER

FIFTYEIGHT

MODELING FOR SHAPES

In this dialog, you select an axial ratio to model component parameters for 4 basic shapes.



Model:	s_{app} (e-01 S)	D_{app} (e+07 cm²/sec)	f (e+08)	f / f0	a (e-02 Å)	b (e-01 Å)	Volume (e-04 Å³)	$s_{20,w}$ (e-01 S)	$D_{20,w}$ (e+07 cm²/sec)
Sphere:	1.2297	8.2456	4.9085	1.0000	0.2600	2.5999	7.3613	1.2297	8.2456
Prolate:	0.7970	5.3441	7.5735	1.5429	1.2068	1.2068	7.3613	0.7970	5.3441
Oblate:	0.8436	5.6569	7.1548	1.4576	0.5601	0.5601	7.3613	0.8436	5.6569
Long Rod:	0.8175	5.4819	7.3832	1.5042	1.0542	1.0542	7.3613	0.8175	5.4819

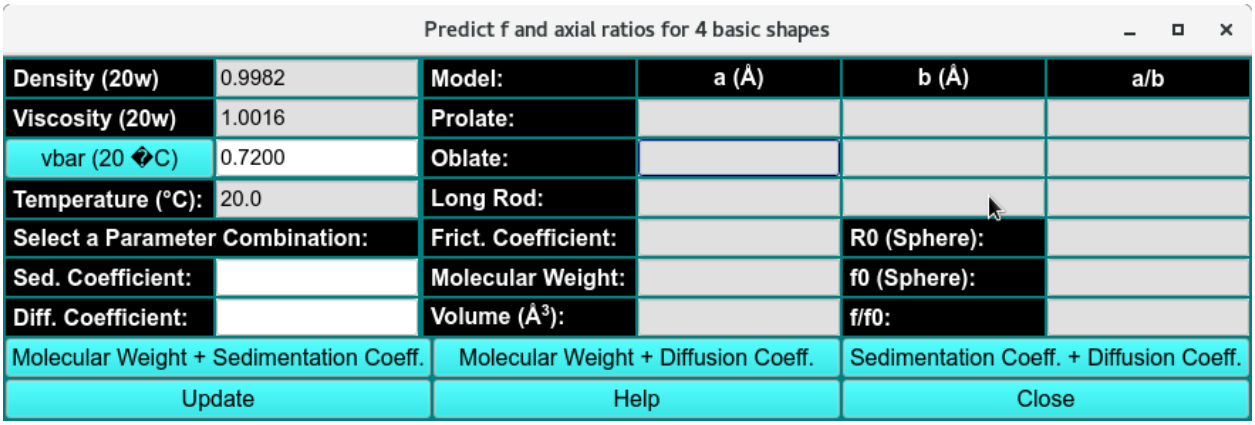
Analyte Shape Prediction

CHAPTER

FIFTYNINE

PREDICT HYDRODYNAMIC PARAMETERS FOR 4 BASIC SHAPES

In this dialog, you can enter a combination of 2 different hydrodynamic parameters and predict the remaining hydrodynamic parameters for 4 basic shapes.



Prediction of Analyte Hydrodynamic Parameters

CHAPTER

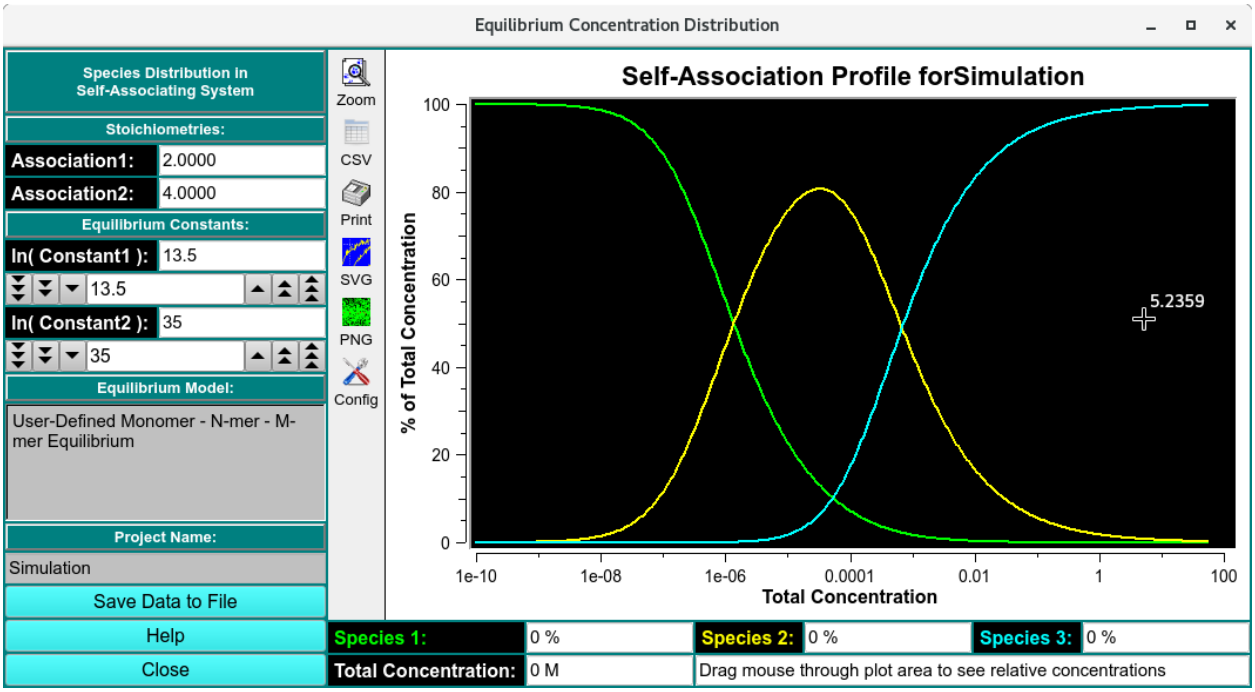
SIXTY

SELF-ASSOCIATION SIMULATOR

Index

- *Self-assoc. Simulator Functions:*

This module is used to simulate binding isotherms for reversibly self-associating systems with up to three species (monomer, N-mer, M-mer). The program is started from the Simulation menu (Self-Association Equilibrium):



Self-Association Simulation Profile

60.1 Self-assoc. Simulator Functions:

Association (1,2):	Enter an integral value to specify the stoichiometry of the association (e.g., 2, 3, 4…)
Equilibrium constants:	Enter the natural log of the association constants for each reaction.
Save Data to File:	Export the currently displayed association profile to a csv file.
Self-Association Profile for Simulation:	Binding isotherms for Monomer, N-mer and M-mer.
Species 1/2/3:	Percentage of species' concentration with respect to the total concentration, requires mouse click on the canvas to be activated.
Total Concentration:	Total concentration of the reacting molecule.
Help	Display this and other documentation.
Close	Close all windows and exit.

CHAPTER

SIXTYONE

TIME STATE VIEWER

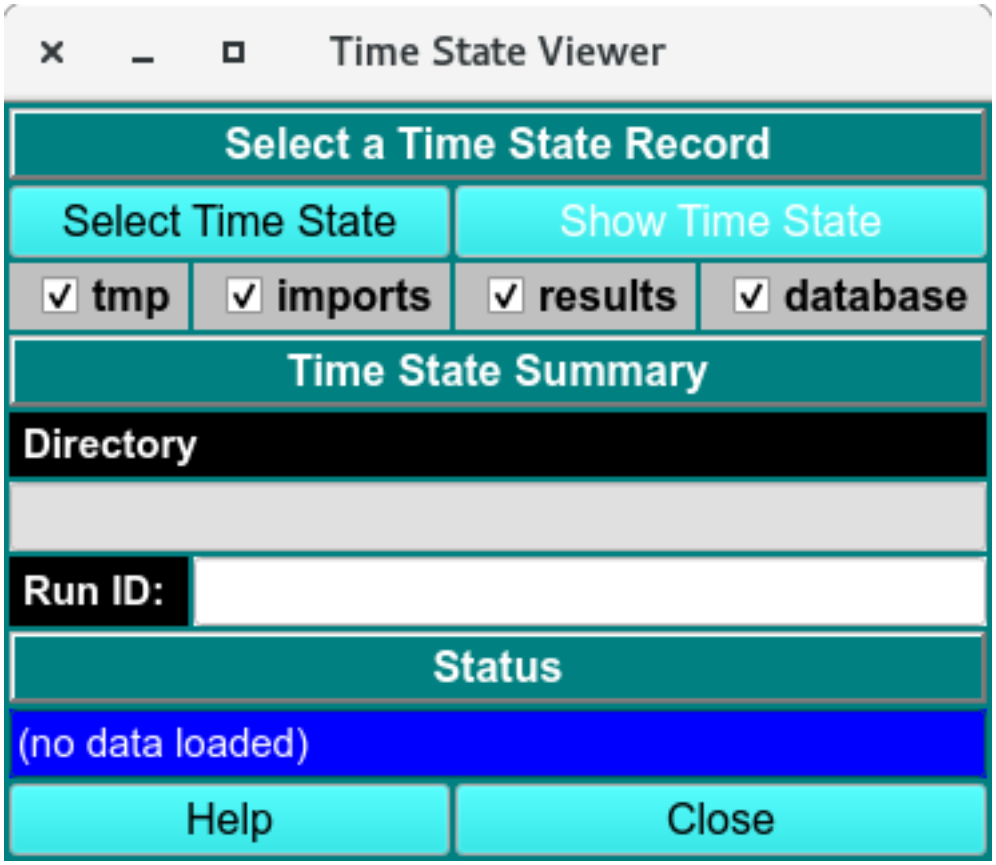
Index

- *Process*

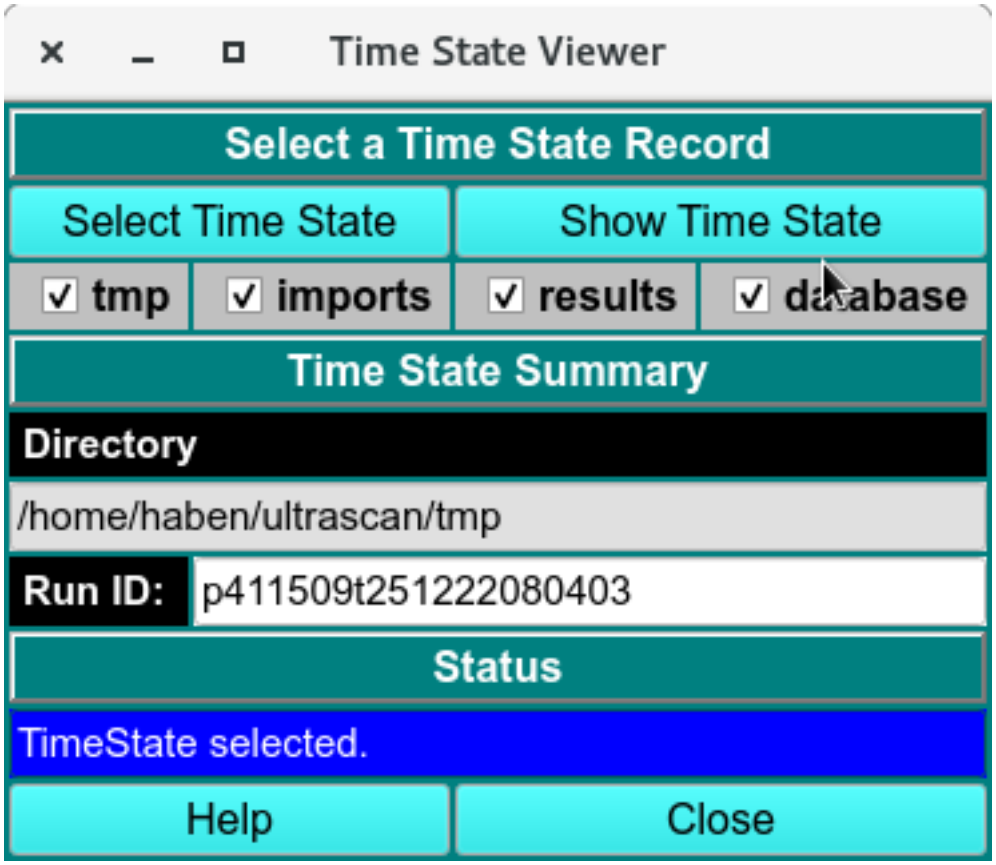
The Time State viewer module provides users with the metadata of only select dataset. Metadata include the RunID, filename, directory location and Instrument parameters (TimeInterval, Time, RawSpeed, SetSpeed, ω^2T, Temperature and step) across the experimental time/scan number.

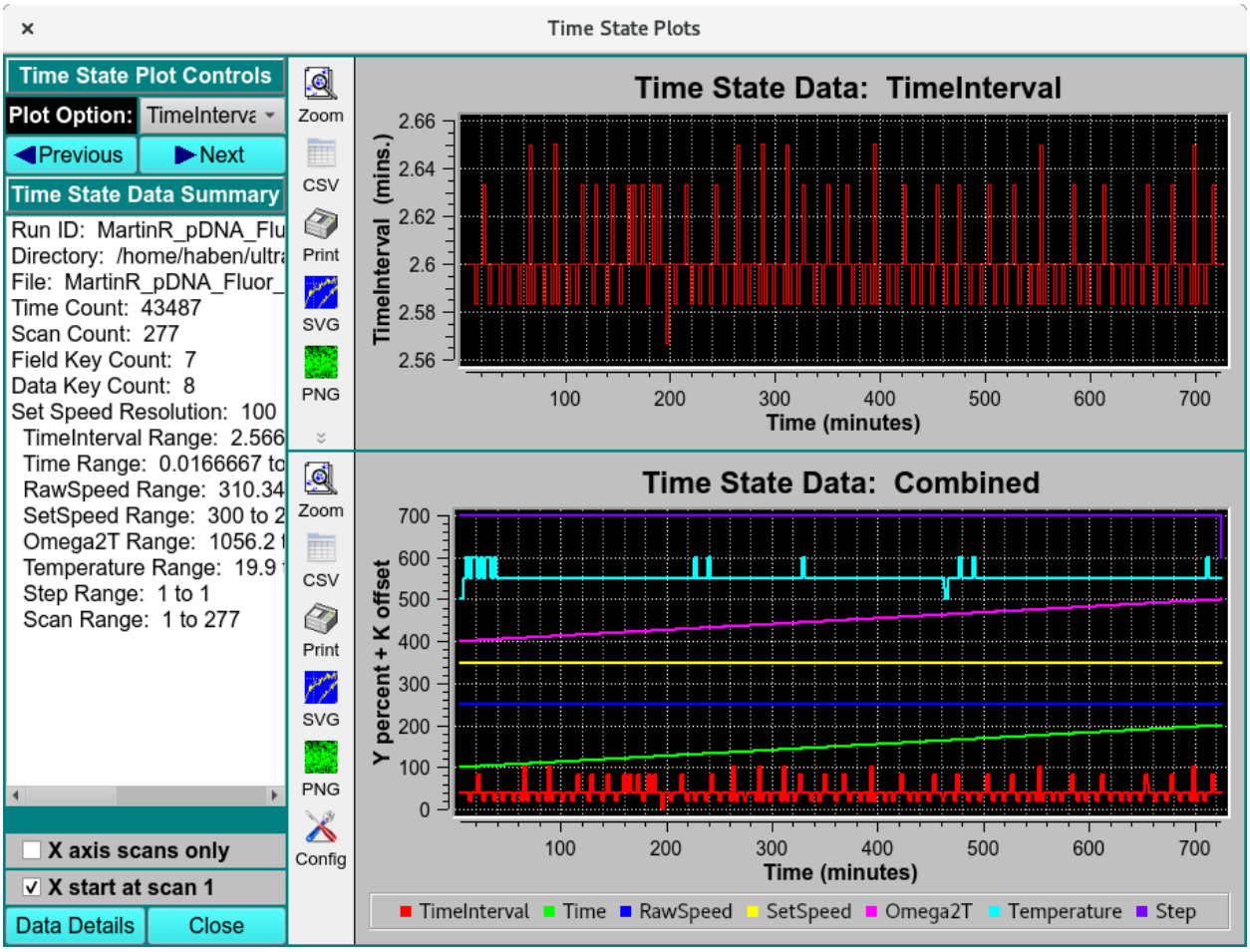
61.1 Process

Step 1. Click **Select Time State** to select a run.

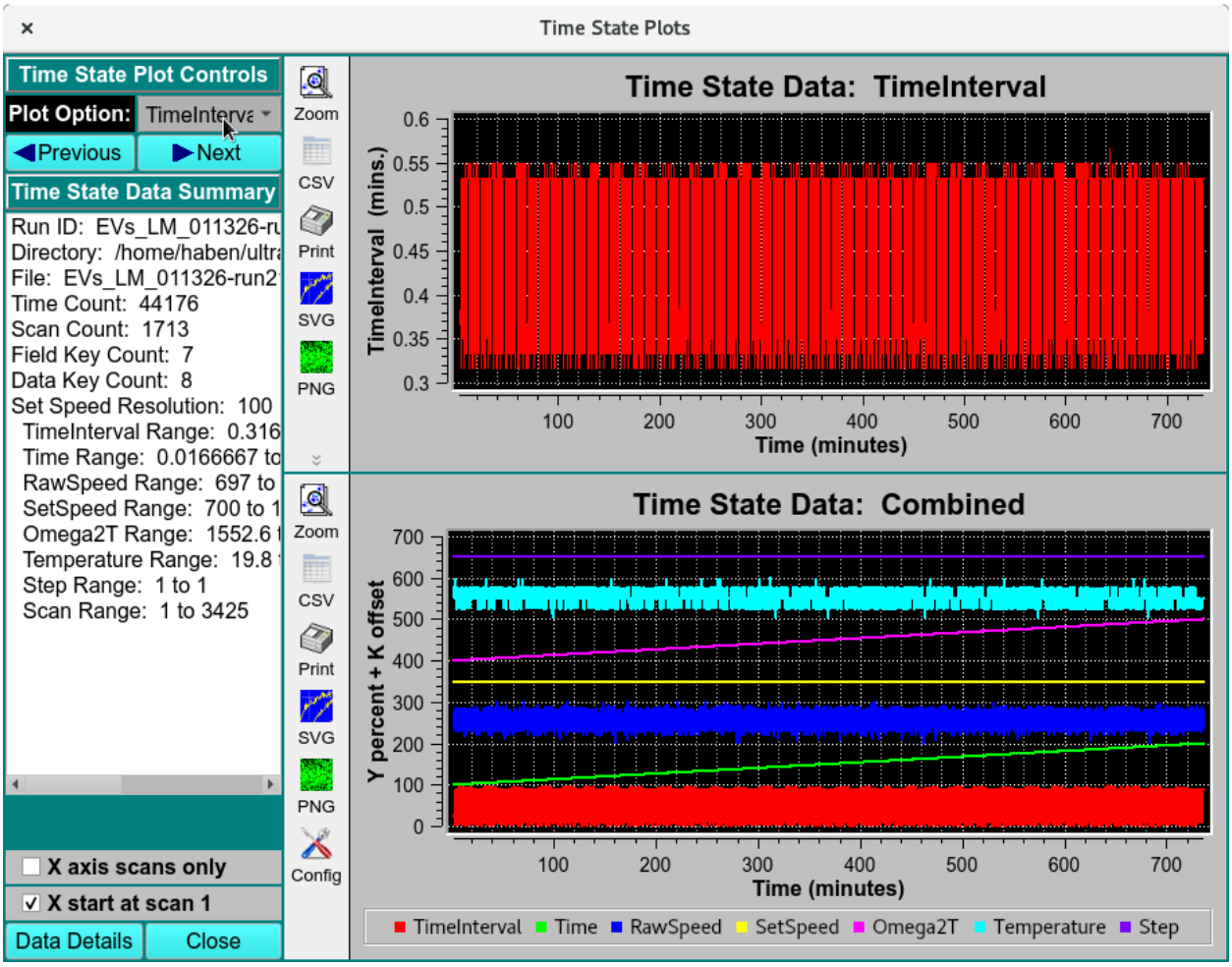


Step 2. **Show Time State** button is enabled and call the **Time State Plots** window.

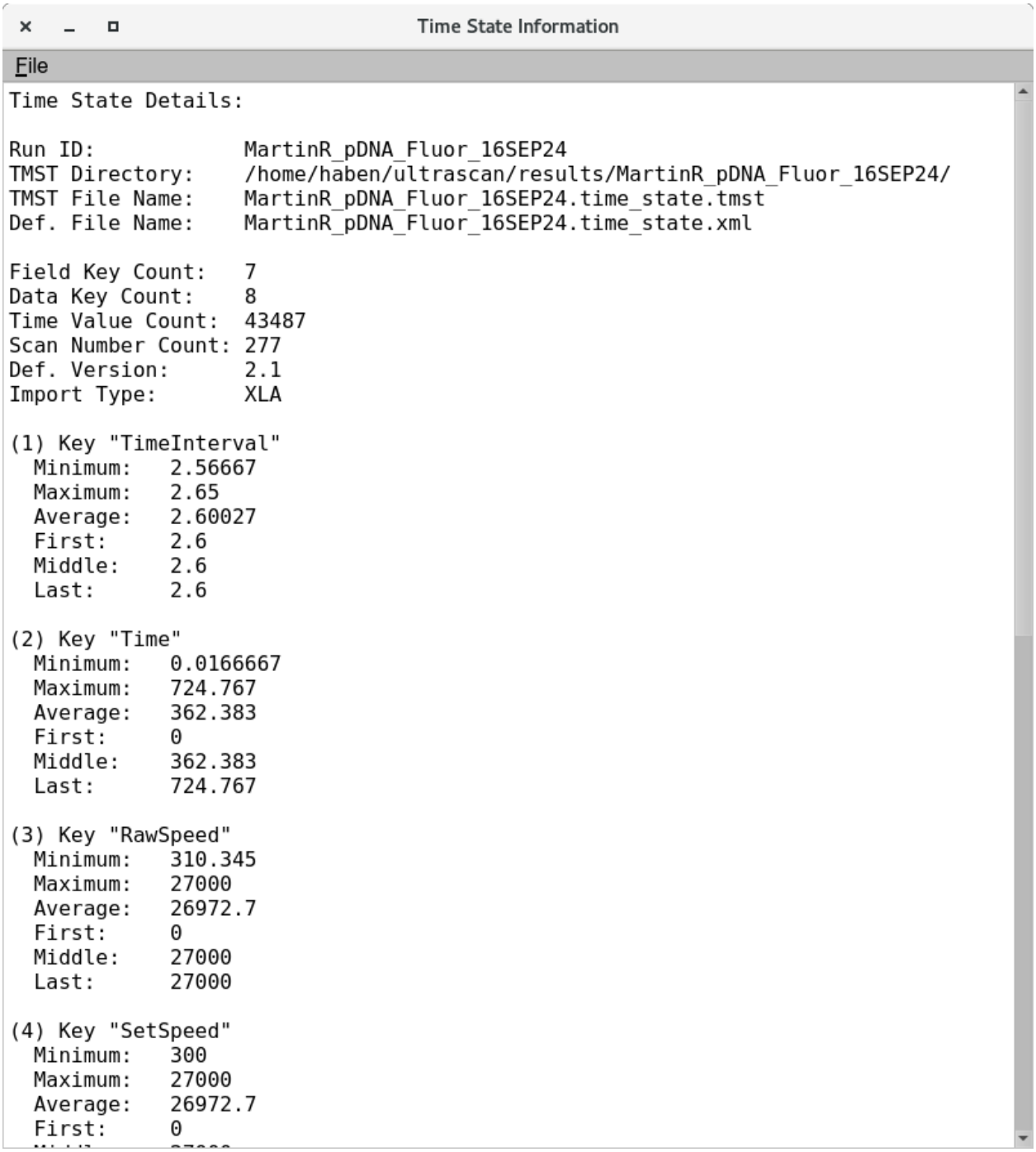

Time State Viewer
Select a Time State Record
Select Time State
Show Time State
tmp
imports
results
database
Time State Summary
Directory
/home/haben/ultrascan/tmp
Run ID:
p411509t251222080403
Status
TimeState selected.
Help
Close



Step 3. Flip through the pulldown Plot Options(TimeInterval, Time, RawSpeed, SetSpeed, ω^2T, Temperature and step) to plot in the individual time state data. The bottom **Time State data: Combined** show the combined curves of TimeInterval, Time, RawSpeed, SetSpeed, ω^2T, Temperature and step across the total scan number.



Step 4. Click **Data Details** to load the **Time State Details:** of the run.



fe-data-loader>`========== Finite Element Model Viewer Flow ==========

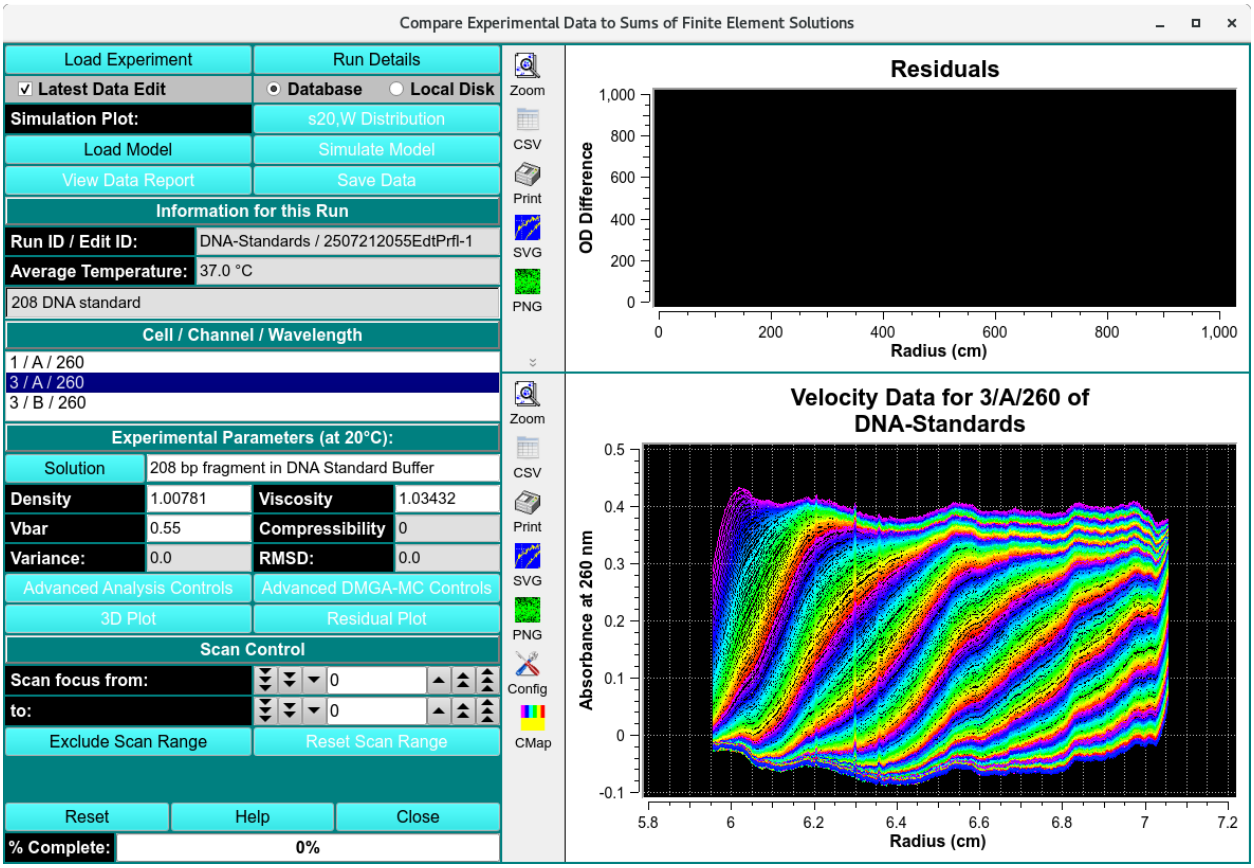
This module is used to display, and export FE solutions fitted to velocity experiments by various methods.

>fe-data-loader>`========== Finite Element Model Viewer Module ==========

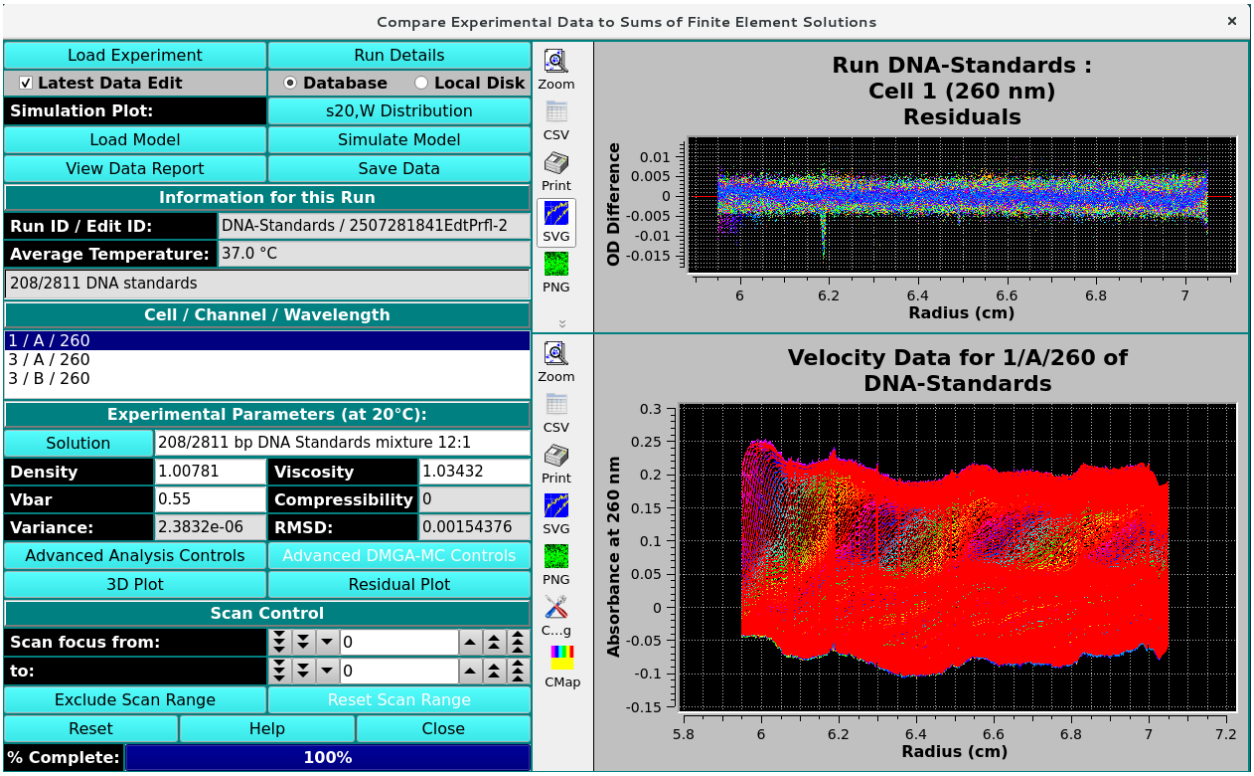
Index

- *FE Viewer Functions*
- *Noise Vector Selection*

This module is used to display, and export FE solutions fitted to velocity experiments by various methods. The program can import models from *finite element nonlinear fitting sessions*, from *2-dimensional spectrum analysis* fits, from *genetic algorithm* fits, and from *C(s)* fits. The program will then compare the fitted model to the experimental data and display residual plots, sedimentation and diffusion coefficient distributions, as well as molecular weight distributions. The model can be displayed as a 3-dimensional plot showing the partial concentrations mapped onto a 2-dimensional grid of any two of the following parameters: s, D, f, f/f_0, and MW. Residual plots including deconvoluted time- and radially-invariant noise plots can also be displayed. The results can be saved for inclusion into a velocity result report.



Compare Experiment to Finite Element Solution



Simulated Finite Element Solution

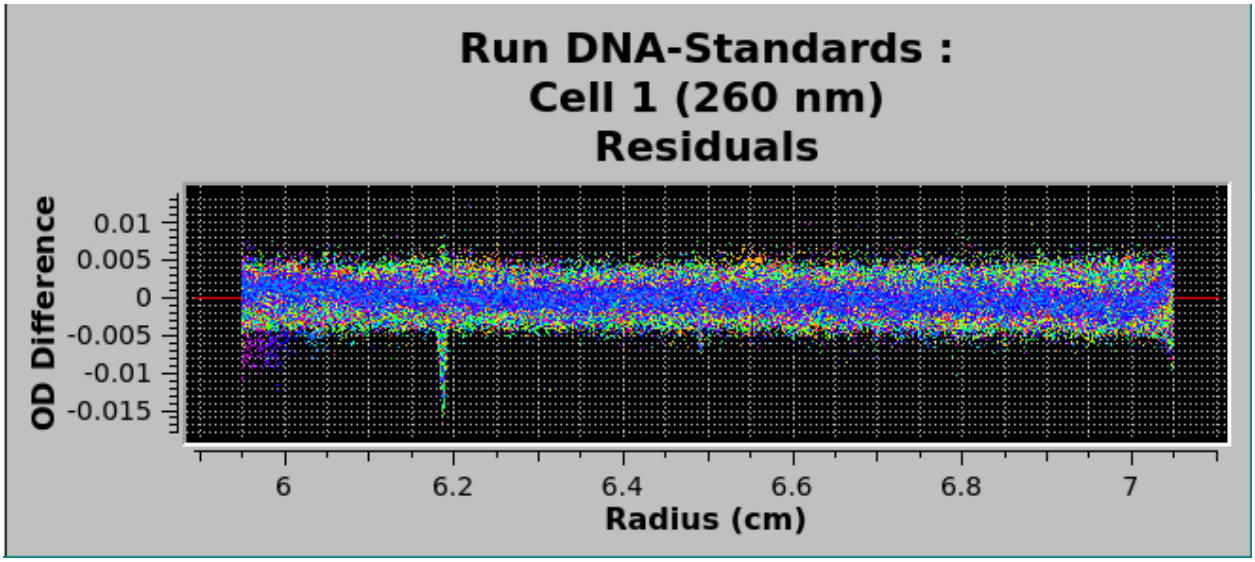
CHAPTER

SIXTYTWO

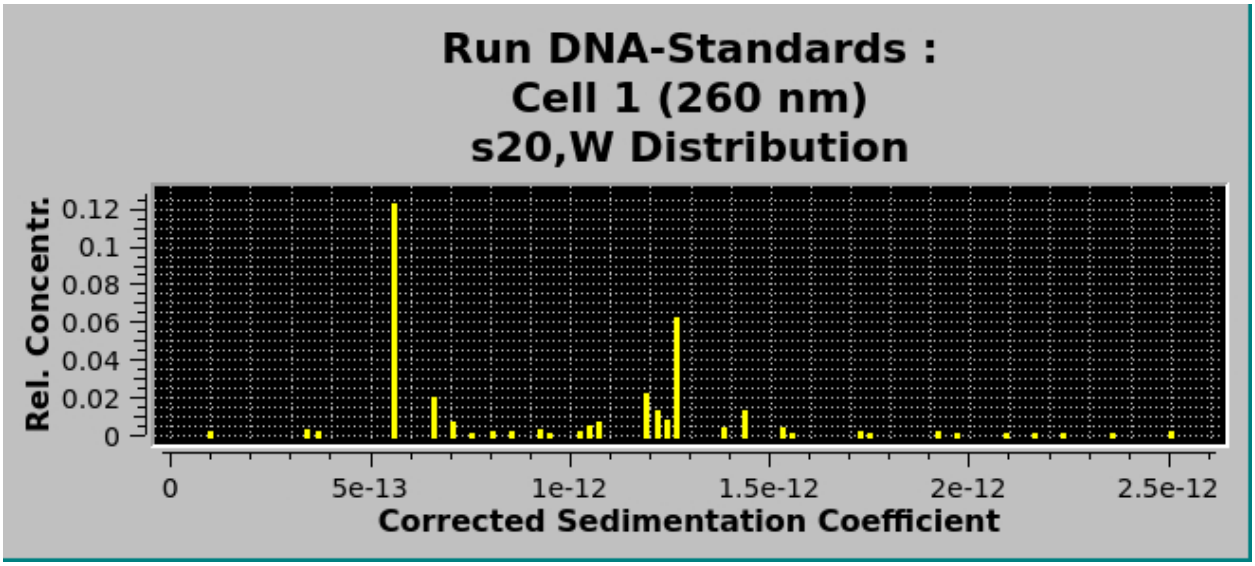
FE VIEWER FUNCTIONS

Load Experiment	Click here and, in the resulting Load Data Dialog, select an edited data set to load.
Run Details	Pop up a dialog showing Run details.
Latest Data Edit	Uncheck to allow choosing an edit other than the latest one for the raw experimental data.
Database	Select to specify data input from the database.
Local Disk	Select to specify data input from local disk.
Load Model	Load model data, choosing a model in a Model Loader dialog. If appropriate noise exists, you may also be presented with a *Noise Loader dialog* to select time-invariant and/or radially-invariant noise vector(s) to load.
Simulate Model	Run a simulation and display results. For details on the results of clicking this button, see Simulate Model Results.
View Data Report	Generate a report file and view it in a dialog.
Save Data	Generate reports and save to file.

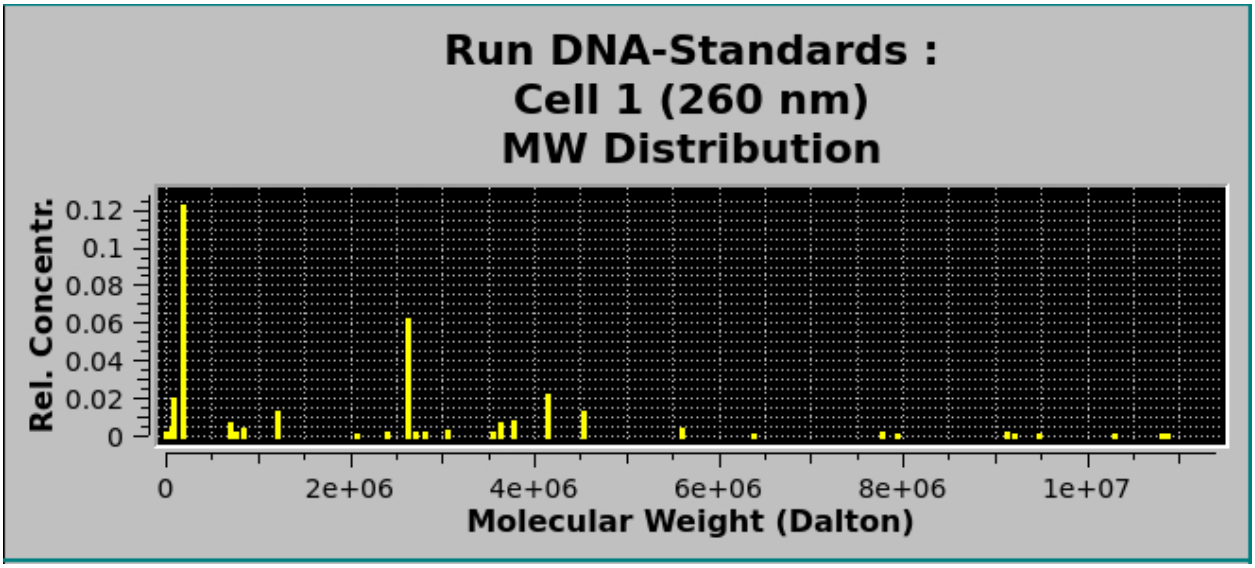
Simulation Plot: Bar plot of the selected distributions. The choices are as follows:



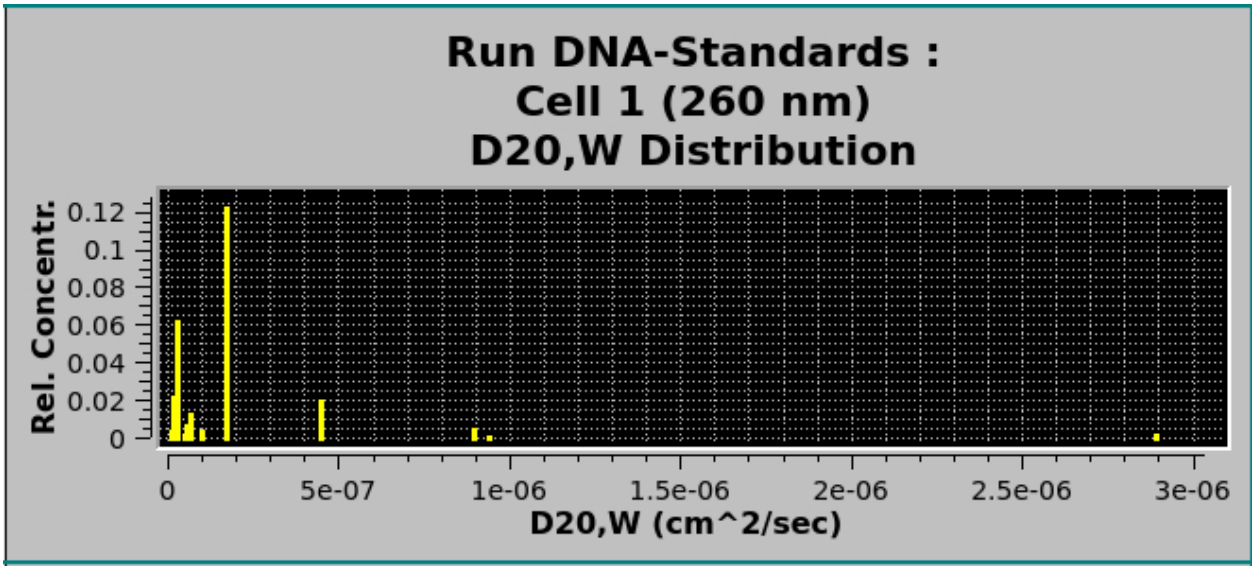
Residuals



$s_{20,w}$ distribution



MW distribution



$D_{20,w}$ distribution

Information for this Run:

Run ID / Edit ID:	The Run identifier string and the Edit identifier (generally a time string) are displayed for loaded edit.
Average Temperature:	The average temperature over all the scans of the loaded data.
(experiment description)	A text string is displayed giving a fairly detailed description of the experiment.

Cell / Channel / Wavelength: One or more rows of data edit Triples. If more than one, click on a row to select it as the data of interest.

Experimental Parameters (at 20°C):

Solution	Click this button to open a Solution Management that allows changes to buffer and analyte characteristics of the data set.
Density	Shows the density value for the loaded experiment. Click the Solution button to open a dialog in which density and other values may be changed.
Viscosity	Shows the viscosity value for the loaded experiment. Click the Solution button to open a dialog in which viscosity and other values may be changed.
Vbar	Shows the vbar value for the loaded experiment. Click the Solution button to open a dialog in which vbar and other values may be changed.
Compressibility	Compressibility value for the loaded experiment. The value shown here is set in Solution/Buffer dialogs, but may also be modified in this text box.
Variance:	Variance value (square of RMSD) for residuals.
RMSD:	Root-Mean-Square-of-Differences for residuals.
Advanced Analysis Controls	Click to reveal an Advanced Analysis Controls, with options to set parameters related to Reversible Associations (RA) models or to show details on model components.
3D Plot	Show a dialog of plot controls for a 3-dimensional model plot. See 3-Dimensional Plot Controls Details.
Residual Plot	Show a plot dialog with a more advanced set of data/residuals/noise plots. See Simulate Model Results.

Window Controls

Reset	Indicate that window is reset and the plots are initiated.
Help	Display this detailed van Holde-Weischet Analysis help.
Close	Close all windows and exit.

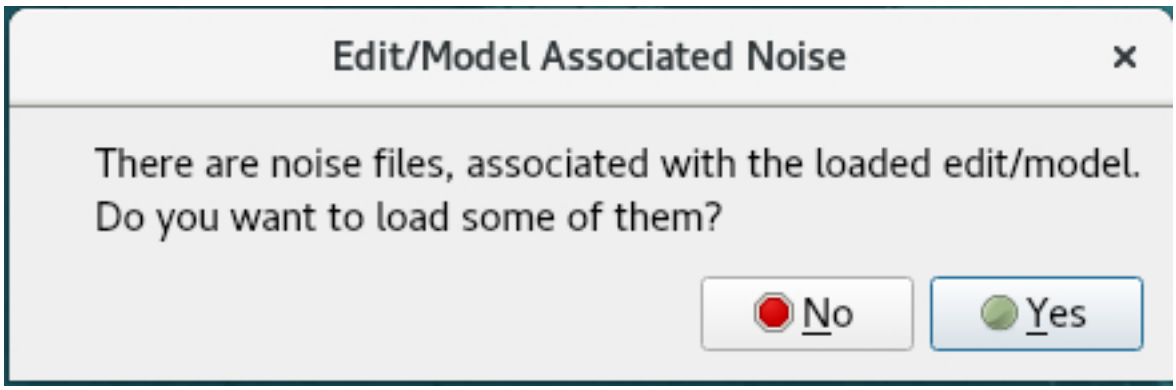
The main **Compare Experiment to Finite Element Solution** window may need additional controls when the model loaded includes Reversible Associations (RA). These controls can be revealed using the Advanced Analysis Controls button to invoke a separate dialog.

CHAPTER

SIXTYTHREE

NOISE VECTOR SELECTION

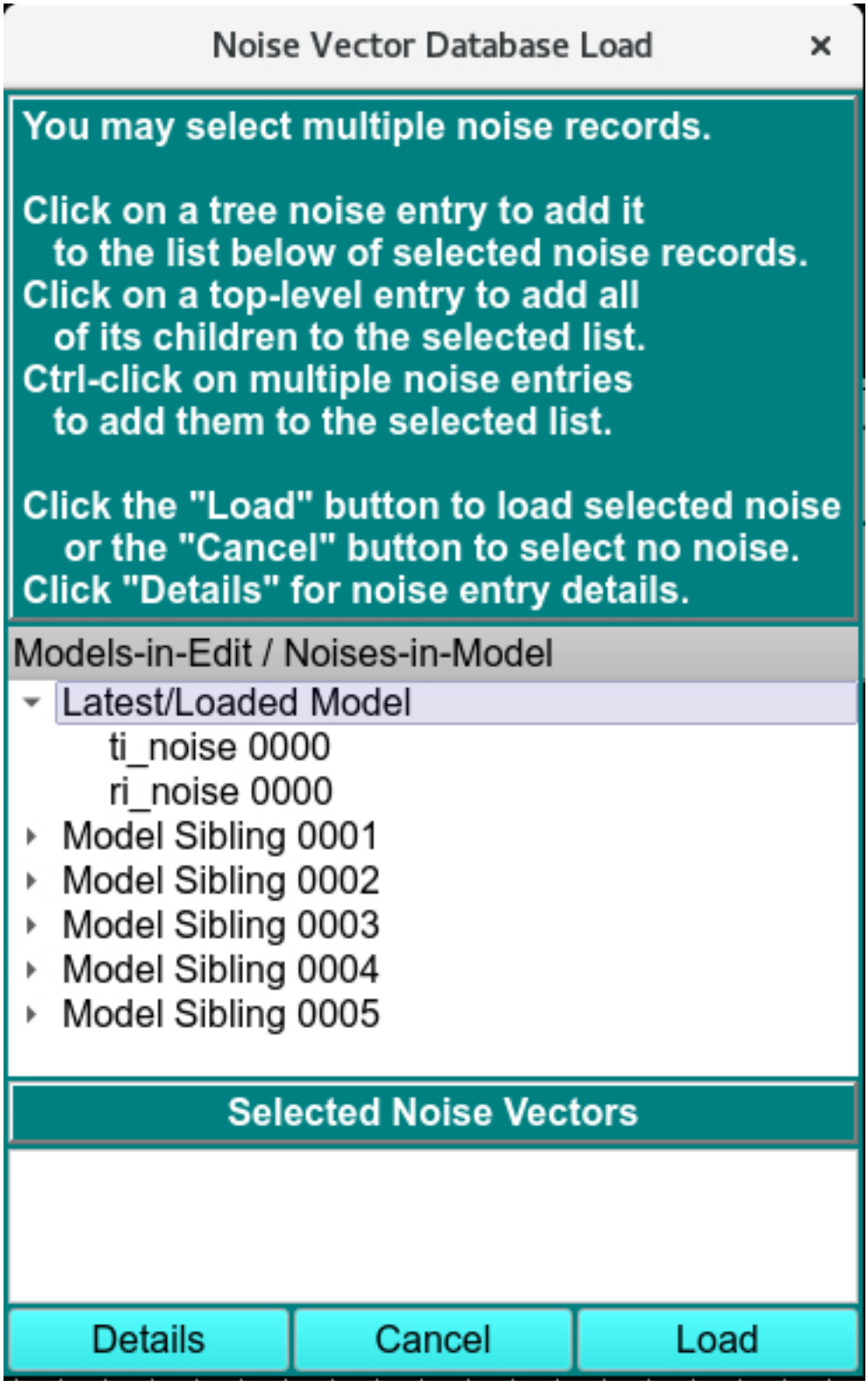
When the **Noise Dialog Level** in the Configuration and Preferences setting is enabled (1), the user can select the noise profiles to simulate the data with.



Noise Vector prompt

No - no noise will be selected, model will be simulated without noise correction,

Yes - User will be prompted to select the noise vector.



Noise Vector Database Load

Note: The model siblings are ordered with the most recently fitted noise first and the earlier noise profiles last.

Select the appropriate noise for the model and click **Load**.

triple========== Finite Element Model Simulation ==========

Index

- *Simulation Product Output Summary*
- *Data/Residuals Viewer*
  - *Data/Residuals Viewer Functions*
  - *Noise Data Analysis*
- *Residual Bit Map*
- *Report: FE Match Model Simulation*
  - *Distribution Information Units*

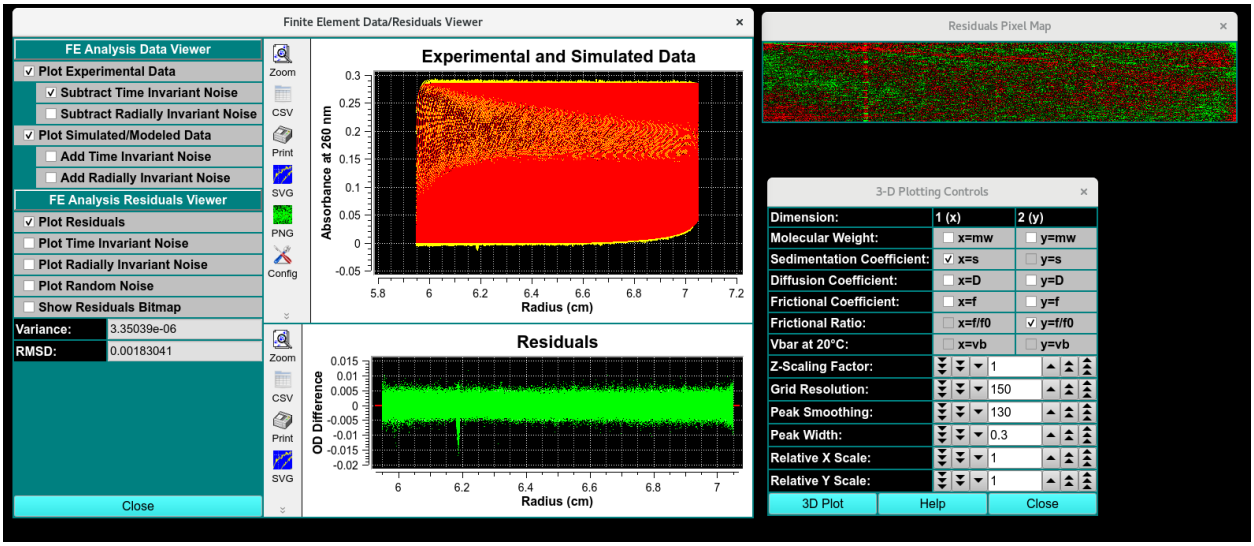
CHAPTER

SIXTYFOUR

SIMULATION PRODUCT OUTPUT SUMMARY

When the **Simulate Model** button is clicked in the **Finite Element Model Viewer** window, a simulation is performed using the Edit data, a Loaded Model, and simulation parameters primarily created using the edit data set. A simulation data set is created that has the same ranges as the edit set, but with readings values that are calculated and compared to the actual experimental data. The comparison spawns a number of new dialogs and options that allow the user to evaluate the quality of the model.

- **Finite Element Data/Residuals Viewer** - Visual comparison of simulated and experimental data. Graphical display of time and radially invariant noises subtracted.
- **Residuals Pixel Map** - Bit map of every scan as a pixel line and the positive or negative difference of the model to the data are highlighted as red (negative) and green(positive) pixels. an Excellent residual map will have a random distribution of red and green pixels.
- :doc:`3-Dimensional Plot Controls <../3d_plot> - 3D plots of the simulated data.
- **Data report File** - Generate a Simulated model Report file and view it in a dialog.



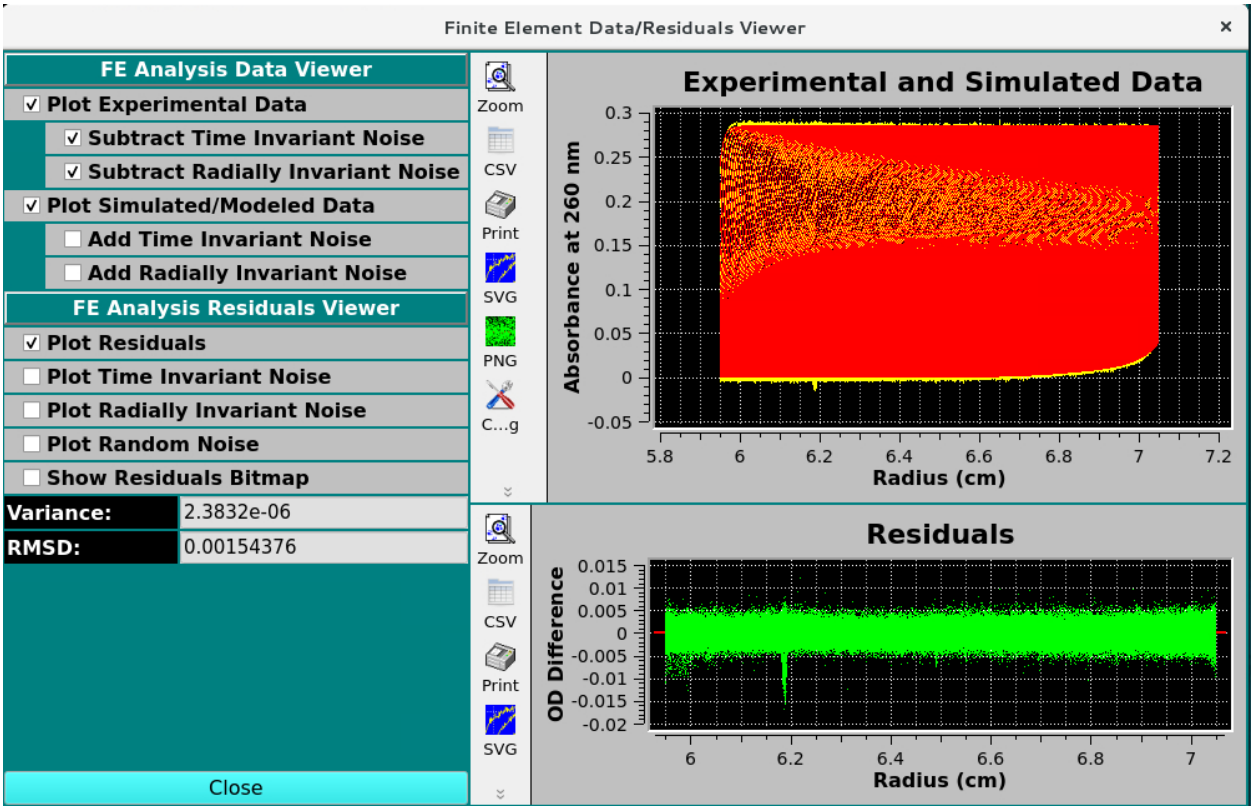
Simulation Results

CHAPTER

SIXTYFIVE

DATA/RESIDUALS VIEWER

The simulation creates a data set with the same ranges as the edit experimental data set. The actual values for scan readings vectors are synthetically produced, as illustrated by the plot below.



Simulated Viewer Result

65.1 Data/Residuals Viewer Functions

FE Analysis Data Viewer

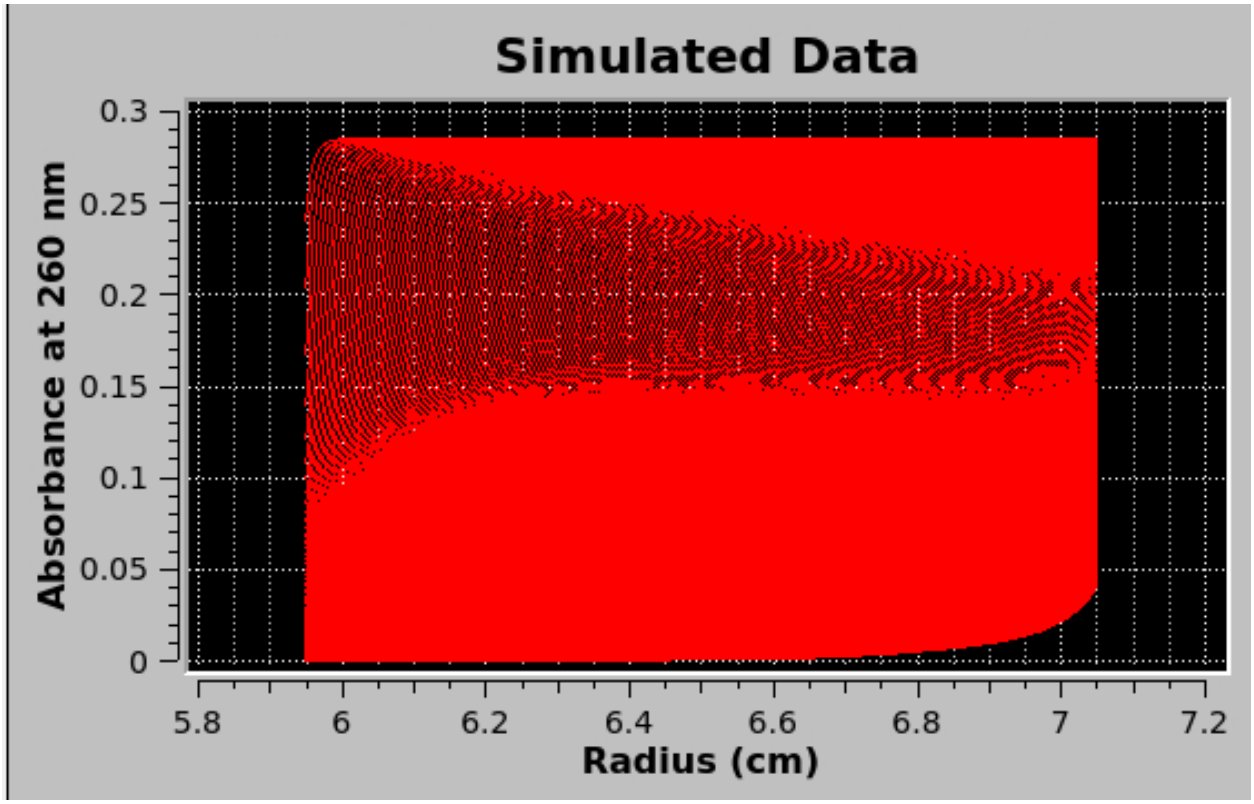
Plot Experimental Data:	View the experimental data in the Experimental and Simulated Data Plot.
Subtract Time Invariant Noise:	Add Time Invariant Noise to the Experimental data in the Experimental and Simulated Data Plot.
Subtract Radially Invariant Noise:	Add Radially Invariant Noise to the Experimental data in the Experimental and Simulated Data Plot.
Plot Simulated/Modeled Data:	View the simulated data in the Experimental and Simulated Data Plot.
Add Time Invariant Noise:	Add Time Invariant Noise to the Simulated data in the Experimental and Simulated Data Plot.
Add Radially Invariant Noise:	Add Radially Invariant Noise to the Simulated data in the Experimental and Simulated Data Plot.

FE Analysis Residual Viewer

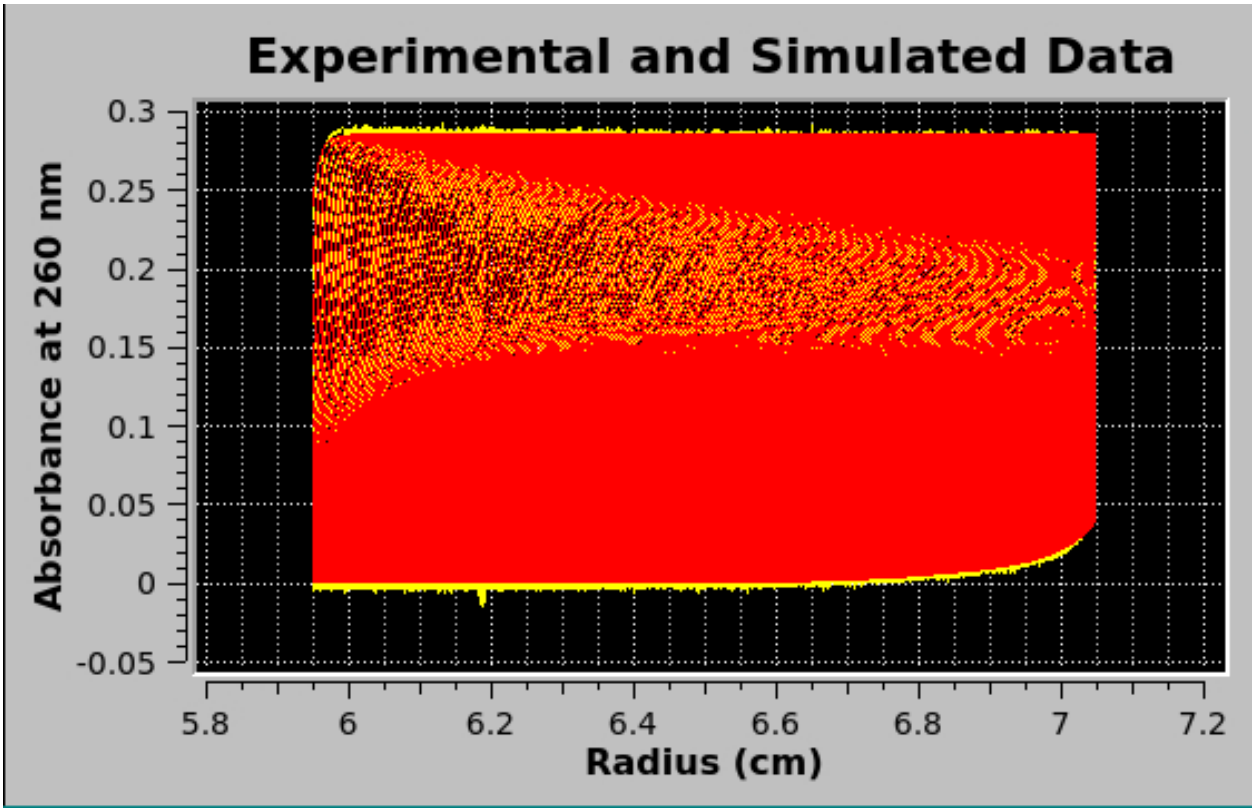
Plot Residuals	click to show the random residuals in the Residuals Plot
Plot Time Invariant Noise	click to show the time invariant noise in the Residuals Plot
Plot Radially Invariant Noise	click to show the Radially invariant noise in the Residuals Plot
Plot Random Noise	click to show the random residuals in the Residuals Plot
Show Residuals Bit map	Show a plot dialog of pixelated bit map.
Variance:	Variance value (square of RMSD) for residuals.
RMSD:	Root-Mean-Square-of-Differences for residuals.

65.2 Noise Data Analysis

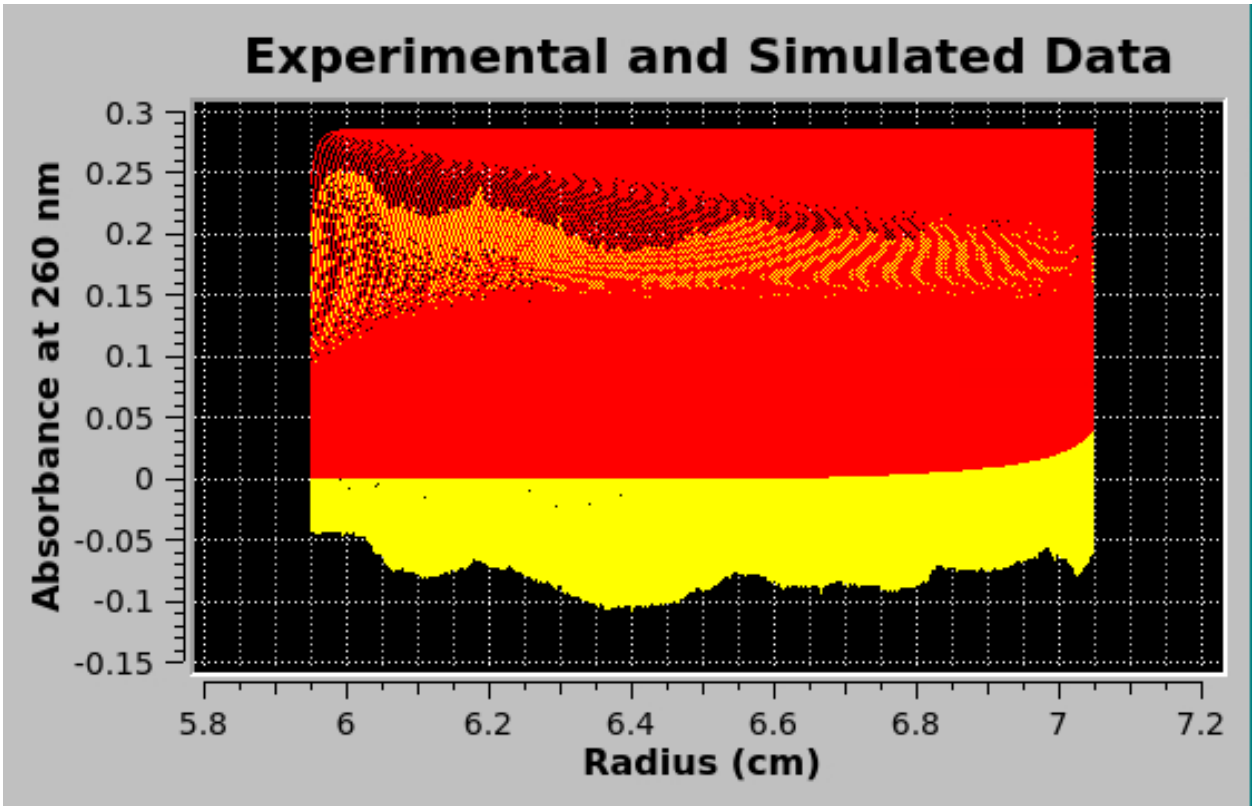
The simulation creates a data set with the same ranges as the edit experimental data set. The actual values for scan readings vectors are synthetically produced, as illustrated by the plot below.	Upon completion of simulation computations the plot of experimental data (in yellow and cyan) is overlaid in the main window lower plot with the simulation data (in red), as shown in the image below.



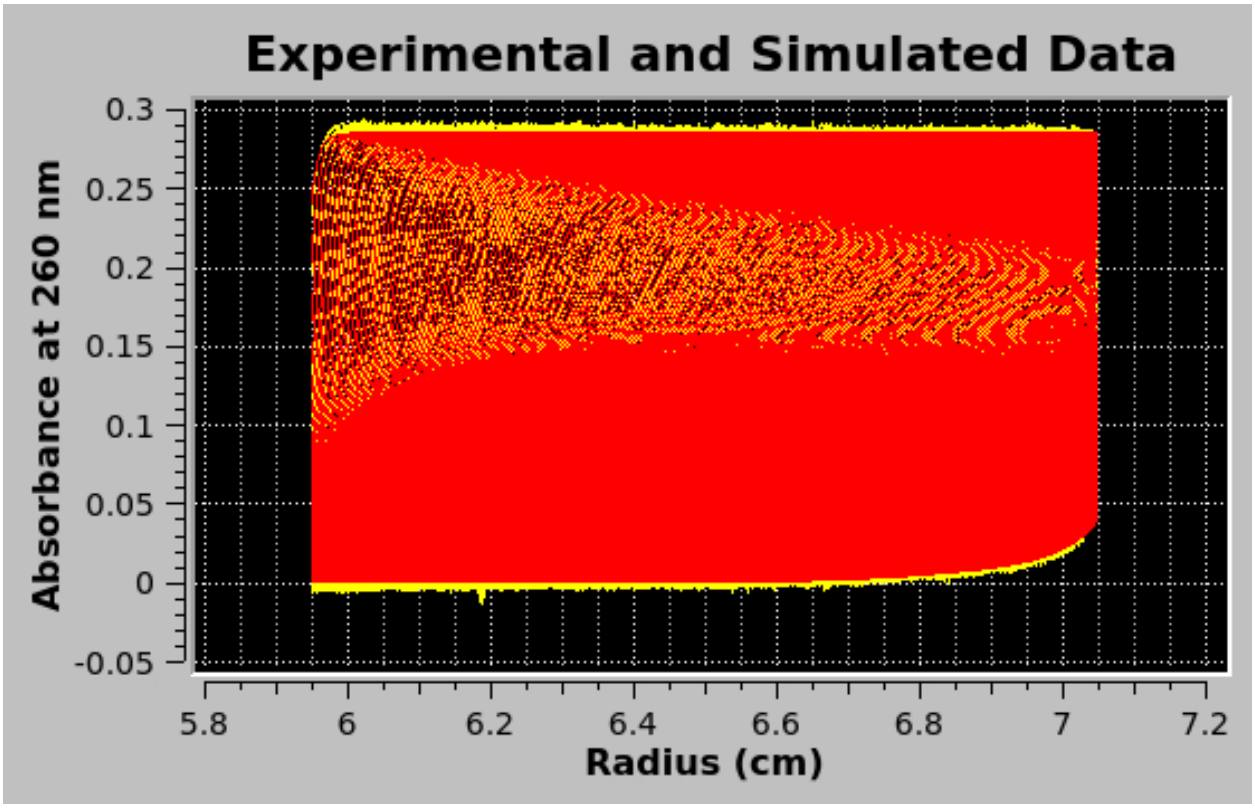
Simulation Model



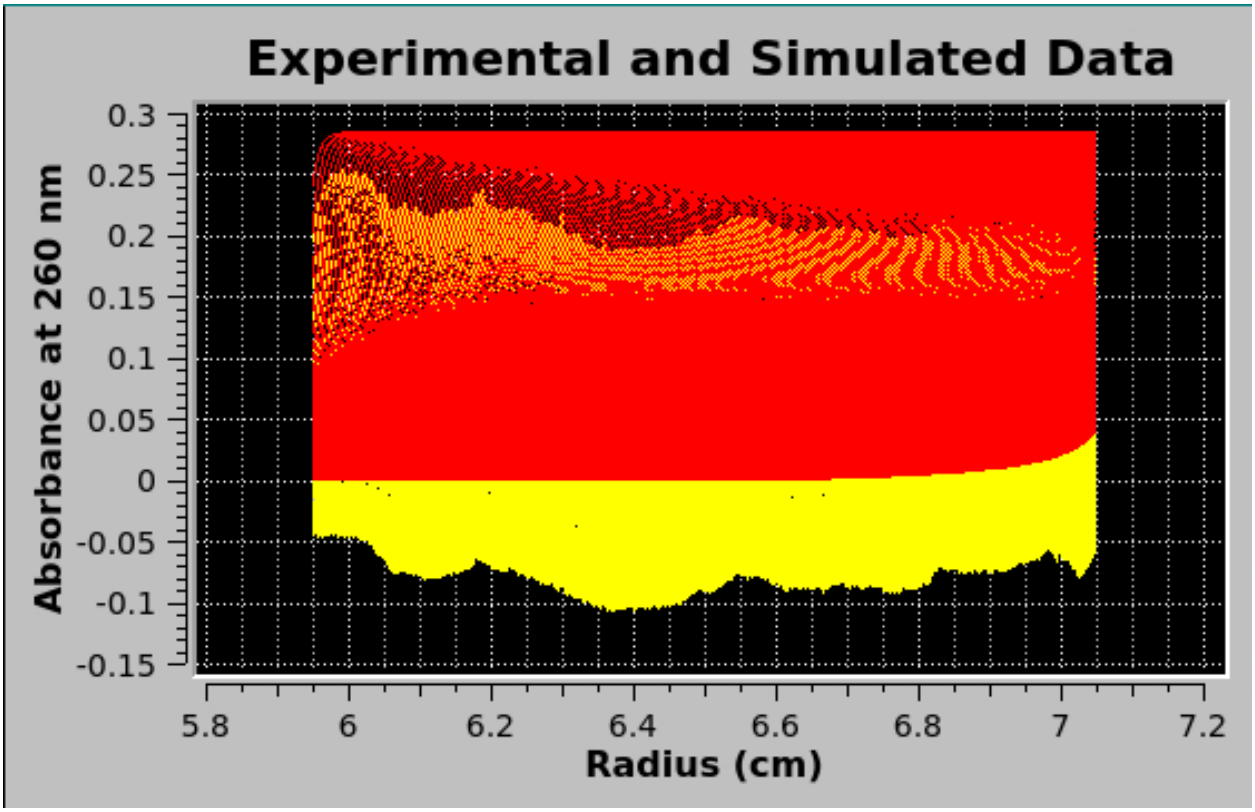
Experimental Data and Simulation Overlay



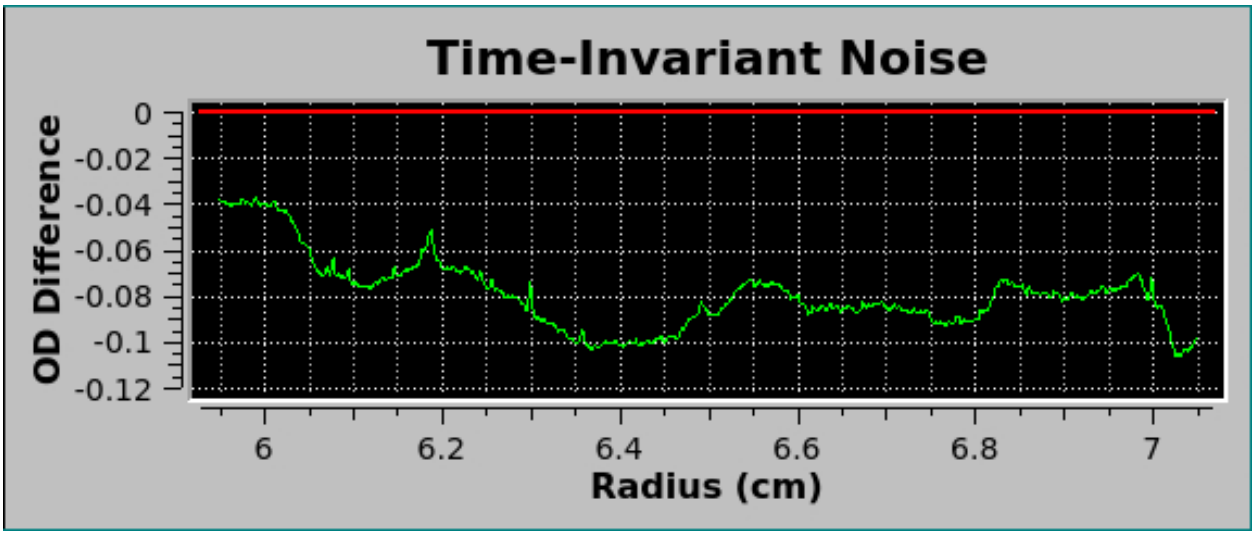
Add time-invariant noise to data



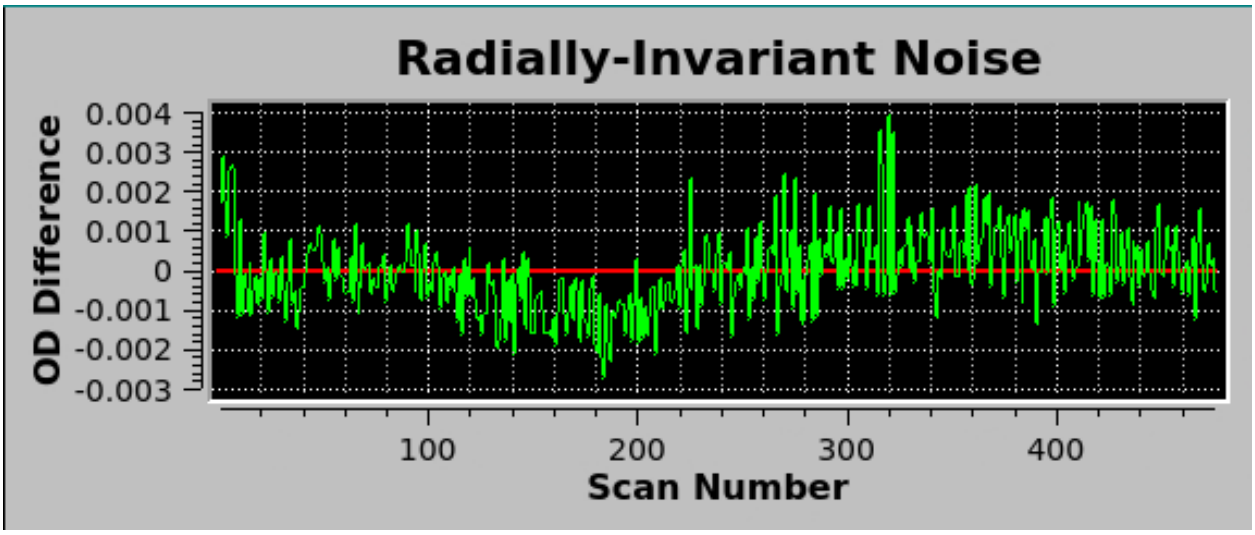
Add radially-invariant noise to data



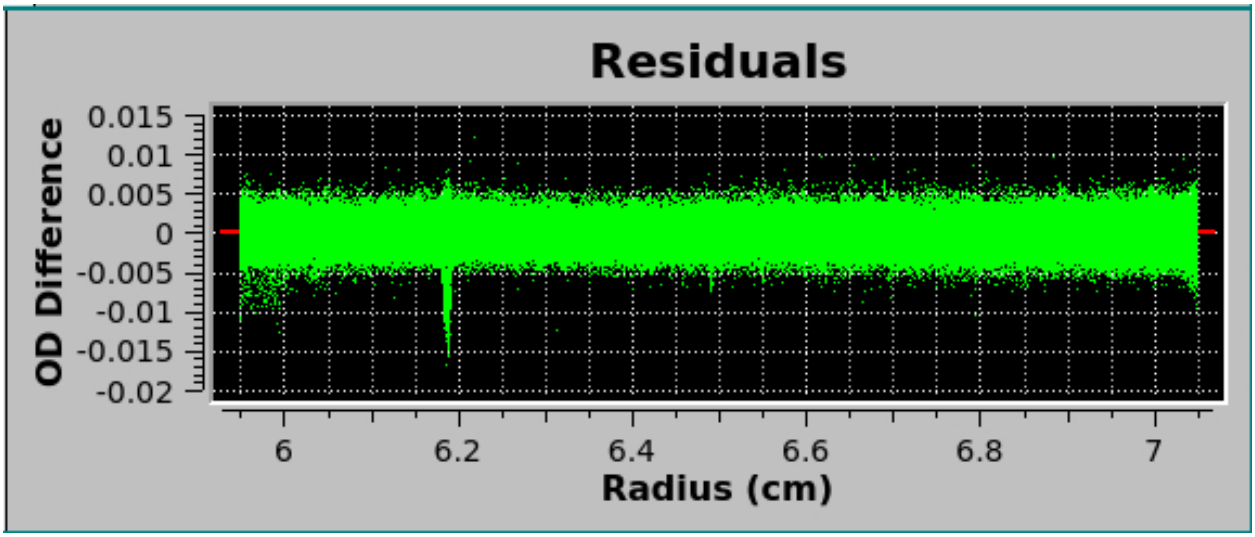
Add time-invariant noise and radially-invariant to data



Time-invariant noise plot



Radially-invariant noise



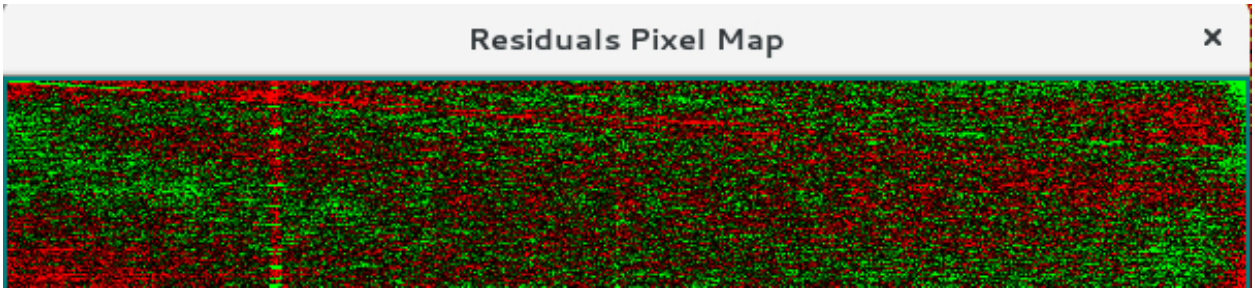
Random noise

CHAPTER

SIXTYSIX

RESIDUAL BIT MAP

Experimental-Simulation residuals are plotted in another way in a bit map. This small window represents each residual #Scans x #Readings point as a color: green where simulation is greater than experimental; red where experimental is greater. A random distribution of colors throughout the bit map is indication of a good model fit.



Pixel Bit Map

CHAPTER

SIXTYSEVEN

REPORT: FE MATCH MODEL SIMULATION

The **Save Data** button produces a set of report files. One of these is displayable via the **View Report** button, which produces a dialog that shows the contents of a report. A dialog sample follows.

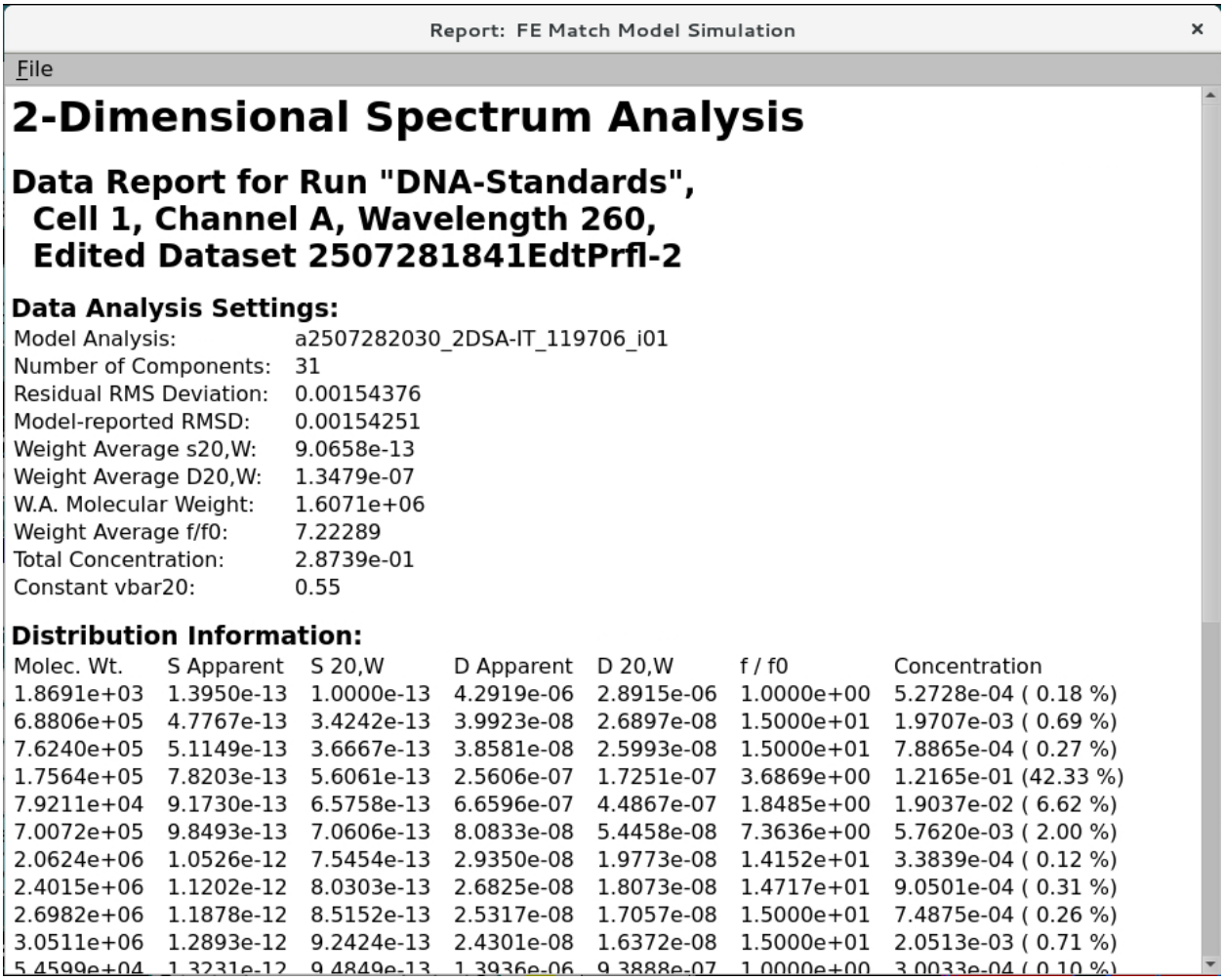


Table 1: **Detailed 2DSA Report**

Heading 1:	Analysis type
Heading 2:	This heading indicates the "dataset named in the Import module", the cell number, channel, and wavelength triple, the edit profile processed in the Edit data module,
Model Analysis:	The name of the model is determined by the analysis number_the type of analysis completed_the database request number_number of models in the analysis cohort
Number of components:	Number of components with unique s, and D values.
Residual RMS Deviation:	The RMSD determined by re-simulating the model in the finite element viewer module.
Model-reported RMSD:	The RMSD of the simulated model determined by message passing interface (MPI) or the 2DSA GUI.
Weight Averaged s20,W, D20,W, Molecular weight, and f/f0:	Describes the weight averaged (using % of the total concentration) to calculate the s20,W, D20,W, Molecular weight, and f/f0 coefficients of each component at 20ºC and in water.
Total concentration:	The baseline and noise subtracted optical density at the measured wavelength of the absorbing components as a function of radius.
Constant vbar20:	The analyte vbar at 20ºC.

Note: Difference between the Residual RMSD and model-reported RMSD should be within 3 significant figures. Greater then 3 significant figures indicates an error in analysis.

67.1 Distribution Information Units

- **Molec. Wt.:** Dalton
- **S Apparent:** seconds
- **S 20,W:** Seconds
- **D Apparent:** cm^2/s
- **D 20,W:** cm^2/s
- **f/f0:** unitless
- **Concentration:** optical density (OD) at measured wavelength, (% contribution)

CHAPTER

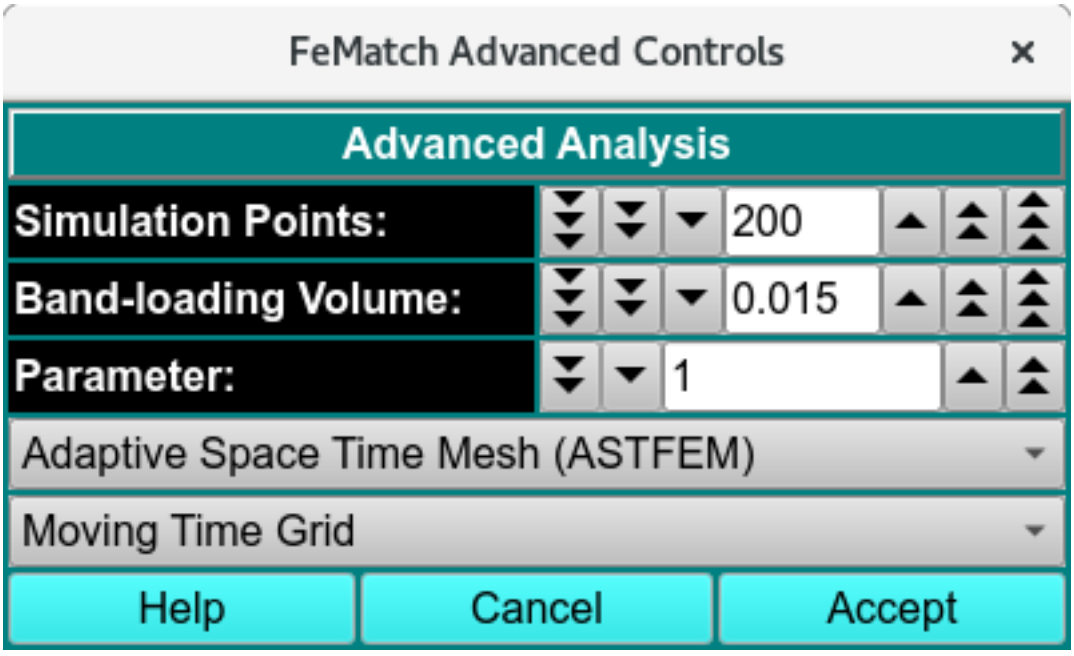
SIXTYEIGHT

ADVANCED ANALYSIS DIALOG

Index

- *Advanced Control Functions*

This dialog may be invoked from US_FeMatch in order to set advanced control parameters.



FEMatch Advanced Controls Dialog

68.1 Advanced Control Functions

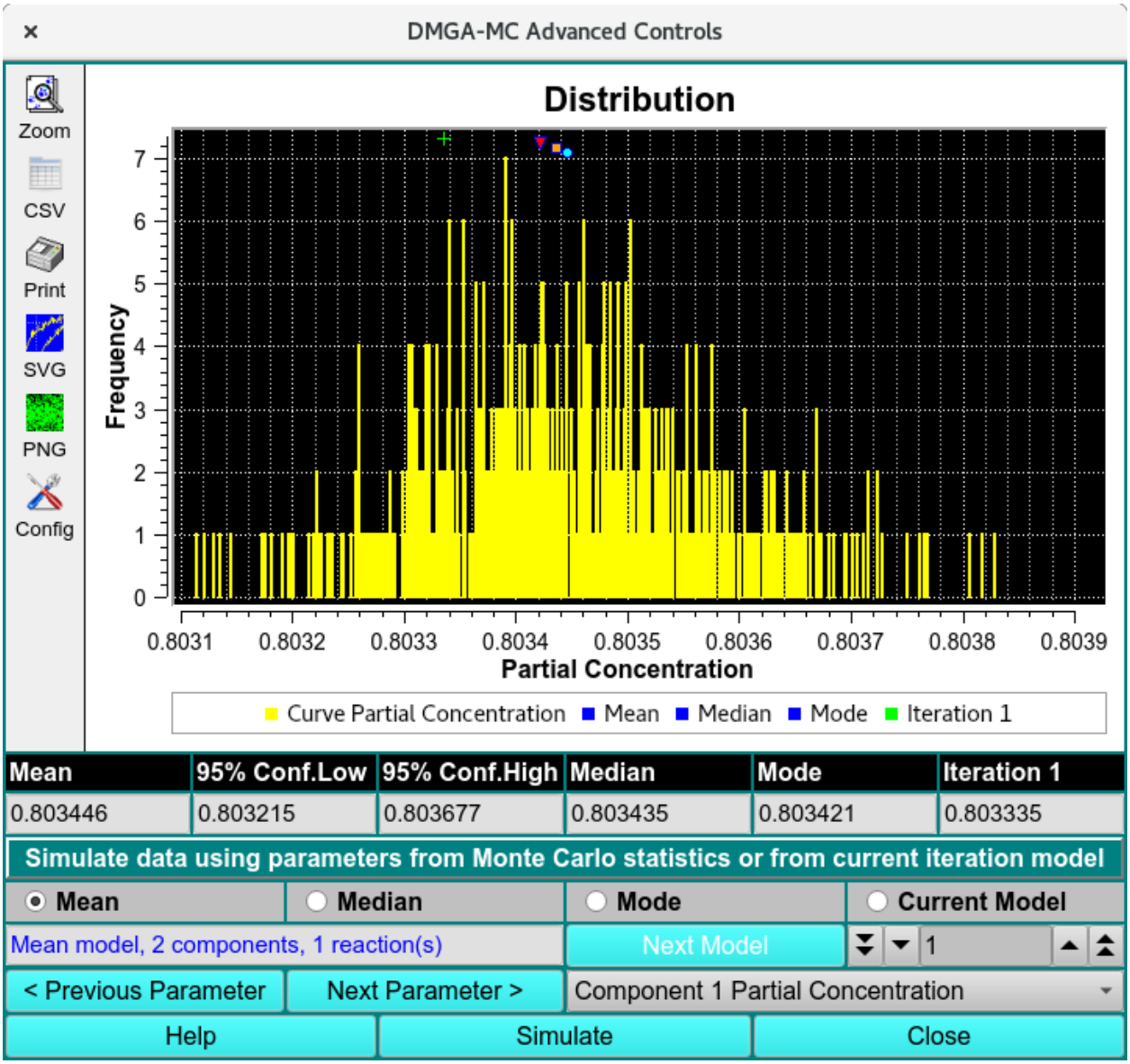
Simulation Points:	Specify simulation points.
Band-loading Volume:	Specify a band-loading volume value.
Parameter:	Choose a parameter value.
(mesh type)	Select from one of several mesh types, including Adaptive Space Time (ASTFEM), Claverie, Moving Hat, user File, or Adaptive Space Volume (ASVFEM).
(grid type)	Select a moving-time or constant-time grid type.
Help	Display this and other documentation.
Cancel	Close this dialog and exit with no selections made.
Accept	Close the dialog and communicate selections made here to the calling dialog.

CHAPTER

SIXTYNINE

DMGA ADVANCED ANALYSIS DIALOG

Introduction



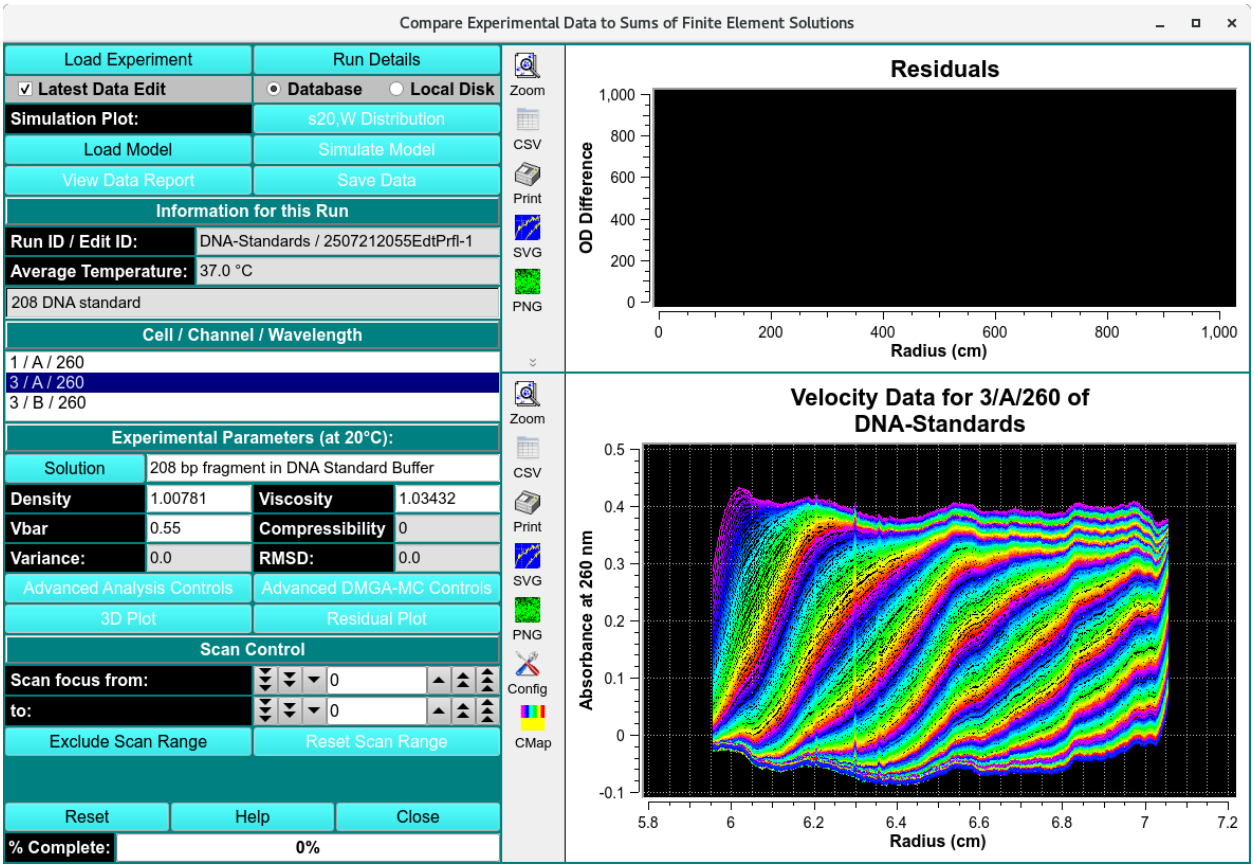
Discrete Genetic Algorithm Results

CHAPTER

SEVENTY

STEPS TO COMPARE EXPERIMENT TO FE MODELS:

- **Step 1:** *Load experimental velocity data.* Click on Load Data to select an edited velocity data set from the database or from local disk.
- **Step 2:** *Select a model from database or disk.* Simply click on Load Model and choose a model in the resulting dialog. If they exist, you will be given the choice of also loading noise vectors.
- **Step 3:** *Simulate a model.* Simulate the loaded model with a finite element solution by clicking **Simulate Model**.
- **Step 4:** *Examine the results.* After simulation, a variety of options are available for displaying :doc:`simulation results <fe_match_simulate>, residuals, and distributions. Report text files and graphics plot files can also be generated.



Compare Experiment to Finite Element Solution

CHAPTER

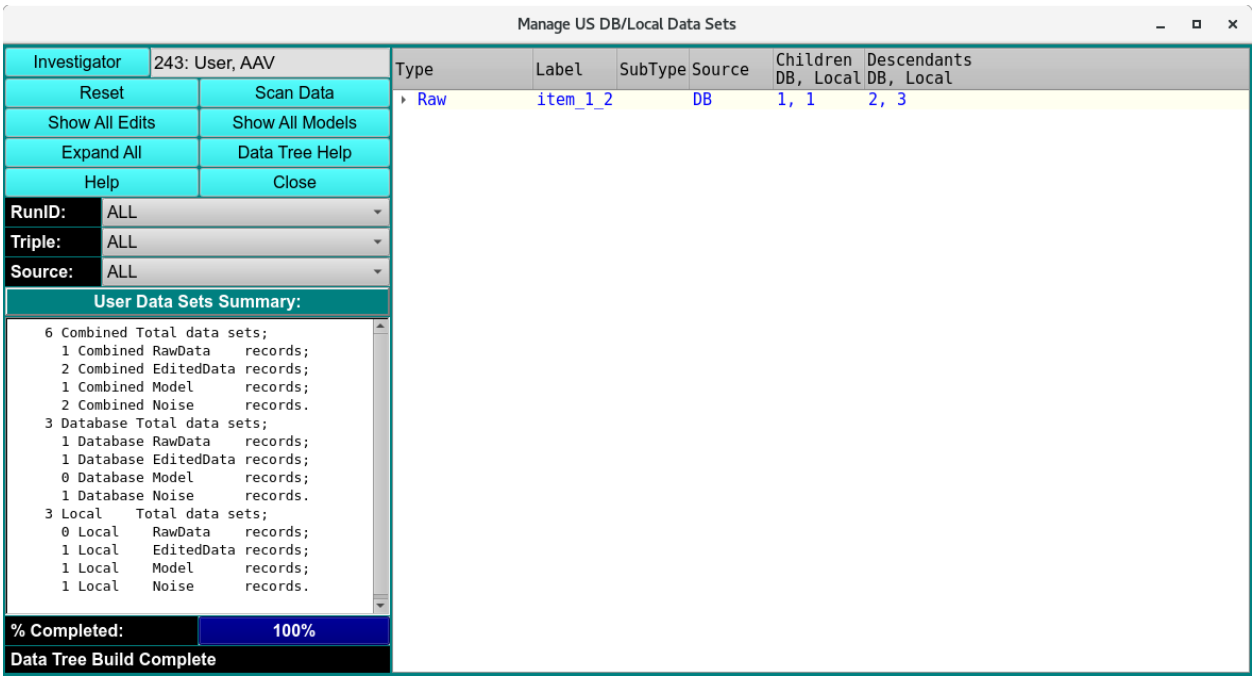
SEVENTYONE

MANAGE DATA IN DATABASE AND ON LOCAL DISK

Index

- *Manage Data Functions:*
- *Item Row Context Menu Options:*
  - *Dialog items:*

This module is launched from the US main window using the Utilities > Report Generator menu item. It enables the user to create and view a composite report generated from selections of individual analysis program reports and plots. Upon launch, selection of a Run (or **all** runs) populates a data tree with reports from which to select the components of the composite report. A click on the **View** or the **Save** button generates the report, which can be viewed in a browser and printed. Both the composite report HTML and all the component reports and plots are copied to a new folder so that the page is fully self-contained, can be archived, and can be viewed in the future.



71.1 Manage Data Functions:

Runs:	Select a run identifier to present reports for that run or select **all** to show all available reports of all runs.
(data tree selections)	Select reports and plots for the composite by checking the box to the left of an item. A check or uncheck of an item above the final (report/plot) tree level selects or de-selects all lower-level items. A check box with a shaded mark indicates that some (not all) of the item's descendant items have been selected. Click on the **+** or **-** boxes to expand or collapse tree portions.
View	Once selections are as desired, click on this button to generate the composite report and view it in the configured browser. See a *Sample composite report view*. A print within the browser can be used to generate hardcopy or a PDF version. Component reports and plots will start on page boundaries.
Save	Click this button to generate the report, but without being immediately taken to a view of it. You will receive a Notice of report save..
Load Profile	Make component report selections by means of a previously saved selection profile. Click this button to initiate a Profile Load Dialog.
Save Profile	Save component report selections for later use by clicking this button to initiate a Profile Save Dialog. Note that you need only specify the base name for the saved file. An **rs_** prefix and **.xml** extension will be automatically added.
Help	Display this and other documentation.
Close	Close all windows and exit.

UltraScan III

Reports from Run demo1_veloc:

demo1_veloc 2 / A / 260
2-Dimensional Spectrum Analysis
Analysis Report

2-Dimensional Spectrum Analysis

Data Report for Run "demo1_veloc", Cell 2, Channel A, Wavelength 260, Edited Dataset 1105311417

Detailed Run Information:

Cell Description:	lysozyme / 208-1 mixed in 0.2M NaCl/TE
Data Directory:	/home/gary/ultrascan/results/demo1_veloc
Rotor Speed:	42000 rpm
Average Temperature:	20.0 °C
Temperature Variation:	Within tolerance
Time Correction:	1 minute 13 seconds
Run Duration:	3 hours 4 minutes 7 seconds
Wavelength:	260 nm
Baseline Absorbance:	0.047647 OD
Meniscus Position:	6.019 cm
Edited Data starts at:	6.046 cm
Edited Data stops at:	7.076 cm

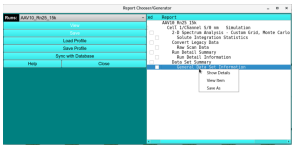
Hydrodynamic Settings:

Viscosity corrected:	1.04774
Viscosity (absolute):	1.04774
Density corrected:	0.999600 g/ccm
Density (absolute):	0.999598 g/ccm
Vbar:	0.7286 ccm/g
Vbar corrected for 20 °C:	0.7286 ccm/g
Buoyancy (Water, 20 °C):	0.272687
Buoyancy (absolute)	0.271693
Correction Factor (s):	1.049537
Correction Factor (D):	1.045711

Scan Information:

71.2 Item Row Context Menu Options:

Right-mouse-button clicking on a data tree row displays a context menu that applies to the row item.



71.2.1 Dialog items:

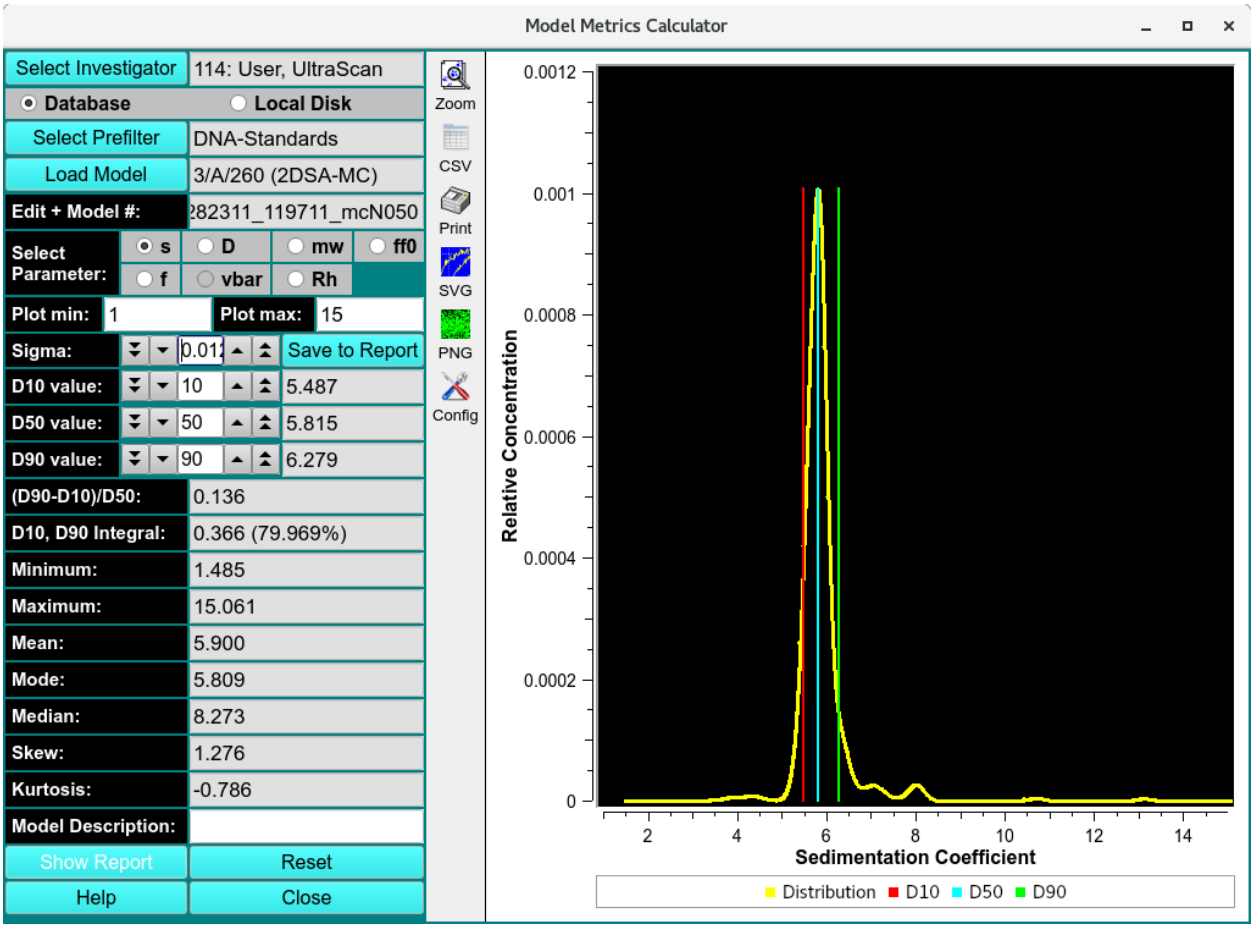
Show	Details Release on this menu item to see an Item Details Window.
View	Item Release on this menu item to view an Item View Window. This is a way to preview individual items that are potential candidates for the composite report.
Save	As Save a copy of the currently selected item. The location and name of the file copy may be chosen in the resulting standard file dialog.

CHAPTER

SEVENTYTWO

FINITE ELEMENT MODEL STATISTICS AND METRICS

This program allows you to obtain statistics for a model computed by any finite element analysis module in UltraScan.



Model Metric Calculator

The following metrics are supported:

- Three distribution limits D_i
- The span computed from the distribution limits
- The concentration integral between the lowest and highest distribution limit

- The *minimum* X-value of the distribution
- The *maximum* X-value of the distribution
- The mean, mode and median of the distribution
- The skew and kurtosis of the distribution

The program is started by selecting a model from the database or disk. When using the database, the investigator needs to be specified, and a prefilter can be applied. Once a model has been loaded, the user can choose the distribution based on any of the hydrodynamic parameters available in the model by checking the radio button next to the parameter. By default, the sedimentation coefficient distribution will be shown. One of the parameters fitted will be fixed and not available for plotting.

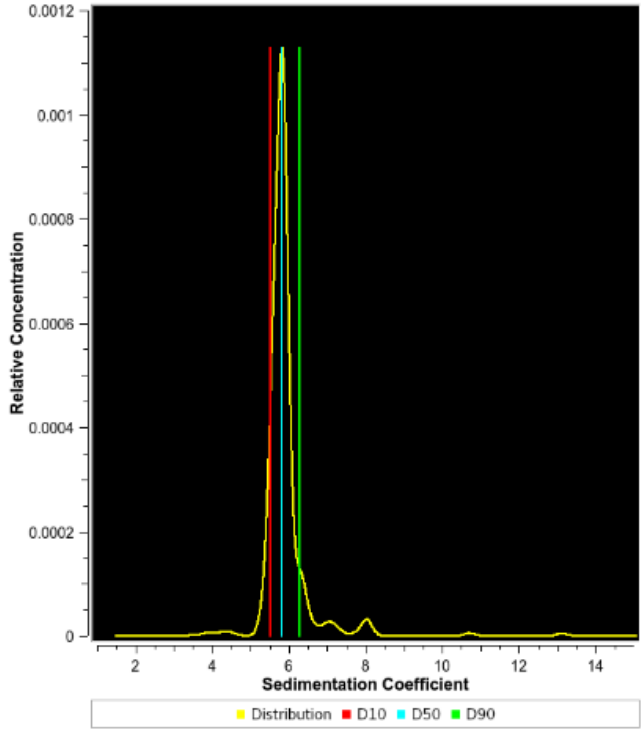
Once the model is loaded and the distribution parameter has been selected, default distributions limits D1 - D3 are set and the corresponding statistical values are calculated. At this point, the user can choose to add the current calculation and plot window to a report by clicking on **Save to Report**. The user can also change the smoothing level of the displayed distribution by changing the standard deviation (**Sigma**) value to non-zero, positive value. The distribution limits can also be adjusted to bracket different regions of the distribution and obtain different spans and centers. The adjustment of the top and bottom distribution limits can be used to integrate individual peaks in the distribution.

The integral value between the top and bottom distribution limits should be close, but will not exactly match the selected difference. For example, a distribution limit of 10% and 90% will not provide exactly 80% difference. The reason for this is the discrete nature of the finite element model distributions, which contain discrete solute points. To obtain integral values closer to the selected percent difference the discrete nature of the distribution can be reduced to a more continuous distribution by increasing the sigma value

For each model to be included in the report, **Save to Report** needs to be clicked once all parameters are set to acceptable values. Once all models have been evaluated, clicking on *Show Report* will call up the default browser on your system and display the cumulative report in html format. Clicking on the **Reset** button will reset all values to their original default values.

UltraScan Model Statistics Report

Information for Model 1 ():



Investigator:	User, UltraScan	**Distribution Type:**	Sedimentation Coefficient Distribution
Experiment:	DNA-Standards	**Analysis Method:**	2DSA-MC (50 MC Iterations)
Dataset:	Cell 3, Channel A, 260 nm	**Edit and Model name:**	e2507281841EdtPrfl-2_a2507282311_119711_mcN050

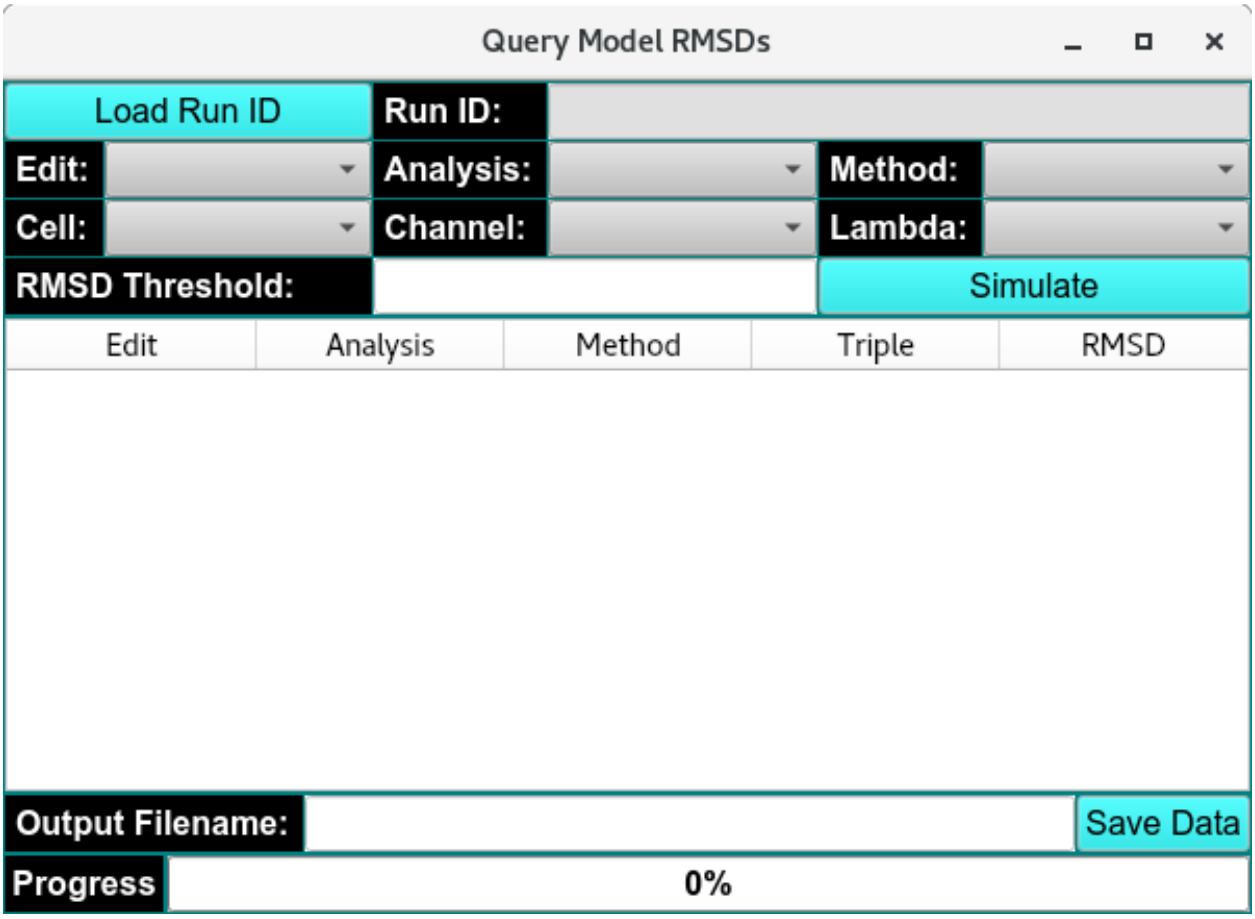
Statistics:

D10	5.510	**D50**	5.815	**D90**	6.269
(D90-D10)/D50:	0.131	**Minimum:**	1.485	**Maximum:**	15.061
D10, D90 Integral:	0.366 (79.935%)	**Total Concentration:**	0.458	**CSV File**	
Mean:	5.900	**Mode:**	5.823	**Median:**	8.273
Sigma:	0.010	**Skew:**	1.276	**Kurtosis:**	-0.786

Model Metric Report

triple========== RMSD Model Query ==========

The RMSD Model Query and Simulation module allows you to retrieve and evaluate RMSD (Root Mean Square Deviation) results for all models fitted to a selected experimental run.

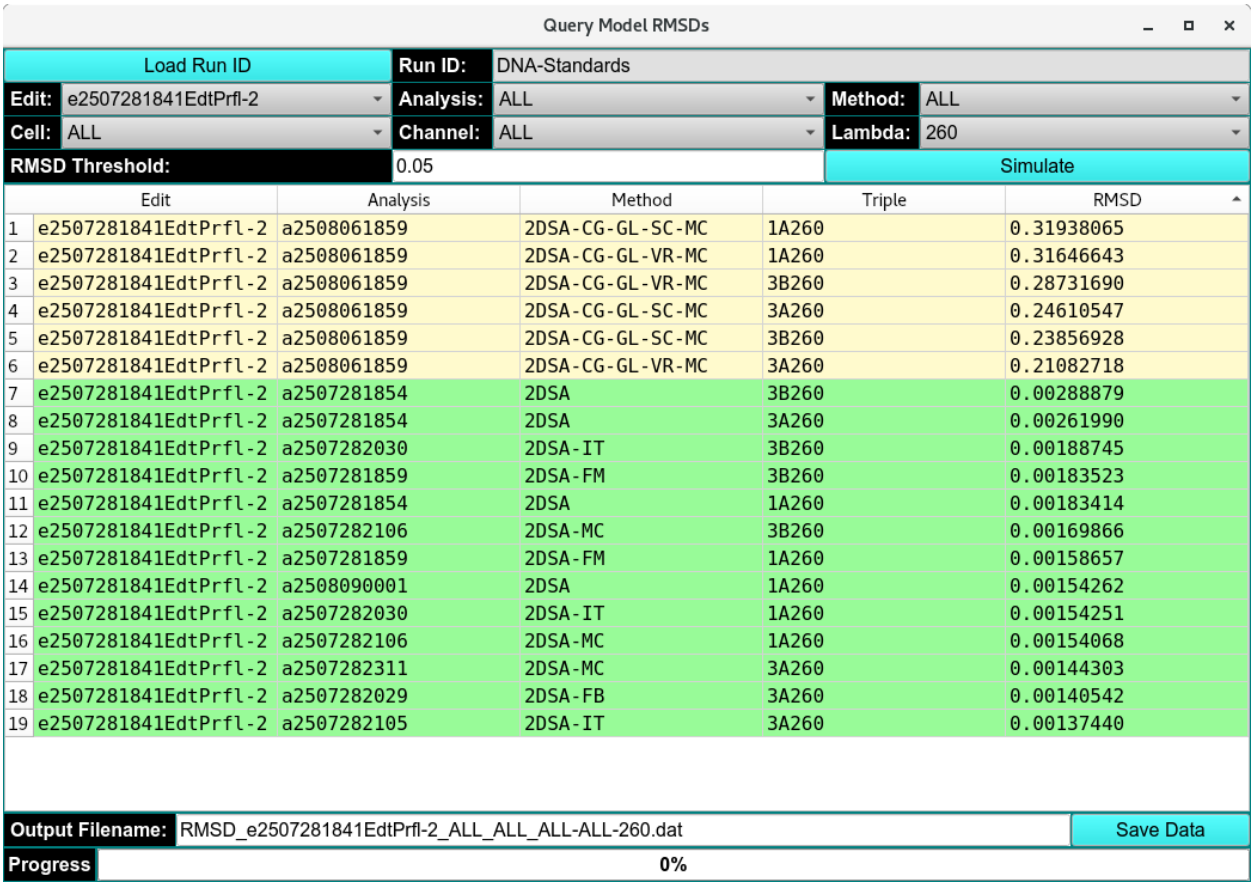


Empty RMSD Model Query Window

When you open the module, click on "Load Data" to loads the chosen experimental run. You can then specify the following parameters:

- **Edit Profile:** Select the data edit profile to be used.
- **Cells and Cell Channels:** Choose the cell(s) and corresponding channel(s).
- **Analysis:** Select the analysis type.
- **Analysis Method:** Choose the analysis method(s) to include.
- **Lambda:** Specify the wavelength(s) at which the sample(s) were measured.
- **RMSD Threshold:** Set the maximum acceptable RMSD value.

Once parameters are selected, you can run the simulation to compute the RMSD for every combination of edit, analysis, method, and sample triple. RMSD results are displayed in a table, where values below the threshold are highlighted in green and values above the threshold in yellow. The table can be exported as a .dat file for further processing.



Loaded RMSD Model Query Window

For deeper inspection, select an item from the results list and click Simulate to generate:

- Finite Element Data / Residual Viewer
- Residual Pixel Map
- *3D Plot Controls*

These tools allow detailed examination of model fits and residual map.

CHAPTER

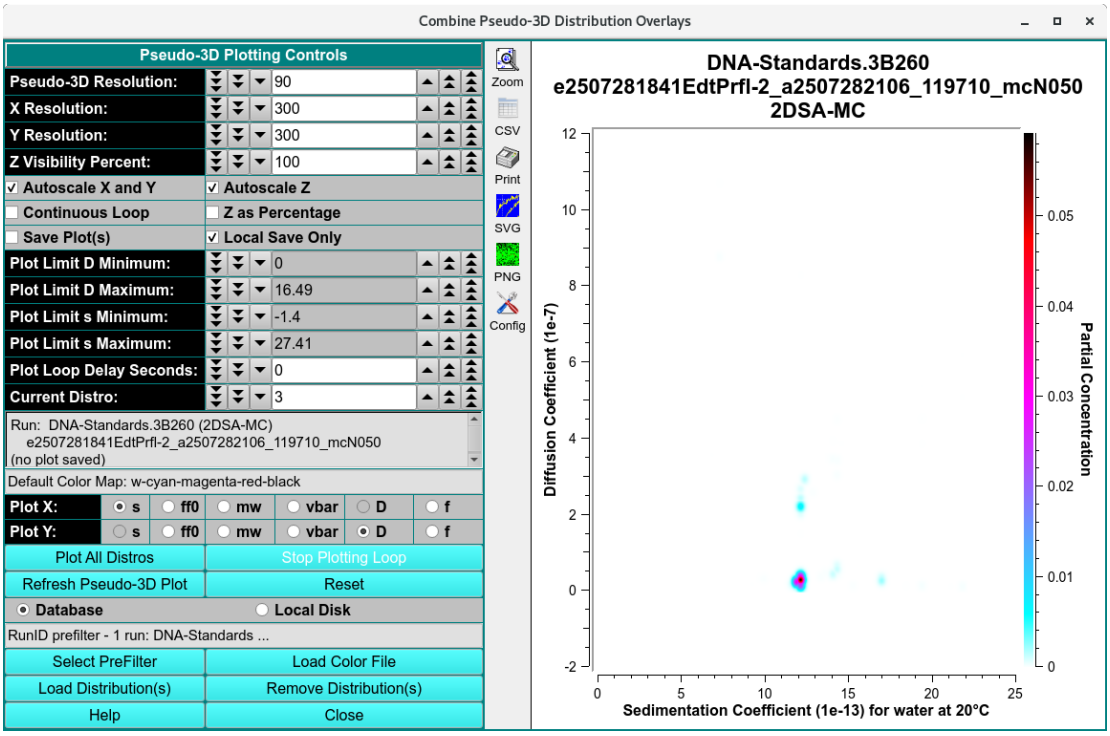
SEVENTYTHREE

PSEUDO 3-DIMENSIONAL DISTRIBUTION PLOTS

Index

- *Pseudo-3D Functions*

Solution distribution data can be displayed in pseudo-3-dimensional form where the Z axis is simulated using colors in a color map. Data from different cells can be combined.



73.1 Pseudo-3D Functions

Pseudo-3D Resolution:	Choose near-100 flattening parameter for Gaussian distribution of frequency points.
X Resolution:	Choose the number of pixels to represent the full X (Sedimentation Coefficient or Molecular Weight) data range.
Y Resolution:	Choose the number of pixels to represent the full Y (Frictional Ratio) data range.
Z Floor Percent:	Choose the percent of the Z (Frequency) range to add below the minimum Z in order to affect display of faint values.
Autoscale X and Y	Select to automatically set X and Y plot range limits or unselect to allow explicitly setting these values.
Autoscale Z	Select to automatically scale Z to the current distribution's range or unselect to scale Z to the global Z range of all distributions.
Continuous Loop	Check this box if you wish to continually loop through plots of distributions. While in a continuous loop, the plots are not automatically saved for reports.
Z as Percentage	Check this box if you wish to have the Z (color-mapped) values plotted as Percent of Total Concentration instead of the default Partial Concentration values.
Plot Limit f/f0 Minimum:	Choose the lower plot limit for f/f_0 (or vbar).
Plot Limit f/f0 Maximum:	Choose upper lower plot limit for f/f_0 (or vbar).
Plot Limit s Minimum:	Choose the lower plot limit for sedimentation (or MW).
Plot Limit s Maximum:	Choose the upper plot limit for sedimentation (or MW).
Plot Loop Delay Seconds:	Choose the number of seconds to delay between plots when all distros are plotted.
Current Distro:	Choose the index (1 to n) of the distribution to make the current one.
Plot f/f0 vs s	Select to plot f/f_0 versus sedimentation coefficient corrected for water at 20°Celsius.
Plot f/f0 vs MW	Select to plot f/f_0 versus molecular weight (Dalton).
Plot vbar vs s	Select to plot vbar versus sedimentation coefficient corrected for water at 20°Celsius.
Plot vbar vs MW	Select to plot vbar versus molecular weight (Dalton).
Plot All Distros	Loop to successively plot all loaded model distributions.
Stop Plotting Loop	Stop the successive plot of all loaded model distributions.
Refresh Pseudo-3D Plot	Refresh the pseudo-3d plot with current settings, current color map and current distribution data.
Reset	Reset parameters to their default settings.
Database	Select to specify data input from the database.
Local Disk	Select to specify data input from local disk.
Select PreFilter(s)	*Select Run(s) as Models Pre-Filter* dialog to select Run IDs. This can significantly reduce model loading time, particularly with large model counts in the database. The text box to the right of this button will display a summary of runs selected as pre-filters.
Load Color File	Load a color map for Z dimension display as specified through the Color Map Load dialog.
Load Distribution(s)	Load model distribution data as specified through a *Model Loader dialog*. Multiple distribution files may be selected.
Remove Distribution(s)	Open a *Remove Model Distributions* dialog to remove selected model distributions from the loaded set.

Window Controls

Help	Display this detailed help.
Close	Close all windows and exit.

CHAPTER

SEVENTYFOUR

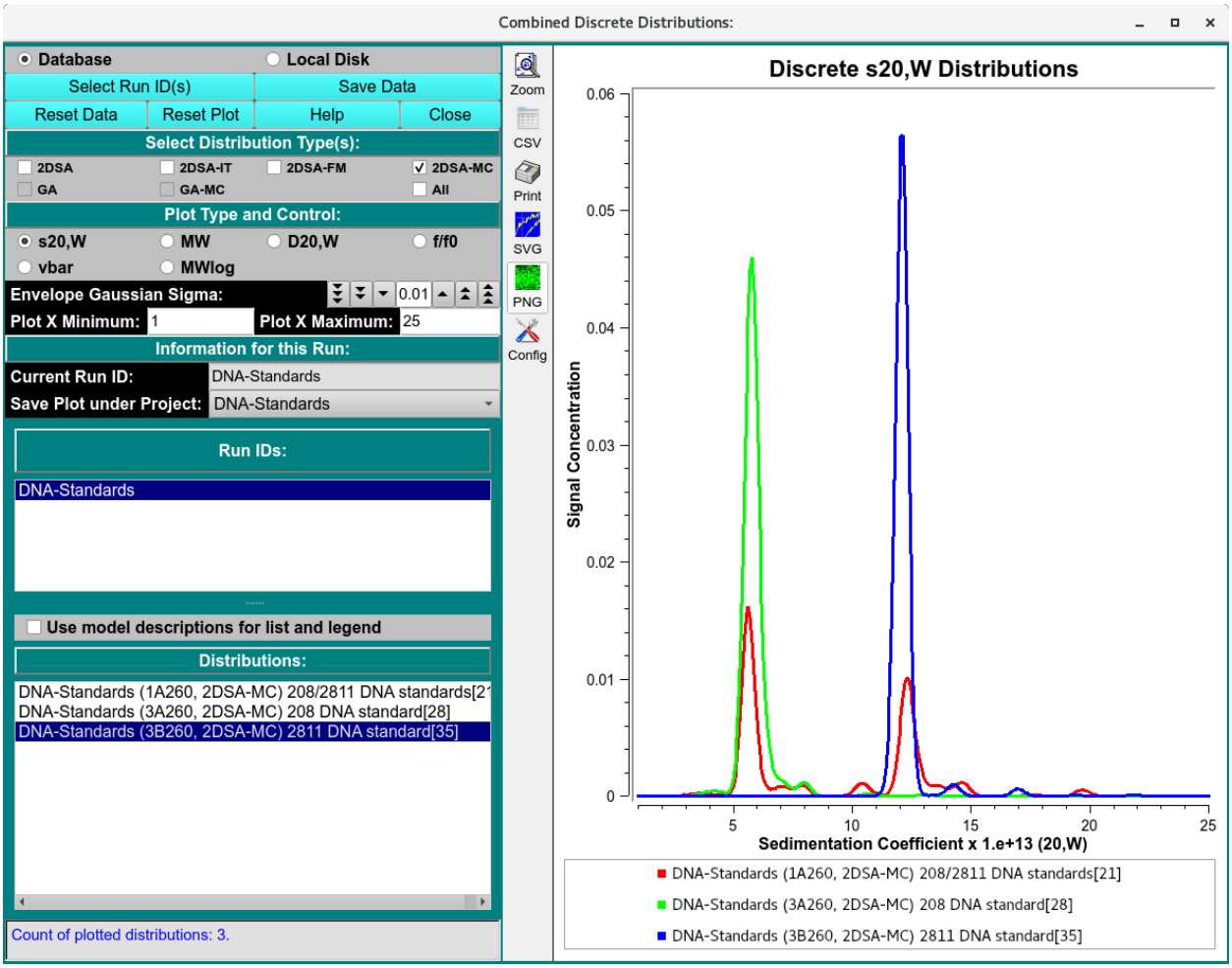
DISCRETE DISTRIBUTIONS COMBINED PLOTS

Index

- *Discrete Dis. Functions:*

To combine model distributions from multiple datasets (either from different cells of the same run or from different runs), use **Combine Discrete Distributions** from the *Velocity* menu. This will allow you to compare bar plots of distributions from different cells to each other.

Note: Distribution plots present relative concentration (Y) for X points where the "X" type may be sedimentation coefficent, molecular weight, diffusion coefficient, frictional ratio or vbar (specific density). Vertical bars are computed at every 2% mark of the X range.



74.1 Discrete Dis. Functions:

Database	Check this radio button to indicate that distribution data to be selected is in the database.
Local Disk	Check this radio button to indicate that distribution data to be selected is on local disk file systems.
Select Run ID(s)	Click on this button to open a *Run ID Select Dialog* in which one or more run IDs may be selected from which distribution data files can be taken.
Save Data	Save combined discrete distribution plots and data to a local data file and, if input source so indicates, to the database.
Reset Data	Clear all data, lists and plots. To choose runs and distributions, a new set of Select Run ID(s) selections must be made.
Reset Plot	Clear all plots.

Window Controls

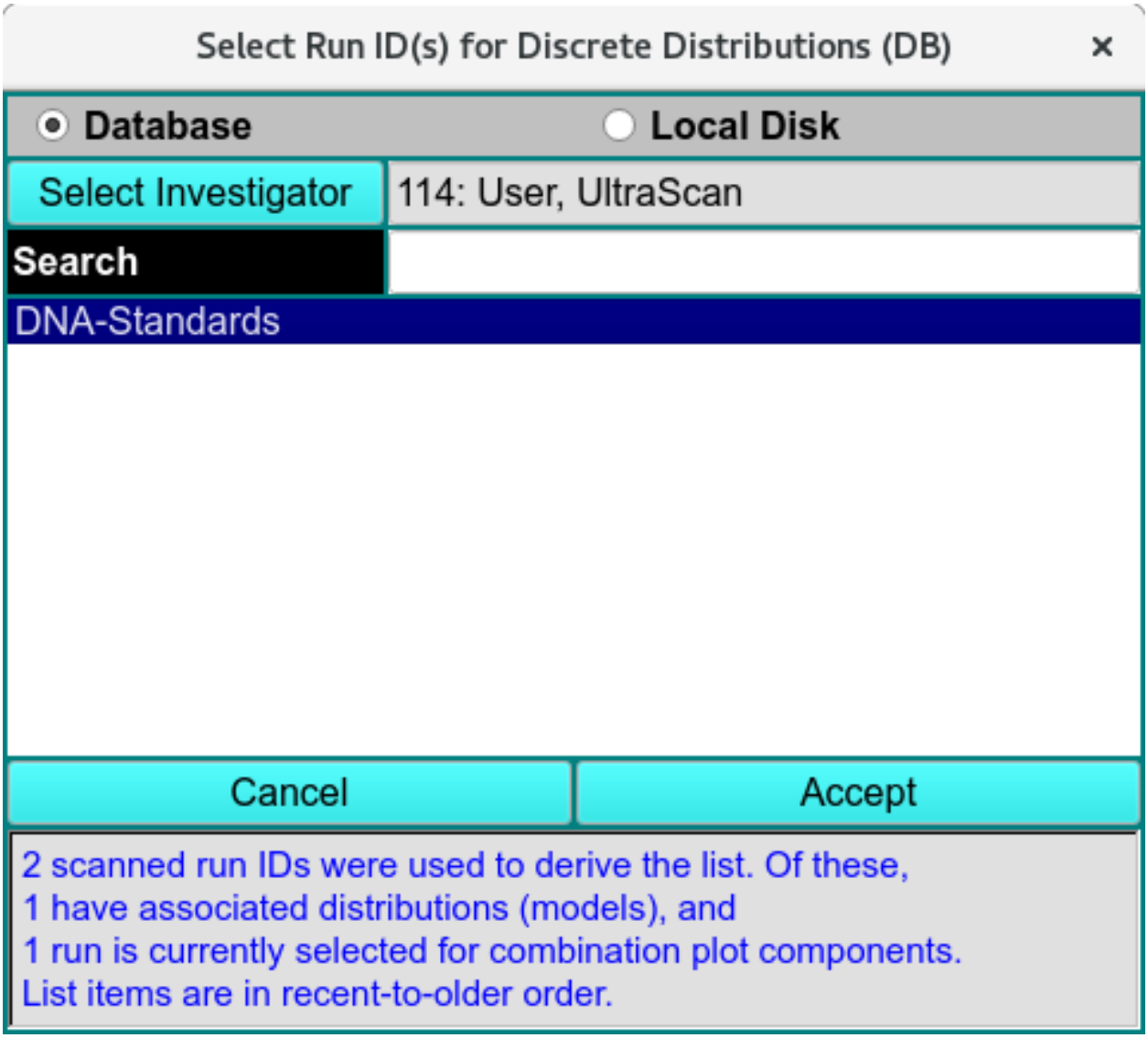
Reset Plot	Indicate that parameters are to be reset and the plots re-displayed based on original parameters. Data is still saved, so a new set of plots can be created by simply selecting runs and distributions (models).
Help	Display this detailed Discrete Distribution Plots help.
Close	Close all windows and exit.

Select Distribution Type(s): Check one or more of the check-box options below this banner to filter the list of Distributions (see below).

Select Plot Type:	Select one of the options below this banner to specify the "X" type of the combined distribution plot.
Current Run ID:	The read-only text to the right of this label indicates the currently selected run.
Save Plot under Project:	The name shown to the right of this label specifies the project subdirectory under which the combination is to be saved. By default, the name shown is the first run selected for plotting, but this can be changed by the user to any listed name. If "All" is selected, plot copies will be saved to all of the component run IDs.

Run IDs: The list below shows all runs selected in Select Run ID(s) dialogs. Click on a list line to reveal the distribution (model) choices for the run.

Distributions:	The list below shows the distributions for the currently selected run. Click on a model line to have its bar plot added to the combined plot.



Run ID Select Dialog

CHAPTER

SEVENTYFIVE

INTEGRAL DISTRIBUTIONS COMBINED PLOTS

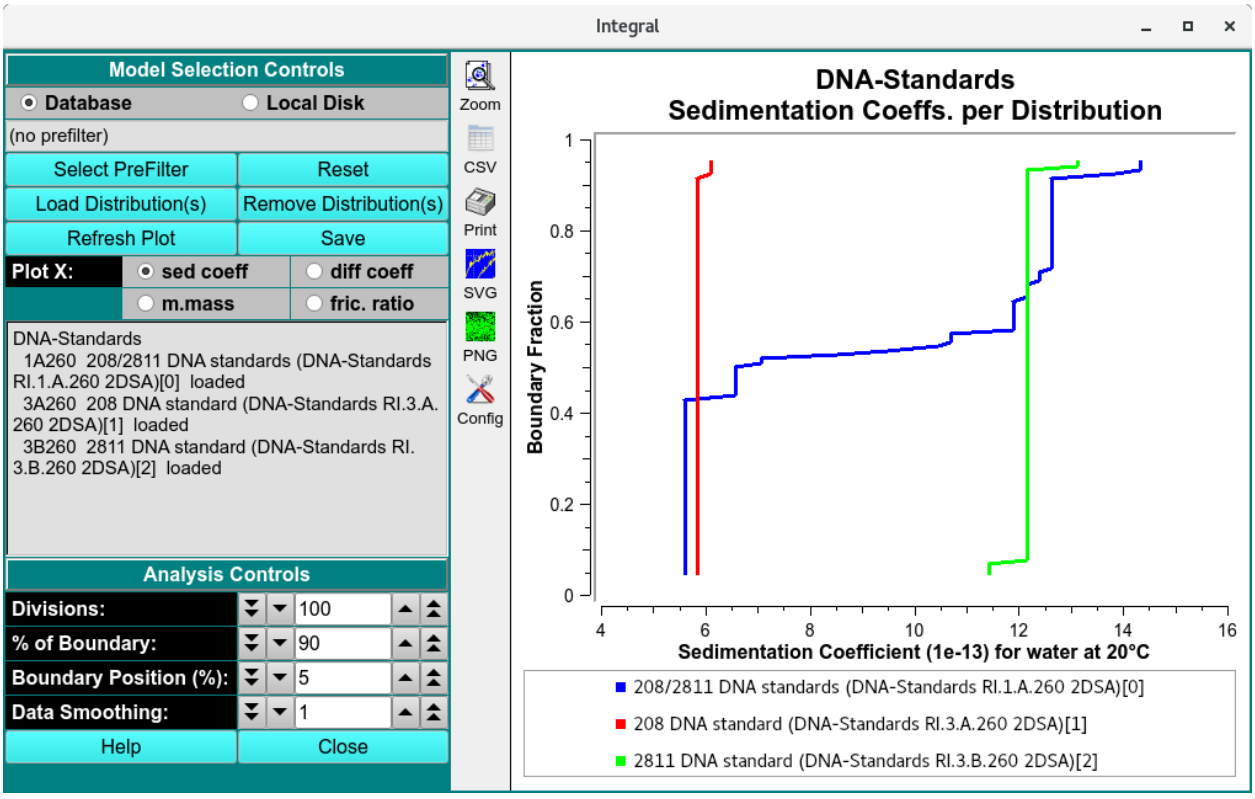
Index

- *Integral Dis. Functions:*

To combine integral model distributions from multiple datasets (either from different cells of the same run or from different runs), use **Combine Integral Distributions** from the *Velocity* menu. This will allow you to compare distributions from different cells to each other.

Note: Distribution plots present relative concentration (Y) for X points where the "X" type may be sedimentation coefficient, molecular weight, diffusion coefficient, frictional ratio or vbar (specific density).

All distributions will be plotted relative to the boundary fraction used in analysis. The default analysis is set to 90%, indicating the distribution will range from 5 % to 95 % of the total boundary, 5% from the top and bottom are excluded and the data is smoothed by 1%.



75.1 Integral Dis. Functions:

Database	Check this radio button to indicate that distribution data to be selected is in the database.
Local Disk	Check this radio button to indicate that distribution data to be selected is on local disk file systems.
Select PreFilter(s)	*Select Run(s) as Models Pre-Filter* dialog to select Run IDs.This can significantly reduce model loading time, particularly with large model counts in the database. The text box to the right of this button will display a summary of runs selected as pre-filters.
Load Distribution(s)	Load model distribution data as specified through a Model Loader dialog. Multiple distribution files may be selected.
Remove Distribution(s)	Open a *Remove Model Distributions* dialog to remove selected model distributions from the loaded set.
Refresh Plot	Resets the plot to the default and automatic plot ranges.
Save	Saves the integral distributions of sedimentation coefficient, Molar mass, Frictional ratio and diffusion coefficient distributions for every model loaded to *ultrascan/reports/experimentname* file directory in csv format.

Plot X: Select the "X" axis variable from

- **sed. coefff** (sedimentation coefficient),

- **diff coeff** (diffusion coefficient),
- **m.mass** (Molar mass),
- **fric. ratio** (Frictional ratio).

Analysis Controls

Divisions:	Choose the number of divisions to use in integral distribution analysis.
% of Boundary:	Choose the percentage of the range from concentration baseline to plateau that is to be used for analysis.
Boundary Position (%):	Choose the percent of the plateau-baseline range that is to be added to the baseline to form the beginning of analysis span. For typical analysis 5% is recommended.
Data Smoothing:	Choose the number of points to use for any smoothing of raw input data.

Window Controls

Reset	Indicate that parameters are to be reset and the plots re-displayed based on original parameters. Data is still saved, so a new set of plots can be created by simply selecting runs and distributions (models).
Help	Display this detailed Discrete Distribution Plots help.
Close	Close all windows and exit.

Triple========== van Holde-Weischet Combined Plots ==========

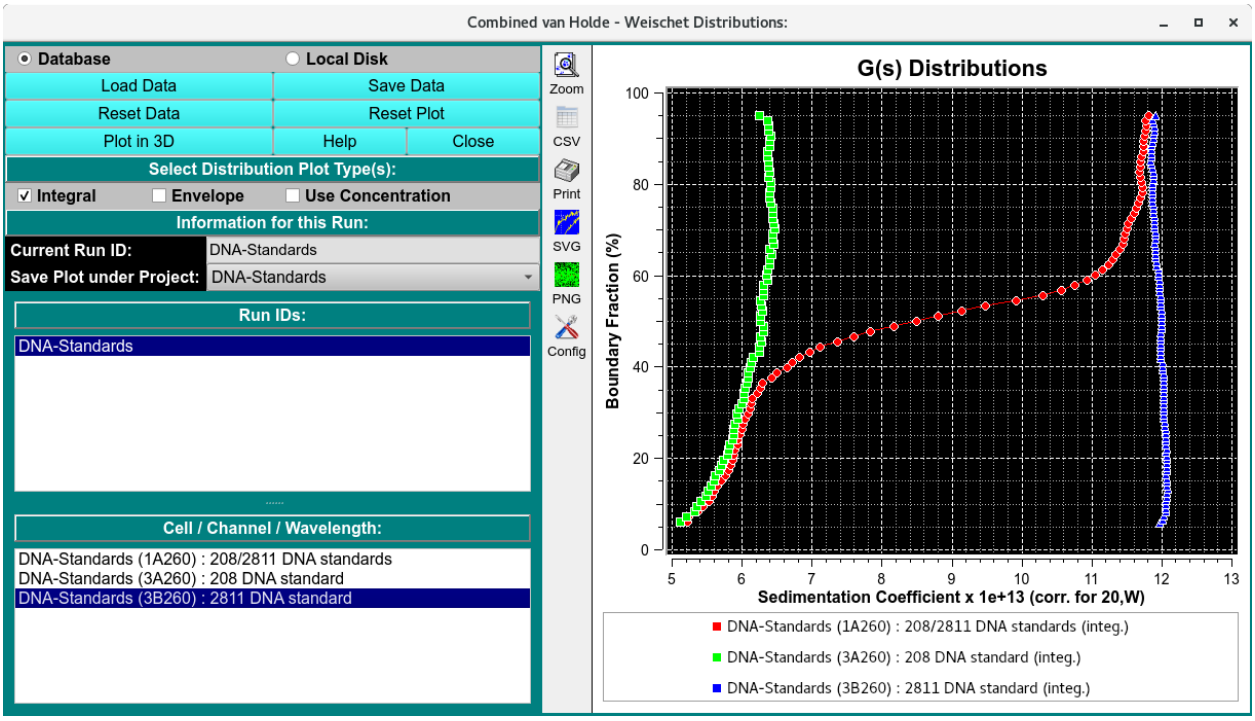
Index

- *Combine vHW Functions:*

To combine integral van Holde-Weischet G(s) distributions from multiple datasets (either from different cells of the same run or from different runs), use **Combine Distribution Plots (vHW)** from the *Velocity* menu. This will allow you to compare integral distributions from different cells to each other. This may be useful if you analyzed the same sample under different conditions; for example, different pH levels or different concentrations. If the samples are the same, the distributions should overlay; if the samples behave differently, the distributions will be shifted relative to each other.

Note: All distributions will be plotted relative to the boundary fraction used in analysis. For example, if you analyzed 100% of the boundary, the distribution will range from 0 % to 100 % of the Y-axis. A distribution that was analyzed for 72 % of the boundary, shifted 8 % from the bottom will be plotted from 8 % to 80 % of the Y-axis. It is important that this distinction be made, since, for example, concentration-dependent samples will behave differently depending on the position in the boundary; or heterogeneous samples will show different distributions depending on which portion of the boundary was analyzed.

When data are loaded, arrays of X,Y distribution and envelope values are read and saved. These are then used to quickly display plots. If you want to change the mix of data plotted, you may click the Reset Plot button; and then quickly make new selections that are plotted without need for recalculations.

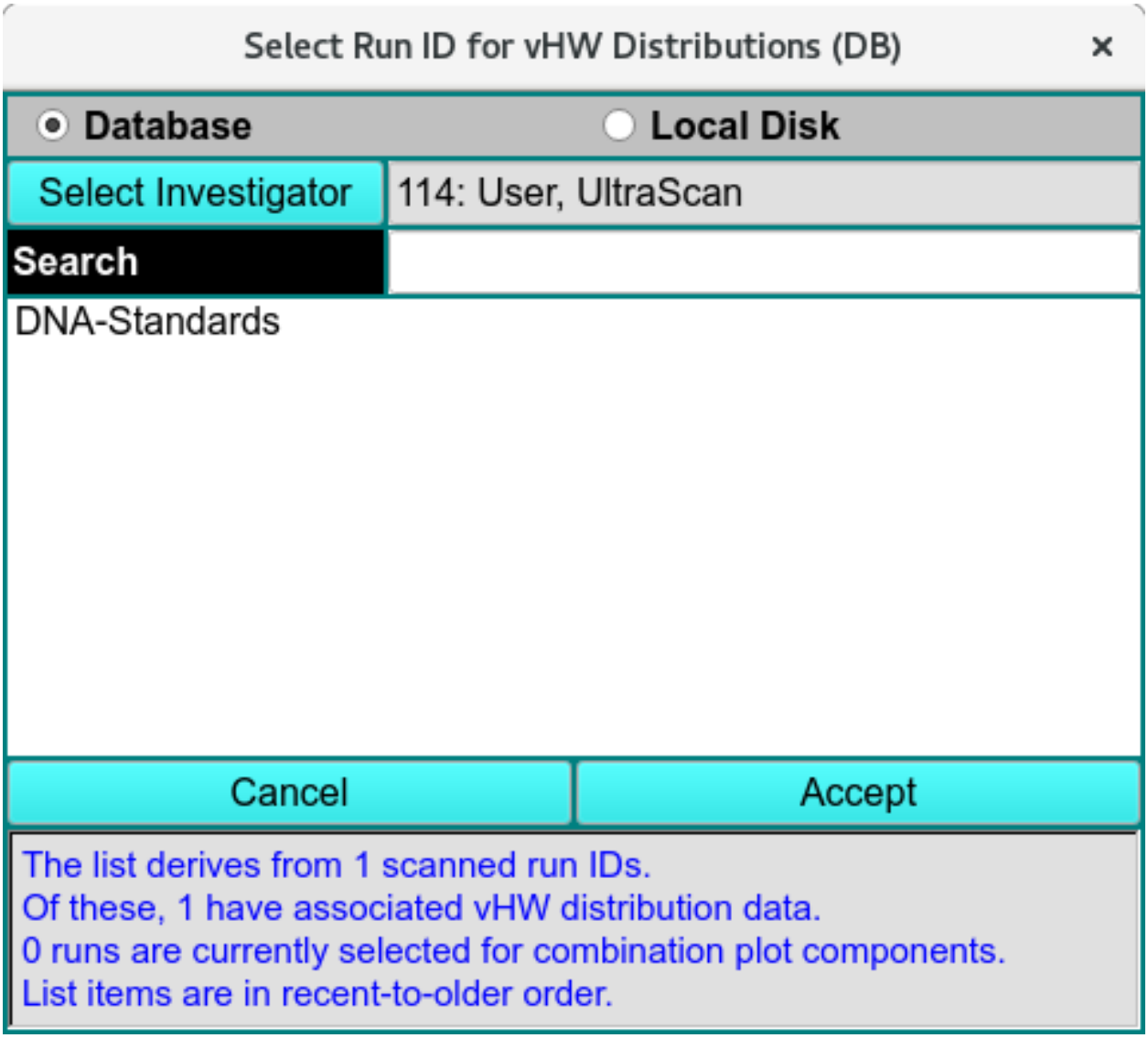


CHAPTER

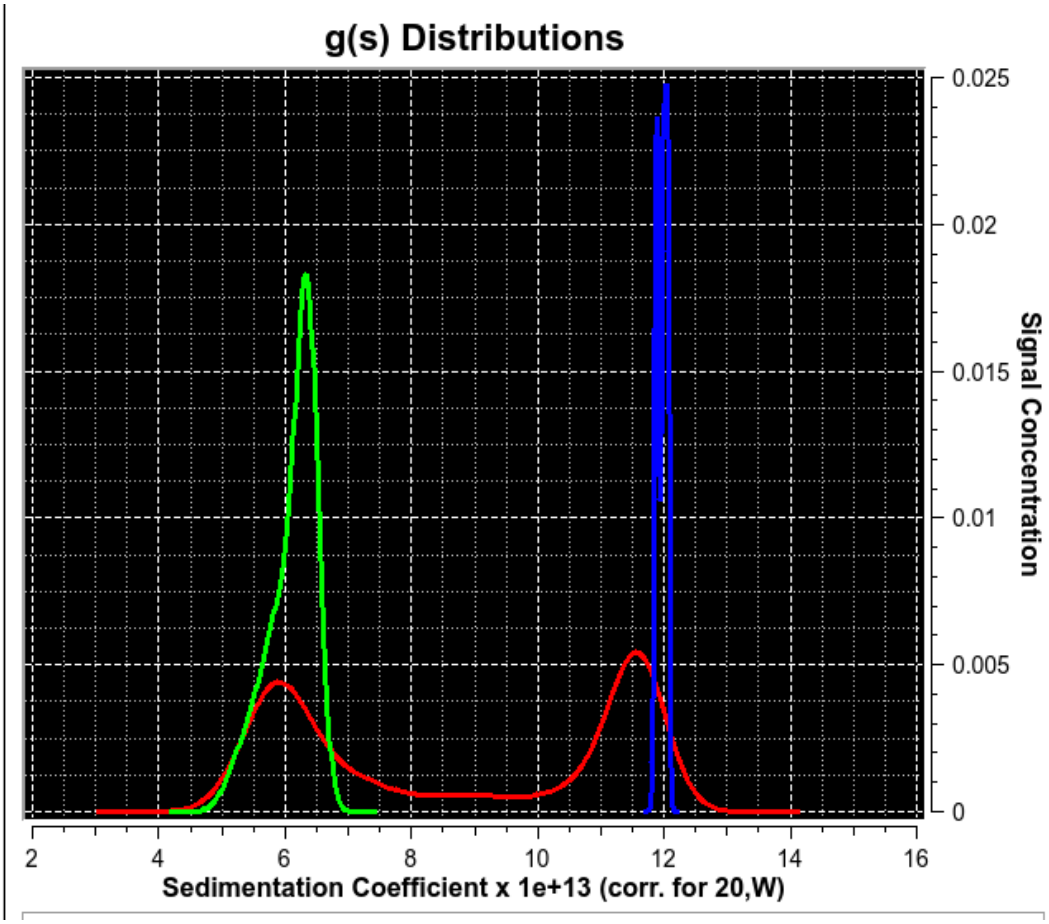
SEVENTYSIX

COMBINE VHW FUNCTIONS:

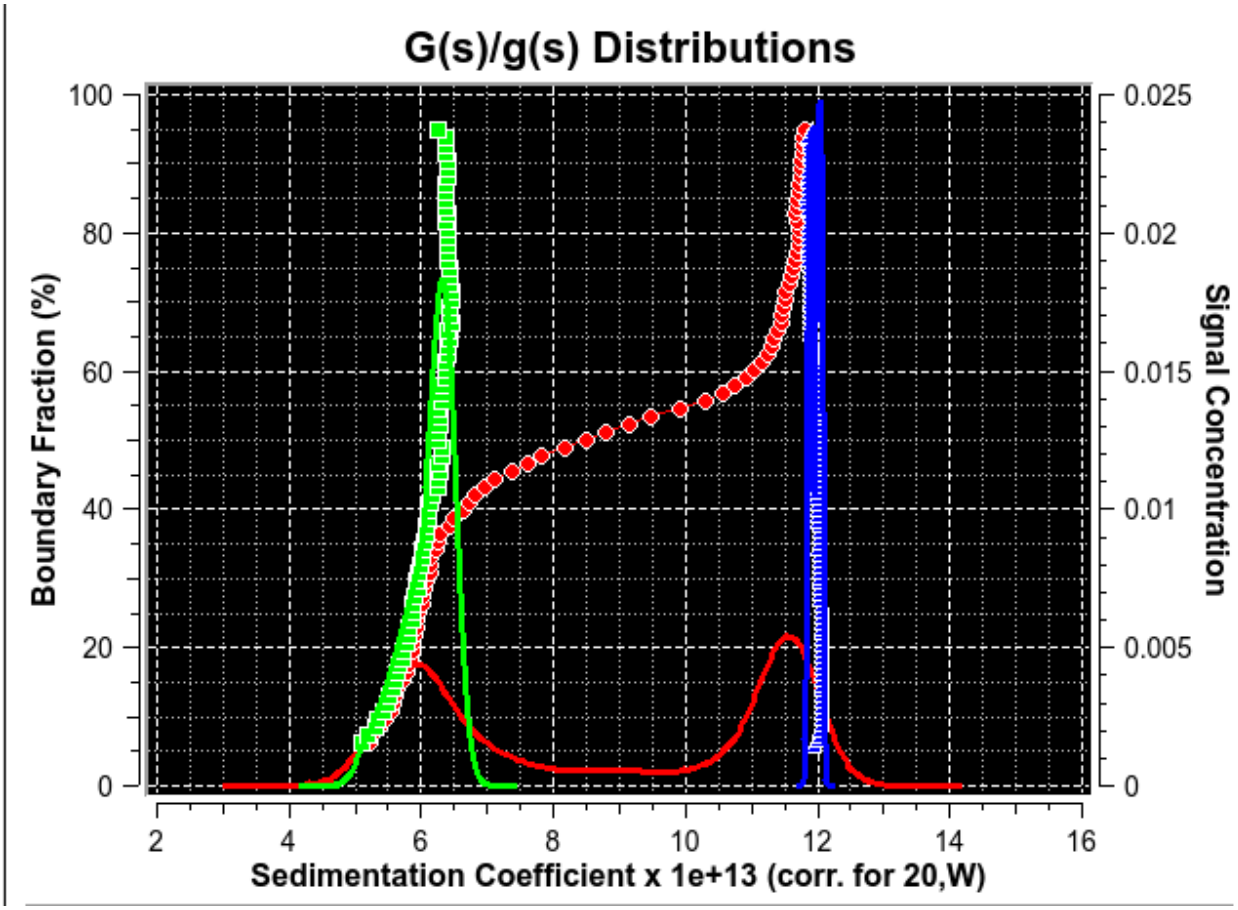
Database	Check this radio button to indicate that distribution data to be selected is in the database.
Local Disk	Check this radio button to indicate that distribution data to be selected is on local disk file systems.
Load Data	Click on this button to open Run ID Select Dialog in which one or more run IDs may be selected from which distribution data files can be taken.
Save Data	Save combined integral distribution and envelope data to a local data file and, if input source so indicates, to the database.
Reset Data	Clear all data, lists and plots. To choose runs and Triples, a new set of Load Data selections must be made.
Reset Plot	Clear all plots. Data is still saved, so a new set of plots can be created by simply selecting runs and Triples.
Help	Display detailed vHW_Combine help.
Close	Close this application.
Integral	Check to specify that plots are to include integral distributions.
Envelope	Check to specify that plots are to include envelopes.
Current Run ID:	The read-only text to the right of this label indicates the currently selected run.
Save Plot under Project:	The name shown to the right of this label specifies the project subdirectory under which the combination is to be saved. By default, the name shown is the first run selected for plotting, but this can be changed by the user to any listed name. If “All” is selected, plot copies will be saved to all of the component run IDs.
Run IDs:	The list below shows all runs selected in Load Data dialogs. Click on a list line to reveal the Cell / Channel / Wavelength Triples for the run.
Cell/Channel/ Wavelength:	The list below shows the Triples for the currently selected run. Click on a Triple line to have its distribution/envelope added to the combined plot.



RunID Select Dialog



Envelope Display



Combined Display

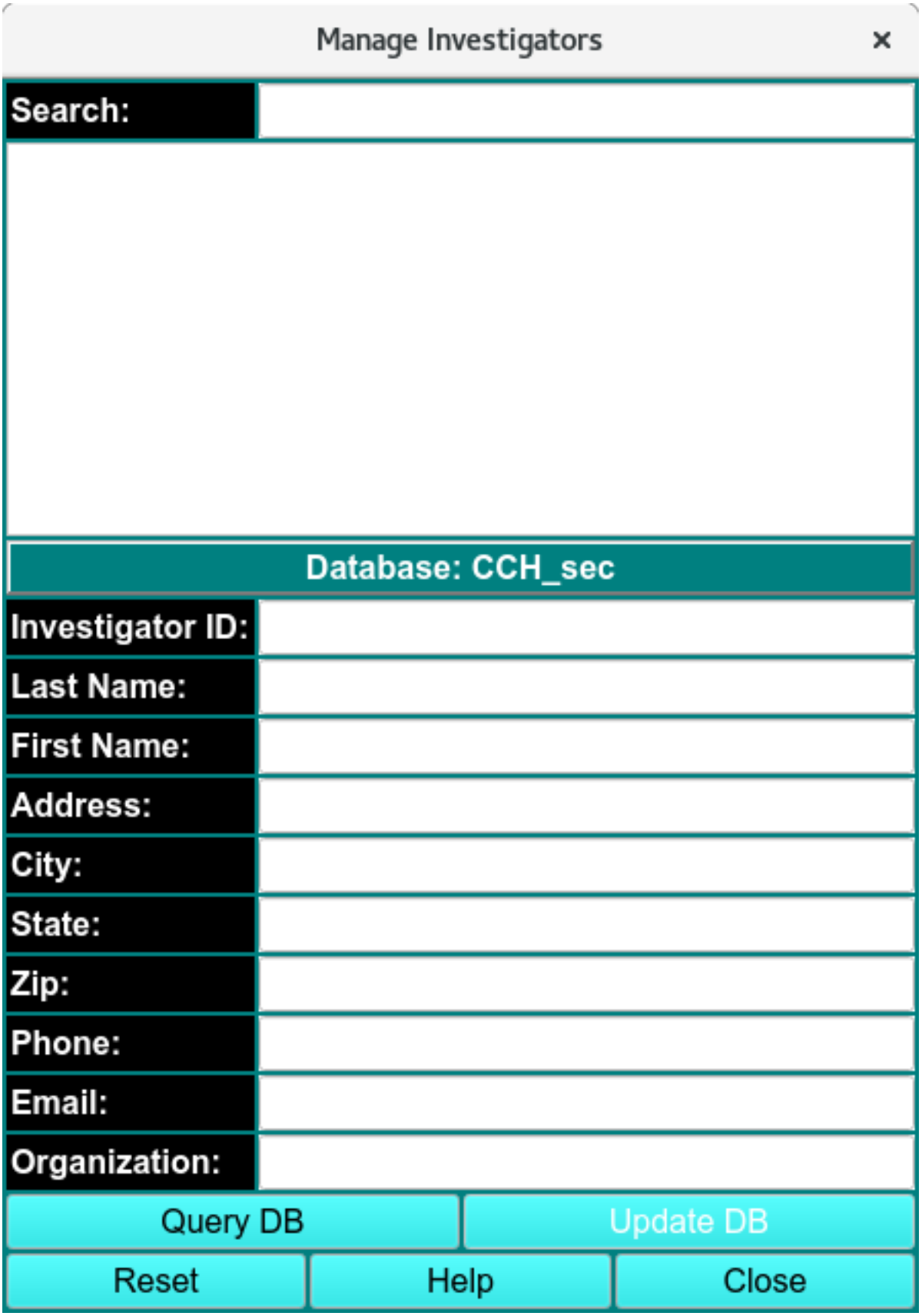
Besides the integral distribution plots shown above, you may display envelope plots. You may also combine both integral and envelope plots in a single plot.

CHAPTER

SEVENTYSEVEN

MANAGING PERSONAL INFORMATION

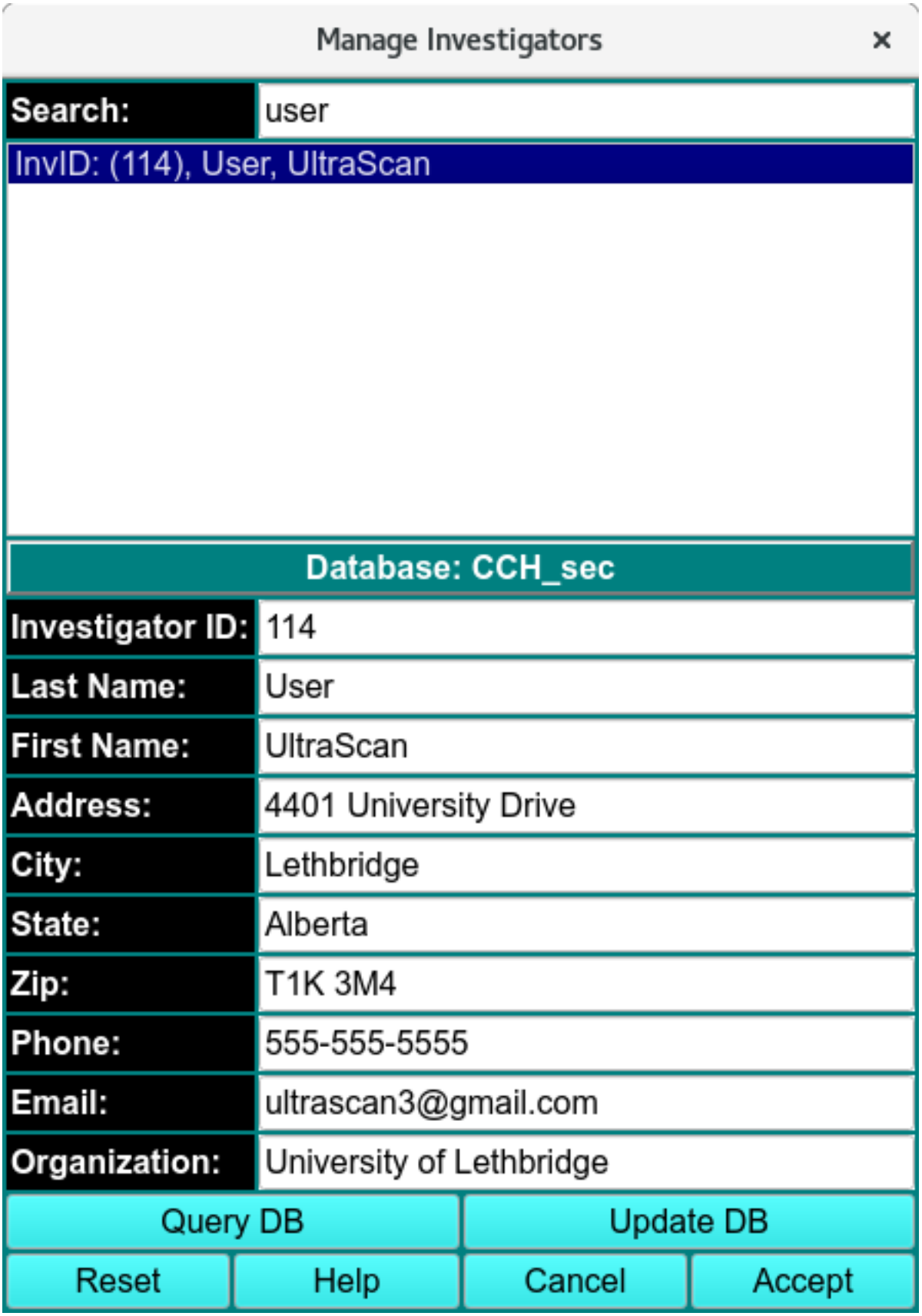
Using this window, you can update any personal information. Your database authorizations must allow update privileges for the record being changed.


Manage Investigators
Search:
Database: CCH_sec
Investigator ID:
Last Name:
First Name:
Address:
City:
State:
Zip:
Phone:
Email:
Organization:
Query DB
Update DB
Reset
Help
Close

Manage Investigators Empty

- **Step 1:** Use **Query DB** to get a list of persons in the current database.
- **Step 2:** Entering text in the **Search** window narrows the number of entries visible in the list of personnel available for selection.
- **Step 3:** Double-clicking on the name of the investigator will populate the line edit windows to allow viewing and changing personal data.
- **Step 4:** Change any data desired.
- **Step 5:** Use **Update DB** to commit the changes to the database.
- **Step 6:** Use **Accept** to pass information back to the calling program and close this dialog or **Cancel** to simply close without changing the investigator data of the caller.

Note: Depending upon how the window is initiated, the Cancel button may be labelled Close with no Accept button present.


Manage Investigators
Search:
user
InvID: (114), User, UltraScan
Database: CCH_sec
Investigator ID:
114
Last Name:
User
First Name:
UltraScan
Address:
4401 University Drive
City:
Lethbridge
State:
Alberta
Zip:
T1K 3M4
Phone:
555-555-5555
Email:
ultrascan3@gmail.com
Organization:
University of Lethbridge
Query DB
Update DB
Reset
Help
Cancel
Accept

Manage Investigators Filled out

CHAPTER

SEVENTYEIGHT

MANAGE BUFFER INFORMATION

This window allows the user to manage buffer information by creating new buffer compositions, editing existing entries, and synchronizing buffer data between your local system and the user database.

In each panel, tabs are visible at the top of the window to enable the user to move to another panel, to perform specific buffer subtasks.

78.1 Select Buffer Tab

Index

- *Select Buffer Panel*
  - *Buffer Sel. Functions:*
  - *Select Buffer Steps:*

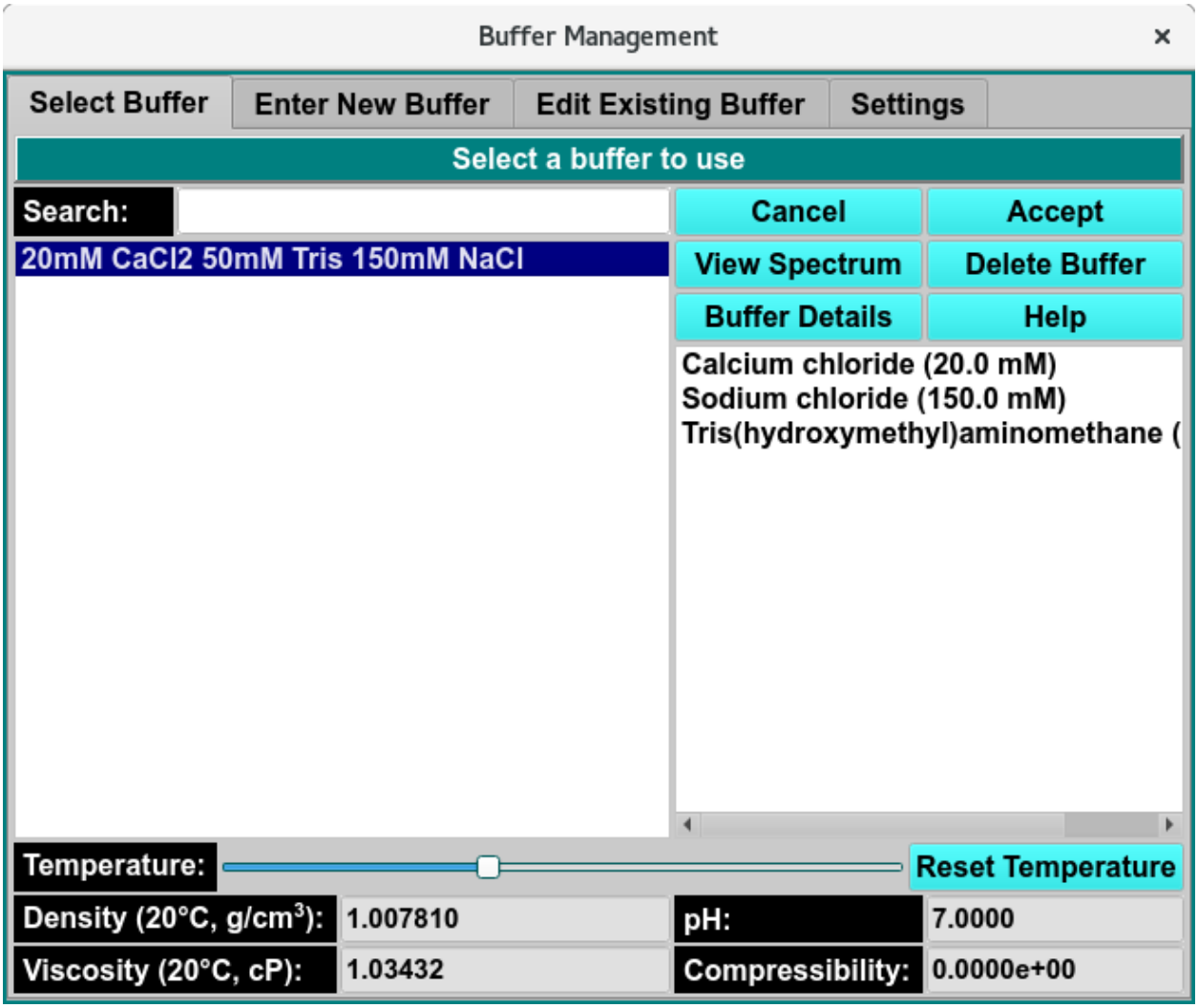
Panel Tab Options:

- *Select Buffer* - A panel whose primary purpose is to select a buffer to return to the caller.
- *Enter New Buffer <buffer_new>* - A panel whose primary purpose is to enter a brand new buffer, defined mostly by specifying components and each one's concentration.
- *Edit Existing Buffer <buffer_edit>* - A panel whose primary purpose is to change non-hydrodynamic characteristics of an already existing buffer.
- *Settings <buffer_settings>`* - A panel whose primary purpose is to set Database-or-Disk input, the investigator; or to synchronize the local buffer components file from the database.

78.1.1 Select Buffer Panel

Using this panel, you can select a buffer in the current database or on the local disk. Most commonly, you select a buffer in the provided list and click the Accept button to return it to the caller.

As with all panels, a set of tabs allows you to navigate to other panels in order to perform specialized subtasks relating to buffer management. Links to and summaries of the panels are given in the final section of this page.



Select Buffer Window

Buffer Sel. Functions:

(buffer list, left-side)	The items listed on the left side are all the descriptions (names) of buffers available or the descriptions as limited by the **Search** text. A single click on a list item selects a buffer.
(component list, right-side)	The items listed on the right side are the components of the currently selected buffer, along with each one's concentration. This list is solely for information purposes.
Search:	Enter a partial description to search for a specific buffer or set of buffers. Clear the text to expand to the full list.
Density	The density of the currently selected buffer.
Viscosity	The viscosity of the currently selected buffer.
pH	The pH of the currently selected buffer.
Compressibility	The compressibility of the currently selected buffer.
Cancel	Close the buffer dialog and return to the caller with no change in buffer selection.
Accept	Close the buffer dialog and return to the caller with the current buffer selection.
View Spectrum	Open a *Manage Spectrum* dialog showing the Wavelength/Extinction profile associated with the currently selected buffer.
Delete Buffer	Permanently delete the currently selected buffer. A confirmation dialog will appear to allow you to cancel the delete request or to proceed with it. Upon confirmation, the removal will proceed regardless of any further action ("Cancel" or "Accept").
Buffer Details	Open a *Buffer Information* dialog showing detailed information about the currently selected buffer.
Help	Show this documentation.
Temperature:	Temperature range slider, select the temperature using the slider and the Density and Viscosity will update.
Reset Temperature:	Reset the Temperature range slider to 20°C, select the temperature using the slider and the Density and Viscosity will update.

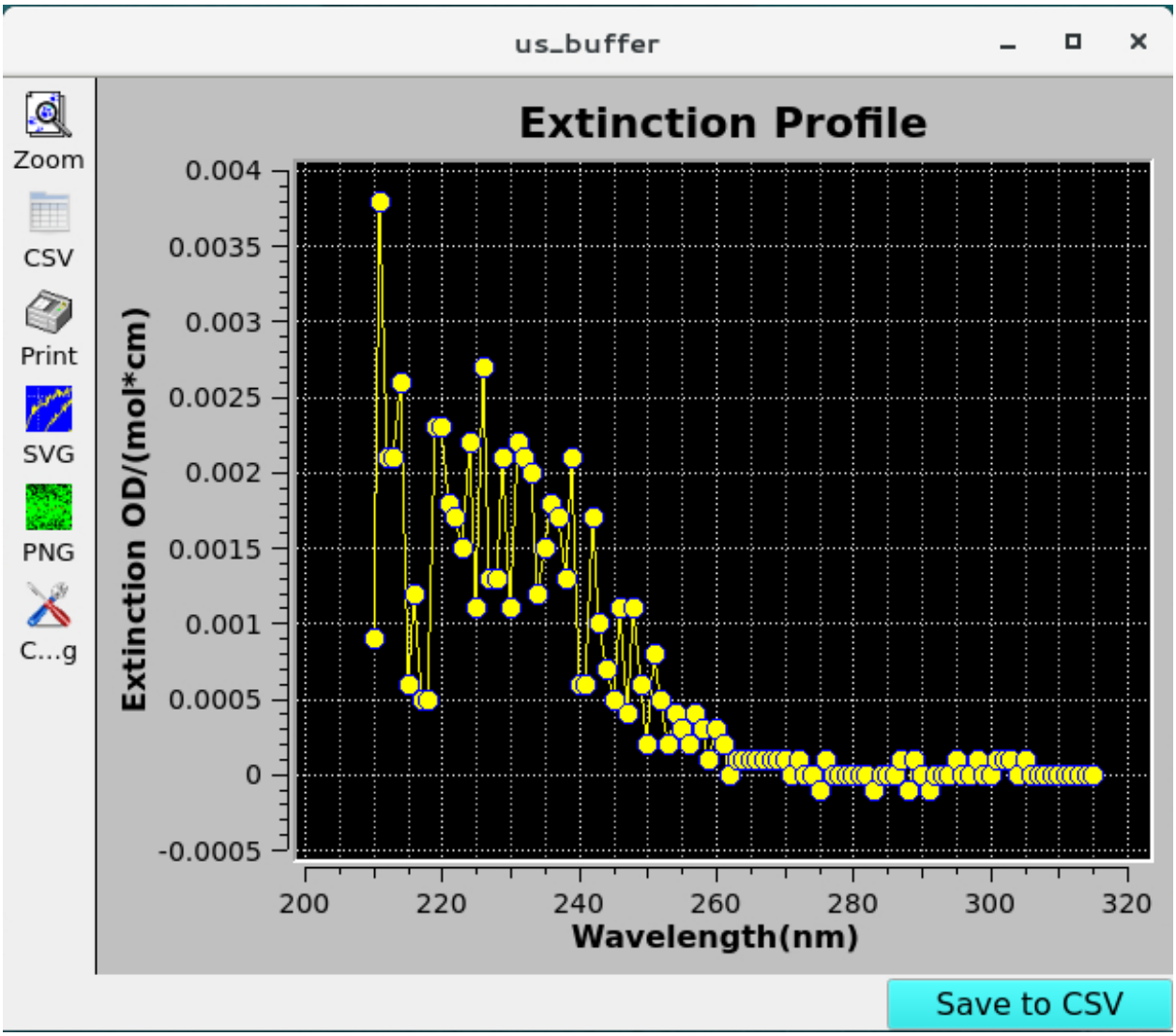
Select Buffer Steps:

As mentioned above, the most common use of this panel is to select a buffer in the provided list and click on the Accept button to return the selected buffer to the caller, such as a Solution Management dialog.

The panel allows actions beyond simple buffer selection, including getting buffer information, deleting a buffer, or choosing the buffer for which non-hydrodynamic modifications are to be made. In summary, the most common actions for this panel are as follows.

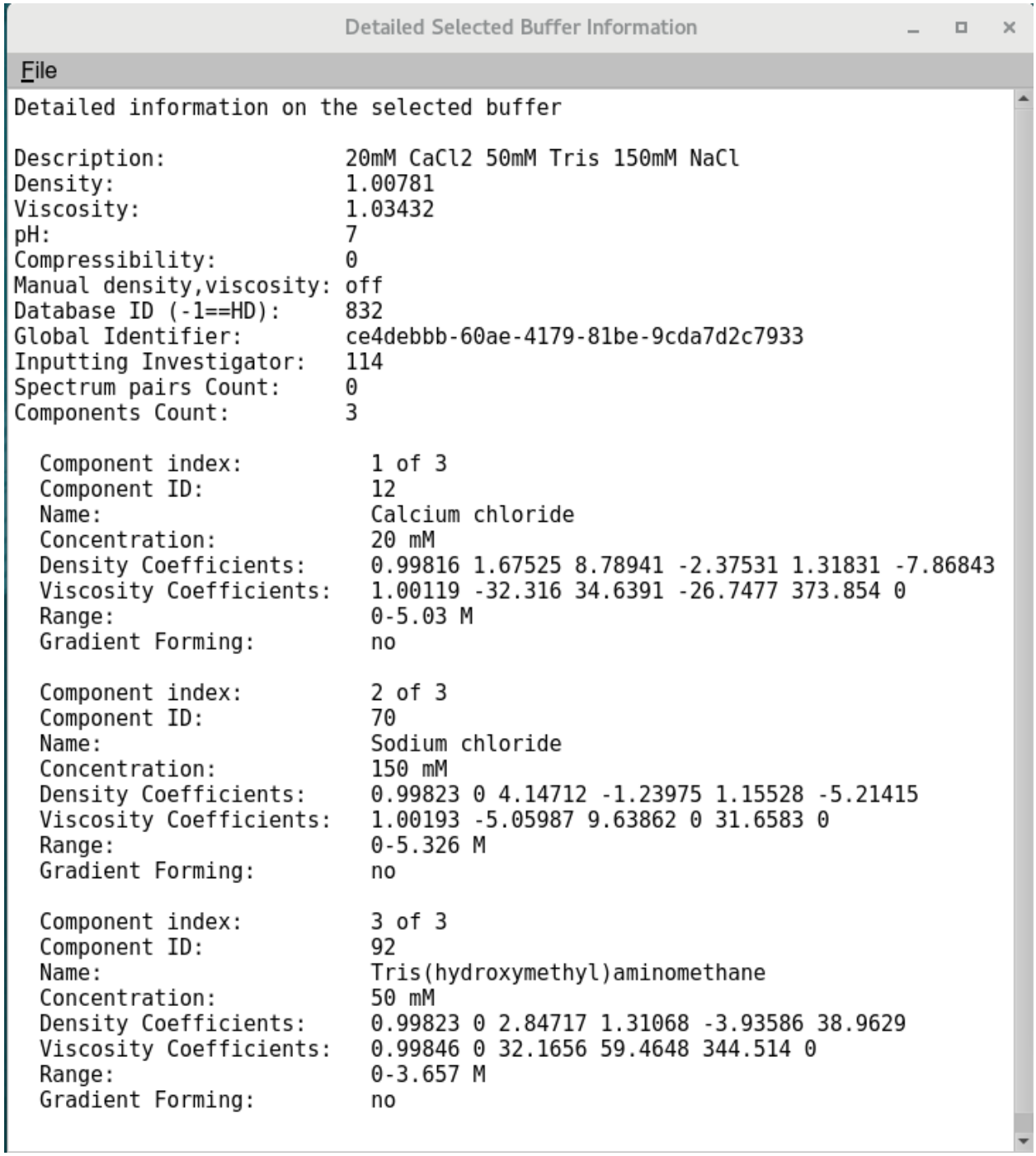
- Select a buffer and **Accept** it for a caller.
- Select a buffer and **modify** its non-hydrodynamic characteristics by selecting the Edit Existing Buffer panel.
- Select a buffer to **obtain information** about it, including the details shown with the *Buffer Details* button.

- Select a buffer in order to **remove it** permanently from the database or local disk, with the Delete Buffer button.



View Buffer Spectrum

Note: Buffer spectra can be added **Managae Spectrum** button found in either the **:ref:`New Buffer <../analyte/analyte_new.html#add-analyte-spectrum>`_** tab or the **:ref:`Edit Analyte <buffer_edit.html#edit-buffer-spectrum>`_** tab.



Buffer Information

78.2 Enter New Buffer Tab

Index

- *Enter New Buffer Panel*
  - *Buffer New Functions:*
- *Add New Buffer Steps*
- *Add Buffer Spectrum*

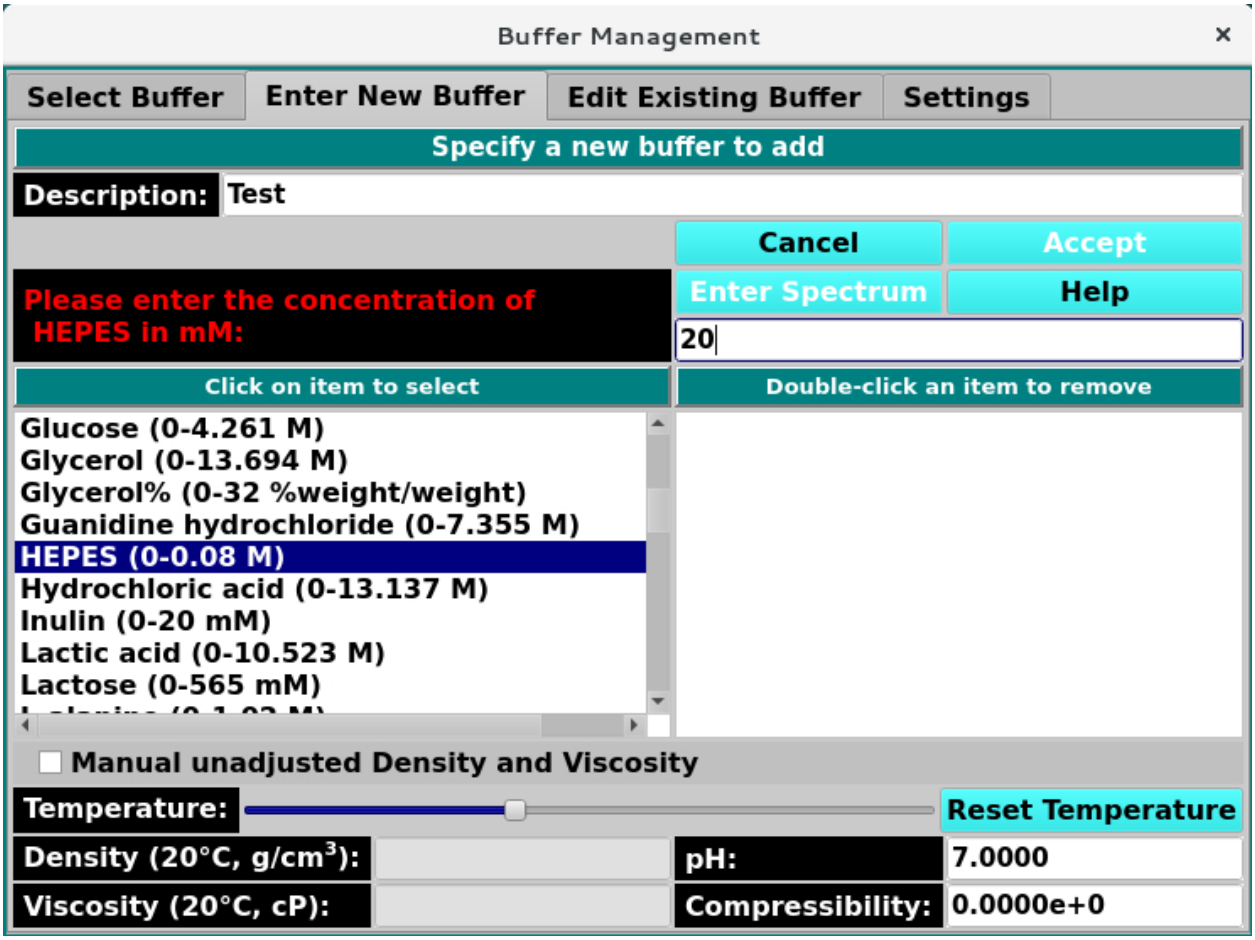
Panel Tab Options:

- `Select Buffer <buffer_select> - A panel whose primary purpose is to select a buffer to return to the caller.
- *Enter New Buffer* - A panel whose primary purpose is to enter a brand new buffer, defined mostly by specifying components and each one's concentration.
- `Edit Existing Buffer <buffer_edit> - A panel whose primary purpose is to change non-hydrodynamic characteristics of an already existing buffer.
- `Settings <buffer_settings> - A panel whose primary purpose is to set Database-or-Disk input, the investigator; or to synchronize the local buffer components file from the database.

78.2.1 Enter New Buffer Panel

Using this panel, you can create a new buffer in the current database or on the local disk. Most commonly, you select a set of buffer components in the provided list. As each is selected, you enter its concentration. After providing a description for the new buffer, you can click on the Accept button to upload the buffer to the DB or to local disk and to return to the Select Buffer panel.

As with all panels, a set of tabs allows you to navigate to other panels in order to perform specialized subtasks relating to buffer management. Links to and summaries of the panels are given in the final section of this page.



Enter New Buffer Panel

Buffer New Functions:

Description Enter	A new and unique description for a buffer to create.
(all components list, left-side)	The items listed on the left are descriptions of all available buffer components. Click on one to select it as a component to add to the new buffer. Then enter its concentration in the box above.
(buffer components list, right-side)	The items listed on the right side are the components added so far to the buffer. An item may be removed by double-clicking on it.
Manual unadjusted Density and Viscosity	Check this box to allow entry of Density and Viscosity values that override any values computed for the sum of buffer components or that define the values for a buffer with no components, such as water only.
Density value set by components choice.	It is also editable here (if "Manual…" is checked).
Viscosity value set by components choice.	It is also editable here (if "Manual…" is checked).
pH	pH value for the new buffer. It may be modified here.
Compressibility	The compressibility of the new buffer. It may be modified here.
Cancel	Close the panel and return to the Select Buffer <buffer_select.html>_ panel with no new buffer created.
Accept	Close the panel and return to the Select Buffer <buffer_select.html>_ panel with the new buffer selected.
Enter Spectrum	This button brings up a *Manage Spectrum dialog* to set a Wavelength/Extinction profile.

Note: Please enter the concentration of (component) in (unit): Enter a concentration for the currently selected component. Choose a value within the component range, then hit the keyboard Return key to add the component.

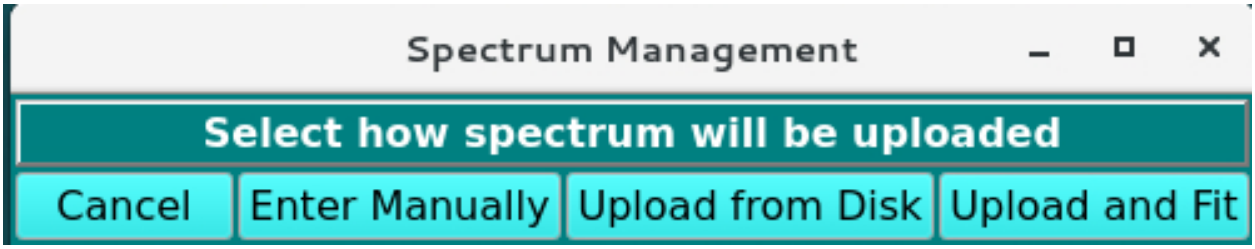
78.2.2 Add New Buffer Steps

Once the new buffer has been sufficiently defined, the **Accept** button becomes enabled. Clicking on it causes the buffer to be added to the DB or the local disk and causes a return to the *Select Buffer* panel with the new buffer selected.

78.2.3 Add Buffer Spectrum

A Spectrum of the buffer can be added to each buffer using the **Enter Spectrum** Button.

- Manually add single wavelength data by clicking **Enter Manually**.
- Upload a new spectrum in CSV, TXT, DAT, or DSP format by clicking **Upload from Disk** and selecting a file from the file manager.
- Open the Spectrum Fitter



Edit Existing Buffer Setting

78.3 Edit Existing Buffer Tab

Index

- *Edit Buffer Panel*
  - *Buffer Edit Functions:*
- *Edit Buffer Spectrum*

Panel Tab Options:

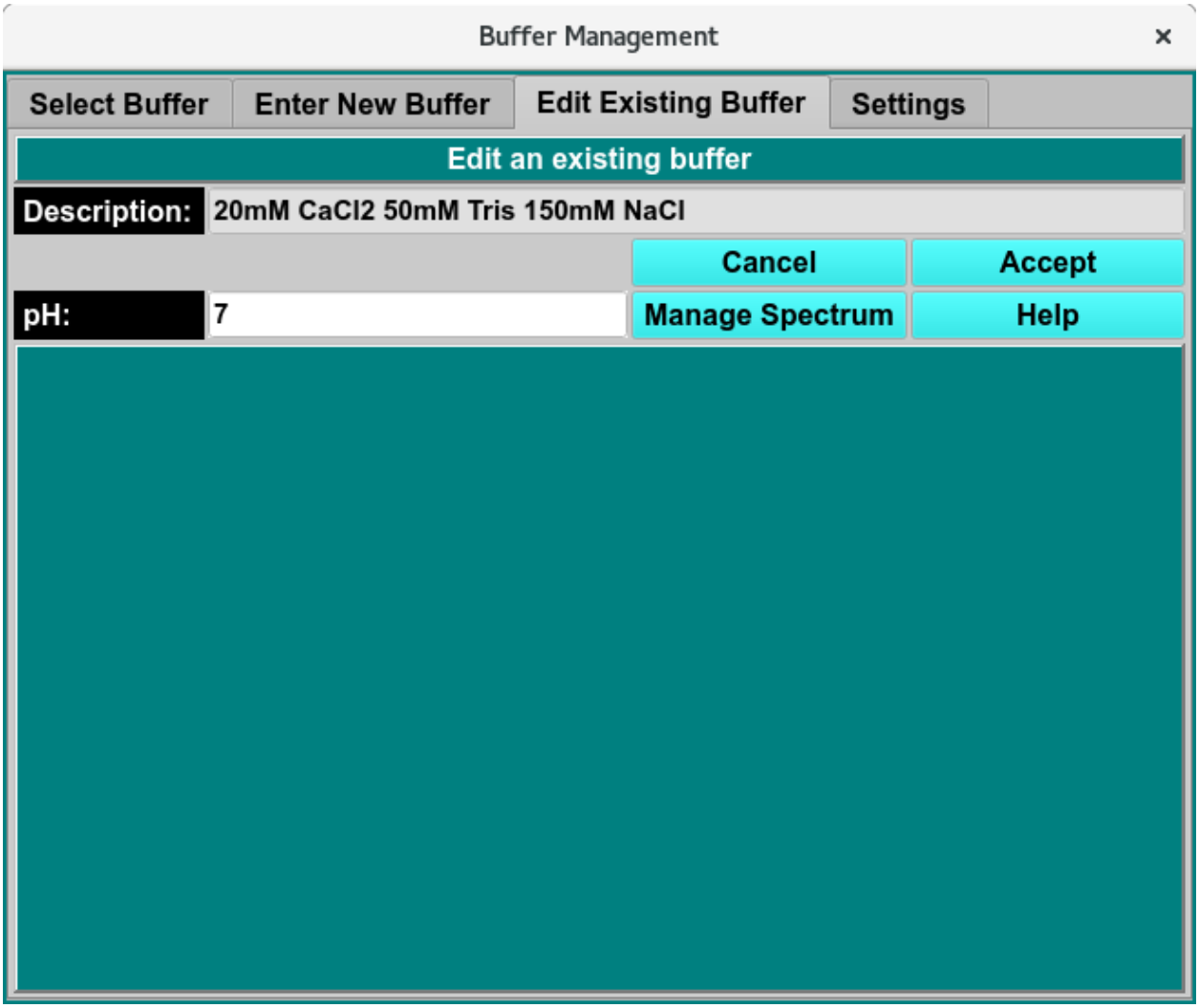
- `Select Buffer <buffer_select> - A panel whose primary purpose is to select a buffer to return to the caller.
- `Enter New Buffer <buffer_new> - A panel whose primary purpose is to enter a brand new buffer, defined mostly by specifying components and each one's concentration.
- *Edit Existing Buffer* - A panel whose primary purpose is to change non-hydrodynamic characteristics of an already existing buffer.
- :ref:`Settings <buffer_settings> - A panel whose primary purpose is to set Database-or-Disk input, the investigator; or to synchronize the local buffer components file from the database.

78.3.1 Edit Buffer Panel

Using this panel, you can change non-hydrodynamic characteristics of a buffer previously selected in the Select Buffer panel.

Once any desired modifications have been made, clicking on the Accept button will return you to the Select Buffer panel with a modified buffer.

As with all panels, a set of tabs allows you to navigate to other panels in order to perform specialized subtasks relating to buffer management. Links to and summaries of the panels are given in the final section of this page.



Edit Buffer Window

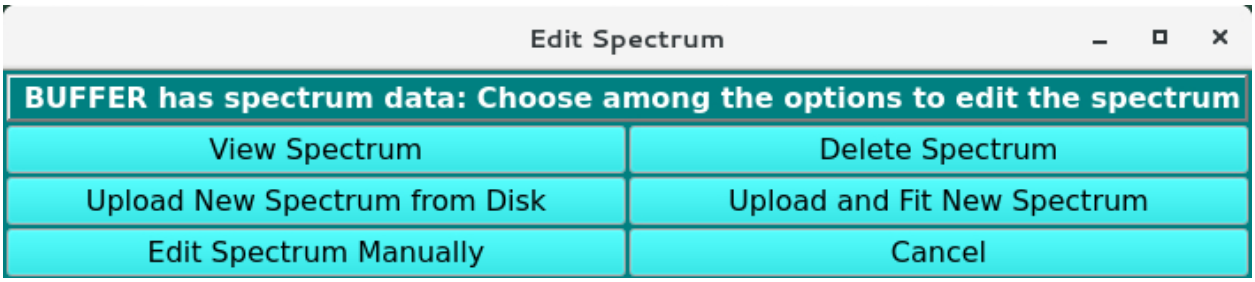
Buffer Edit Functions:

Description	The description of the currently selected buffer for which modification are to be made.
Cancel	Click this button to abandon any changes and return to the :doc:`Select Buffer <buffer_select> panel with the original unchanged buffer.
Accept	Click this button to save any changes and return to the :doc:`Select Buffer <buffer_select> panel with a buffer as modified in this panel.
pH:	Manually change pH to be associated with the current buffer.
Manage Spectrum	This button brings up a *Manage Spectrum* dialog to set a Wavelength / Extinction profile for the buffer.
Help	Show this documentation.

Note: **Important Note** No other parameters than those indicated in this panel may be modified for an existing buffer. This is because existing buffers may be already associated with solutions and runs. Changing hydrodynamic characteristics already associated with models will invalidate those models.

To create a buffer with most characteristics similar to an existing one, the users should note those characteristics; then name and enter a new buffer in the Enter New Buffer panel, with selected parameters changed.

78.3.2 Edit Buffer Spectrum



Edit Buffer - Manage Spectrum Menu

View Spectrum

78.4 Buffer Settings Tab

Index

- *Settings Panel*
  - *Buffer Set. Functions:*

Panel Tab Options:

- :doc:`Select Buffer <select_buffer> - A panel whose primary purpose is to select a buffer to return to the caller.
- :doc:`Enter New Buffer <buffer_new> - A panel whose primary purpose is to enter a brand new buffer, defined mostly by specifying components and each one's concentration.
- Edit Existing Buffer - A panel whose primary purpose is to change non-hydrodynamic characteristics of an already existing buffer.
- *Settings* - A panel whose primary purpose is to set Database-or-Disk input, the investigator; or to synchronize the local buffer components file from the database.

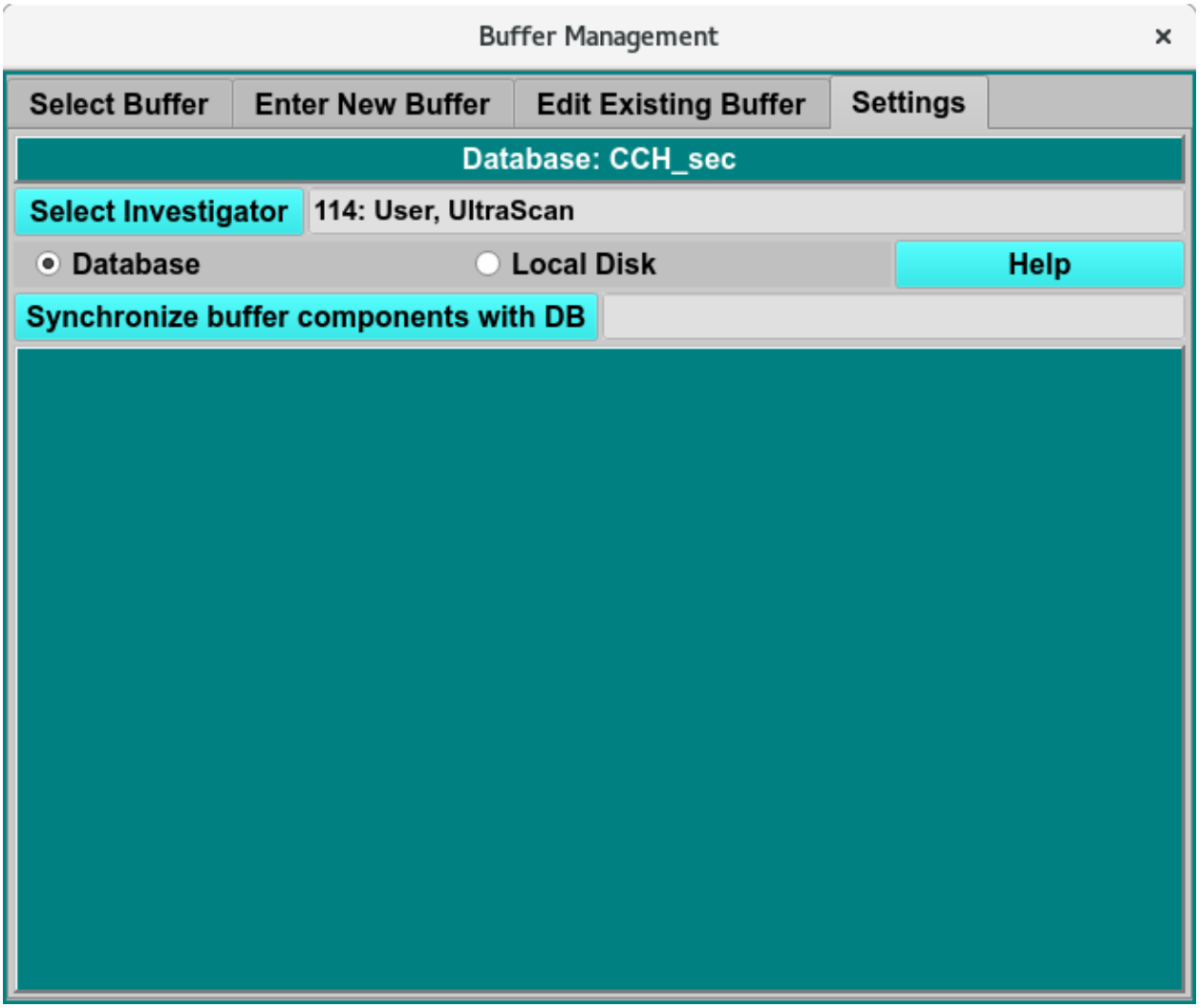
78.4.1 Settings Panel

Using this window, you can manage settings that affect all other buffer panels. The essential environment for buffer management that you can change here involves the following.

- The investigator for database access.
- The choice of Database or Local Disk for input/output.
- A local buffer components file synchronized with the master list in the database.

As with all panels, a set of tabs allows you to navigate to other panels in order to perform specialized subtasks relating to buffer management. Links to and summaries of the panels are given in the final section of this page.

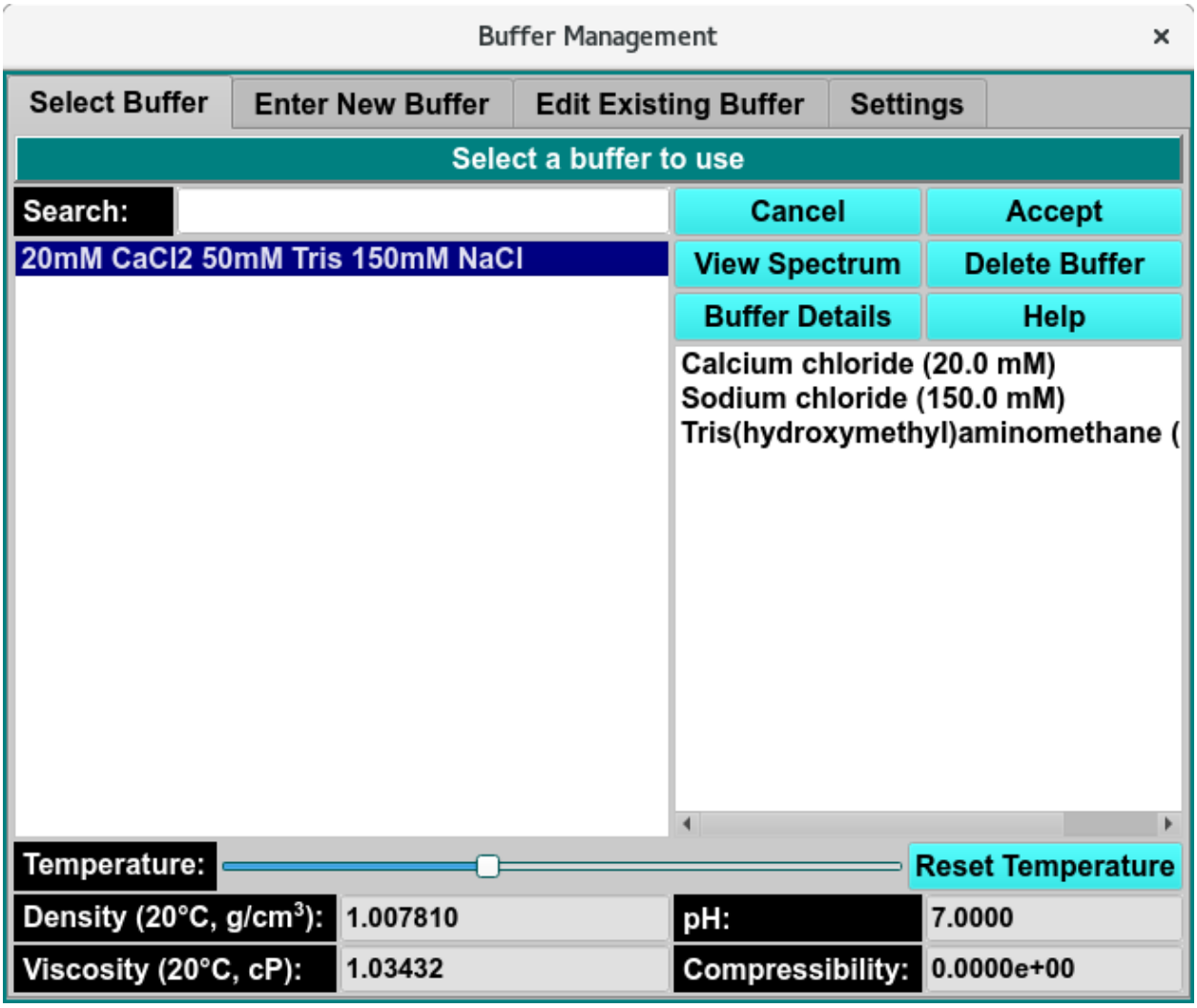
New Analyte



Buffer Settings Window

Buffer Set. Functions:

Select Investigator	This button brings up a window that allows selecting the current investigator for limiting buffer descriptions.
Database	Check to select read or write of the buffer definitions to or from the database.
Local Disk	Check to select read or write of the buffer definition to or from a local disk file.
Synchronize buffer components with DB	Click to obtain a local file listing of all buffer components downloaded from the database. This action is rarely needed, most commonly for new users who have no buffer components file whatsoever.
(status box)	Status of any action, such as synchronizing buffer components.
Help	Show this documentation.



Select buffer A panel whose primary purpose is to select a buffer to return to the caller.

Enter New Buffer A panel whose primary purpose is to enter a brand new buffer, defined mostly by specifying components and each one's concentration.

Edit Existing Buffer A panel whose primary purpose is to change non-hydrodynamic characteristics of an already existing buffer.

Settings A panel whose primary purpose is to set Database-or-Disk input, the investigator; or to synchronize the local buffer components file from the database.

CHAPTER

SEVENTYNINE

MANAGE ANALYTE INFORMATION

79.1 Select Analyte Tab

Index

- *Select Analyte Panel*
  - *Analyte Sel. Functions:*

Panel Tab Options:

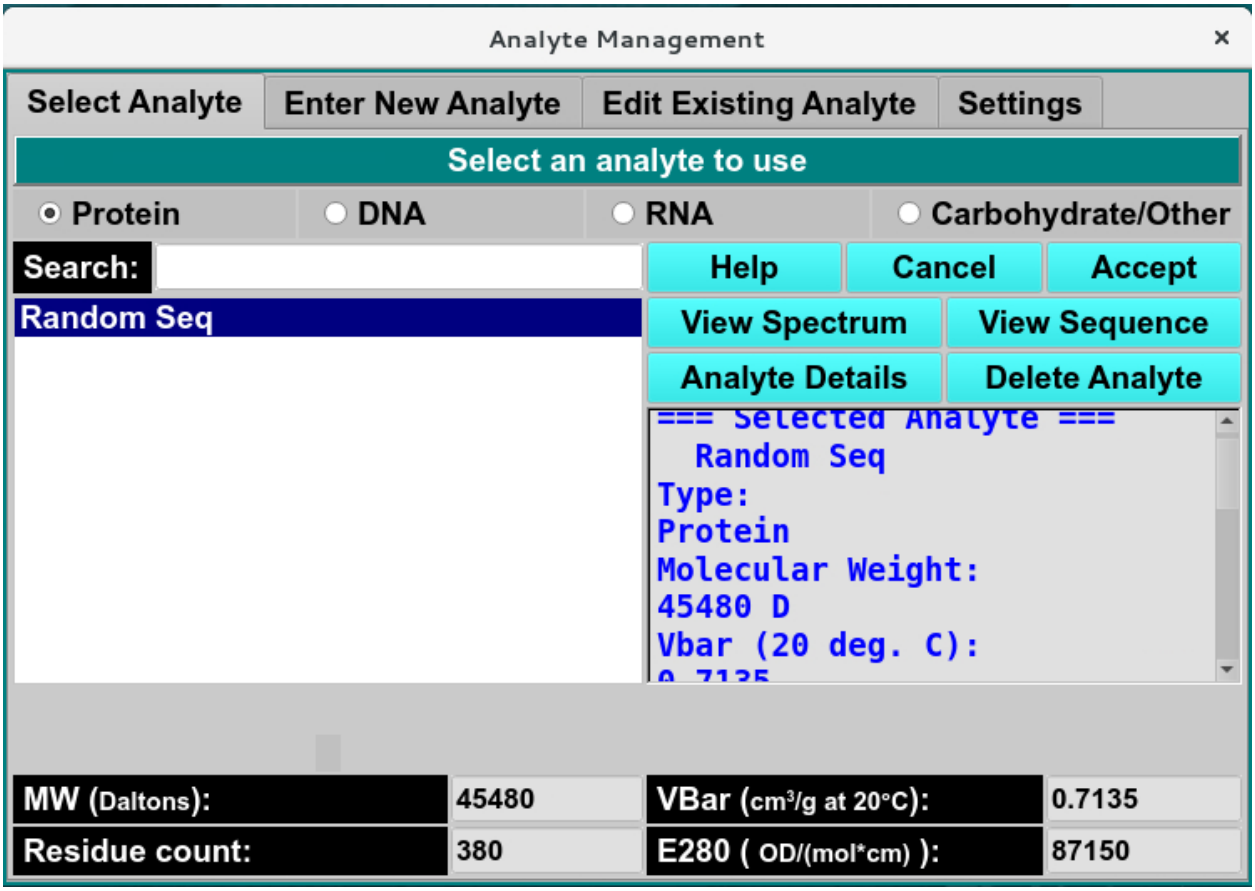
- *Select Analyte* - A panel whose primary purpose is to select an analyte to return to the caller.
- *Enter New Analyte* - A panel whose primary purpose is to enter a brand new analyte, defined by specifying sequence for protein and nucleic components or manually enter the molecular weight (Da), Vbar and other required parameters of proteins, RNA, DNA, Carbohydrate/other analytes.
- *Edit Existing Analyte* - A panel whose primary purpose is to add spectral characteristics of an already existing analyte.
- *Analyte Settings* - A panel whose primary purpose is to set Database-or-Disk input, the investigator; or to synchronize the local analyte components file from the database.

79.1.1 Select Analyte Panel

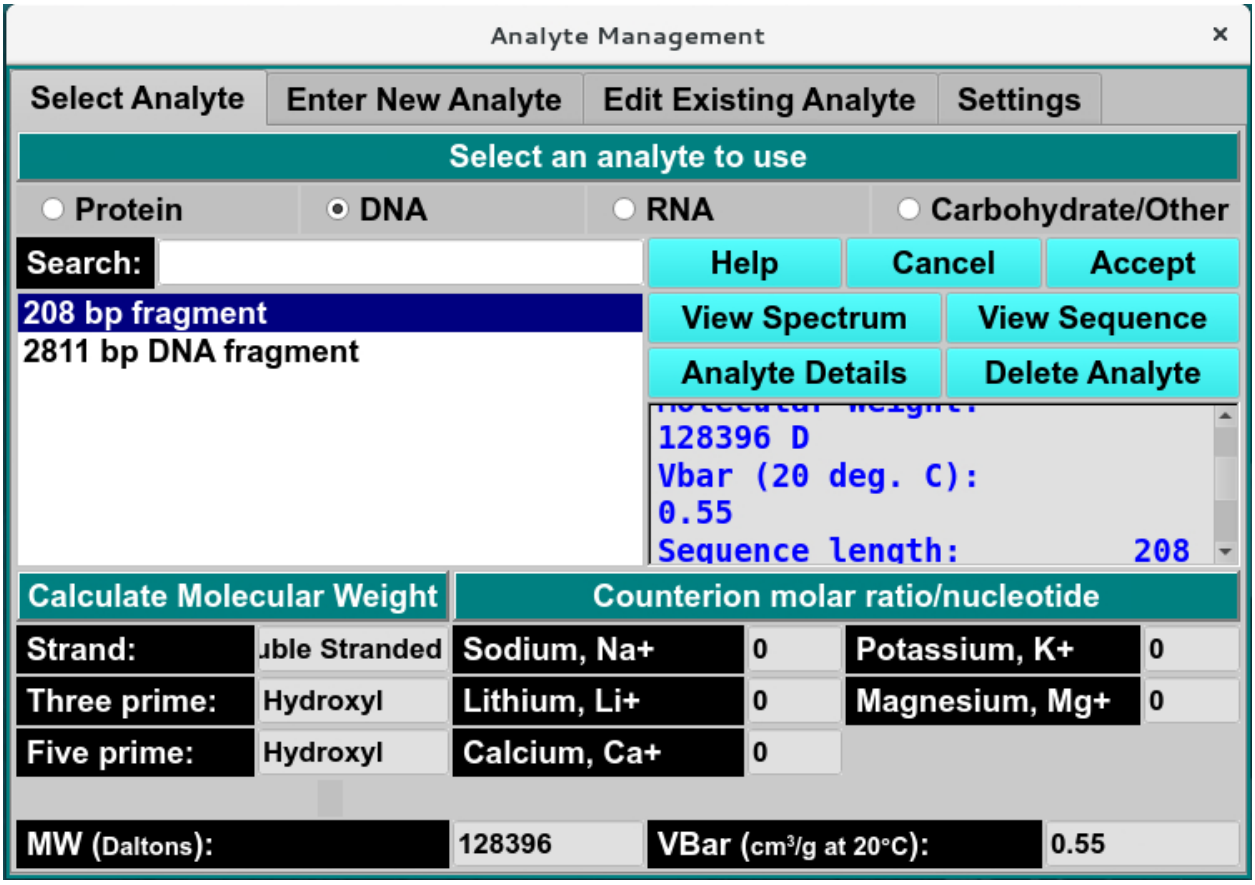
The Select Analyte tab provides an interface for managing analytes saved in the selected database pr local disk. Users can select from existing analytes by clicking on the analyte and clicking **Accept** and returning it to the caller for analytes in the protein, DNA, RNA, and carbohydrates/other lists. This panel can also be used to view the analyte details such as molecular weight, partial specific volume (Vbar), and, for proteins, the extinction coefficient at 280 nm.

This tab also allows users to view spectral data or analyte properties and to delete analytes that are **not used** in any model.

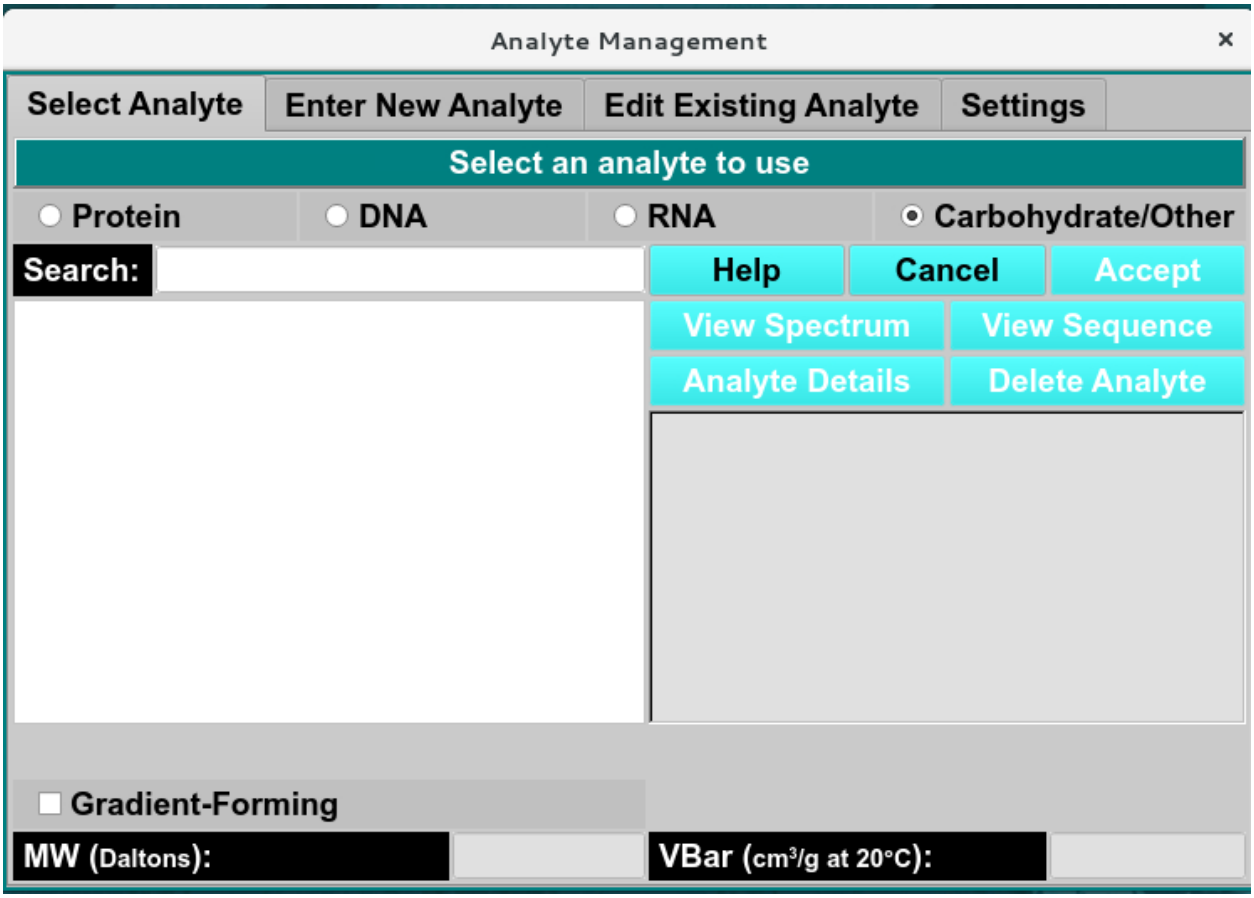
Note: Analytes associated with models can not be deleted.



Select Analyte Window - Protein View



Select Analyte Window - Nucliec acids View



Select Analyte Window - Carbohydrates/Other View

Analyte Sel. Functions:

Protein

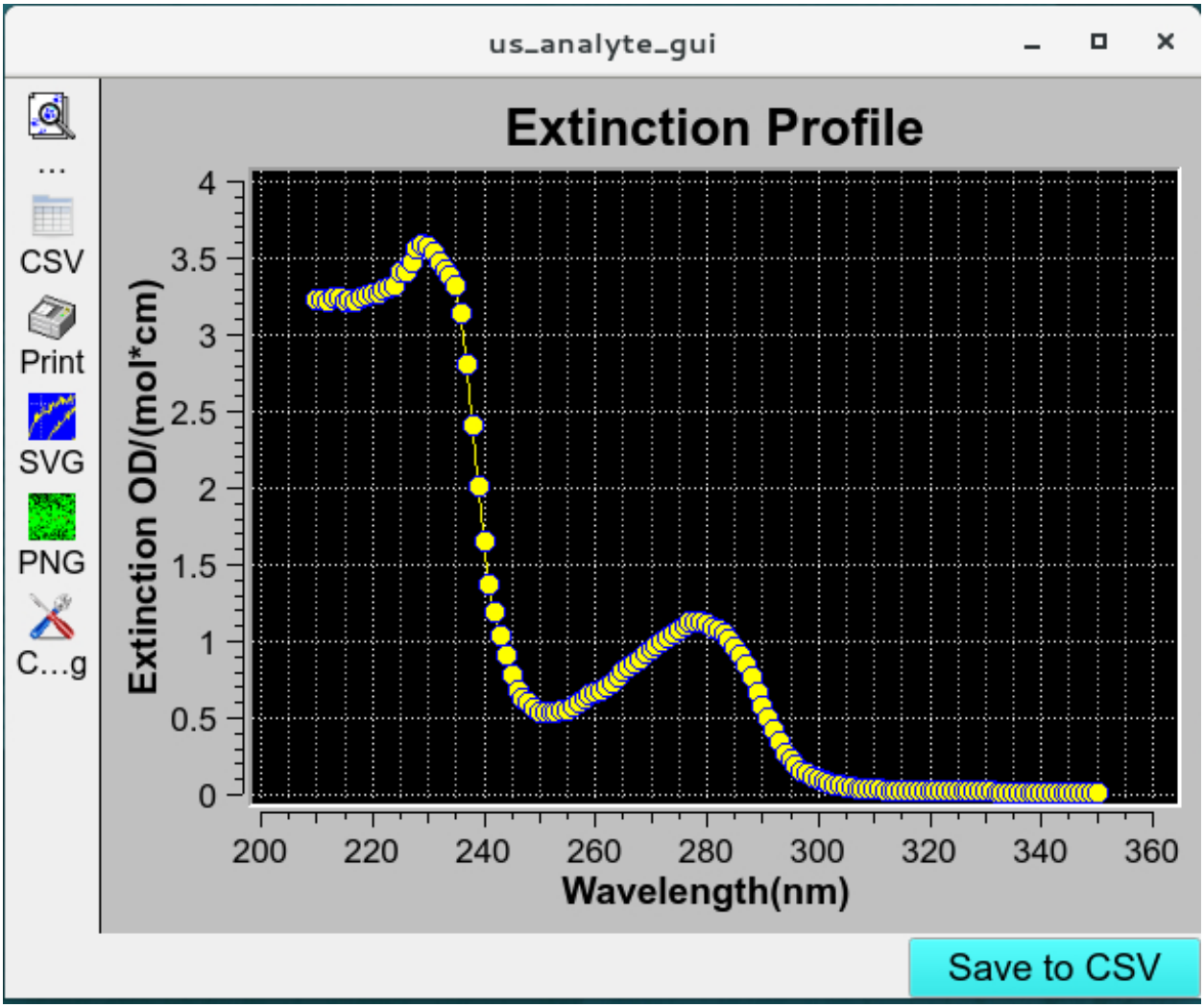
(analyte list, left-side)	The items listed on the left side are all the descriptions (names) of analytes available or the descriptions as limited by the **Search** text. A single click on a list item selects a analytes.
(analyte details, right-side)	The view on the right side is the details of the currently selected analyte.
Search:	Enter a partial description to search for a specific analyte. Clear the text to expand to the full list.
MW (Dalton):	The molecular weight in Dalton of the currently selected analyte.
Vbar (cm^3/g at 20°C):	The Vbar of the currently selected analyte.
Residue Count:	The protein residue count of the currently selected analyte.
E280 (OD/(mol*cm)):	The extinction coefficient of the currently selected analyte at 280 nm in optical density/(mole*cm).
Help	Show this documentation.
Cancel	Close the analyte dialog and return to the caller with no change in analyte selection.
Accept	Close the analyte dialog and return to the caller with the current analyte selection.
View Spectrum	Open a *Manage Spectrum* dialog to show the Wavelength/Extinction profile associated with the currently selected analyte.
View Sequence	Open a *Manage Sequence* dialog to show the available sequence profile associated with the currently selected analyte.
Analyte Details	Open a *Analyte Information* dialog to show the detailed information about the currently selected analyte.
Delete Analyte	Permanently delete the currently selected analyte. A confirmation dialog will appear to allow you to cancel the delete request or to proceed with it. Upon confirmation, the removal will proceed regardless of any further action ("Cancel" or "Accept").

DNA/RNA

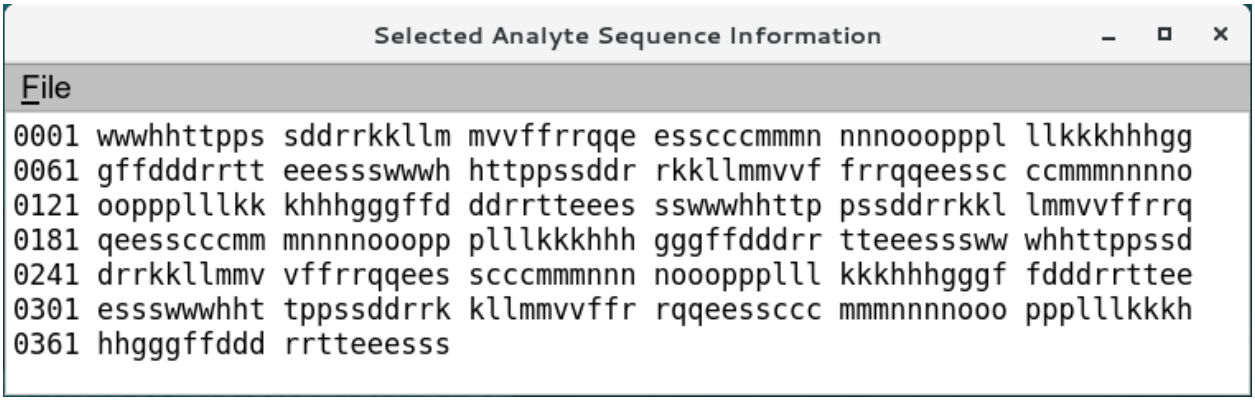
Strand	Show the Complementary or Double stranded.
Three prime	Show the three prime Hydroxyl or phosphate ends.
Five prime	Show the five prime Hydroxyl or phosphate ends.
Counterion molar ratio/nucleotide	Show the Counterion molar ratio of Sodium (Na+), Lithium (Li+), Calcium (Ca+), Potassium (K+), or Magnesium (Mg+)

Carbohydrate/Other

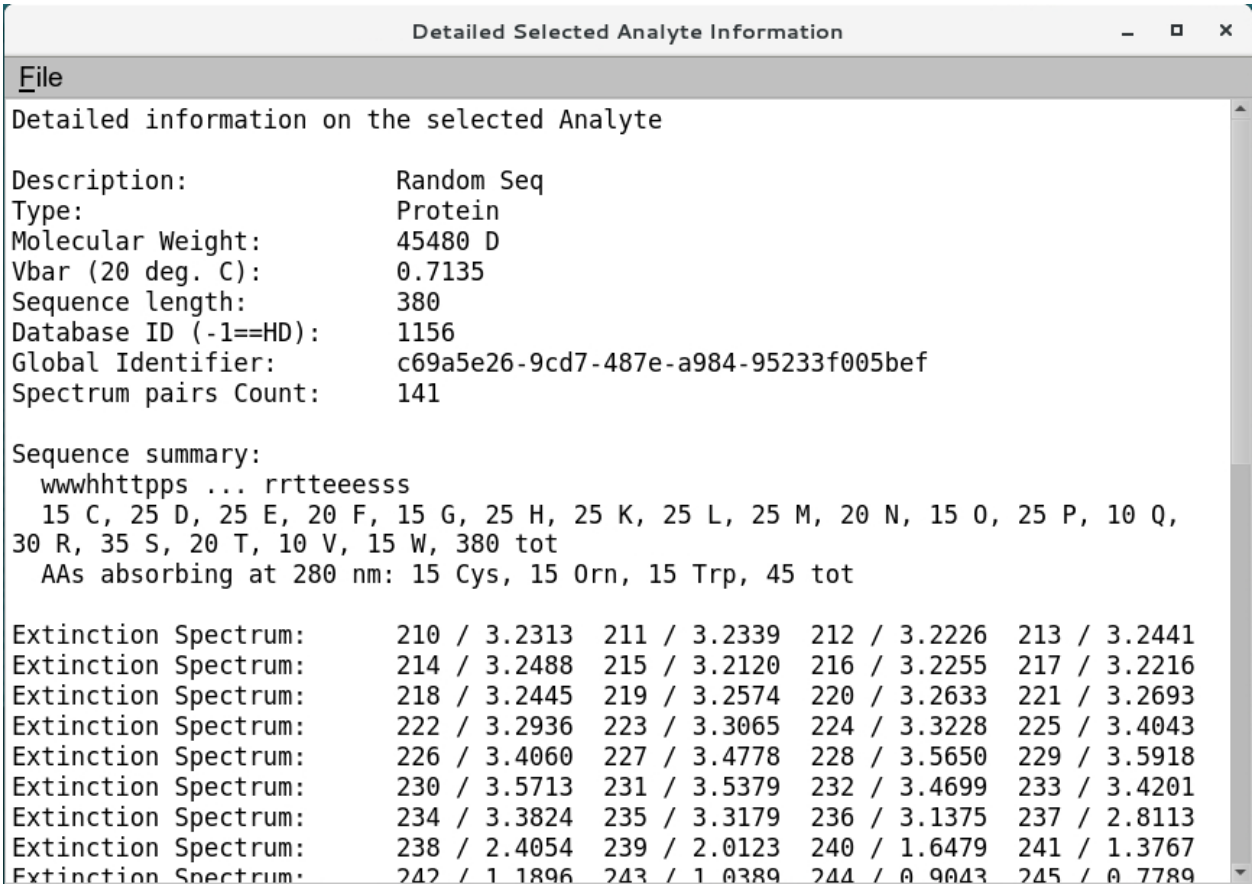
Gradient-Forming	Select analyte as Gradient-Forming.



View Analyte Spectrum



View Analyte Sequence file



View Analyte Detailed Information

79.2 Enter New Analyte Tab

Index

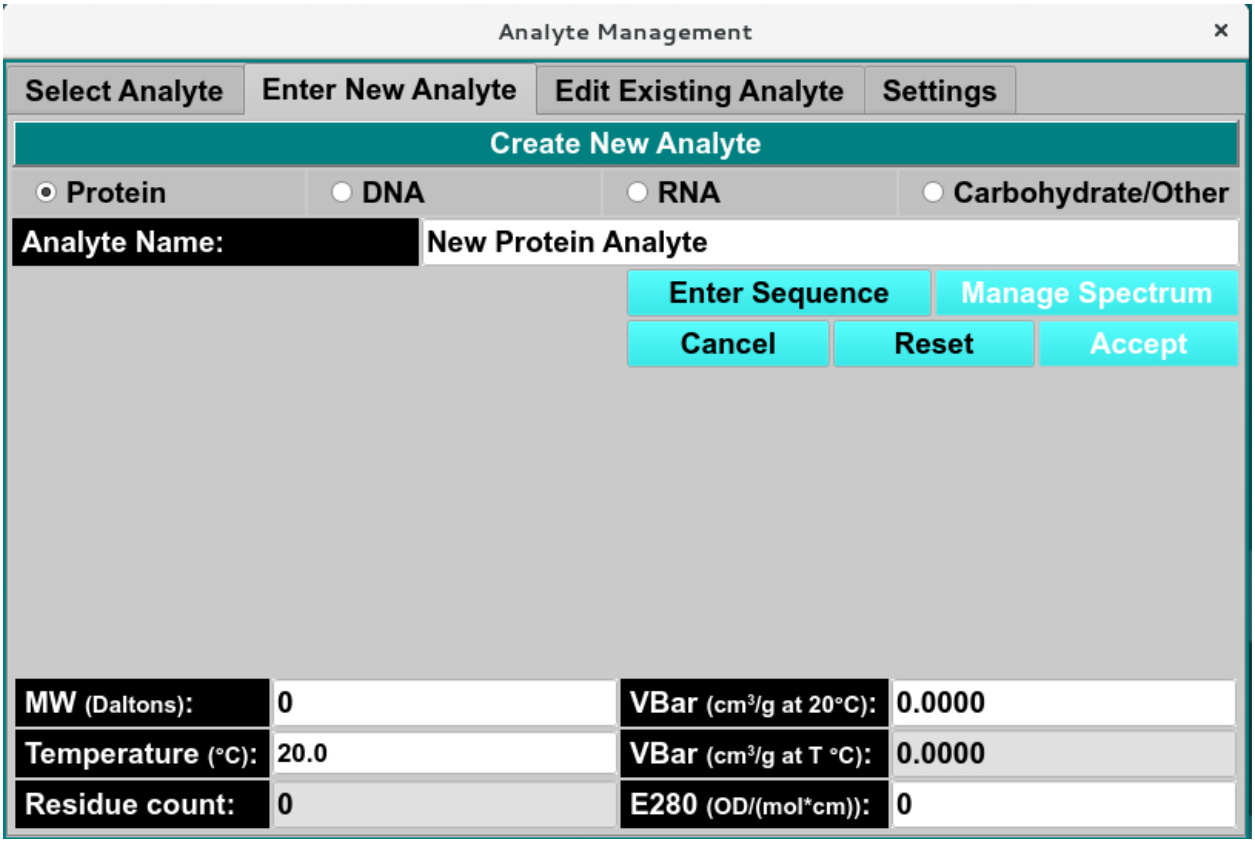
- *Enter New Analyte Panel*
- *Add Analyte Spectrum*

Panel Tab Options:

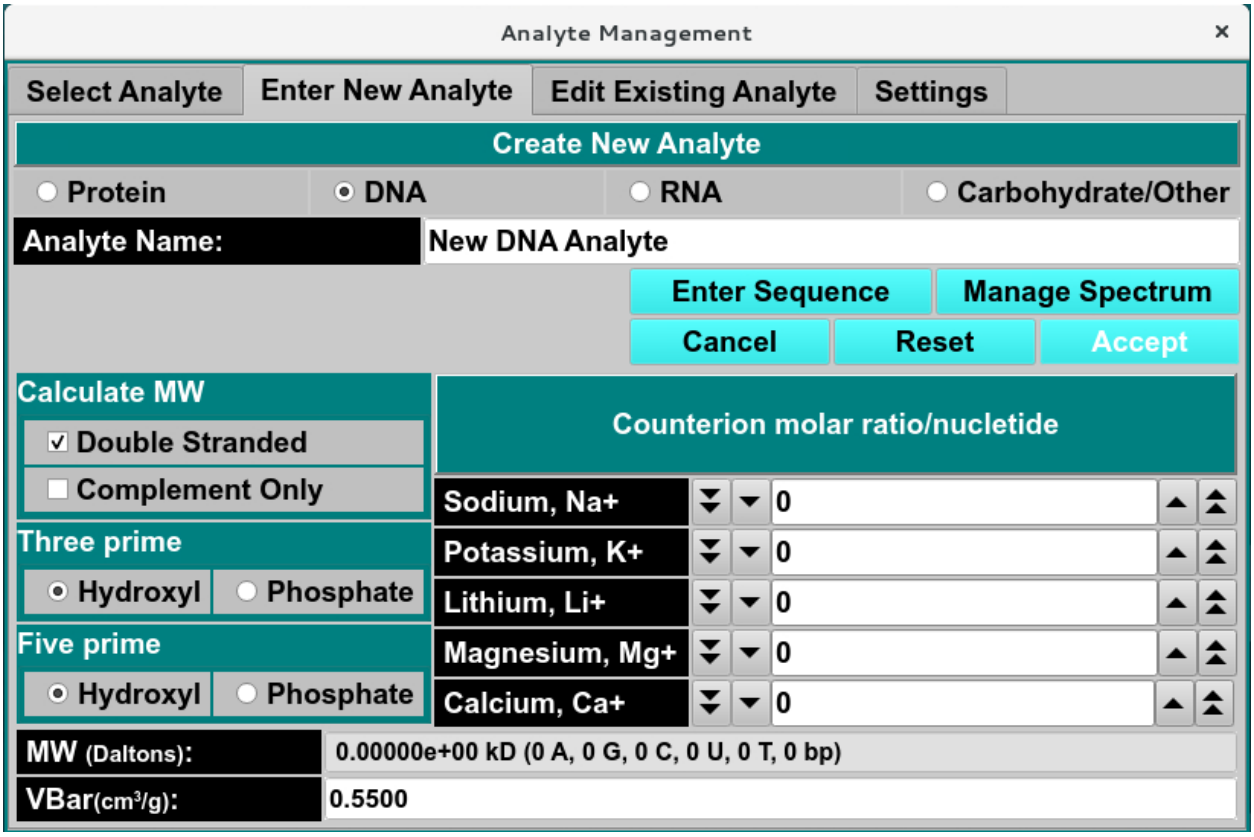
- :doc:`Select Analyte <analyte_select> - A panel whose primary purpose is to select an analyte to return to the caller.
- *Enter New Analyte* - A panel whose primary purpose is to enter a brand new analyte, defined by specifying sequence for protein and nucleic components or manually enter the molecular weight (Da), Vbar and other required parameters of proteins, RNA, DNA, Carbohydrate/other analytes.
- :doc:`Edit Existing Analyte <analyte_edit> - A panel whose primary purpose is to add spectral characteristics of an already existing analyte.
- :doc:`Analyte Settings <analyte_settings> - A panel whose primary purpose is to set Database-or-Disk input, the investigator; or to synchronize the local analyte components file from the database.

79.2.1 Enter New Analyte Panel

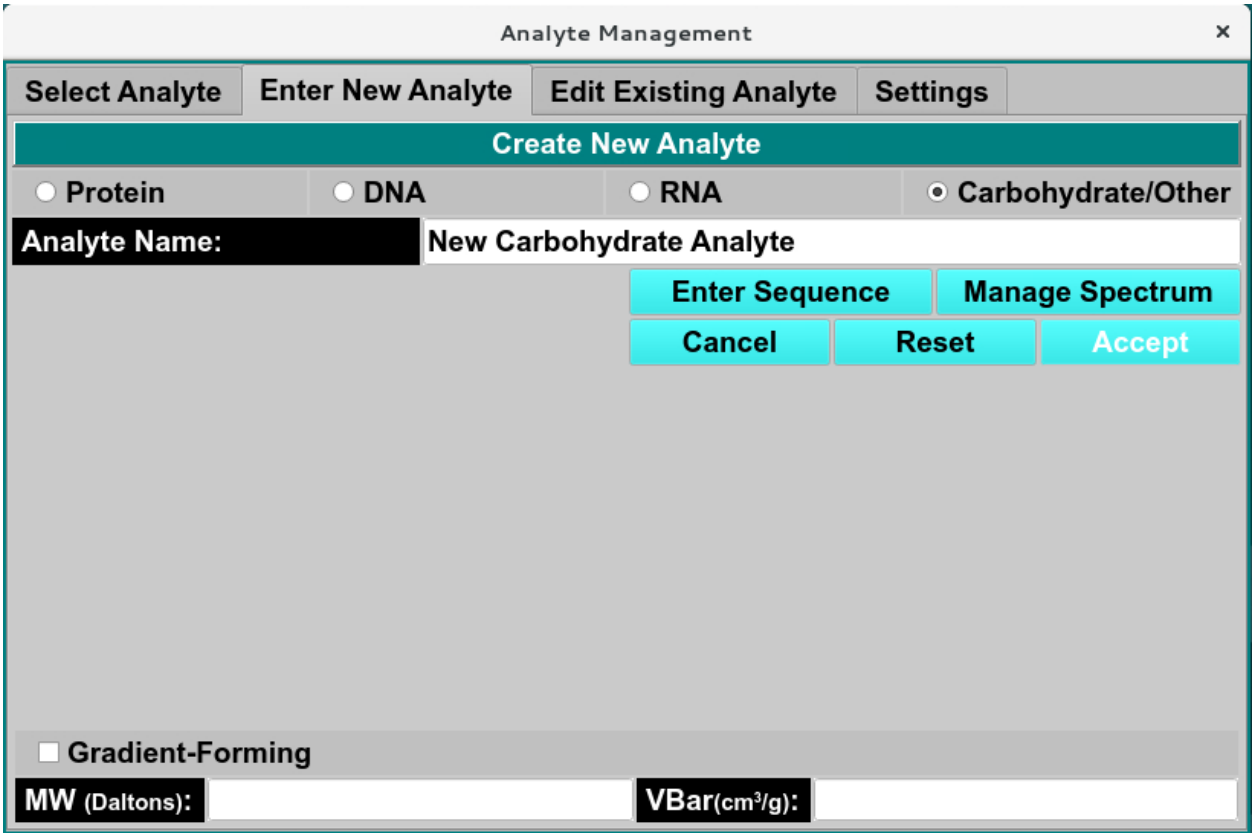
Using this panel, the user can create a new analyte in the current database or in the local disk. A new analyte can be entered by clicking the *Enter Sequence* and pasting the sequence. The MW, Vbar and E280 (for Proteins) will be automatically updated once the sequence is saved.



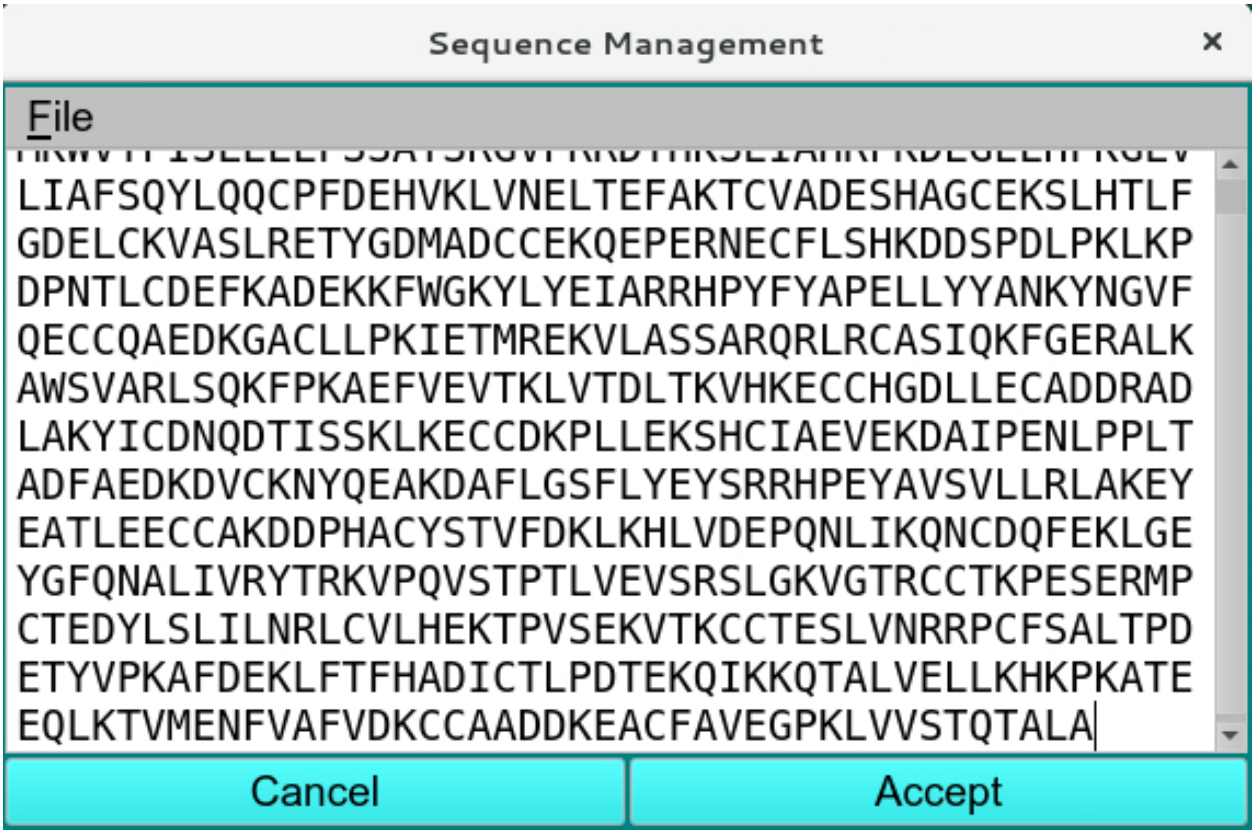
Add Analyte Window - Protein View



Add Analyte Window - DNA/RNA View



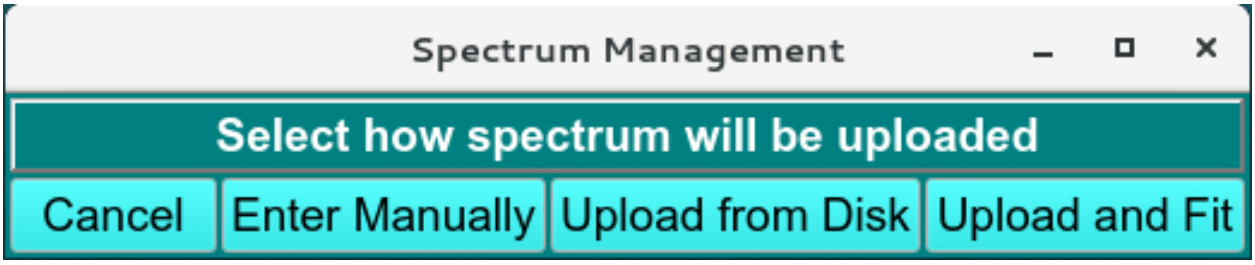
Add Analyte Window - Carbohydrate/Other View



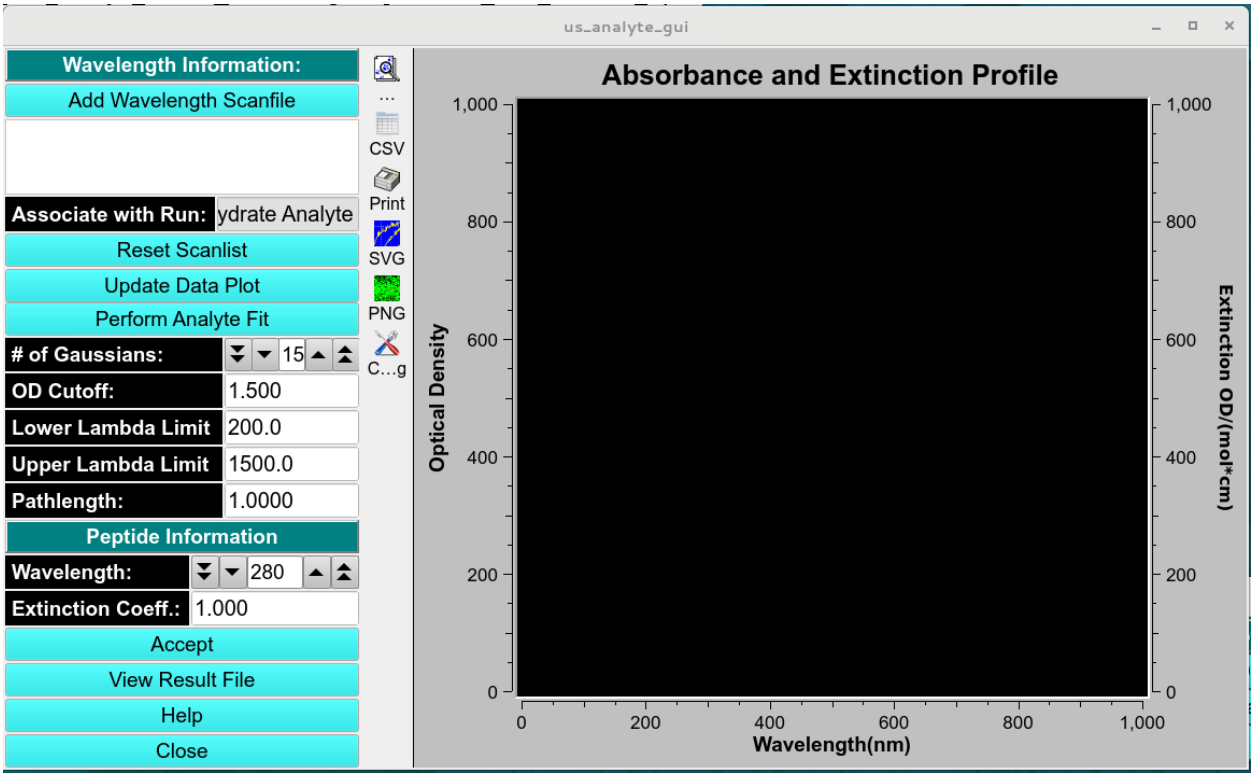
Add Analyte Sequence Window

79.2.2 Add Analyte Spectrum

The panel can also add a spectrum of the analyte by clicking and opening the *Manage Spectrum menu* and manually entering the extinction coefficient at a specified wavelength (**Enter Manually**), by loading a csv file from local disk (**Upload from disk**), or by loading a dilution spectrum series and fitting to determine the extinction coefficient in the *Analyte Spectrum Fitter* module.



Select Analyte Window - Protein View



Select Analyte Window - Protein View

>`========== Edit Existing Analyte Tab ==========

Index

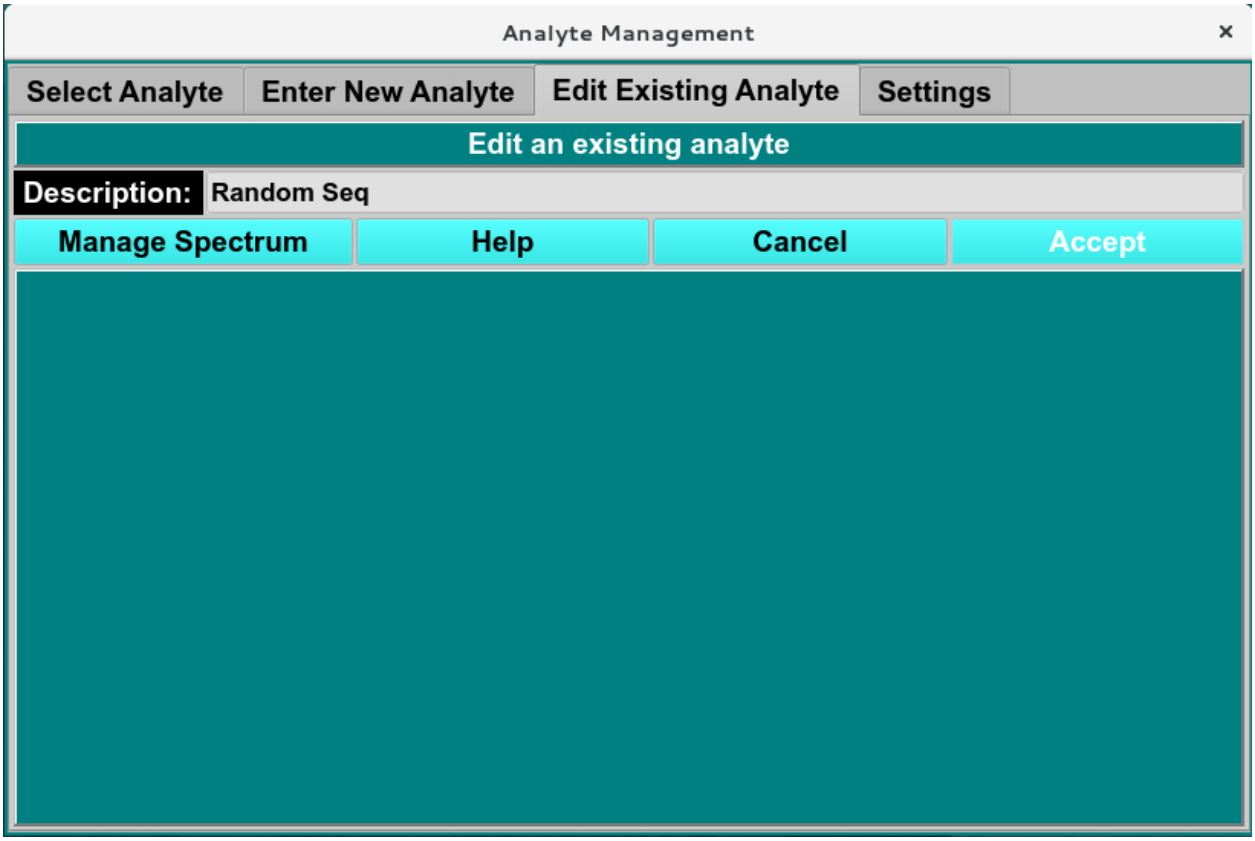
- *Edit Existing Analyte Panel*
- *Edit Analyte Spectrum*

Panel Tab Options:

- Select Analyte - A panel whose primary purpose is to select an analyte to return to the caller.
- Enter New Analyte - A panel whose primary purpose is to enter a brand new analyte, defined by specifying sequence for protein and nucleic components or manually enter the molecular weight (Da), Vbar and other required parameters of proteins, RNA, DNA, Carbohydrate/other analytes.
- *Edit Existing Analyte* - A panel whose primary purpose is to add spectral characteristics of an already existing analyte.
- Analyte Settings - A panel whose primary purpose is to set Database-or-Disk input, the investigator; or to synchronize the local analyte components file from the database.

79.3 Edit Existing Analyte Panel

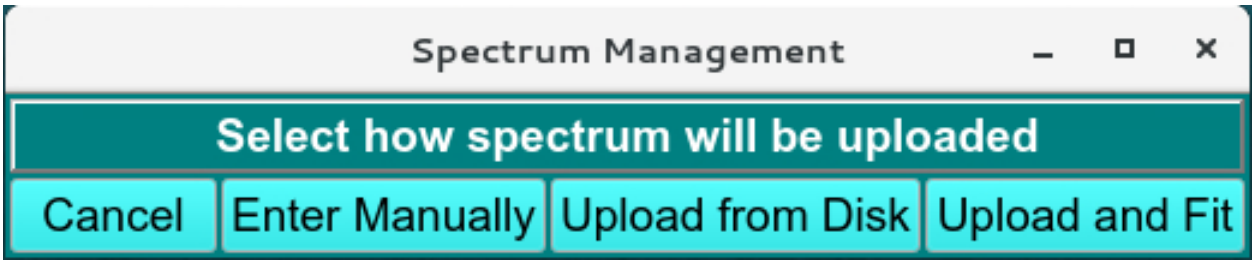
This panel allows the user to modify non-hydrodynamic characteristics.



Edit Existing Analyte Window

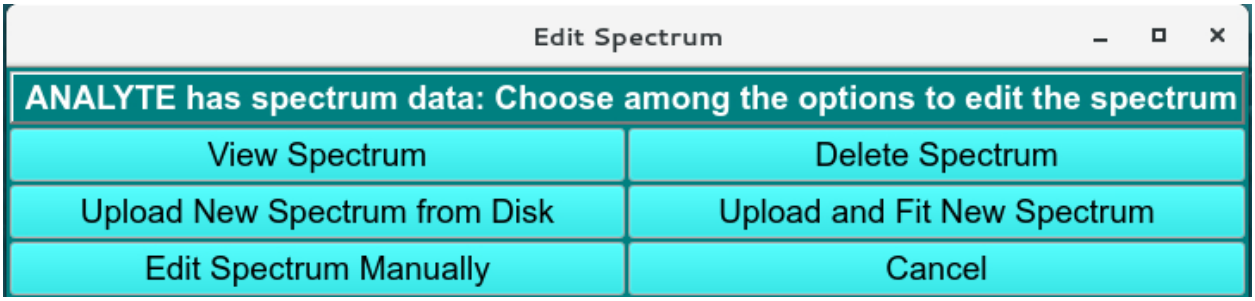
79.4 Edit Analyte Spectrum

The **Manage Spectrum** menu window allows the user to upload from local disk, upload and fit the spectrum using **Analyte Spectrum Fitter** window to determine the extinction coefficient at a wavelength range or add manually absorbance at specific wavelength.

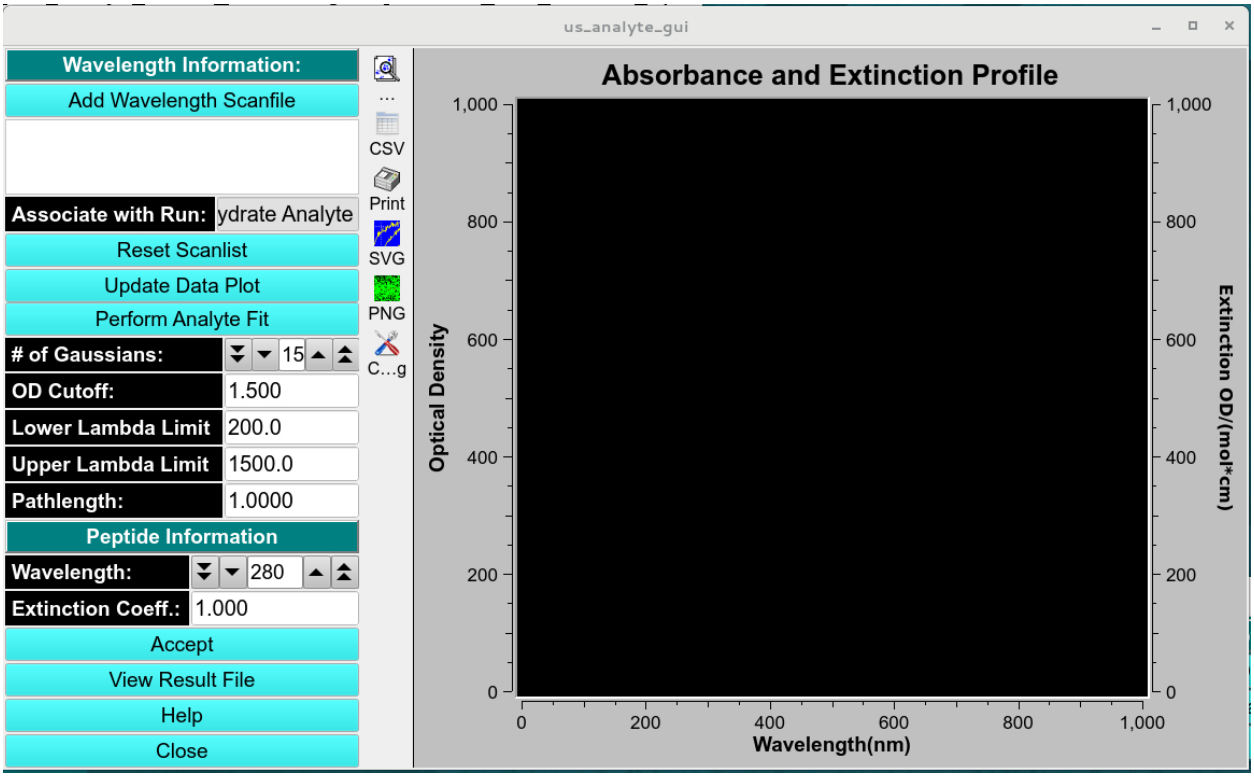


Edit Existing Spectrum Menu

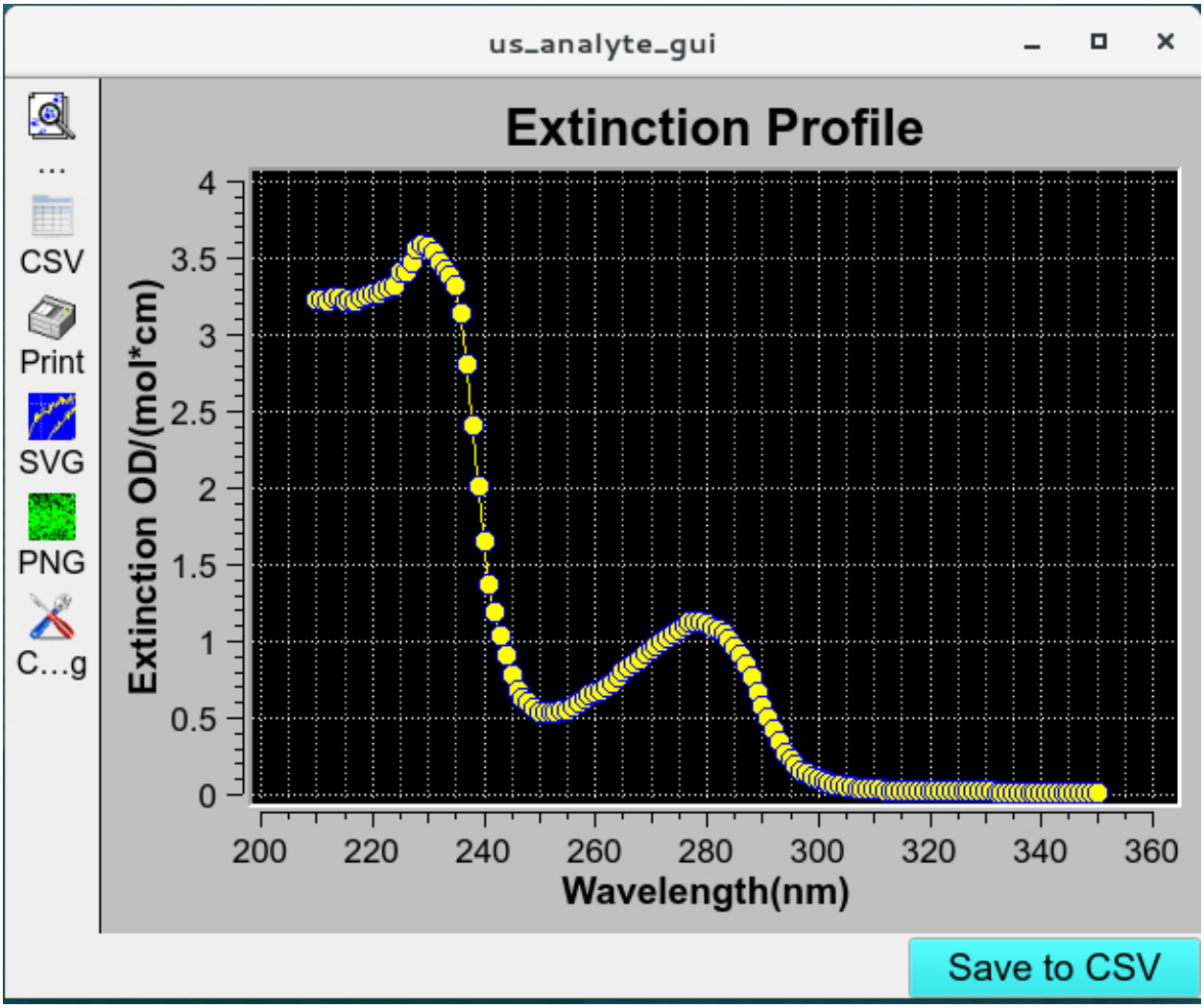
Once an analyte has an associated spectrum, the **Manage Spectrum** menu window updates to include view Spectrum and delete spectrum buttons.



Analyte Spectrum Fitter



Analyte Spectrum Fitter



Analyte Spectrum

79.5 Analyte Settings Tab

Index

- *Analyte Settings Panel*

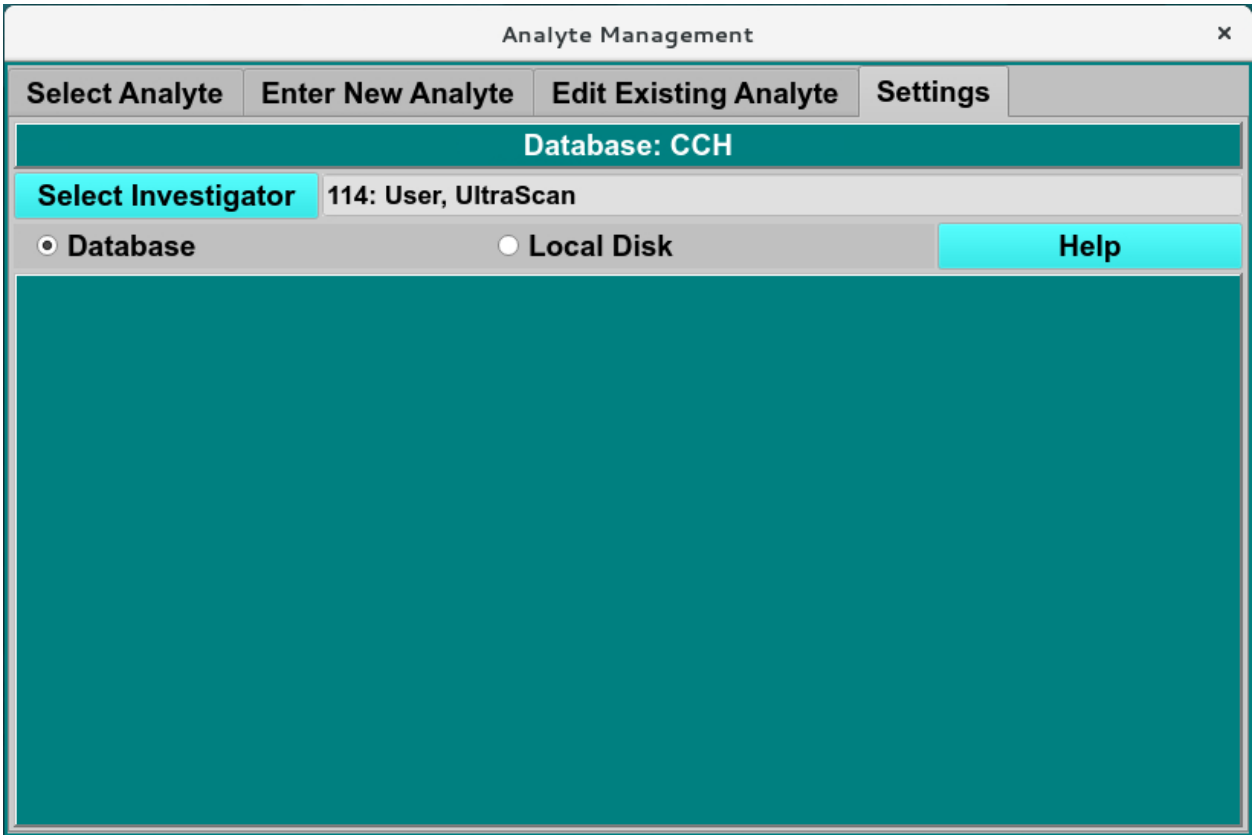
Panel Tab Options:

- `Select Analyte <analyte_select> - A panel whose primary purpose is to select an analyte to return to the caller.
- `Enter New Analyte <analyte_new> - A panel whose primary purpose is to enter a brand new analyte, defined by specifying sequence for protein and nucliec components or manually enter the molecular weight (Da), Vbar and other required parameters of proteins, RNA, DNA, Carbohydrate/other analytes.

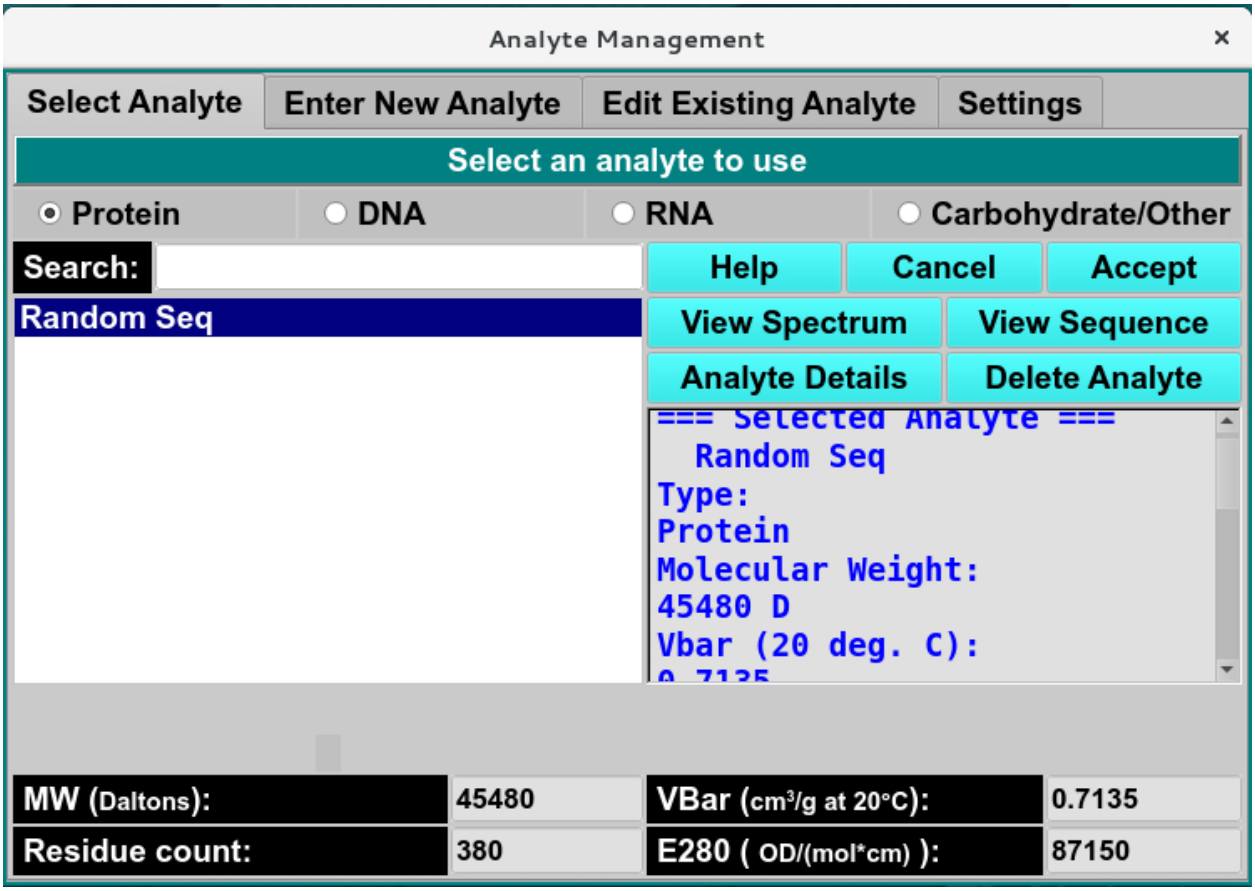
- `Edit Existing Analyte <analyte_edit> - A panel whose primary purpose is to add spectral characteristics of an already existing analyte.
- *Analyte Settings* - A panel whose primary purpose is to set Database-or-Disk input, the investigator; or to synchronize the local analyte components file from the database.

79.5.1 Analyte Settings Panel

The Analyte Settings panel allows user to select the investigator, and set the data input from Database or local disk.



Analyte Specrum



Select Analyte: A panel whose primary purpose is to select a analyte to return to the caller.

Enter New Analyte: A panel whose primary purpose is to enter a brand new analyte, defined mostly by entering the sequence of proteins and nucleic acids and each one's concentration.

Edit Existing Analyte: A panel whose primary purpose is to change non-hydrodynamic characteristics of an already existing analyte.

Settings: A panel whose primary purpose is to set Database-or-Disk input, the investigator; or to synchronize the local analyte components file from the database.

CHAPTER

EIGHTY

MANAGE SOLUTION INFORMATION

This window allows the users to specify the analyte(s) measured and the buffer condition. Once created, the solution module allows the user to manage the solution information by creating new solutions, editing existing entries, and synchronizing solution data between your local system and the user database.

In each panel, tabs are visible at the top of the window to enable the user to move to another panel, to perform specific solution subtasks.

80.1 Select Solution Tab

Index

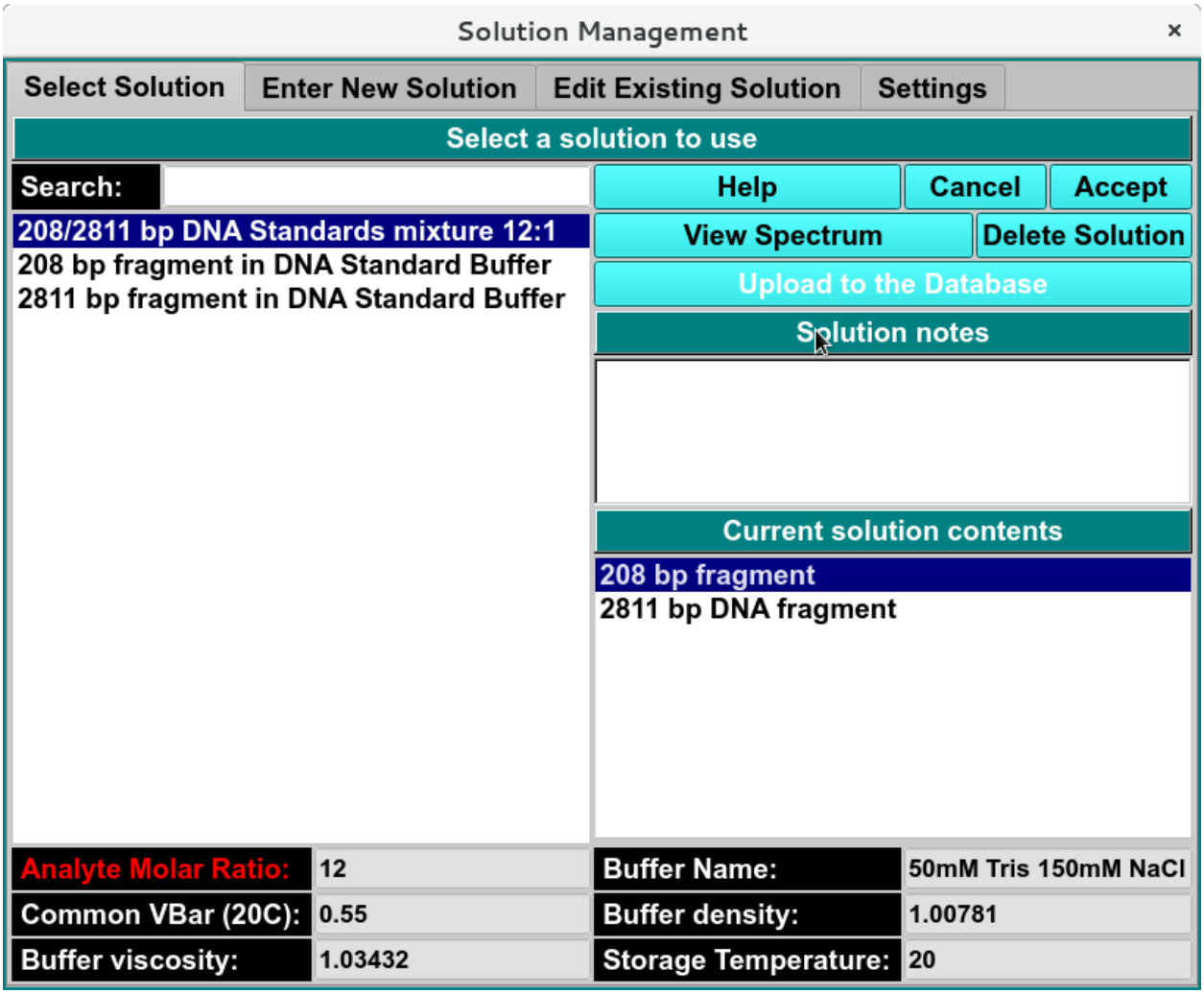
- *Select Solution Panel:*
- *Select Solution Functions:*

Panel Tab Options:

- *Select Solution* - A panel whose primary purpose is to select a Solution to return to the caller.
- *Enter New Solution* - A panel whose primary purpose is to enter a brand new Solution, defined mostly by specifying components and each one's concentration.
- *Edit Existing Solution* - A panel whose primary purpose is to change non-hydrodynamic characteristics of an already existing Solution.
- *Settings* - A panel whose primary purpose is to set Database-or-Disk input and to select the investigator.

80.1.1 Select Solution Panel:

Using this panel, you can select a solution in the current database or on the local disk. Most commonly, you select a solution in the provided list and click the **Accept** button to return it to the caller.



Select Solution Window

Note: local solutions can be uploaded by clicking **upload to the Database** button and view solution notes.

80.1.2 Select Solution Functions:

Search	use to search by enter all or portion of the title.
Help	Access this help documentation for the Solution Management module
Cancel	Close window and return to previous page.
Accept	Return to previous page with selected solution.
Delete Solution	Delete selected solution.
Solution Notes	display notes added to solution
Current Solution Contents	Describe the analyte component(s) of solution.
Analyte Molar Ratio	Once a solution has been selected and a solution contents item (analyte) has been chosen, use this to enter the comparative amount of that analyte that is in the solution.
Buffer Name:	selected buffer name.
Common VBar(20C):	The common vbar at a temperature of 20 degrees C.
Buffer Density	The density of the currently selected buffer.
Buffer viscosity:	The viscosity of the currently selected buffer.
Storage Temperature:	The average temperature of all the runs.

80.2 Enter New Solution Tab

Index

- *Enter New Solution Panel*
- *New Solution Functions:*
- *Add New Solution Steps:*

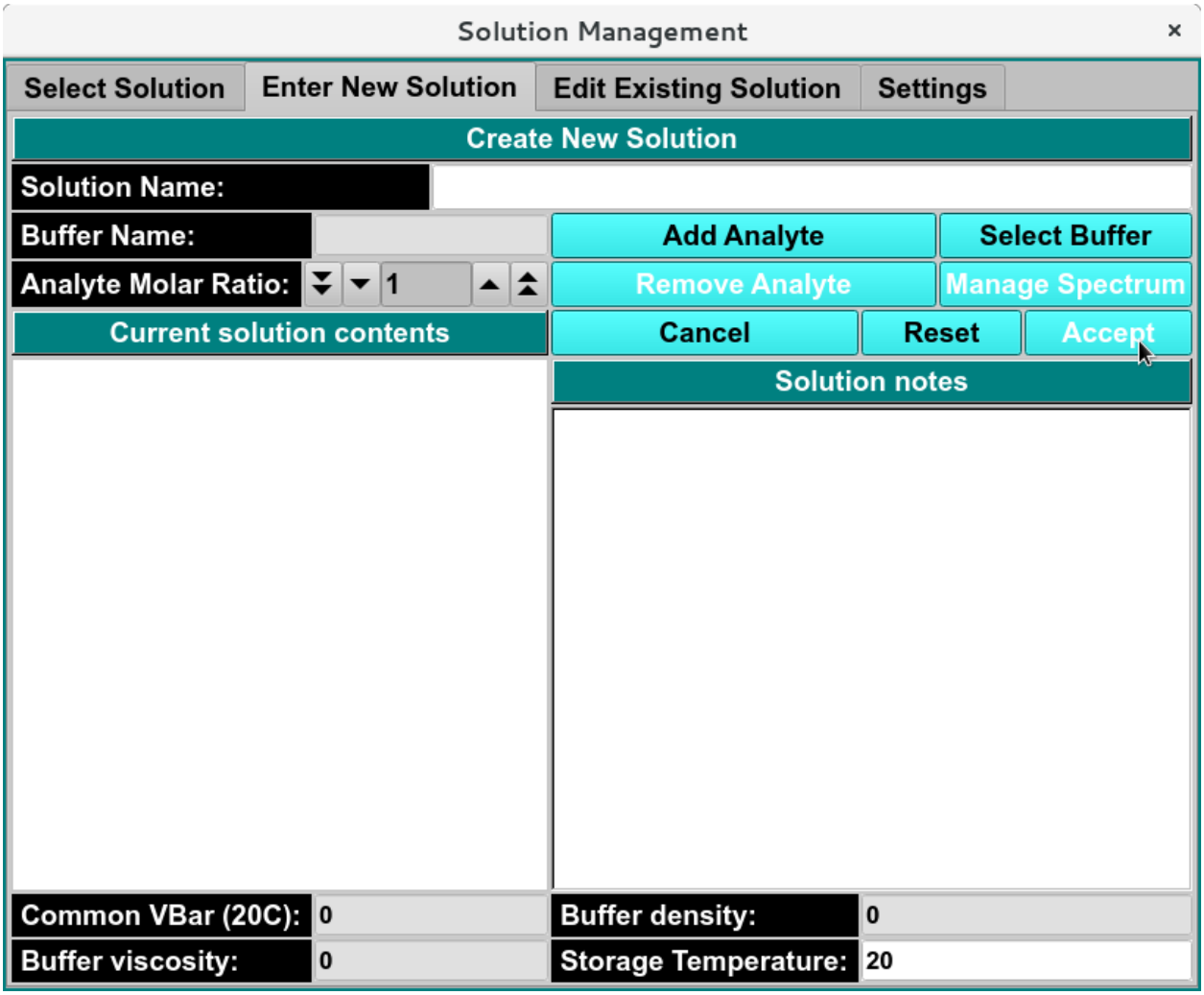
Panel Tab Options:

- `Select Solution <solution_select> - A panel whose primary purpose is to select a Solution to return to the caller.
- *Enter New Solution* - A panel whose primary purpose is to enter a brand new Solution, defined mostly by specifying components and each one's concentration.
- `Edit Existing Solution <solution_edit> - A panel whose primary purpose is to change non-hydrodynamic characteristics of an already existing Solution.
- `Settings <solution_settings> - A panel whose primary purpose is to set Database-or-Disk input and to select the investigator.

80.2.1 Enter New Solution Panel

Using this panel, you can create a new solution in the current database or on the local disk. Most commonly, you select or create analytes in the solution using the Analyte Management or you can click on the Accept button to upload the solution to the DB or to local disk and to return to the Select Solution panel.

As with all panels, a set of tabs allows you to navigate to other panels in order to perform specialized subtasks relating to solution management.



Enter New solution Panel

80.2.2 New Solution Functions:

Search	Use to search by enter all or portion of the title.
Help	Access this help documentation for the Solution Management module
Cancel	Close window and return to previous page.
Accept	Return to previous page with selected solution.
Delete Solution	Delete selected solution.
Solution Notes	Display notes added to solution
Current Solution Contents	Describe the analyte component(s) of solution.
Analyte Molar Ratio	Once a solution has been selected and a solution contents item (analyte) has been chosen, use this to enter the comparative amount of that analyte that is in the solution.
Buffer Name:	Selected buffer name.
Common VBar(20C):	The common vbar at a temperature of 20 degrees C.
Buffer Density	The density of the currently selected buffer.
Buffer viscosity:	The viscosity of the currently selected buffer.
Storage Temperature:	The average temperature of all the runs.

80.2.3 Add New Solution Steps:

- **Step 1:** - Enter the :doc:`Analyte Management <../analyte/index> module to select or create an analyte by clicking **Add Analyte** button.
- **Step 2:** - For solutions with multiple analytes, enter the Analyte Molar Ratio integer of each Analyte.
- **Step 3:** - Enter the :doc:`Buffer Managenemt <../buffer/index> module to call or create a buffer by clicking **Select Buffer**.
- **Step 4:** - Add a descriptive title in the **Solution Name:**
- **Optional/Additional:** - Enter Solution note, enter a solution spectrum profile.

Once the new solution has been sufficiently defined, the **Accept** button becomes enabled.

80.3 Edit Existing Solution Tab

Index

- *Edit Solution Panel:*

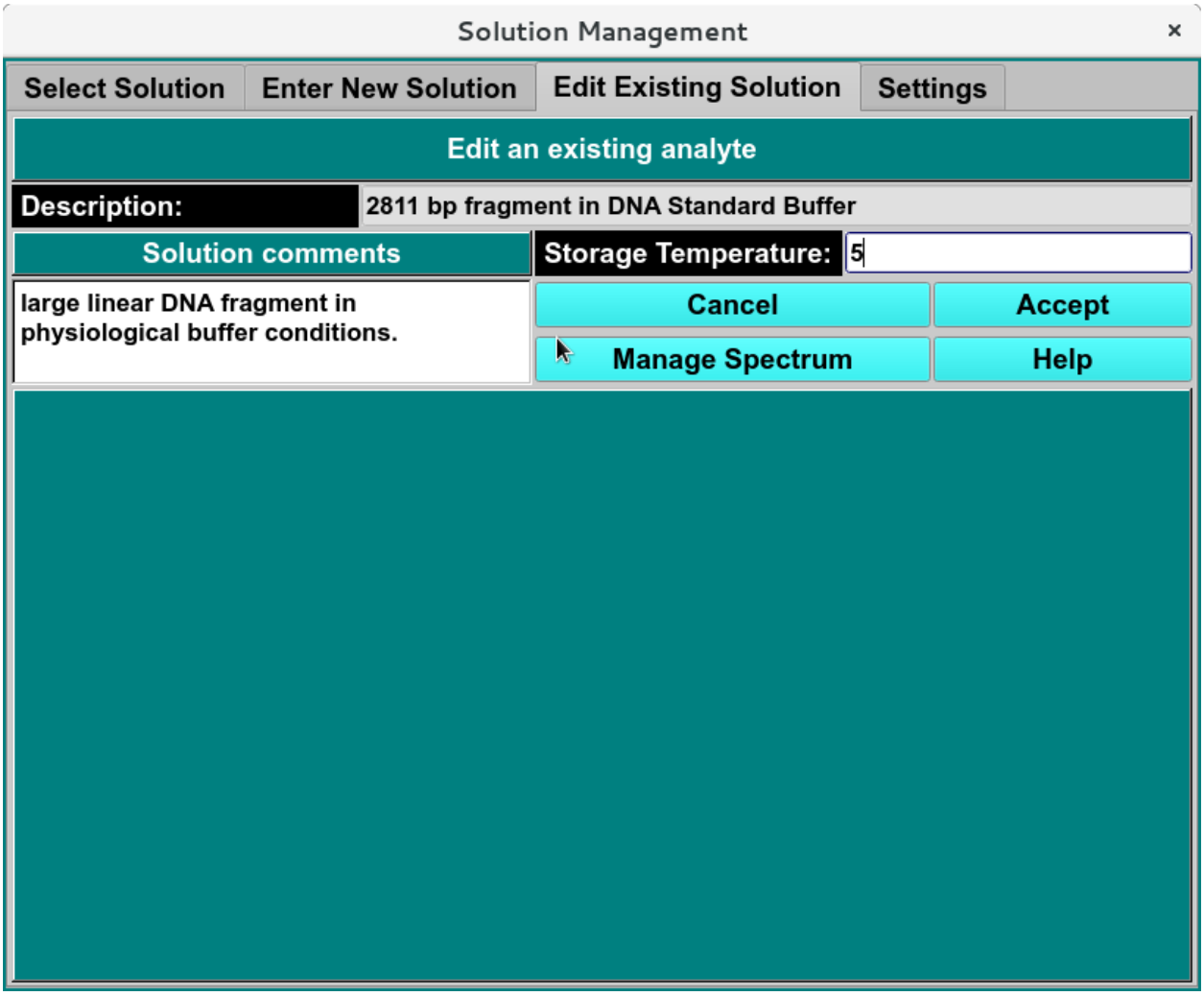
Panel Tab Options:

- Select Solution - A panel whose primary purpose is to select a Solution to return to the caller.
- *Enter New Solution* - A panel whose primary purpose is to enter a brand new Solution, defined mostly by specifying components and each one's concentration.

- *Edit Existing Solution* - A panel whose primary purpose is to change non-hydrodynamic characteristics of an already existing Solution.
- *Settings* - A panel whose primary purpose is to set Database-or-Disk input and to select the investigator.

80.3.1 Edit Solution Panel:

Using this panel, you can add a spectrum profile, change the storage temperature and add or edit the solution comments.

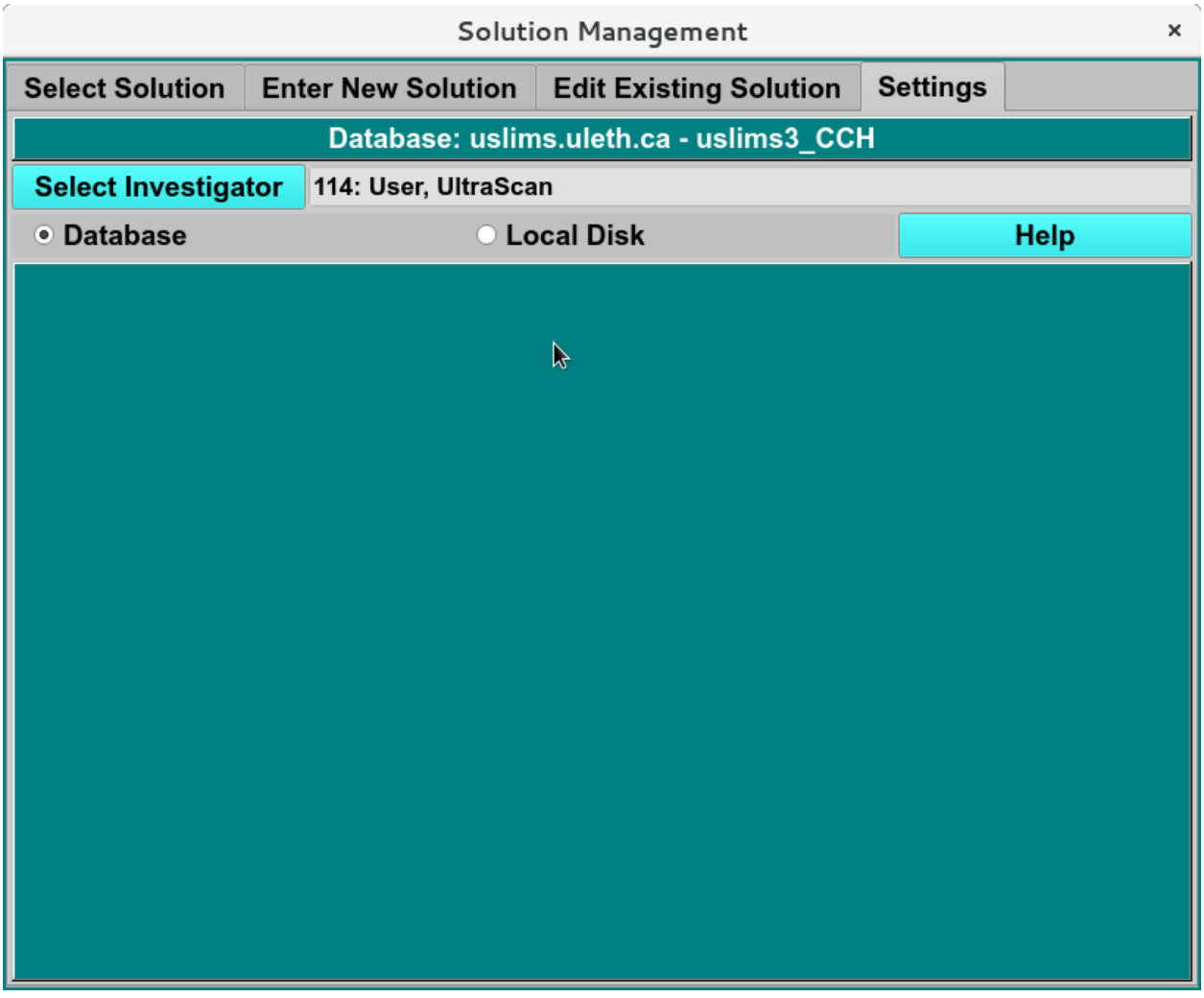


Edit solution Window

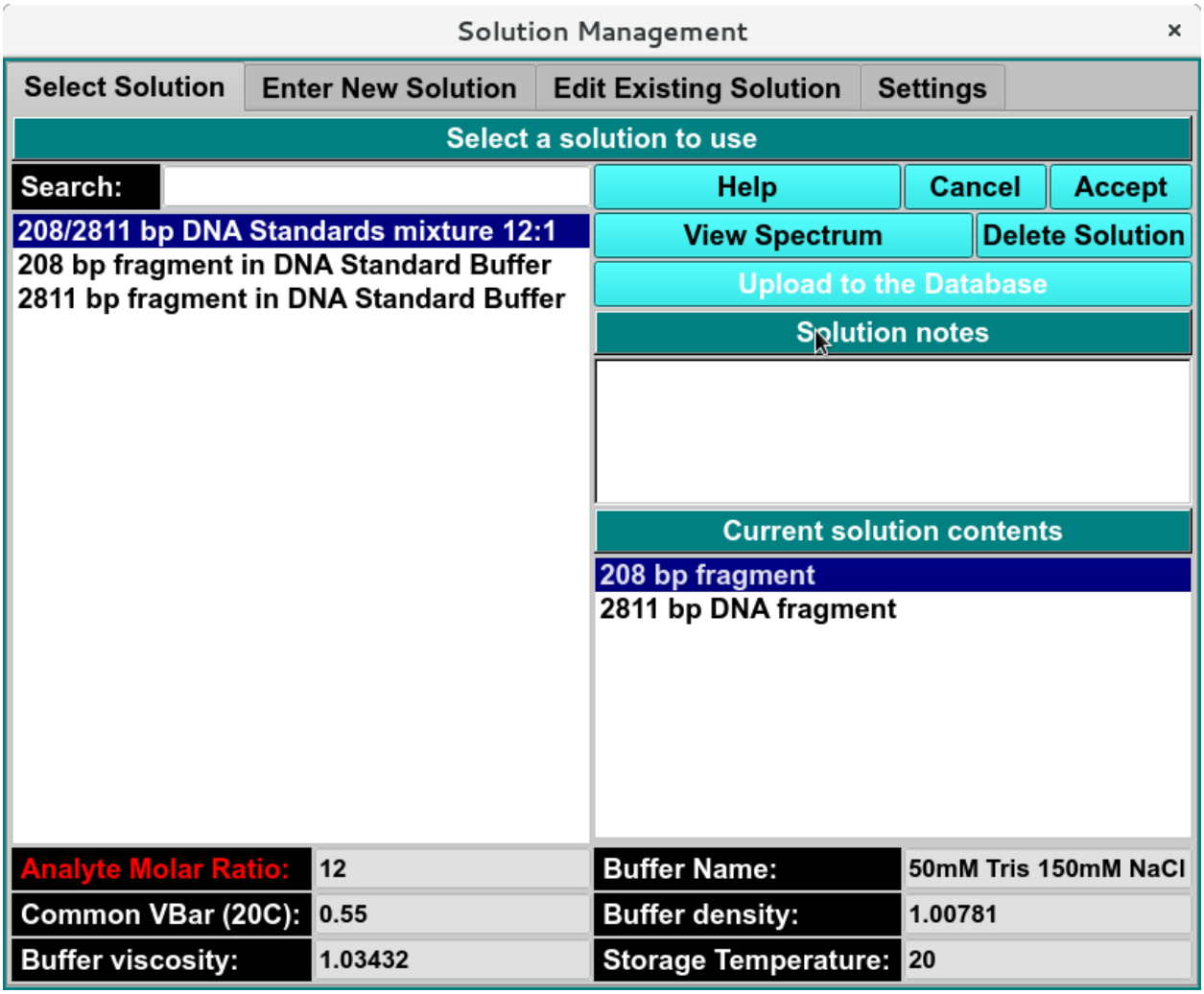
80.4 Solution Settings Tab

Panel Tab Options:

- :doc:`Select Solution <solution_select> - A panel whose primary purpose is to select a Solution to return to the caller.
- :doc:`Enter New Solution <solution_new> - A panel whose primary purpose is to enter a brand new Solution, defined mostly by specifying components and each one's concentration.
- :doc:`Edit Existing Solution <solution_edit> - A panel whose primary purpose is to change non-hydrodynamic characteristics of an already existing Solution.
- **Settings Solutions** - A panel whose primary purpose is to set Database-or-Disk input and to select the investigator.



Solution Settings Window



- Select Solution - A panel whose primary purpose is to select a solution to return to the caller.
- Enter New Solution - A panel whose primary purpose is to enter a brand new solution, defined mostly by specifying components and each one's concentration.
- Edit Existing Solution - A panel whose primary purpose is to change non-hydrodynamic characteristics of an already existing solution.
- Settings - A panel whose primary purpose is to set Database-or-Disk input, the investigator; or to synchronize the local solution components file from the database.

CHAPTER

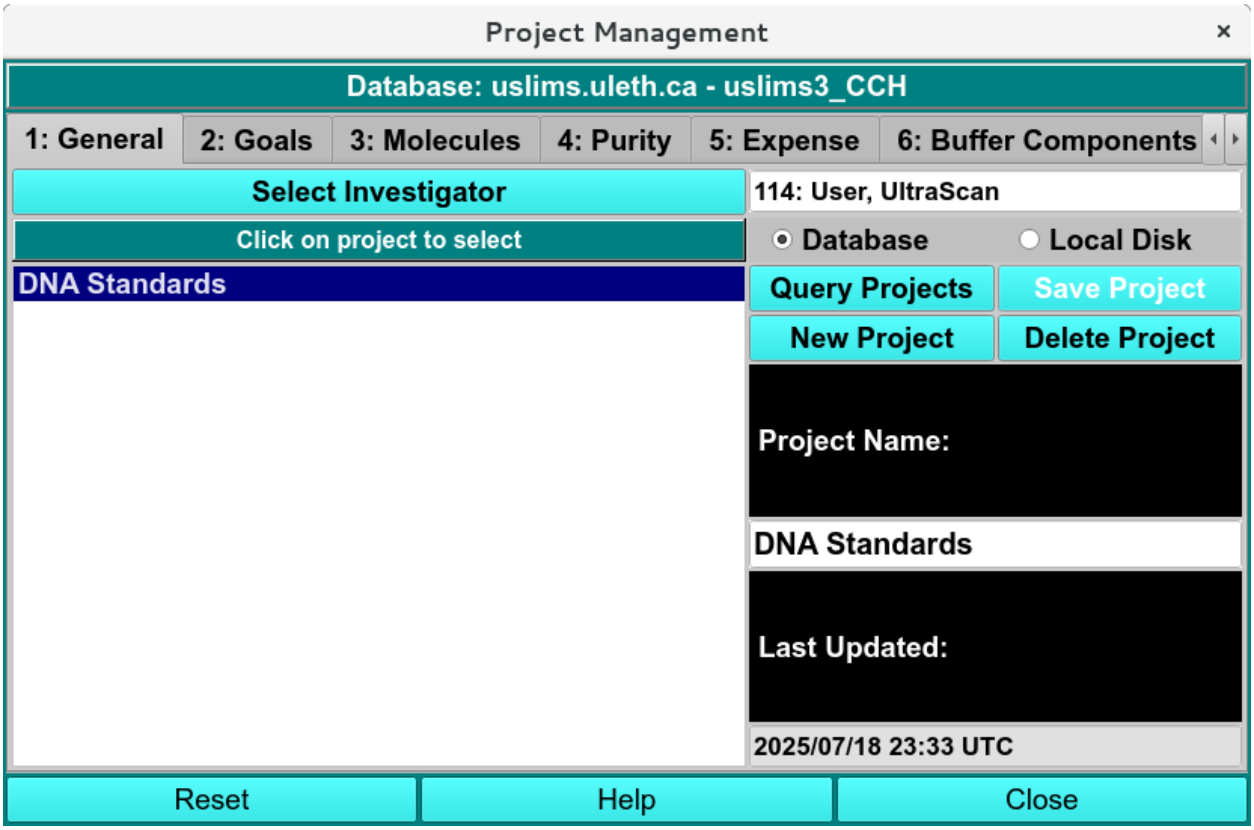
EIGHTYONE

PROJECT MANAGEMENT

Index

- *Create Project Process:*
- *Projects Functions:*
- *Database Project tab*

Using this window and in the database "Project" tab accessible after you log in, you can add new project by describing the goals, experimental design and sample components of the project or editing an existing project.



Project Management Module

81.1 Create Project Process:

Using this window, you can manage project information on your local disk or in the current database. You must be identified as the investigator to delete or update a project description in the database.

- **Project Navigation** To load project information into the current window, first select the Use Database or the Use Local Disk radio button. Then click the Query Projects button. To display information about a single project, click on the project name listed in the click on project to select widget.
- **Create New Project** To create a new project, click **New Project** and add a descriptive title. Edit the project goals, molecules, purity, expense, buffer components, salt information, AUC Questions and other notes. Then Click **Save Project** to create the project. You can edit an existing project by clicking on the Project Name and editing the appropriate box in the lower right-hand part of the window. the project goals, molecules, purity, expense, buffer components, salt information, AUC Questions and other notes can also be edited by clicking on the appropriate tab at the top of the window. Click on the **Save Project** button to save any changes.

Note: All of the fields are required to be completed before saving the project information.

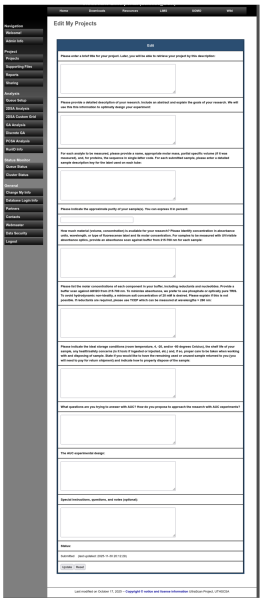
- **Delete Project** To delete a project, select the appropriate project in the click on project to select widget, and then click the **Delete Project** button. After making the appropriate changes, be sure to click the **Save Project** button to save your changes.

81.2 Projects Functions:

Click on project to select	After clicking on the Query Projects button, this widget will contain a list of projects available in the current location (database or local disk). Click on a project in the widget to select.
Use Database	Check to select read or write of the project definition to or from the database.
Use Local Disk	Check to select read or write of the project definition to or from the hard disk.
Query Projects	This button initiates a read of project definitions and population of the project list widget.
Save Project	Saves the current project definition to the hard disk or database.
New Project	Creates a new project by returning all values to their defaults. Work through each tab to change the current contents as appropriate, and then click on the Save Project button.
Delete Project	Deletes the current project from the database or hard disk.
Project Name	The name of the project currently selected in the Click on project to select list widget, or a new name when creating a new project definition in the database or on the hard disk.
Global Identifier	Read-only global identifier of the project.
Reset	Reset all project values to default setting.
Help	Show this documentation.
Close	Close the dialog and possibly return project selections to the caller.

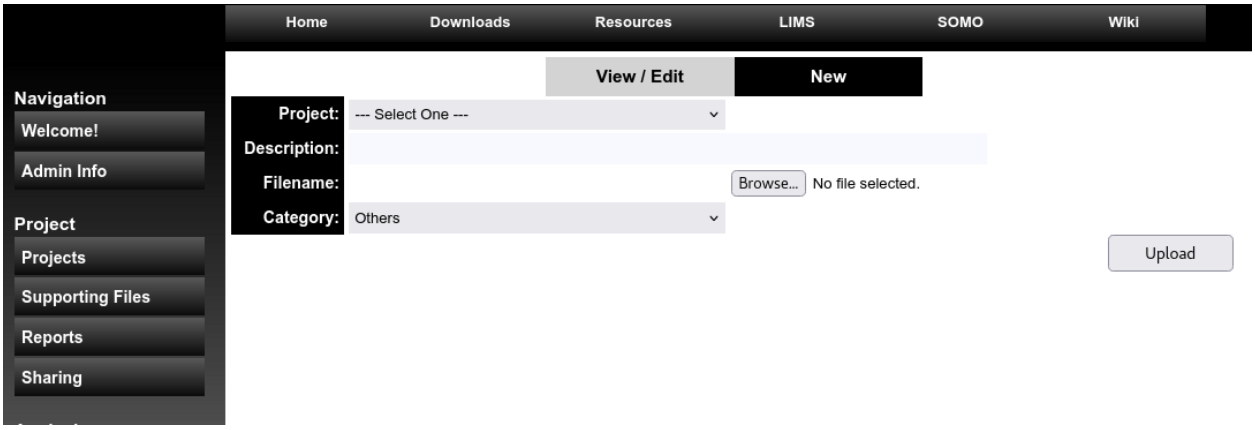
81.3 Database Project tab

An existing project can be edited or a new project created using the Project tab.



Project Management Website Tab

Supporting data or documents can be uploaded and associated with the project.



Add Supporting Data/Document to Existing Project

CHAPTER

EIGHTYTWO

MANAGE DATA IN DATABASE AND ON LOCAL DISK

Index

- *Data Management Process:*
- *Data Management Functions:*

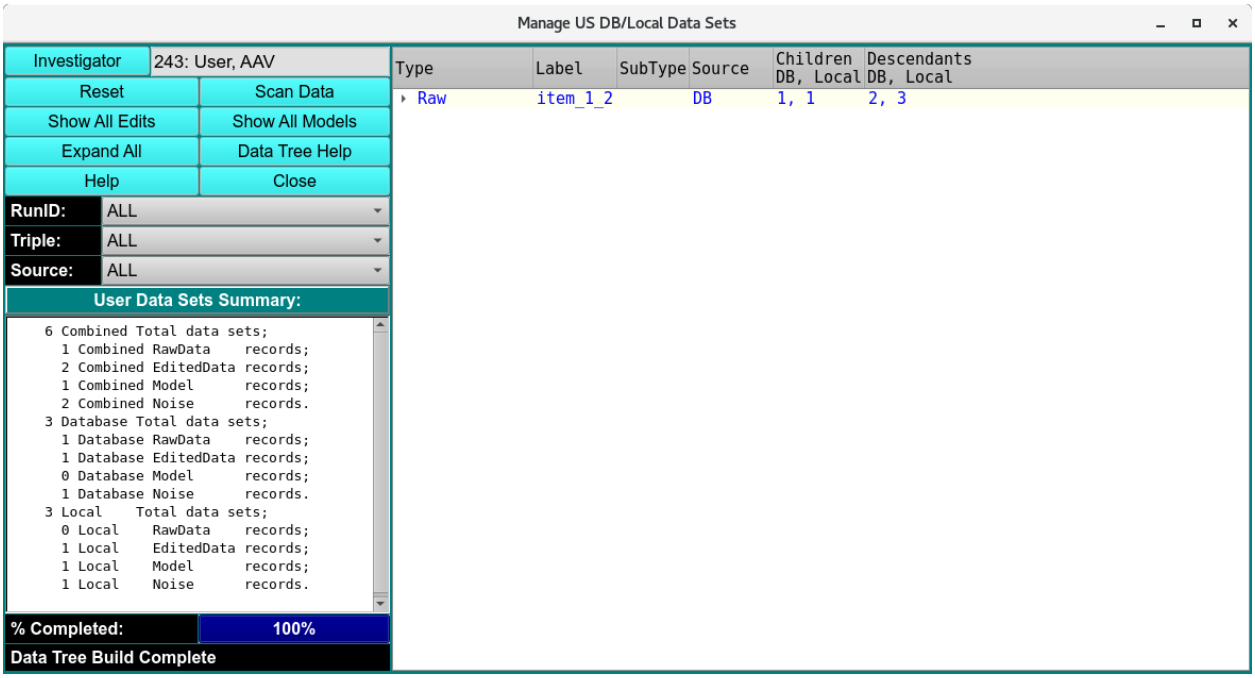
This module is used to display the four main types of data that may exist for the user in the database and/or on local disk. They are: (1) Raw experiment data; (2) Edited experiment data; (3) Model; and (4) Noise. Besides presenting a tree view of the user's data and their relationships, this module provides a simple means of performing processing on data: (1) upload to DB; (2) download to local disk; (3) remove from DB or local.

The primary end of any processing of the data is to achieve a synchronizing of data in the database and on local disk. Particularly when preparing for off-network work or when returning to the network after such work, it is desirable to insure that any new data produced has a presence on local disk for off-network work and in the database for normal day-to-day processing. Instances of any data need to be in sync on the two media. This module provides a means to accomplish that.

82.1 Data Management Process:

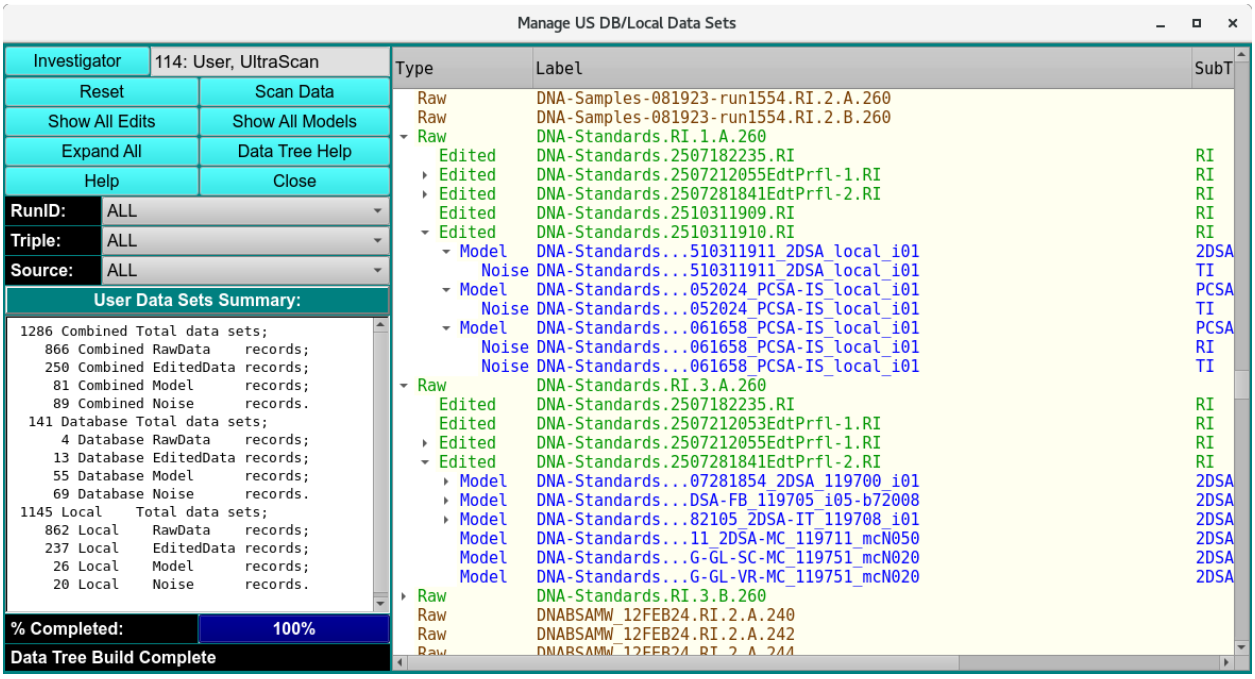
- **Database Password:** Upon opening, US_ManageData requires you to sign on to the database with your DB password.
- **Investigator:** For database work, you must specify your investigator name so you are limited to data that you have permission to access and modify.
- **Scan Data:** Initiate a scan of all your data on the database and local disk. You will then be presented a full tree view of that data.
- **Navigate and Process:** Once you have a tree view of your data, you may navigate it using expand/collapse buttons or the normal **+** and **-** mechanism for specific branch expand and collapse. Context menus at each row allow upload/download/remove/details.

Initially you are presented with a small window to enter your database password. Then you will see the main window with a sample data tree that shows the kinds of states that data may be in (see tree help below).



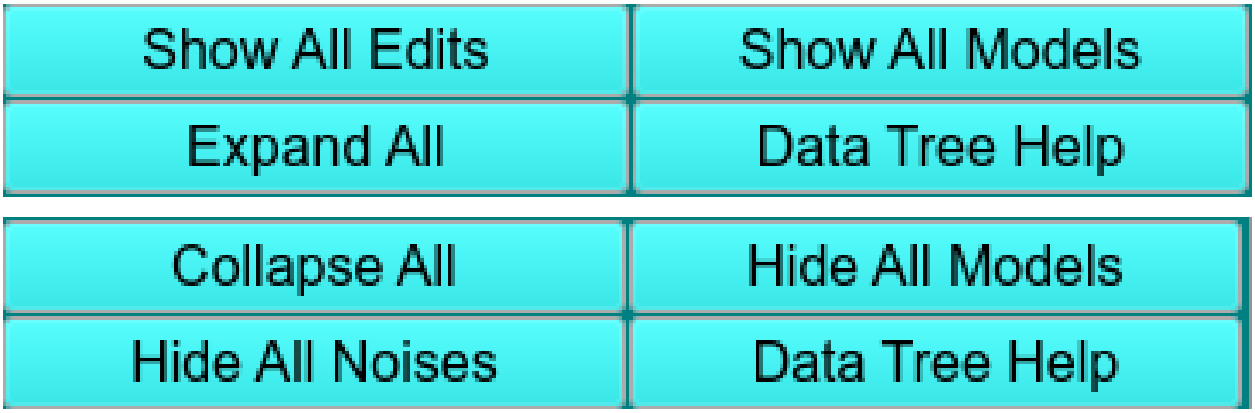
Data Main Window

After clicking on the **Scan Data** button, the tree will be populated with your actual data.



Populated Data Manager

The tree menu will expand and collapse with buttons like these.



Tree Menu Buttons

82.2 Data Management Functions:

Investigator:	Often the investigator text field will already be correctly filled out if your home directory name is the same as the first or last of your investigator name. If not, you may enter all or a portion of the first or last name and hit the **Enter** key to have the investigator found. If that, too, fails, you may click the button to enter a full *Investigator dialog*.
Reset	Reset the tree to the default sample data.
Scan Data	Click this button to initiate a full scan of all your data in the database and on local disk. You should re-initiate a scan after any series of processes (upload|download|remove) on the data.
Show All Edits	Expand the tree view to insure all rows at the level of Edited data are revealed. The button will then be relabelled **Collapse All** so you can hide edits and their descendants.
Show All Models	Expand the tree view to insure all rows at the level of Model data are revealed. The button will then be relabelled **Hide All Models** so you can hide models and their descendants.
Expand All	Expand the tree view to insure all rows at the level of Noise data and their ancestors are revealed. The button will then be relabelled **Hide All Noises** so you can hide noise record rows.
Data Tree Help	This button lets you pop up a Tree Help Text Window with some helpful notes on use of the tree view and on the color legend for its rows.
Tree Navigation and Context Menus	Individual rows in the tree view may be expanded ("+") to show children and other descendants or may be collapsed ("-"). A right-mouse-button click on any row brings up a Manage Data Context Menu that allows you to perform processes on the data record or show details about it.
Help	Display this and other documentation.
Close	Close all windows and exit.

CHAPTER

EIGHTYTHREE

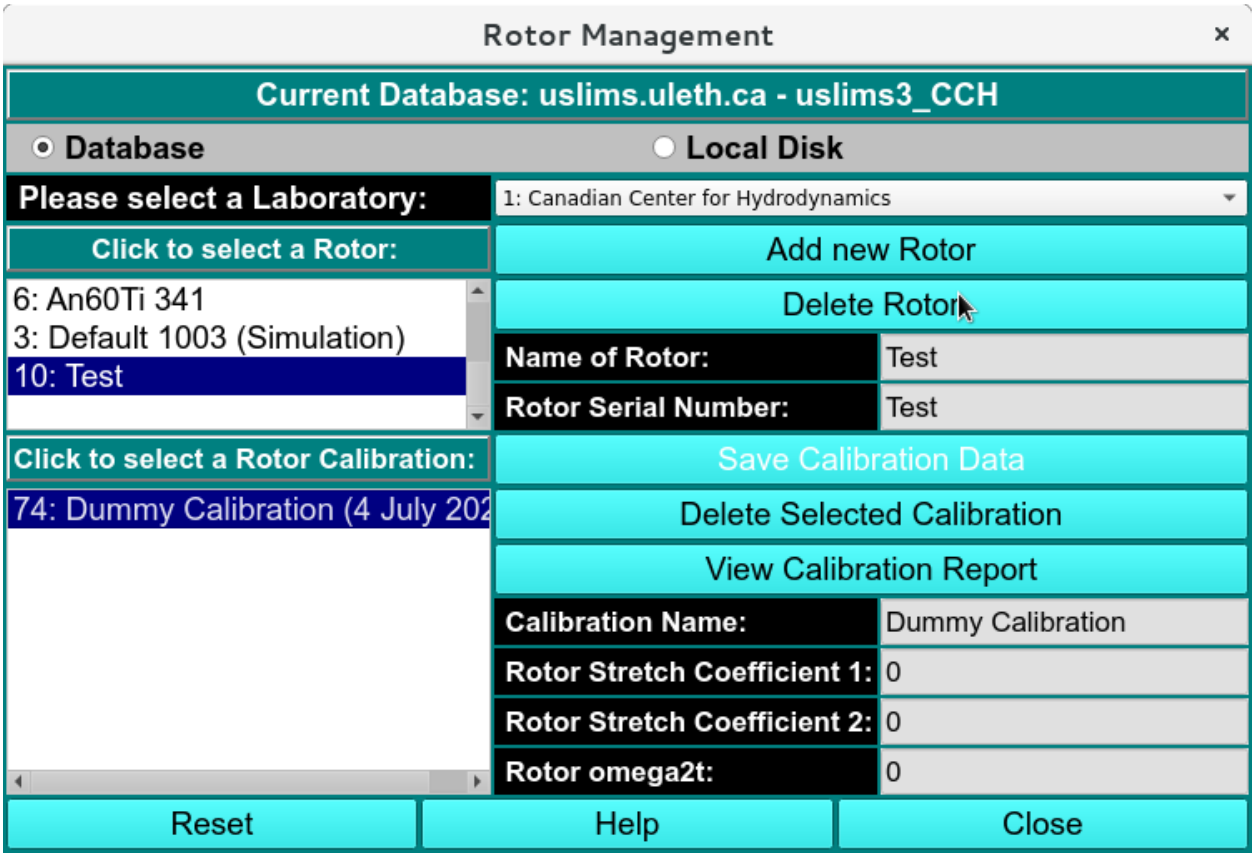
MANAGE ROTOR INFORMATION

Index

- *How to Manage Rotors and Rotor Calibration files*
  - *Select and View Rotor information*
  - *Add Rotor Calibration*
  - *Add New Rotor*
- *Rotor Calibration:*

83.1 How to Manage Rotors and Rotor Calibration files

Using this window, you can manage rotor information on your local disk or in the current database.

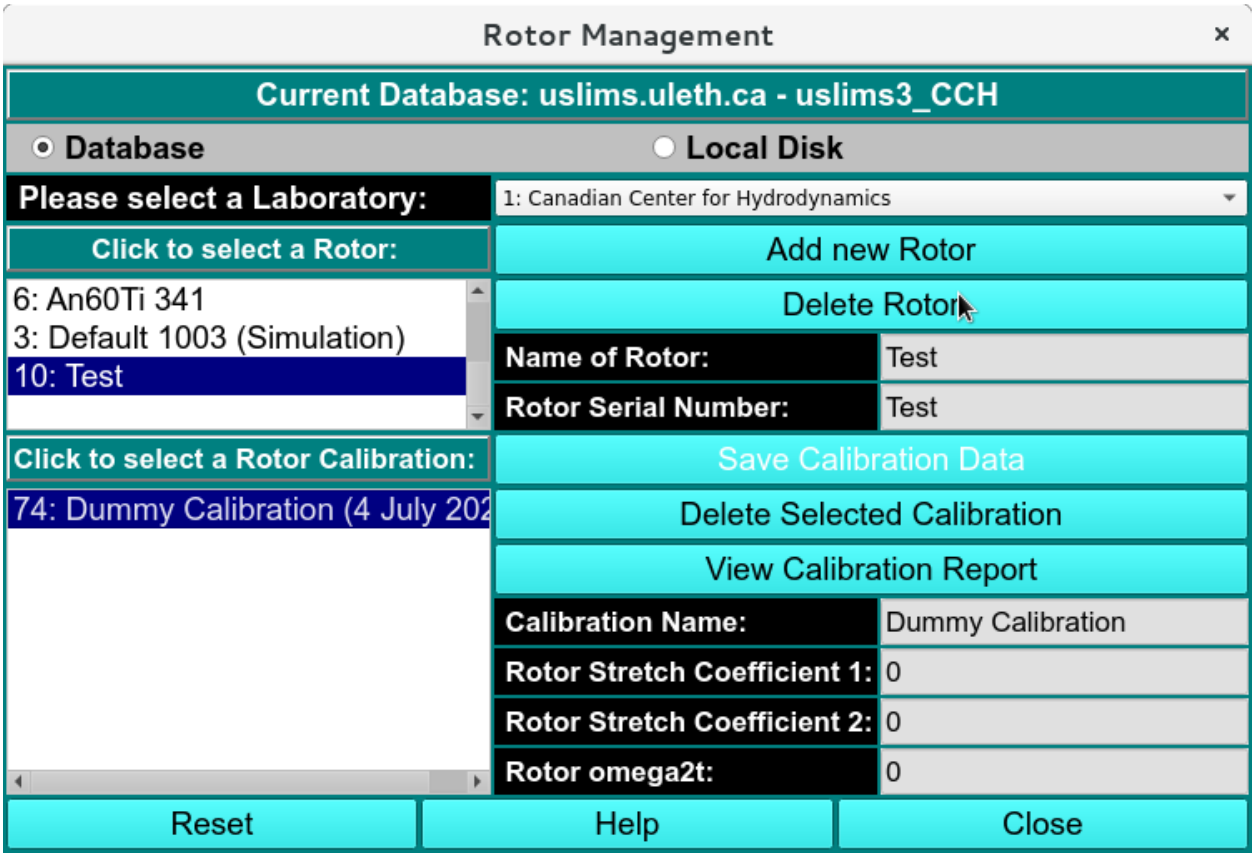


Edit solution Window

To load rotor information into the current window, first select the Use Database or the Use Local Disk radio button. Then select the appropriate lab in the lab drop-down box. Follow the sequence below that fits your particular situation. Note: most likely you will be using Sequence 1.

83.1.1 Select and View Rotor information

In this sequence you have come to this window from US_Convert, and you would like to select the rotor and rotor calibration that were in place during your experiment. First, select the appropriate rotor from the top listbox, and then select the correct rotor calibration profile. The rotor calibration profiles are sorted in reverse order by date, so that the most recent one is on top. This is the most likely choice. Once the lab, rotor, and rotor calibration have been selected, you may click on the *View Calibration Report <rotor_calibration>* button to view the report, or view other related information in the stretch coefficient boxes or the ω^2T box. Click the **Accept** button to confirm your choices and return to the previous window.

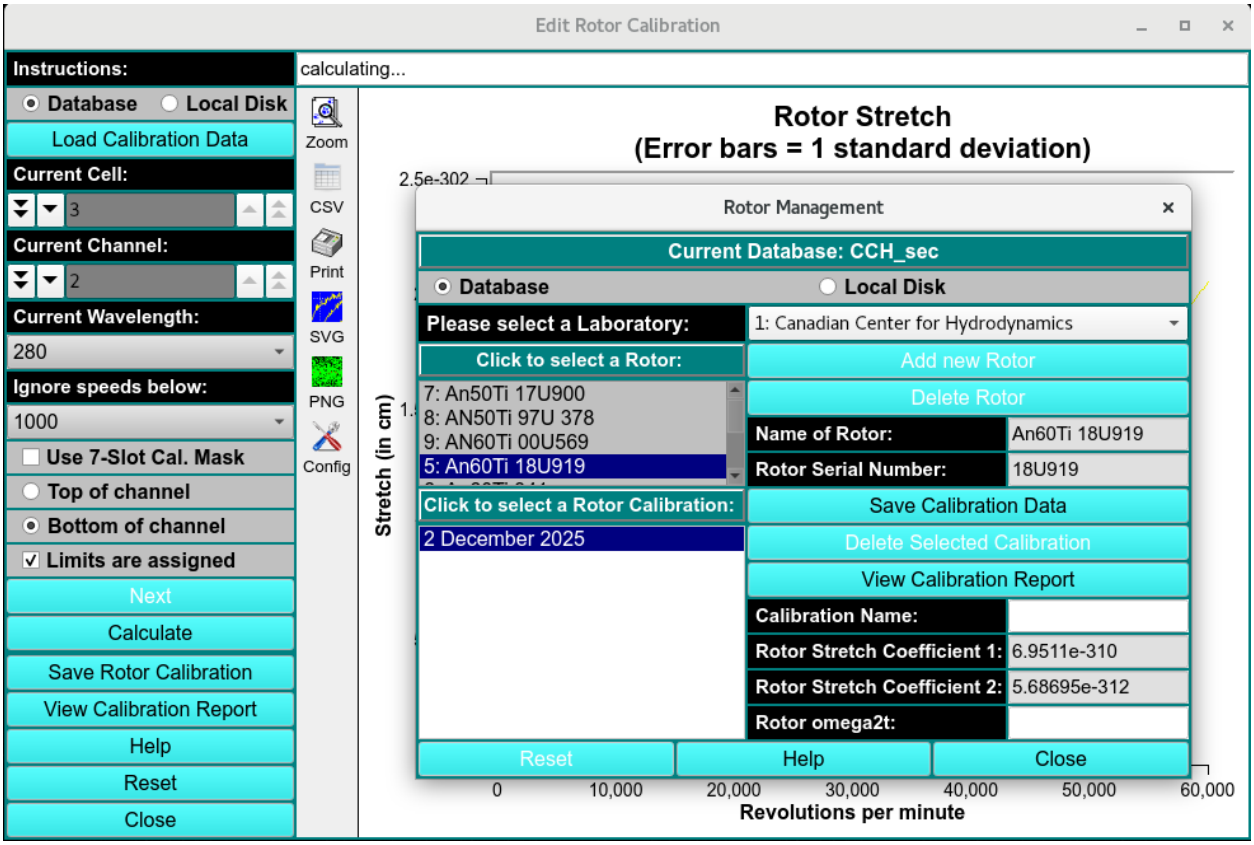


Rotor Management

83.1.2 Add Rotor Calibration

Note: Administrators only.

In this sequence you have come to this window from *Rotor Calibration* module and you would like to associate the calibration with an existing rotor. Click on the rotor you would like to associate and then click on **Save Calibration Data.**

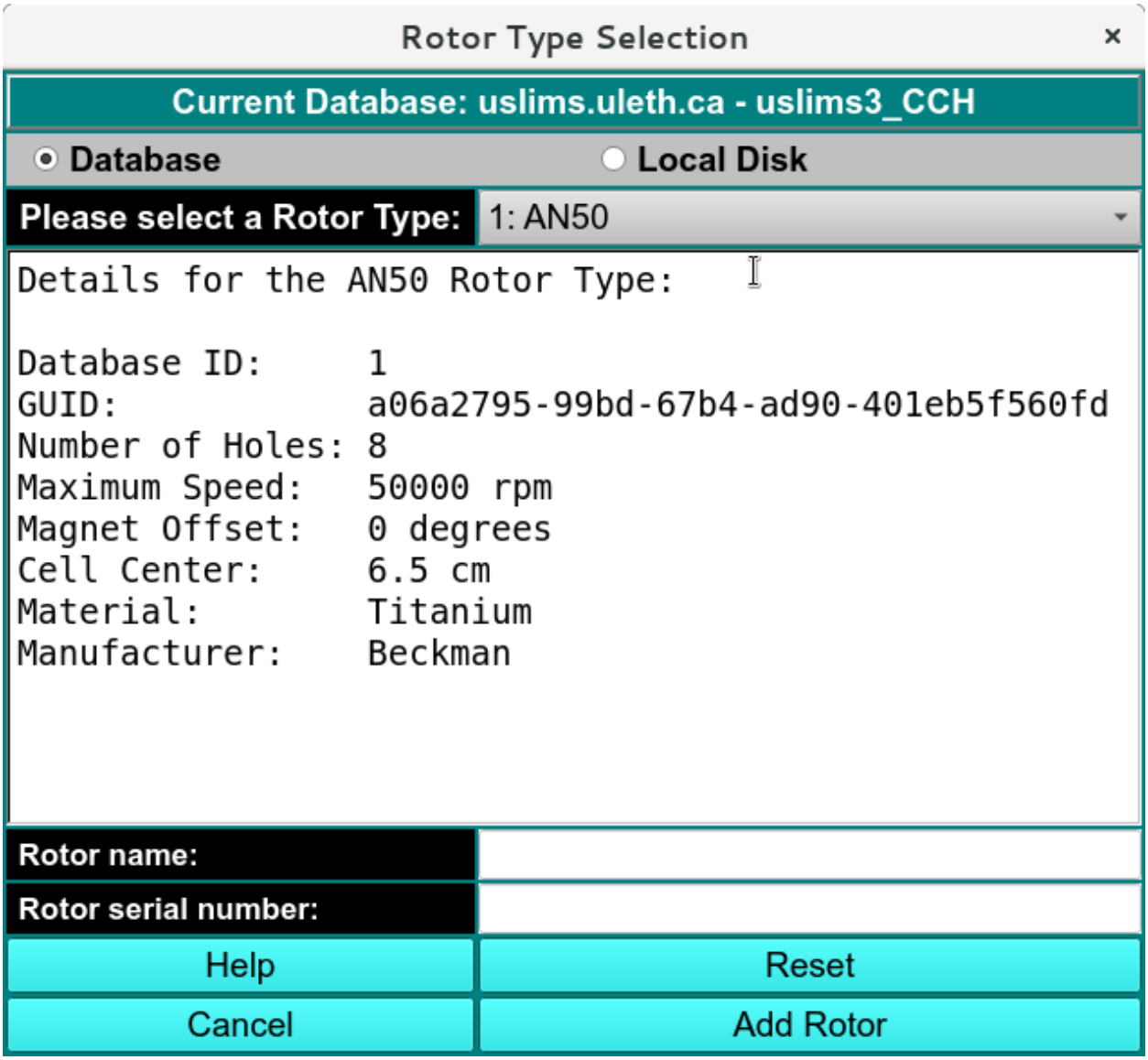


Add Rotor Calibration

83.1.3 Add New Rotor

Note: Administrators only.

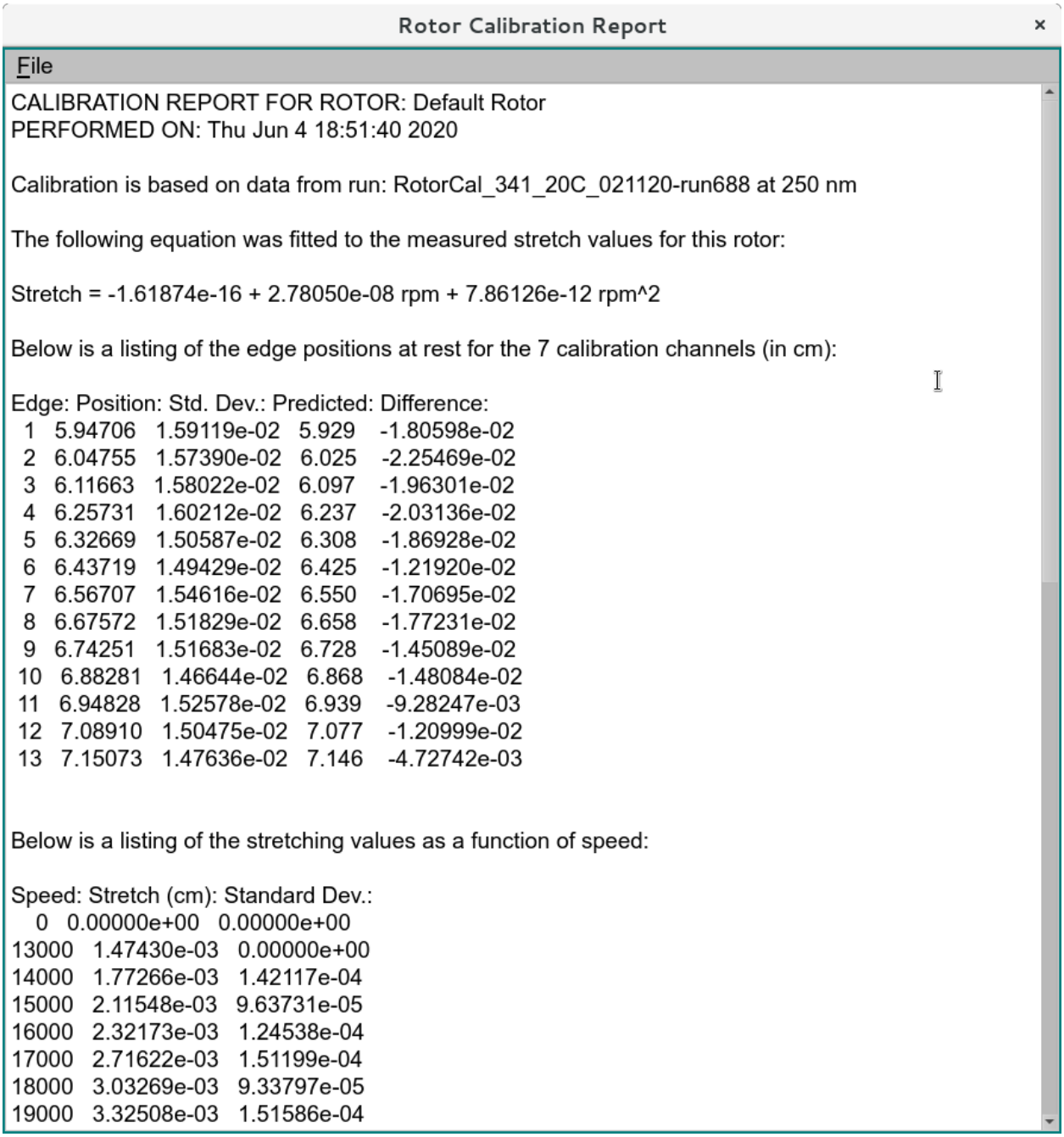
Click on **Add Rotor** to add a new rotor.



Add New Rotor

83.2 Rotor Calibration:

Calibration reports can be loaded to each Rotor in this window.



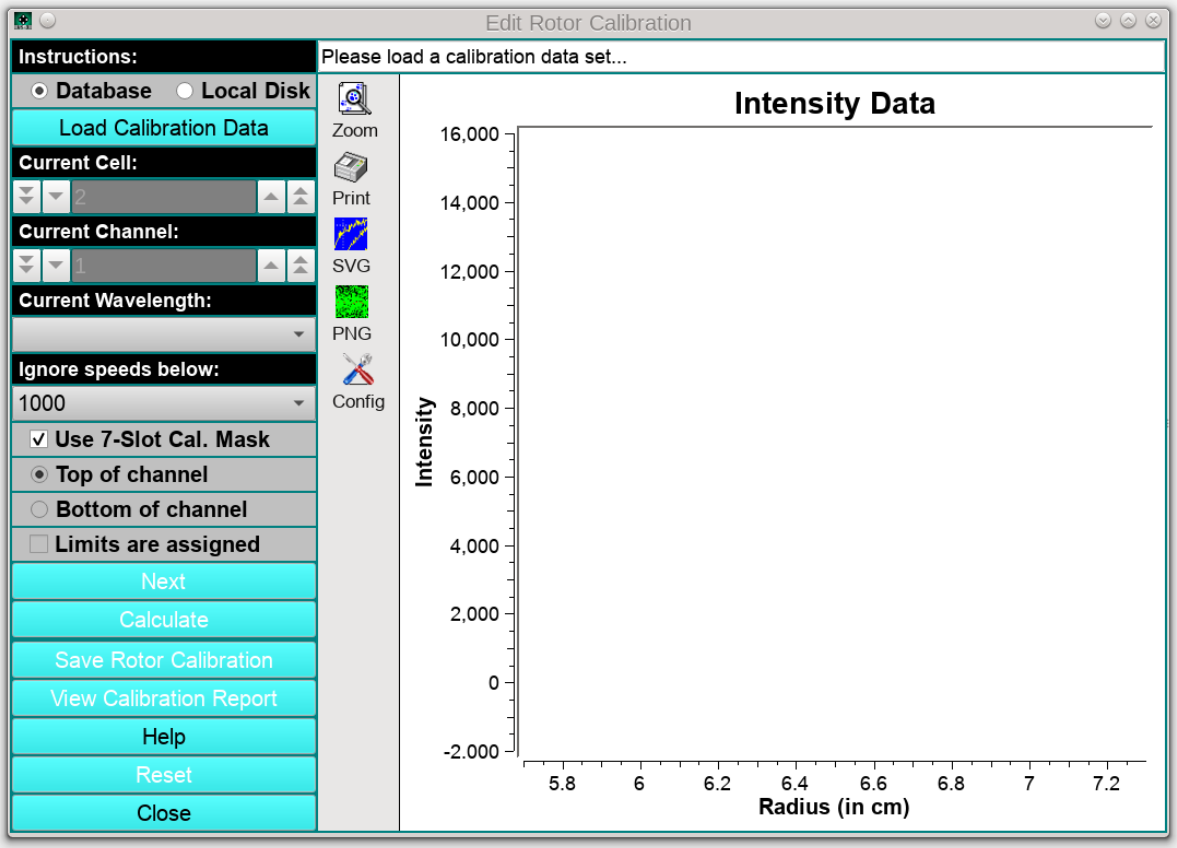
Rotor Calibration Report

The Rotor Calibration reports can be generated in *Edit Rotor Calibration*

CHAPTER

EIGHTYFOUR

ROTOR STRETCH CALIBRATION UTILITY



Edit Calibration Data

The purpose of this program is to analyze rotor stretch profile data acquired with the analytical ultracentrifuge, and to create a rotor stretch calibration profile for different rotors in use at a laboratory.

In combination with the centerpiece geometry, the rotor stretch calibration serves to inform all analysis protocols relying on finite element modeling about the position of the cell bottom, which is one of the boundary conditions required for the solution of the Lamm equation. During acceleration, the rotor will stretch and move the bottom of the celll position by the amount of the stretch outward. The stretch is a function of the rotor speed, which depends on the square of the angular velocity. The rotor stretch calibration profile is stored in

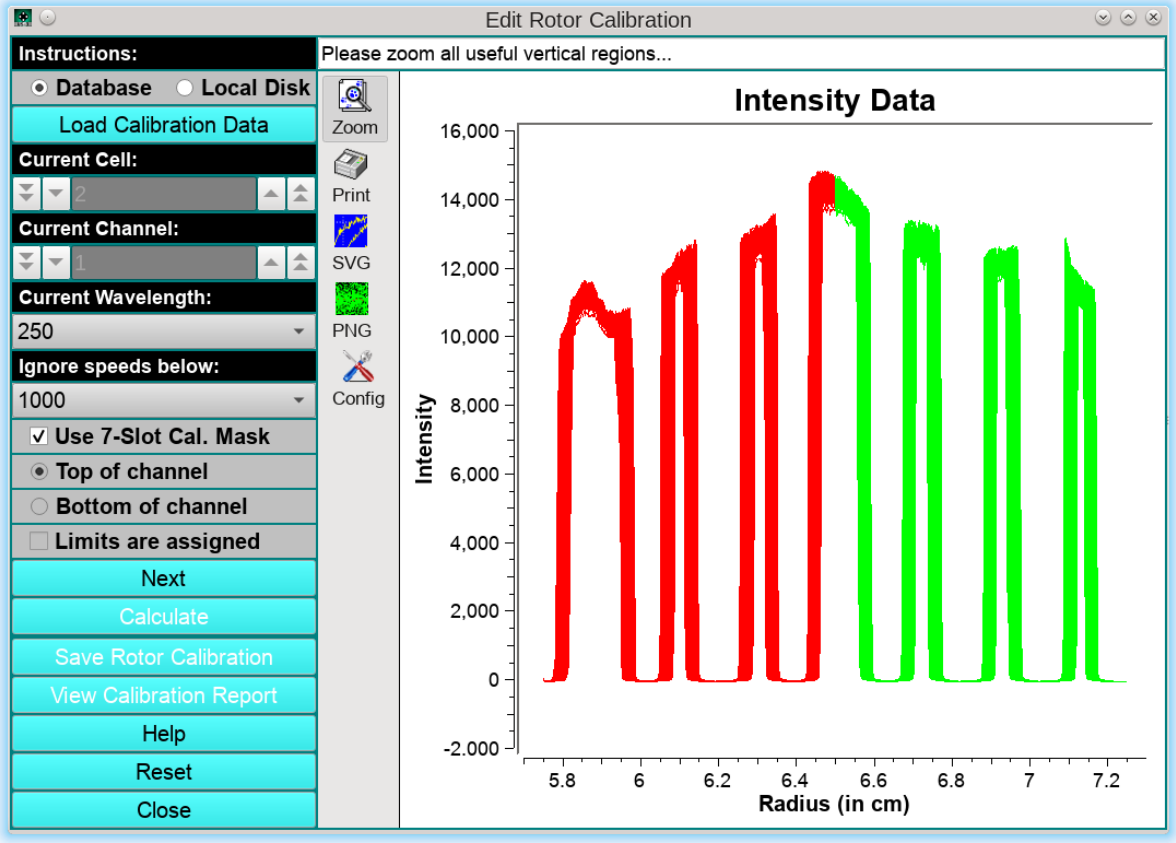
the LIMS database for each rotor in use and retrieved on the fly to calculate boundary conditions. This value is calculated based on a second-order polynomial that describes the amount of stretch observed for each rotor.

The amount by which each rotor stretches as a function of rotor speed depends on the rotor's type and age. Hence it is advisable to individually calibrate each rotor in use periodically. The rotor stretch is mostly independent of temperature and can be measured at any temperature.

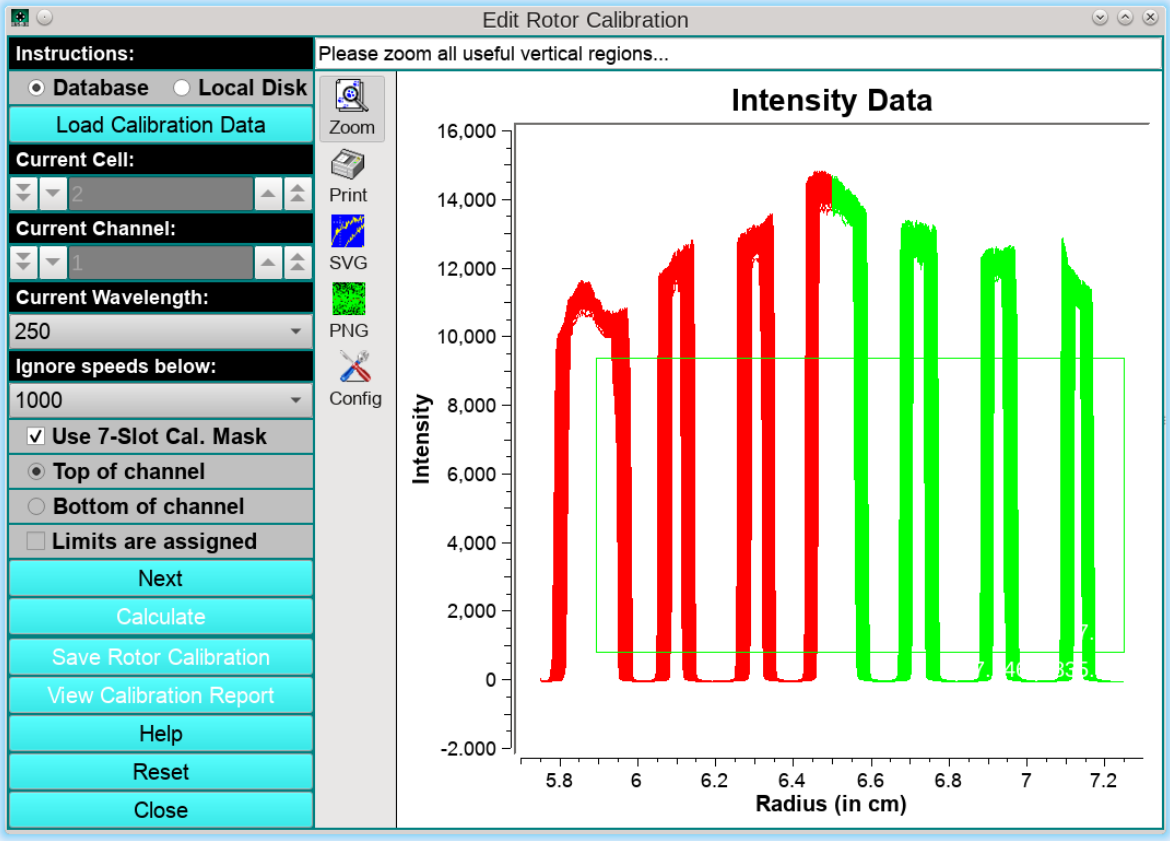
The rotor stretch can be measured with either the Proteomelab XLA/XLI instrument or the Optima AUC. The Optima is preferred, because it offers higher radial resolution, and hence more reproducible data. The recommended protocol is to acquire the calibration data by scanning the 7-channel calibration disk **(available from AUC Solutions, 15826 Los Altos, Houston, Texas 77083, USA Telephone: (210) 920-4AUC)** which provides the highest statistical significance by measuring displacement replicates from 13 edges.

The scanning protocol prescribes to spin up the rotor to 3,000 rpm and to wait 15 minutes, then to scan the calibration disk five times in intensity mode. When scanning is complete, the rotor is accelerated by 1,000 rpm to 4,000 rpm, and, following another 15 minute equilibration period, the calibration disk is scanned again five times in intensity mode. This cycle is repeated until the maximum speed of the rotor has been reached.

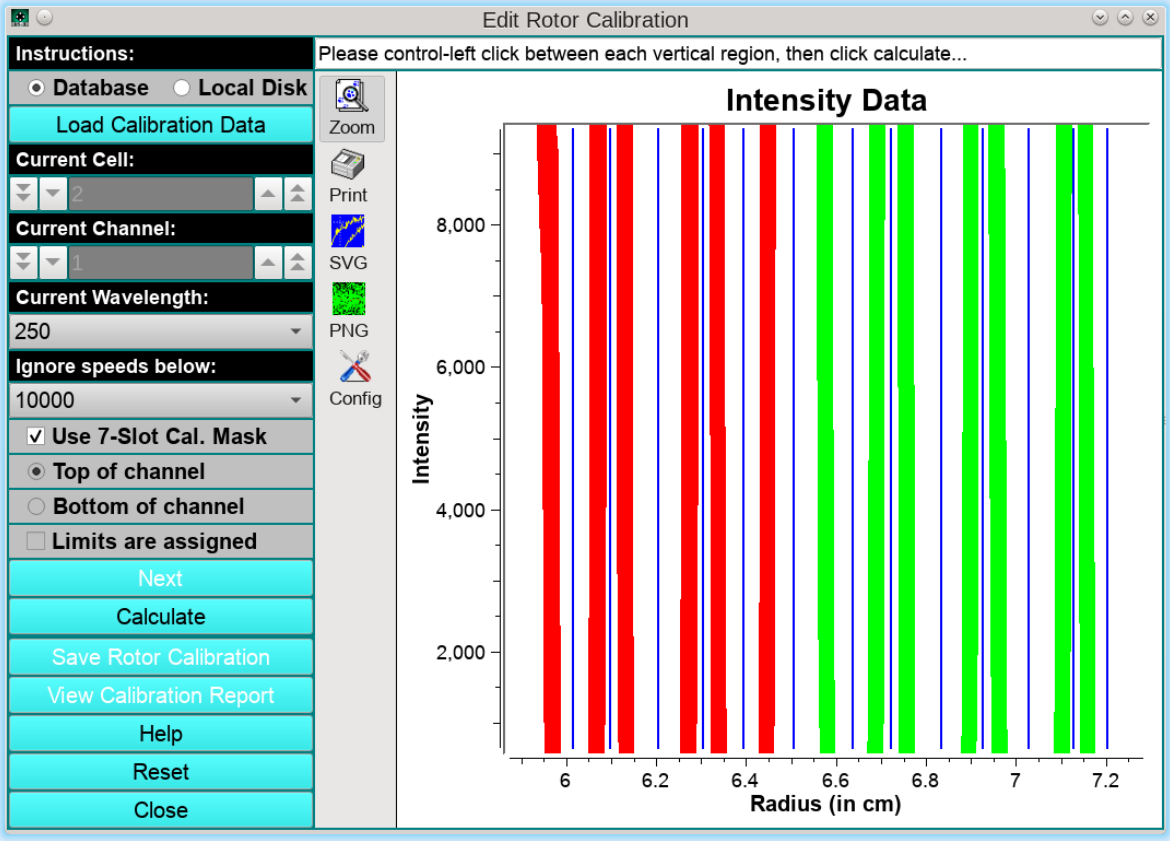
The first step in the program is to load the calibration data from a 7-channel calibration disk by clicking on the **Load Calibration Data** button. Once the data are loaded, the user can select a different cell, channel, or wavelength from this data set by manipulating the respective counters or drop-down lists. The intensity data will be shown graphically in the main window. The left half of the cell will be drawn in red, the right half in green. Make sure the checkbox **Use 7-Slot Cal. Mask** is checked!, and select the lowest speed to be included in the fit. We recommend to set the slowest speed to 10,000 rpm, because the rotor stretch below 10 krpm is less than the radial resolution of the instrument and therefore those measurements are not useful. In the status bar the message: "Please zoom all useful vertical regions…" will appear:



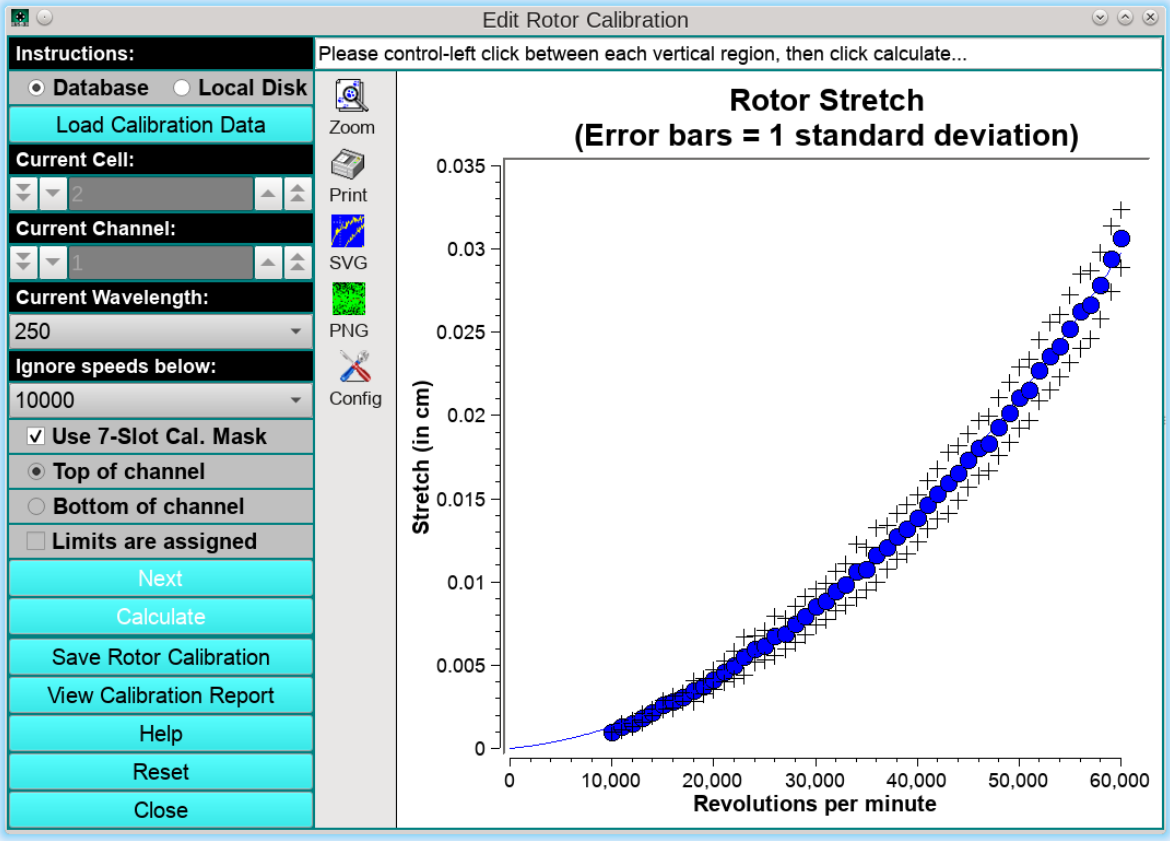
In the next step, the user needs to zoom all vertical regions, marking the edge positions in the graph as shown by the green rectangle below:



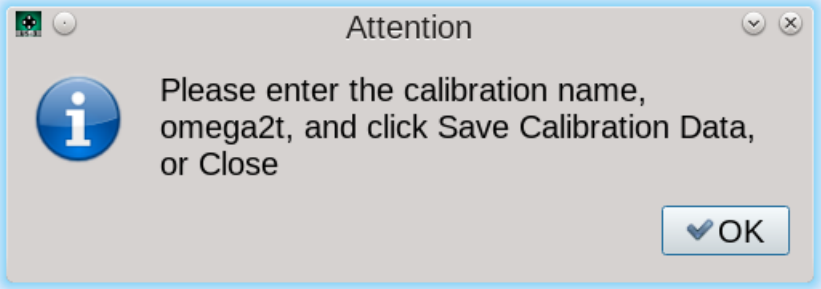
Here it is recommended to exclude the left-most vertical region since it typically is partially obliterated by the screwring of the cell housing, as well as any noisy sections of the image, because they distort the averaged data. After releasing the mouse, just the vertical regions will be shown, and need to be separated by the lines draw in blue, which can be created by clicking in the middle between each vertical region. (Windows/Linux users: Hold control key while clicking, Mac users: Hold command key):



Next, the user should click **Calculate** to process the data. The resulting fit will be shown on the next screen:



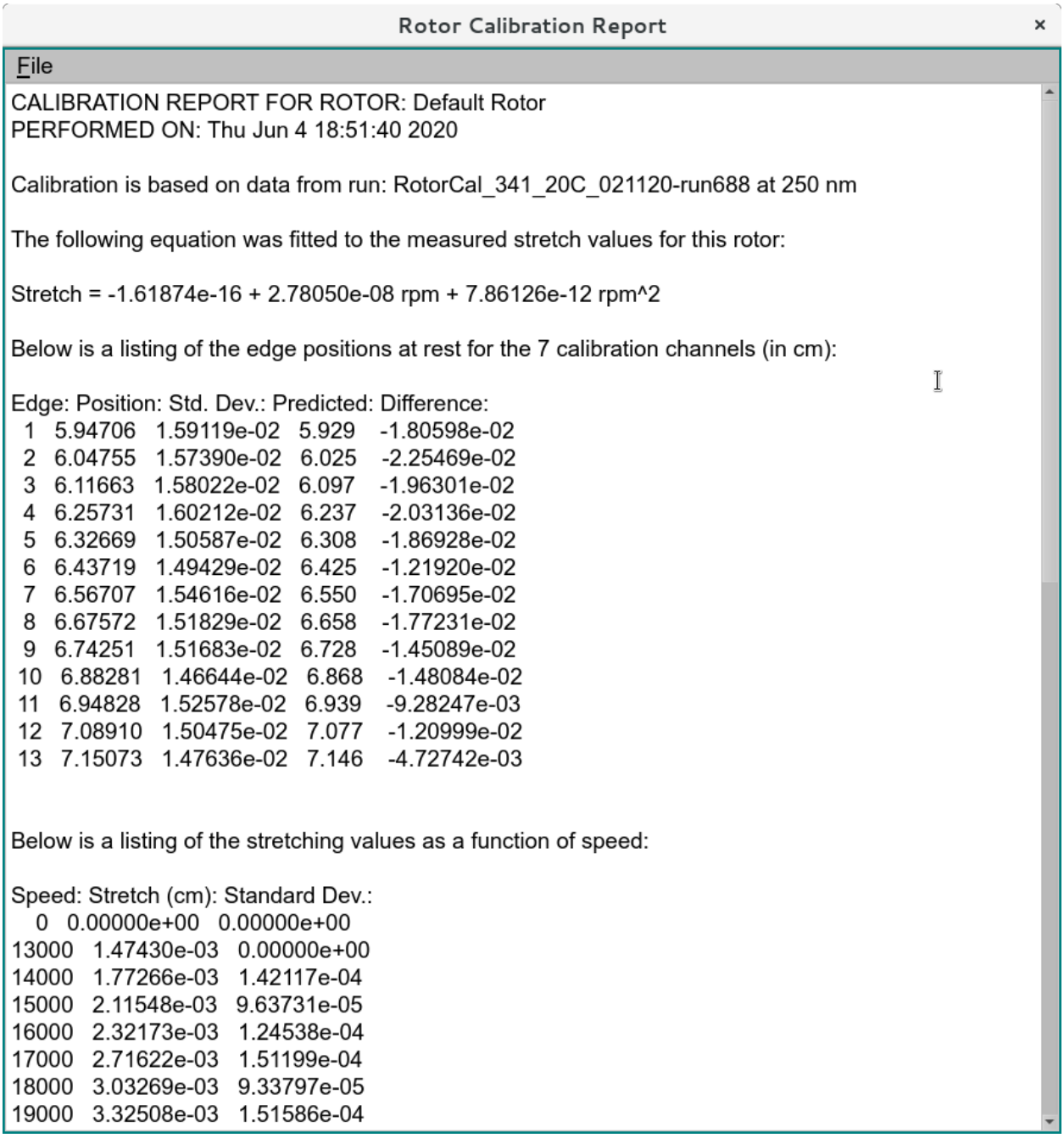
Absolute stretch values are plotted in blue as a function of speed, with black crosses depicting the standard deviations of each stretch value, and the blue line is a second-order polynomial fit that characterizes the stretch of the rotor as a function of rotor speed. At this point, the user can save the calibration profile by clicking on Save Rotor Calibration, which will bring up a new dialog:



Clicking OK, the user is prompted with the *Rotor Management* program where a new stretch profile can be stored.

It is important to note that the appropriate rotor is pre-selected, based on the rotor information used to store the data originally in the LIMS database when the data was committed to the database. The user can add a name for the calibration as well as a w2t value for the rotor, if known. The date on which the rotor data are

committed is automatically added to each rotor's profile in the LIMS system. The final report can be printed out by clicking on **View Calibration Report**:



Triple========== Common Dialogs of Ultrascan-III ==========

Index

- *Select Pre-Filter*

- *Data Loader*
  - *Data Set Selection*
  - *Load Run Functions:*
- *Load Distribution Model*
  - *Advanced Model List Options*
- *Remove Model Distributions*
  - *Remove Functions:*
- *Select to Delete*
- *Common Plot Controls*
  - *Plot Functions*

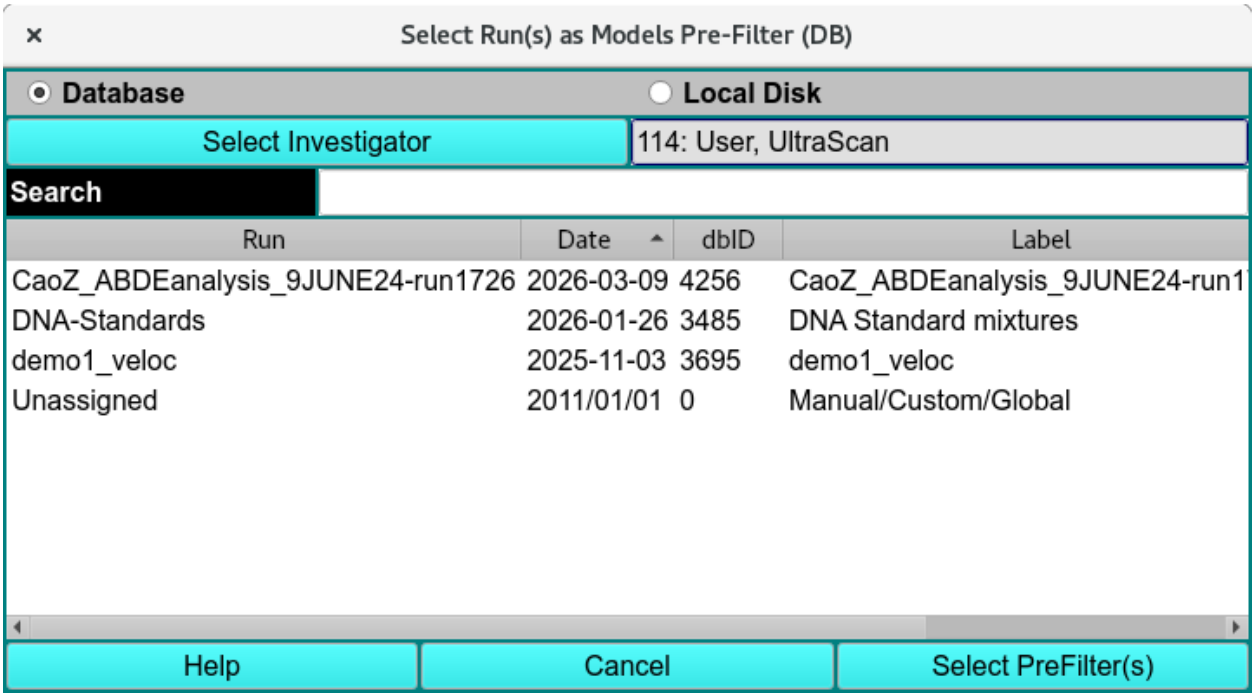
UltraScan-III provides several dialogs for inputting data for analysis or viewing, including options to load run data or load/remove distribution/fitted model data.

CHAPTER

EIGHTYFIVE

SELECT PRE-FILTER

A number of UltraScan-III applications load model distribution data for processing. For large-volume data bases, it helps the speed of model list preparation to do a pre-filter of the model data by run ID. These applications use the US_SelectRuns dialog class to allow the user to choose runs with which to filter model listings. The dialog presented when a US_SelectRuns is executed allows a list of run ID data choices from database or local disk. A Search text field allows the list to be pared down further to those of interest. After one or more data sets are selected in the list, a button labelled **Select PreFilter(s)** passes experiment data to the caller which passes selections to the *Load Model Dialog*.



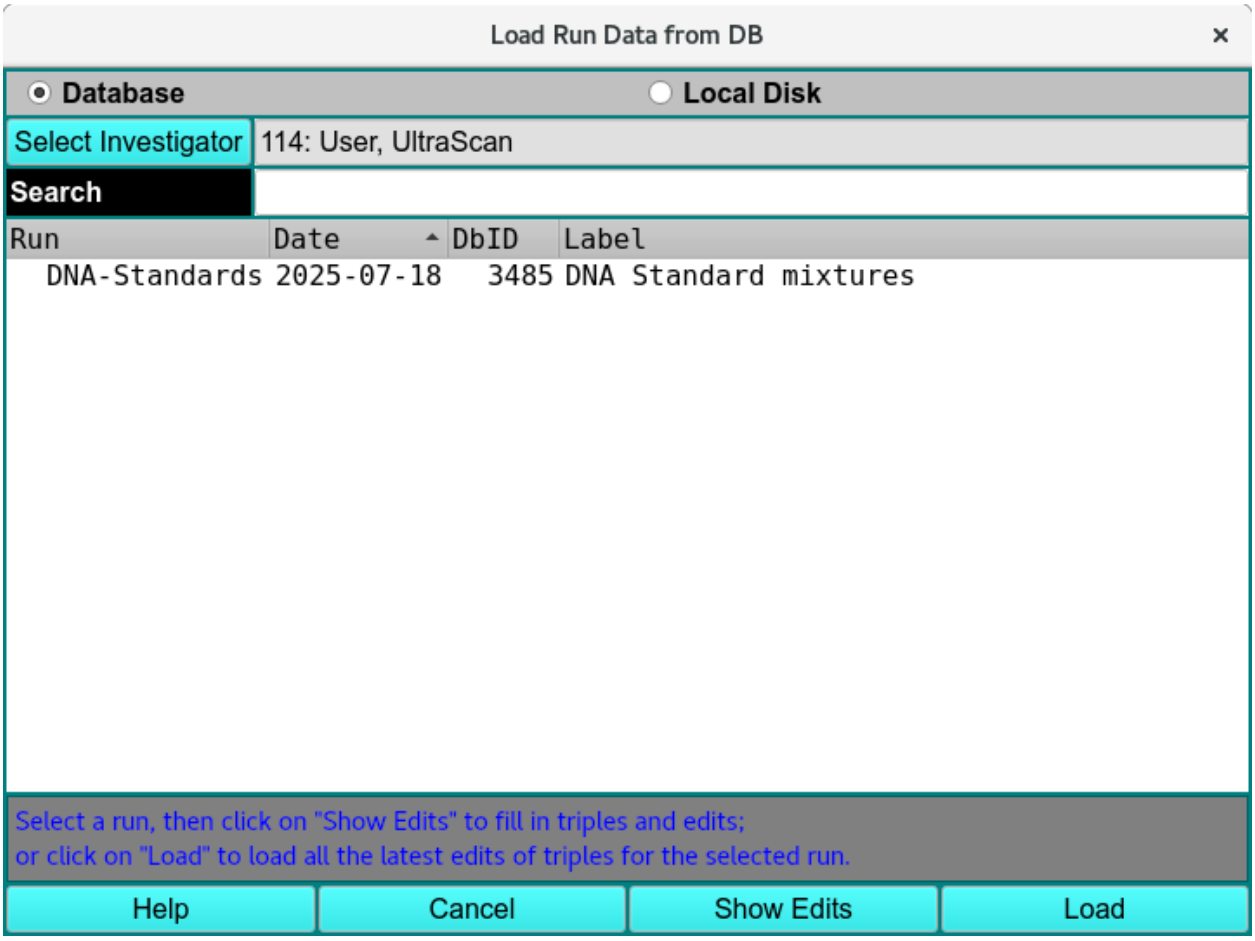
Load Run Data from DB

CHAPTER

EIGHTYSIX

DATA LOADER

Ultrascan-III modules can load models by first loading the Run as **Experiment**, **PreFilter** or **Run Data**.

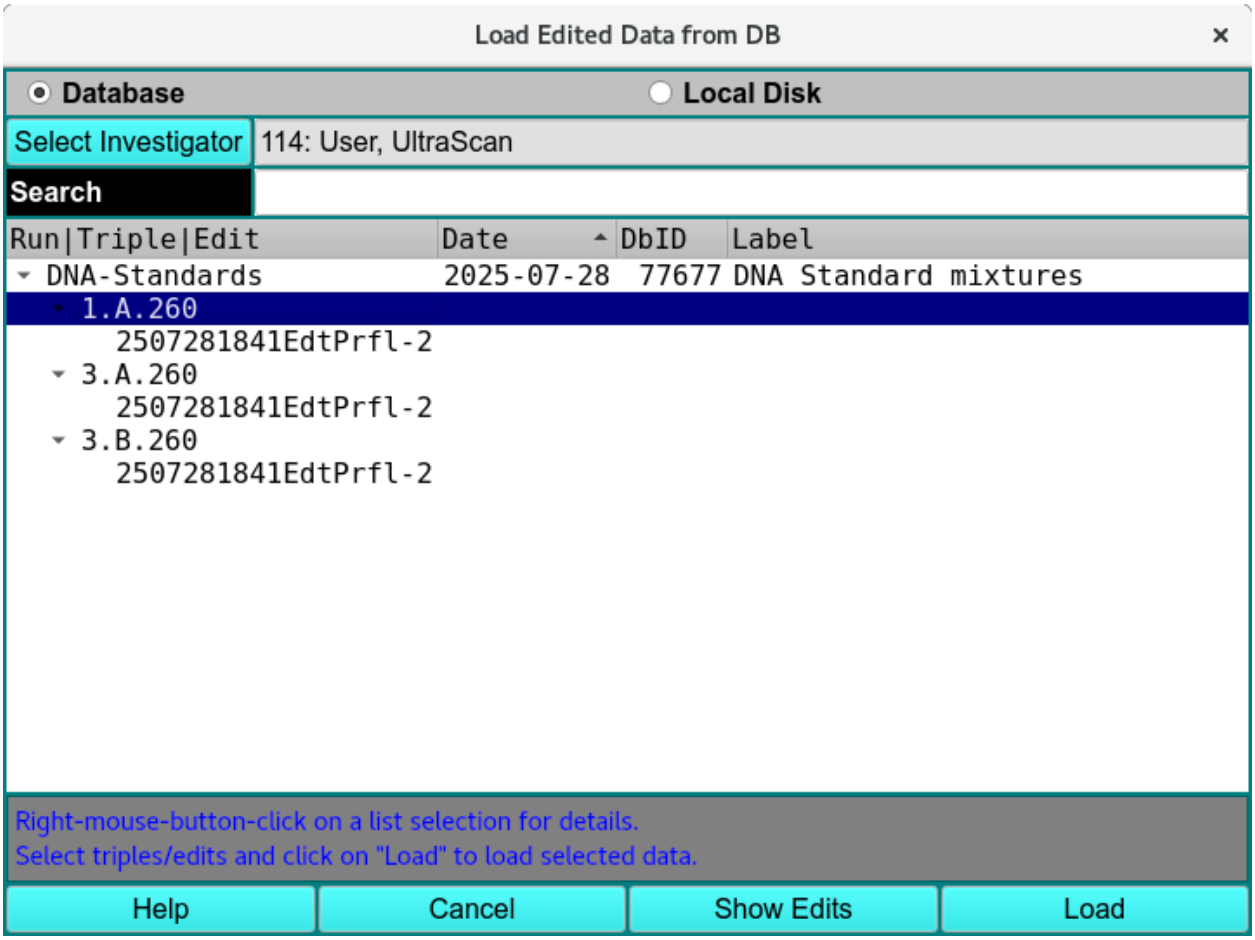


Load Run Data from DB

86.1 Data Set Selection

Most often, you need not expand the list tree and need only select the top level run ID description. This effectively selects all Triples for the run and selects the first listed edit for each Triple. In the default case where Latest Data Edit was checked in the calling dialog, there will be only one edit (the latest) for each Triple. If Latest Data Edit was not checked, there may be multiple edits for Triples. In this case, selecting a Run description or a Triple description effectively selects the earliest edit(s).

You may, of course, select specific Triples of a run. Similarly, in the case where Lastest Data Edit was not checked, you may select any edit of a Triple.



Select Specific Data Sets and Edits

86.2 Load Run Functions:

Database	Check this radio button to generate a list from the database.
Local Disk	Check this radio button to generate an edit list from local disk */ultrascan/results* files.
Select Investigator	Click on this button to open an Investigator Dialog that enables you to browse potential database investigators and select one. This button is disabled for normal non-privileged users and whenever Local Disk is selected.
(investigator text)	This read-only text box displays the current investigator ID, last name, and first name.
Search	As characters are entered in the text box to the right of the Search label, the list of data sets is modified to consist of matching entries. A blank field restores the full list. Note that the search is case-insensitive.
Run	The Run ID given in the Import stage.
date	The date the data was imported to Ultrascan or generated using connected instruments.
Database identification (DbID)	The Database ID number.
Run	The Run Label given in the remove_mod_dis: Import stage.
Show Edits	Click to transform the data list from showing only runs to showing a data tree with Triple and edit children. If no Run List entry is selected, the entire data tree will be transformed. If a Run is selected, only that run with its Triple and edit descendants will be shown.
Load	Click to accept the choice(s) highlighted in the data list and pass data back to the caller.

Window Controls

Help	Click to bring up this documentation.
Cancel	Click to close the dialog with no model list modifications made.

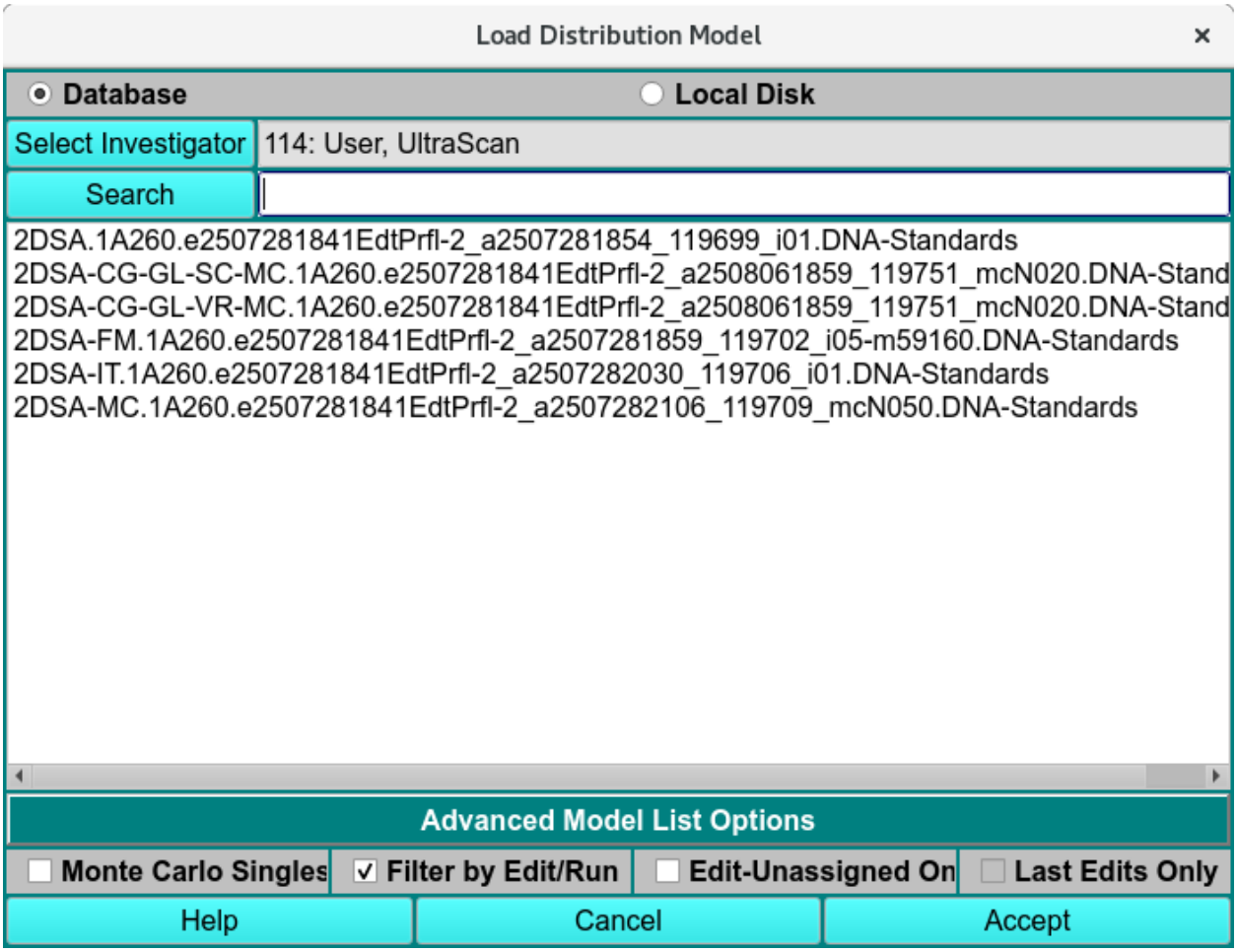
Note: Note that multiple selections must all be from the same run. Only one selection is allowed for each Triple.

CHAPTER

EIGHTYSEVEN

LOAD DISTRIBUTION MODEL

After the Run has been loaded the triple analysis model is loaded using *Load Distribution Model*.



Load Distribution Model

87.1 Advanced Model List Options

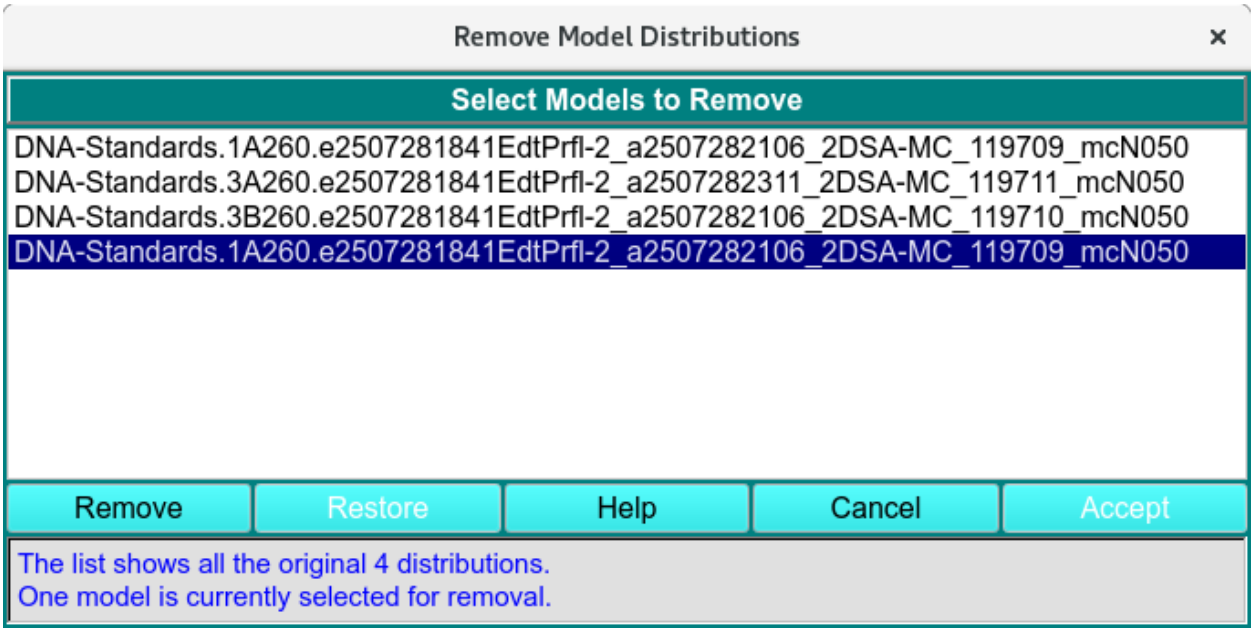
Monte Carlo Singles	Checking the Monte Carlo Singles box expands the list to include all the Monte Carlo singles of any triple with Monte Carlo analysis done.
Filter by Edit/Run	Filters the list by the experimental run initially selected and the latest Edit of the experiment selected.
Edit-Unassigned one	
Last Edit only	

CHAPTER

EIGHTYEIGHT

REMOVE MODEL DISTRIBUTIONS

The dialog is presented when **Remove Distribution(s)** is clicked. The dialog shows a list of loaded models, by selecting items in this list and clicking on a Remove button, you can remove models from the original distribution list. After all the desired removals have been performed, an Accept button passes the reduced model data to the caller.



Remove Distribution Model

88.1 Remove Functions:

(list)	The list box is initially populated with a passed list of all loaded models. Individual or multiple item lines can be selected for removal.
Remove	After making selections, click this button to remove selections from the list.
Restore	Click on this button to restore the list to its original full-list state.
Help	Click to bring up this documentation.
Cancel	Click to close the dialog with no model list modifications made.
Accept	Click to accept the modified list as it appears and pass it back to the caller.
(status)	The ongoing status of the list and current selections is documented here.

CHAPTER

EIGHTYNINE

SELECT TO DELETE

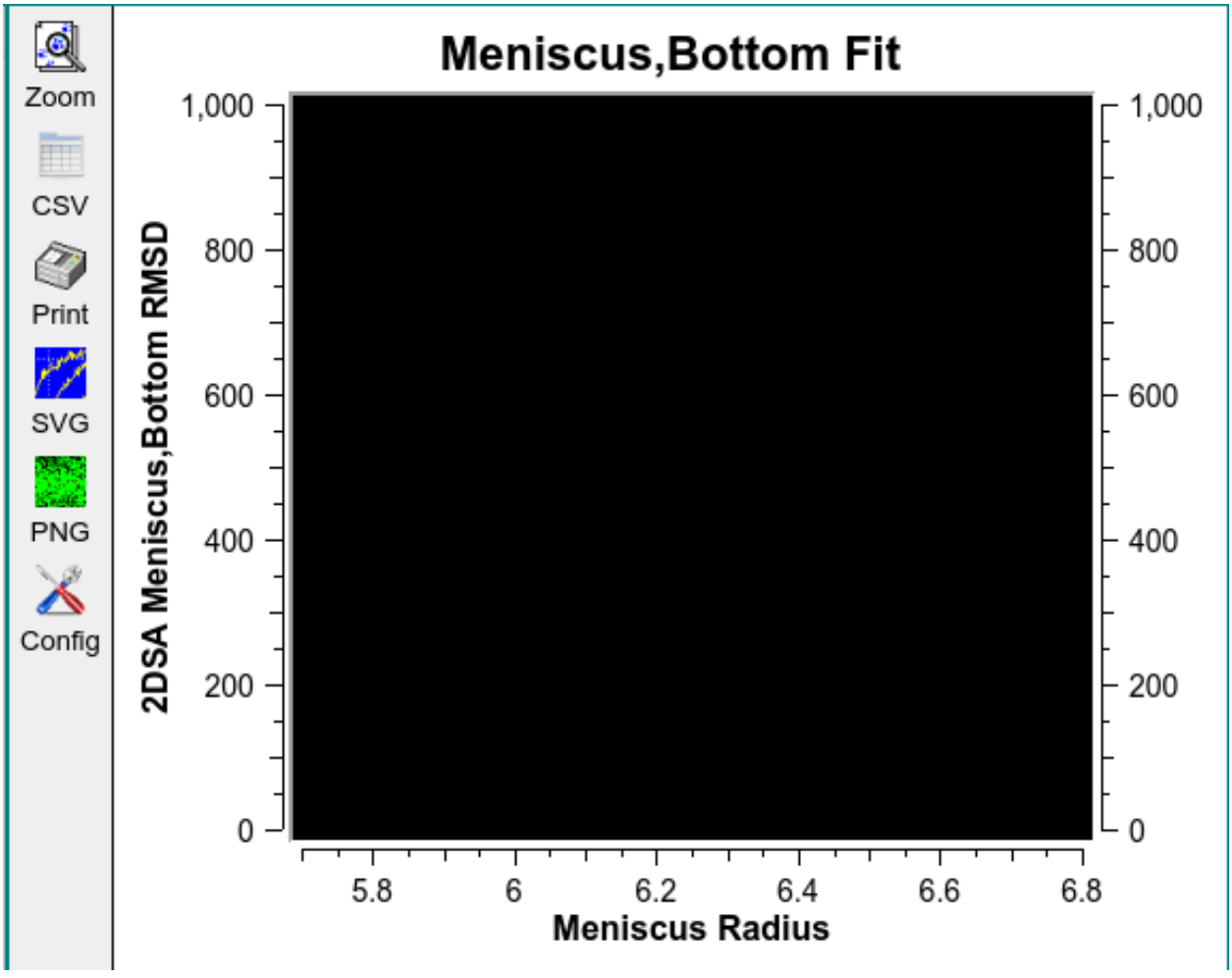
Using this dialog window, you can select triples and wavelengths to be deleted from the selection pool. For Triplies, the final state of included and excluded triples determines what data will be output. The dialog consists of two list windows. The one on the left initially contains the included dataset triples at invocation. The right-side list is initially empty. Delete selection consists of clicking on one or more left-side-list triples and moving them to the **Excluded** list on the right, via the **Add ===>** button. You may remove triples from the excluded list by clicking on their entries in the "Excluded" list and moving them back to the "Included" list via the button. Once the lists of included and excluded triples is as desired, the Accept button closes the dialog and communicates the selected triple deletions to the caller.

CHAPTER

NINETY

COMMON PLOT CONTROLS

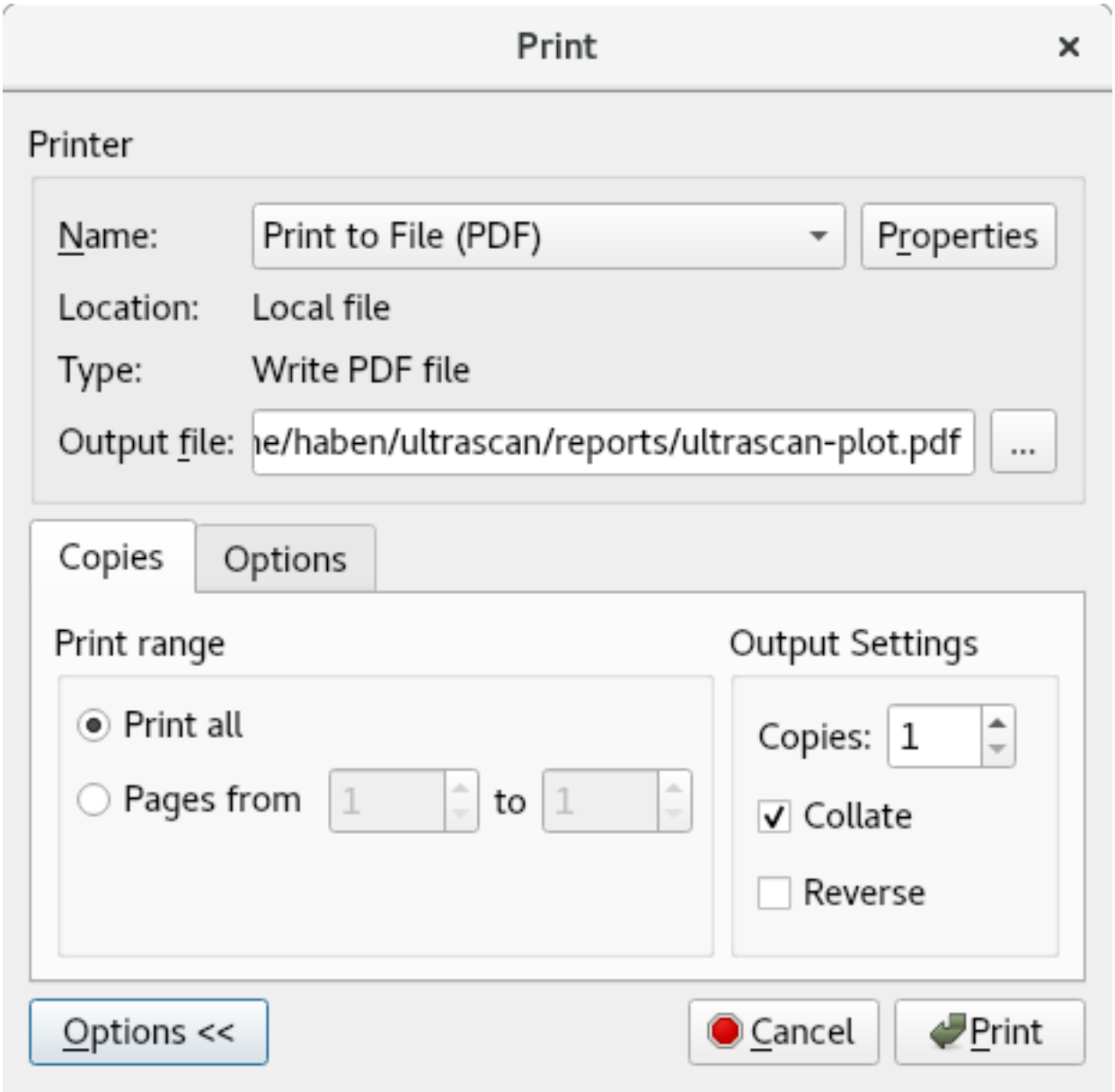
Ultrascan-III uses common functions to modify and save displayed plots.



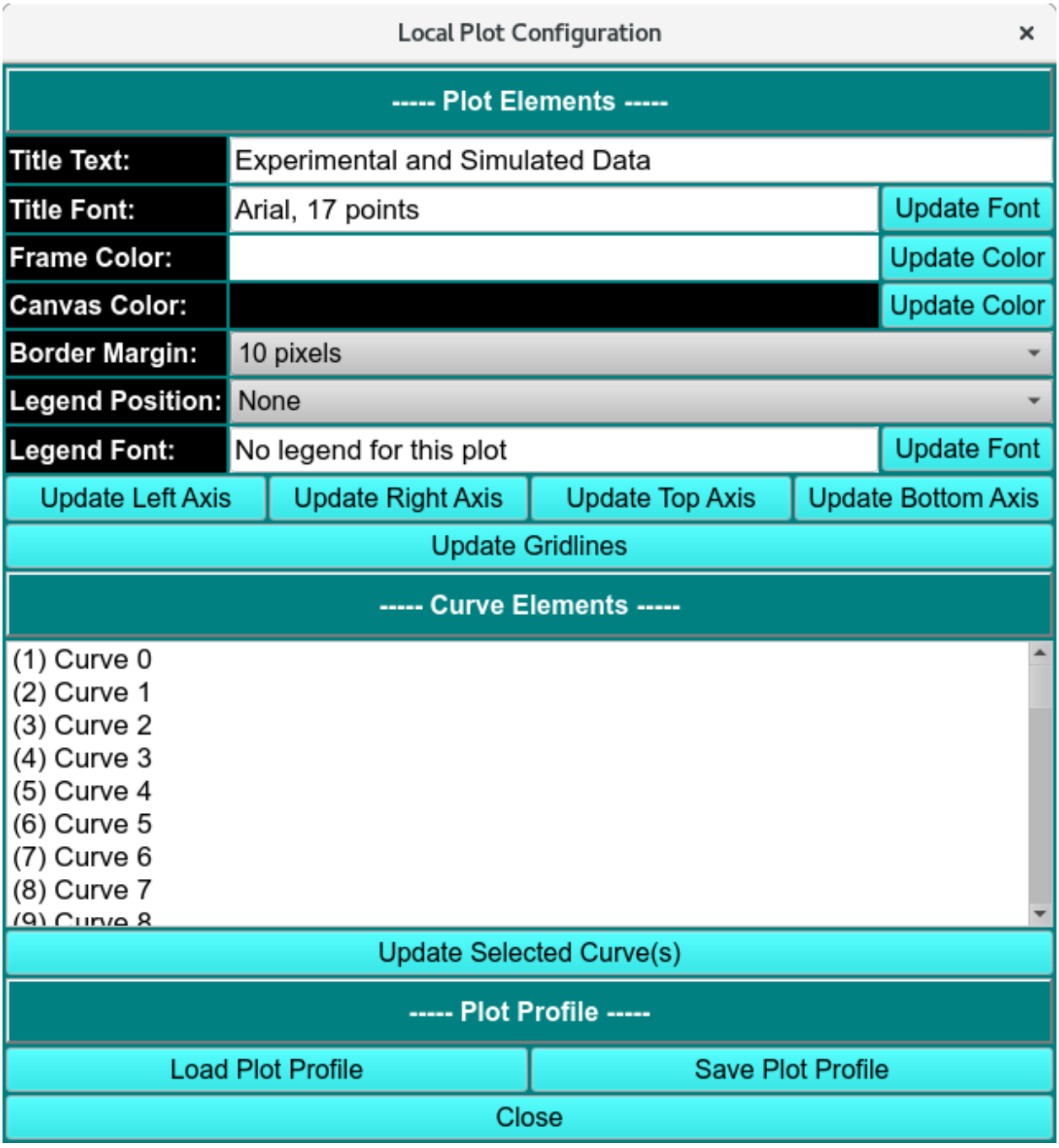
Plot Controls

90.1 Plot Functions

Zoom	Click Zoom and select the area to zoom in by holding left mouse key and dragging across the area of interest to zoom. Right to click to reset to previous zoom until the original plot is displayed.
CSV	Export the plot displayed in a CSV format in *ultrascan/report*.
Print	Print the *displayed plot* as a PDF in *ultrascan/report*.
SVG	Export the displayed plot as an SVG image format.
PNG	Export the displayed plot as an PNG image format.
Config	The :*configuration panel* <*config_plot*> of the local plot. The panel is used to change the plot profile.



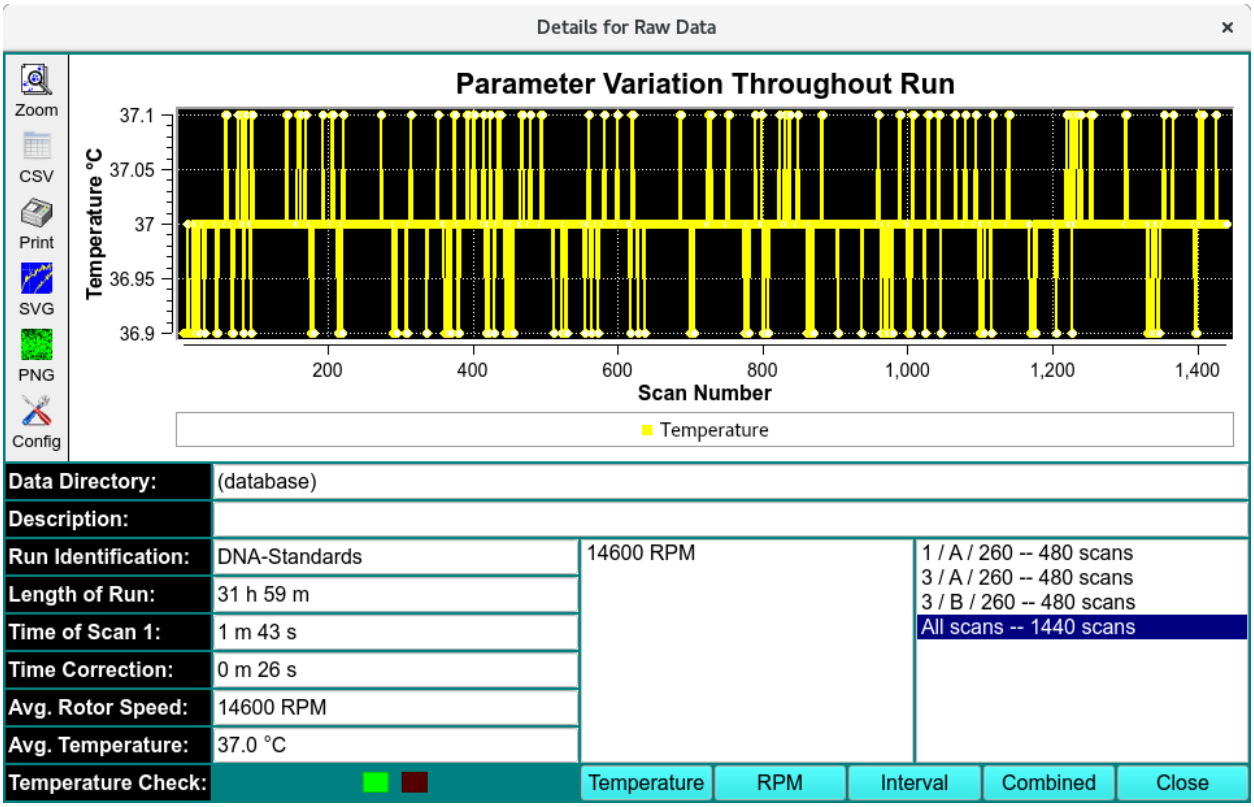
Print Controls



Local Plot Configuration Controls

tripletriple========== Run Detail of Raw Data ==========

As raw data is collected, the temperature (°C), rotor speed (RPM), and time interval between scans (min) are recorded. The Run Details module enables users to view variations in temperature, rotor speed, and scan intervals-both for individual triple scans and across the entire run.



Details for Raw Data

- Temperature
- RPM
- Interval
- Combined

Run Details

Parameter Variation Throughout Run	Plot window to view the variations in temperature, speed, scan time interval and all combined.
Data Directory:	Name of data directory if local.
Description:	Run label or selected triple identification.
Run Identification:	Run label
Length of Run:	Total scan time of run.
Time of Scan 1:	Time of scan 1 after protocol specified rotor speed and temperature are reached.
Time Correction:	Scan time corrections
Avg. Rotor Speed:	average rotor speed across the total run time
Avg. Temperature:	average temperate across the total run time
Speed text box	average rotor speed
triple and number of scans - used for Scan Number in plot	

CHAPTER

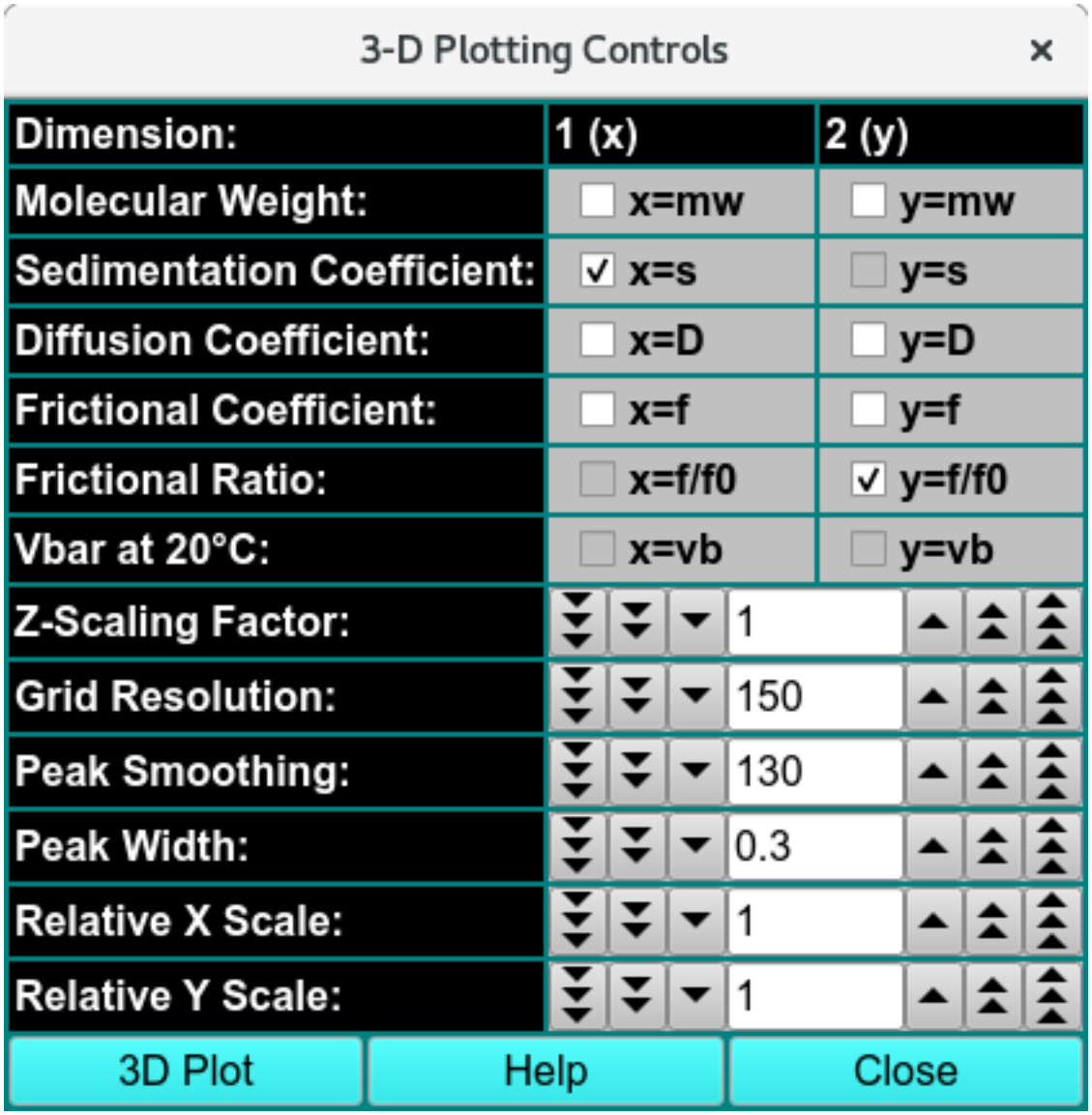
NINETYONE

3-DIMENSIONAL PLOT CONTROLS

Index

- *3D Plot Control Functions:*
- *Model Solute 3-Dimensional Viewer*
  - ***Keys to control view:***

Upon completion of FE Match simulation, a pair of enhanced plot controls dialogs appears. The **3D Plot** button in the first of these launches a 3-dimensional plot window whose primary control resides in this dialog. There are size, orientation, and other controls that are in the 3D dialog itself. A second dialog displays a set of data and residuals plots that are more refined and controllable than those in the main FE Match window.



3-D Plotting Controls

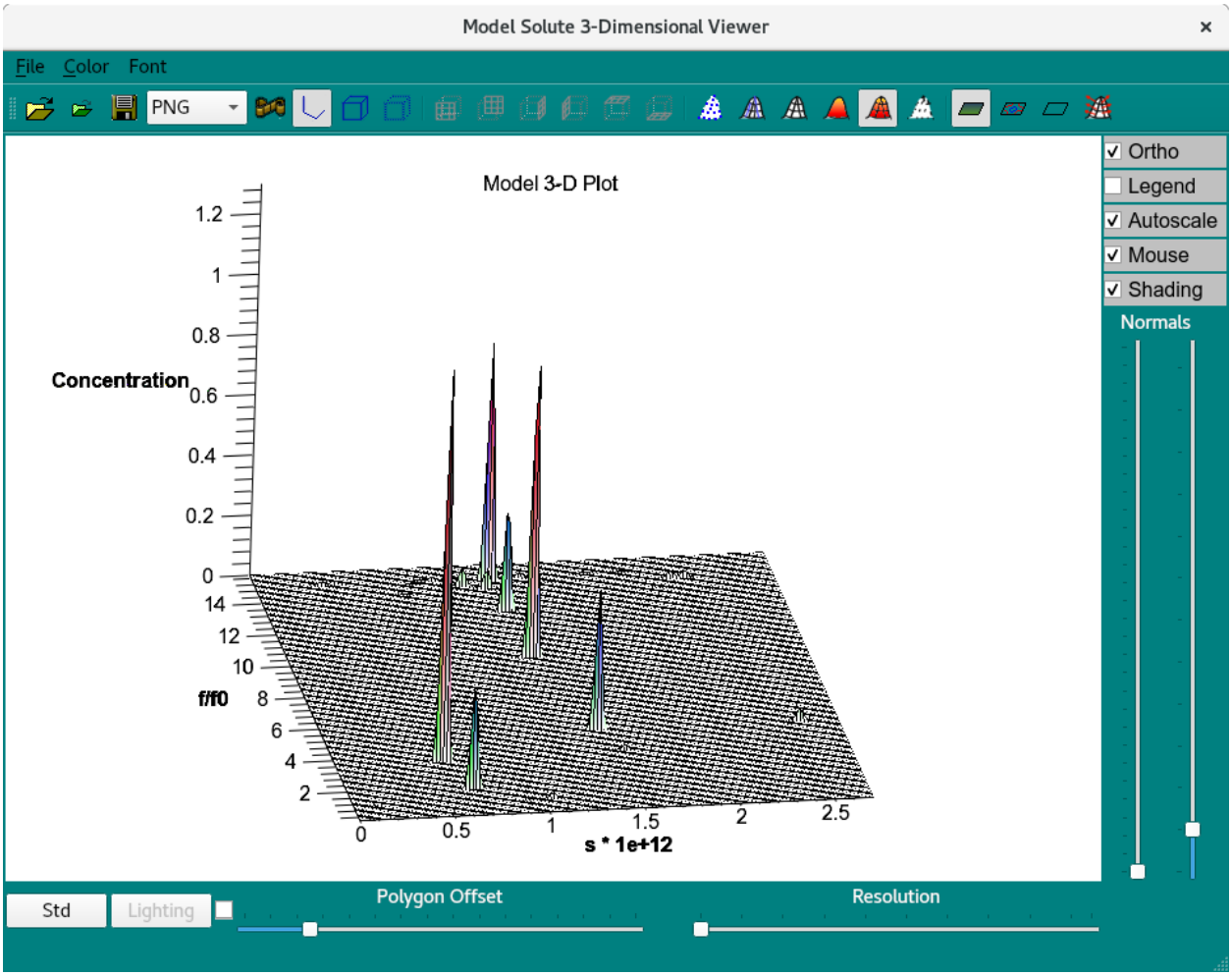
91.1 3D Plot Control Functions:

Dimensions:	The five pairs of check boxes allow the user a way to set the 3D plot X and Y dimension value types: MW; $s_{20,w}$; $D_{20,w}$; f; and f/f_0.
Z-Scaling Factor:	Set the scale factor for 3D Z values, to increase/decrease peak extents relative to X and Y.
Grid Resolution:	Set the number of pixels in X and Y for which Z raster values are calculated.
Peak Smoothing:	Set the smoothing factor applied to peaks.
Peak Width:	Set the factor that governs the spread of peaks.
3D Plot	Click this button initially to launch the 3D-plot window. Thereafter, when dimension value or other settings are changed, click this button for a re-draw of the 3D plot.

Window Controls

Help	Display this detailed help.
Close	Close all windows and exit.

91.2 Model Solute 3-Dimensional Viewer



Model Solute 3-Dimensional Viewer

91.2.1 Keys to control view:

- **Shft + Mouse scroll wheel** - expand the z dimension (Concentration in the 3-D Viewer)
- **Alt + hold right arrow key** - expand X axis (sedimentation - s*1e+12)
- **Alt + hold left arrow key** - contract X axis
- **Alt + hold up arrow key** - expand Y axis (Fricitional ratio - f/f0)
- **Alt + hold down arrow key** - expand Y axis
- **Mouse scroll wheel** - zoom in on all Dimensions
- **Hold left mouse key and move up/down** - rotate X axis
- **Hold left mouse key and move left/right** - rotate along Y axis

CHAPTER

NINETYTWO

UNKNOWN PAGE

Referenced in /ultrascan3/programs/us_analysis_profile/us_analysis_profile.h:221:

CHAPTER

NINETYTHREE

GLOBAL EQUILIBRIUM ANALYSIS - UNLINKED

Index

- *Model Parameters*
- *Model Controls*
- *Fit Controls*
- *Glob. Eq. Functions*

93.1 Model Parameters

93.2 Model Controls

93.3 Fit Controls

93.4 Glob. Eq. Functions

Error: /ultrascan3/programs/us_globalequil/us_model_adpars.h:46: Failure Error: Context: { showHelp.show_help(“global_equil-modapar.html”); };

Error: /ultrascan3/programs/us_globalequil/us_eqmodel_control.h:210: Failure Error: Context: void help(void) { showHelp.show_help(“global_equil-modctrl.html”); };

Error: /ultrascan3/programs/us_globalequil/us_eqfit_control.h:186: Failure Error: Context: { showHelp.show_help(“global_equil-fitctrl.html”); };

Error: /ultrascan3/programs/us_globalequil/us_globalequil.h:268: Failure Error: Context: showHelp.show_help(“global_equil.html”);

Error: /ultrascan3/programs/us_globalequil/us_model_select.h:70: Failure Error: Context: { showHelp.show_help(“global_equil-modsel.html”); };

CHAPTER

NINETYFOUR

IMPORT RAMP DATA - TO REMOVE PROGRAM

CHAPTER

NINETYFIVE

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